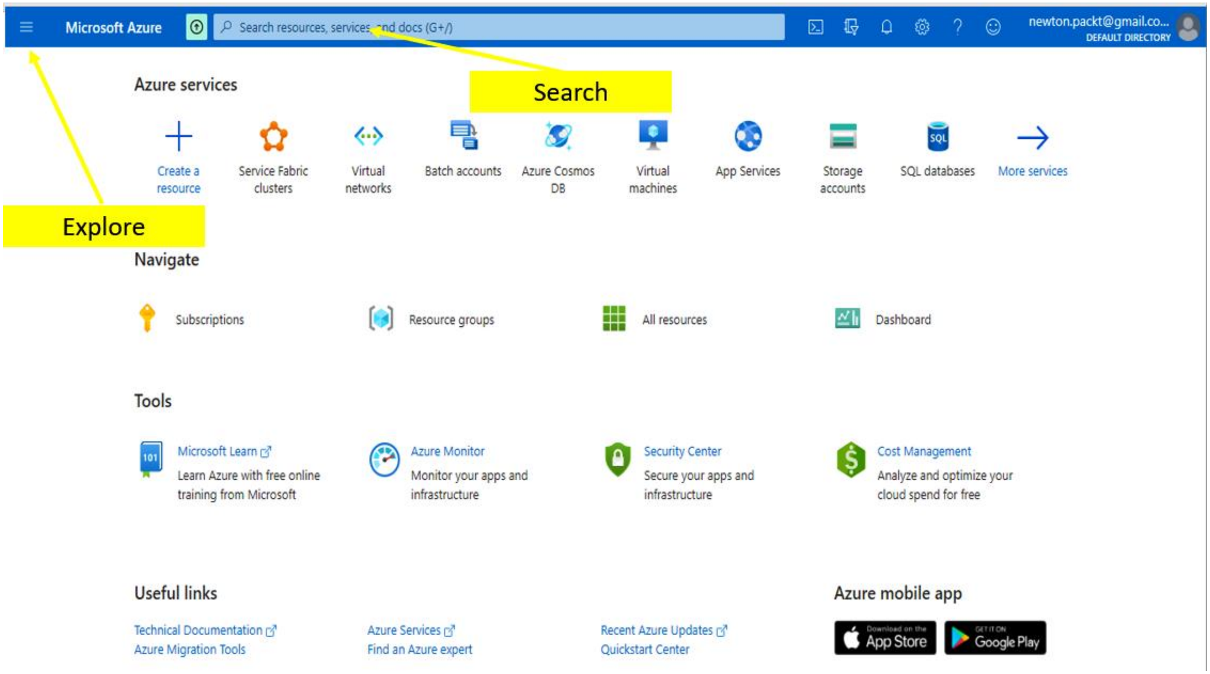
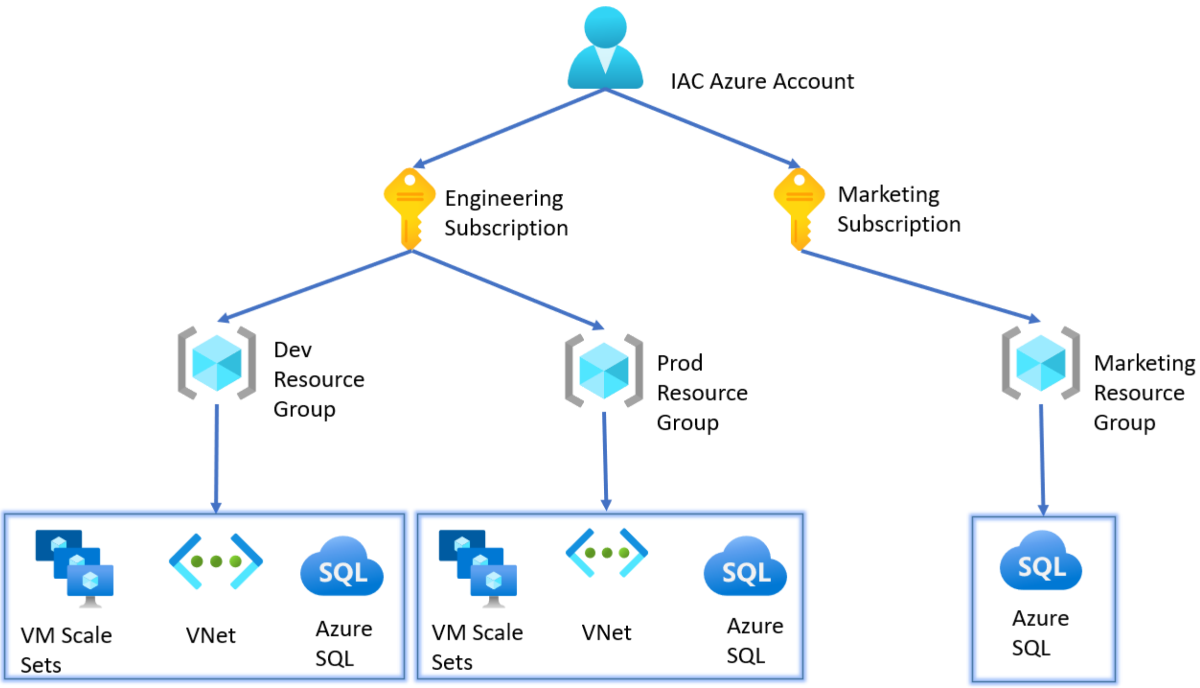
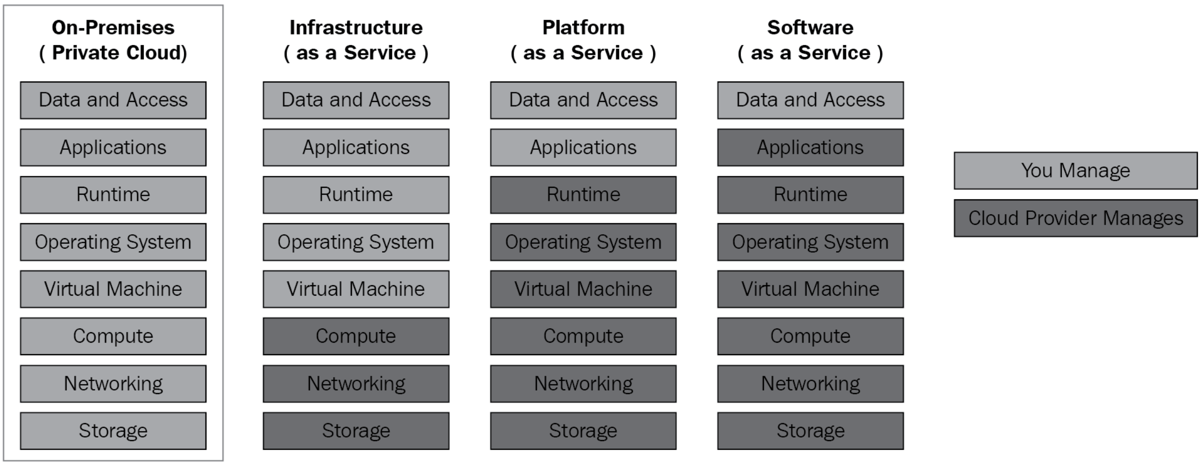
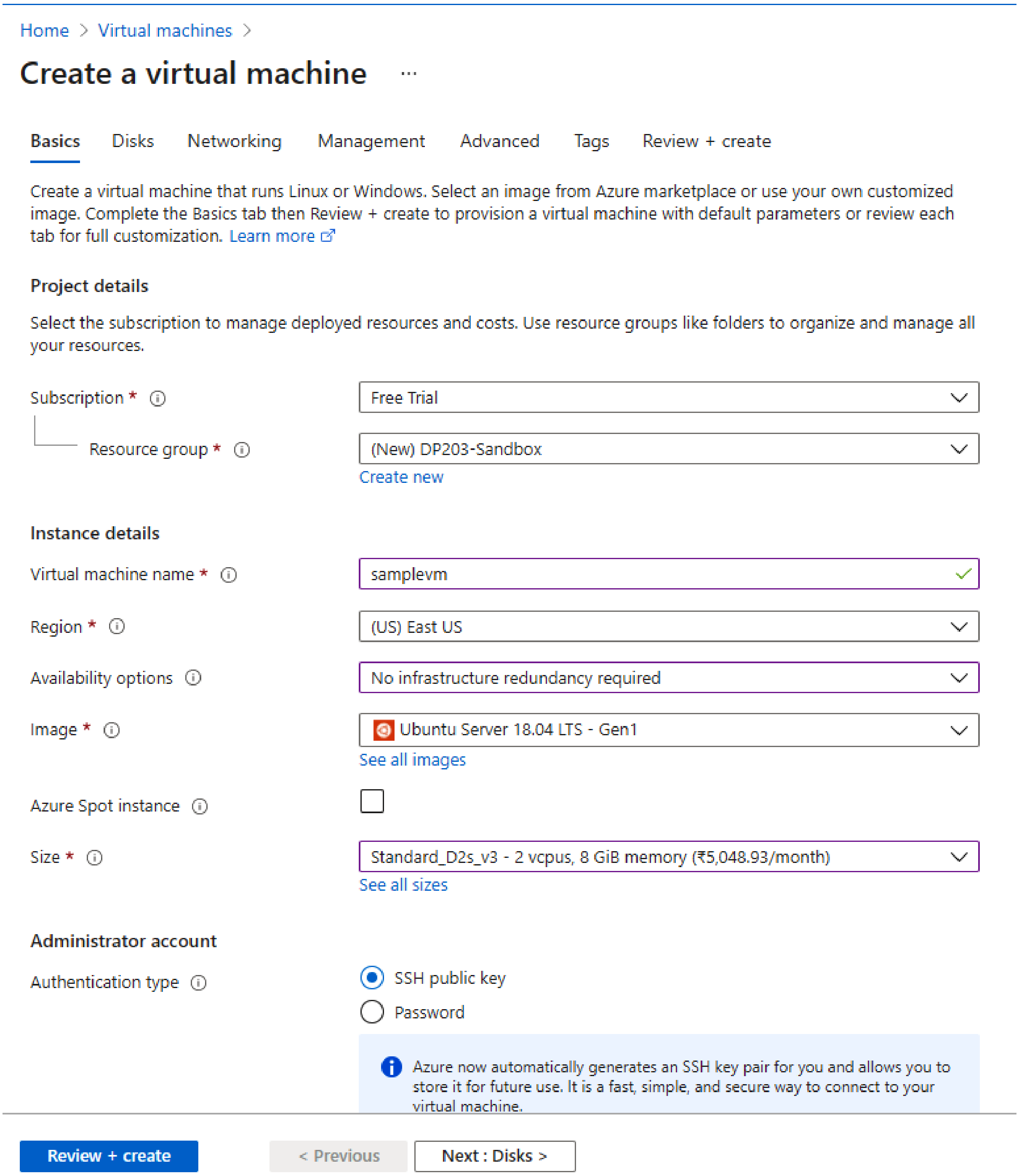
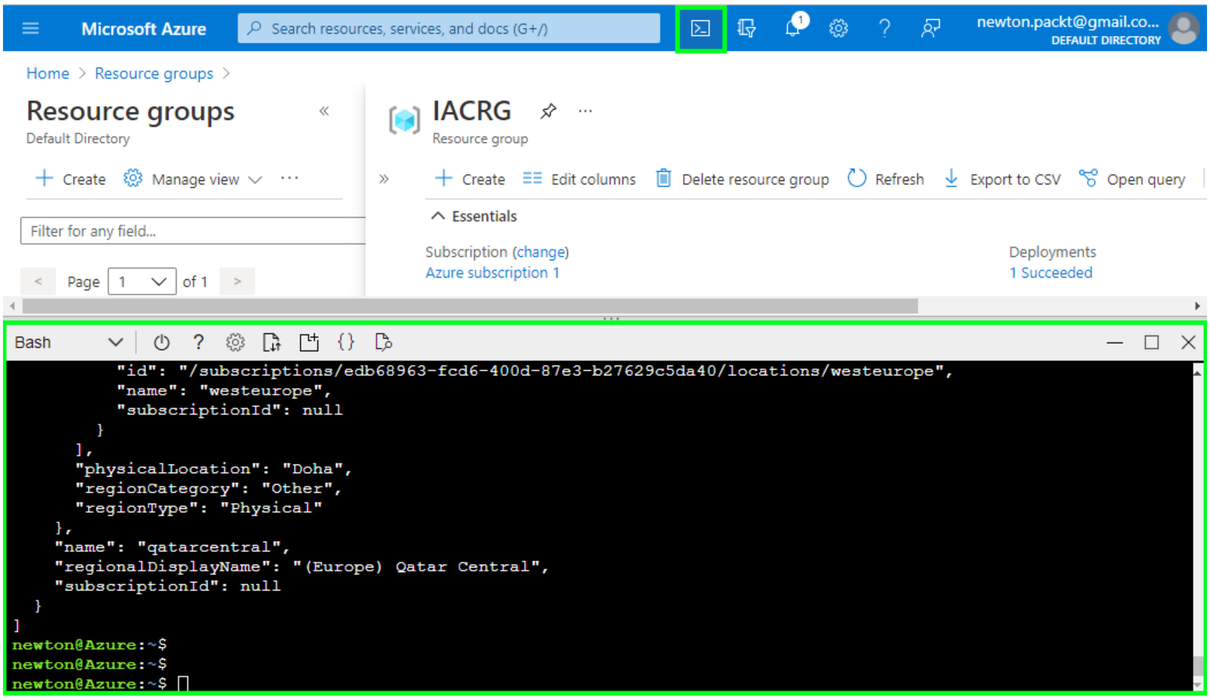
Chapter 1: Introducing Azure Basics

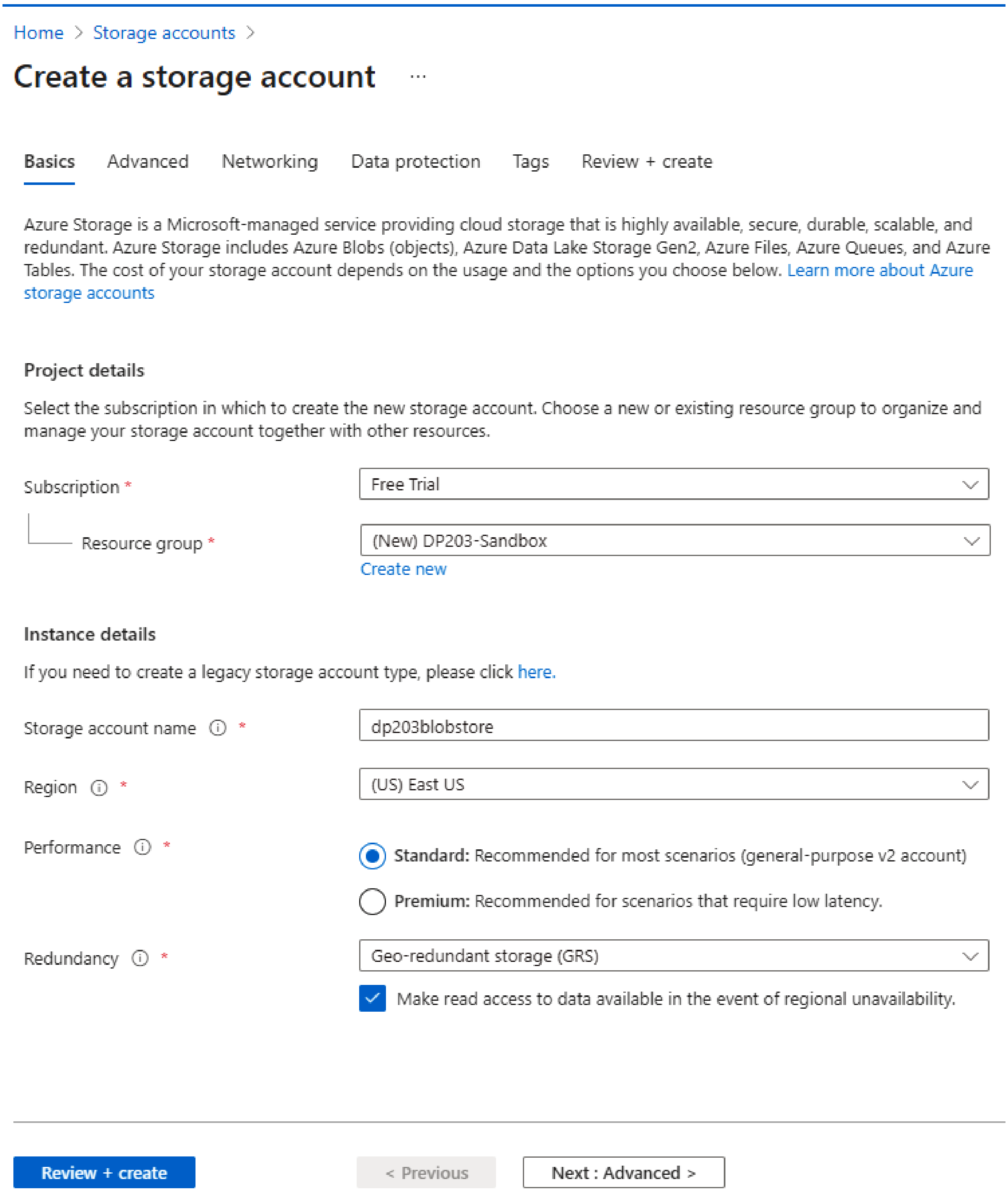


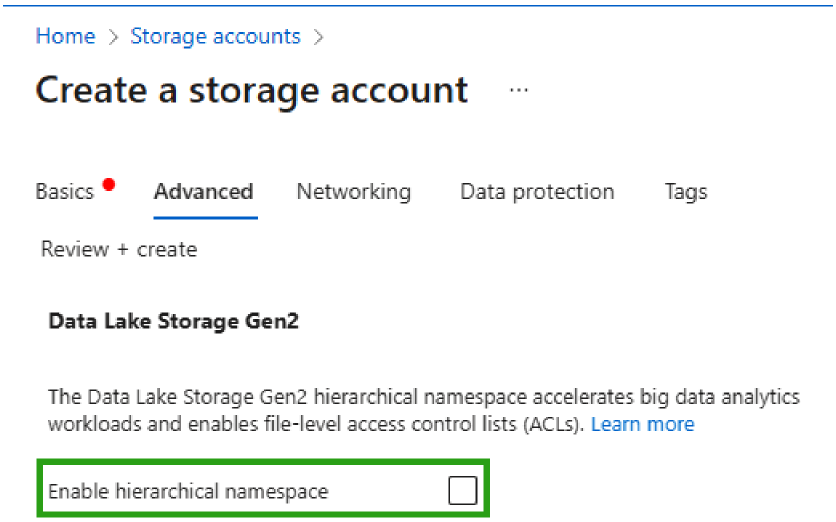
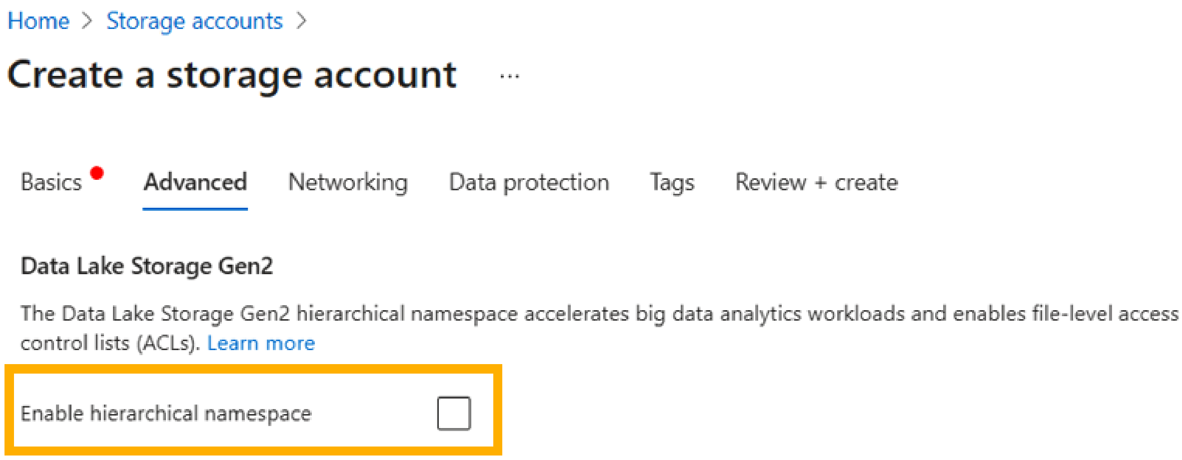



On-Premises
(Private Cloud)
Data and Access
Applications
Runtime
Operating System
Virtual Machine
Compute
Networking
Storage
Infrastructure
(as a Service)
Data and Access
Applications
Runtime
Operating System
Virtual Machine
Compute
Networking
Storage
Platform
(as a Service)
Data and Access
Applications
Runtime
Operating System
Virtual Machine
Compute
Networking
Storage
Software
(as a Service)
Data and Access
Applications
Runtime
Operating System
Virtual Machine
Compute
Networking
Storage
You Manage
Cloud Provider Manages

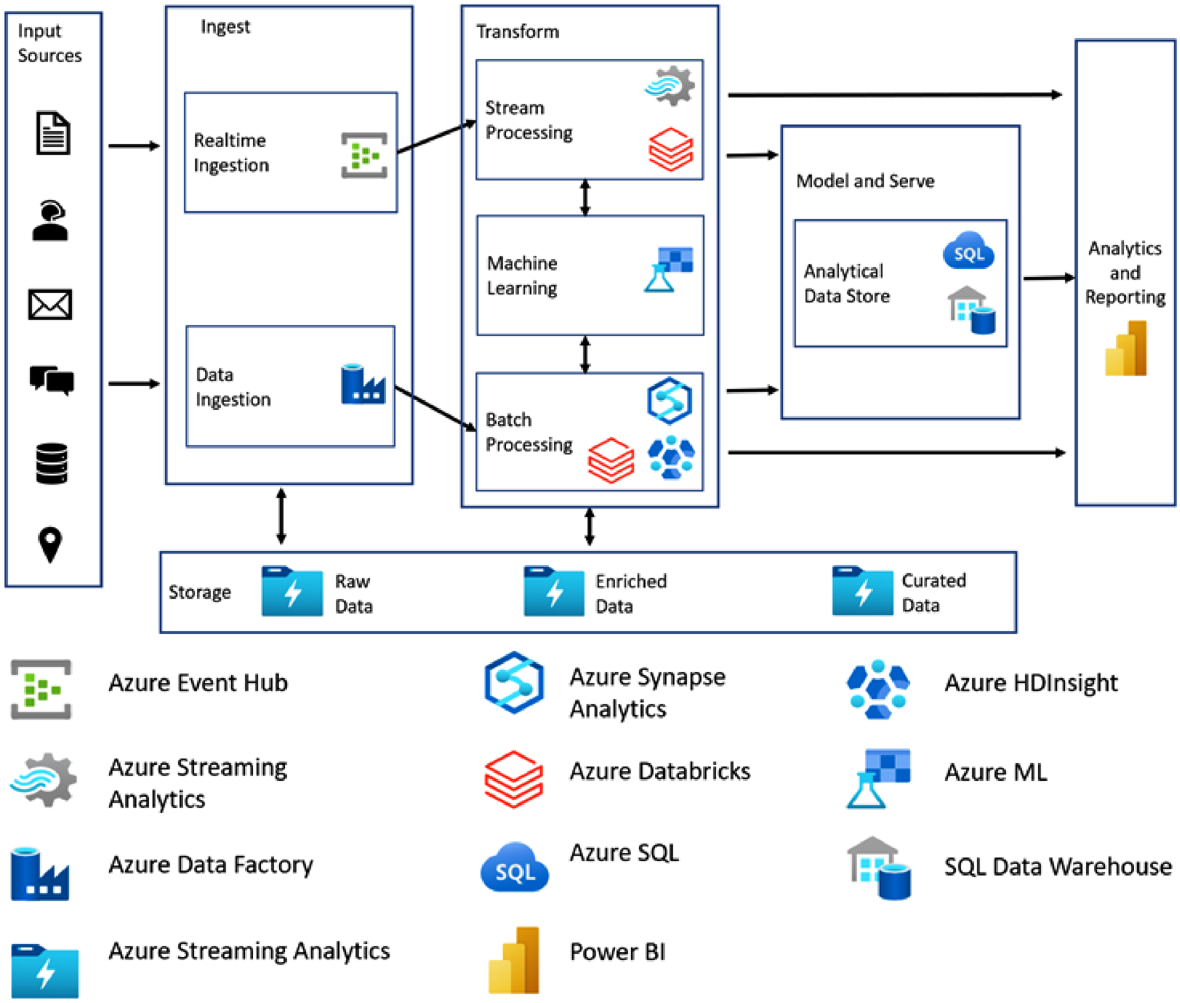

Home > Virtual machines >
Create a virtual machine ...
Basics
Disks
Networking
Management
Advanced
Tags
Review + create
Create a virtual machine that runs Linux or Windows. Select an image from Azure marketplace or use your own customized image. Complete the Basics tab then Review + create to provision a virtual machine with default parameters or review each tab for full customization. Learn more
Project details
Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.
Subscription *
Free Trial
Resource group *
(New) DP203-Sandbox
Create new
Instance details
Virtual machine name *
samplevm
Region *
(US) East US
Availability options
No infrastructure redundancy required
Image *
Ubuntu Server 18.04 LTS - Gen1
See all images
Azure Spot instance
Size *
Standard_D2s_v3 - 2 vcpus, 8 GiB memory (₹5,048.93/month)
See all sizes
Administrator account
Authentication type
SSH public key
Password
Azure now automatically generates an SSH key pair for you and allows you to store it for future use. It is a fast, simple, and secure way to connect to your virtual machine.
Review + create
< Previous
Next : Disks >


Microsoft Azure
Search resources, services, and docs (G+/)
newton.packt@gmail.co...
DEFAULT DIRECTORY
Home > Resource groups >
Resource groups
Default Directory
Create
Manage view
Filter for any field...
Page 1 of 1
IACRG
Resource group
Create
Edit columns
Delete resource group
Refresh
Export to CSV
Open query
Essentials
Subscription (change)
Azure subscription 1
Deployments
1 Succeeded
Bash
"id": "/subscriptions/edb68963-fcd6-400d-87e3-b27629c5da40/locations/westeurope",
"name": "westeurope",
"subscriptionId": null
}
],
"physicalLocation": "Doha",
"regionCategory": "Other",
"regionType": "Physical"
},
"name": "qatarcentral",
"regionalDisplayName": "(Europe) Qatar Central",
"subscriptionId": null
}
]
newton@Azure:~$
newton@Azure:~$
newton@Azure:~$


Home > Storage accounts >
Create a storage account ...
Basics
Advanced
Networking
Data protection
Tags
Review + create
Azure Storage is a Microsoft-managed service providing cloud storage that is highly available, secure, durable, scalable, and redundant. Azure Storage includes Azure Blobs (objects), Azure Data Lake Storage Gen2, Azure Files, Azure Queues, and Azure Tables. The cost of your storage account depends on the usage and the options you choose below. Learn more about Azure storage accounts
Project details
Select the subscription in which to create the new storage account. Choose a new or existing resource group to organize and manage your storage account together with other resources.
Subscription *
Free Trial
Resource group *
(New) DP203-Sandbox
Create new
Instance details
If you need to create a legacy storage account type, please click here.
Storage account name *
dp203blobstore
Region *
(US) East US
Performance *
Standard: Recommended for most scenarios (general-purpose v2 account)
Premium: Recommended for scenarios that require low latency.
Redundancy *
Geo-redundant storage (GRS)
Make read access to data available in the event of regional unavailability.
Review + create
< Previous
Next : Advanced >

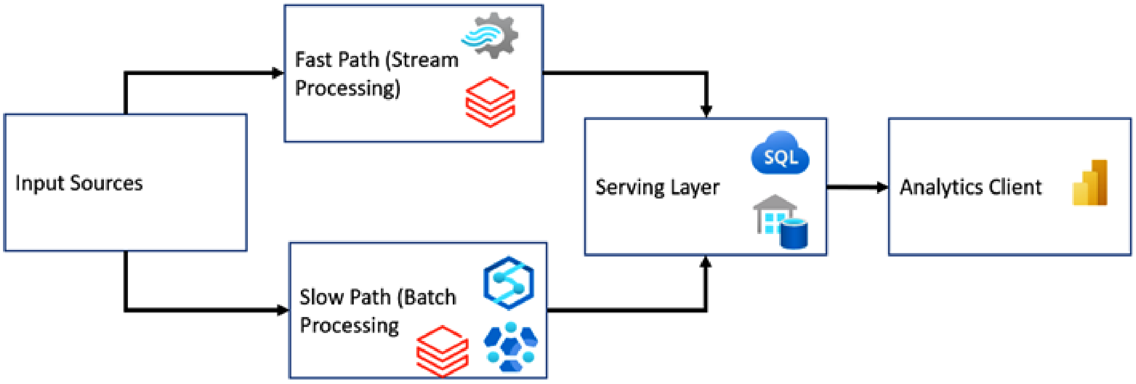

Home > Storage accounts >
Create a storage account ...
Basics
Advanced
Networking
Data protection
Tags
Review + create
Data Lake Storage Gen2
The Data Lake Storage Gen2 hierarchical namespace accelerates big data analytics workloads and enables file-level access control lists (ACLs). Learn more
Enable hierarchical namespace

Home > Storage accounts >
Create a storage account ...
Basics
Advanced
Networking
Data protection
Tags
Review + create
Data Lake Storage Gen2
The Data Lake Storage Gen2 hierarchical namespace accelerates big data analytics workloads and enables file-level access control lists (ACLs). Learn more
Enable hierarchical namespace

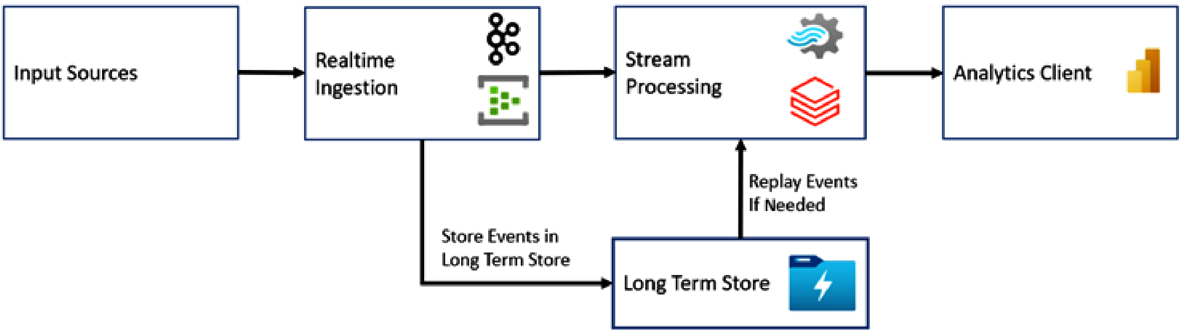
Chapter 2: Designing a Data Storage Structure



Storage Technologies	Azure Data Lake Gen2 Azure Blob Storage Azure CosmosDB Azure SQL Database
Data Transformation Technologies	Spark (via Azure Synapse, Azure HDInsight or Azure Databricks) Apache Hive (via Azure HDInsight) Apache Pig (via Azure HDInsight)
Analytical Datastore	Synapse SQL Warehouse (via Azure HDInsight) Apache HBase (via Azure HDInsight) Apache Hive (via Azure HDInsight)

Ingestion Tools	Azure Event Hub Azure IoT Hub Apache Kafka (via Azure HDInsight)
Stream Processing Tools	Azure Stream Analytics Spark Streaming (via Azure HDInsight or Azure Databricks) Apache Storm (via Azure HDInsight)
Analytical Datastore	Synapse SQL Warehouse Apache HBase (via Azure HDInsight) Apache Hive (via Azure HDInsight)



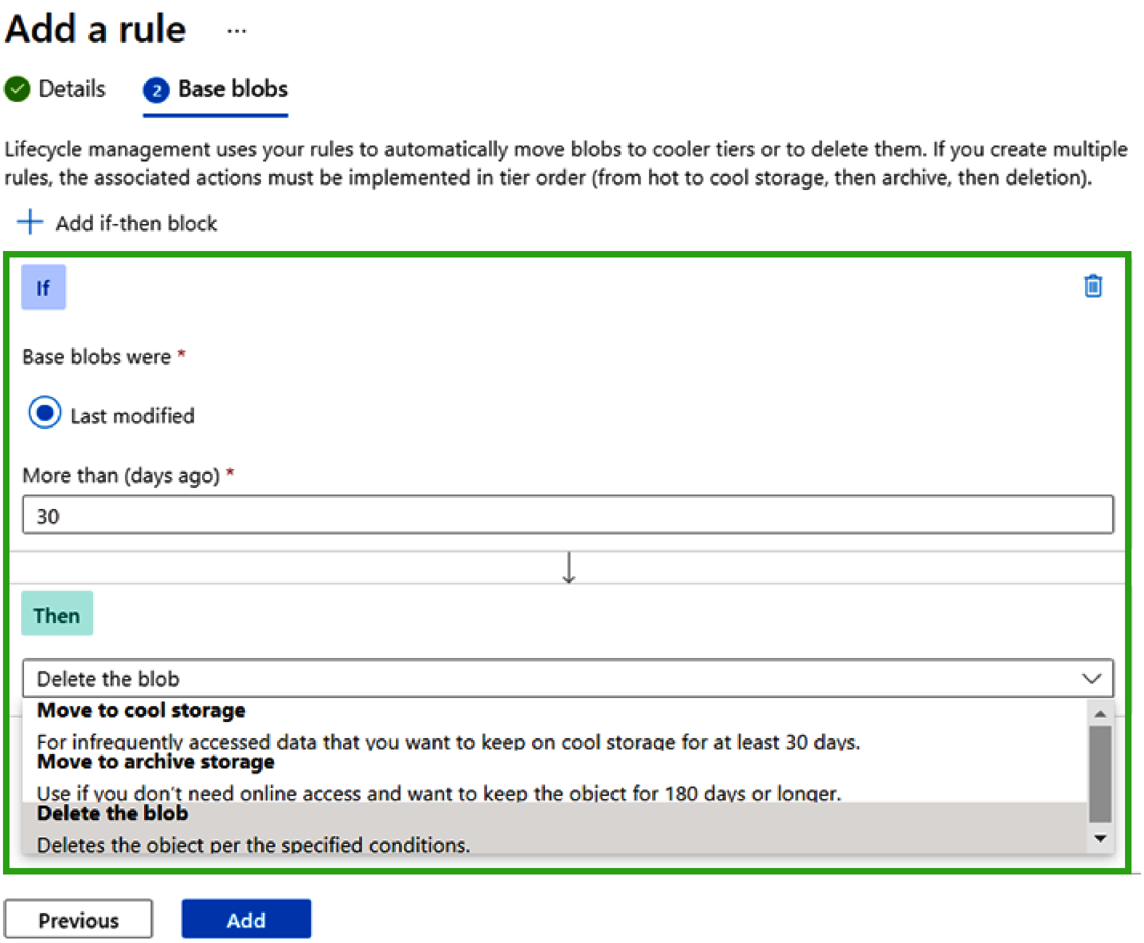


Driver ID	Name	License Number
111	Annie	A1234
222	Brian	B5678
333	Charlie	C3456

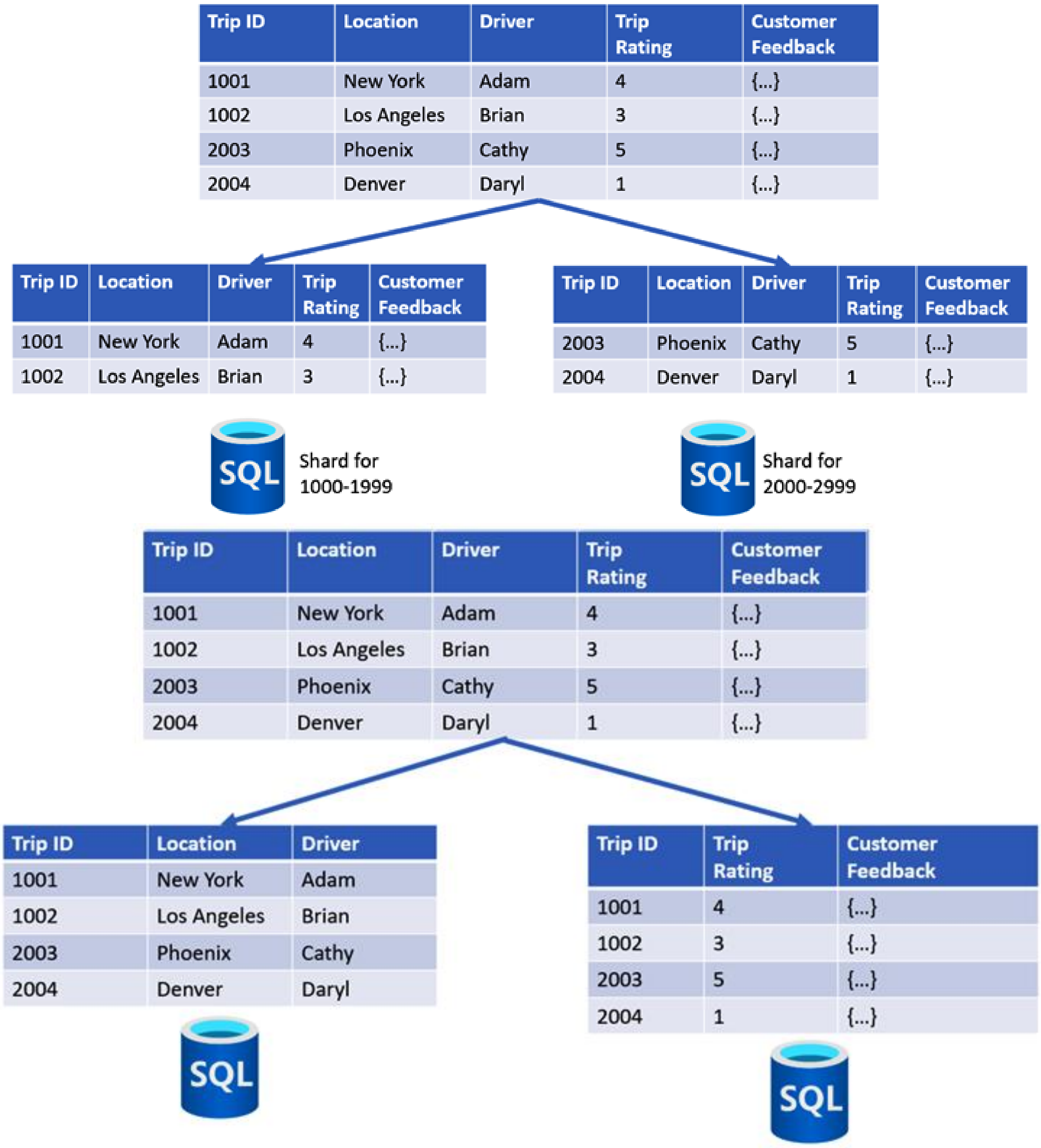
111	**Annie**	**A1234**	**222**	**Brian**	**B5678**	**333**	**Charlie**	**C3456**

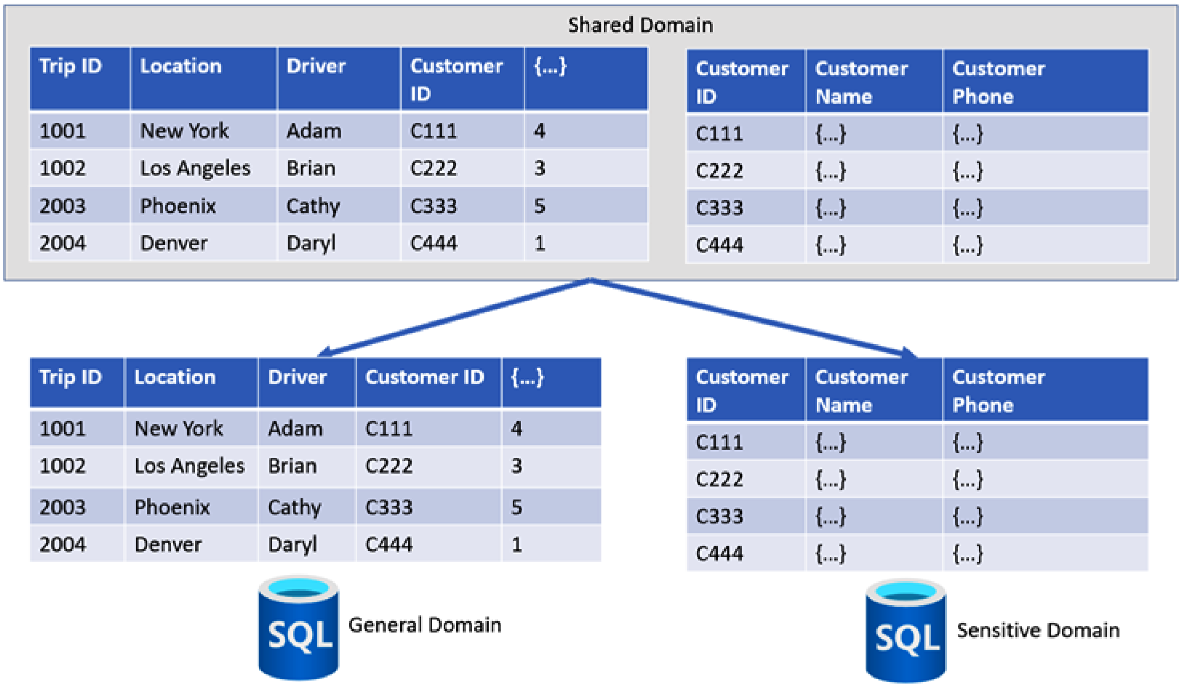
111	**222**	**333**	**Annie**	**Brian**	**Charlie**	**A1234**	**B5678**	**C3456**

Features	AVRO	Parquet	ORC
READ Performance	Low	High	High
WRITE Performance	High	Low	Low
Ability to Split files for Parallel Processing	Yes	Yes	Yes (Best)
Schema Evolution Support	Yes (Best)	Yes	Yes



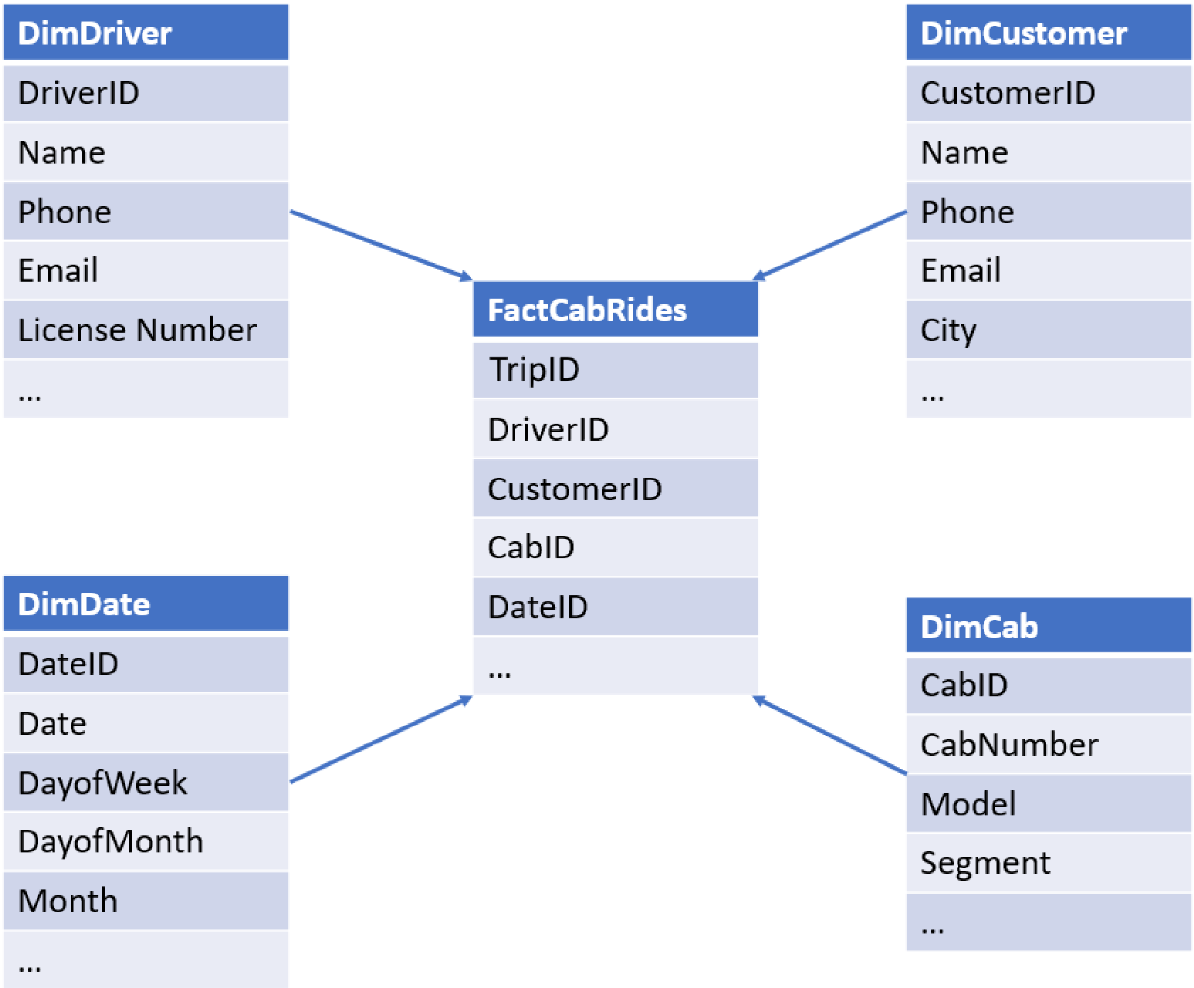
Chapter 3: Designing a Partition Strategy





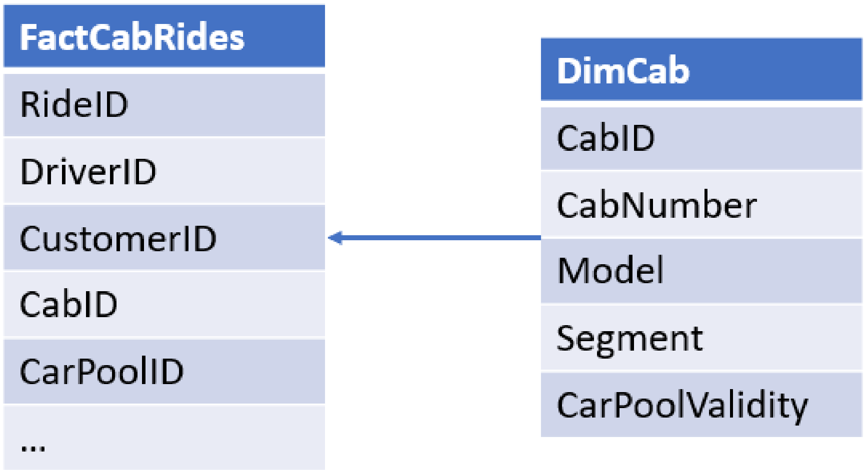
Resource	Limit
Number of storage accounts per region per subscription, including standard, and premium storage accounts.	250
Maximum storage account capacity	5 PiB (can be increased by calling Azure Support)
Maximum request rate[1] per storage account	20,000 requests per second
Maximum ingress[1] per storage account (US, Europe regions)	10 Gbps
Maximum ingress[1] per storage account (regions other than US and Europe)	5 Gbps if RA-GRS/GRS is enabled, 10 Gbps for LRS/ZRS
Maximum egress for general-purpose v2 and Blob storage accounts (all regions)	50 Gbps
Maximum number of IP address rules per storage account	200
Maximum number of virtual network rules per storage account	200

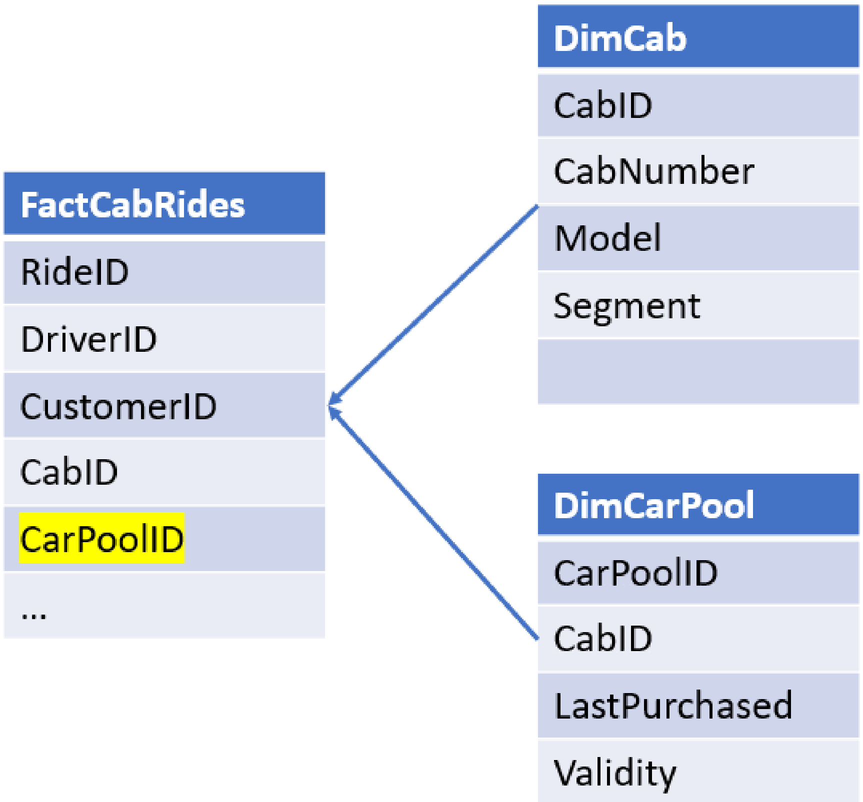
Chapter 4: Designing the Serving Layer



SurrogateID	CustomerID	Name	City	Version	...
1	1	Adam	New York	0	...

SurrogateID	CustomerID	Name	City	Version	...
1	1	Adam	New York	0	...
2	1	Adam	New Jersey	1	...
3	1	Adam	Miami	2	...





Surrogate ID	Customer ID	Name	CurrCity	PrevCity	StartDate	EndDate	isActive
1	1	Adam	Miami	NULL	01-Jan-2020	25-Mar-2020	False
2	1	Adam	Miami	New York	25-Mar-2020	01-Dec-2020	False
3	1	Adam	Miami	New Jersey	01-Dec-2020	NULL	True

```sql
SELECT [customerId]
    , [name]
    , [address]
    , [validFrom]
    , [validTo]
    , IIF (YEAR(validTo) = 9999, 1, 0) AS IsActual
FROM [dbo].[Customer]
FOR SYSTEM_TIME BETWEEN '2022-01-22' AND '2022-01-24'
WHERE CustomerId = 101
ORDER BY validFrom DESC;
```

Results Messages

Search to filter items...

customerId	name	address	validFrom	validTo	IsActual
101	Alan Li	111 Updated Lane, LA	2022-01-23T07:38:32....	9999-12-31T23:59:59....	1
101	Alan Li	101 Test Lane, LA	2022-01-23T07:36:26....	2022-01-23T07:38:32....	0

```sql
CREATE TABLE DimEmployee (
    [employeeId] VARCHAR(20) NOT NULL,
    [name] VARCHAR(100),
    [department] VARCHAR(50),
    [title] VARCHAR(50),
    [parentEmployeeId] VARCHAR(20)
)
```

Results Messages

Search to filter items...

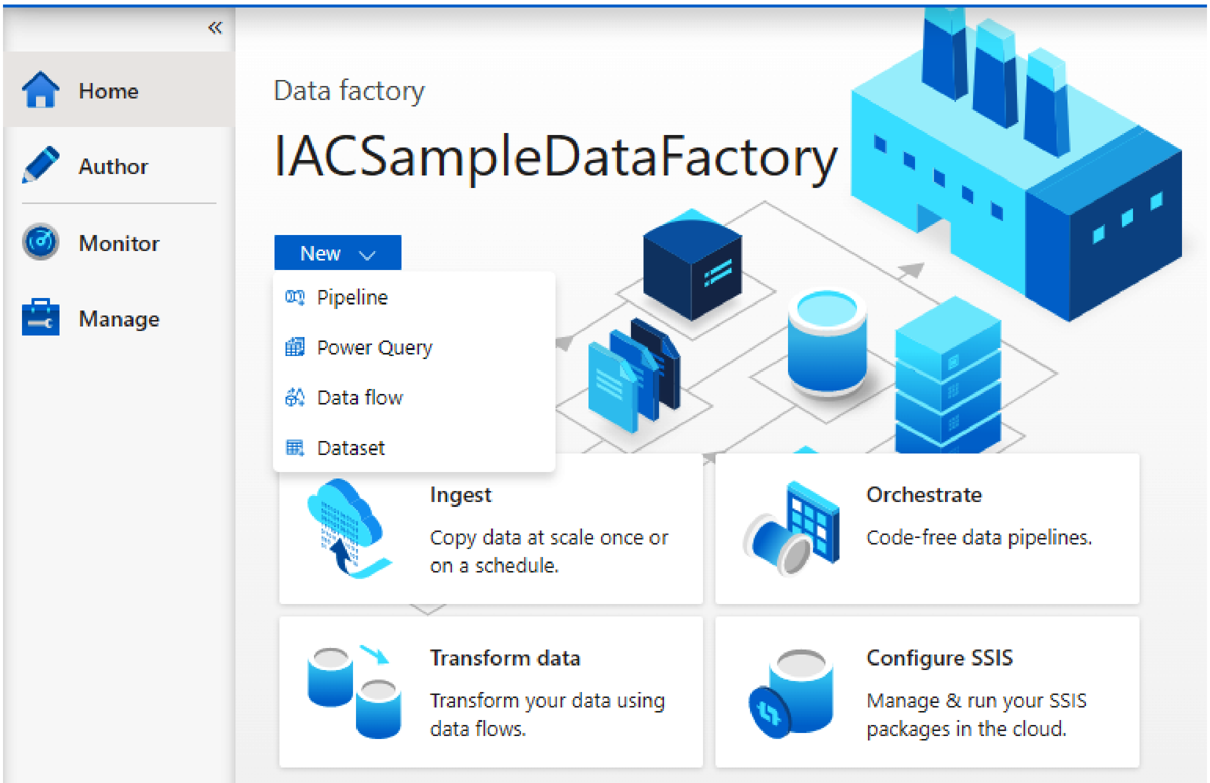
employeeId	name	department	title	parentEmployeeId
100	Alan Li	Manufacturing	Manager	
200	Brenda Jackman	Manufacturing	Supervisor	100
300	David Hood	Manufacturing	Machine operator	200

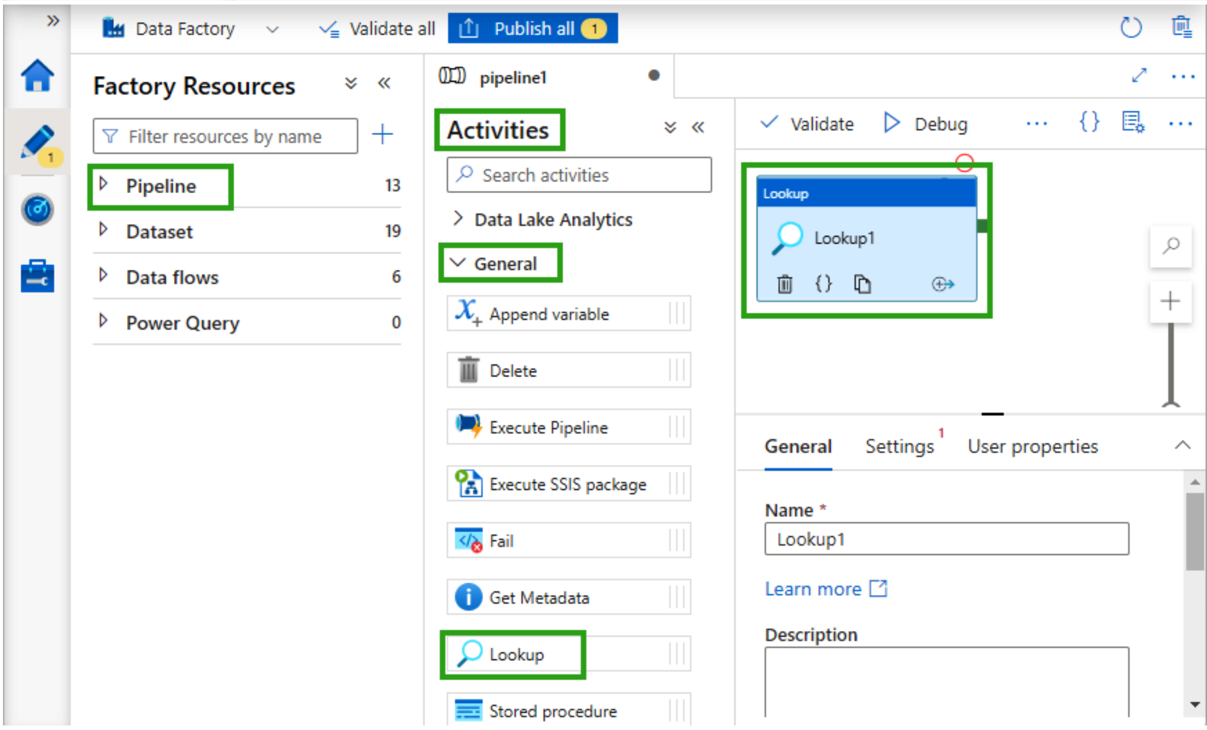
```sql
CREATE TABLE DimEmployee (
    [EmployeeID] nvarchar(20) NOT NULL,
    [Name] nvarchar(100),
    [Department] nvarchar(50),
    [Title] nvarchar(50),
    [ParentEmployeeID] nvarchar(20)
)

SELECT * FROM [dbo].[DimEmployee];
```

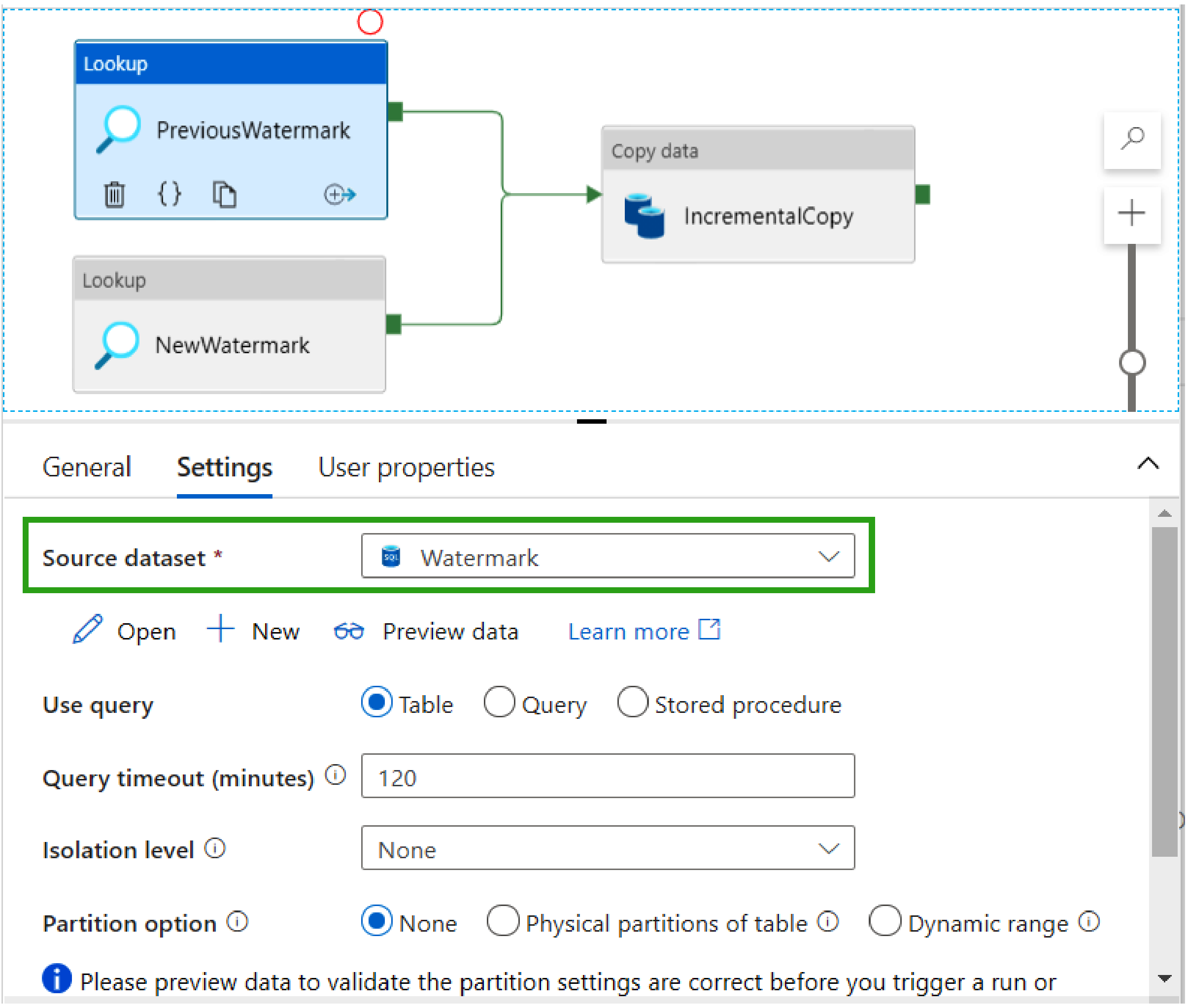
Results Messages

Search to filter items...

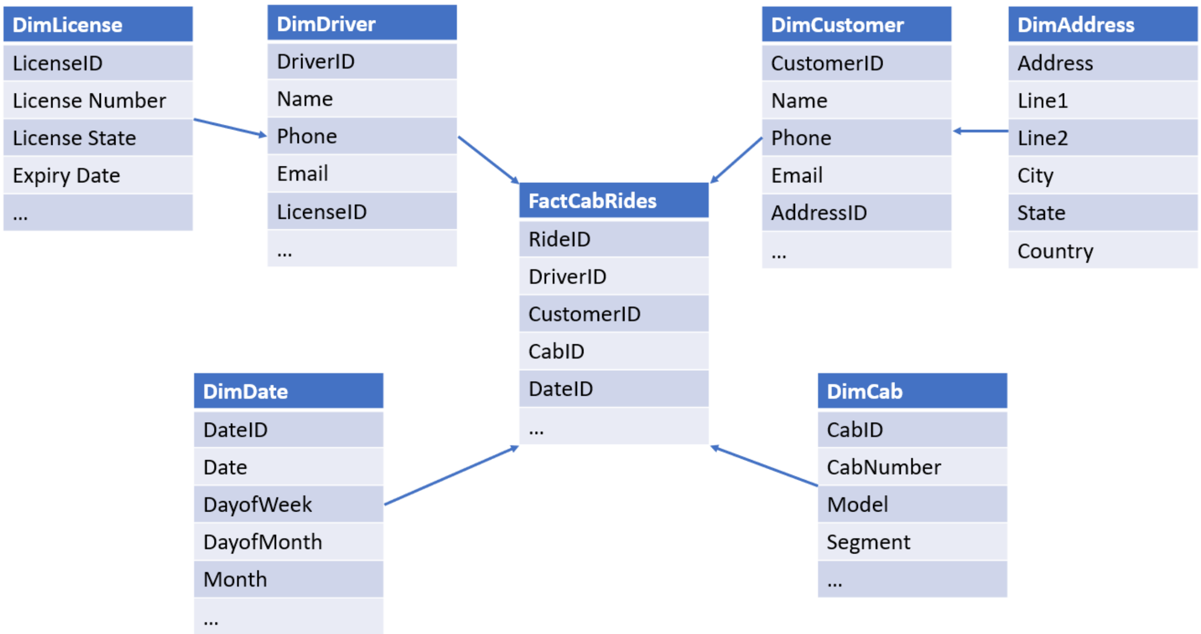
EmployeeID	Name	Department	Title	ParentEmployeeID
100	Alan Li	Manufacturing	Manager	
200	Brenda Jackman	Manufacturing	Supervisor	100
300	David Hood	Manufacturing	Machine operator	200


Home
Author
Monitor
Manage
Data factory
IACSampleDataFactory
New
Pipeline
Power Query
Data flow
Dataset
Ingest
Copy data at scale once or on a schedule.
Orchestrate
Code-free data pipelines.
Transform data
Transform your data using data flows.
Configure SSIS
Manage & run your SSIS packages in the cloud.


Data Factory
Validate all
Publish all 1
Factory Resources
Filter resources by name
Pipeline 13
Dataset 19
Data flows 6
Power Query 0
pipeline1
Activities
Search activities
Data Lake Analytics
General
Append variable
Delete
Execute Pipeline
Execute SSIS package
Fail
Get Metadata
Lookup
Stored procedure
Validate
Debug
Lookup
Lookup1
General
Settings
User properties
Name *
Lookup1
Learn more
Description


Lookup
PreviousWatermark
Lookup
NewWatermark
Copy data
IncrementalCopy
General
Settings
User properties
Source dataset *
Watermark
Open
New
Preview data
Learn more
Use query
Table
Query
Stored procedure
Query timeout (minutes)
120
Isolation level
None
Partition option
None
Physical partitions of table
Dynamic range
Please preview data to validate the partition settings are correct before you trigger a run or

Service	Authentication options	Supports data encryption at rest	Supports row-level security	Supports dynamic data masking	Supports firewalls
Synapse Analytics	SQL/Azure Active Directory (Azure AD)	Yes	Yes	Yes	Yes
Cosmos DB	Azure AD	Yes	No	No	Yes
Azure SQL	SQL/Azure AD	Yes	Yes	Yes	Yes
HBase/Phoenix	Local/Azure AD	Yes	Yes	Yes	Yes (with Azure virtual networks (VNets))
Hive LLAP	Local/Azure AD	Yes	Yes	Yes	Yes (with Azure VNets)
Azure Data Explorer	Azure AD	Yes	No	Yes	Yes
Azure Analysis Services	Azure AD	Yes	Yes	No	Yes



CustomerID	Name	City	Email	...
1	Adam	New York	adam@....	...

CustomerID	Name	City	Email	...
1	Adam	New Jersey	adam@....	...

SurrogateID	CustomerID	Name	City	isActive	...
1	1	Adam	New York	True	...

SurrogateID	CustomerID	Name	City	isActive	...
1	1	Adam	New York	False	...
2	1	Adam	New Jersey	False	...
3	1	Adam	Miami	True	...

SurrogateID	CustomerID	Name	City	StartDate	EndDate
1	1	Adam	New York	01-Jan-2020	NULL

SurrogateID	CustomerID	Name	City	StartDate	EndDate
1	1	Adam	New York	01-Jan-2020	25-Mar-2020
2	1	Adam	New Jersey	25-Mar-2020	01-Dec-2020
3	1	Adam	Miami	01-Dec-2020	NULL

SurrogateID	CustomerID	Name	City	StartDate	EndDate	isActive
1	1	Adam	New York	01-Jan-2020	25-Mar-2020	False
2	1	Adam	New Jersey	25-Mar-2020	01-Dec-2020	False
3	1	Adam	Miami	01-Dec-2020	NULL	True

CustomerID	Name	City	PrevCity	...
1	Adam	New York	NULL	

CustomerID	Name	City	PrevCity	...
1	Adam	New Jersey	New York	

Column	Data Type
DateKey	int
Date	varchar
DayName	varchar
DayOfWeek	int
DayOfMonth	int
MonthName	varchar
MonthOfYear	int
Year	int
FiscalYear	int
FiscalQuarter	Int
USHoliday	varchar
UKHoliday	varchar

DateKey	Date	DayName	DayOfWeek	DayOfMonth	MonthName	...
1	01012000	Saturday	6	1	January	...
2	02012000	Sunday	7	2	January	....
....	...	...	...	...	...	...
32	01022000	Tuesday	2	1	February	...
...	...	...	...	...	....	...

RideId	CustomerID	DriverID	CabID	TimeID	...
R1	C1	D1	Cab1	1	
R2	C2	D2	Cab2	1	
R3	C3	D3	Cab3	32	

RideId	CustomerID	DriverID	CabID	TimeID	...
R1	C1	D1	Cab1	1	
R2	C2	D2	Cab2	1	

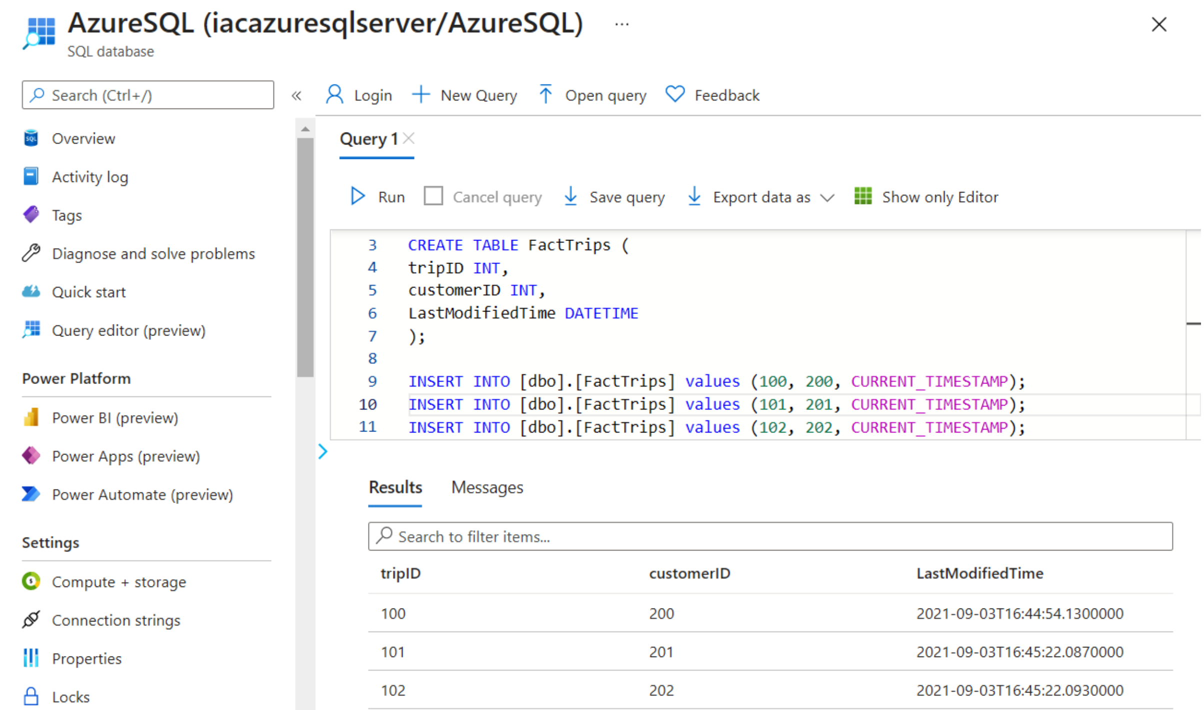
Column	Data Type
DateID	int
Date	varchar
DayName	varchar
DayOfWeek	int
DayOfMonth	int

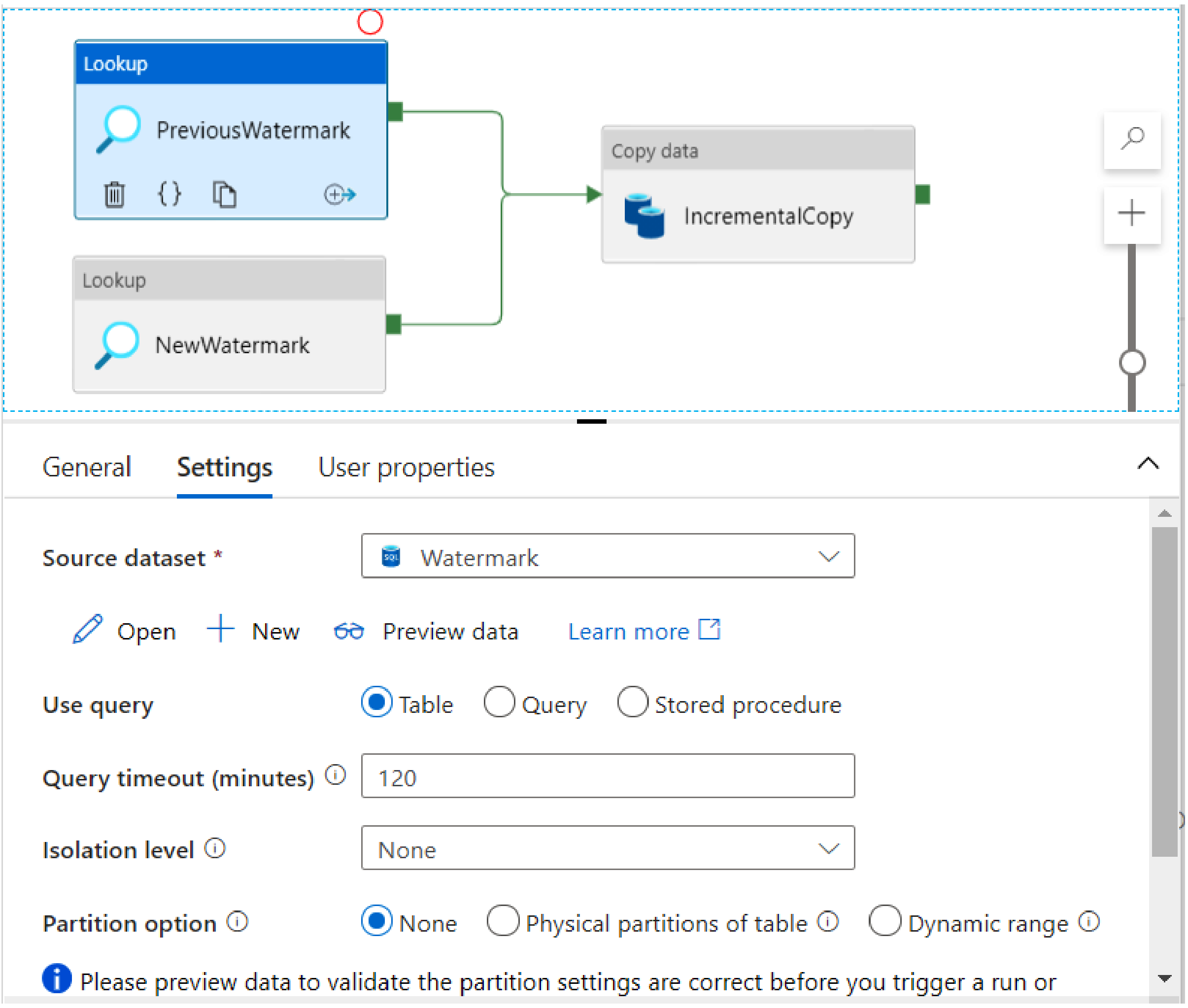
Column	Data Type
...	...
...	...
...	...

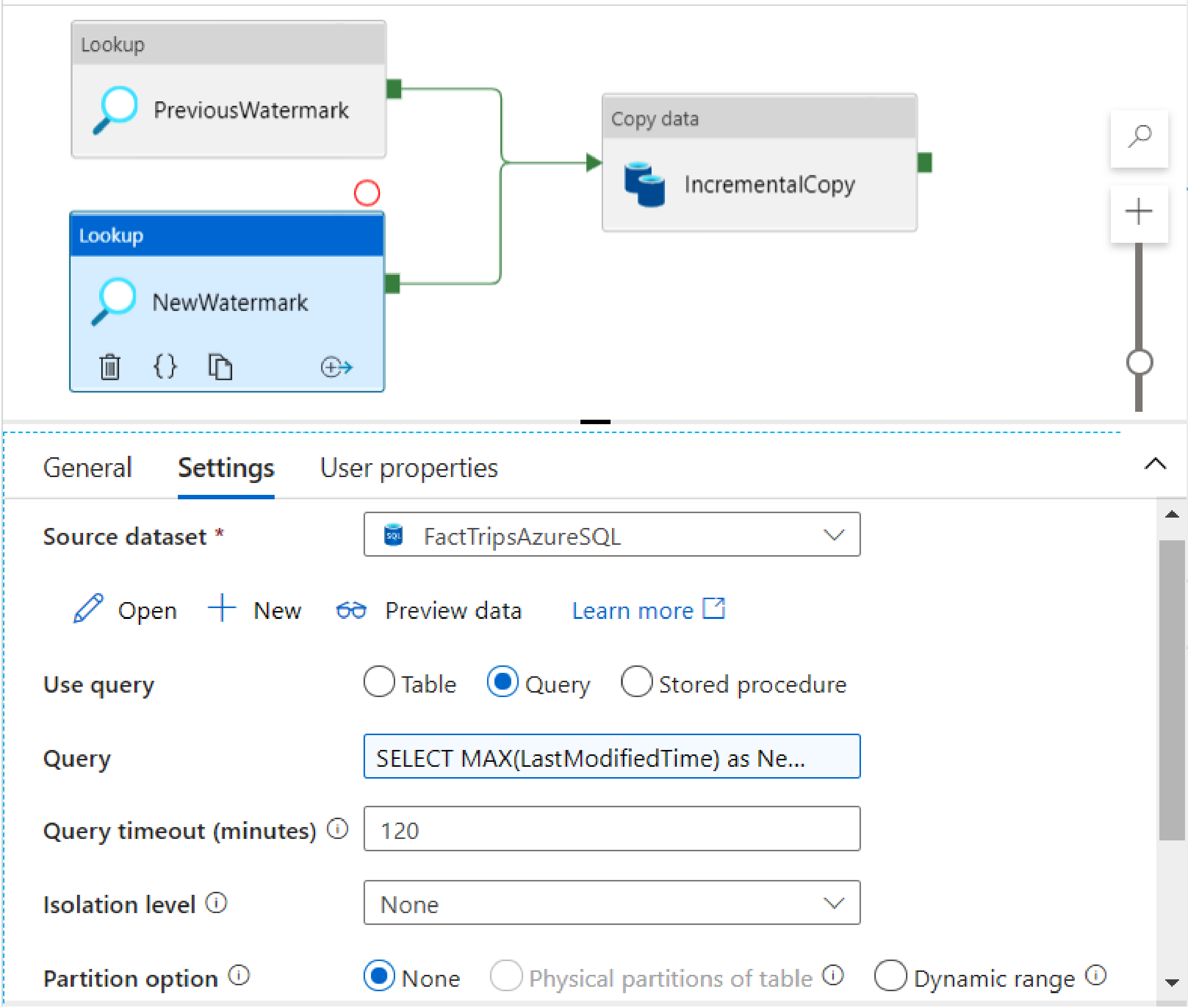
Column	Data Type
DateKey	int
Date ID	varchar
MonthID	int
YearID	int
FiscalID	int
...	...

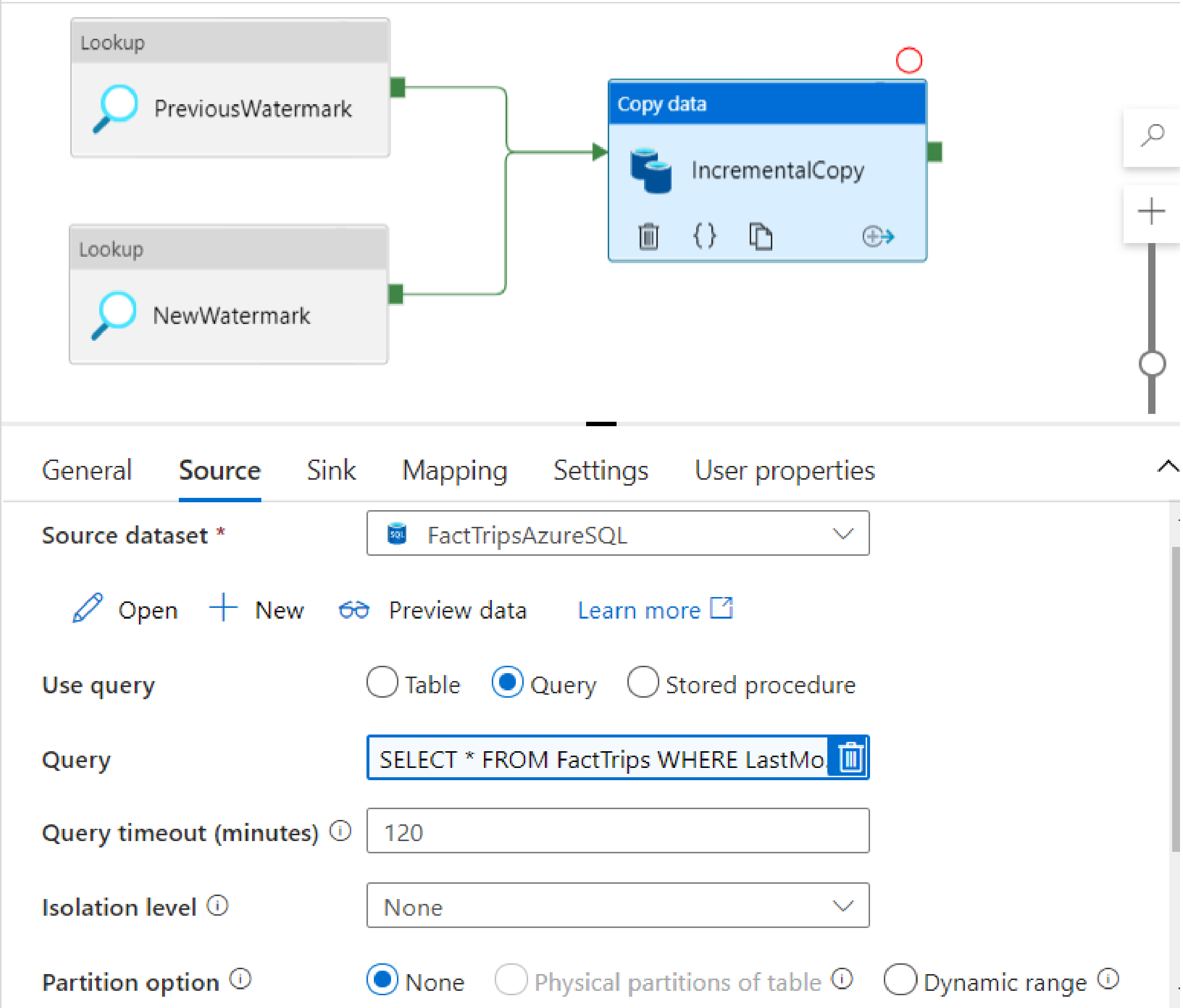
Column	Data Type
MonthID	int
MonthName	varchar
MonthOfYear	int

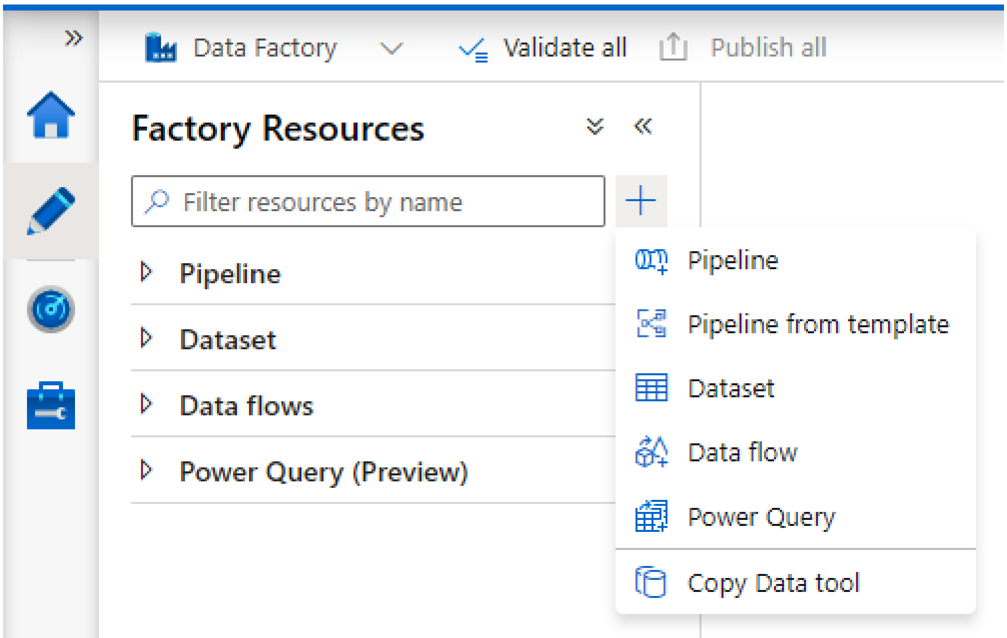
Column	Data Type
FiscalID	int
Fiscal Quarter	int
Fiscal Year	int

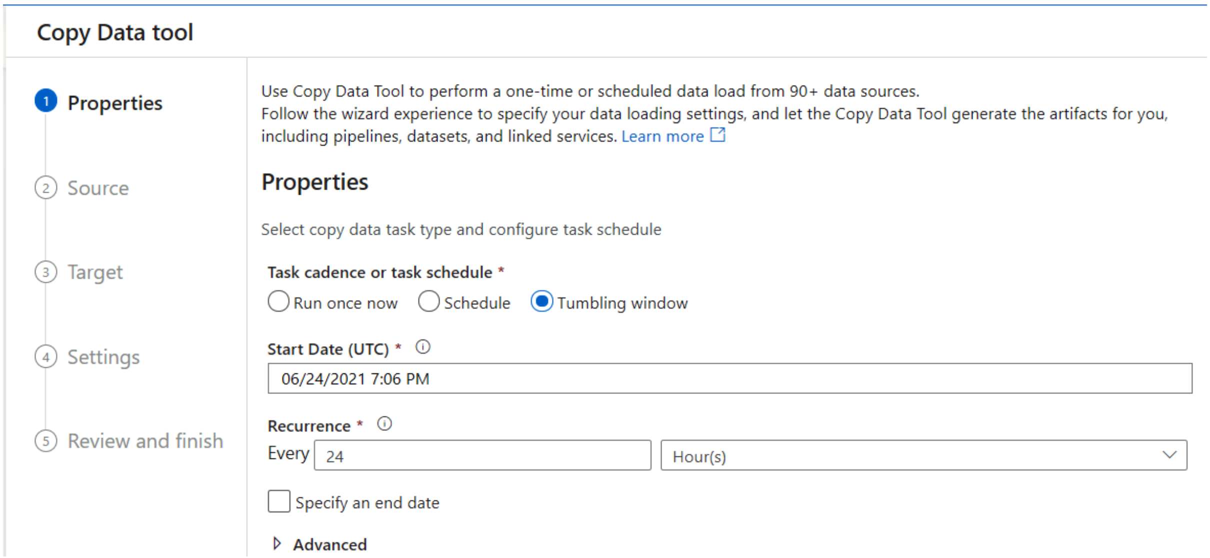


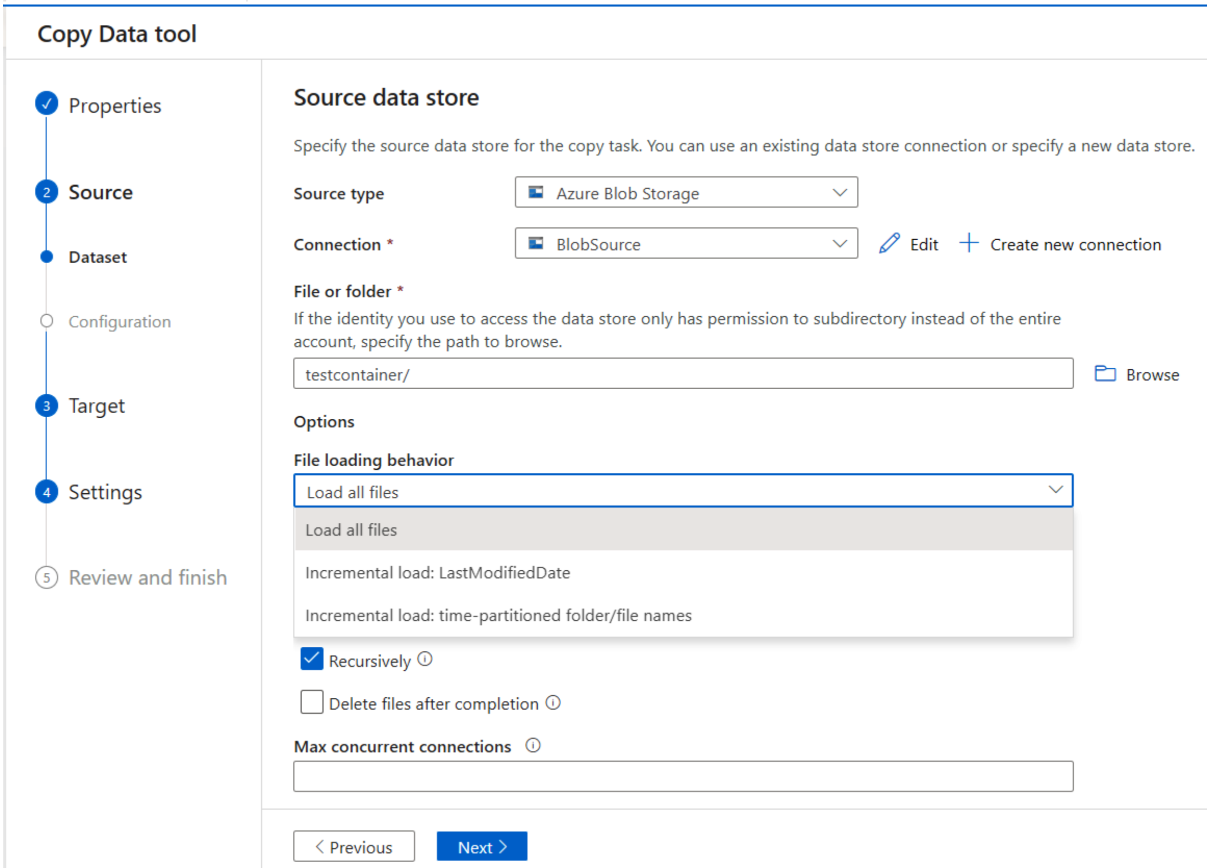

Lookup
PreviousWatermark
Lookup
NewWatermark
Copy data
IncrementalCopy
General
Settings
User properties
Source dataset *
Watermark
Open
New
Preview data
Learn more
Use query
Table
Query
Stored procedure
Query timeout (minutes)
120
Isolation level
None
Partition option
None
Physical partitions of table
Dynamic range
Please preview data to validate the partition settings are correct before you trigger a run or

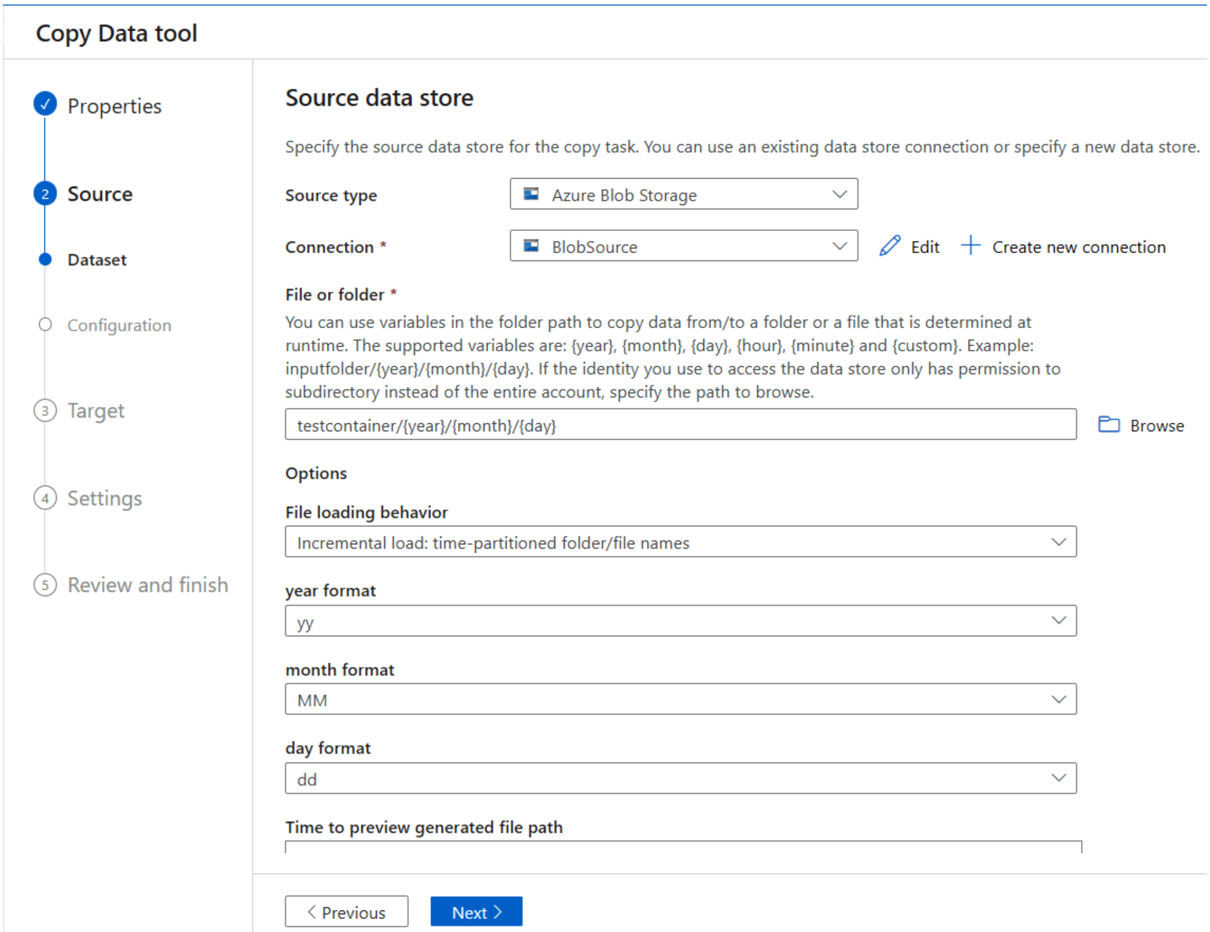

Lookup
PreviousWatermark
Lookup
NewWatermark
Copy data
IncrementalCopy
General
Settings
User properties
Source dataset *
FactTripsAzureSQL
Open
New
Preview data
Learn more
Use query
Table
Query
Stored procedure
Query
SELECT MAX(LastModifiedTime) as Ne...
Query timeout (minutes)
120
Isolation level
None
Partition option
None
Physical partitions of table
Dynamic range


Lookup
PreviousWatermark
Lookup
NewWatermark
Copy data
IncrementalCopy
General
Source
Sink
Mapping
Settings
User properties
Source dataset *
FactTripsAzureSQL
Open
New
Preview data
Learn more
Use query
Table
Query
Stored procedure
Query
SELECT * FROM FactTrips WHERE LastMo
Query timeout (minutes)
120
Isolation level
None
Partition option
None
Physical partitions of table
Dynamic range


Data Factory
Validate all
Publish all
Factory Resources
Filter resources by name
Pipeline
Dataset
Data flows
Power Query (Preview)
Pipeline
Pipeline from template
Dataset
Data flow
Power Query
Copy Data tool


Copy Data tool
1 Properties
2 Source
3 Target
4 Settings
5 Review and finish
Use Copy Data Tool to perform a one-time or scheduled data load from 90+ data sources.
Follow the wizard experience to specify your data loading settings, and let the Copy Data Tool generate the artifacts for you, including pipelines, datasets, and linked services. Learn more
Properties
Select copy data task type and configure task schedule
Task cadence or task schedule *
Run once now
Schedule
Tumbling window
Start Date (UTC) *
06/24/2021 7:06 PM
Recurrence *
Every
24
Hour(s)
Specify an end date
Advanced


Copy Data tool
Properties
2 Source
Dataset
Configuration
3 Target
4 Settings
5 Review and finish
Source data store
Specify the source data store for the copy task. You can use an existing data store connection or specify a new data store.
Source type
Azure Blob Storage
Connection *
BlobSource
Edit
Create new connection
File or folder *
If the identity you use to access the data store only has permission to subdirectory instead of the entire account, specify the path to browse.
testcontainer/
Browse
Options
File loading behavior
Load all files
Load all files
Incremental load: LastModifiedDate
Incremental load: time-partitioned folder/file names
Recursively
Delete files after completion
Max concurrent connections
Previous
Next

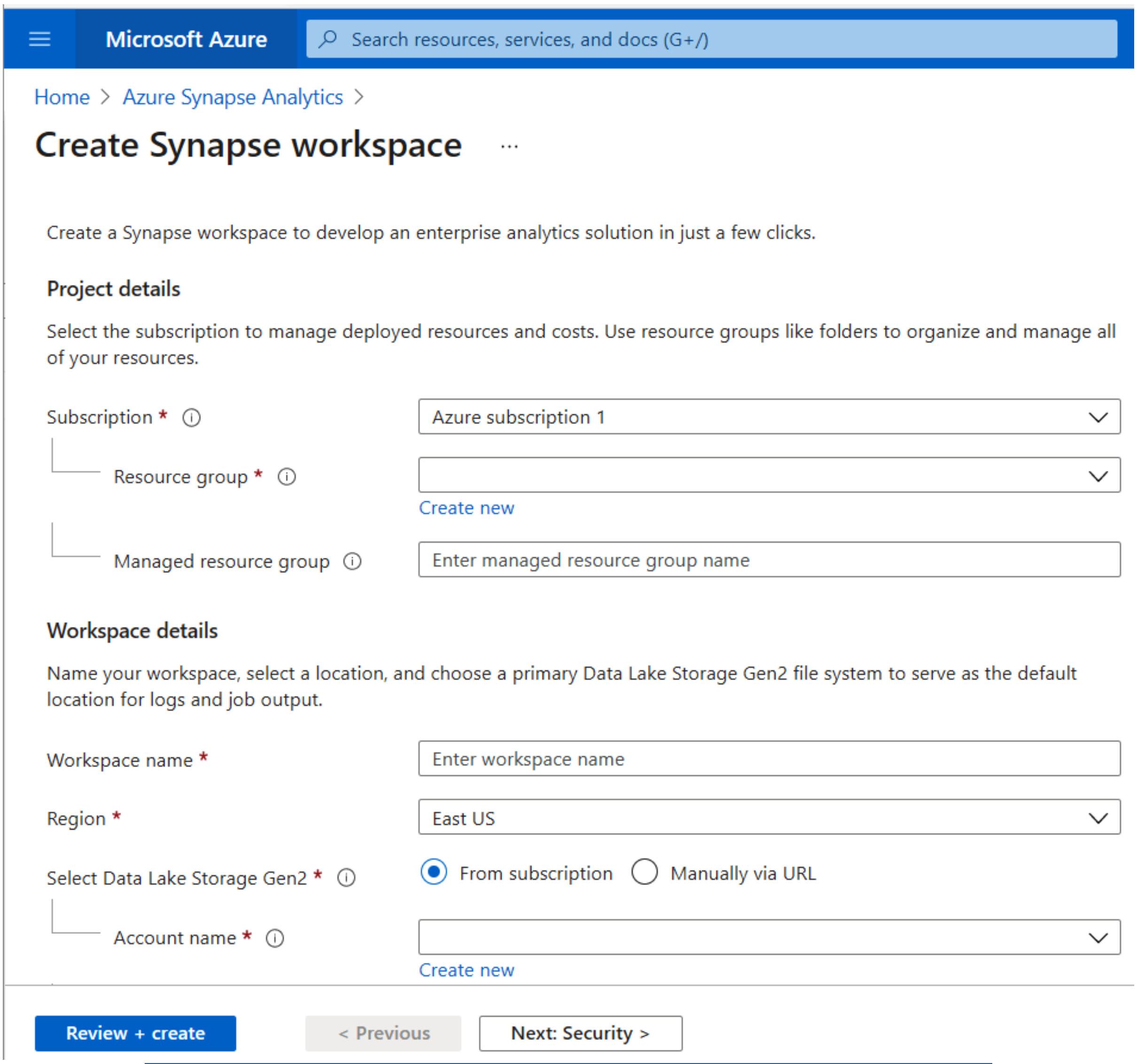

Copy Data tool
Properties
Source
Dataset
Configuration
Target
Settings
Review and finish
Source data store
Specify the source data store for the copy task. You can use an existing data store connection or specify a new data store.
Source type
Azure Blob Storage
Connection *
BlobSource
Edit
Create new connection
File or folder *
You can use variables in the folder path to copy data from/to a folder or a file that is determined at runtime. The supported variables are: {year}, {month}, {day}, {hour}, {minute} and {custom}. Example: inputfolder/{year}/{month}/{day}. If the identity you use to access the data store only has permission to subdirectory instead of the entire account, specify the path to browse.
testcontainer/{year}/{month}/{day}
Browse
Options
File loading behavior
Incremental load: time-partitioned folder/file names
year format
yy
month format
MM
day format
dd
Time to preview generated file path
Previous
Next

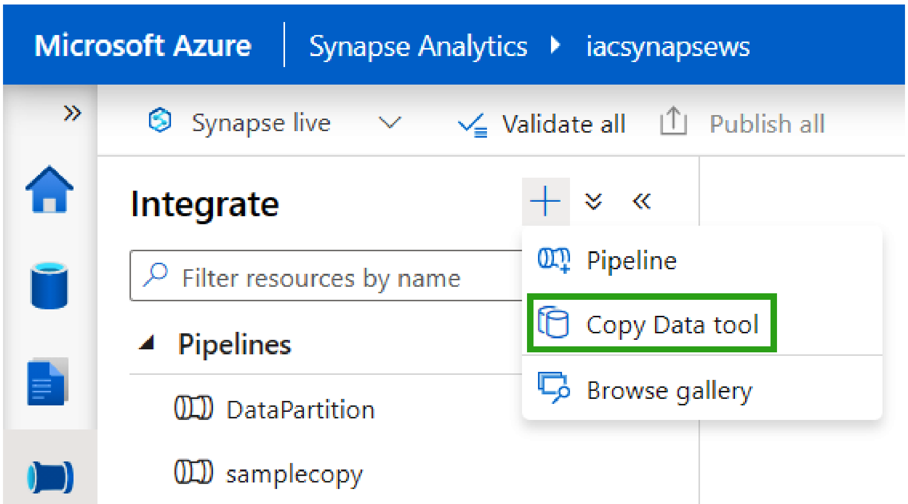
Service	**Authentication options**	**Supports data encryption at rest**	**Supports row-level security**	**Supports dynamic data masking**	**Supports firewalls**
Synapse Analytics	SQL/Azure Active Directory (Azure AD)	Yes	Yes	Yes	Yes
Cosmos DB	Azure AD	Yes	No	No	Yes
Azure SQL	SQL/Azure AD	Yes	Yes	Yes	Yes
HBase/Phoenix	Local/Azure AD	Yes	Yes	Yes	Yes (with Azure virtual networks (VNets))
Hive LLAP	Local/Azure AD	Yes	Yes	Yes	Yes (with Azure VNets)
Azure Data Explorer	Azure AD	Yes	No	Yes	Yes
Azure Analysis Services	Azure AD	Yes	Yes	No	Yes

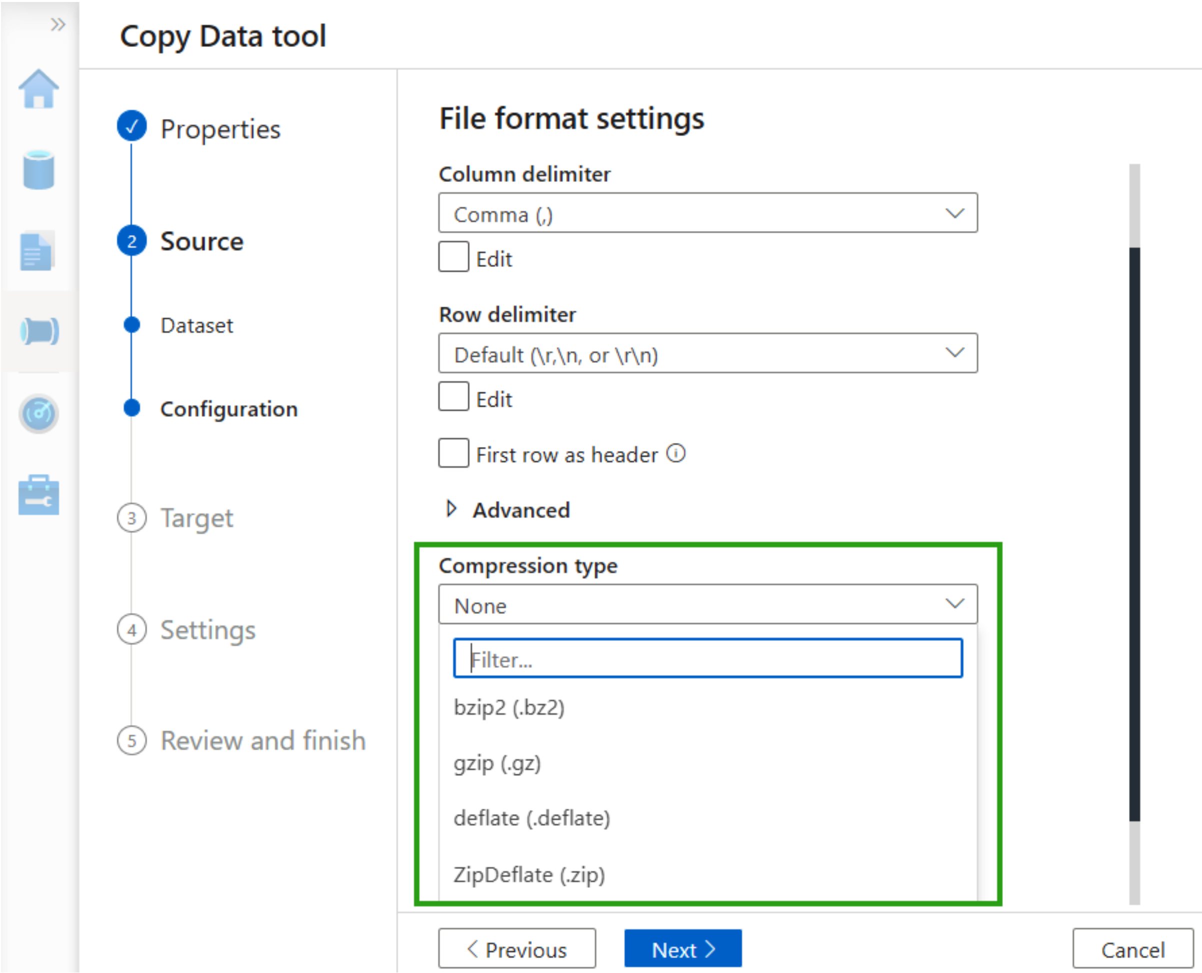
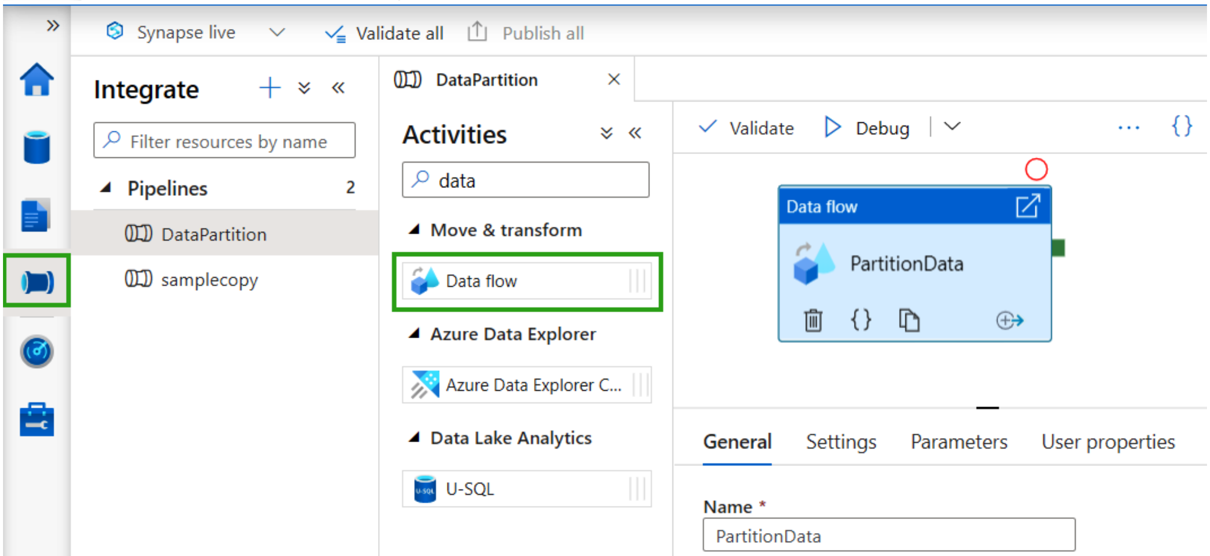
Service	Supports redundant regional servers for high availability (HA)	Supports query scale-out	Supports scale-up	Supports in-memory caching
Synapse Analytics	No	Yes	Yes	Yes
Cosmos DB	Yes	Yes	Yes	No
Azure SQL	Yes	No	Yes	Yes
HBase/Phoenix	Yes	Yes	No	No
Hive LLAP	No	Yes	No	Yes
Azure Data Explorer	Yes	Yes	Yes	Yes
Azure Analysis Services	No	Yes	Yes	Yes

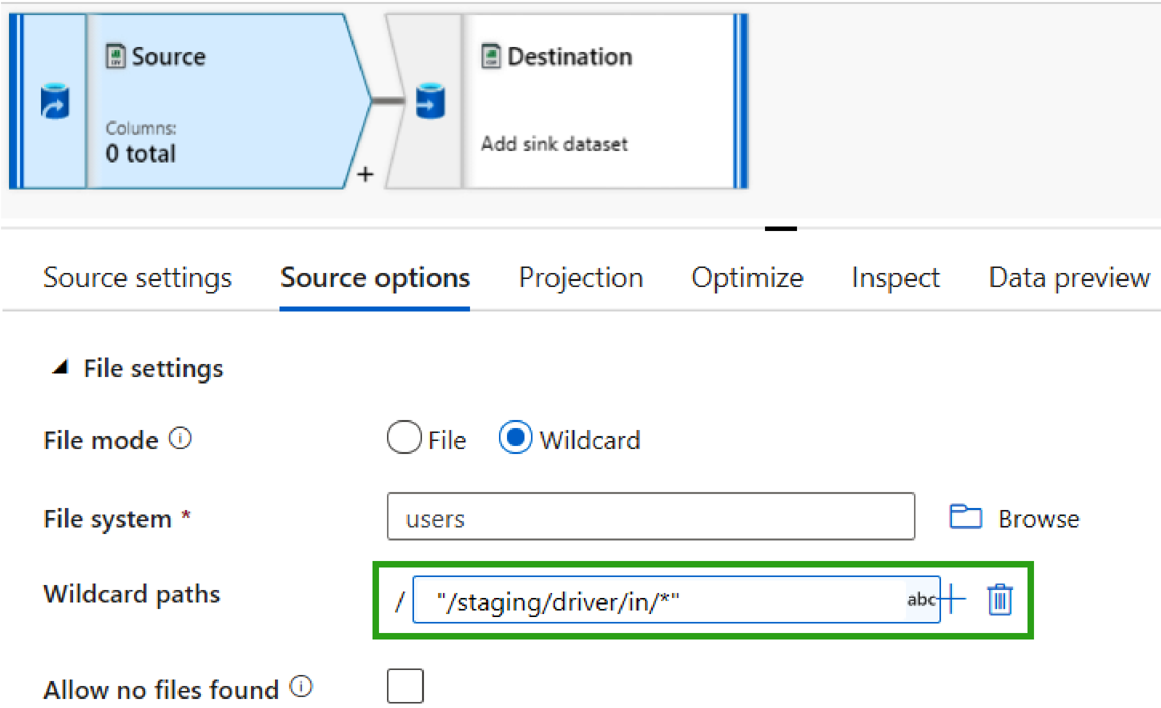
Service	Authentication options	Supports data encryption at rest	Supports row-level security	Supports dynamic data masking	Supports firewalls
Synapse Analytics	SQL/**Azure Active Directory (Azure AD)**	Yes	Yes	Yes	Yes
Cosmos DB	Azure AD	Yes	No	No	Yes
Azure SQL	SQL/Azure AD	Yes	Yes	Yes	Yes
HBase/Phoenix	Local/Azure AD	Yes	Yes	Yes	Yes (with Azure **virtual networks** (**VNets**))
Hive LLAP	Local/Azure AD	Yes	Yes	Yes	Yes (with Azure VNets)
Azure Data Explorer	Azure AD	Yes	No	Yes	Yes
Azure Analysis Services	Azure AD	Yes	Yes	No	Yes

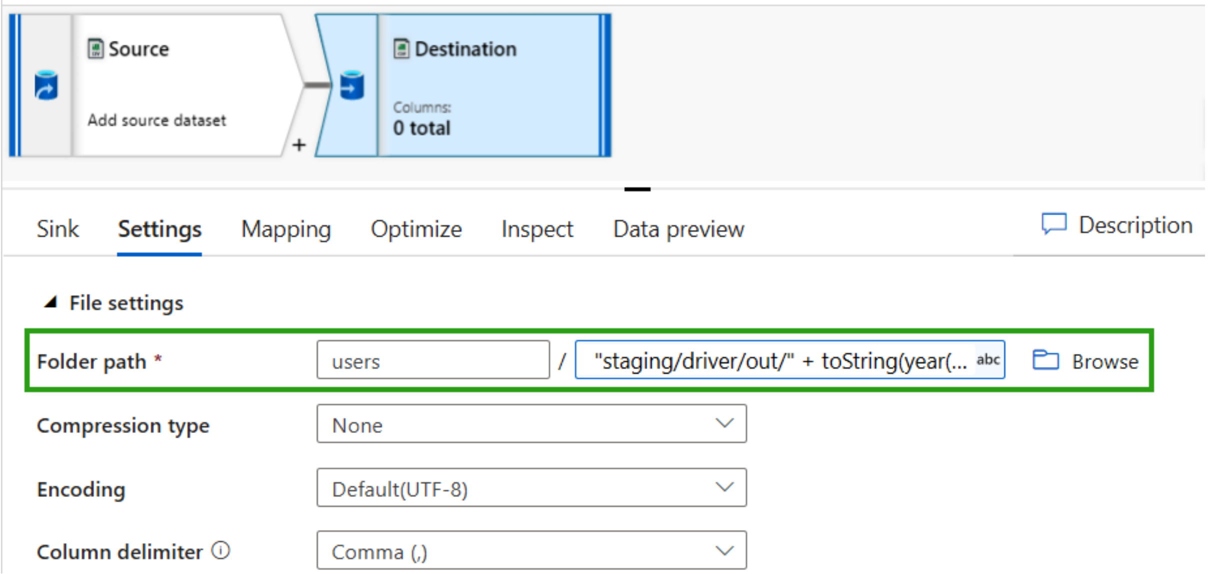
Service	Supports redundant regional servers for high availability (HA)	Supports query scale-out	Supports scale-up	Supports in-memory caching
Synapse Analytics	No	Yes	Yes	Yes
Cosmos DB	Yes	Yes	Yes	No
Azure SQL	Yes	No	Yes	Yes
HBase/Phoenix	Yes	Yes	No	No
Hive LLAP	No	Yes	No	Yes
Azure Data Explorer	Yes	Yes	Yes	Yes
Azure Analysis Services	No	Yes	Yes	Yes

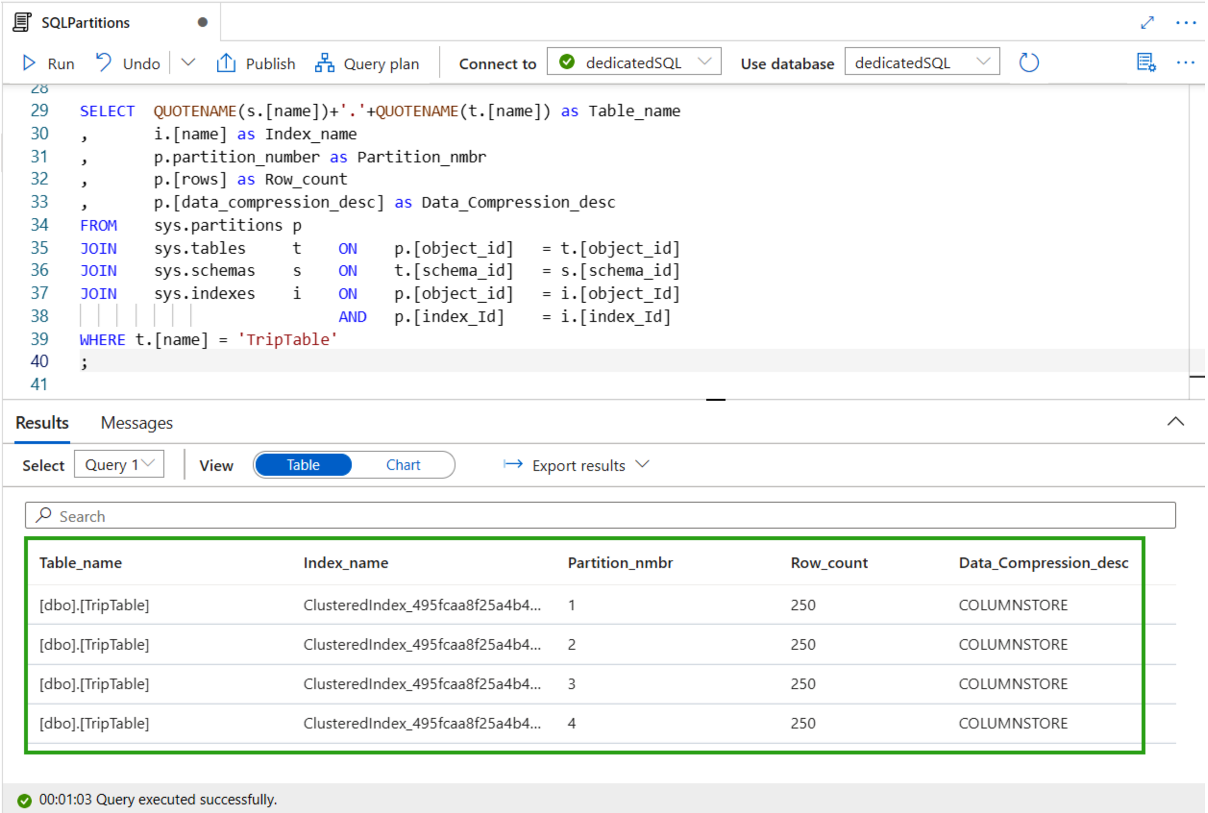

Microsoft Azure
Search resources, services, and docs (G+/)
Home > Azure Synapse Analytics >
Create Synapse workspace ...
Create a Synapse workspace to develop an enterprise analytics solution in just a few clicks.
Project details
Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all of your resources.
Subscription *
Azure subscription 1
Resource group *
Create new
Managed resource group
Enter managed resource group name
Workspace details
Name your workspace, select a location, and choose a primary Data Lake Storage Gen2 file system to serve as the default location for logs and job output.
Workspace name *
Enter workspace name
Region *
East US
Select Data Lake Storage Gen2 *
From subscription
Manually via URL
Account name *
Create new
Review + create
< Previous
Next: Security >

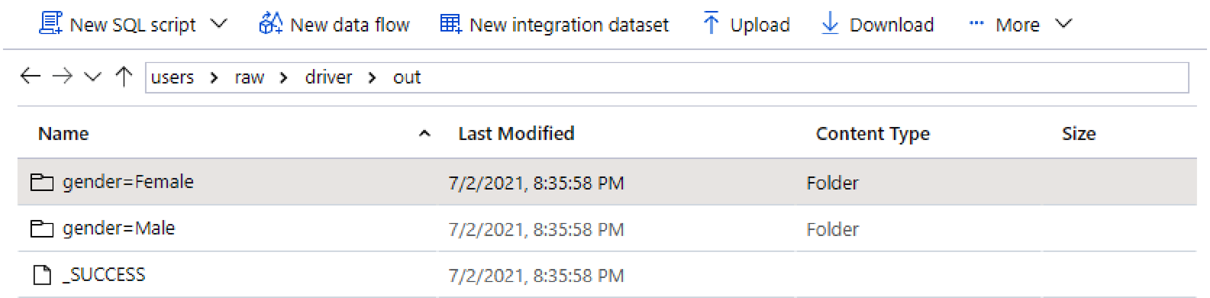

Microsoft Azure
Synapse Analytics
iacsynapsews
Synapse live
Validate all
Publish all
Integrate
Filter resources by name
Pipelines
DataPartition
samplecopy
Pipeline
Copy Data tool
Browse gallery

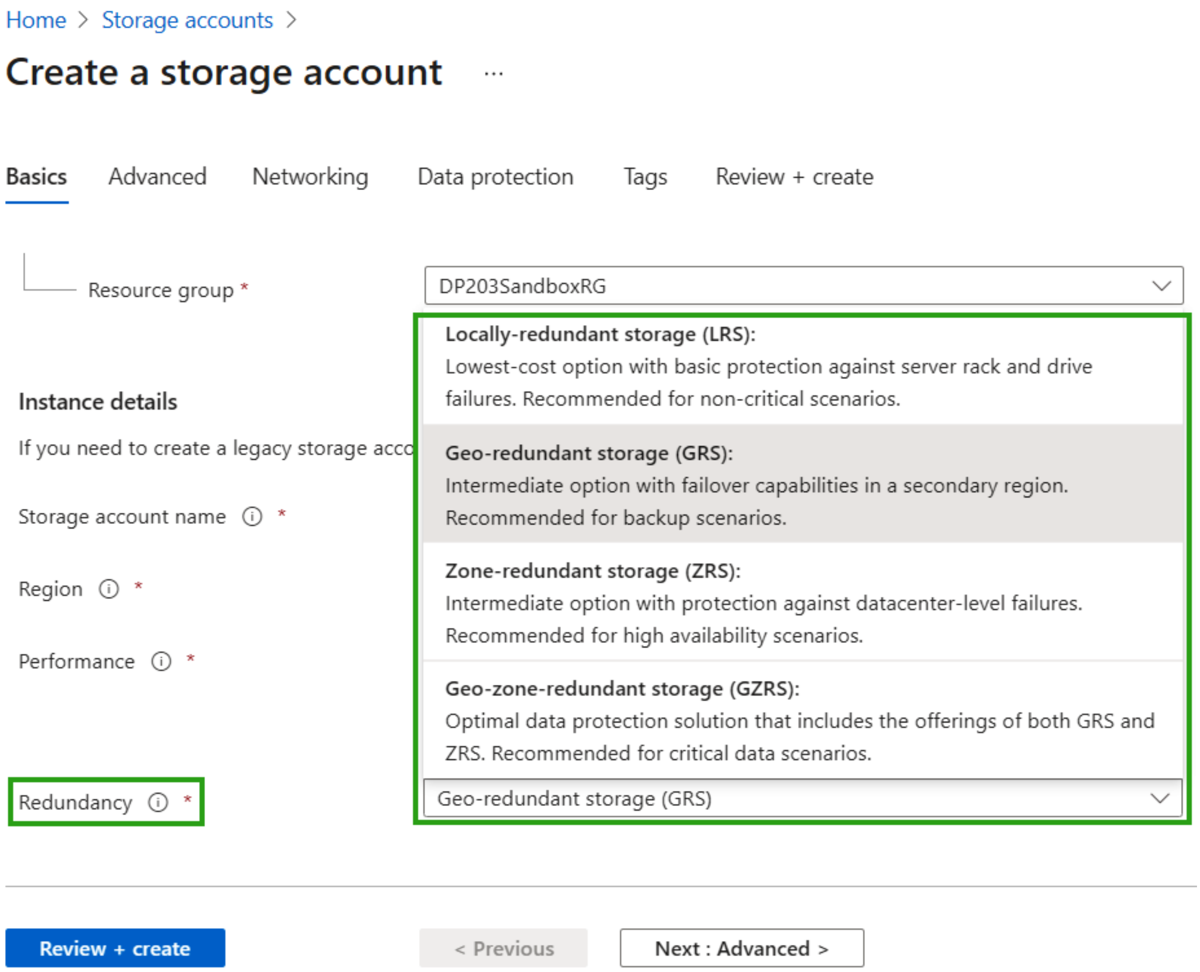
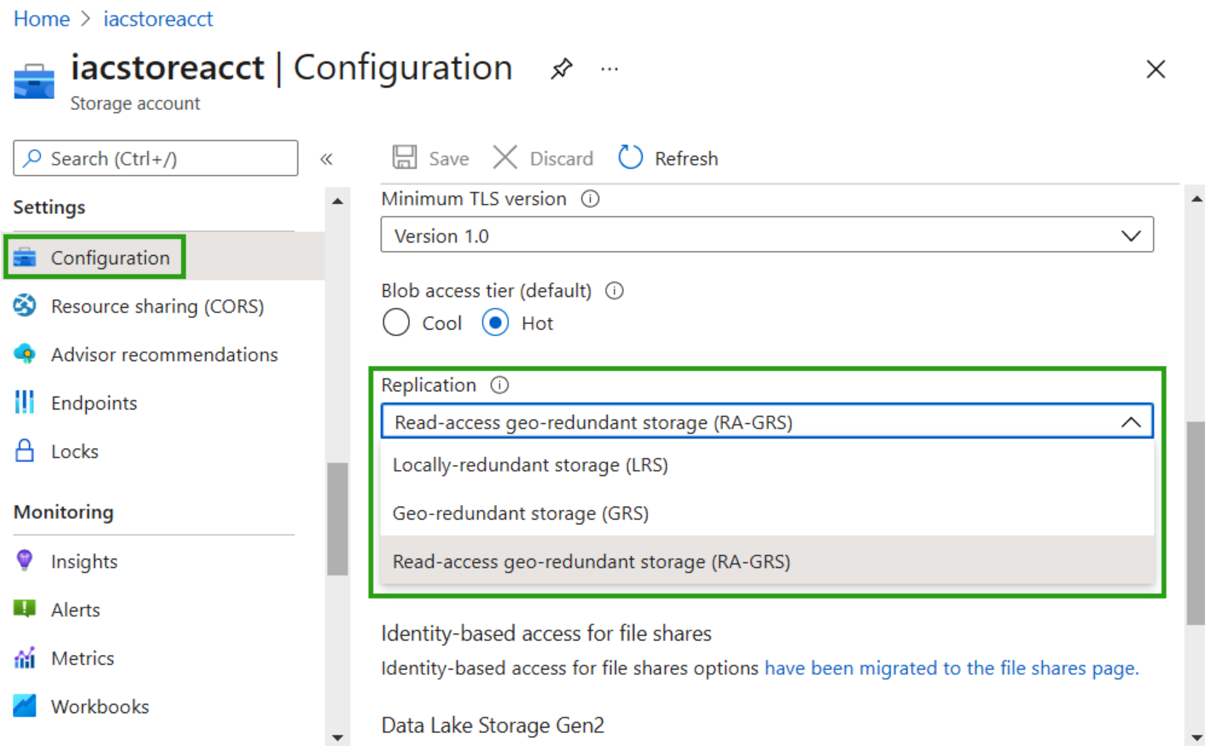

Copy Data tool
Properties
Source
Dataset
Configuration
Target
Settings
Review and finish
File format settings
Column delimiter
Comma (,)
Edit
Row delimiter
Default (\r,\n, or \r\n)
Edit
First row as header
Advanced
Compression type
None
Filter...
bzip2 (.bz2)
gzip (.gz)
deflate (.deflate)
ZipDeflate (.zip)
Previous
Next
Cancel

Synapse live
Validate all
Publish all
Integrate
Filter resources by name
Pipelines
2
DataPartition
samplecopy
DataPartition
Activities
data
Move & transform
Data flow
Azure Data Explorer
Azure Data Explorer C...
Data Lake Analytics
U-SQL
Validate
Debug
Data flow
PartitionData
General
Settings
Parameters
User properties
Name *
PartitionData


Source
Columns:
0 total
Destination
Add sink dataset
Source settings
Source options
Projection
Optimize
Inspect
Data preview
File settings
File mode
File
Wildcard
File system *
users
Browse
Wildcard paths
/ "/staging/driver/in/*"
abc
Allow no files found

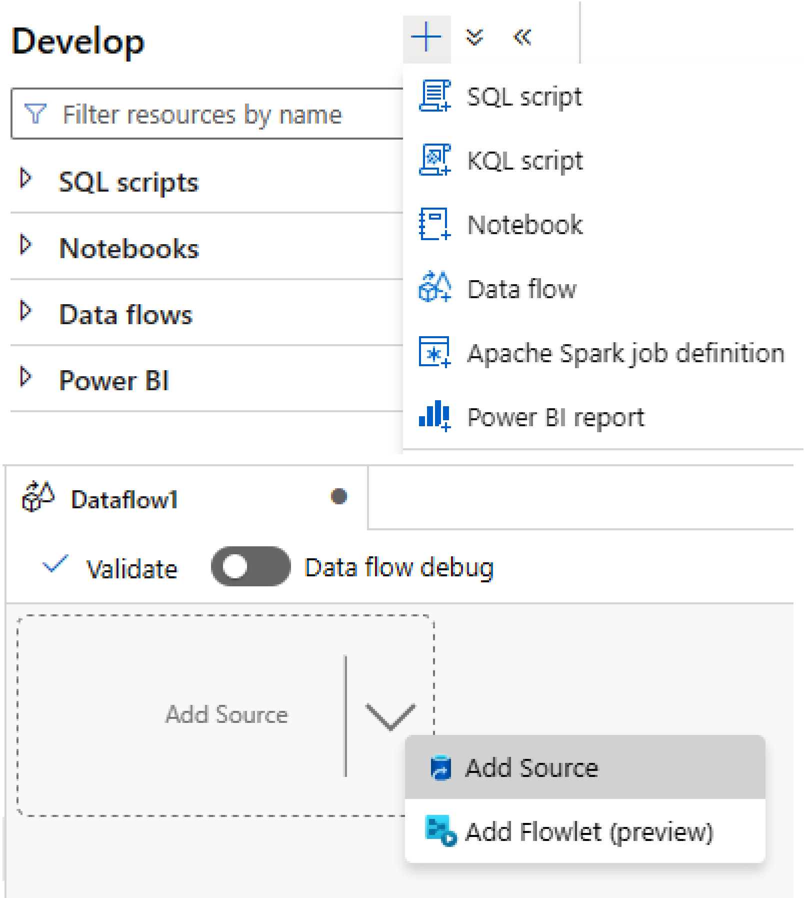

Source
Add source dataset
Destination
Columns:
0 total
Sink
Settings
Mapping
Optimize
Inspect
Data preview
Description
File settings
Folder path *
users
/ "staging/driver/out/" + toString(year(...
abc
Browse
Compression type
None
Encoding
Default(UTF-8)
Column delimiter
Comma (,)

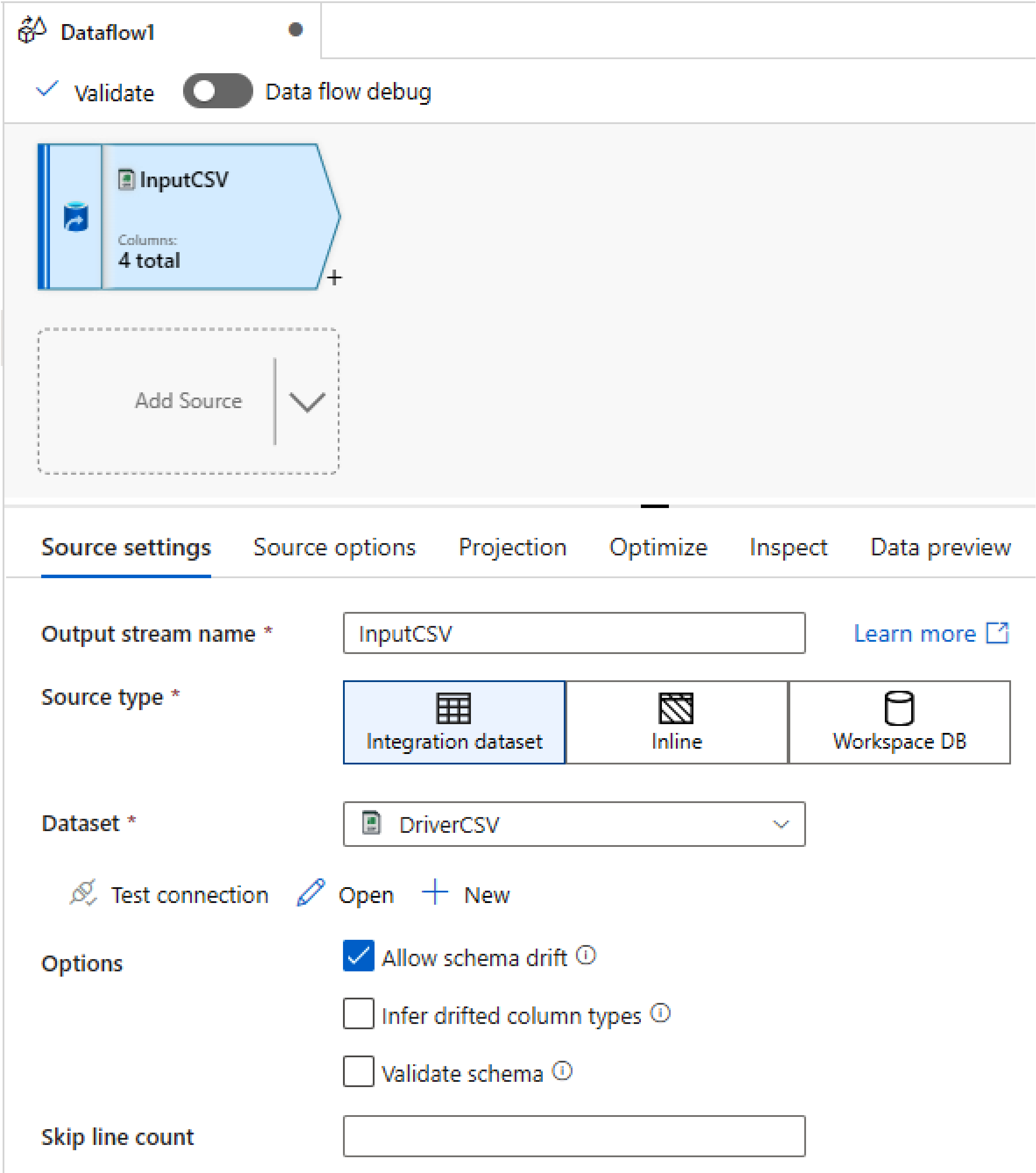

SQLPartitions
Run
Undo
Publish
Query plan
Connect to
dedicatedSQL
Use database
dedicatedSQL
SELECT QUOTENAME(s.[name])+'.'+QUOTENAME(t.[name]) as Table_name
, i.[name] as Index_name
, p.partition_number as Partition_nmbr
, p.[rows] as Row_count
, p.[data_compression_desc] as Data_Compression_desc
FROM sys.partitions p
JOIN sys.tables t ON p.[object_id] = t.[object_id]
JOIN sys.schemas s ON t.[schema_id] = s.[schema_id]
JOIN sys.indexes i ON p.[object_id] = i.[object_Id]
AND p.[index_Id] = i.[index_Id]
WHERE t.[name] = 'TripTable'
;
Results
Messages
Select
Query 1
View
Table
Chart
Export results
Search
Table_name
Index_name
Partition_nmbr
Row_count
Data_Compression_desc
[dbo].[TripTable]
ClusteredIndex_495fcaa8f25a4b4...
1
250
COLUMNSTORE
[dbo].[TripTable]
ClusteredIndex_495fcaa8f25a4b4...
2
250
COLUMNSTORE
[dbo].[TripTable]
ClusteredIndex_495fcaa8f25a4b4...
3
250
COLUMNSTORE
[dbo].[TripTable]
ClusteredIndex_495fcaa8f25a4b4...
4
250
COLUMNSTORE
00:01:03 Query executed successfully.

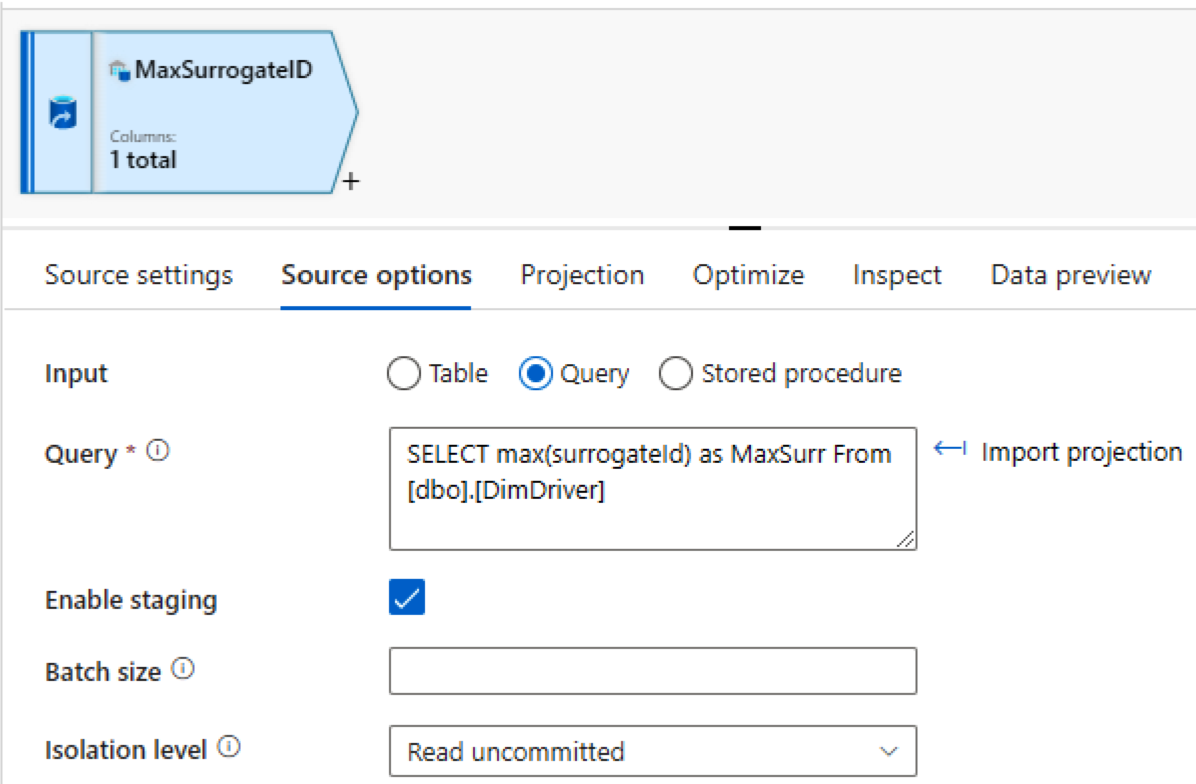

New SQL script
New data flow
New integration dataset
Upload
Download
More
users > raw > driver > out
Name
Last Modified
Content Type
Size
gender=Female
7/2/2021, 8:35:58 PM
Folder
gender=Male
7/2/2021, 8:35:58 PM
Folder
_SUCCESS
7/2/2021, 8:35:58 PM

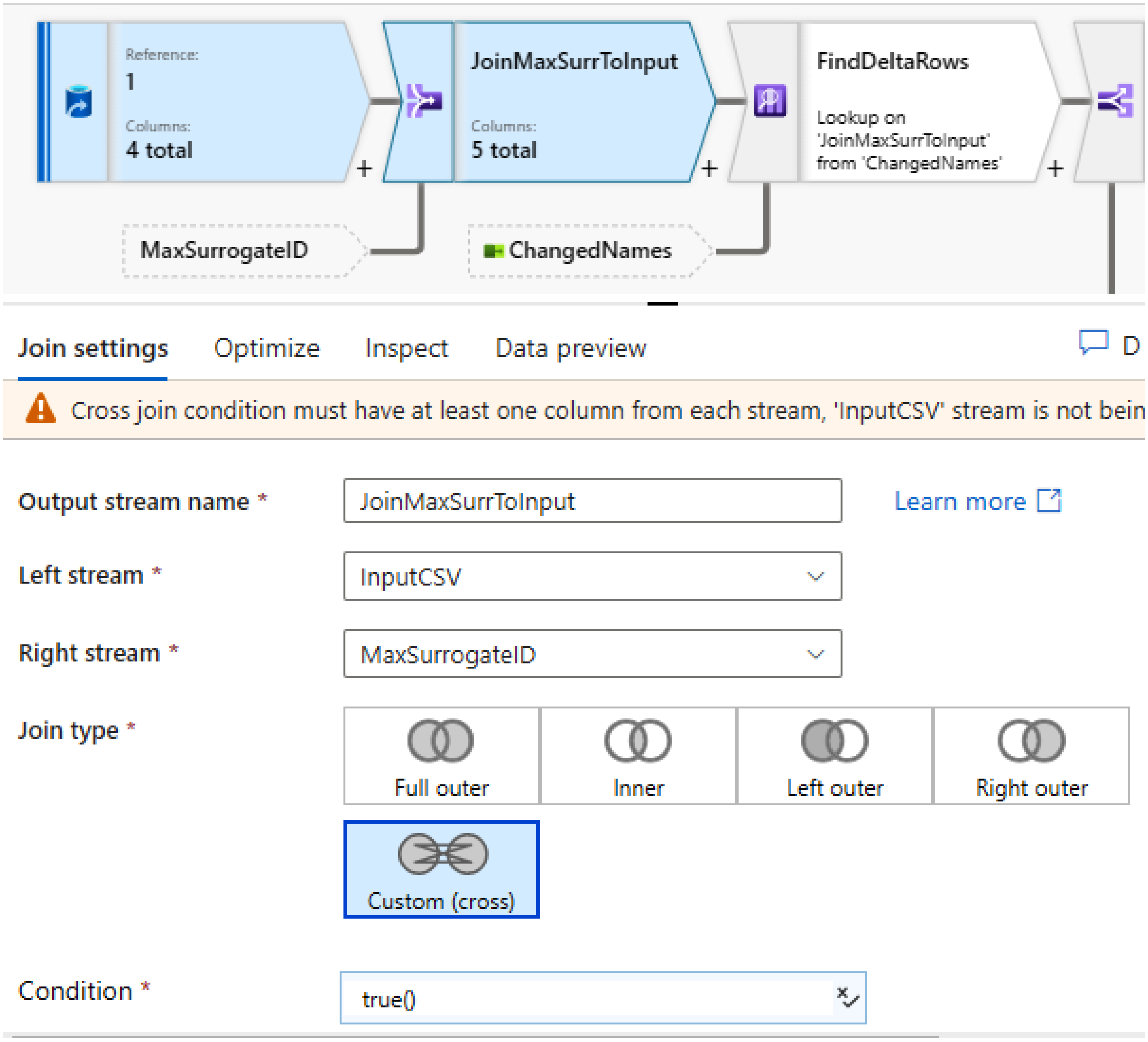

Home > Storage accounts >
Create a storage account ...
Basics
Advanced
Networking
Data protection
Tags
Review + create
Resource group *
DP203SandboxRG
Locally-redundant storage (LRS):
Lowest-cost option with basic protection against server rack and drive failures. Recommended for non-critical scenarios.
Instance details
If you need to create a legacy storage acco
Geo-redundant storage (GRS):
Intermediate option with failover capabilities in a secondary region. Recommended for backup scenarios.
Storage account name *
Region *
Zone-redundant storage (ZRS):
Intermediate option with protection against datacenter-level failures. Recommended for high availability scenarios.
Performance *
Geo-zone-redundant storage (GZRS):
Optimal data protection solution that includes the offerings of both GRS and ZRS. Recommended for critical data scenarios.
Redundancy *
Geo-redundant storage (GRS)
Review + create
< Previous
Next : Advanced >

Home > iacstoreacct
iacstoreacct | Configuration
Storage account
Search (Ctrl+/)
Save
Discard
Refresh
Settings
Configuration
Resource sharing (CORS)
Advisor recommendations
Endpoints
Locks
Monitoring
Insights
Alerts
Metrics
Workbooks
Minimum TLS version
Version 1.0
Blob access tier (default)
Cool
Hot
Replication
Read-access geo-redundant storage (RA-GRS)
Locally-redundant storage (LRS)
Geo-redundant storage (GRS)
Read-access geo-redundant storage (RA-GRS)
Identity-based access for file shares
Identity-based access for file shares options have been migrated to the file shares page.
Data Lake Storage Gen2

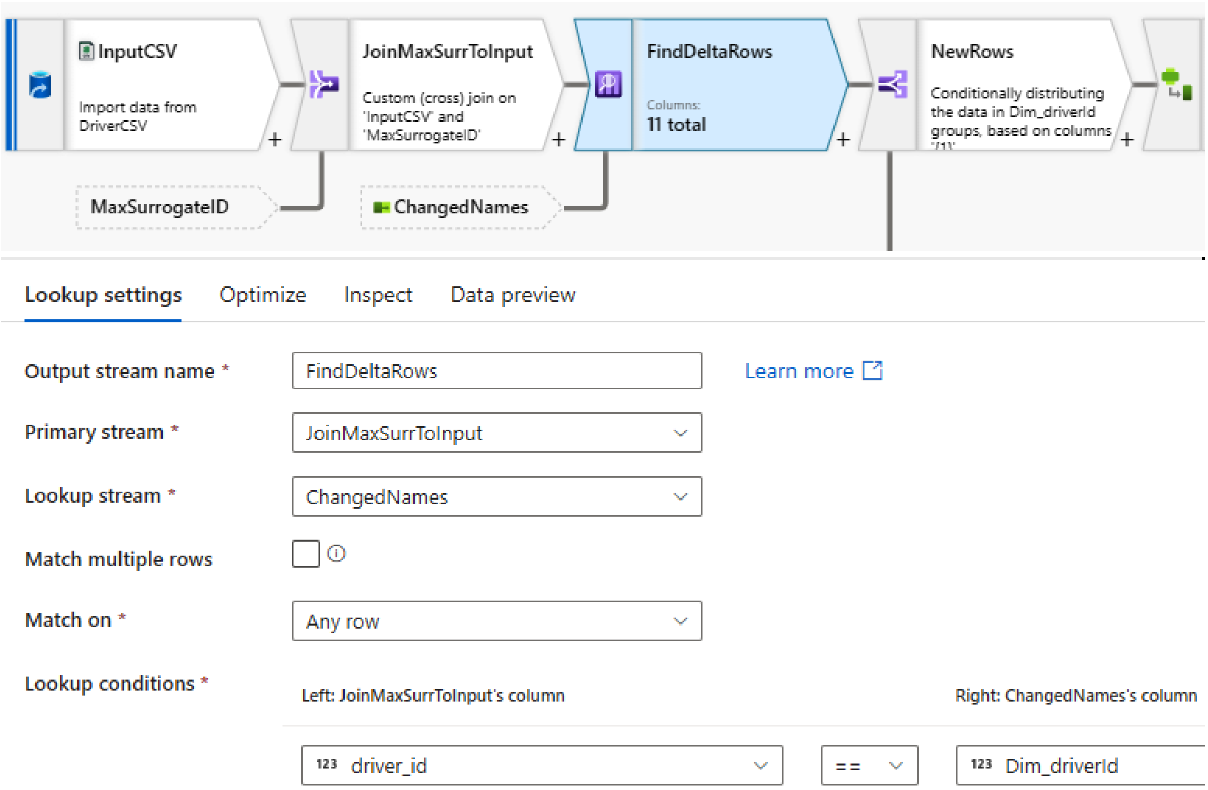
Chapter 6: Implementing Logical Data Structures

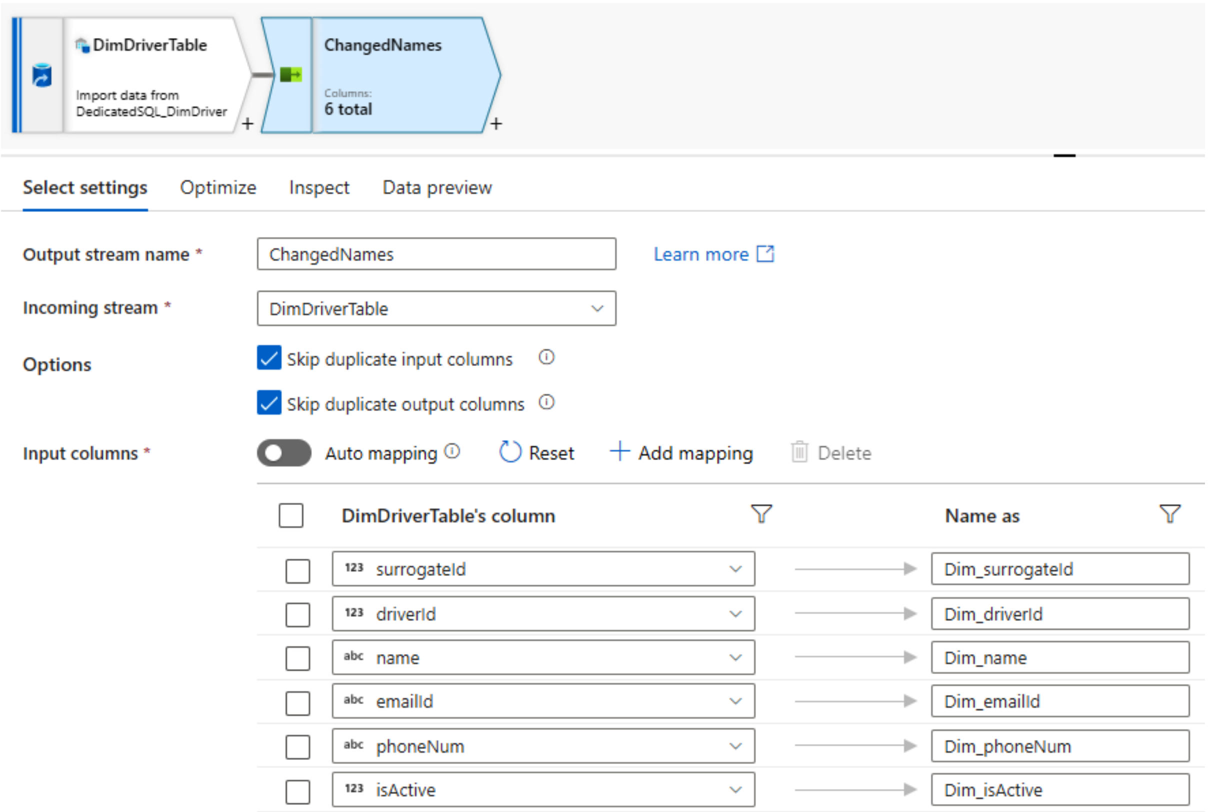


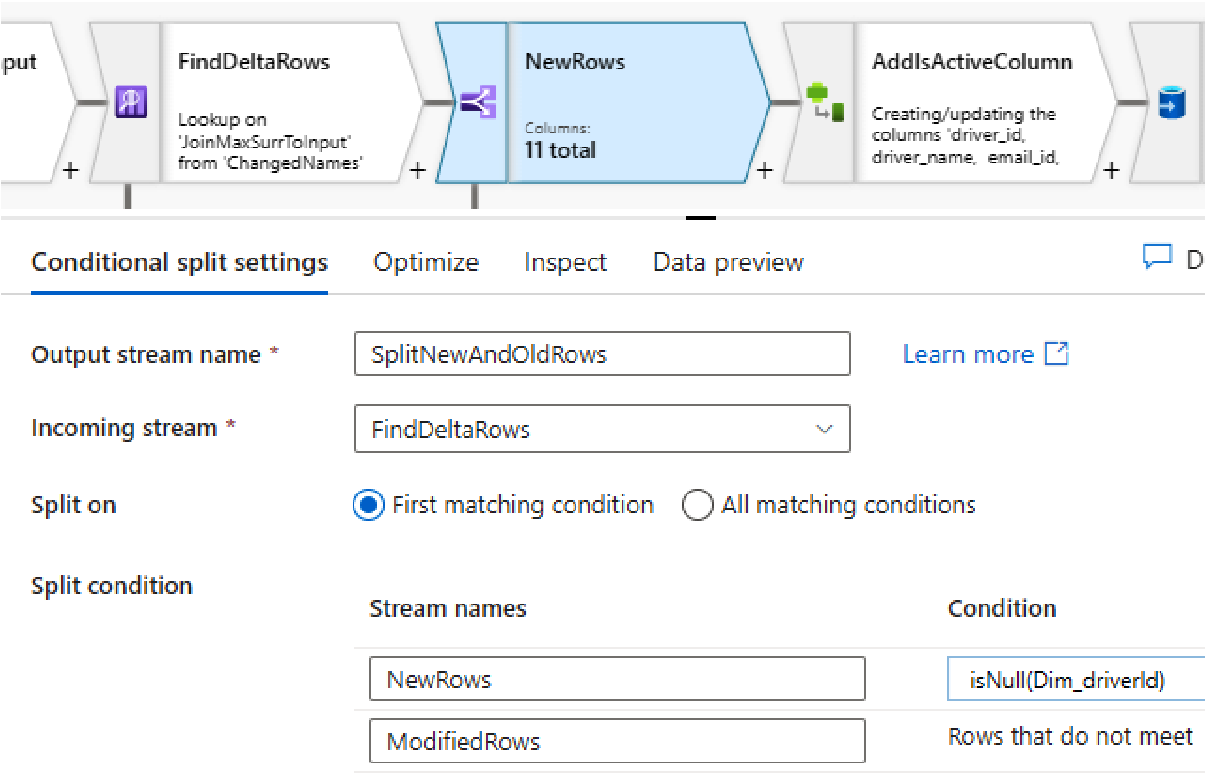

Dataflow1
Validate
Data flow debug
InputCSV
Columns:
4 total
Add Source
Source settings
Source options
Projection
Optimize
Inspect
Data preview
Output stream name *
InputCSV
Learn more
Source type *
Integration dataset
Inline
Workspace DB
Dataset *
DriverCSV
Test connection
Open
New
Options
Allow schema drift
Infer drifted column types
Validate schema
Skip line count

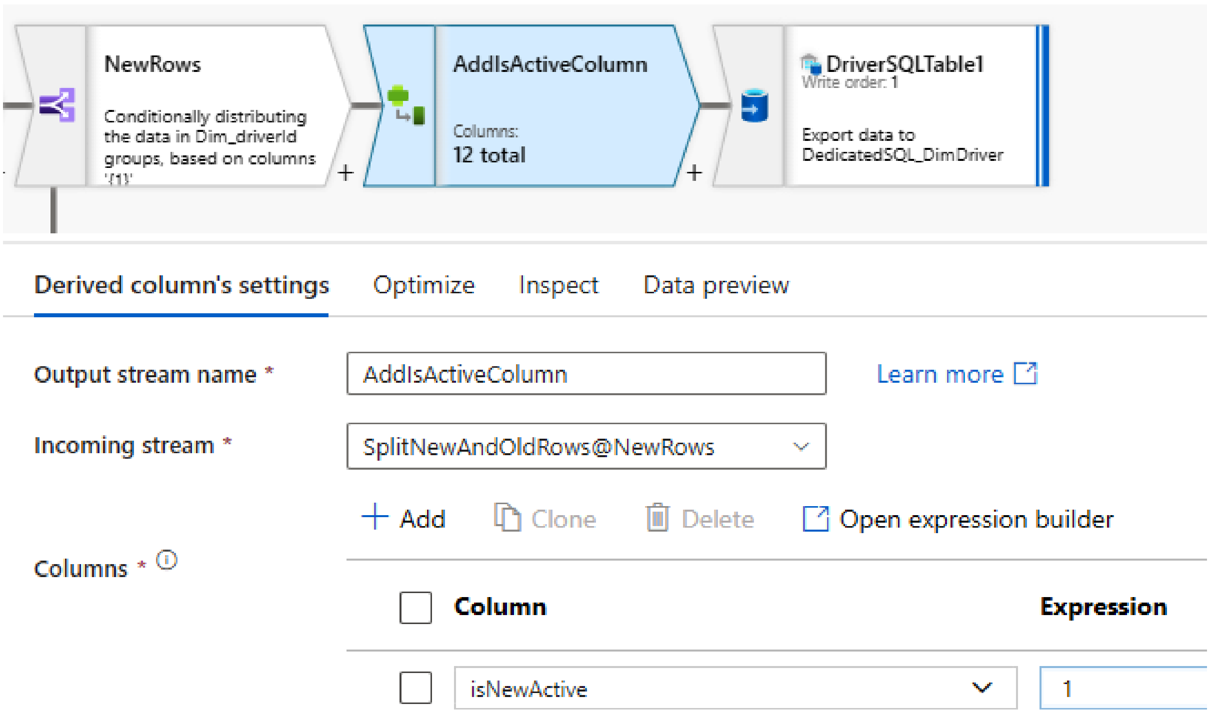

MaxSurrogateID
Columns:
1 total
Source settings
Source options
Projection
Optimize
Inspect
Data preview
Input
Table
Query
Stored procedure
Query *
SELECT max(surrogateId) as MaxSurr From [dbo].[DimDriver]
Import projection
Enable staging
Batch size
Isolation level
Read uncommitted

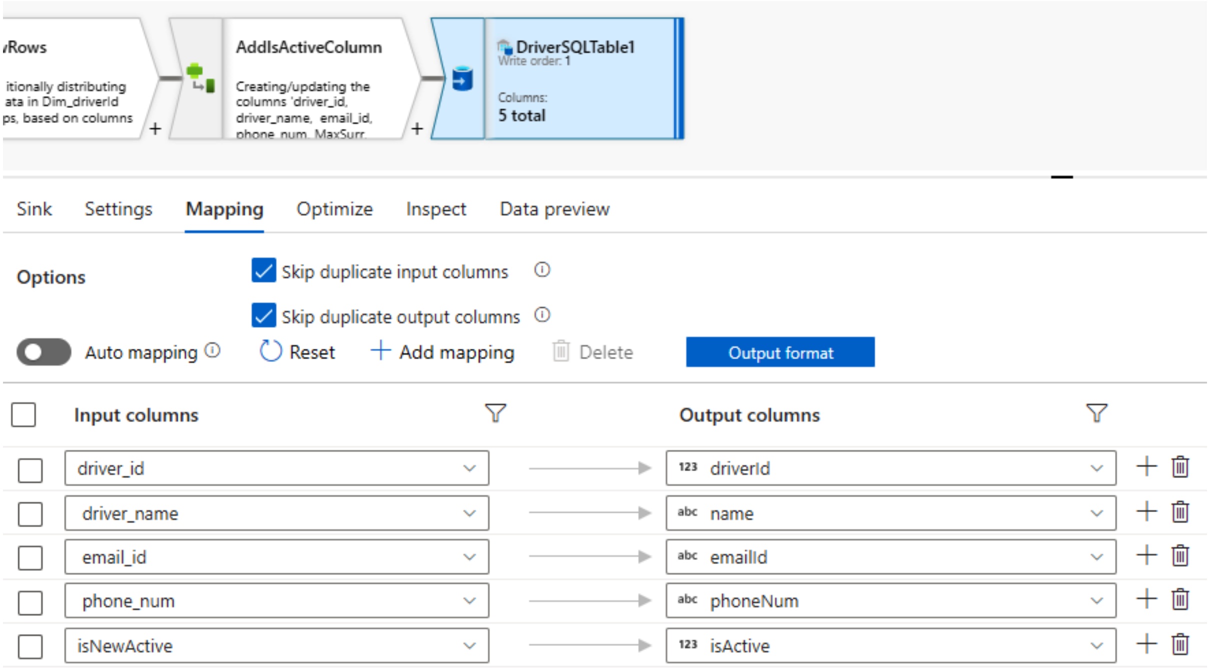

Reference:
1
Columns:
4 total
JoinMaxSurrToInput
Columns:
5 total
FindDeltaRows
Lookup on 'JoinMaxSurrToInput' from 'ChangedNames'
MaxSurrogateID
ChangedNames
Join settings
Optimize
Inspect
Data preview
Cross join condition must have at least one column from each stream, 'InputCSV' stream is not bein
Output stream name *
JoinMaxSurrToInput
Learn more
Left stream *
InputCSV
Right stream *
MaxSurrogateID
Join type *
Full outer
Inner
Left outer
Right outer
Custom (cross)
Condition *
true()

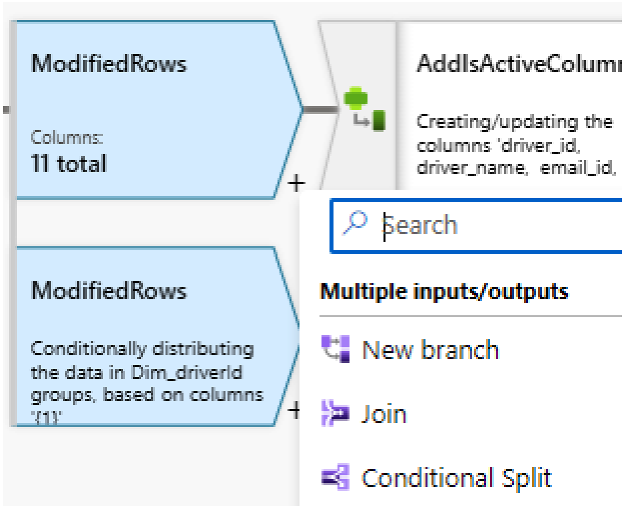

InputCSV
Import data from DriverCSV
JoinMaxSurrToInput
Custom (cross) join on 'InputCSV' and 'MaxSurrogateID'
FindDeltaRows
Columns:
11 total
NewRows
Conditionally distributing the data in Dim_driverId groups, based on columns
MaxSurrogateID
ChangedNames
Lookup settings
Optimize
Inspect
Data preview
Output stream name *
FindDeltaRows
Learn more
Primary stream *
JoinMaxSurrToInput
Lookup stream *
ChangedNames
Match multiple rows
Match on *
Any row
Lookup conditions *
Left: JoinMaxSurrToInput's column
Right: ChangedNames's column
123 driver_id
==
123 Dim_driverId

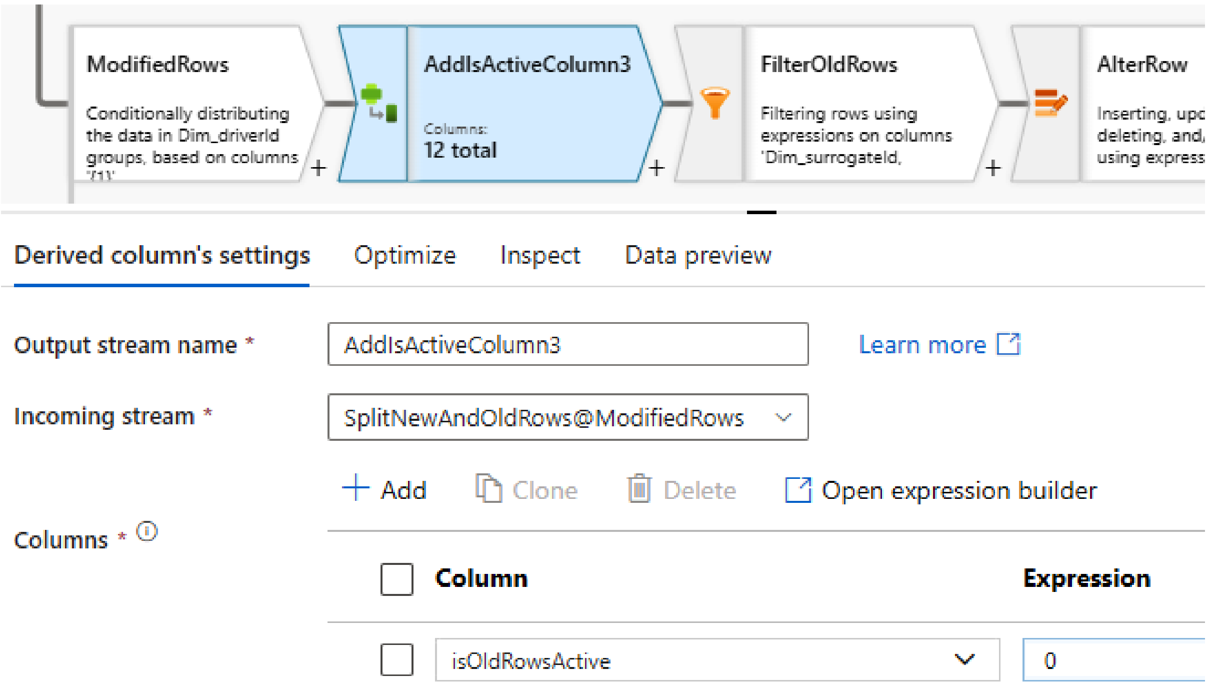

DimDriverTable
Import data from DedicatedSQL_DimDriver
ChangedNames
Columns:
6 total
Select settings
Optimize
Inspect
Data preview
Output stream name *
ChangedNames
Learn more
Incoming stream *
DimDriverTable
Options
Skip duplicate input columns
Skip duplicate output columns
Input columns *
Auto mapping
Reset
Add mapping
Delete
DimDriverTable's column
Name as
123 surrogateId
Dim_surrogateId
123 driverId
Dim_driverId
abc name
Dim_name
abc emailId
Dim_emailId
abc phoneNum
Dim_phoneNum
123 isActive
Dim_isActive

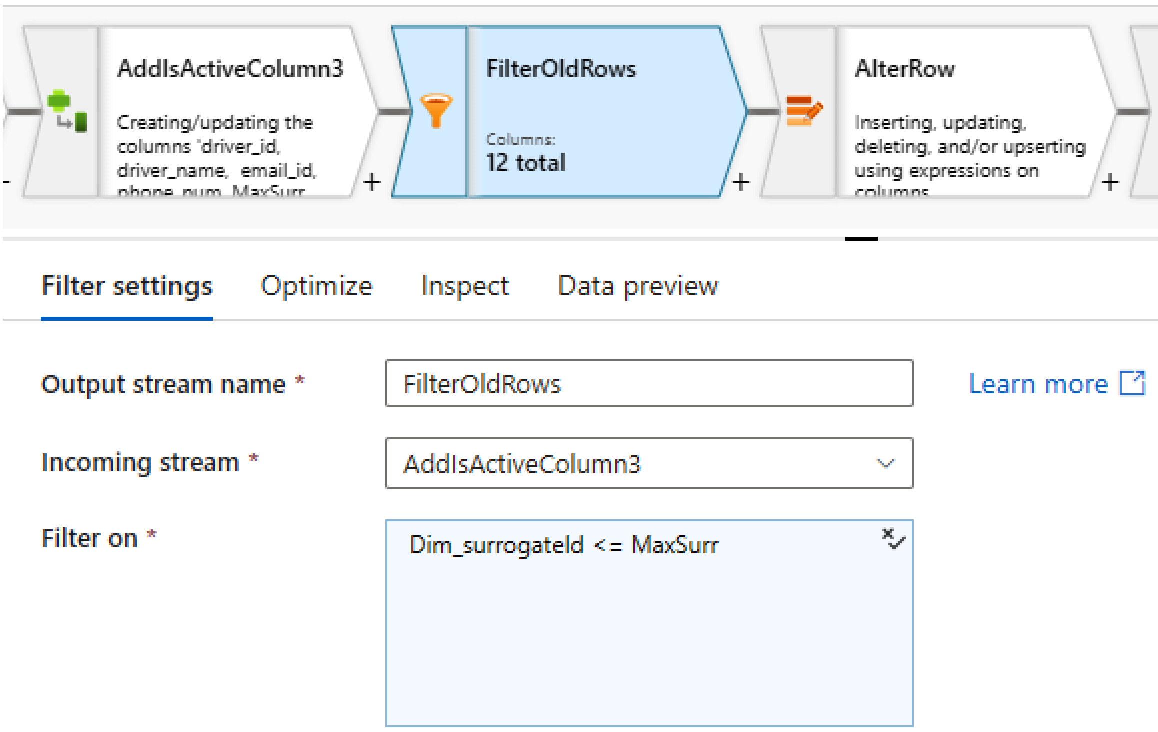

put
FindDeltaRows
Lookup on 'JoinMaxSurrToInput' from 'ChangedNames'
NewRows
Columns:
11 total
AddIsActiveColumn
Creating/updating the columns 'driver_id, driver_name, email_id,
Conditional split settings
Optimize
Inspect
Data preview
Output stream name *
SplitNewAndOldRows
Learn more
Incoming stream *
FindDeltaRows
Split on
First matching condition
All matching conditions
Split condition
Stream names
Condition
NewRows
isNull(Dim_driverId)
ModifiedRows
Rows that do not meet

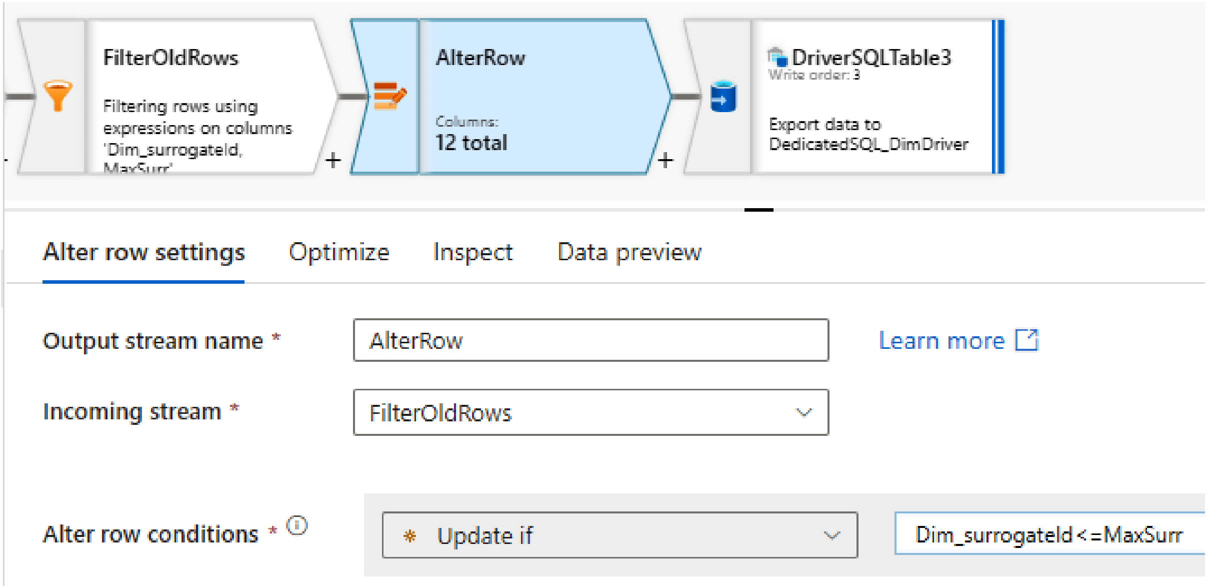

NewRows
Conditionally distributing the data in Dim_driverId groups, based on columns
AddIsActiveColumn
Columns:
12 total
DriverSQLTable1
Write order: 1
Export data to DedicatedSQL_DimDriver
Derived column's settings
Optimize
Inspect
Data preview
Output stream name *
AddIsActiveColumn
Learn more
Incoming stream *
SplitNewAndOldRows@NewRows
Add
Clone
Delete
Open expression builder
Columns *
Column
Expression
isNewActive
1

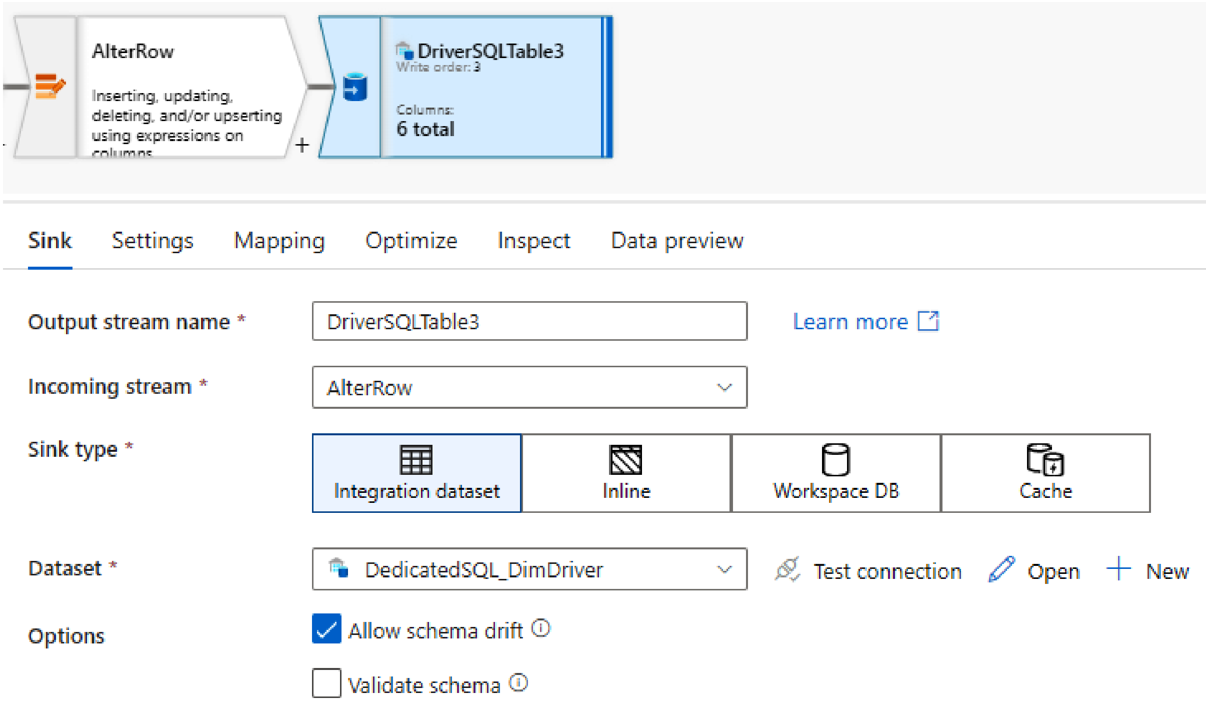

AddIsActiveColumn
DriverSQLTable1
Write order: 1
Columns:
5 total
Sink
Settings
Mapping
Optimize
Inspect
Data preview
Options
Skip duplicate input columns
Skip duplicate output columns
Auto mapping
Reset
Add mapping
Delete
Output format
Input columns
Output columns
driver_id
driverId
driver_name
name
email_id
emailId
phone_num
phoneNum
isNewActive
isActive

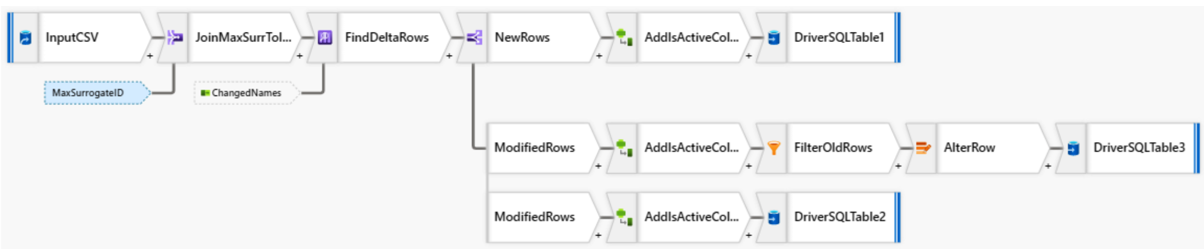

ModifiedRows
Columns:
11 total
AddIsActiveColum
Creating/updating the columns 'driver_id, driver_name, email_id,
ModifiedRows
Conditionally distributing the data in Dim_driverId groups, based on columns '{1}'
Search
Multiple inputs/outputs
New branch
Join
Conditional Split

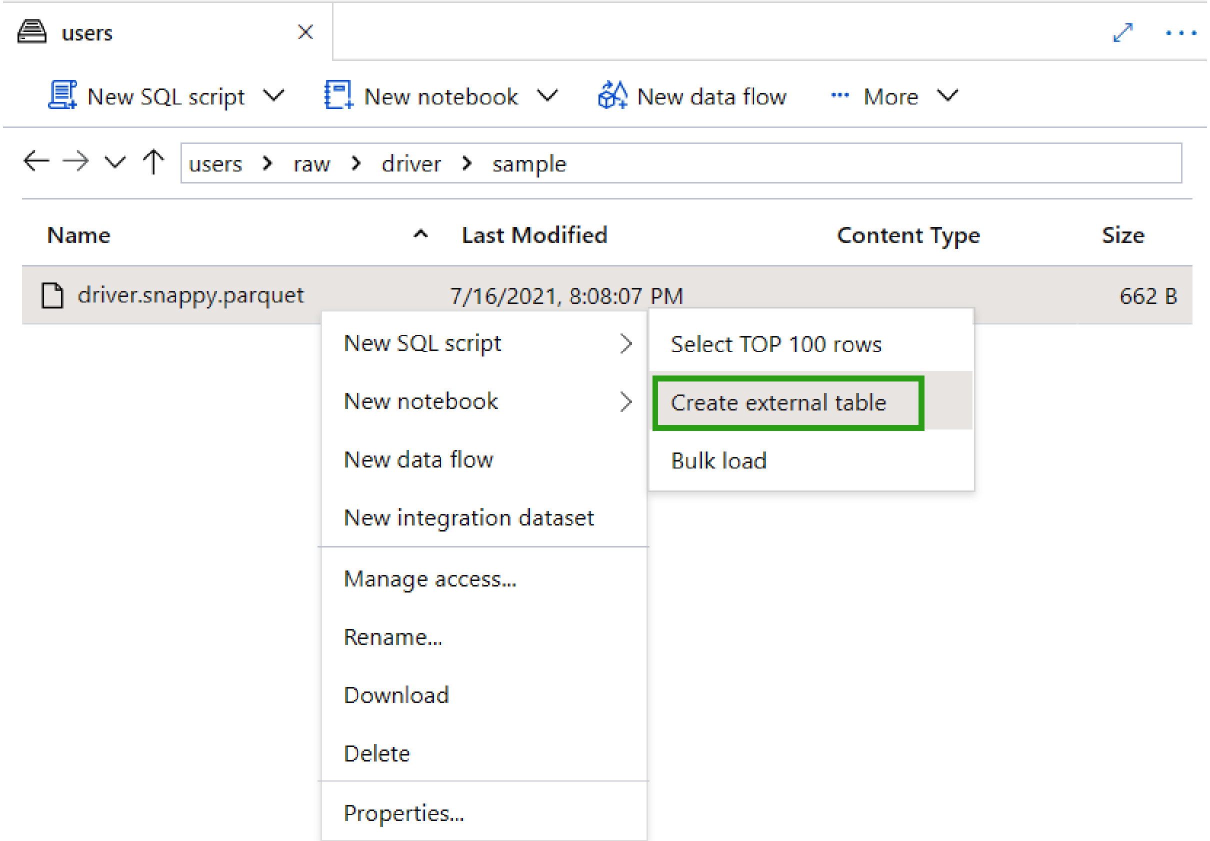

ModifiedRows
Conditionally distributing the data in Dim_driverId groups, based on columns
AddIsActiveColumn3
Columns:
12 total
FilterOldRows
Filtering rows using expressions on columns 'Dim_surrogateId,
AlterRow
Derived column's settings
Optimize
Inspect
Data preview
Output stream name *
AddIsActiveColumn3
Learn more
Incoming stream *
SplitNewAndOldRows@ModifiedRows
Add
Clone
Delete
Open expression builder
Columns *
Column
Expression
isOldRowsActive
0

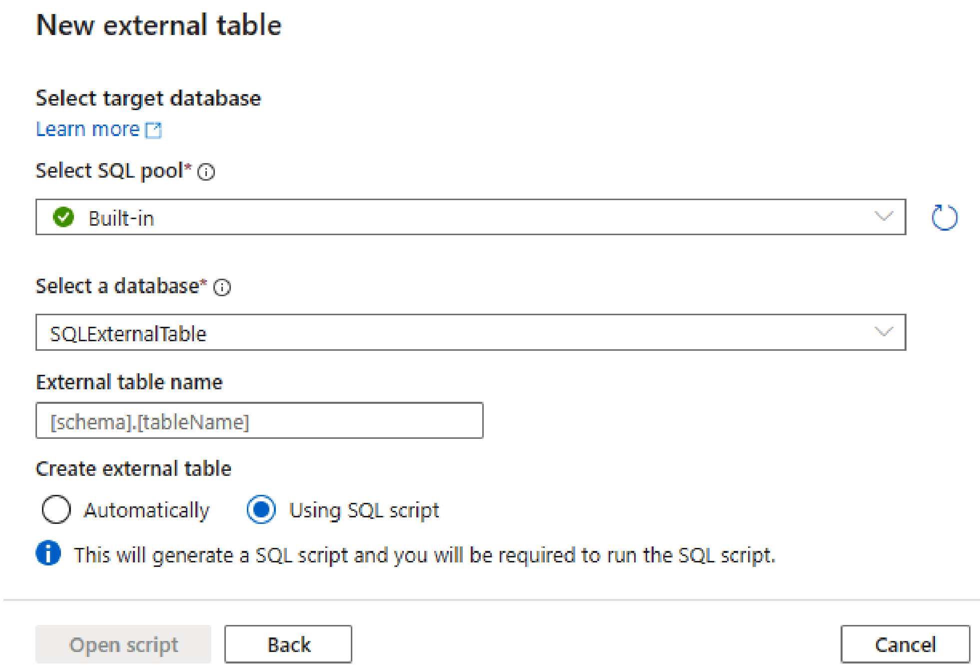

AddIsActiveColumn3
Creating/updating the columns 'driver_id, driver_name, email_id,
FilterOldRows
Columns:
12 total
AlterRow
Inserting, updating, deleting, and/or upserting using expressions on
Filter settings
Optimize
Inspect
Data preview
Output stream name *
FilterOldRows
Learn more
Incoming stream *
AddIsActiveColumn3
Filter on *
Dim_surrogateId <= MaxSurr


FilterOldRows
Filtering rows using expressions on columns 'Dim_surrogateId, MaxSurr'
AlterRow
Columns:
12 total
DriverSQLTable3
Write order: 3
Export data to DedicatedSQL_DimDriver
Alter row settings
Optimize
Inspect
Data preview
Output stream name *
AlterRow
Learn more
Incoming stream *
FilterOldRows
Alter row conditions *
Update if
Dim_surrogateId<=MaxSurr

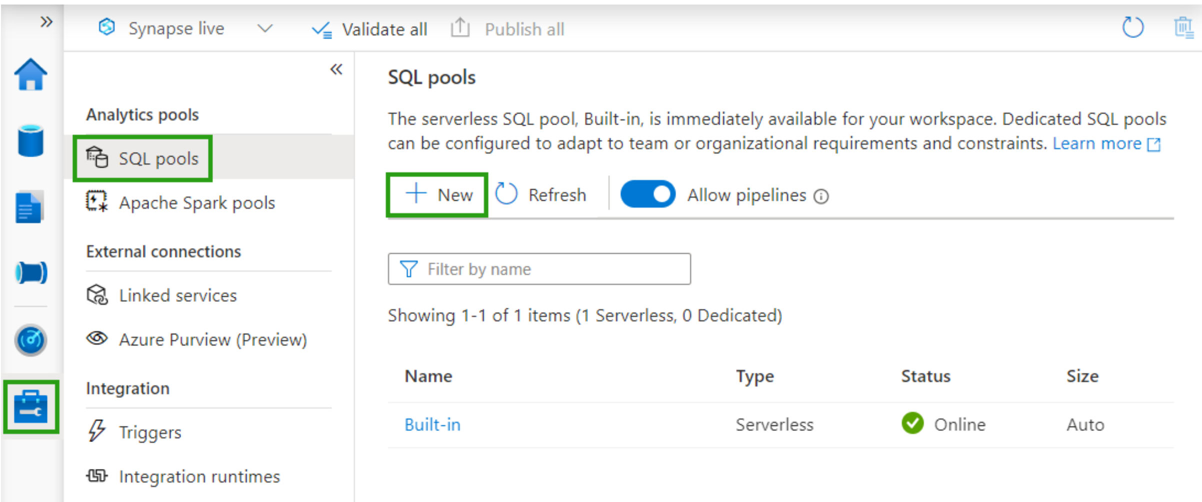

AlterRow
Inserting, updating, deleting, and/or upserting using expressions on columns
DriverSQLTable3
Write order: 3
Columns:
6 total
Sink
Settings
Mapping
Optimize
Inspect
Data preview
Output stream name *
DriverSQLTable3
Learn more
Incoming stream *
AlterRow
Sink type *
Integration dataset
Inline
Workspace DB
Cache
Dataset *
DedicatedSQL_DimDriver
Test connection
Open
New
Options
Allow schema drift
Validate schema

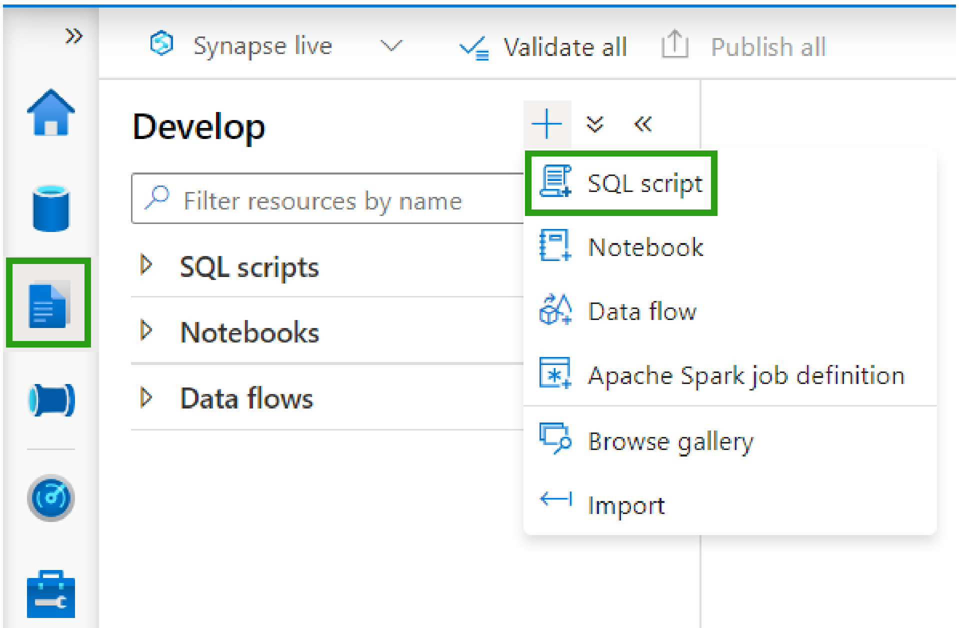

InputCSV
JoinMaxSurrTol...
FindDeltaRows
NewRows
AddIsActiveCol...
DriverSQLTable1
MaxSurrogateID
ChangedNames
ModifiedRows
AddIsActiveCol...
FilterOldRows
AlterRow
DriverSQLTable3
ModifiedRows
AddIsActiveCol...
DriverSQLTable2


users
New SQL script
New notebook
New data flow
More
users › raw › driver › sample
Name
Last Modified
Content Type
Size
driver.snappy.parquet
7/16/2021, 8:08:07 PM
662 B
New SQL script
New notebook
New data flow
New integration dataset
Manage access...
Rename...
Download
Delete
Properties...
Select TOP 100 rows
Create external table
Bulk load


New external table
Select target database
Learn more
Select SQL pool*
Built-in
Select a database*
SQLExternalTable
External table name
[schema].[tableName]
Create external table
Automatically
Using SQL script
This will generate a SQL script and you will be required to run the SQL script.
Open script
Back
Cancel

Chapter 7: Implementing the Serving Layer





Run | Undo | Publish | Query plan | Connect to: Built-in

```sql
CREATE EXTERNAL TABLE TripsExtTable (
    [tripId] VARCHAR (10),
    [driverId] VARCHAR (10),
    [customerId] VARCHAR (10),
    [cabId] VARCHAR (10),
    [tripDate] VARCHAR (10),
    [startLocation] VARCHAR (50),
    [endLocation] VARCHAR (50)
)
WITH (
    LOCATION = '/parquet/trips/*.parquet',
    DATA_SOURCE = [Dp203DataSource],
    FILE_FORMAT = [Dp203ParquetFormat]
)
```

Results | Messages

View: Table | Chart | Export results

Search

tripId	driverId	customerId	cabId	tripDate	startLocation	endLocation
107	207	307	407	20220203	Los Angeles	San Diego
108	208	308	408	20220301	Phoenix	Las Vegas
100	200	300	400	20220101	New York	New Jersey
101	201	301	401	20220102	Tempe	Phoenix
111	211	311	411	20220303	New York	New Jersey

SQLTripsExtTable

Run | Undo | Publish | Query plan | Connect to: Built-in | Use database: SQLExternalTable

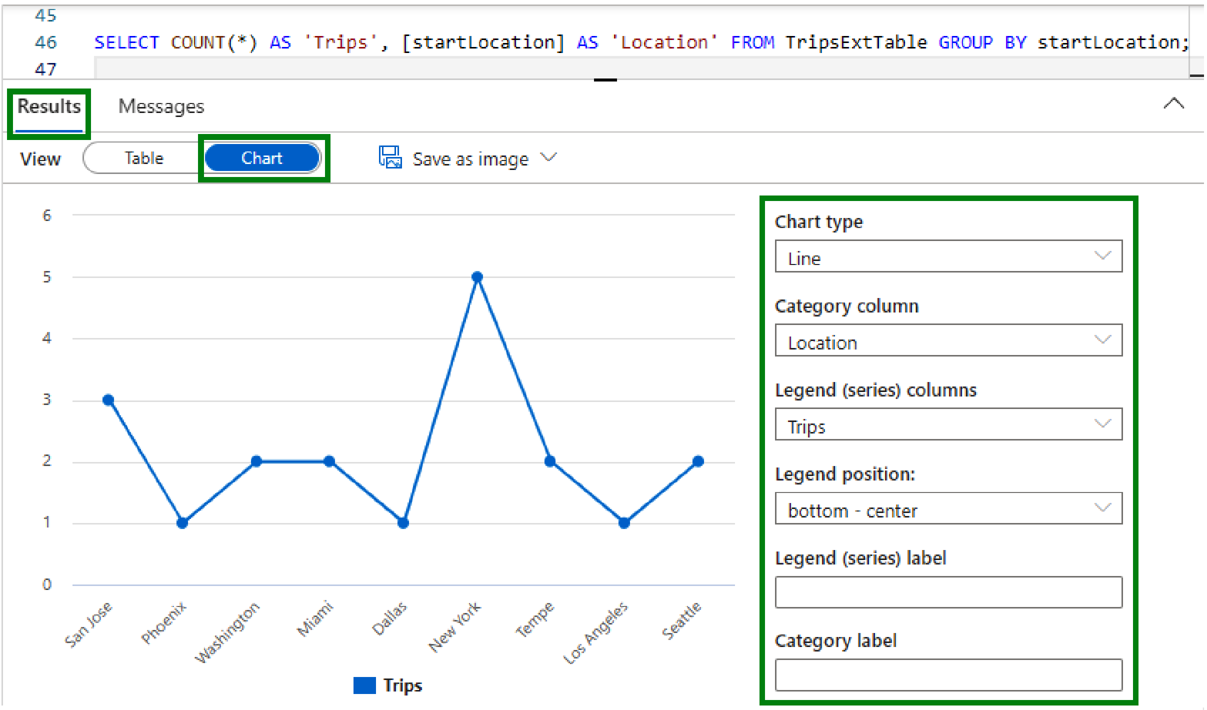
```sql
CREATE EXTERNAL TABLE TripsExtTable (
    [RideID] varchar(50),
    [DriverID] varchar(50),
    [CustomerID] varchar(50),
    [cabID] varchar(50),
    [DateID] datetime2(7),
    [StartLocation] varchar(50),
    [EndLocation] varchar(50)
    )
    WITH (
    LOCATION = 'raw/trips/out/year=2022/*/*/*.parquet',
    DATA_SOURCE = [users_iacstoreacct_dfs_core_windows_net],
    FILE_FORMAT = [SynapseParquetFormat]
    )
GO
```

Results | Messages

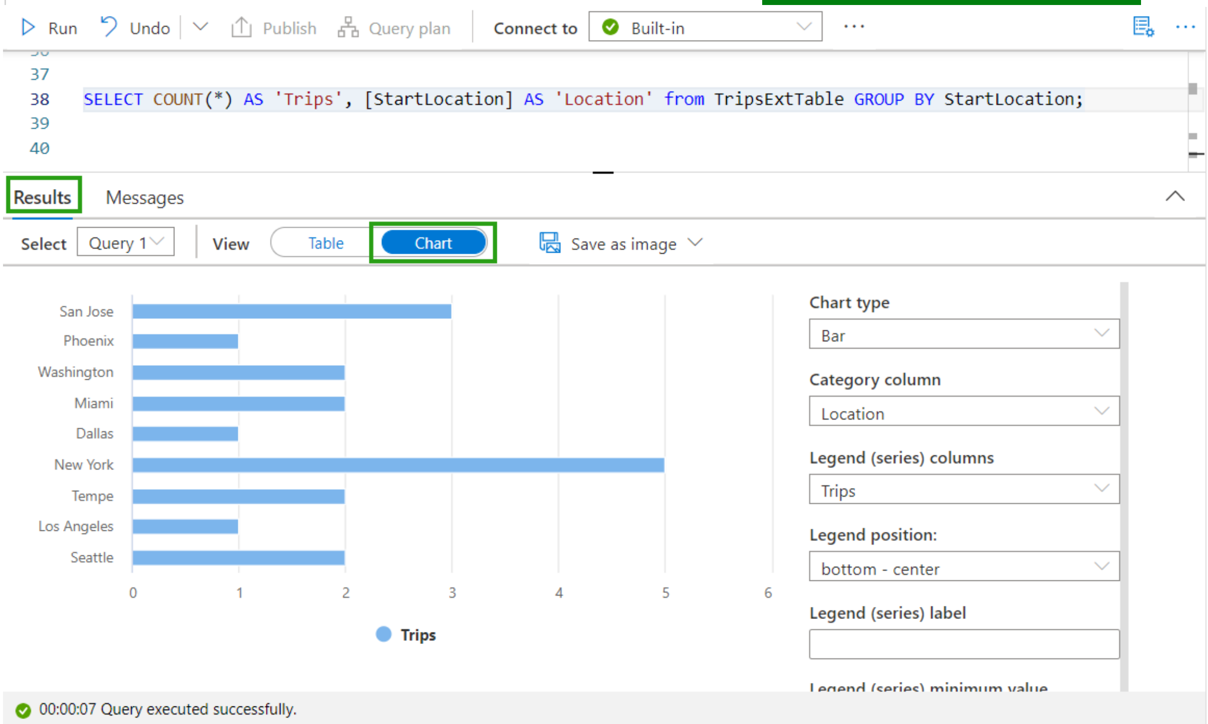
View: Table | Chart | Export results

Search

RideID	DriverID	CustomerID	cabID	DateID	StartLocation	EndLocation
104	204	304	404	2022-01-03T00:...	New York	Washington
106	206	306	406	2022-02-02T00:...	Seattle	Redmond
110	210	310	410	2022-03-03T00:...	Dallas	Austin



45
46 SELECT COUNT(*) AS 'Trips', [startLocation] AS 'Location' FROM TripsExtTable GROUP BY startLocation;
47
Results
Messages
View
Table
Chart
Save as image
6
5
4
3
2
1
0
San Jose
Phoenix
Washington
Miami
Dallas
New York
Tempe
Los Angeles
Seattle
Trips
Chart type
Line
Category column
Location
Legend (series) columns
Trips
Legend position:
bottom - center
Legend (series) label
Category label



Run
Undo
Publish
Query plan
Connect to
Built-in
37
38 SELECT COUNT(*) AS 'Trips', [StartLocation] AS 'Location' from TripsExtTable GROUP BY StartLocation;
39
40
Results
Messages
Select
Query 1
View
Table
Chart
Save as image
San Jose
Phoenix
Washington
Miami
Dallas
New York
Tempe
Los Angeles
Seattle
0
1
2
3
4
5
6
Trips
Chart type
Bar
Category column
Location
Legend (series) columns
Trips
Legend position:
bottom - center
Legend (series) label
00:00:07 Query executed successfully.

```
%%pyspark
df = spark.read.load('abfss://users@iacstoreacct.dfs.core.windows.net/raw/trips/out/year=2022/*/*/*.parquet', format='parquet')

spark.sql("CREATE DATABASE IF NOT EXISTS TripsDatabase")
df.write.mode("overwrite").saveAsTable("TripsTable")
display(df)
df = spark.sql("""
   SELECT StartLocation AS Location,
   COUNT(*) AS Trips
   from TripsTable
   GROUP BY StartLocation
""")
df.orderBy("Trips", ascending=False).show(5)
```

Command executed in 6 sec 210 ms by newton.packt on 6:19:50 PM, 7/21/21

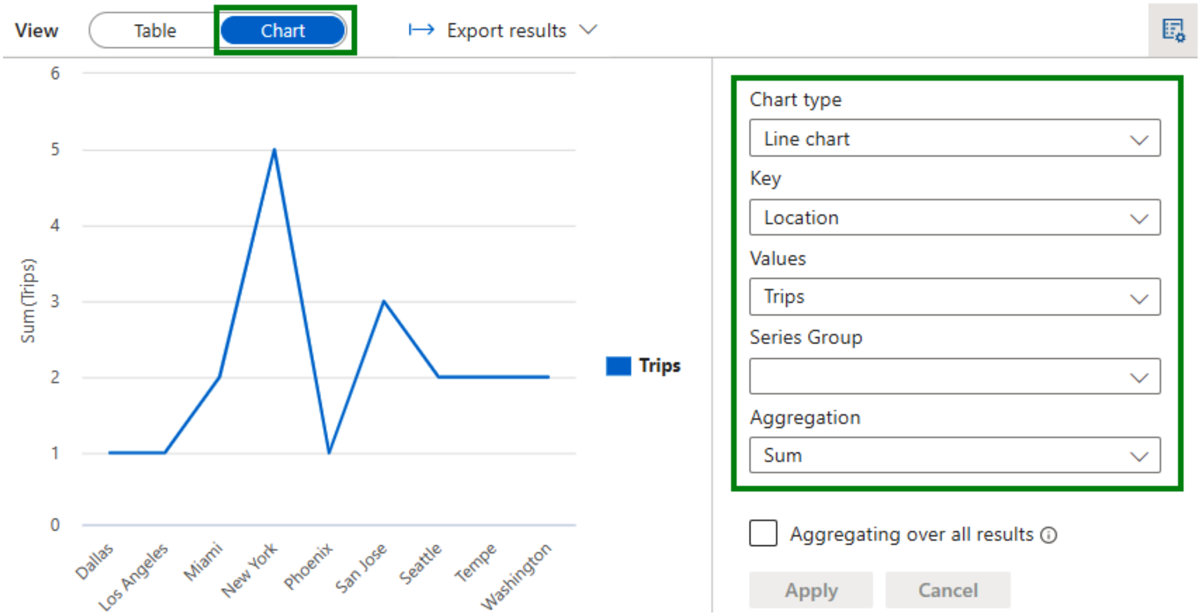
Job execution Succeeded Spark 2 executors 8 cores View in monitoring Open Spark UI

```
+--------+-----+
|Location|Trips|
+--------+-----+
|New York|    5|
|San Jose|    3|
| Seattle|    2|
|   Tempe|    2|
|   Miami|    2|
+--------+-----+
only showing top 5 rows
```

```
df = spark.read.load('abfss://[redacted].dfs.core.windows.net/parquet/trips/*.parquet',
   format='parquet')
spark.sql("CREATE DATABASE IF NOT EXISTS TripsDatabase")
df.write.mode("overwrite").option("overwriteSchema", "true").saveAsTable("TripsTable")
sqldf = spark.sql("""
   SELECT COUNT(*) AS Trips,
   startLocation AS Location
   FROM TripsTable
   GROUP BY startLocation """)
display(sqldf)
```

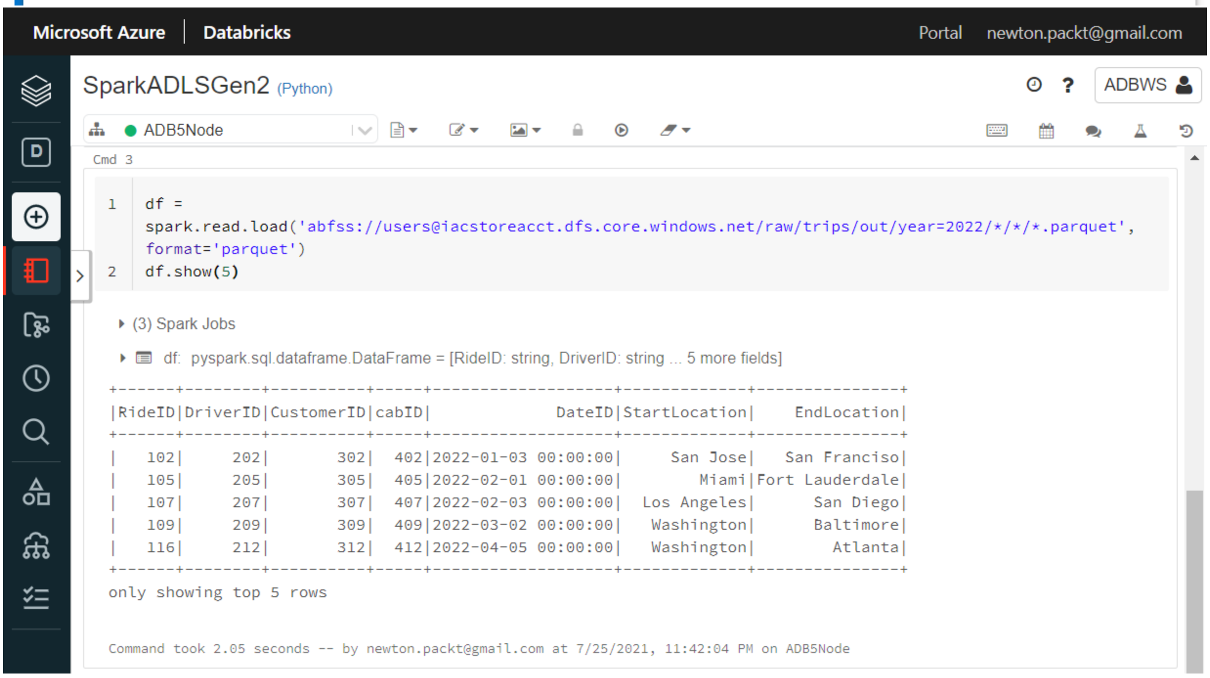
✓ 3 min 54 sec - Apache Spark session started in 2 min 58 sec 614 ms. Command executed in 55 sec 882 ms by newton.pac...

Job execution Succeeded Spark 2 executors 8 cores View in monitoring Open Spark UI

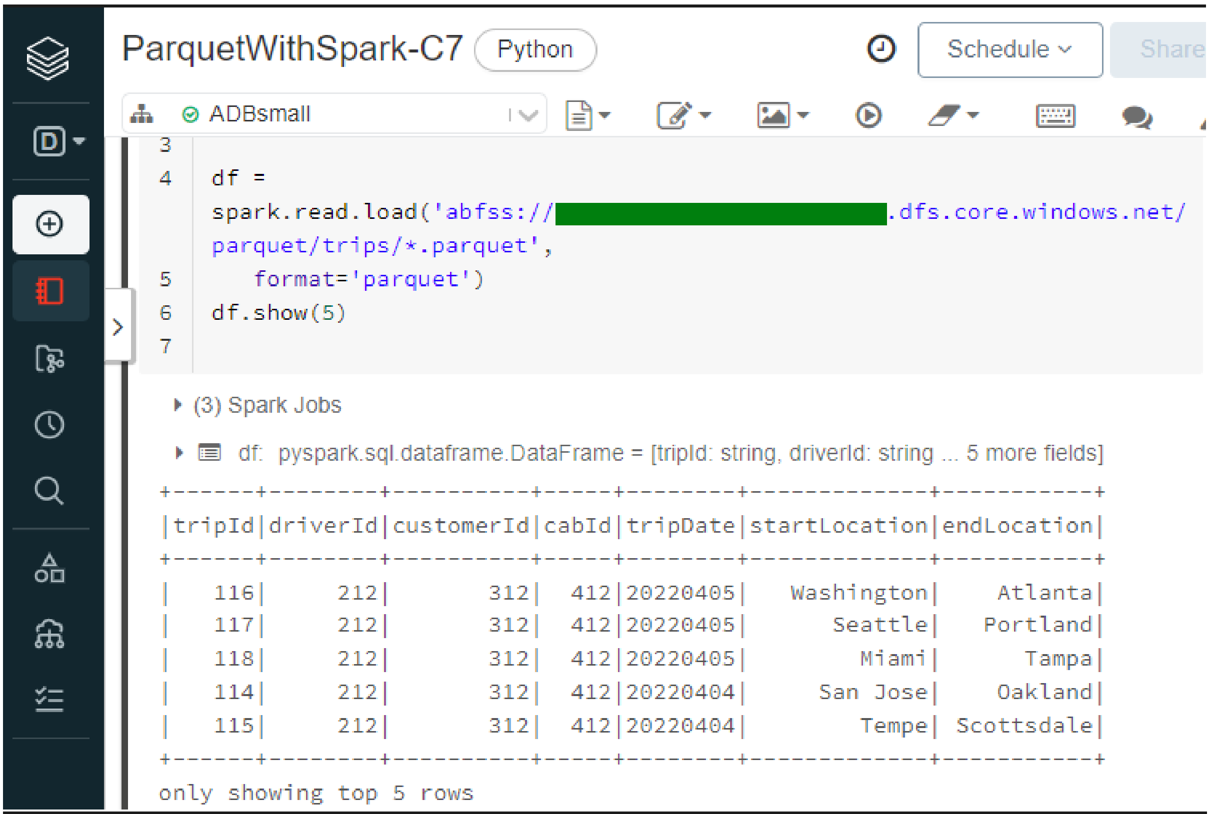


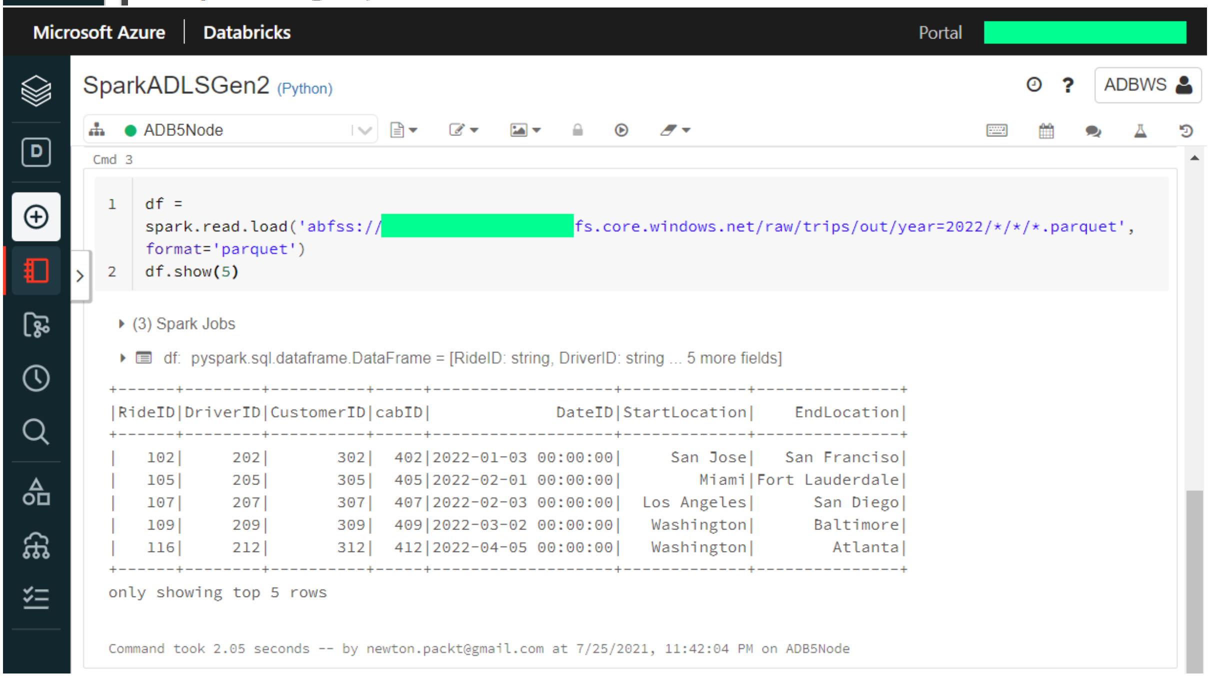
%%pyspark
df = spark.read.load('abfss://.core.windows.net/raw/trips/out/year=2022/*/*/*.parquet', format='parquet')

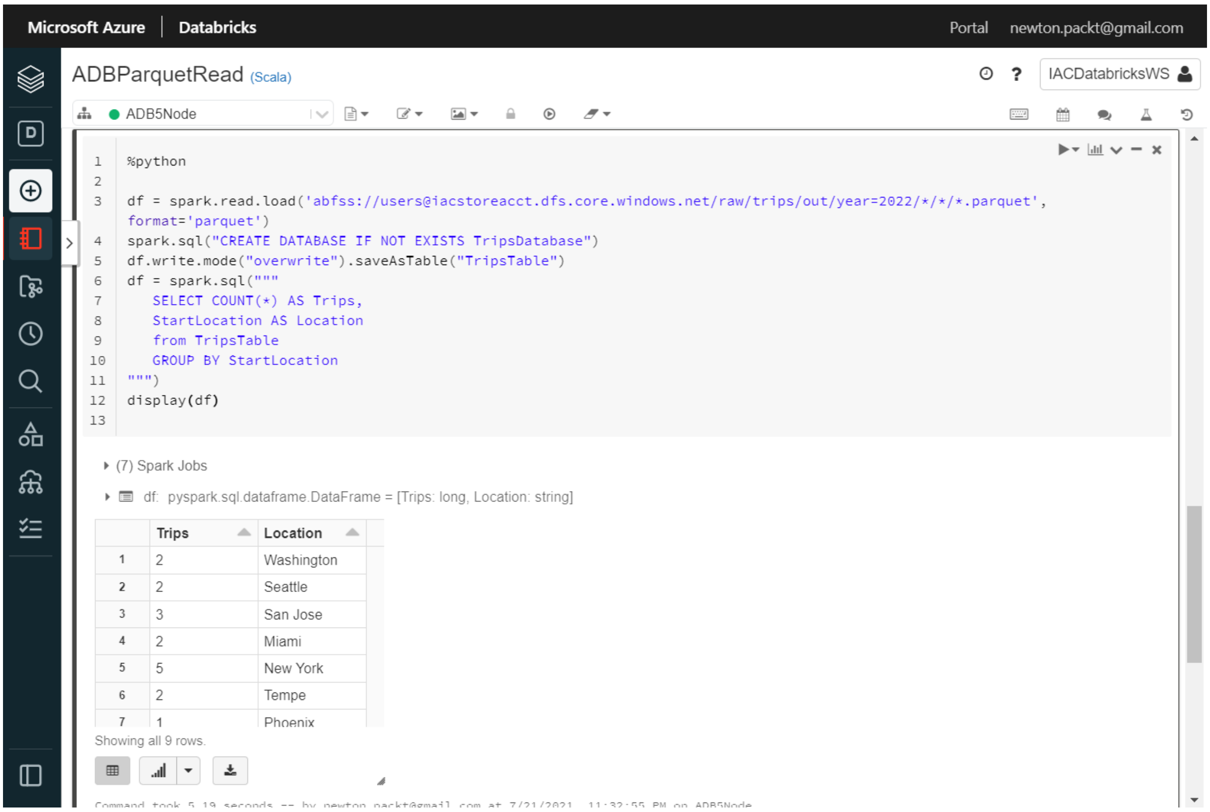
spark.sql("CREATE DATABASE IF NOT EXISTS TripsDatabase")
df.write.mode("overwrite").saveAsTable("TripsTable")
display(df)
df = spark.sql("""
SELECT StartLocation AS Location,
COUNT(*) AS Trips
from TripsTable
GROUP BY StartLocation
""")
df.orderBy("Trips", ascending=False).show(5)
Command executed in 6 sec 210 ms by newton.packt on 6:19:50 PM, 7/21/21
Job execution Succeeded
Spark 2 executors 8 cores
View in monitoring
Open Spark UI
+--------+-----+
|Location|Trips|
+--------+-----+
|New York| 5|
|San Jose| 3|
Seattle	2
Tempe	2
Miami	2
+--------+-----+
only showing top 5 rows



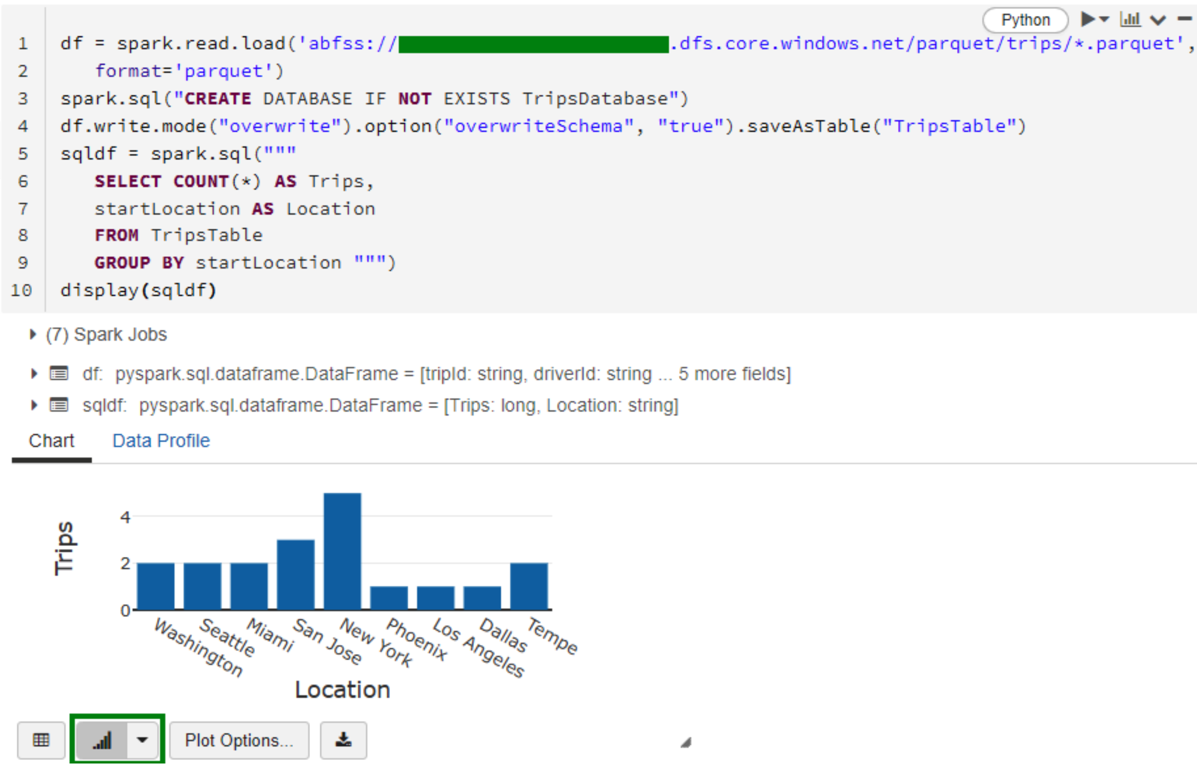
Microsoft Azure
Databricks
Portal
newton.packt@gmail.com
SparkADLSGen2 (Python)
ADBWS
ADB5Node
Cmd 3
df =
spark.read.load('abfss://users@iacstoreacct.dfs.core.windows.net/raw/trips/out/year=2022/*/*/*.parquet',
format='parquet')
df.show(5)
(3) Spark Jobs
df: pyspark.sql.dataframe.DataFrame = [RideID: string, DriverID: string ... 5 more fields]
+------+--------+----------+-----+-------------------+-------------+---------------+
|RideID|DriverID|CustomerID|cabID| DateID|StartLocation| EndLocation|
+------+--------+----------+-----+-------------------+-------------+---------------+
102	202	302	402	2022-01-03 00:00:00	San Jose	San Francisco
105	205	305	405	2022-02-01 00:00:00	Miami	Fort Lauderdale
107	207	307	407	2022-02-03 00:00:00	Los Angeles	San Diego
109	209	309	409	2022-03-02 00:00:00	Washington	Baltimore
116	212	312	412	2022-04-05 00:00:00	Washington	Atlanta
+------+--------+----------+-----+-------------------+-------------+---------------+
only showing top 5 rows
Command took 2.05 seconds -- by newton.packt@gmail.com at 7/25/2021, 11:42:04 PM on ADB5Node


ParquetWithSpark-C7
Python
Schedule
Share
ADBsmall
3
4 df =
spark.read.load('abfss://.dfs.core.windows.net/parquet/trips/*.parquet',
5 format='parquet')
6 df.show(5)
7
(3) Spark Jobs
df: pyspark.sql.dataframe.DataFrame = [tripId: string, driverId: string ... 5 more fields]
+------+--------+----------+-----+--------+-------------+-----------+
|tripId|driverId|customerId|cabId|tripDate|startLocation|endLocation|
+------+--------+----------+-----+--------+-------------+-----------+
116	212	312	412	20220405	Washington	Atlanta
117	212	312	412	20220405	Seattle	Portland
118	212	312	412	20220405	Miami	Tampa
114	212	312	412	20220404	San Jose	Oakland
115	212	312	412	20220404	Tempe	Scottsdale
+------+--------+----------+-----+--------+-------------+-----------+
only showing top 5 rows

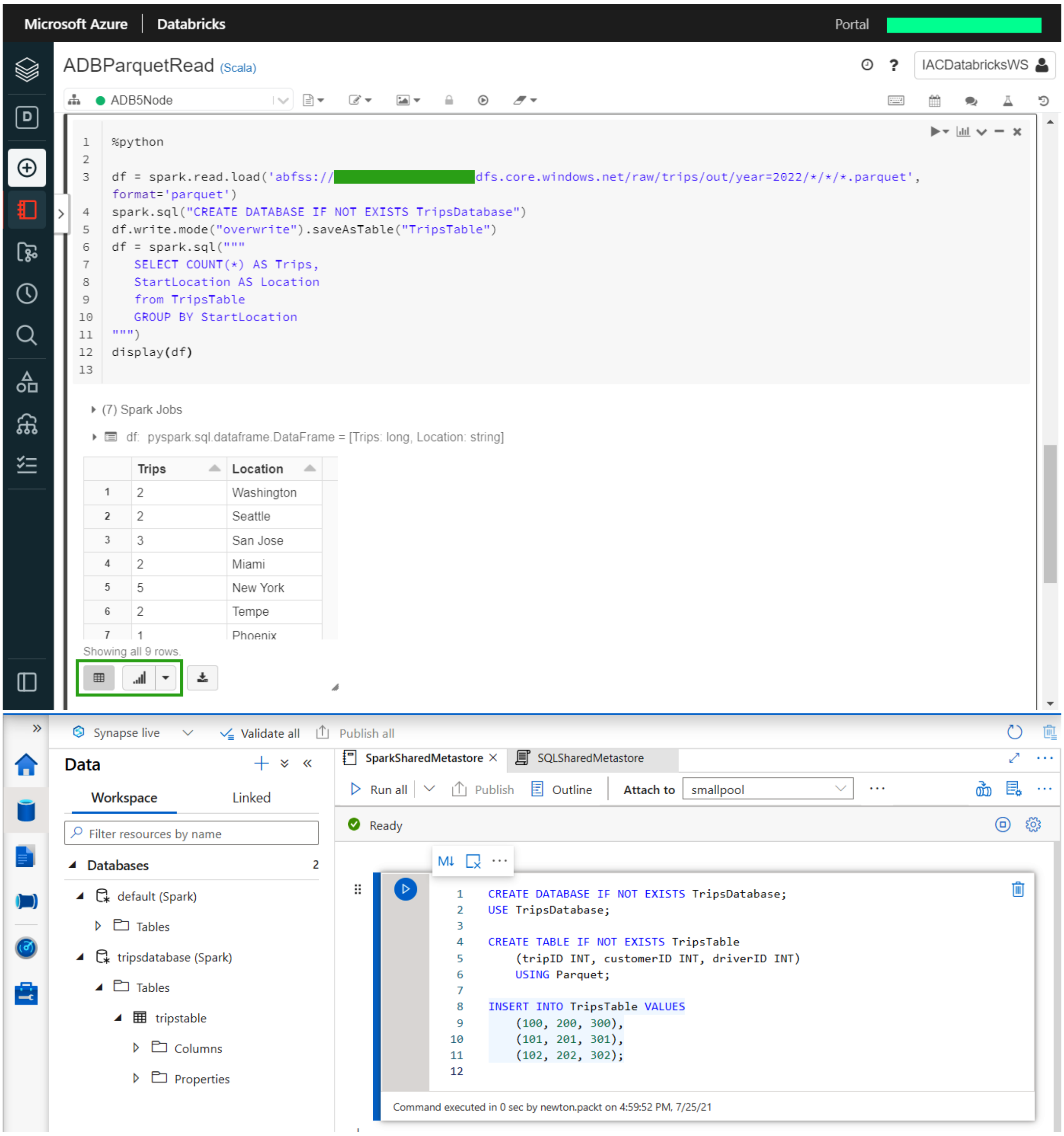

Microsoft Azure
Databricks
Portal
SparkADLSGen2 (Python)
ADBWS
ADB5Node
Cmd 3
1 df =
spark.read.load('abfss://fs.core.windows.net/raw/trips/out/year=2022/*/*/*.parquet',
format='parquet')
2 df.show(5)
(3) Spark Jobs
df: pyspark.sql.dataframe.DataFrame = [RideID: string, DriverID: string ... 5 more fields]
+------+--------+----------+-----+-------------------+-------------+--------------+
|RideID|DriverID|CustomerID|cabID| DateID|StartLocation| EndLocation|
+------+--------+----------+-----+-------------------+-------------+--------------+
102	202	302	402	2022-01-03 00:00:00	San Jose	San Franciso
105	205	305	405	2022-02-01 00:00:00	Miami	Fort Lauderdale
107	207	307	407	2022-02-03 00:00:00	Los Angeles	San Diego
109	209	309	409	2022-03-02 00:00:00	Washington	Baltimore
116	212	312	412	2022-04-05 00:00:00	Washington	Atlanta
+------+--------+----------+-----+-------------------+-------------+--------------+
only showing top 5 rows
Command took 2.05 seconds -- by newton.packt@gmail.com at 7/25/2021, 11:42:04 PM on ADB5Node

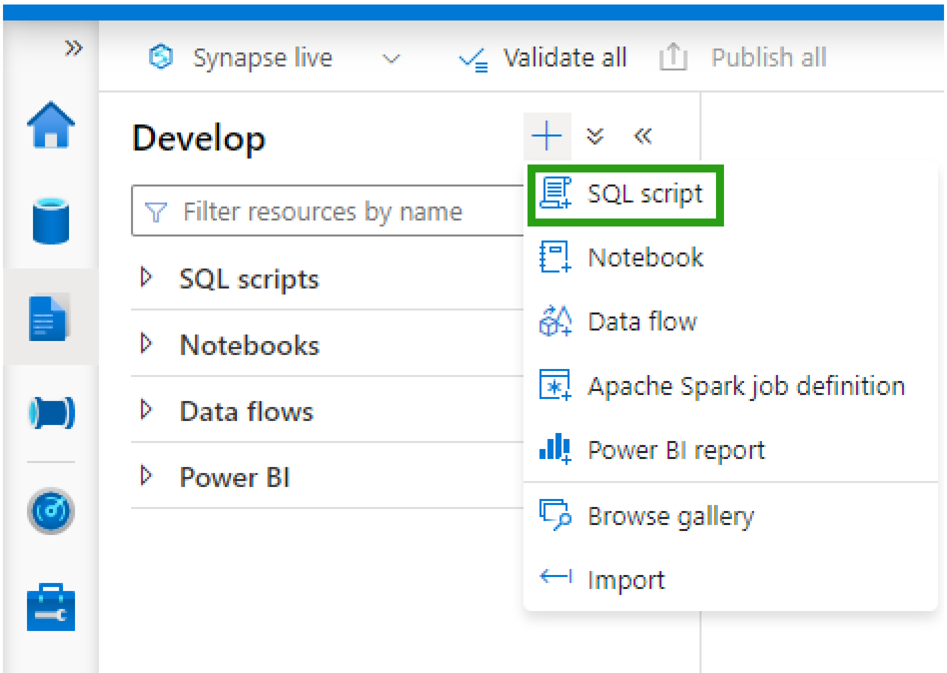


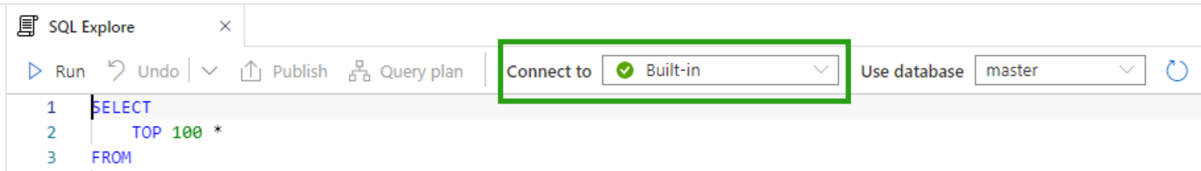
Microsoft Azure | Databricks
Portal newton.packt@gmail.com
ADBParquetRead (Scala)
IACDatabricksWS
ADB5Node
%python
df = spark.read.load('abfss://users@iacstoreacct.dfs.core.windows.net/raw/trips/out/year=2022/*/*/*.parquet',
format='parquet')
spark.sql("CREATE DATABASE IF NOT EXISTS TripsDatabase")
df.write.mode("overwrite").saveAsTable("TripsTable")
df = spark.sql("""
SELECT COUNT(*) AS Trips,
StartLocation AS Location
from TripsTable
GROUP BY StartLocation
""")
display(df)
(7) Spark Jobs
df: pyspark.sql.dataframe.DataFrame = [Trips: long, Location: string]
Trips
Location
1 2 Washington
2 2 Seattle
3 3 San Jose
4 2 Miami
5 5 New York
6 2 Tempe
7 1 Phoenix
Showing all 9 rows.

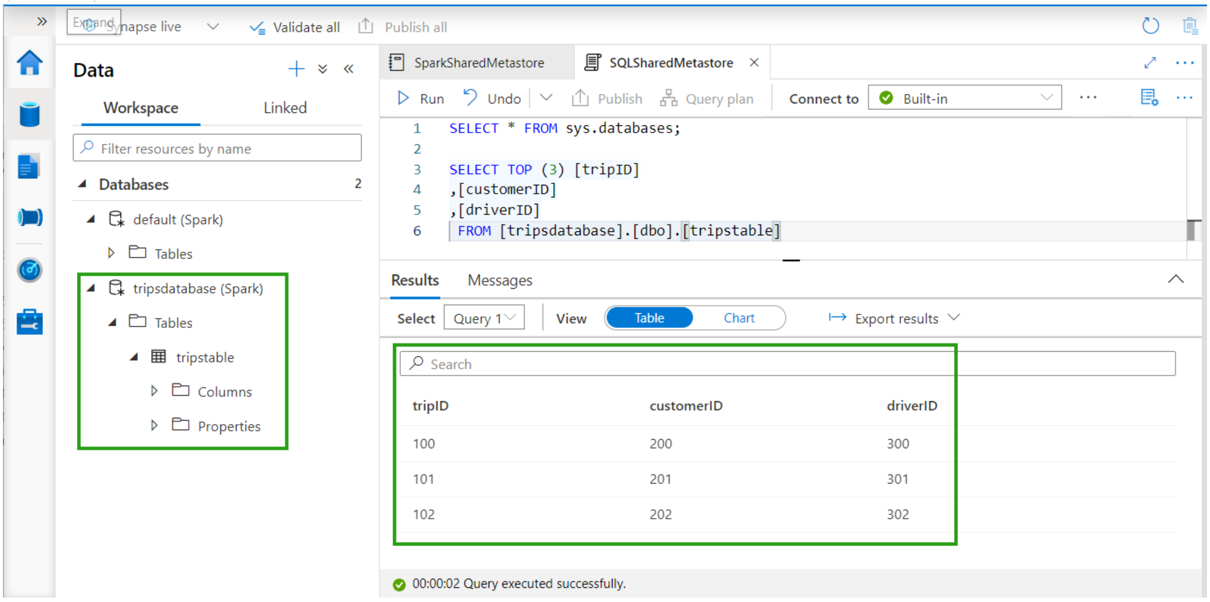


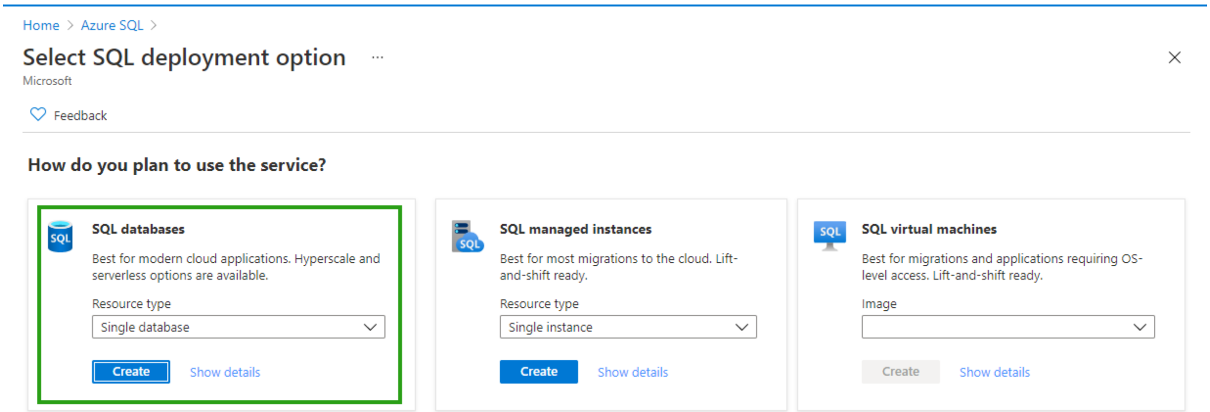
Python
df = spark.read.load('abfss://[redacted].dfs.core.windows.net/parquet/trips/*.parquet',
format='parquet')
spark.sql("CREATE DATABASE IF NOT EXISTS TripsDatabase")
df.write.mode("overwrite").option("overwriteSchema", "true").saveAsTable("TripsTable")
sqldf = spark.sql("""
SELECT COUNT(*) AS Trips,
startLocation AS Location
FROM TripsTable
GROUP BY startLocation """)
display(sqldf)
(7) Spark Jobs
df: pyspark.sql.dataframe.DataFrame = [tripId: string, driverId: string ... 5 more fields]
sqldf: pyspark.sql.dataframe.DataFrame = [Trips: long, Location: string]
Chart
Data Profile
Trips
4
2
0
Washington
Seattle
Miami
San Jose
New York
Phoenix
Los Angeles
Dallas
Tempe
Location
Plot Options...

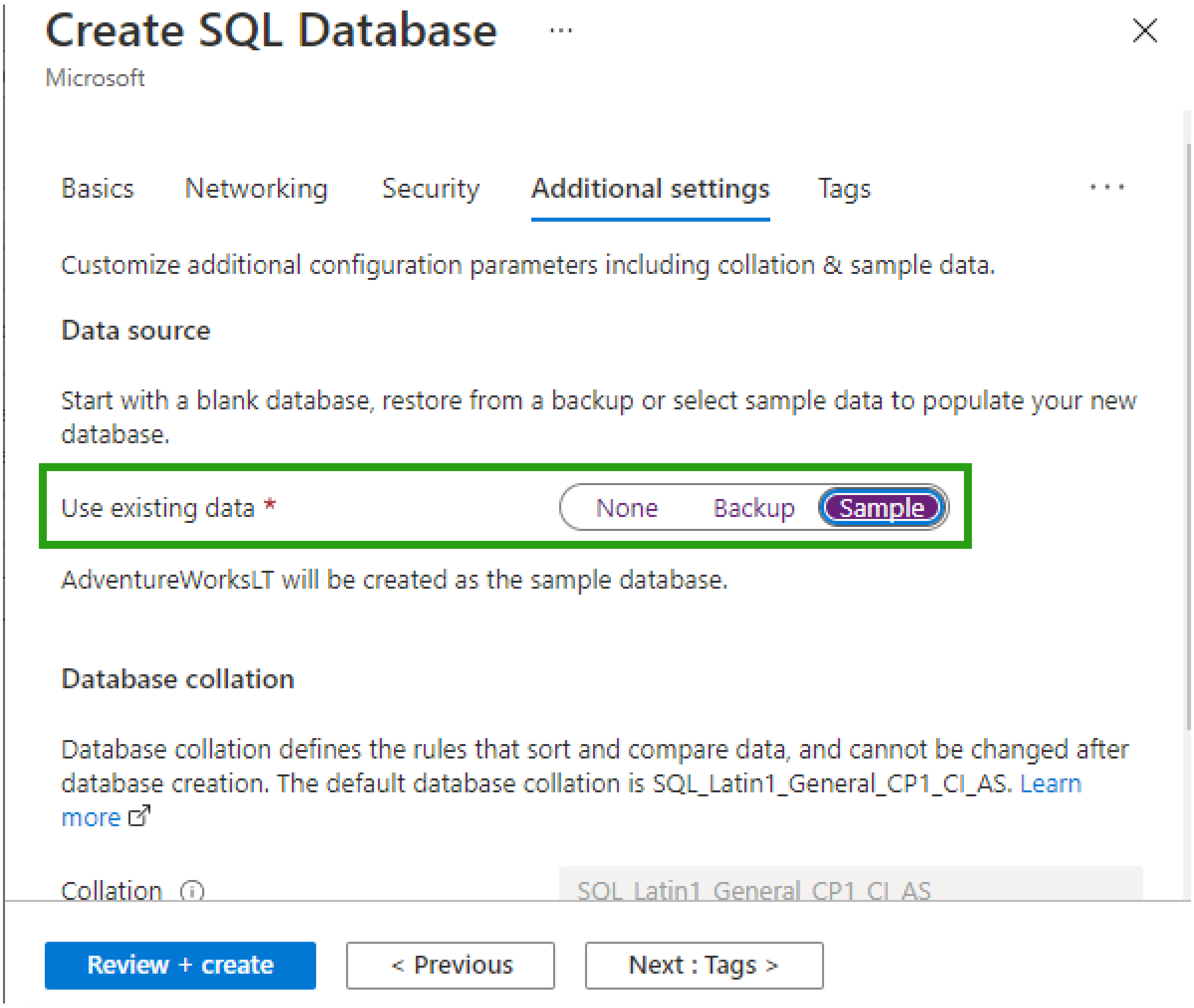

Microsoft Azure
Databricks
Portal
ADBParquetRead (Scala)
IACDatabricksWS
ADB5Node
%python
df = spark.read.load('abfss://dfs.core.windows.net/raw/trips/out/year=2022/*/*/*.parquet', format='parquet')
spark.sql("CREATE DATABASE IF NOT EXISTS TripsDatabase")
df.write.mode("overwrite").saveAsTable("TripsTable")
df = spark.sql("""
SELECT COUNT(*) AS Trips,
StartLocation AS Location
from TripsTable
GROUP BY StartLocation
""")
display(df)
(7) Spark Jobs
df: pyspark.sql.dataframe.DataFrame = [Trips: long, Location: string]
Trips
Location
2 Washington
2 Seattle
3 San Jose
2 Miami
5 New York
2 Tempe
1 Phoenix
Showing all 9 rows.
Synapse live
Validate all
Publish all
Data
Workspace
Linked
Filter resources by name
Databases
default (Spark)
Tables
tripsdatabase (Spark)
Tables
tripstable
Columns
Properties
SparkSharedMetastore
SQLSharedMetastore
Run all
Publish
Outline
Attach to
smallpool
Ready
CREATE DATABASE IF NOT EXISTS TripsDatabase;
USE TripsDatabase;
CREATE TABLE IF NOT EXISTS TripsTable
(tripID INT, customerID INT, driverID INT)
USING Parquet;
INSERT INTO TripsTable VALUES
(100, 200, 300),
(101, 201, 301),
(102, 202, 302);
Command executed in 0 sec by newton.packt on 4:59:52 PM, 7/25/21


Synapse live
Validate all
Publish all
Develop
Filter resources by name
SQL scripts
Notebooks
Data flows
Power BI
SQL script
Notebook
Data flow
Apache Spark job definition
Power BI report
Browse gallery
Import

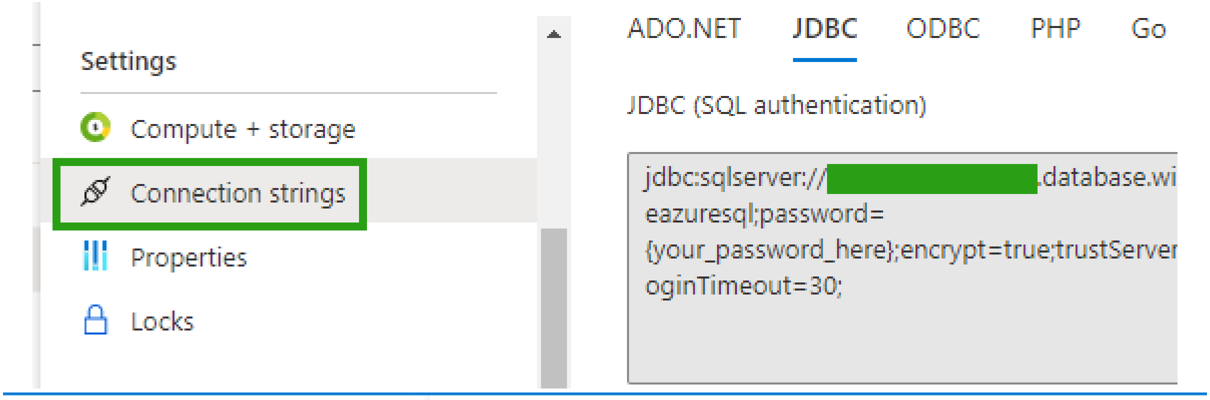

SQL Explore
Run
Undo
Publish
Query plan
Connect to
Built-in
Use database
master
1 SELECT
2 TOP 100 *
3 FROM

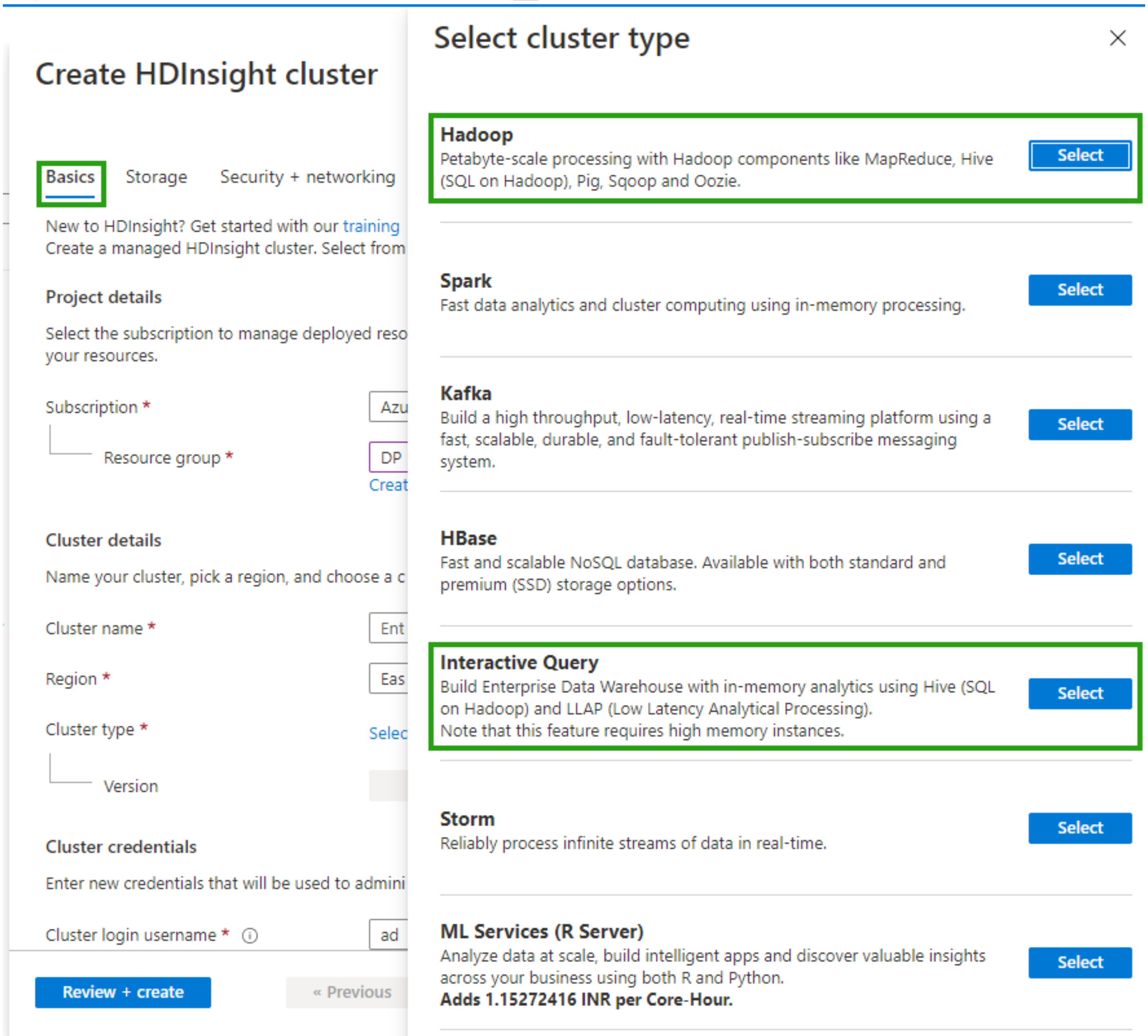

Synapse live
Validate all
Publish all
Data
Workspace
Linked
Filter resources by name
Databases 2
default (Spark)
Tables
tripsdatabase (Spark)
Tables
tripstable
Columns
Properties
SparkSharedMetastore
SQLSharedMetastore
Run
Undo
Publish
Query plan
Connect to
Built-in
1 SELECT * FROM sys.databases;
2
3 SELECT TOP (3) [tripID]
4 ,[customerID]
5 ,[driverID]
6 FROM [tripsdatabase].[dbo].[tripstable]
Results
Messages
Select
Query 1
View
Table
Chart
Export results
Search
tripID
customerID
driverID
100
200
300
101
201
301
102
202
302
00:00:02 Query executed successfully.

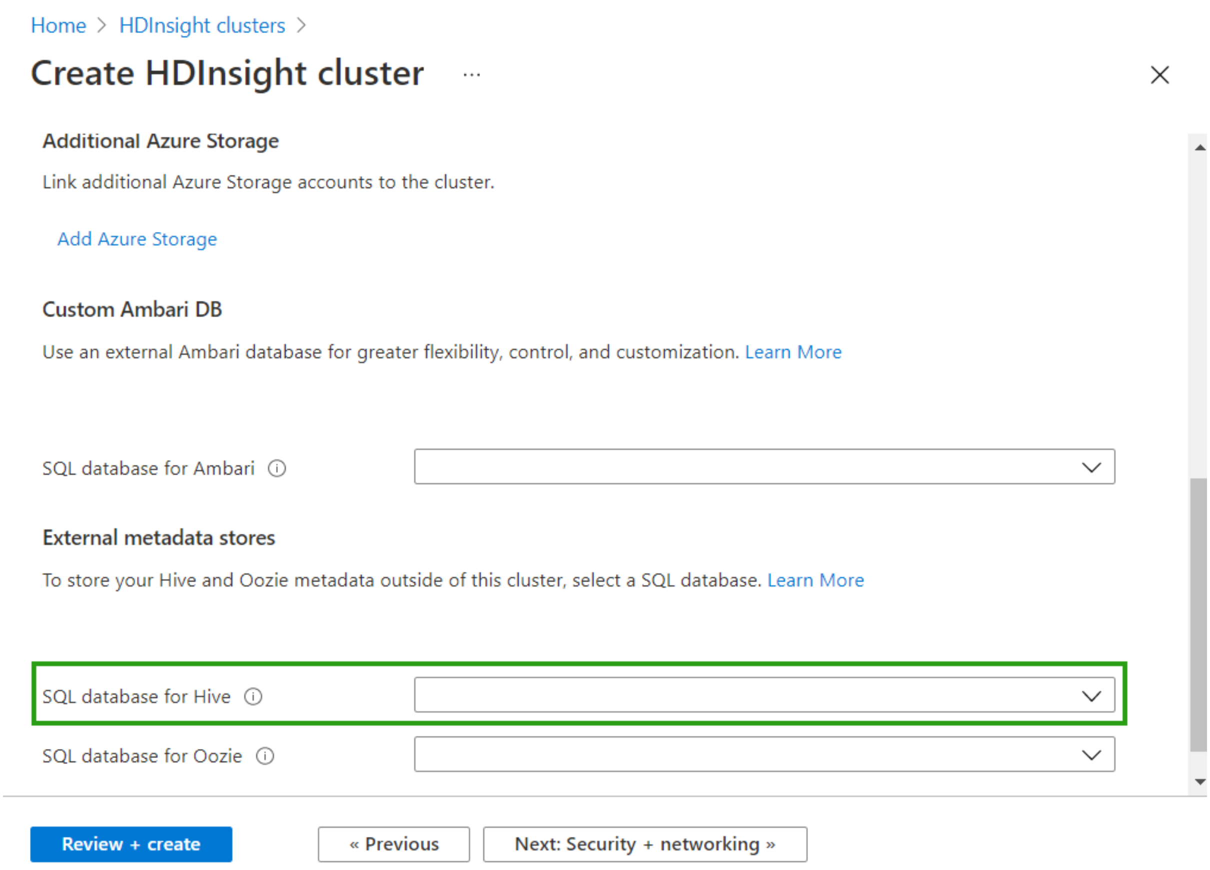

Home > Azure SQL >
Select SQL deployment option
Microsoft
Feedback
How do you plan to use the service?
SQL databases
Best for modern cloud applications. Hyperscale and serverless options are available.
Resource type
Single database
Create
Show details
SQL managed instances
Best for most migrations to the cloud. Lift-and-shift ready.
Resource type
Single instance
Create
Show details
SQL virtual machines
Best for migrations and applications requiring OS-level access. Lift-and-shift ready.
Image
Create
Show details

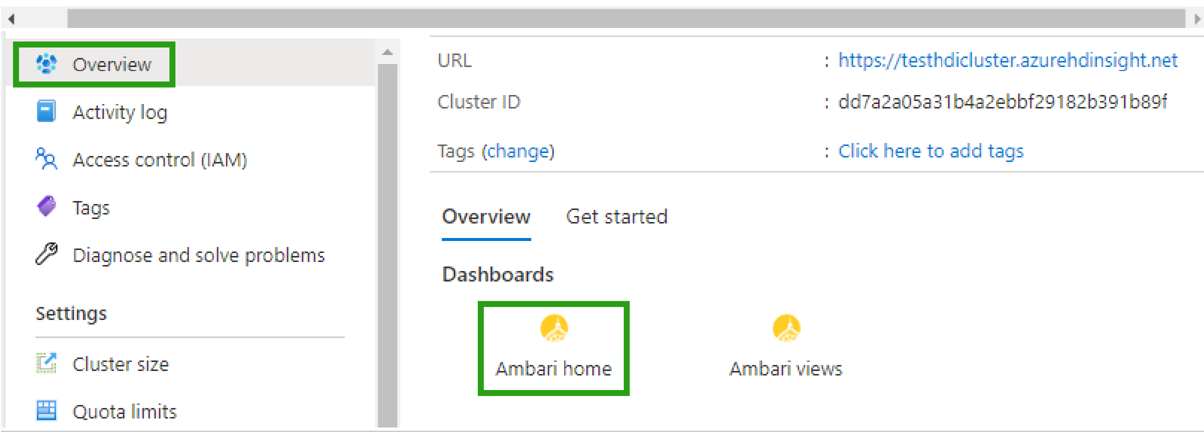

Create SQL Database
Microsoft
Basics
Networking
Security
Additional settings
Tags
Customize additional configuration parameters including collation & sample data.
Data source
Start with a blank database, restore from a backup or select sample data to populate your new database.
Use existing data *
None
Backup
Sample
AdventureWorksLT will be created as the sample database.
Database collation
Database collation defines the rules that sort and compare data, and cannot be changed after database creation. The default database collation is SQL_Latin1_General_CP1_CI_AS. Learn more
Collation
SQL_Latin1_General_CP1_CI_AS
Review + create
< Previous
Next : Tags >

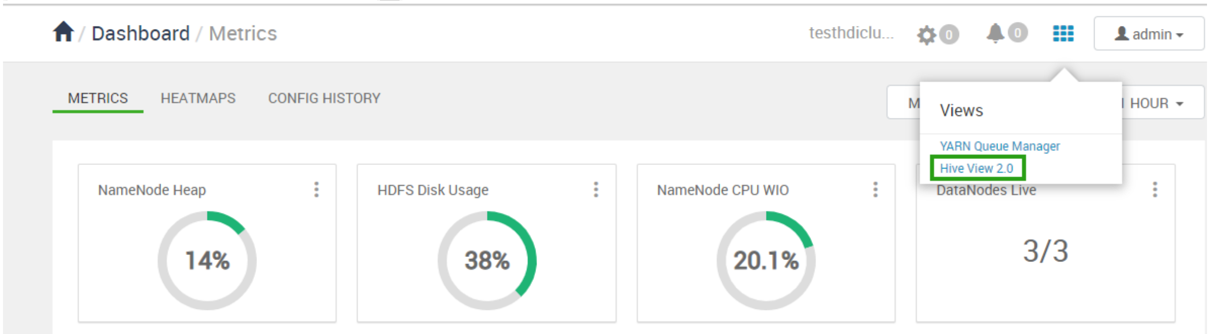

Settings
Compute + storage
Connection strings
Properties
Locks
ADO.NET
JDBC
ODBC
PHP
Go
JDBC (SQL authentication)
jdbc:sqlserver://iacfinanceazuresql.database.wi
eazuresql;password=
{your_password_here};encrypt=true;trustServer
oginTimeout=30;

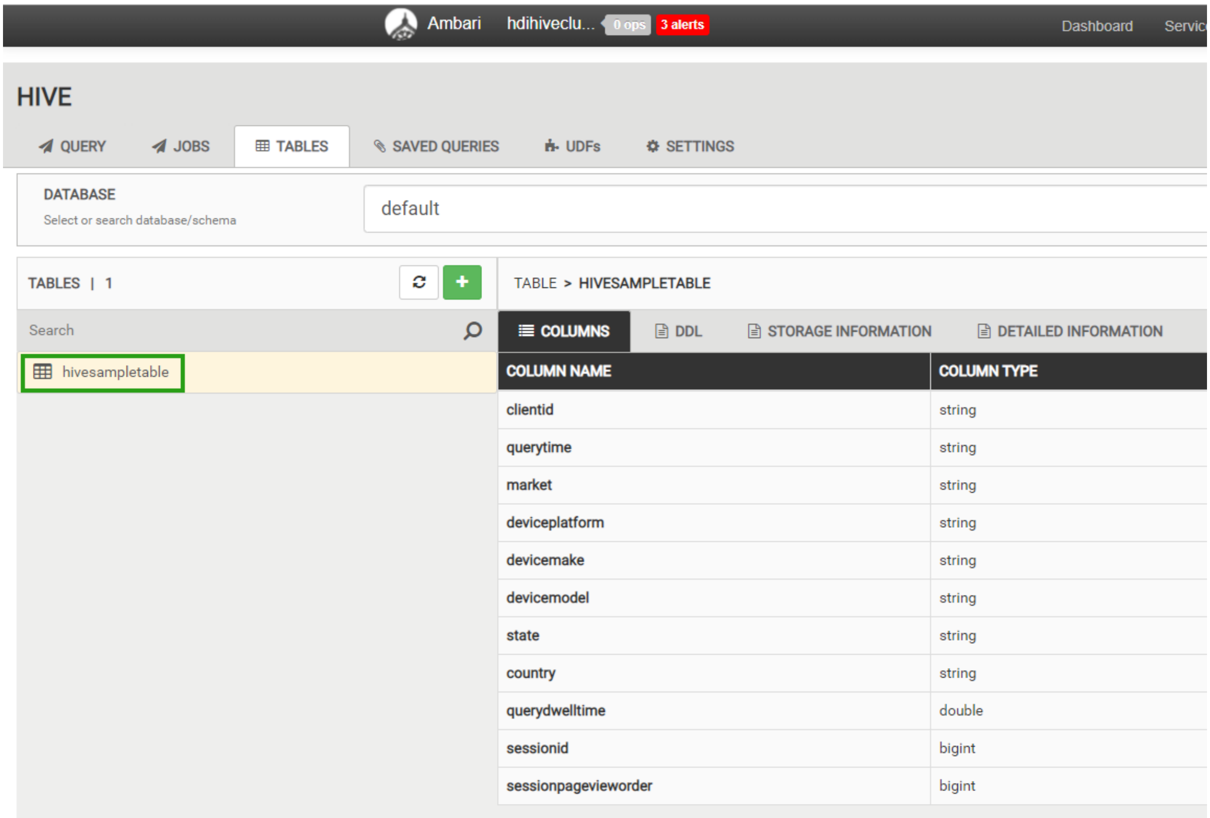

Settings
Compute + storage
Connection strings
Properties
Locks
ADO.NET
JDBC
ODBC
PHP
Go
JDBC (SQL authentication)
jdbc:sqlserver://[redacted].database.wi
eazuresql;password=
{your_password_here};encrypt=true;trustServer
oginTimeout=30;

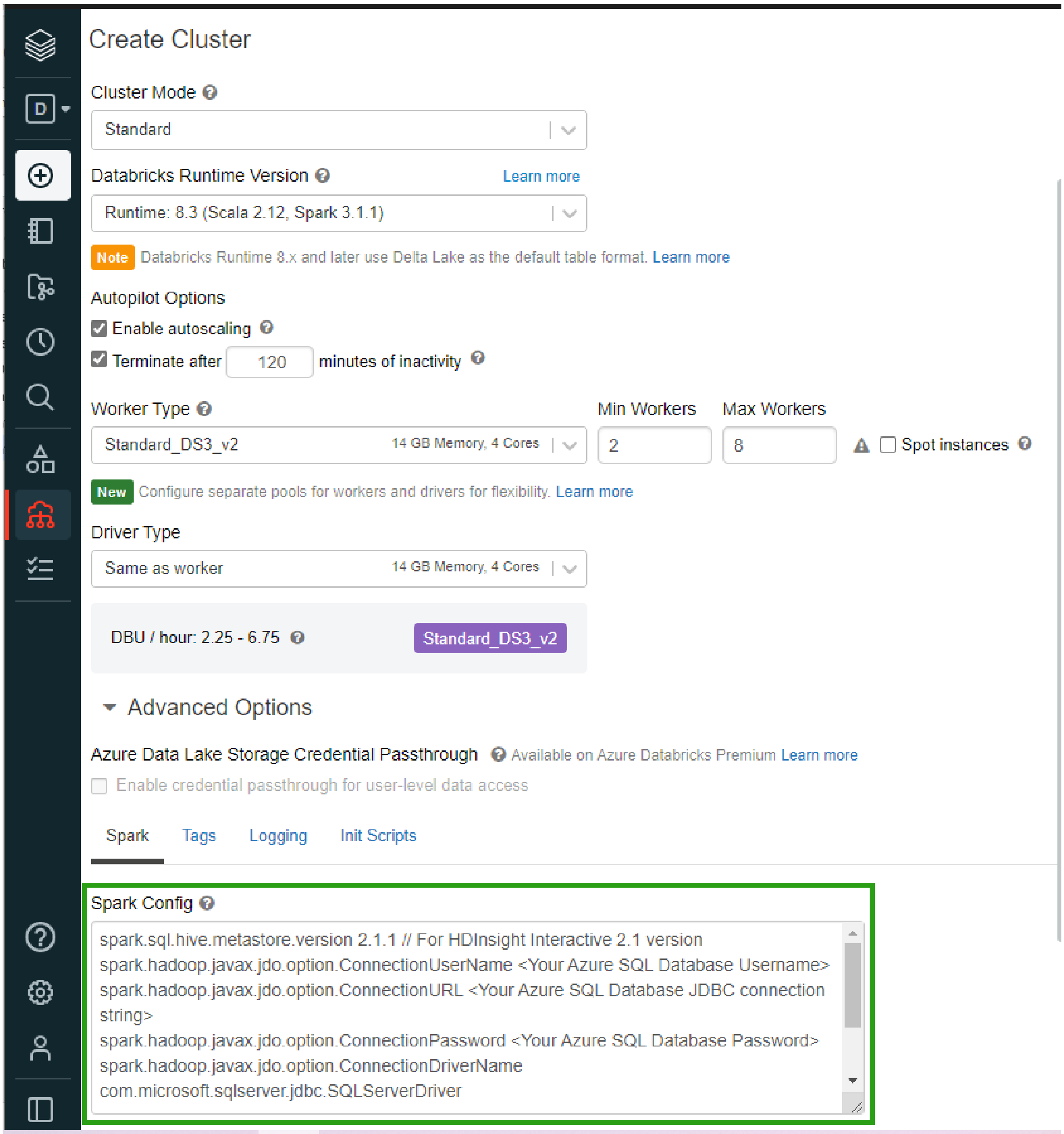

Create HDInsight cluster
Basics
Storage
Security + networking
New to HDInsight? Get started with our training
Create a managed HDInsight cluster. Select from
Project details
Select the subscription to manage deployed reso
your resources.
Subscription *
Azu
Resource group *
DP
Creat
Cluster details
Name your cluster, pick a region, and choose a c
Cluster name *
Ent
Region *
Eas
Cluster type *
Selec
Version
Cluster credentials
Enter new credentials that will be used to admini
Cluster login username *
ad
Review + create
« Previous
Select cluster type
Hadoop
Petabyte-scale processing with Hadoop components like MapReduce, Hive (SQL on Hadoop), Pig, Sqoop and Oozie.
Select
Spark
Fast data analytics and cluster computing using in-memory processing.
Select
Kafka
Build a high throughput, low-latency, real-time streaming platform using a fast, scalable, durable, and fault-tolerant publish-subscribe messaging system.
Select
HBase
Fast and scalable NoSQL database. Available with both standard and premium (SSD) storage options.
Select
Interactive Query
Build Enterprise Data Warehouse with in-memory analytics using Hive (SQL on Hadoop) and LLAP (Low Latency Analytical Processing).
Note that this feature requires high memory instances.
Select
Storm
Reliably process infinite streams of data in real-time.
Select
ML Services (R Server)
Analyze data at scale, build intelligent apps and discover valuable insights across your business using both R and Python.
Adds 1.15272416 INR per Core-Hour.
Select

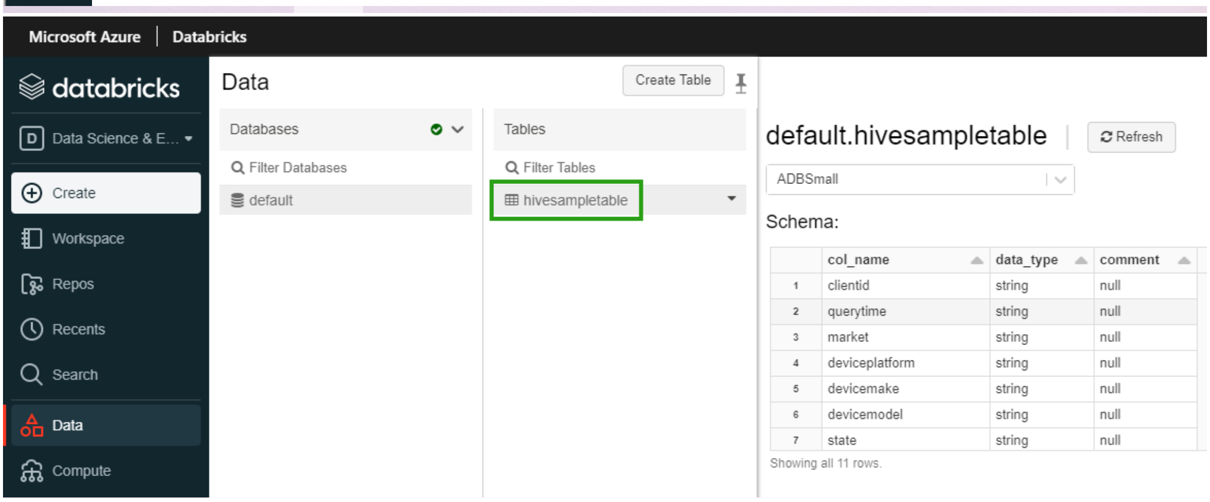

Home > HDInsight clusters >
Create HDInsight cluster ...
Additional Azure Storage
Link additional Azure Storage accounts to the cluster.
Add Azure Storage
Custom Ambari DB
Use an external Ambari database for greater flexibility, control, and customization. Learn More
SQL database for Ambari
External metadata stores
To store your Hive and Oozie metadata outside of this cluster, select a SQL database. Learn More
SQL database for Hive
SQL database for Oozie
Review + create
« Previous
Next: Security + networking »


Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Settings
Cluster size
Quota limits
URL
: https://testhdicluster.azurehdinsight.net
Cluster ID
: dd7a2a05a31b4a2ebbf29182b391b89f
Tags (change)
: Click here to add tags
Overview
Get started
Dashboards
Ambari home
Ambari views

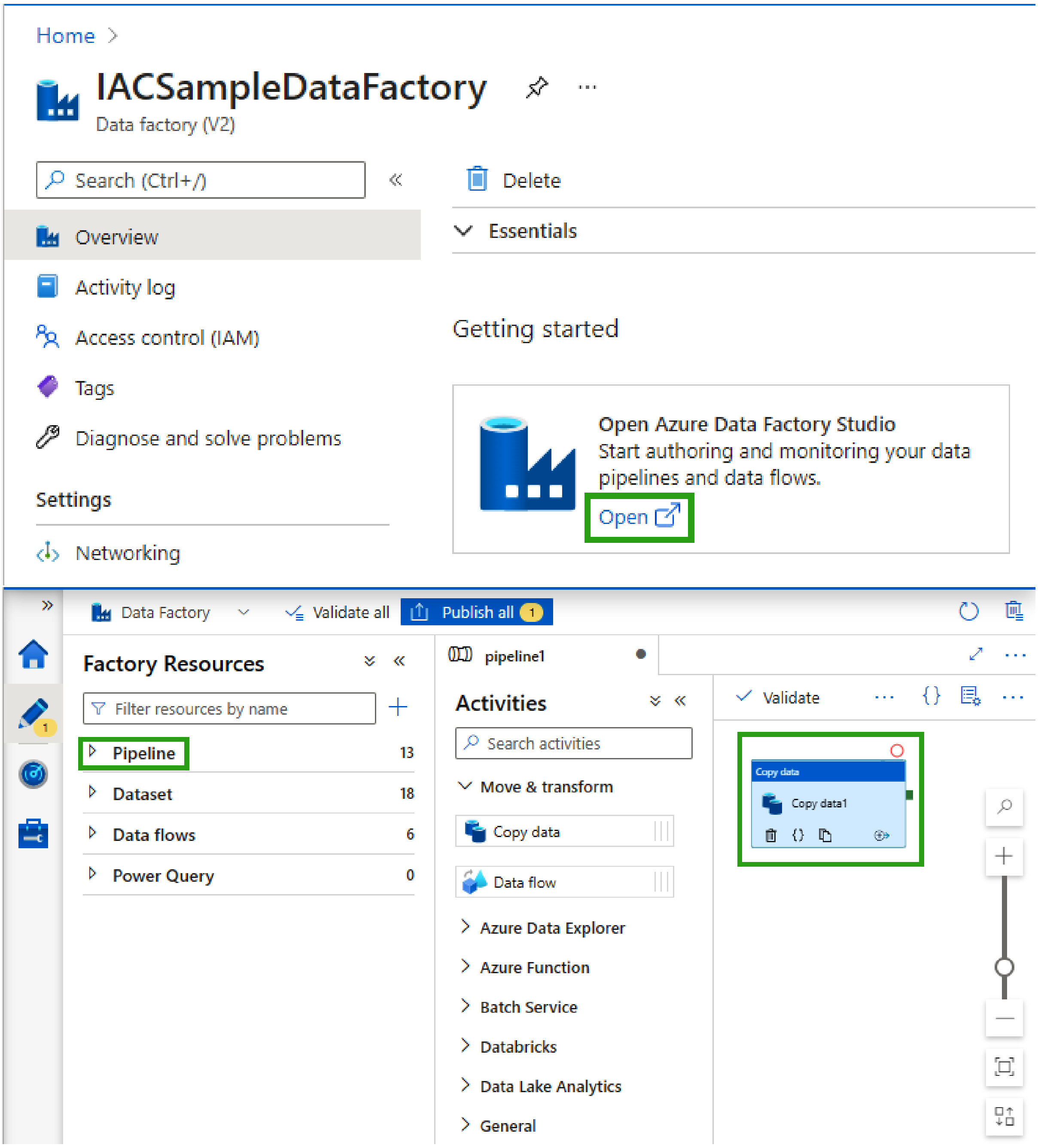

Dashboard / Metrics
testhdiclu...
admin
METRICS
HEATMAPS
CONFIG HISTORY
Views
YARN Queue Manager
Hive View 2.0
HOUR
NameNode Heap
14%
HDFS Disk Usage
38%
NameNode CPU WIO
20.1%
DataNodes Live
3/3

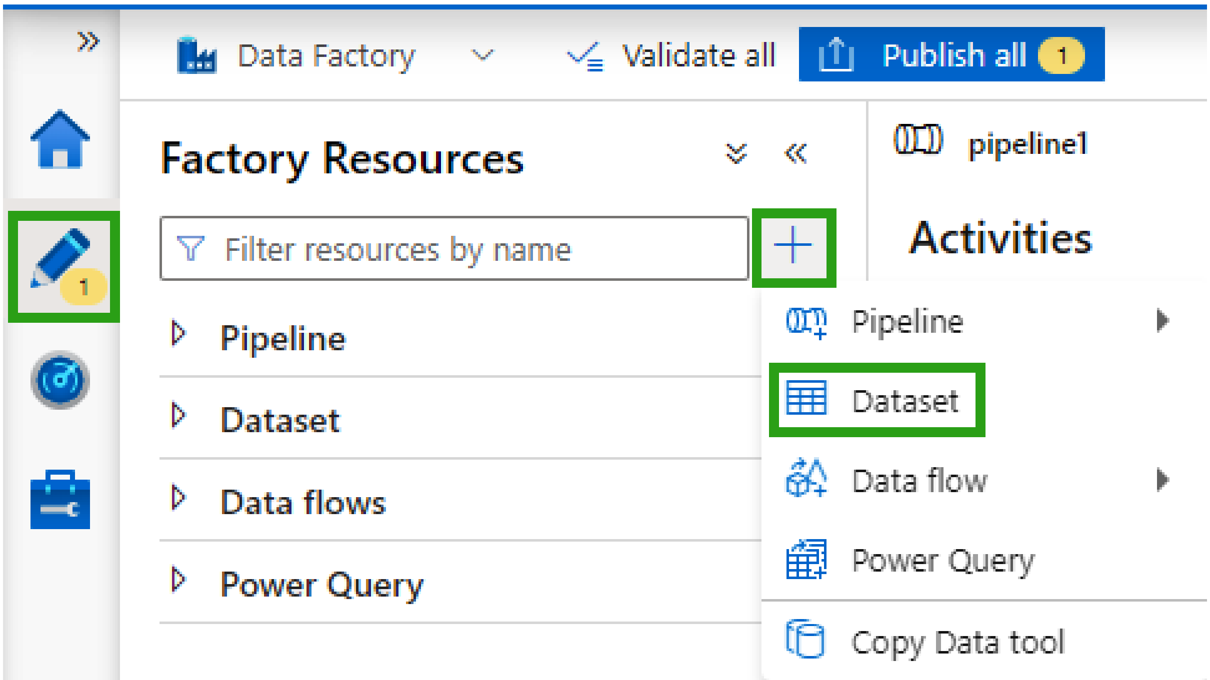

Ambari
hdihiveclu...
0 ops
3 alerts
Dashboard
Servic
HIVE
QUERY
JOBS
TABLES
SAVED QUERIES
UDFs
SETTINGS
DATABASE
Select or search database/schema
default
TABLES | 1
Search
hivesampletable
TABLE > HIVESAMPLETABLE
COLUMNS
DDL
STORAGE INFORMATION
DETAILED INFORMATION
COLUMN NAME
COLUMN TYPE
clientid
string
querytime
string
market
string
deviceplatform
string
devicemake
string
devicemodel
string
state
string
country
string
querydwelltime
double
sessionid
bigint
sessionpagevieworder
bigint

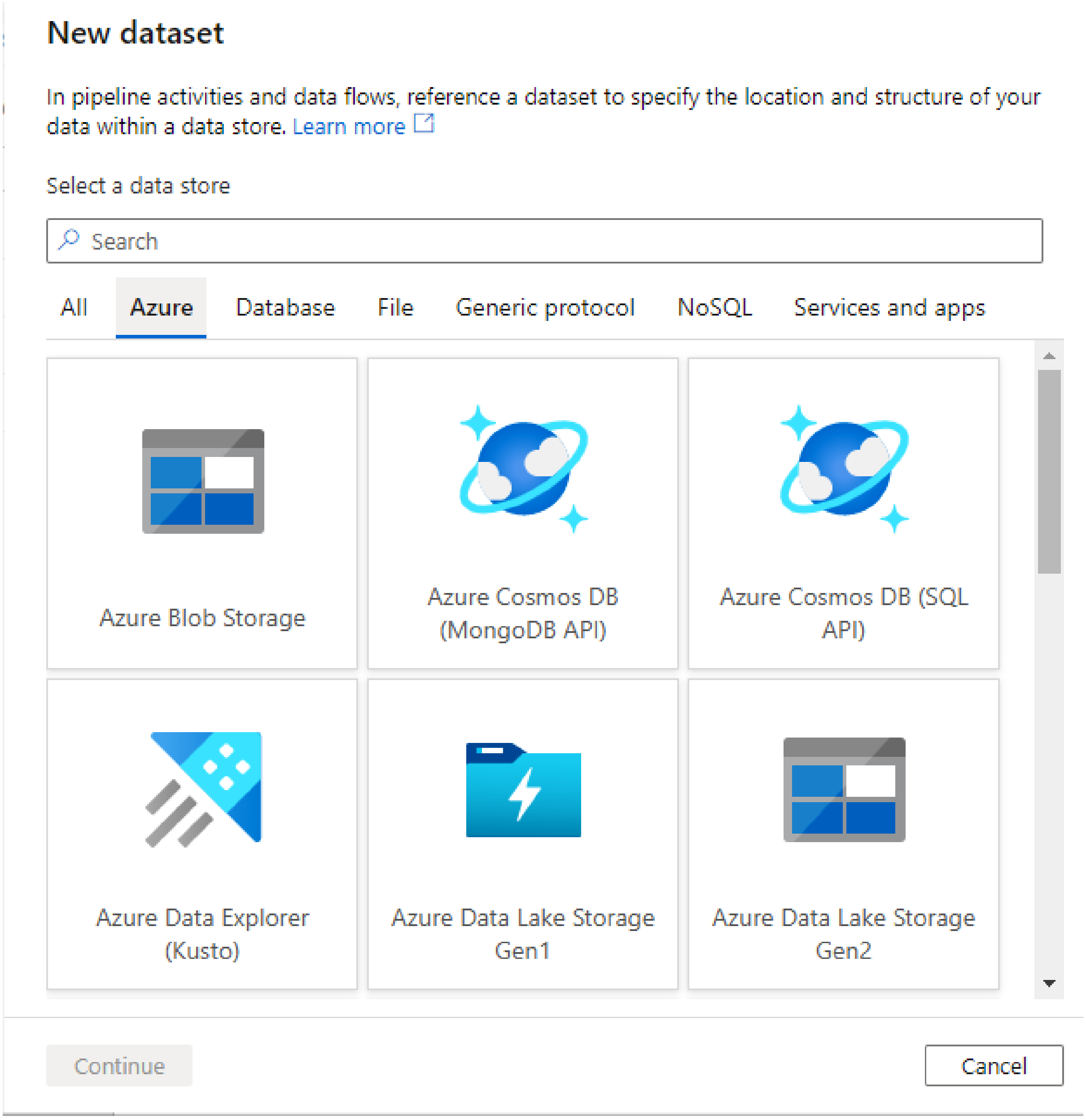

Create Cluster
Cluster Mode
Standard
Databricks Runtime Version
Learn more
Runtime: 8.3 (Scala 2.12, Spark 3.1.1)
Note Databricks Runtime 8.x and later use Delta Lake as the default table format. Learn more
Autopilot Options
Enable autoscaling
Terminate after 120 minutes of inactivity
Worker Type
Standard_DS3_v2 14 GB Memory, 4 Cores
Min Workers 2
Max Workers 8
Spot instances
New Configure separate pools for workers and drivers for flexibility. Learn more
Driver Type
Same as worker 14 GB Memory, 4 Cores
DBU / hour: 2.25 - 6.75
Standard_DS3_v2
Advanced Options
Azure Data Lake Storage Credential Passthrough
Available on Azure Databricks Premium Learn more
Enable credential passthrough for user-level data access
Spark
Tags
Logging
Init Scripts
Spark Config
spark.sql.hive.metastore.version 2.1.1 // For HDInsight Interactive 2.1 version
spark.hadoop.javax.jdo.option.ConnectionUserName <Your Azure SQL Database Username>
spark.hadoop.javax.jdo.option.ConnectionURL <Your Azure SQL Database JDBC connection string>
spark.hadoop.javax.jdo.option.ConnectionPassword <Your Azure SQL Database Password>
spark.hadoop.javax.jdo.option.ConnectionDriverName com.microsoft.sqlserver.jdbc.SQLServerDriver

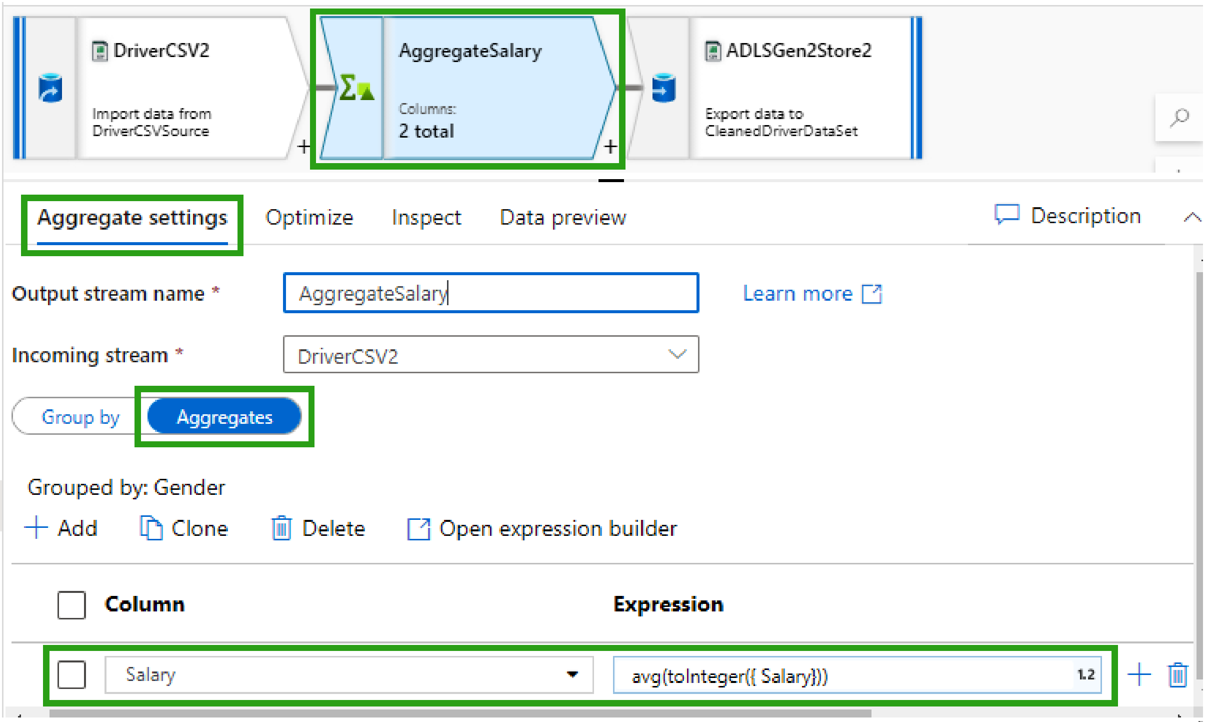

Microsoft Azure | Databricks
databricks
Data Science & E...
Create
Workspace
Repos
Recents
Search
Data
Compute
Data
Create Table
Databases
Filter Databases
default
Tables
Filter Tables
hivesampletable
default.hivesampletable
Refresh
ADBSmall
Schema:
col_name data_type comment
1 clientid string null
2 querytime string null
3 market string null
4 deviceplatform string null
5 devicemake string null
6 devicemodel string null
7 state string null
Showing all 11 rows.

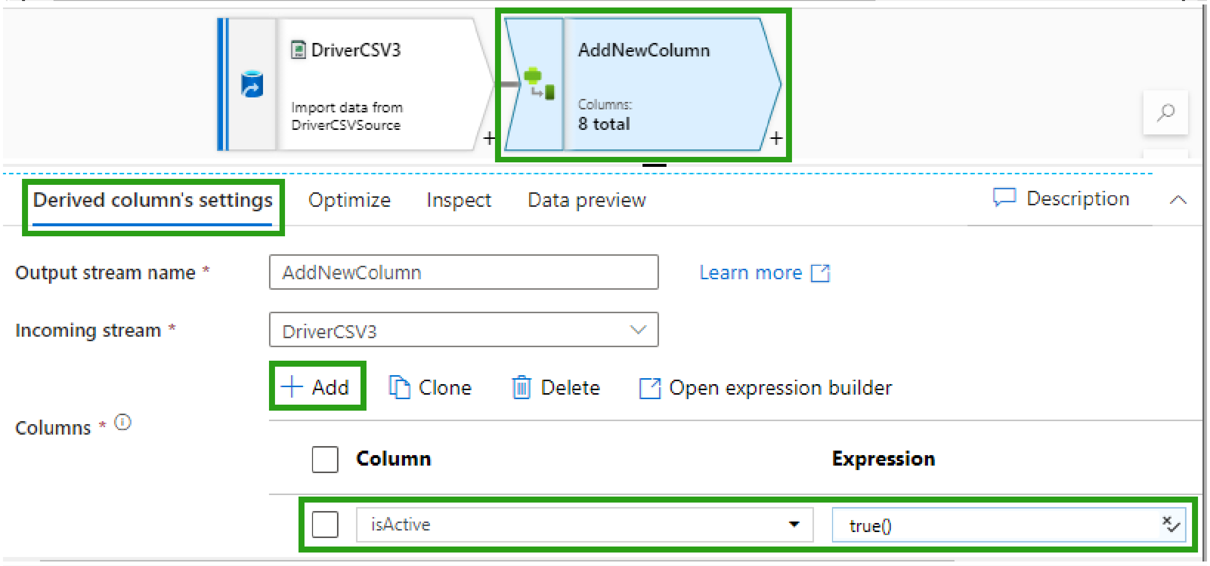
Chapter 8: Ingesting and Transforming Data

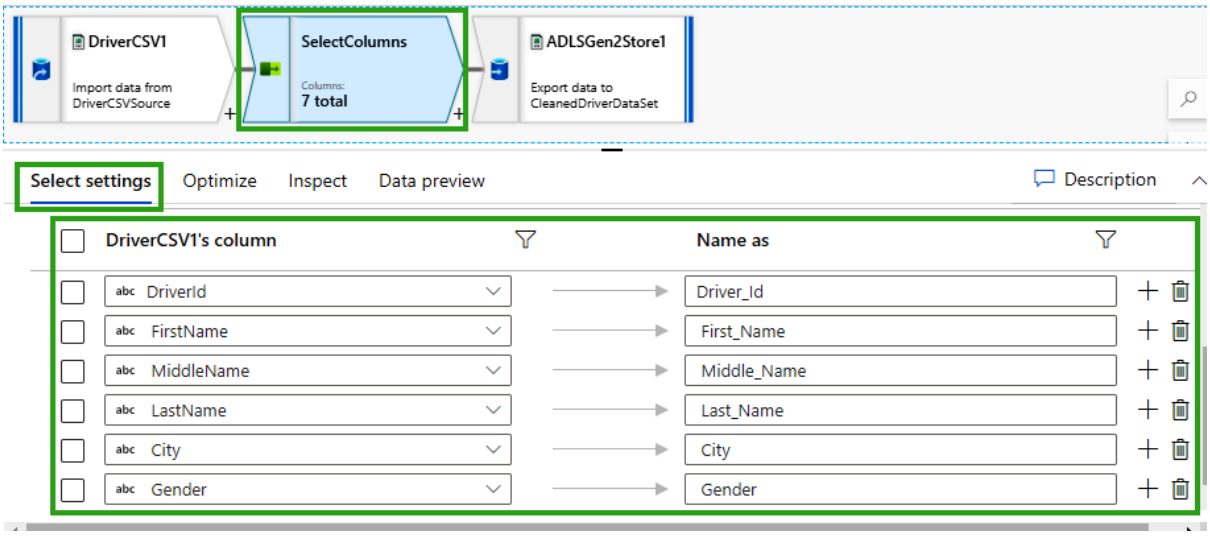


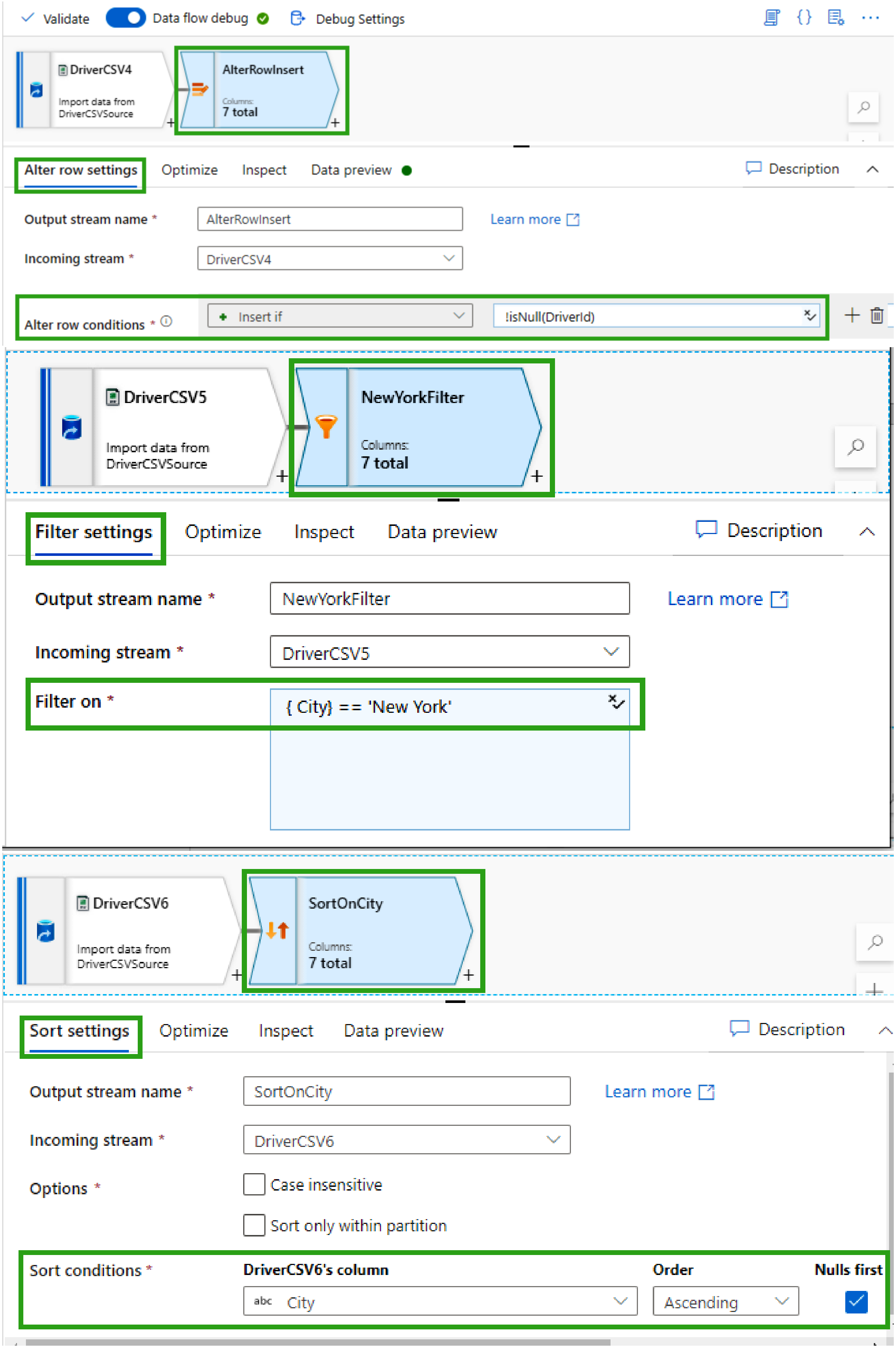

Data Factory
Validate all
Publish all 1
Factory Resources
Filter resources by name
Pipeline
Dataset
Data flows
Power Query
pipeline1
Activities
Pipeline
Dataset
Data flow
Power Query
Copy Data tool

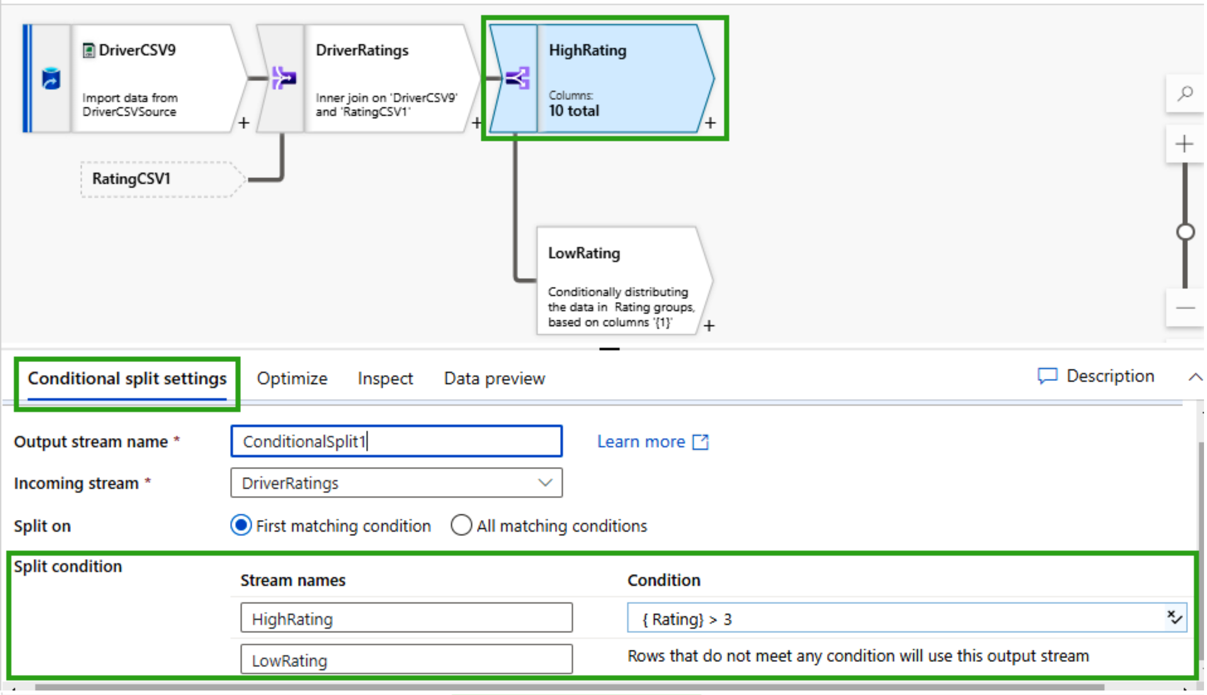

New dataset
In pipeline activities and data flows, reference a dataset to specify the location and structure of your data within a data store. Learn more
Select a data store
Search
All
Azure
Database
File
Generic protocol
NoSQL
Services and apps
Azure Blob Storage
Azure Cosmos DB (MongoDB API)
Azure Cosmos DB (SQL API)
Azure Data Explorer (Kusto)
Azure Data Lake Storage Gen1
Azure Data Lake Storage Gen2
Continue
Cancel

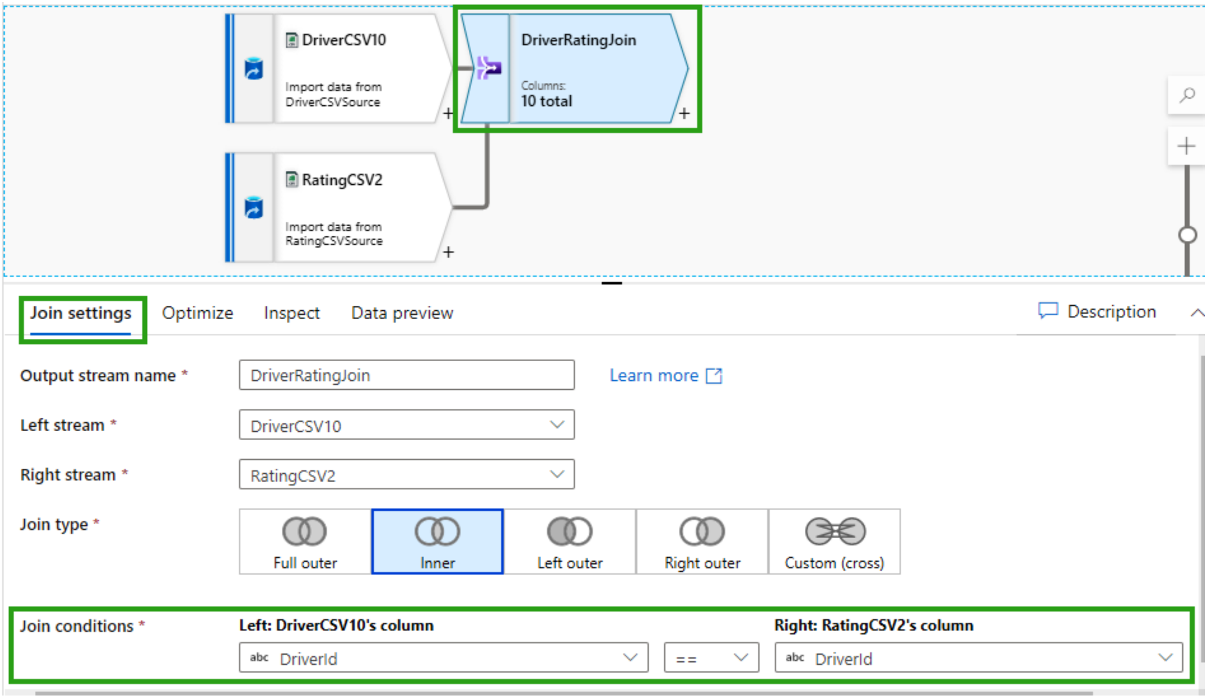

DriverCSV2
Import data from DriverCSVSource
AggregateSalary
Columns:
2 total
ADLSGen2Store2
Export data to CleanedDriverDataSet
Aggregate settings
Optimize
Inspect
Data preview
Description
Output stream name *
AggregateSalary
Learn more
Incoming stream *
DriverCSV2
Group by
Aggregates
Grouped by: Gender
Add
Clone
Delete
Open expression builder
Column
Expression
Salary
avg(toInteger({ Salary}))

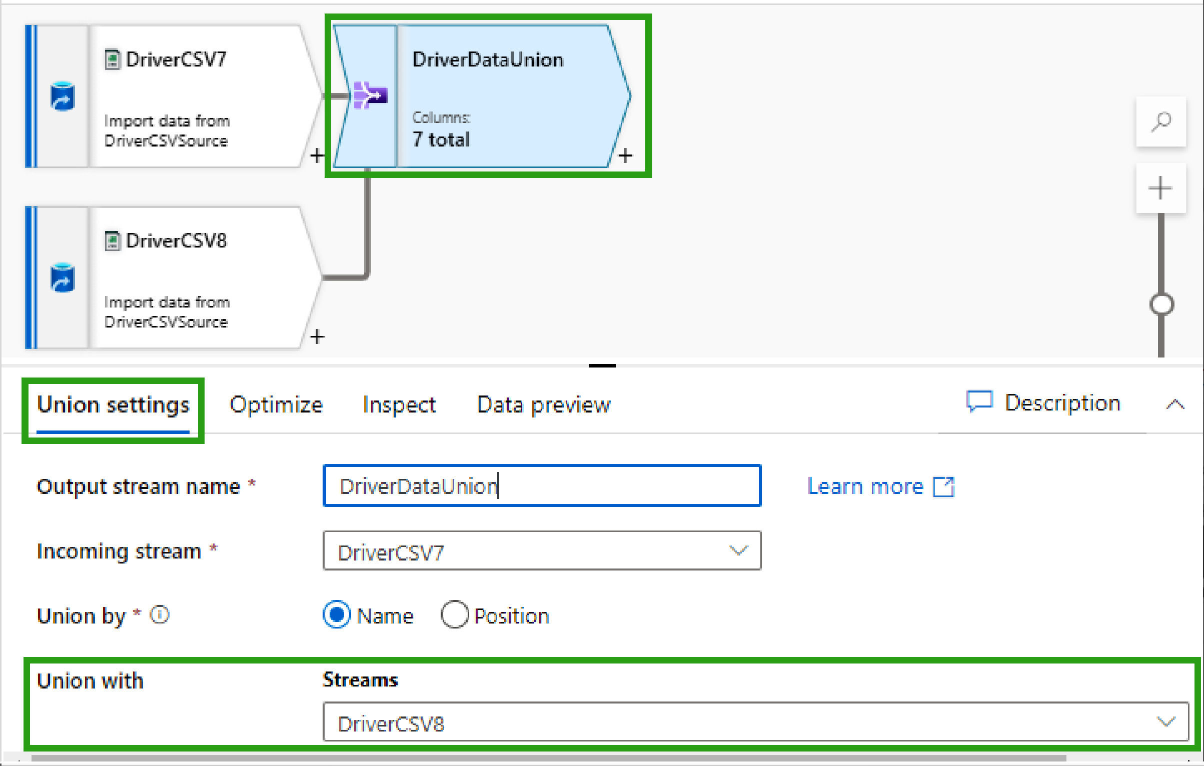

DriverCSV3
Import data from DriverCSVSource
AddNewColumn
Columns:
8 total
Derived column's settings
Optimize
Inspect
Data preview
Description
Output stream name *
AddNewColumn
Learn more
Incoming stream *
DriverCSV3
Add
Clone
Delete
Open expression builder
Columns *
Column
Expression
isActive
true()

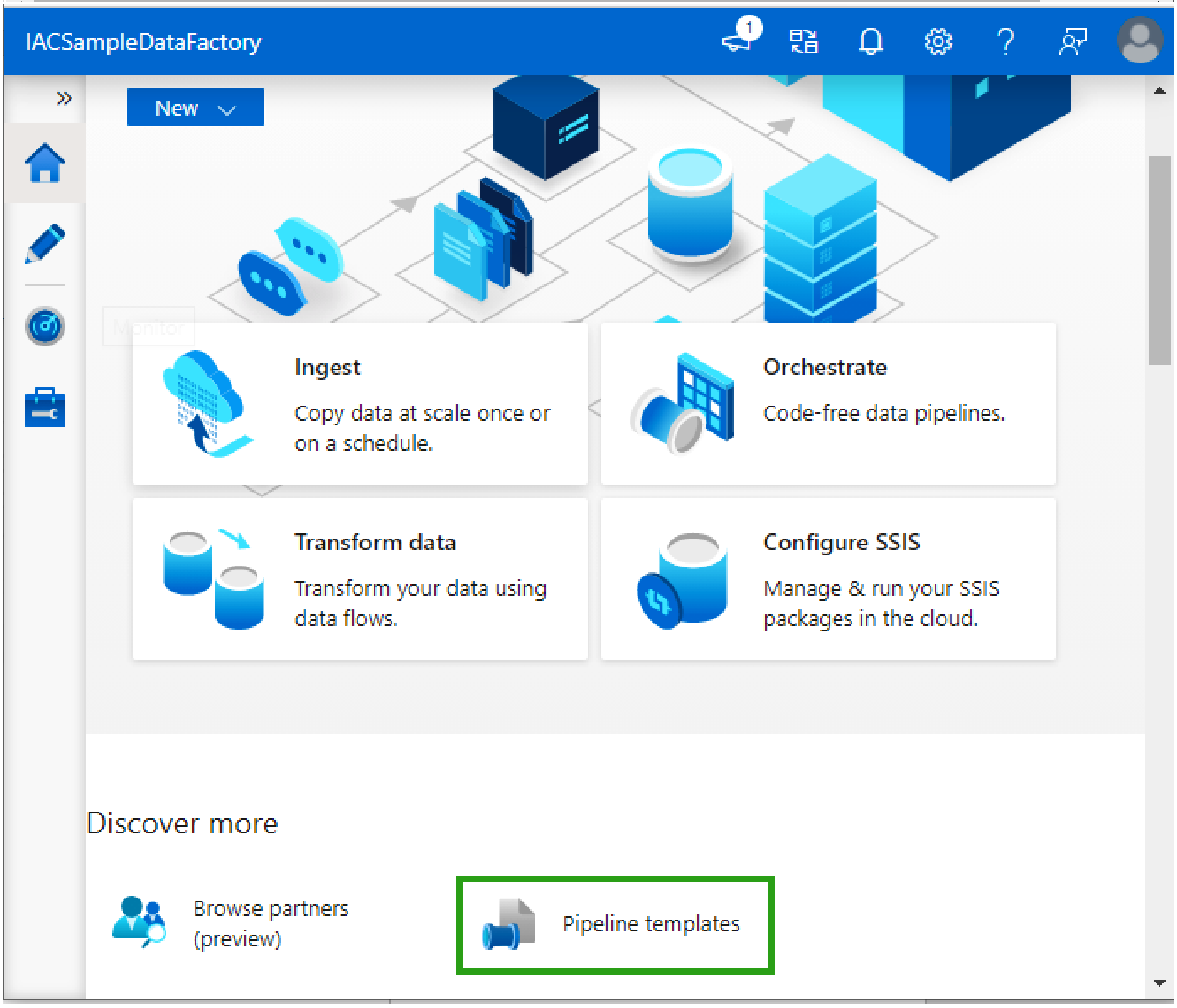

DriverCSV1
Import data from DriverCSVSource
SelectColumns
Columns:
7 total
ADLSGen2Store1
Export data to CleanedDriverDataSet
Select settings
Optimize
Inspect
Data preview
Description
DriverCSV1's column
Name as
DriverId
Driver_Id
FirstName
First_Name
MiddleName
Middle_Name
LastName
Last_Name
City
City
Gender
Gender

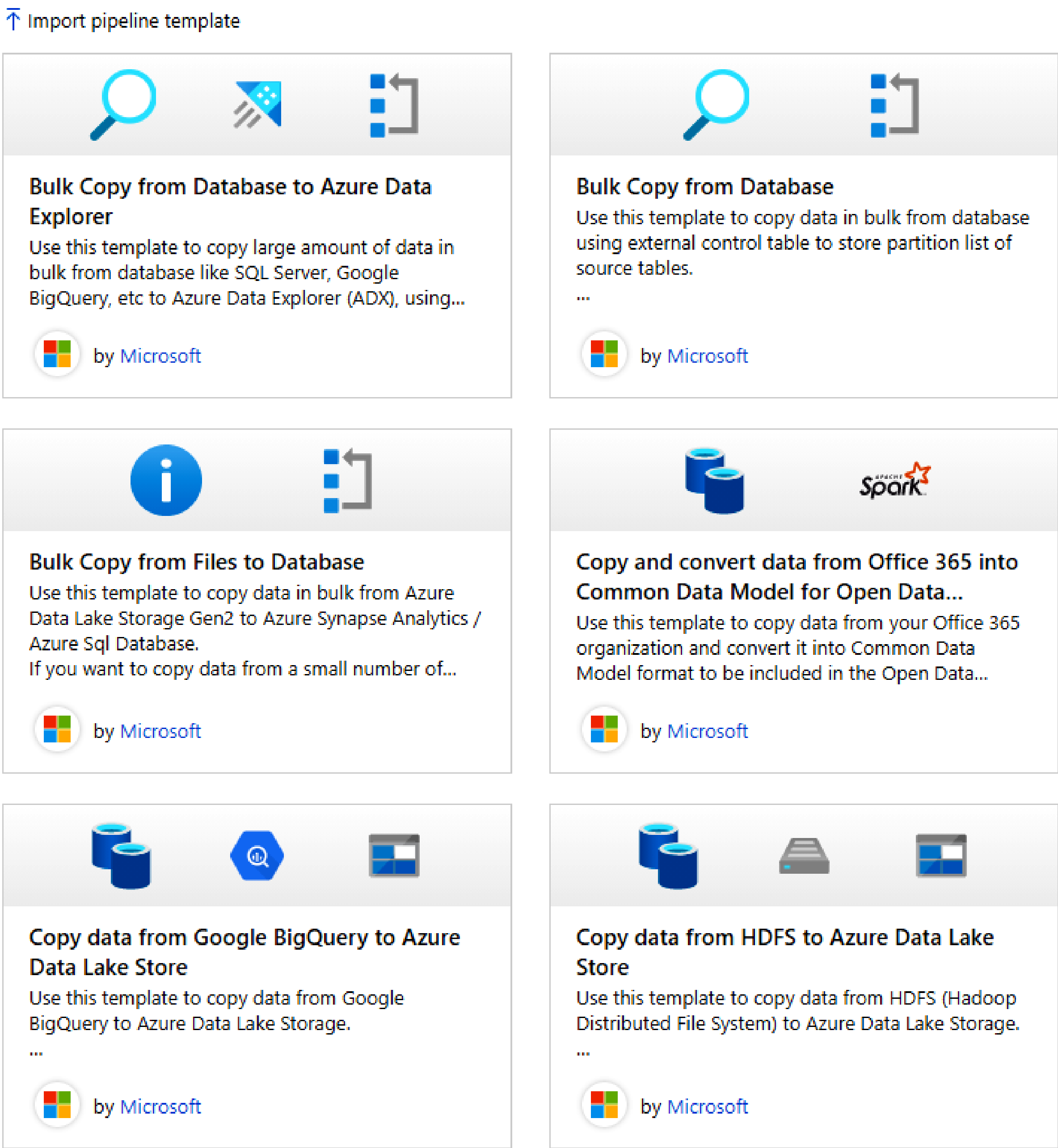

Validate
Data flow debug
Debug Settings
DriverCSV4
Import data from DriverCSVSource
AlterRowInsert
Columns:
7 total
Alter row settings
Optimize
Inspect
Data preview
Description
Output stream name *
AlterRowInsert
Learn more
Incoming stream *
DriverCSV4
Alter row conditions *
Insert if
!isNull(DriverId)
DriverCSV5
Import data from DriverCSVSource
NewYorkFilter
Columns:
7 total
Filter settings
Optimize
Inspect
Data preview
Description
Output stream name *
NewYorkFilter
Learn more
Incoming stream *
DriverCSV5
Filter on *
{ City} == 'New York'
DriverCSV6
Import data from DriverCSVSource
SortOnCity
Columns:
7 total
Sort settings
Optimize
Inspect
Data preview
Description
Output stream name *
SortOnCity
Learn more
Incoming stream *
DriverCSV6
Options *
Case insensitive
Sort only within partition
Sort conditions *
DriverCSV6's column
Order
Nulls first
abc City
Ascending

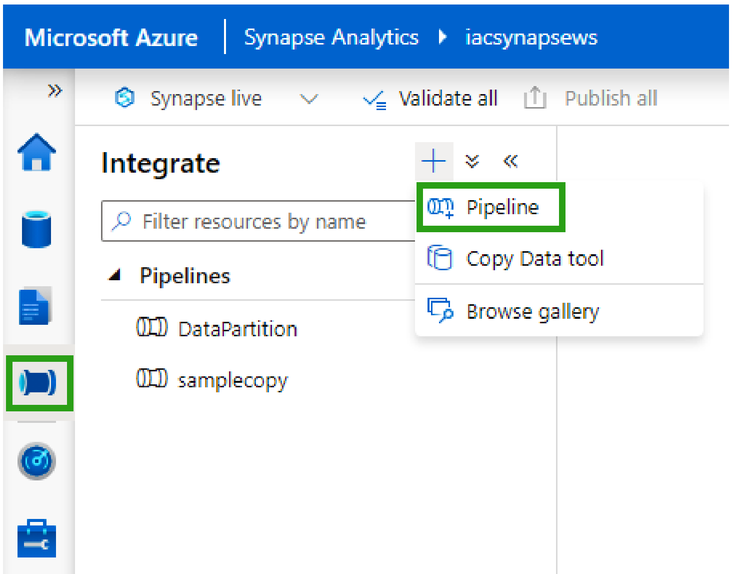

DriverCSV9
Import data from DriverCSVSource
DriverRatings
Inner join on 'DriverCSV9' and 'RatingCSV1'
HighRating
Columns:
10 total
RatingCSV1
LowRating
Conditionally distributing the data in Rating groups, based on columns '{1}'
Conditional split settings
Optimize
Inspect
Data preview
Description
Output stream name *
ConditionalSplit1
Learn more
Incoming stream *
DriverRatings
Split on
First matching condition
All matching conditions
Split condition
Stream names
Condition
HighRating
{ Rating} > 3
LowRating
Rows that do not meet any condition will use this output stream

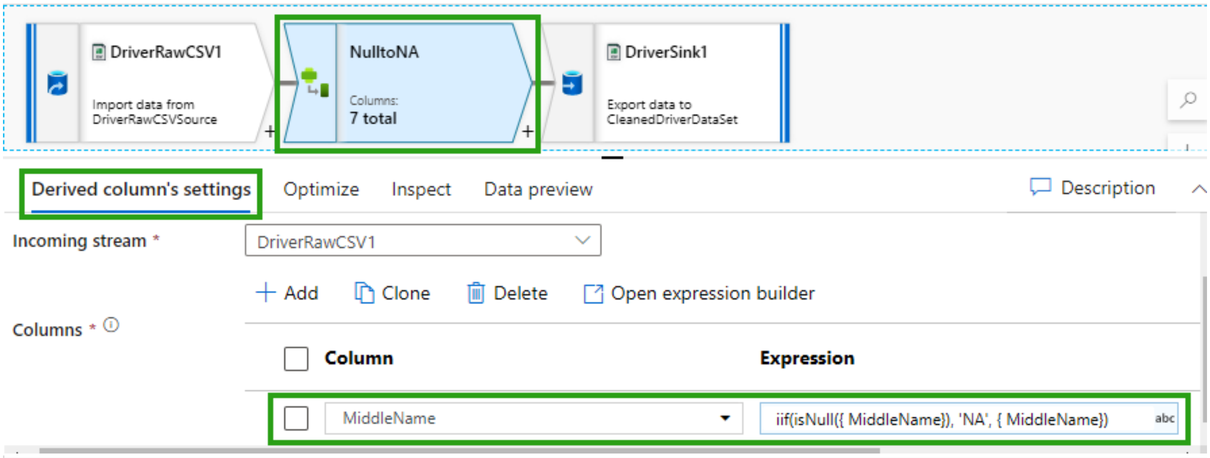

DriverCSV10
Import data from DriverCSVSource
DriverRatingJoin
Columns:
10 total
RatingCSV2
Import data from RatingCSVSource
Join settings
Optimize
Inspect
Data preview
Description
Output stream name *
DriverRatingJoin
Learn more
Left stream *
DriverCSV10
Right stream *
RatingCSV2
Join type *
Full outer
Inner
Left outer
Right outer
Custom (cross)
Join conditions *
Left: DriverCSV10's column
abc DriverId
==
Right: RatingCSV2's column
abc DriverId

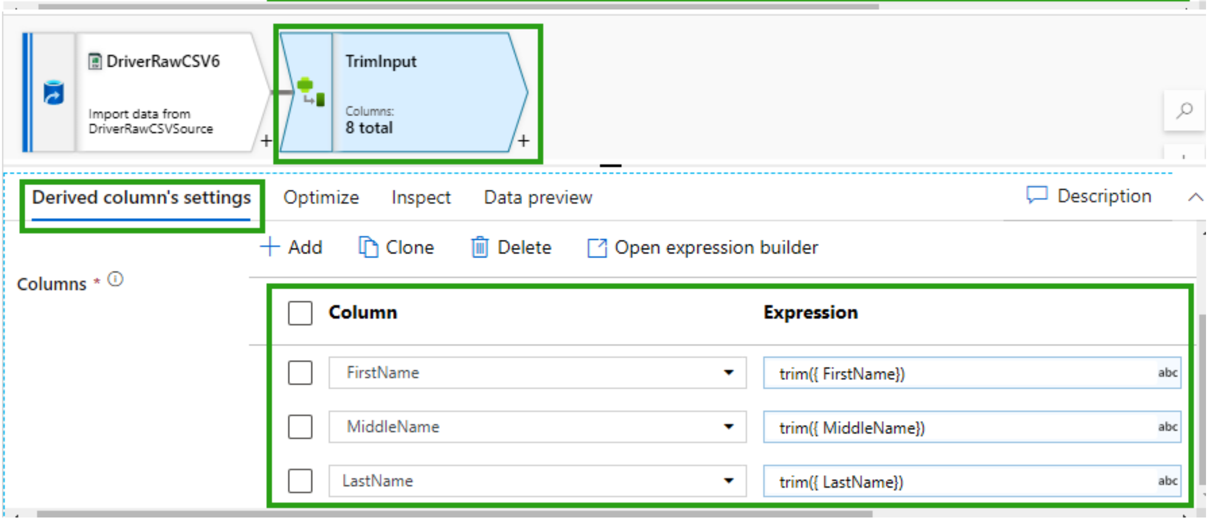

DriverCSV7
Import data from DriverCSVSource
DriverDataUnion
Columns: 7 total
DriverCSV8
Import data from DriverCSVSource
Union settings
Optimize
Inspect
Data preview
Description
Output stream name *
DriverDataUnion
Learn more
Incoming stream *
DriverCSV7
Union by *
Name
Position
Union with
Streams
DriverCSV8

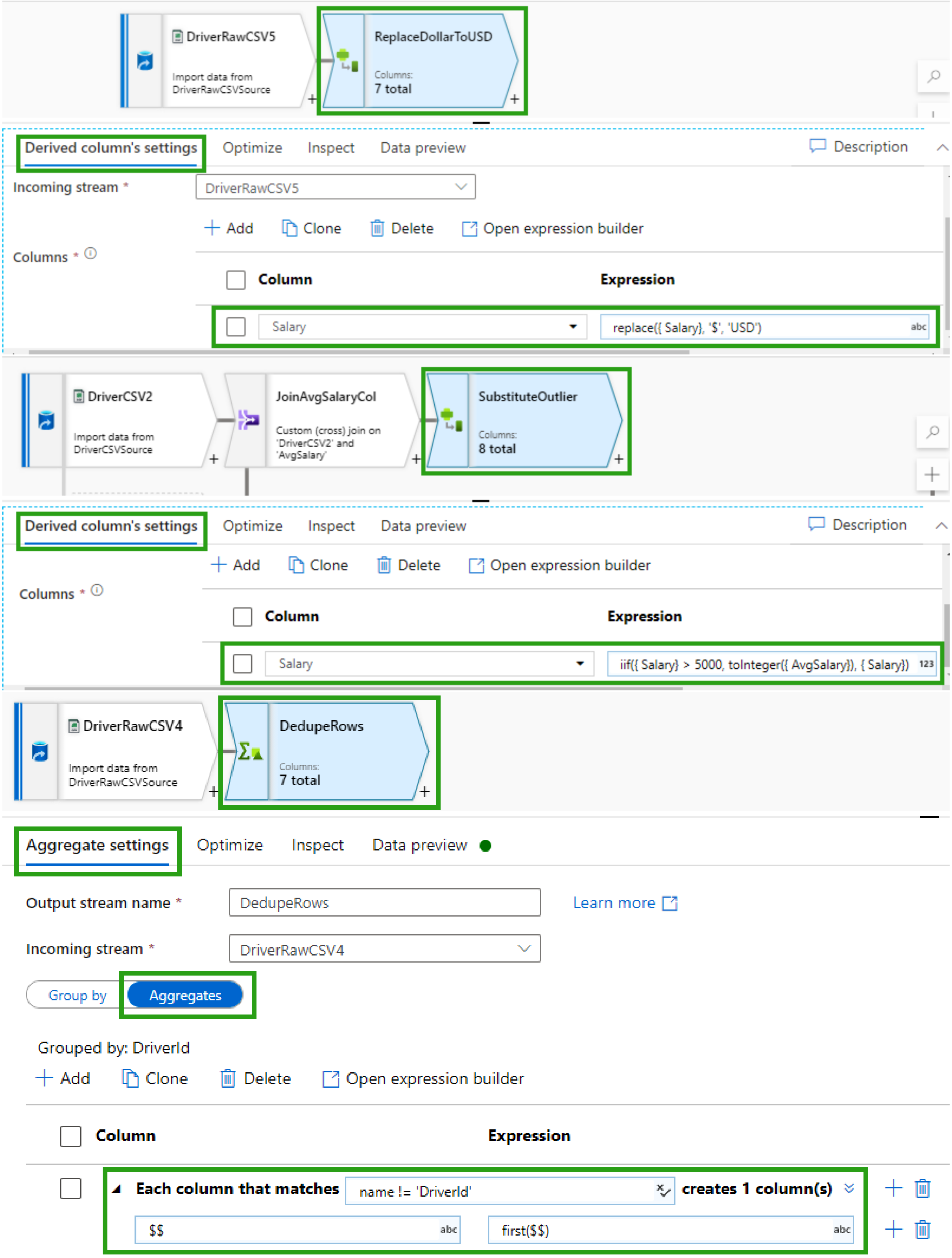

IACSampleDataFactory
New
Ingest
Copy data at scale once or on a schedule.
Orchestrate
Code-free data pipelines.
Transform data
Transform your data using data flows.
Configure SSIS
Manage & run your SSIS packages in the cloud.
Discover more
Browse partners (preview)
Pipeline templates

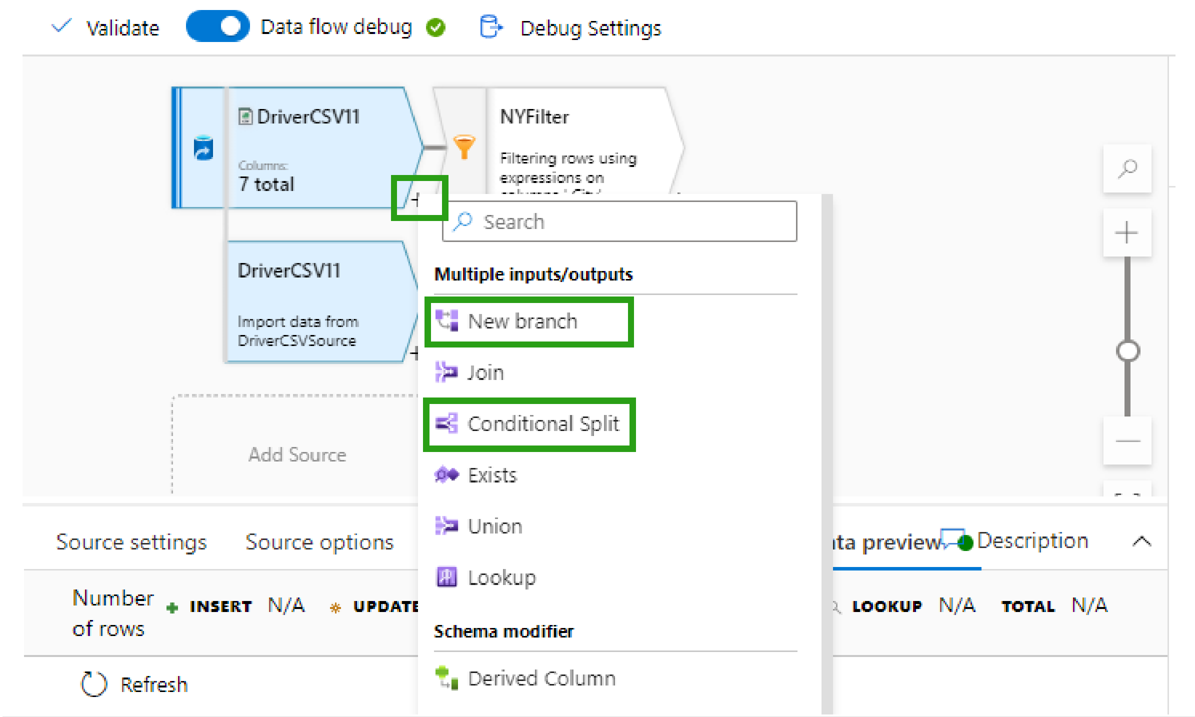

Import pipeline template
Bulk Copy from Database to Azure Data Explorer
Use this template to copy large amount of data in bulk from database like SQL Server, Google BigQuery, etc to Azure Data Explorer (ADX), using...
by Microsoft
Bulk Copy from Database
Use this template to copy data in bulk from database using external control table to store partition list of source tables.
...
by Microsoft
Bulk Copy from Files to Database
Use this template to copy data in bulk from Azure Data Lake Storage Gen2 to Azure Synapse Analytics / Azure Sql Database.
If you want to copy data from a small number of...
by Microsoft
Spark
Copy and convert data from Office 365 into Common Data Model for Open Data...
Use this template to copy data from your Office 365 organization and convert it into Common Data Model format to be included in the Open Data...
by Microsoft
Copy data from Google BigQuery to Azure Data Lake Store
Use this template to copy data from Google BigQuery to Azure Data Lake Storage.
...
by Microsoft
Copy data from HDFS to Azure Data Lake Store
Use this template to copy data from HDFS (Hadoop Distributed File System) to Azure Data Lake Storage.
...
by Microsoft

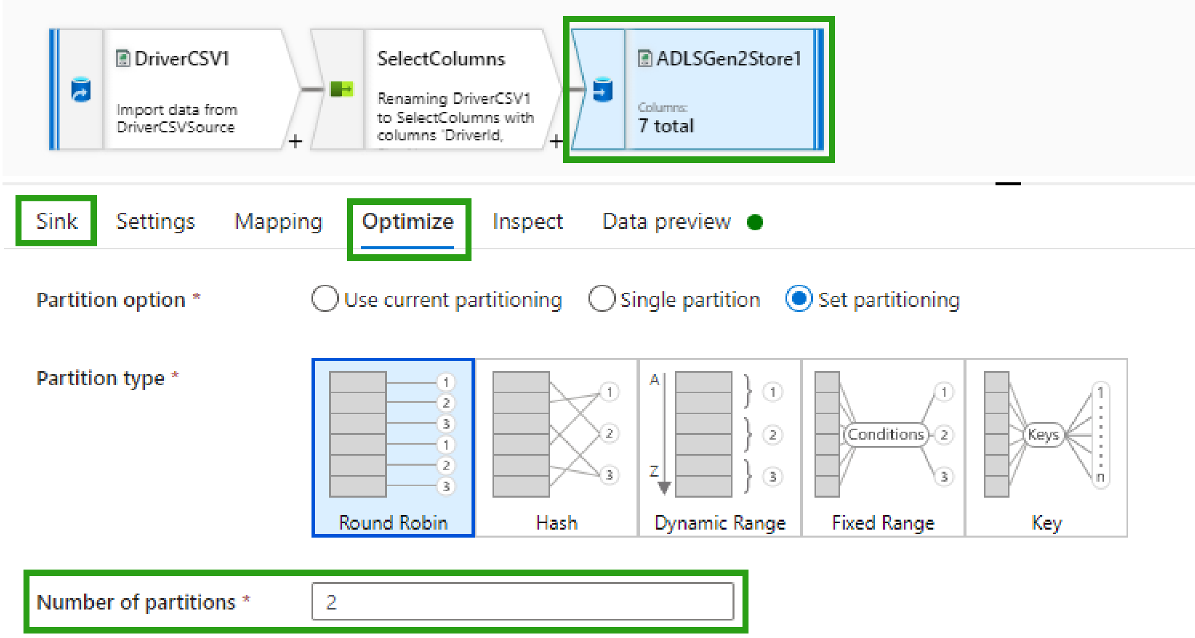

Microsoft Azure
Synapse Analytics
iacsynapsews
Synapse live
Validate all
Publish all
Integrate
Filter resources by name
Pipelines
DataPartition
samplecopy
Pipeline
Copy Data tool
Browse gallery


DriverRawCSV1
Import data from DriverRawCSVSource
NulltoNA
Columns:
7 total
DriverSink1
Export data to CleanedDriverDataSet
Derived column's settings
Optimize
Inspect
Data preview
Description
Incoming stream *
DriverRawCSV1
Add
Clone
Delete
Open expression builder
Columns *
Column
Expression
MiddleName
iif(isNull({ MiddleName}), 'NA', { MiddleName})


DriverRawCSV6
Import data from DriverRawCSVSource
TrimInput
Columns:
8 total
Derived column's settings
Optimize
Inspect
Data preview
Description
Add
Clone
Delete
Open expression builder
Columns *
Column
Expression
FirstName
trim({ FirstName})
MiddleName
trim({ MiddleName})
LastName
trim({ LastName})


DriverRawCSV5
Import data from DriverRawCSVSource
ReplaceDollarToUSD
Columns: 7 total
Derived column's settings
Optimize
Inspect
Data preview
Description
Incoming stream *
DriverRawCSV5
Add
Clone
Delete
Open expression builder
Columns *
Column
Expression
Salary
replace({ Salary}, '$', 'USD')
DriverCSV2
Import data from DriverCSVSource
JoinAvgSalaryCol
Custom (cross) join on 'DriverCSV2' and 'AvgSalary'
SubstituteOutlier
Columns: 8 total
Derived column's settings
Optimize
Inspect
Data preview
Description
Add
Clone
Delete
Open expression builder
Columns *
Column
Expression
Salary
iif({ Salary} > 5000, toInteger({ AvgSalary}), { Salary})
DriverRawCSV4
Import data from DriverRawCSVSource
DedupeRows
Columns: 7 total
Aggregate settings
Optimize
Inspect
Data preview
Output stream name *
DedupeRows
Learn more
Incoming stream *
DriverRawCSV4
Group by
Aggregates
Grouped by: DriverId
Add
Clone
Delete
Open expression builder
Column
Expression
Each column that matches
name != 'DriverId'
creates 1 column(s)
$$
first($$)


Validate
Data flow debug
Debug Settings
DriverCSV11
Columns:
7 total
NYFilter
Filtering rows using expressions on
DriverCSV11
Import data from DriverCSVSource
Add Source
Search
Multiple inputs/outputs
New branch
Join
Conditional Split
Exists
Union
Lookup
Schema modifier
Derived Column
Source settings
Source options
Description
Number of rows
INSERT
N/A
LOOKUP
N/A
TOTAL
N/A
Refresh


DriverCSV1
Import data from DriverCSVSource
SelectColumns
Renaming DriverCSV1 to SelectColumns with columns 'DriverId,
ADLSGen2Store1
Columns:
7 total
Sink
Settings
Mapping
Optimize
Inspect
Data preview
Partition option *
Use current partitioning
Single partition
Set partitioning
Partition type *
Round Robin
Hash
Dynamic Range
Fixed Range
Key
Conditions
Keys
Number of partitions *
2

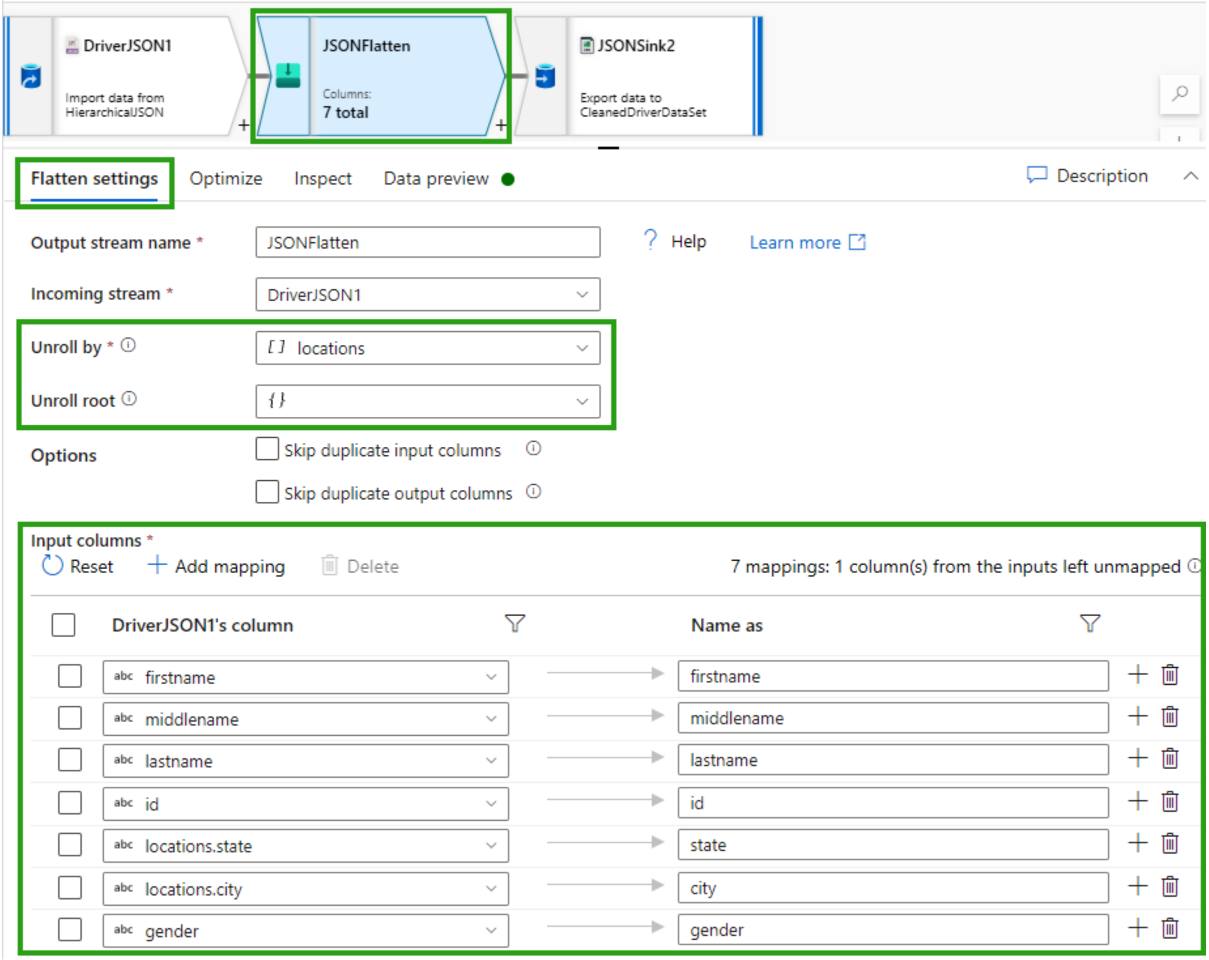
```
root
 |-- firstname: string (nullable = true)
 |-- gender: string (nullable = true)
 |-- id: string (nullable = true)
 |-- lastname: string (nullable = true)
 |-- location: string (nullable = true)
 |-- middlename: string (nullable = true)
 |-- salary: long (nullable = true)

+---------+------+---+--------+----------+----------+------+
|firstname|gender|id |lastname|location  |middlename|salary|
+---------+------+---+--------+----------+----------+------+
|Catherine|Female|102|        |California|Goodwin   |4300  |
|Jenny    |Female|104|Simons  |Arizona   |Anne      |3400  |
|Bryan    |Male  |101|Williams|New York  |M         |4000  |
|Alice    |Female|100|Hood    |New York  |          |4100  |
|Daryl    |Male  |103|Jones   |Florida   |          |5500  |
|Daryl    |Male  |103|Jones   |Florida   |          |5500  |
+---------+------+---+--------+----------+----------+------+
```

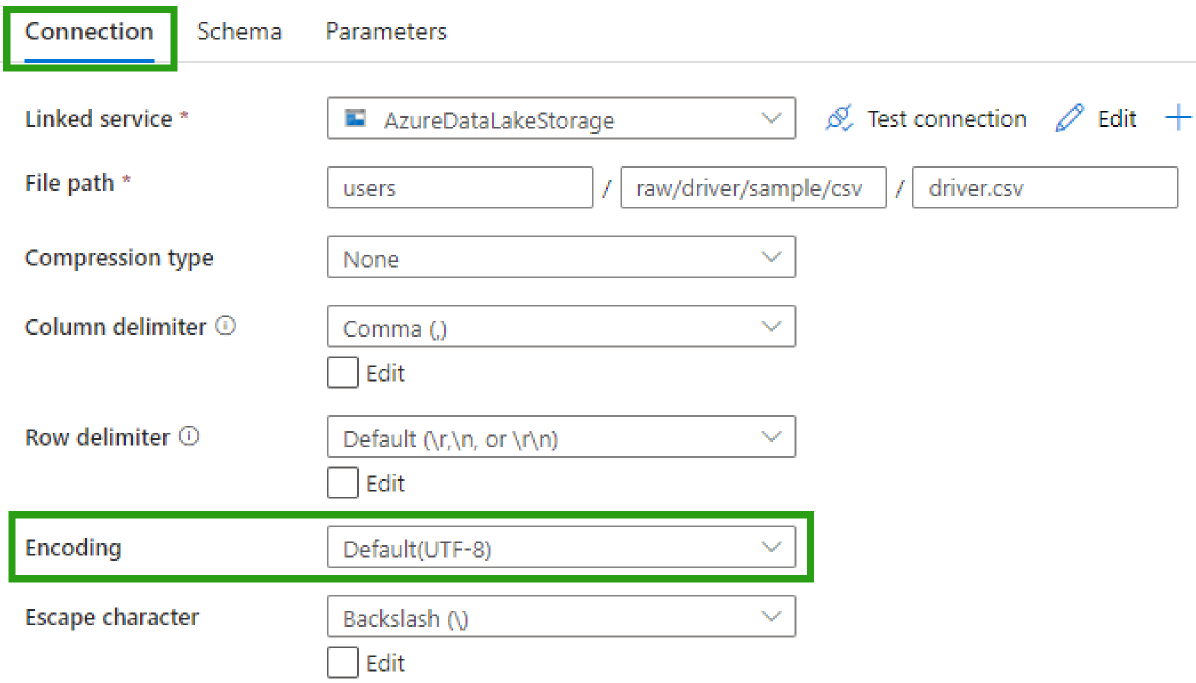
Results Messages

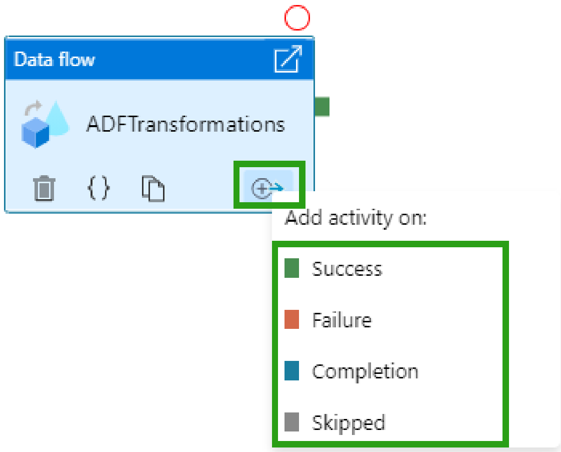
View Table Chart Export results

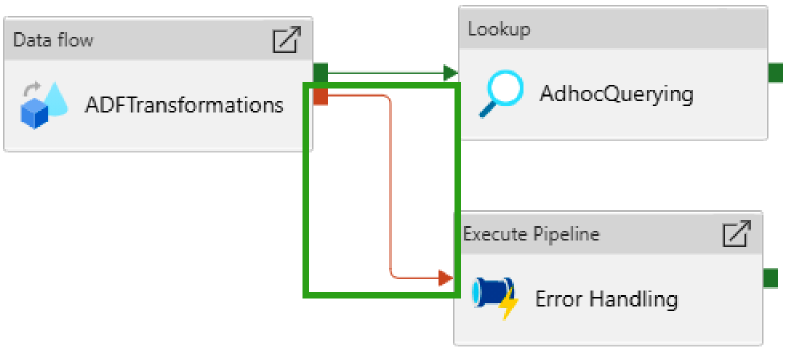
Search

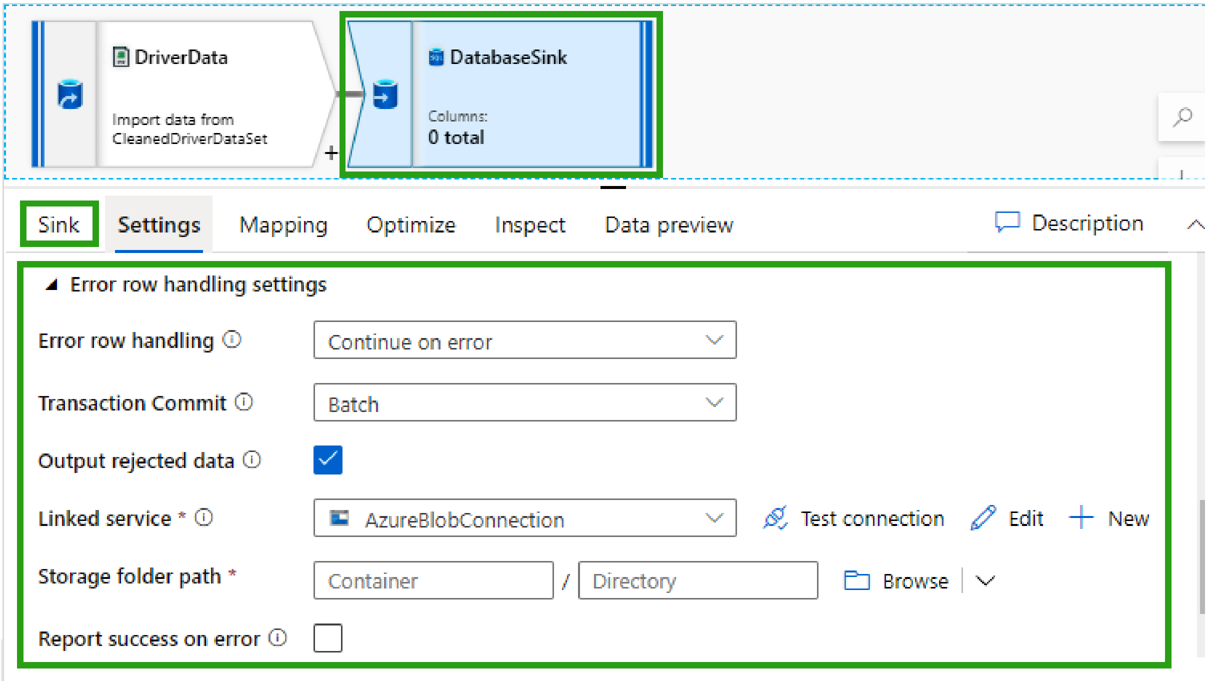
firstname	lastname	driverid	salary
Alice	Hood	100	4100
Bryan	Williams	101	4000
Daryl	Jones	103	5500
Daryl	Jones	103	5500
Jenny	Simons	104	3400
Catherine		102	4300


DriverJSON1
Import data from HierarchicalJSON
JSONFlatten
Columns:
7 total
JSONSink2
Export data to CleanedDriverDataSet
Flatten settings
Optimize
Inspect
Data preview
Description
Output stream name *
JSONFlatten
Help
Learn more
Incoming stream *
DriverJSON1
Unroll by *
[ ] locations
Unroll root
{ }
Options
Skip duplicate input columns
Skip duplicate output columns
Input columns *
Reset
Add mapping
Delete
7 mappings: 1 column(s) from the inputs left unmapped
DriverJSON1's column
Name as
abc firstname
firstname
abc middlename
middlename
abc lastname
lastname
abc id
id
abc locations.state
state
abc locations.city
city
abc gender
gender

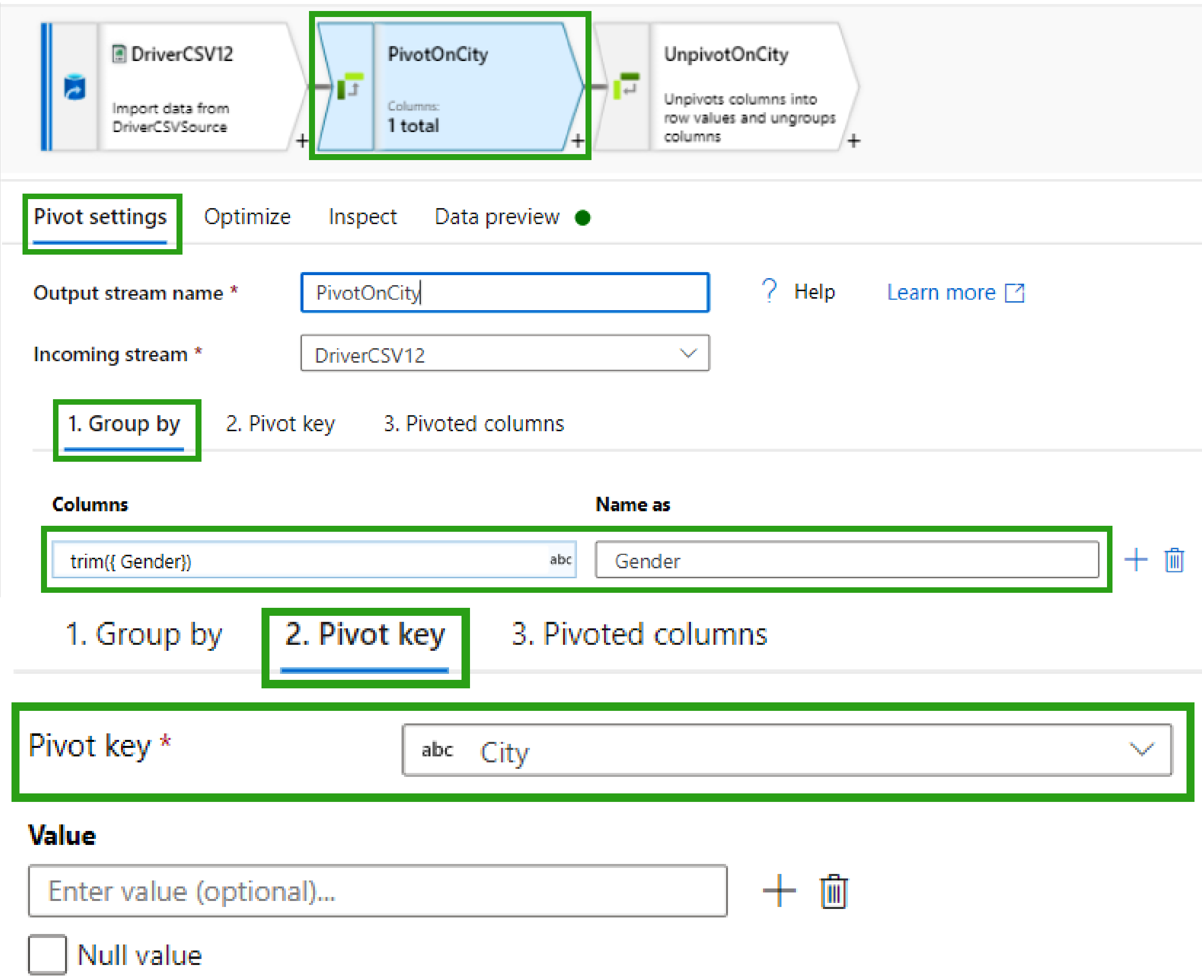

Connection
Schema
Parameters
Linked service *
AzureDataLakeStorage
Test connection
Edit
File path *
users
/
raw/driver/sample/csv
/
driver.csv
Compression type
None
Column delimiter
Comma (,)
Edit
Row delimiter
Default (\r,\n, or \r\n)
Edit
Encoding
Default(UTF-8)
Escape character
Backslash (\)
Edit

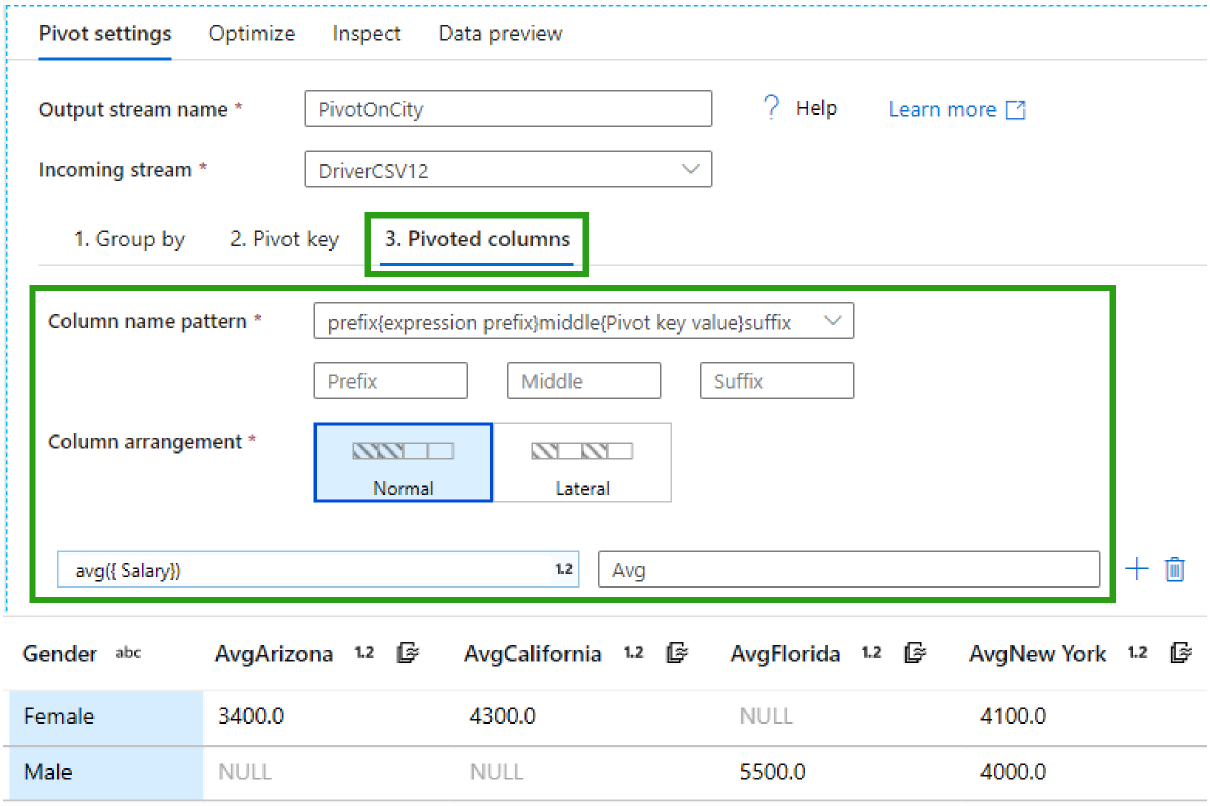

Data flow
ADFTransformations
Add activity on:
Success
Failure
Completion
Skipped

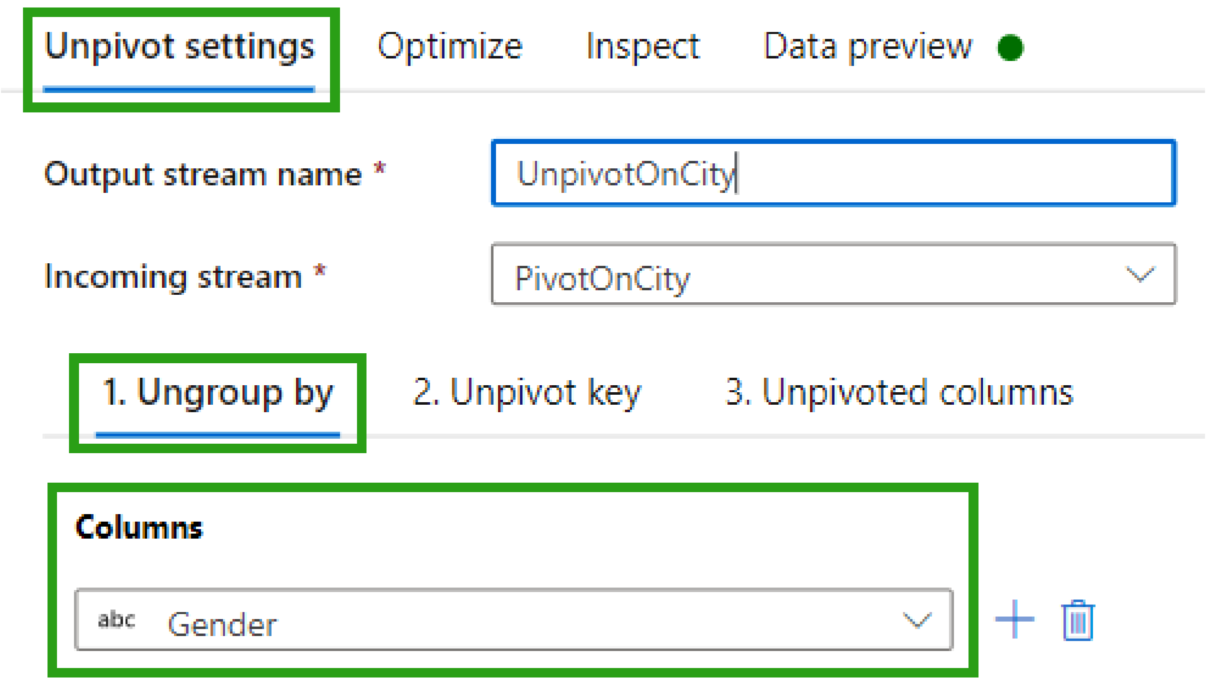

Data flow
ADFTransformations
Lookup
AdhocQuerying
Execute Pipeline
Error Handling

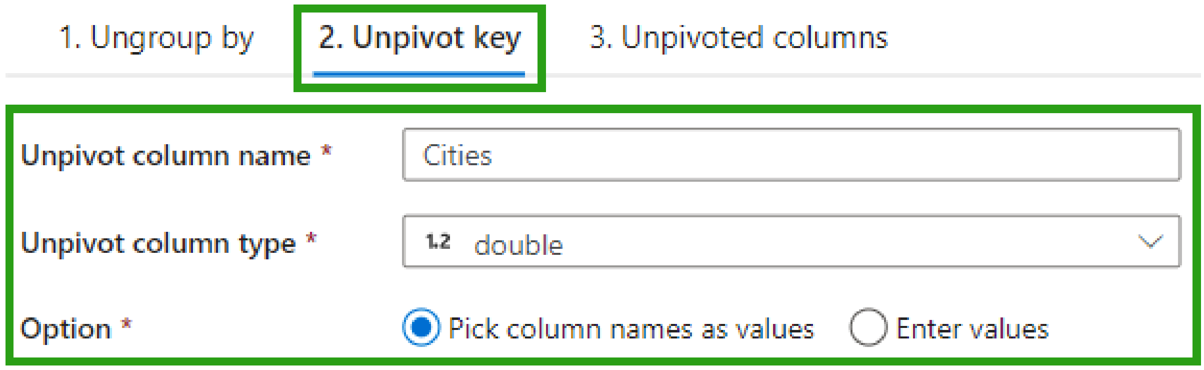

DriverData
Import data from CleanedDriverDataSet
DatabaseSink
Columns:
0 total
Sink
Settings
Mapping
Optimize
Inspect
Data preview
Description
Error row handling settings
Error row handling
Continue on error
Transaction Commit
Batch
Output rejected data
Linked service *
AzureBlobConnection
Test connection
Edit
New
Storage folder path *
Container
Directory
Browse
Report success on error

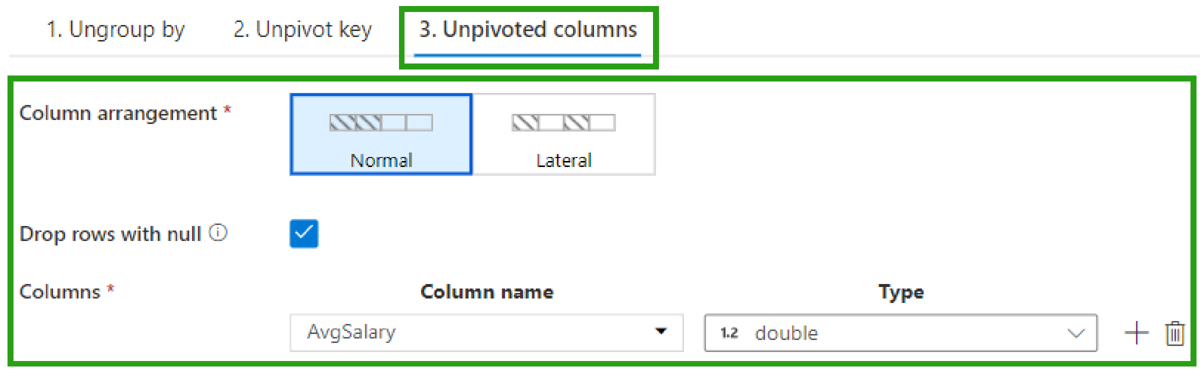
DriverId 123	FirstName abc	MiddleName abc	LastName abc	City abc	Gender abc	Salary 123
200	Alice	NULL	Hood	New York	Female	4100
201	Bryan	M	Williams	New York	Male	4000
202	Catherine	Goodwin	NULL	California	Female	4300
203	Daryl	NULL	Jones	Florida	Male	5500
204	Jenny	Anne	Simons	Arizona	Female	3400
203	Daryl	NULL	Jones	Florida	Male	5500



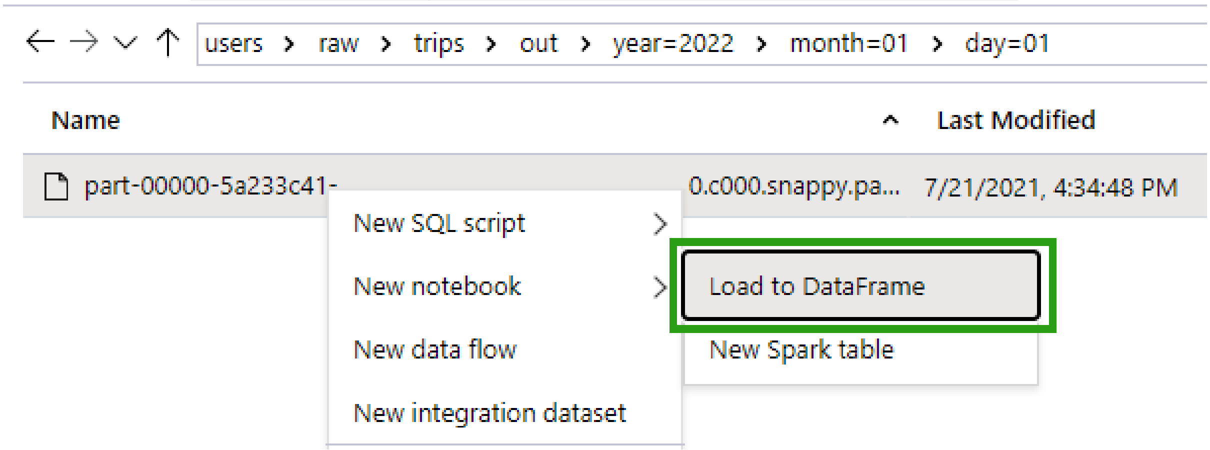

Pivot settings
Optimize
Inspect
Data preview
Output stream name *
PivotOnCity
Help
Learn more
Incoming stream *
DriverCSV12
1. Group by
2. Pivot key
3. Pivoted columns
Column name pattern *
prefix{expression prefix}middle{Pivot key value}suffix
Prefix
Middle
Suffix
Column arrangement *
Normal
Lateral
avg({ Salary})
1.2
Avg
Gender abc
AvgArizona 1.2
AvgCalifornia 1.2
AvgFlorida 1.2
AvgNew York 1.2
Female 3400.0 4300.0 NULL 4100.0
Male NULL NULL 5500.0 4000.0

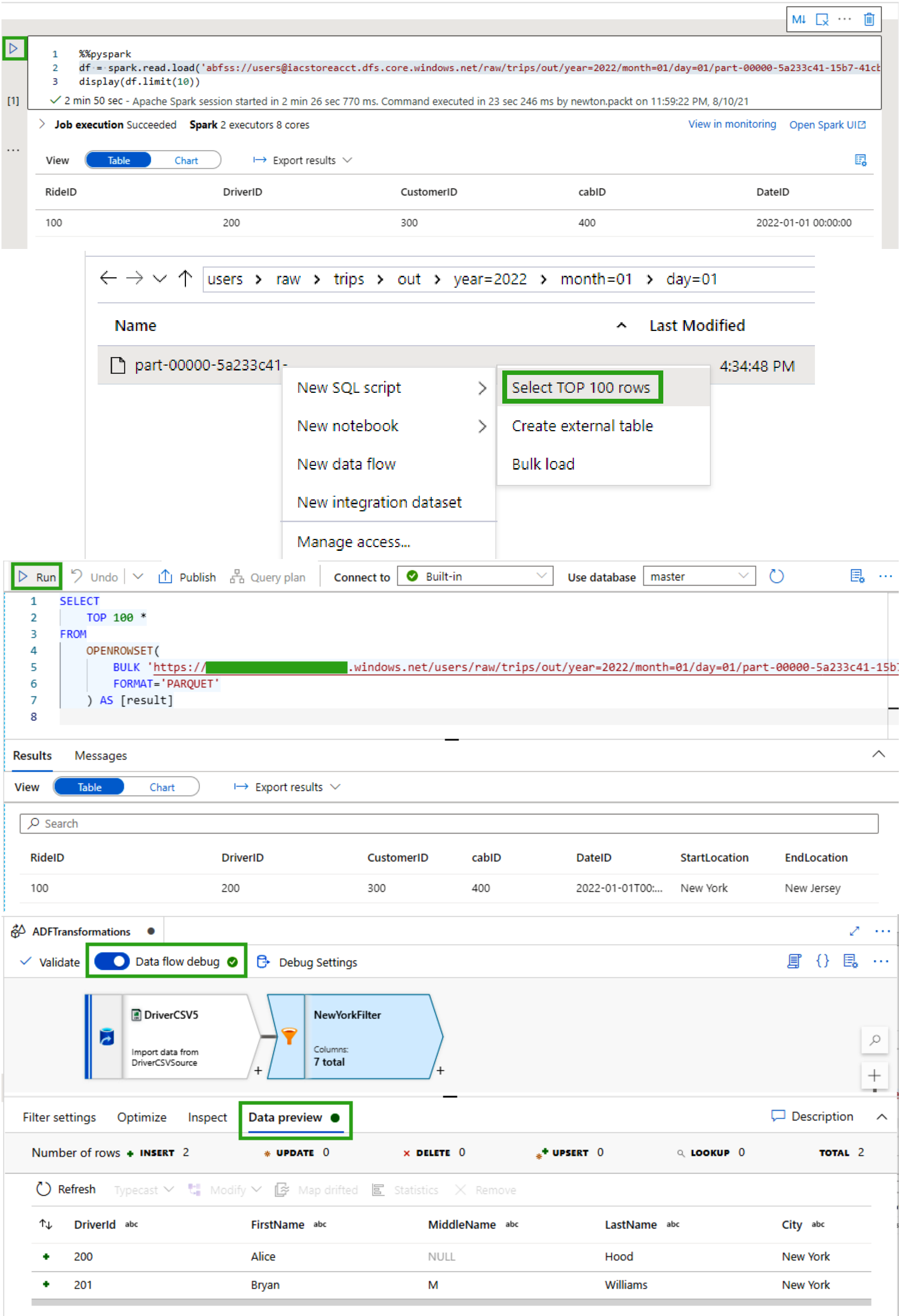

Unpivot settings
Optimize
Inspect
Data preview
Output stream name *
UnpivotOnCity
Incoming stream *
PivotOnCity
1. Ungroup by
2. Unpivot key
3. Unpivoted columns
Columns
abc Gender





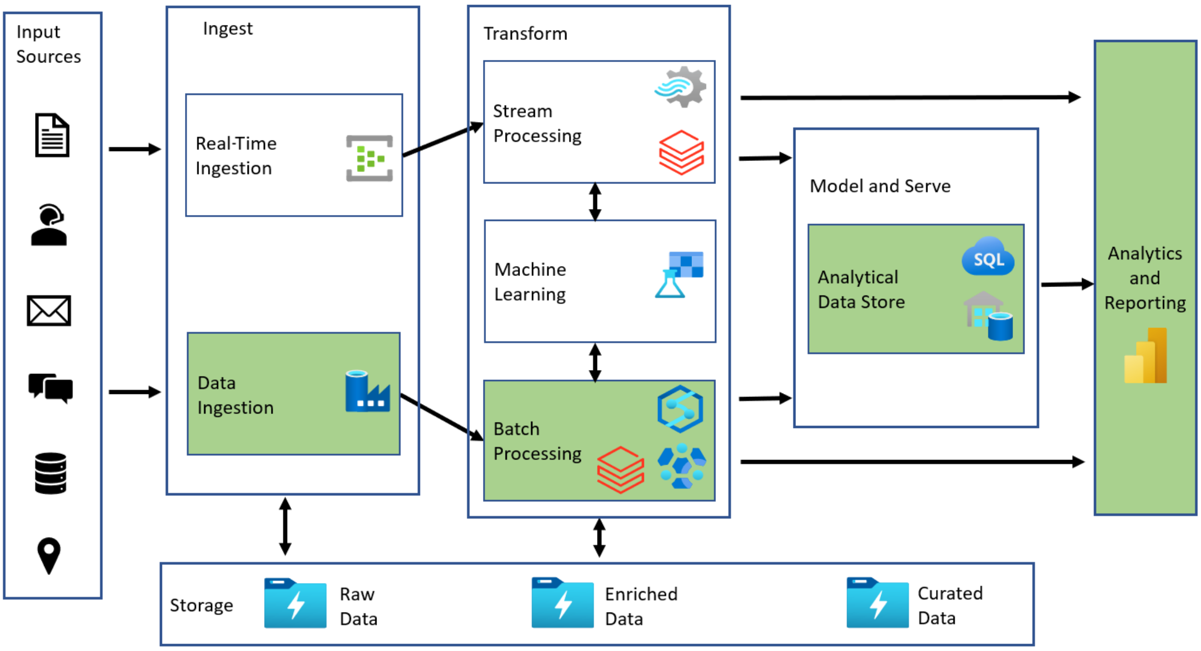
Gender abc	Cities abc	AvgSalary 1.2
Female	AvgArizona	3400.0
Female	AvgCalifornia	4300.0
Female	AvgNew York	4100.0
Male	AvgFlorida	5500.0
Male	AvgNew York	4000.0

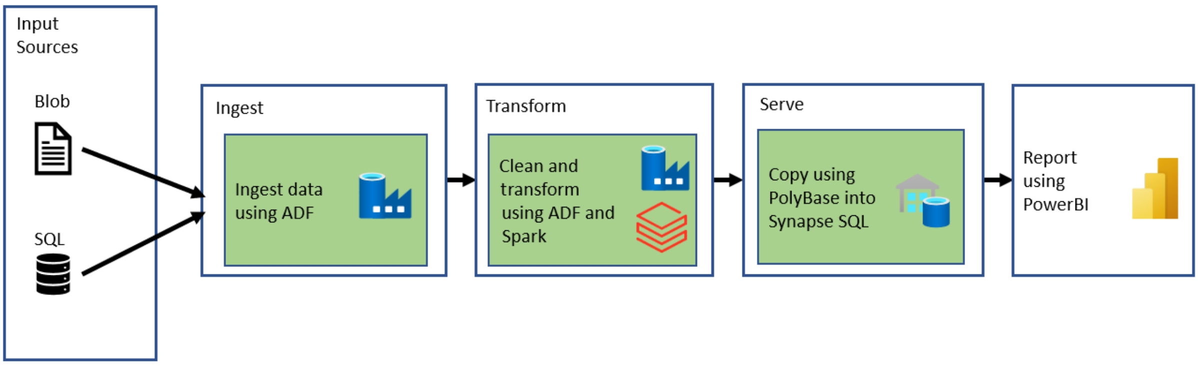


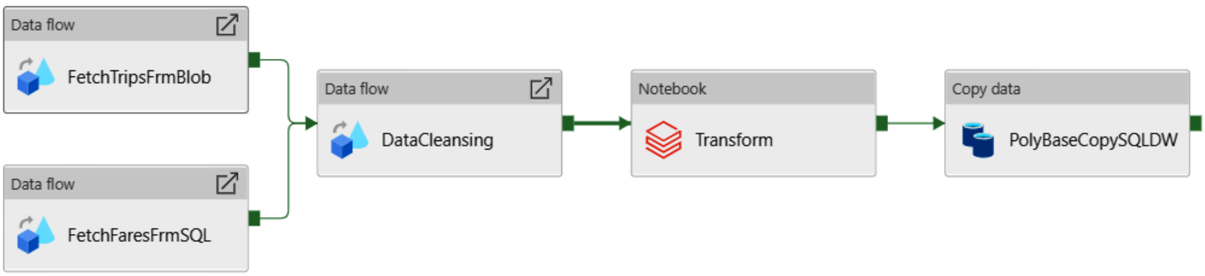


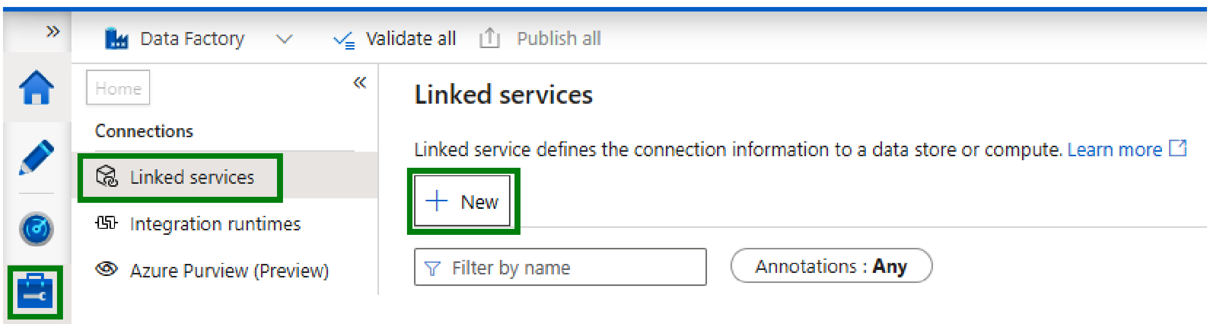
%%pyspark
df = spark.read.load('abfss://users@iacstoreacct.dfs.core.windows.net/raw/trips/out/year=2022/month=01/day=01/part-00000-5a233c41-15b7-41cb
display(df.limit(10))
[1]
2 min 50 sec - Apache Spark session started in 2 min 26 sec 770 ms. Command executed in 23 sec 246 ms by newton.packt on 11:59:22 PM, 8/10/21
Job execution Succeeded Spark 2 executors 8 cores
View in monitoring
Open Spark UI
View
Table
Chart
Export results
RideID DriverID CustomerID cabID DateID
100 200 300 400 2022-01-01 00:00:00
users raw trips out year=2022 month=01 day=01
Name
Last Modified
part-00000-5a233c41-
4:34:48 PM
New SQL script
New notebook
New data flow
New integration dataset
Manage access...
Select TOP 100 rows
Create external table
Bulk load
Run
Undo
Publish
Query plan
Connect to
Built-in
Use database
master
SELECT
TOP 100 *
FROM
OPENROWSET(
BULK 'https://
.windows.net/users/raw/trips/out/year=2022/month=01/day=01/part-00000-5a233c41-15b
FORMAT='PARQUET'
) AS [result]
Results
Messages
View
Table
Chart
Export results
Search
RideID DriverID CustomerID cabID DateID StartLocation EndLocation
100 200 300 400 2022-01-01T00:... New York New Jersey
ADFTransformations
Validate
Data flow debug
Debug Settings
DriverCSV5
Import data from DriverCSVSource
NewYorkFilter
Columns:
7 total
Filter settings
Optimize
Inspect
Data preview
Description
Number of rows
INSERT 2
UPDATE 0
DELETE 0
UPSERT 0
LOOKUP 0
TOTAL 2
Refresh
Typecast
Modify
Map drifted
Statistics
Remove
DriverId FirstName MiddleName LastName City
200 Alice NULL Hood New York
201 Bryan M Williams New York

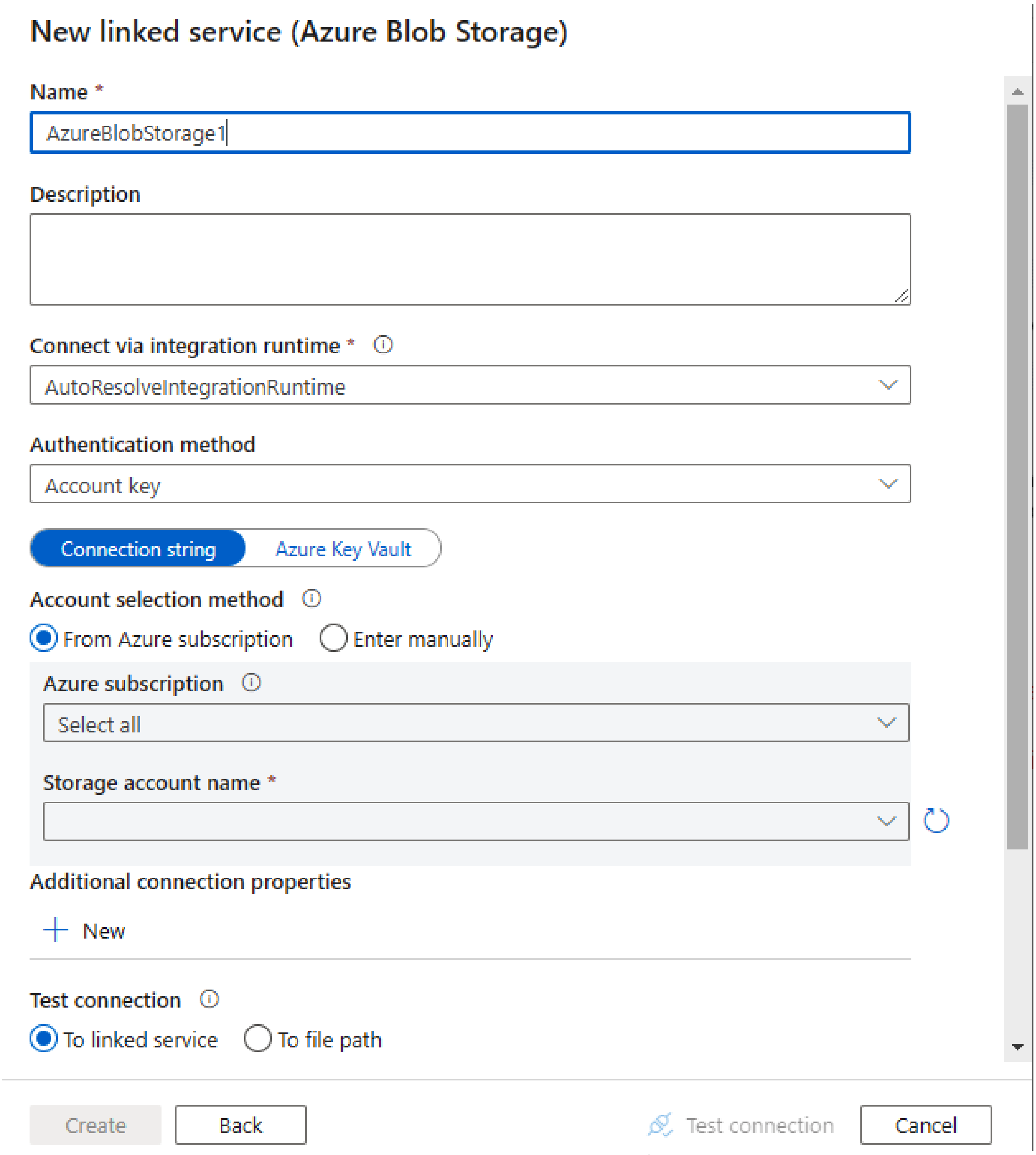
Chapter 9: Designing and Developing a Batch Processing Solution

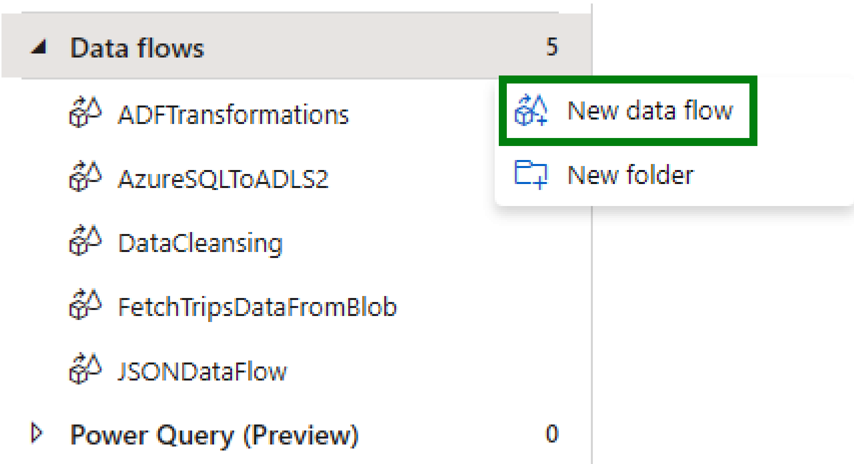


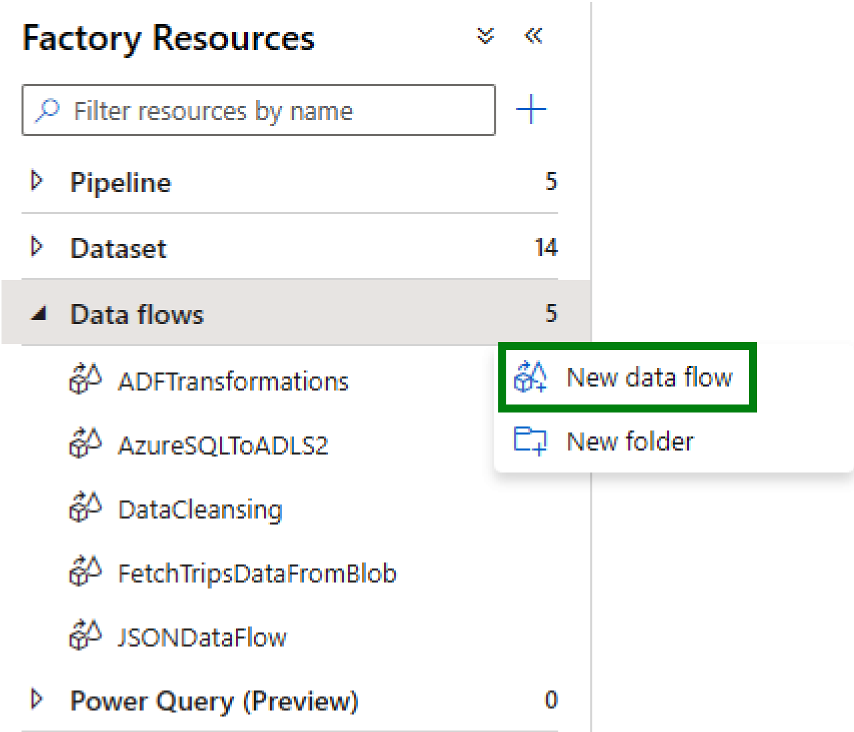


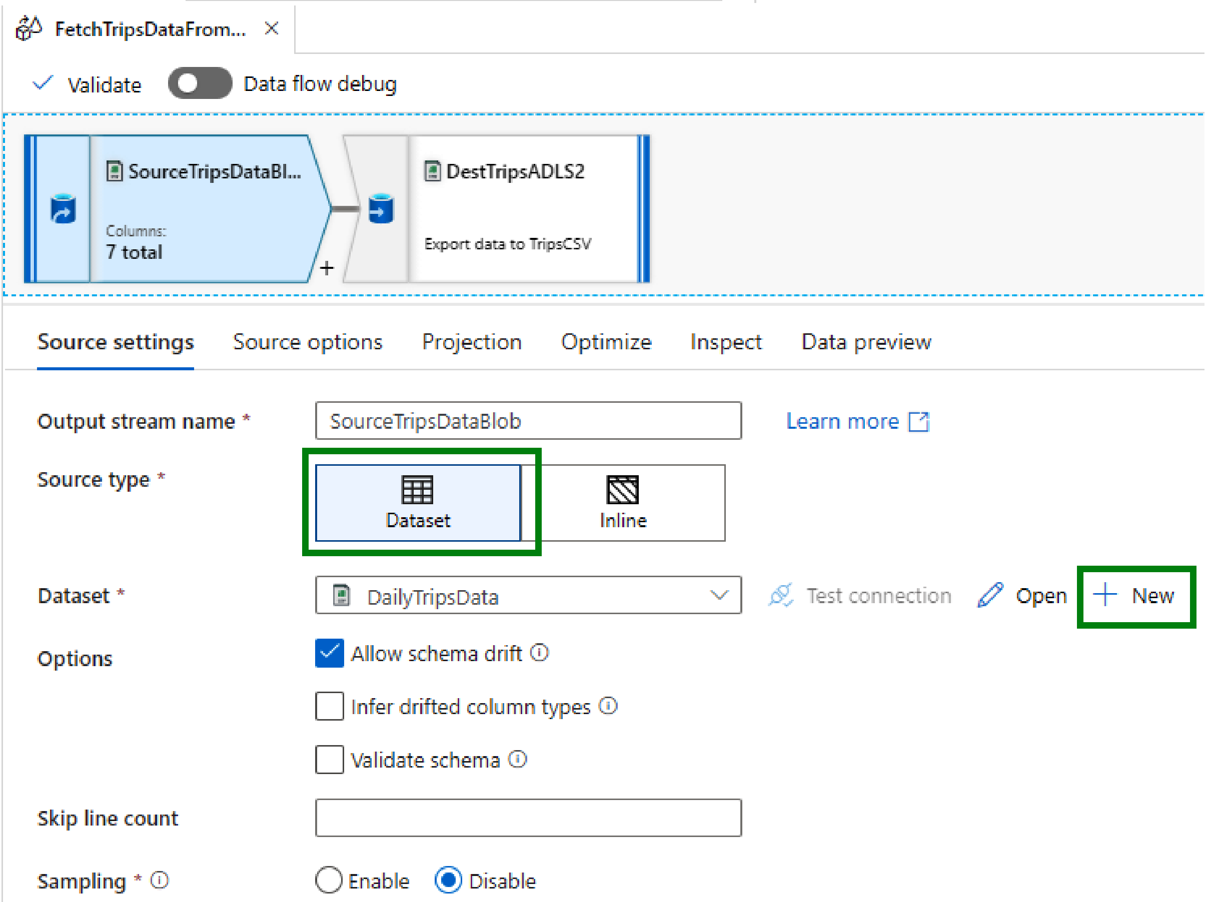


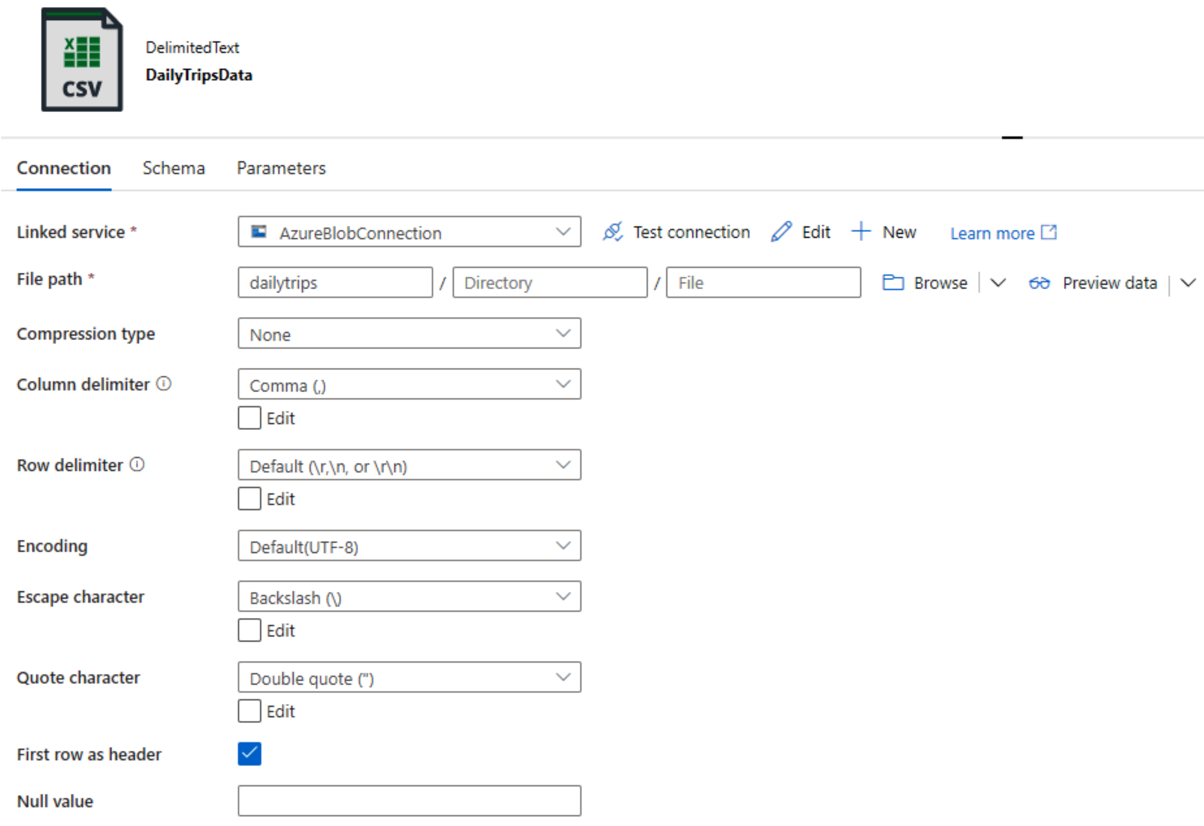


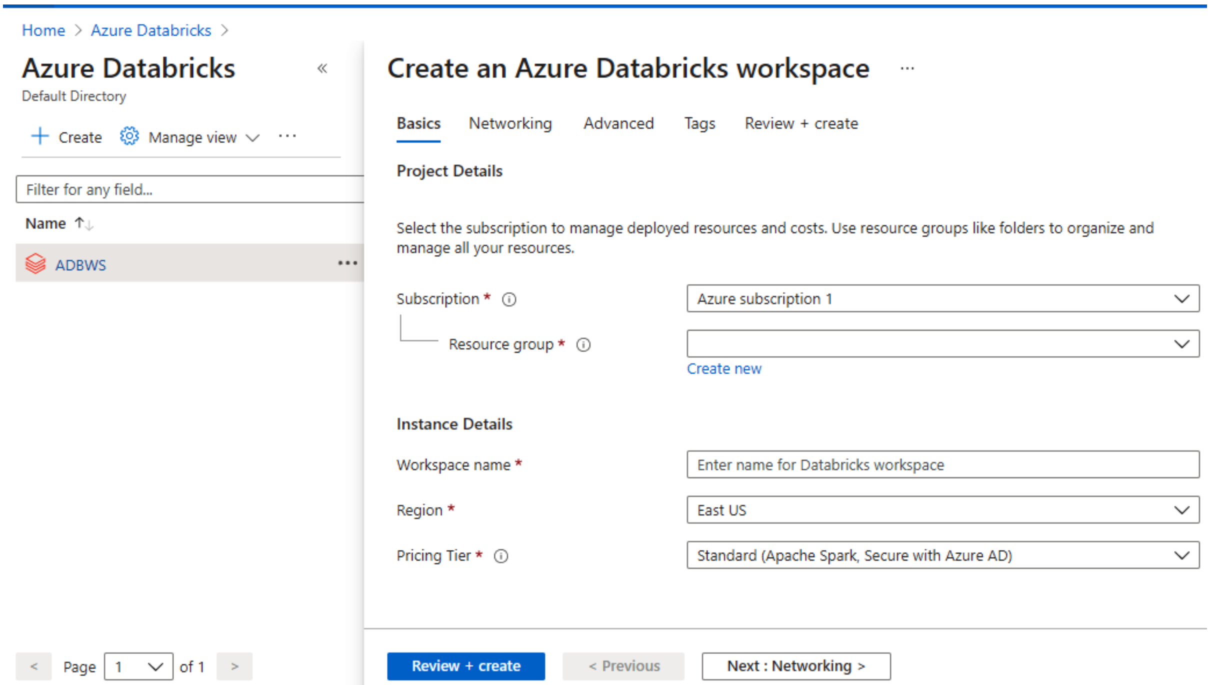

New linked service (Azure Blob Storage)
Name *
AzureBlobStorage1
Description
Connect via integration runtime *
AutoResolveIntegrationRuntime
Authentication method
Account key
Connection string
Azure Key Vault
Account selection method
From Azure subscription
Enter manually
Azure subscription
Select all
Storage account name *
Additional connection properties
New
Test connection
To linked service
To file path
Create
Back
Test connection
Cancel

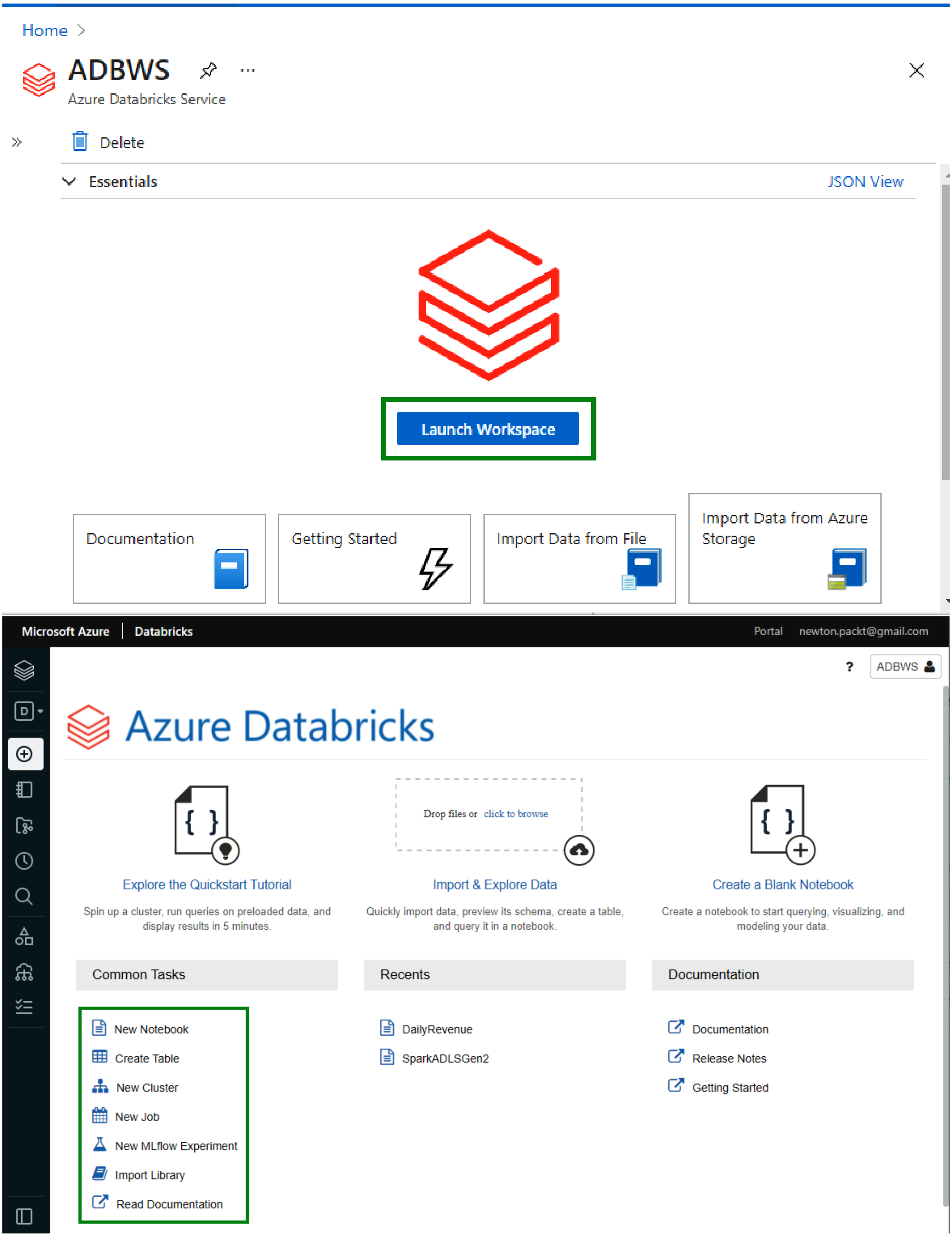

Data flows
5
ADFTransformations
AzureSQLToADLS2
DataCleansing
FetchTripsDataFromBlob
JSONDataFlow
Power Query (Preview)
0
New data flow
New folder


Factory Resources
Filter resources by name
Pipeline 5
Dataset 14
Data flows 5
ADFTransformations
AzureSQLToADLS2
DataCleansing
FetchTripsDataFromBlob
JSONDataFlow
Power Query (Preview) 0
New data flow
New folder

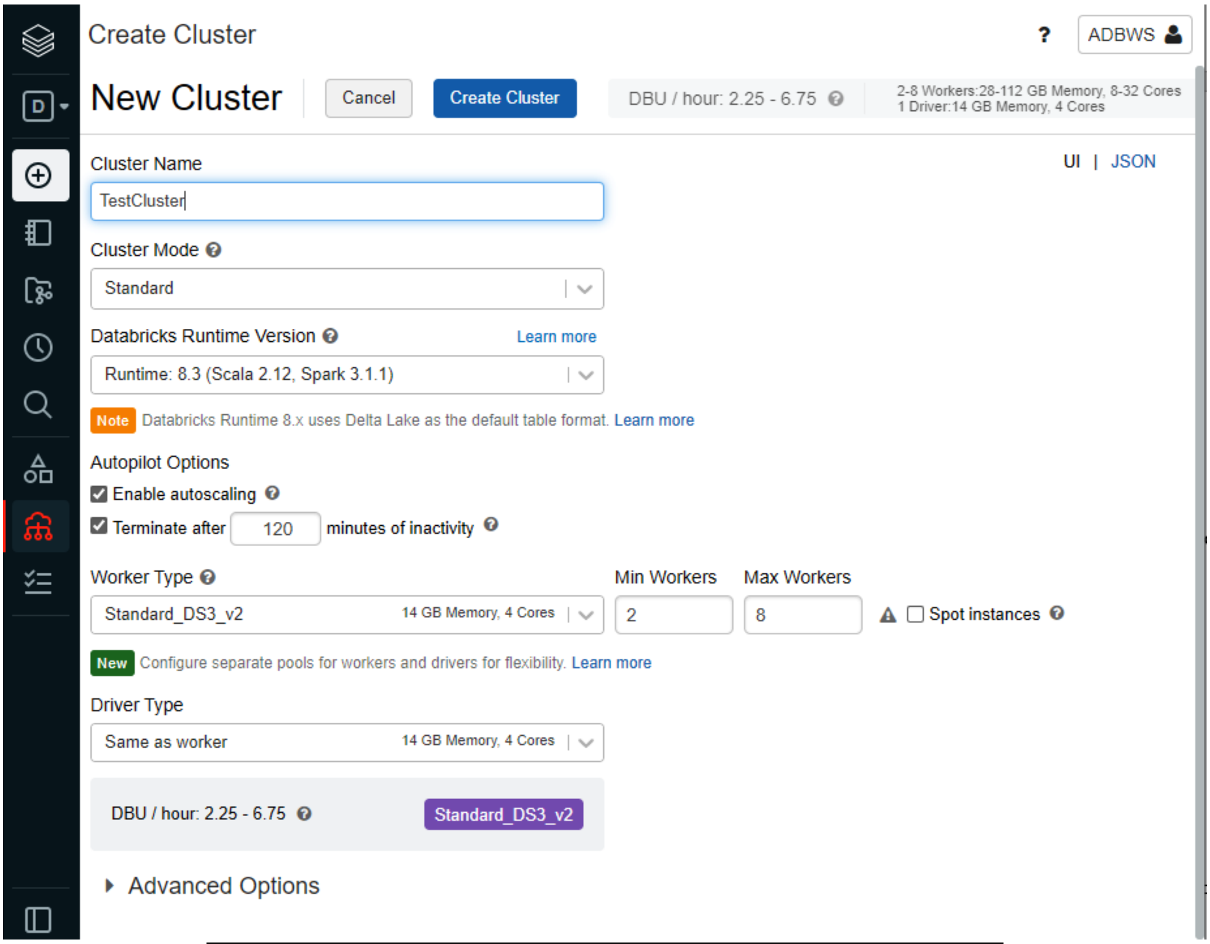

FetchTripsDataFrom...
Validate
Data flow debug
SourceTripsDataBl...
Columns:
7 total
DestTripsADLS2
Export data to TripsCSV
Source settings
Source options
Projection
Optimize
Inspect
Data preview
Output stream name *
SourceTripsDataBlob
Learn more
Source type *
Dataset
Inline
Dataset *
DailyTripsData
Test connection
Open
New
Options
Allow schema drift
Infer drifted column types
Validate schema
Skip line count
Sampling *
Enable
Disable

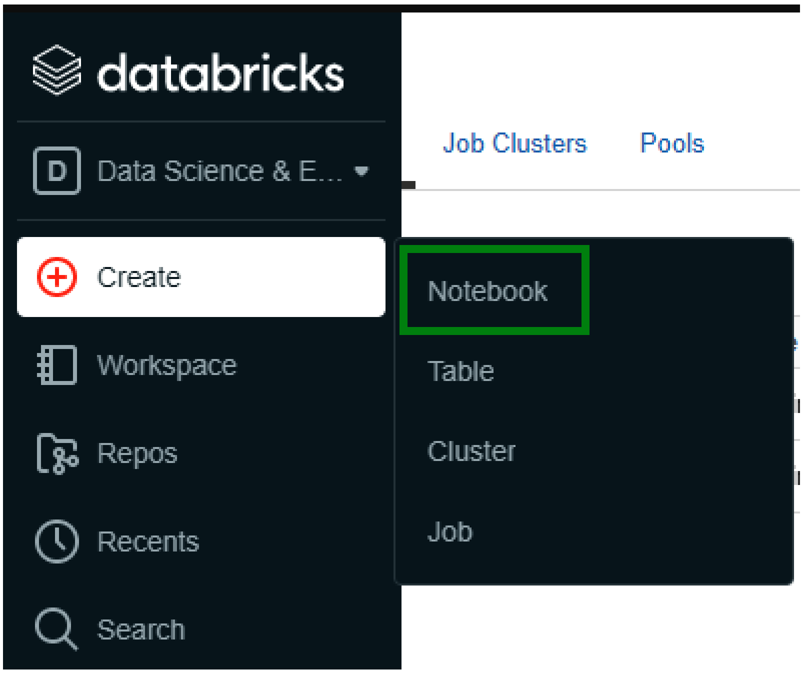

DelimitedText
DailyTripsData
CSV
Connection
Schema
Parameters
Linked service *
AzureBlobConnection
Test connection
Edit
New
Learn more
File path *
dailytrips
Directory
File
Browse
Preview data
Compression type
None
Column delimiter
Comma (,)
Edit
Row delimiter
Default (\r,\n, or \r\n)
Edit
Encoding
Default(UTF-8)
Escape character
Backslash (\)
Edit
Quote character
Double quote (")
Edit
First row as header
Null value

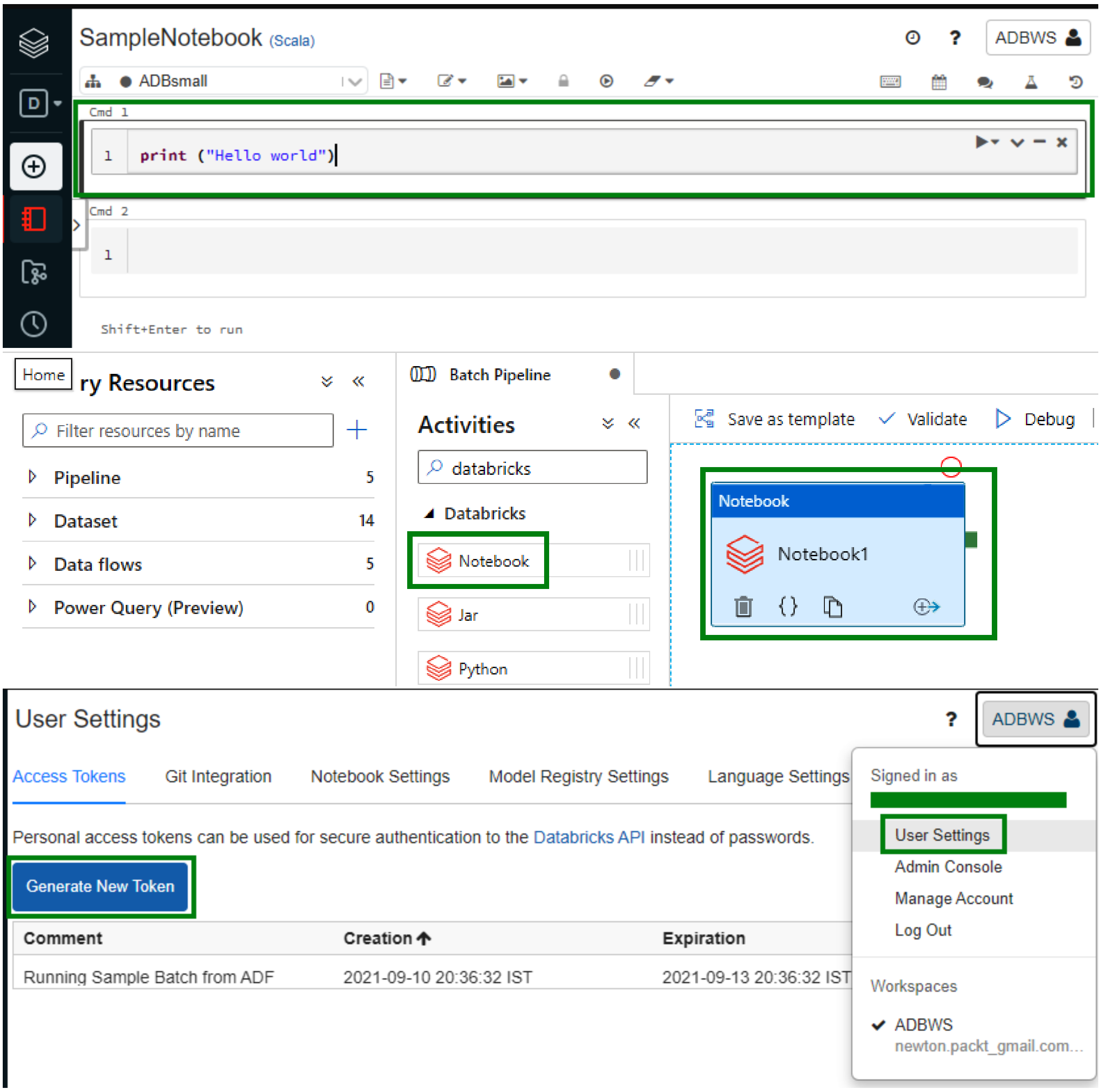

Home > Azure Databricks >
Azure Databricks
Default Directory
Create
Manage view
Filter for any field...
Name
ADBWS
Page 1 of 1
Create an Azure Databricks workspace
Basics
Networking
Advanced
Tags
Review + create
Project Details
Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.
Subscription *
Azure subscription 1
Resource group *
Create new
Instance Details
Workspace name *
Enter name for Databricks workspace
Region *
East US
Pricing Tier *
Standard (Apache Spark, Secure with Azure AD)
Review + create
< Previous
Next : Networking >

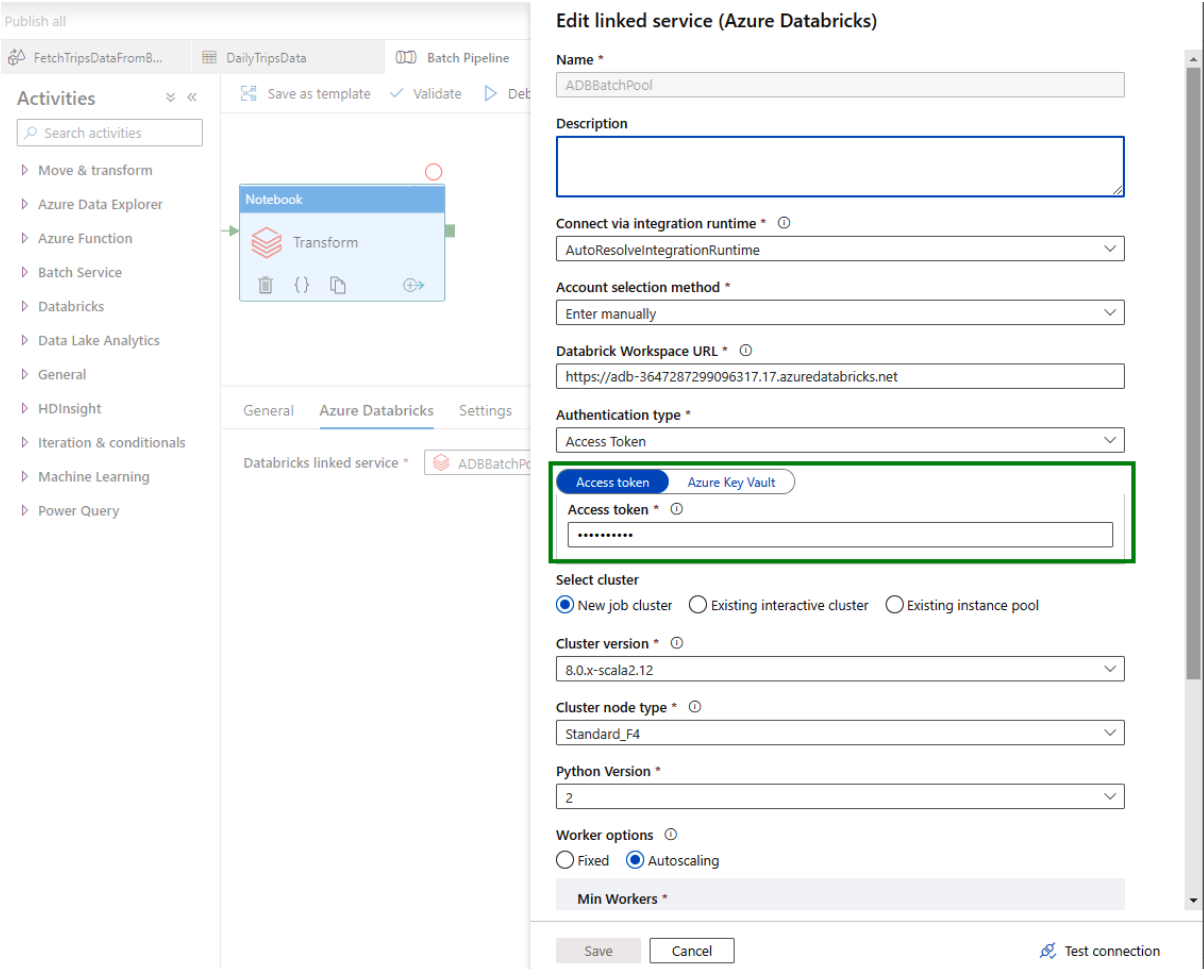


Home
ADBWS
Azure Databricks Service
Delete
Essentials
JSON View
Launch Workspace
Documentation
Getting Started
Import Data from File
Import Data from Azure Storage
Microsoft Azure
Databricks
Portal
newton.packt@gmail.com
ADBWS
Azure Databricks
Drop files or click to browse
Explore the Quickstart Tutorial
Spin up a cluster, run queries on preloaded data, and display results in 5 minutes.
Import & Explore Data
Quickly import data, preview its schema, create a table, and query it in a notebook.
Create a Blank Notebook
Create a notebook to start querying, visualizing, and modeling your data.
Common Tasks
New Notebook
Create Table
New Cluster
New Job
New MLflow Experiment
Import Library
Read Documentation
Recents
DailyRevenue
SparkADLSGen2
Documentation
Documentation
Release Notes
Getting Started

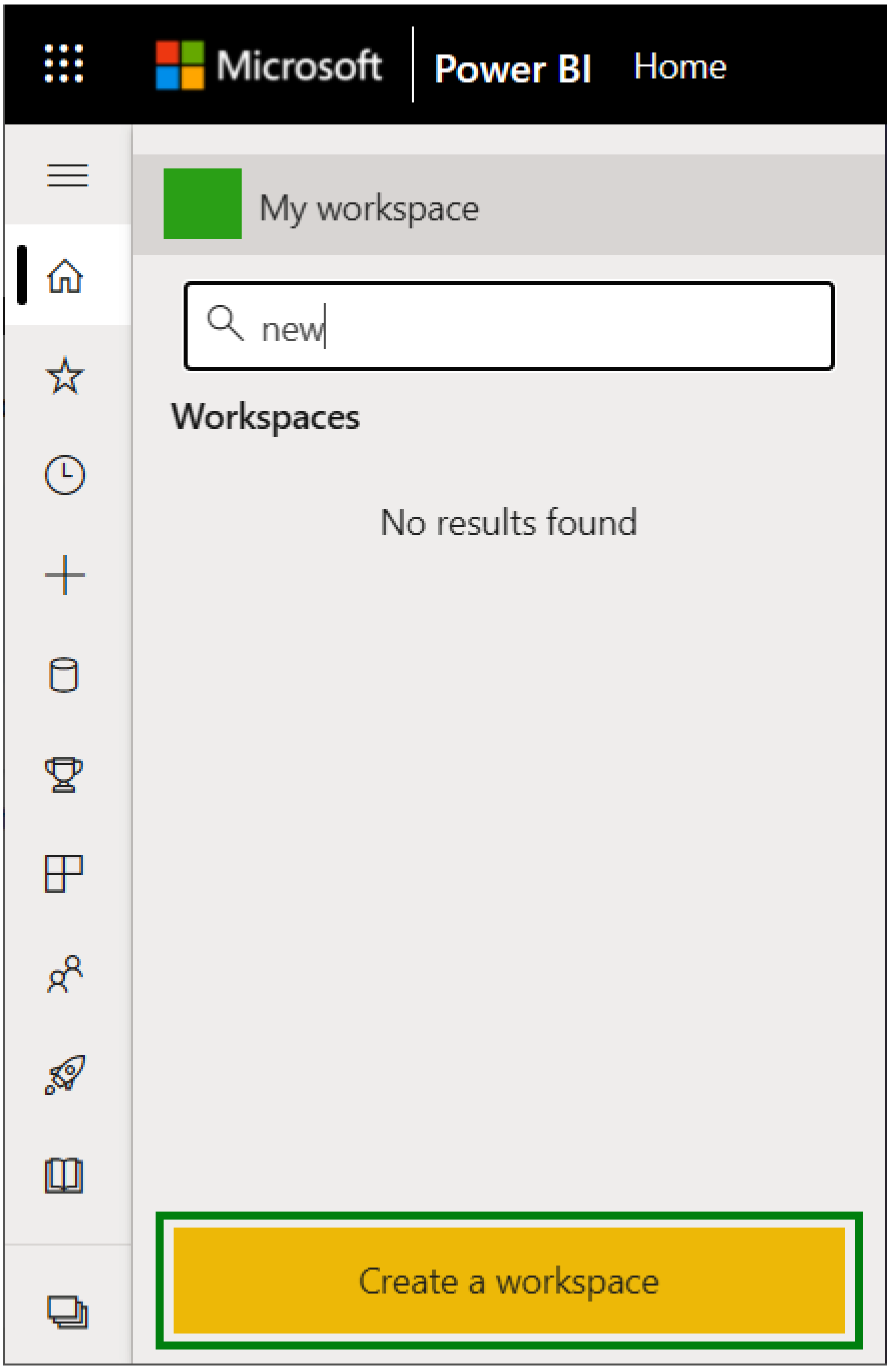

Create Cluster
ADBWS
New Cluster
Cancel
Create Cluster
DBU / hour: 2.25 - 6.75
2-8 Workers:28-112 GB Memory, 8-32 Cores
1 Driver:14 GB Memory, 4 Cores
Cluster Name
UI | JSON
TestCluster
Cluster Mode
Standard
Databricks Runtime Version
Learn more
Runtime: 8.3 (Scala 2.12, Spark 3.1.1)
Note Databricks Runtime 8.x uses Delta Lake as the default table format. Learn more
Autopilot Options
Enable autoscaling
Terminate after 120 minutes of inactivity
Worker Type
Min Workers
Max Workers
Standard_DS3_v2
14 GB Memory, 4 Cores
2
8
Spot instances
New Configure separate pools for workers and drivers for flexibility. Learn more
Driver Type
Same as worker
14 GB Memory, 4 Cores
DBU / hour: 2.25 - 6.75
Standard_DS3_v2
Advanced Options

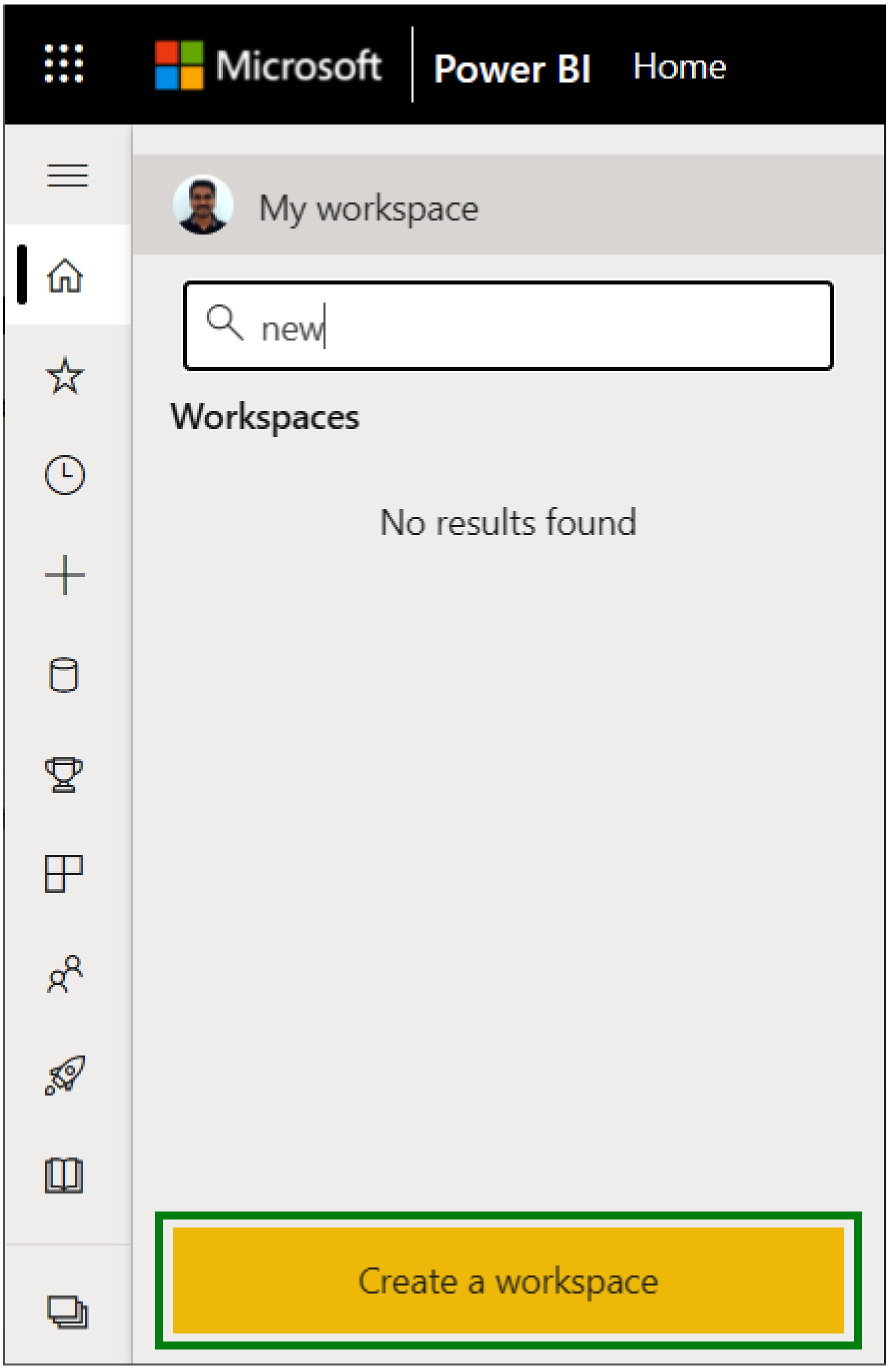

databricks
Job Clusters
Pools
Data Science & E...
Create
Notebook
Workspace
Table
Repos
Cluster
Recents
Job
Search

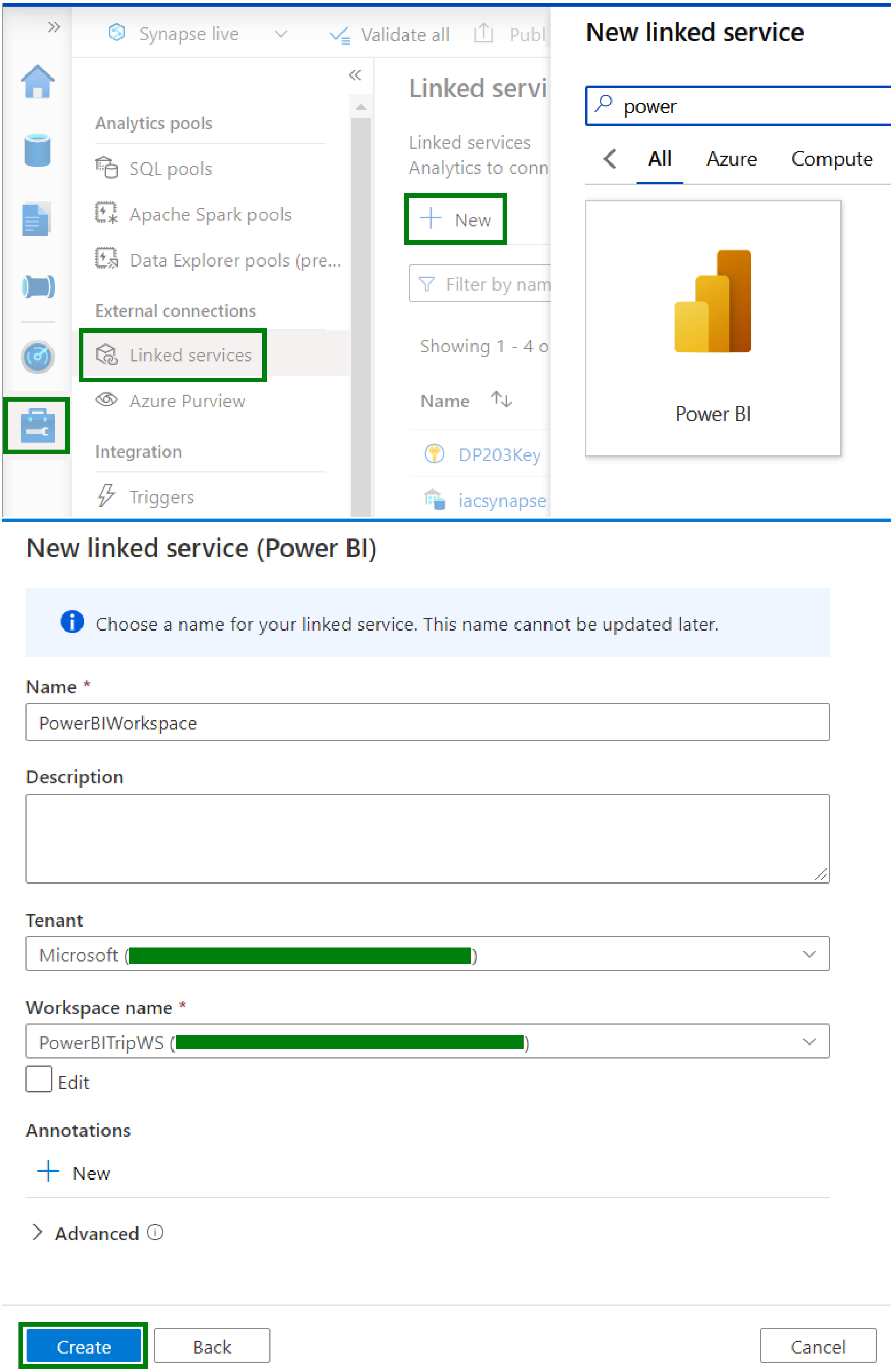

SampleNotebook (Scala)
ADBWS
ADBsmall
Cmd 1
print ("Hello world")
Cmd 2
Shift+Enter to run
Home
ry Resources
Filter resources by name
Pipeline 5
Dataset 14
Data flows 5
Power Query (Preview) 0
Batch Pipeline
Activities
databricks
Databricks
Notebook
Jar
Python
Save as template
Validate
Debug
Notebook
Notebook1
User Settings
Access Tokens
Git Integration
Notebook Settings
Model Registry Settings
Language Settings
Personal access tokens can be used for secure authentication to the Databricks API instead of passwords.
Generate New Token
Comment
Creation
Expiration
Running Sample Batch from ADF
2021-09-10 20:36:32 IST
2021-09-13 20:36:32 IST
Signed in as
User Settings
Admin Console
Manage Account
Log Out
Workspaces
ADBWS
newton.packt_gmail.com...

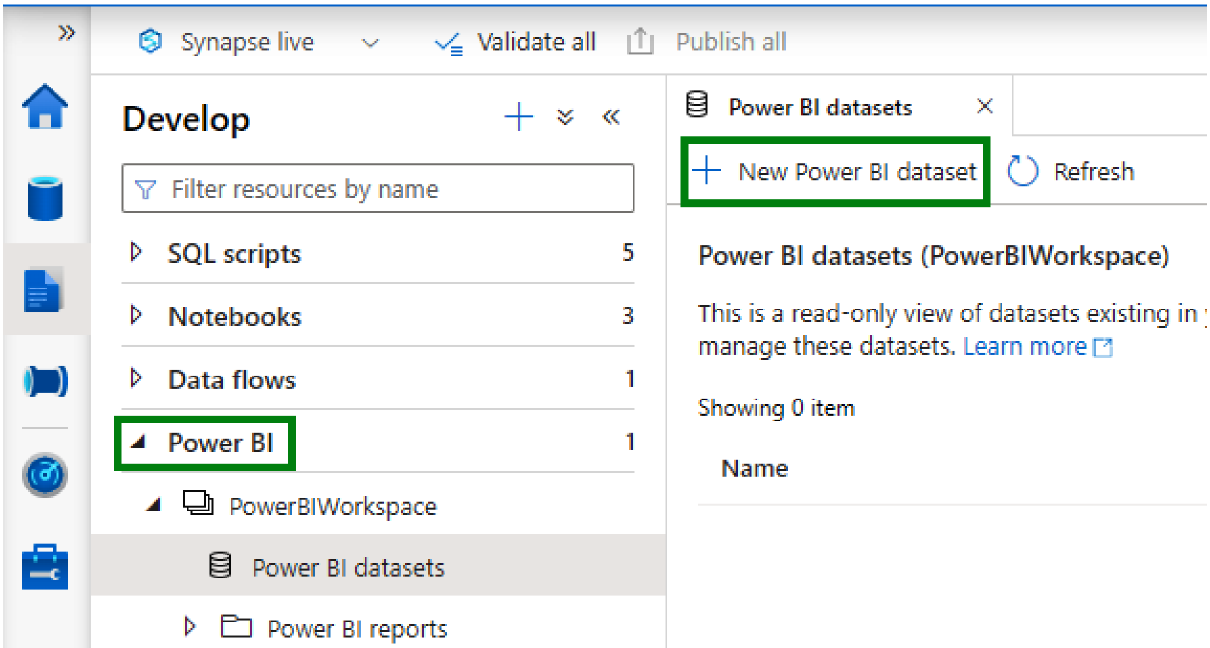


Capability	Azure Data Lake Analytics	Azure Synapse	HDInsight with Spark	HDInsight with Hive	HDInsight with Hive LLAP	Azure Databricks
Autoscaling	No	No	Yes	Yes	Yes	Yes
Scale-out granularity	Per job	Per cluster	Per cluster	Per cluster	Per cluster	Per cluster
In-memory caching of data	No	Yes	Yes	No	Yes	Yes
Query from external relational stores	Yes	No	Yes	No	No	Yes
Authentication	Azure AD	SQL / Azure AD	No	Azure AD[1]	Azure AD[1]	Azure AD
Auditing	Yes	Yes	No	Yes[1]	Yes[1]	Yes
Row-level security	No	Yes[2]	No	Yes[1]	Yes[1]	No
Supports firewalls	Yes	Yes	Yes	Yes[3]	Yes[3]	No
Dynamic data masking	No	Yes	No	Yes[1]	Yes[1]	No


Microsoft
Power BI
Home
My workspace
new
Workspaces
No results found
Create a workspace


Microsoft
Power BI
Home
My workspace
new
Workspaces
No results found
Create a workspace


Synapse live
Validate all
Publ
Linked servi
Linked services
Analytics to conn
New
Filter by nam
Showing 1 - 4 o
Name
DP203Key
iacsynapse
Analytics pools
SQL pools
Apache Spark pools
Data Explorer pools (pre...
External connections
Linked services
Azure Purview
Integration
Triggers
New linked service
power
All
Azure
Compute
Power BI
New linked service (Power BI)
Choose a name for your linked service. This name cannot be updated later.
Name *
PowerBIWorkspace
Description
Tenant
Microsoft ([redacted])
Workspace name *
PowerBITripWS ([redacted])
Edit
Annotations
New
Advanced
Create
Back
Cancel

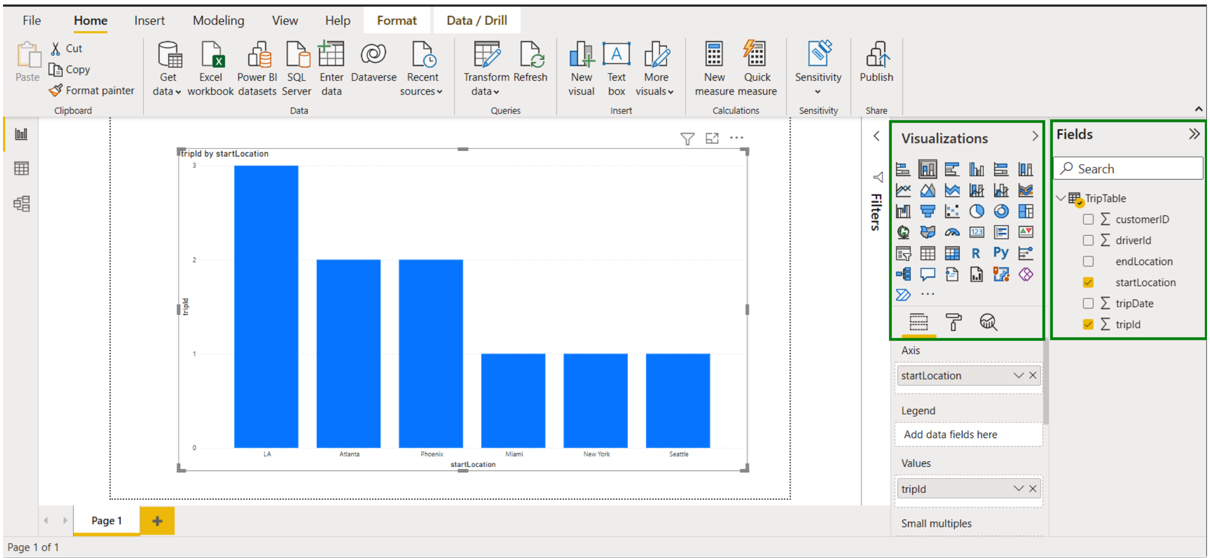


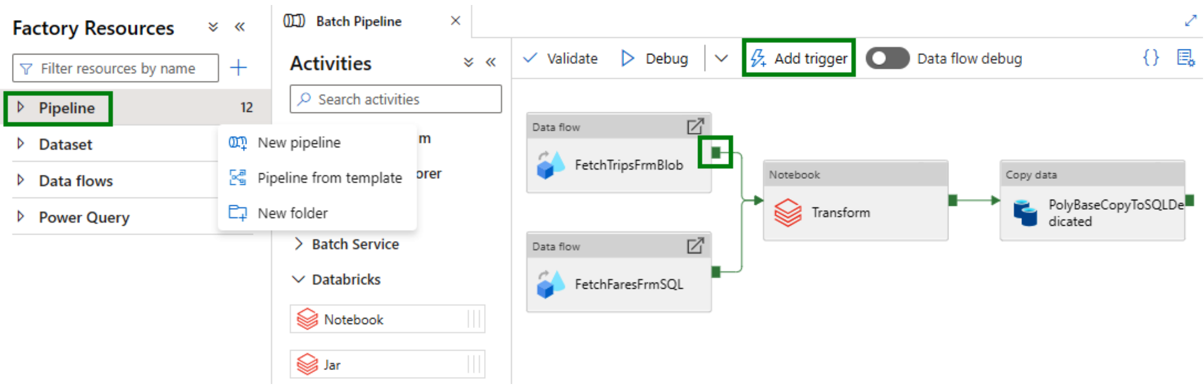
Download .pbids file

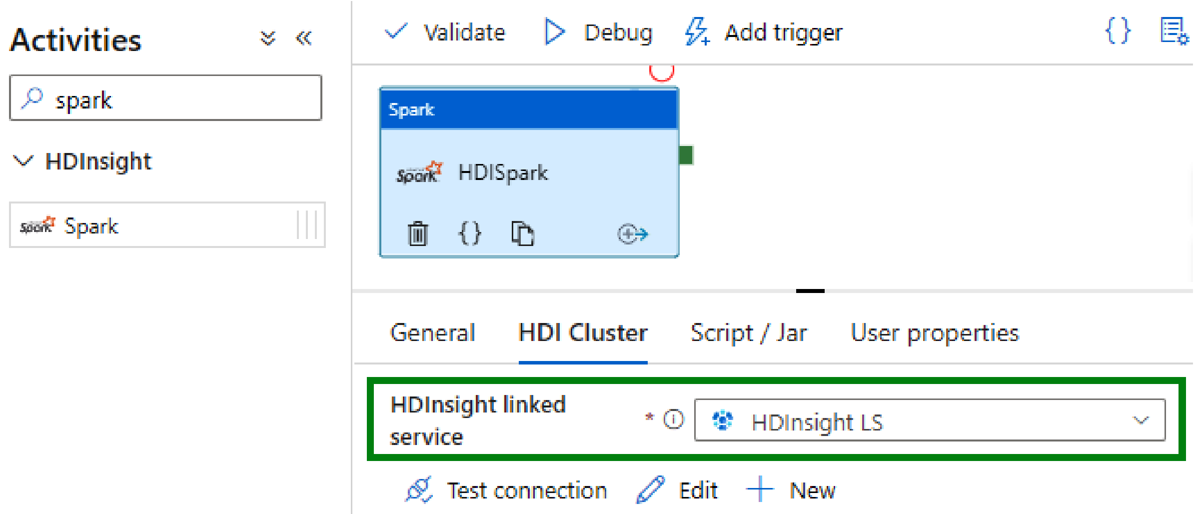
Download the .pbids file below and save it to your local drive.

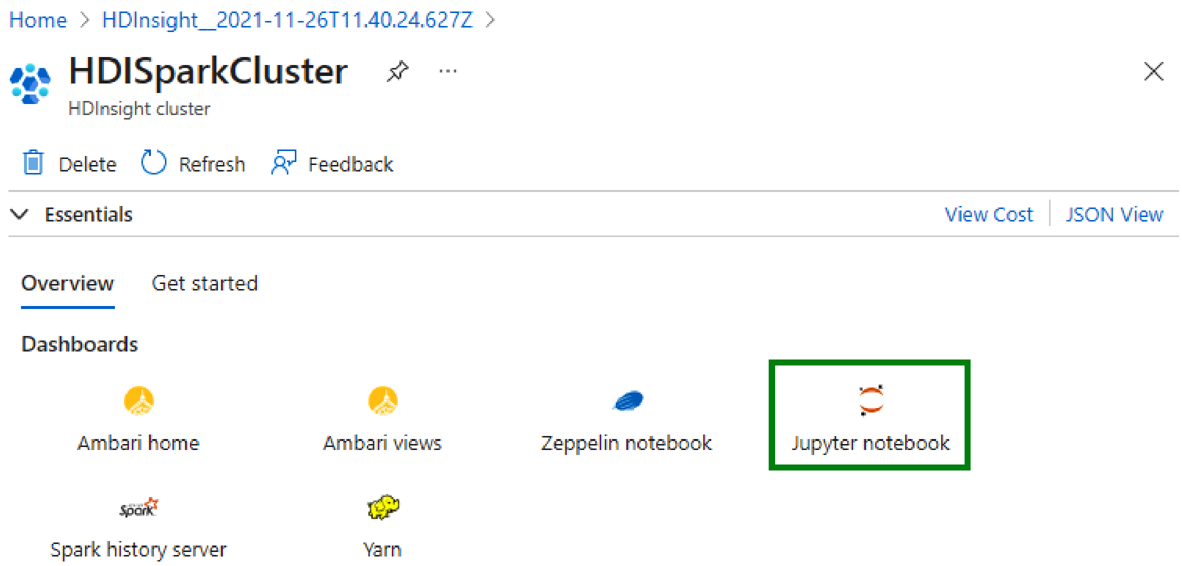
[redacted]sqlpool.pbids

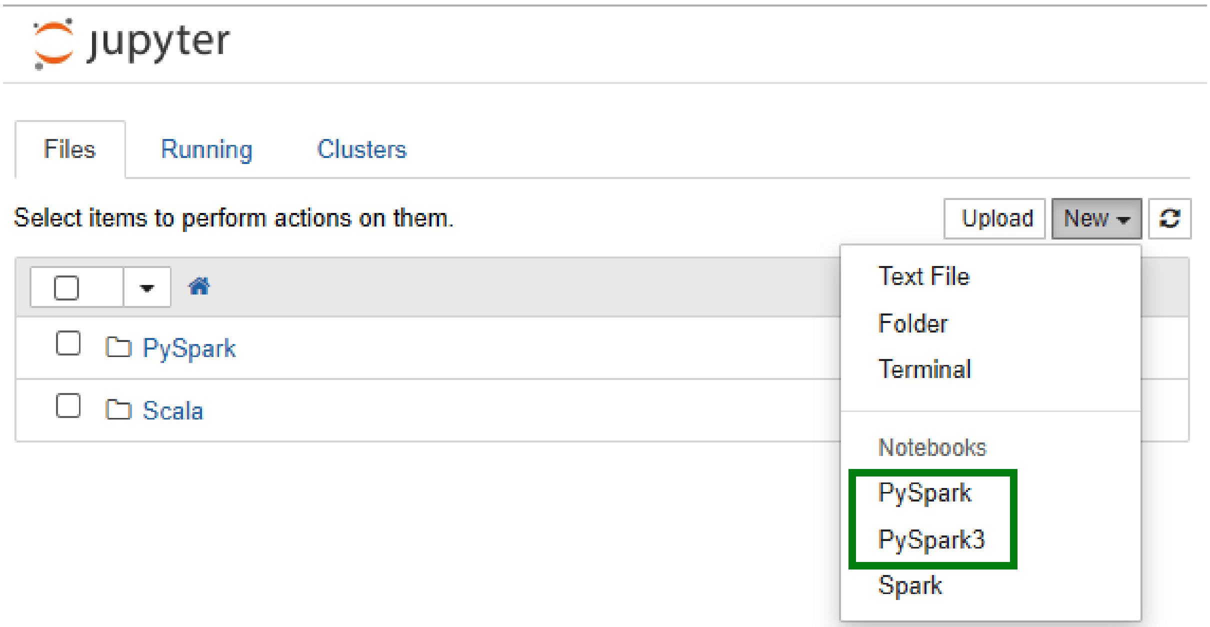
Download

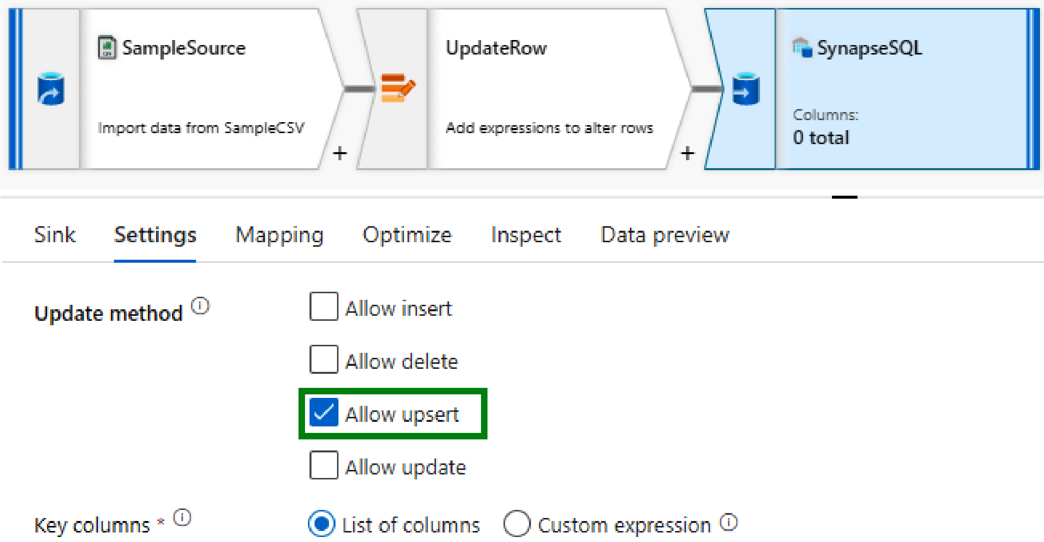


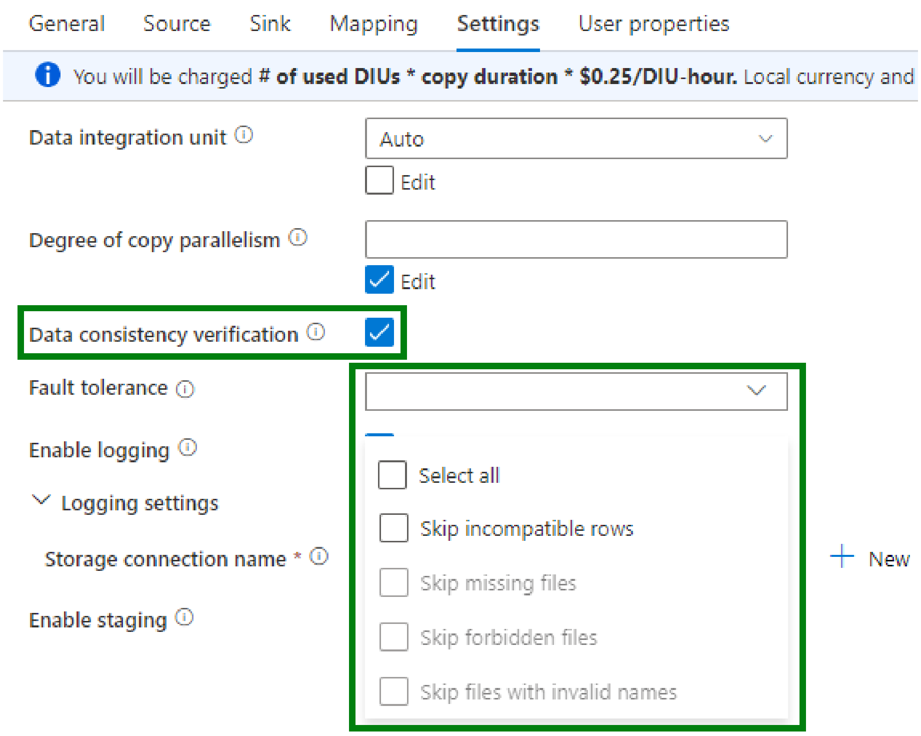

Factory Resources
Filter resources by name
Pipeline 12
Dataset
Data flows
Power Query
New pipeline
Pipeline from template
New folder
Batch Pipeline
Activities
Search activities
Batch Service
Databricks
Notebook
Jar
Validate
Debug
Add trigger
Data flow debug
Data flow
FetchTripsFrmBlob
Data flow
FetchFaresFrmSQL
Notebook
Transform
Copy data
PolyBaseCopyToSQLDedicated

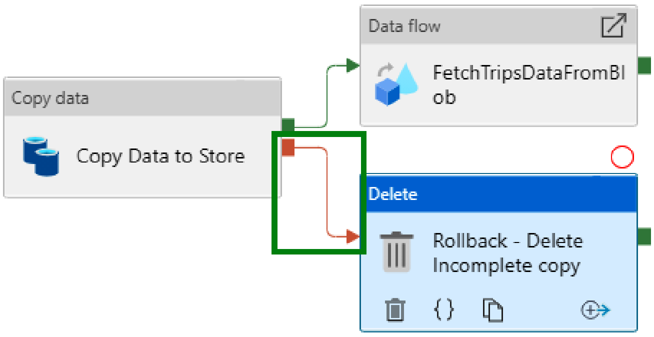

Activities
spark
HDInsight
Spark
Validate
Debug
Add trigger
Spark
HDISpark
General
HDI Cluster
Script / Jar
User properties
HDInsight linked service
HDInsight LS
Test connection
Edit
New

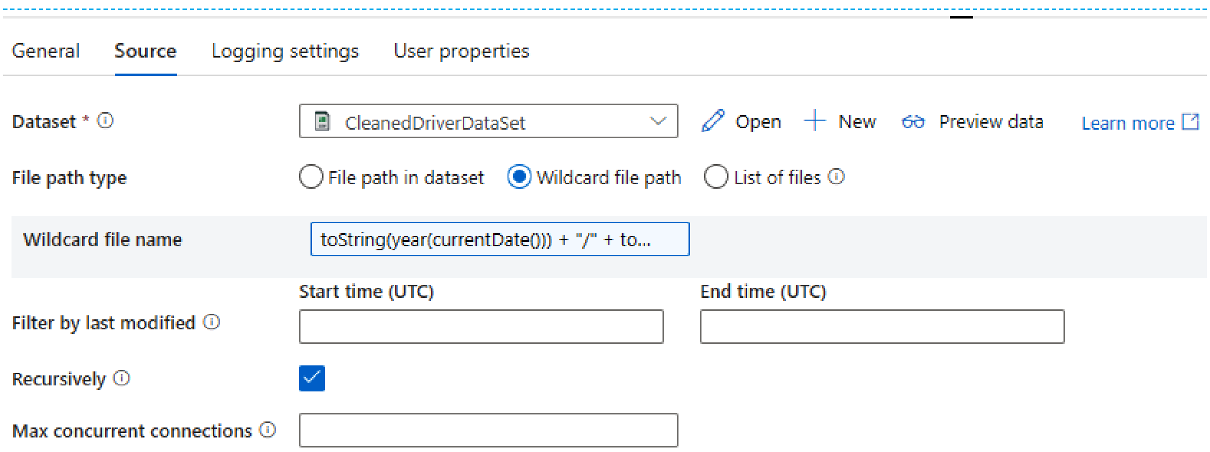

Home > HDInsight_2021-11-26T11.40.24.627Z >
HDISparkCluster
HDInsight cluster
Delete
Refresh
Feedback
Essentials
View Cost
JSON View
Overview
Get started
Dashboards
Ambari home
Ambari views
Zeppelin notebook
Jupyter notebook
Spark history server
Yarn

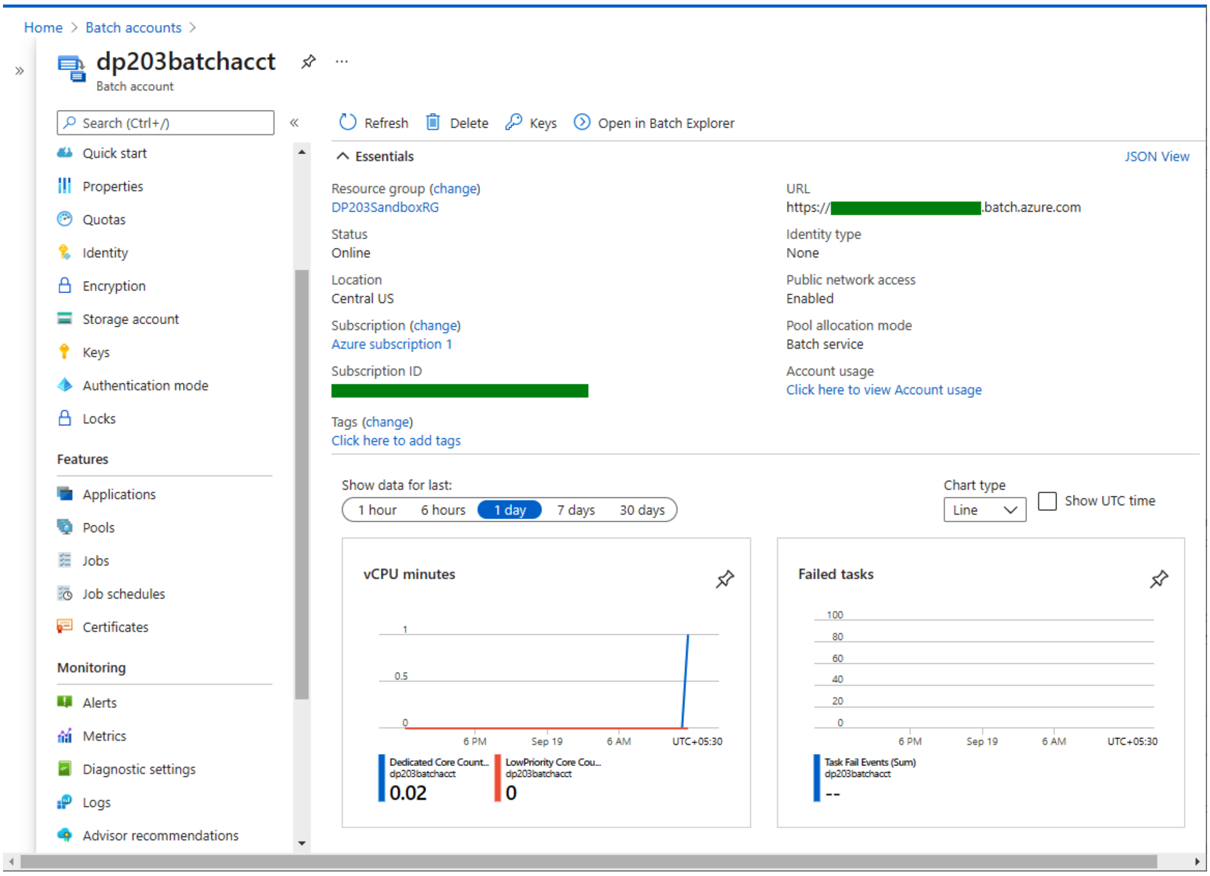

jupyter
Files
Running
Clusters
Select items to perform actions on them.
Upload
New
PySpark
Scala
Text File
Folder
Terminal
Notebooks
PySpark
PySpark3
Spark

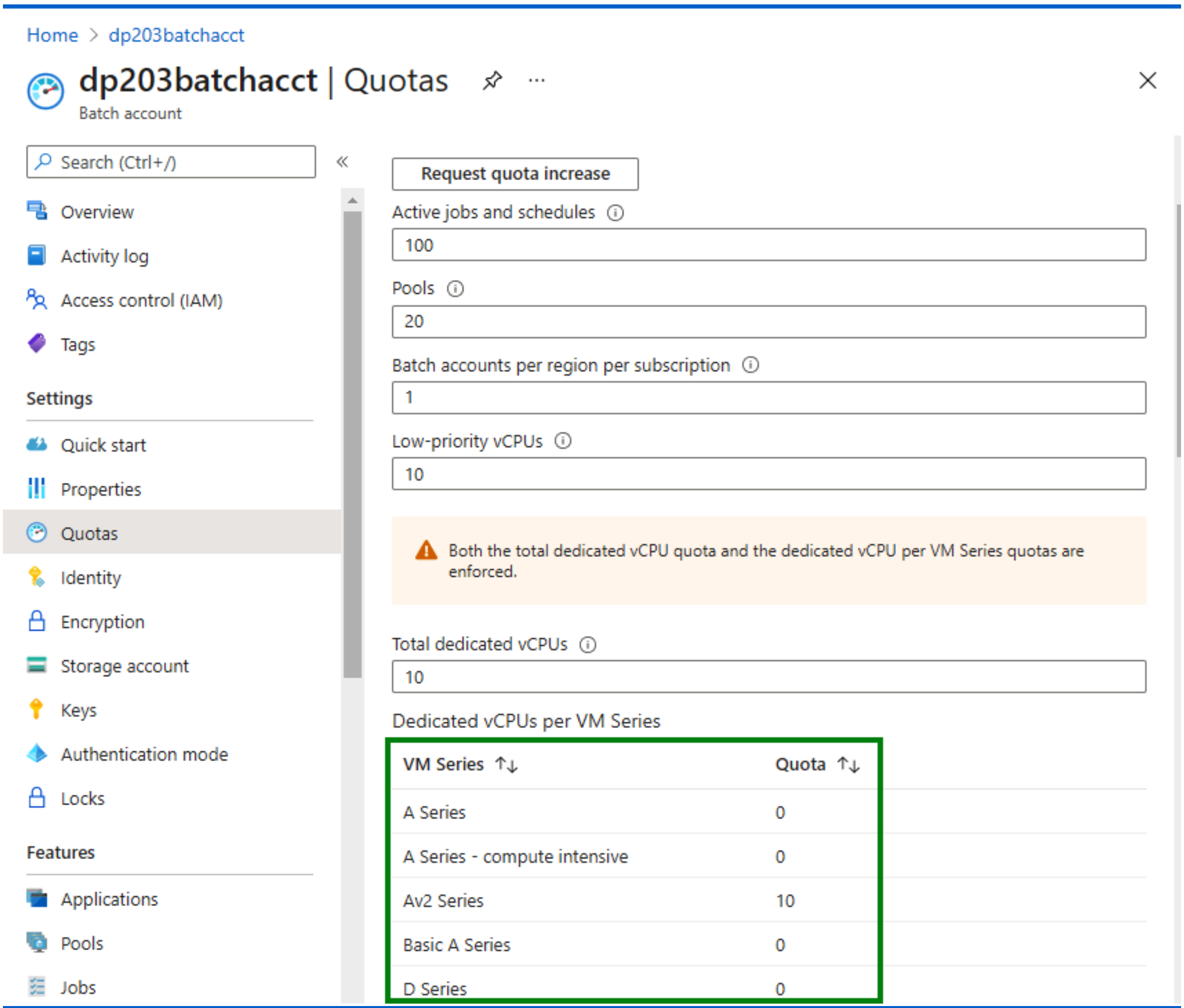

SampleSource
Import data from SampleCSV
UpdateRow
Add expressions to alter rows
SynapseSQL
Columns:
0 total
Sink
Settings
Mapping
Optimize
Inspect
Data preview
Update method
Allow insert
Allow delete
Allow upsert
Allow update
Key columns *
List of columns
Custom expression

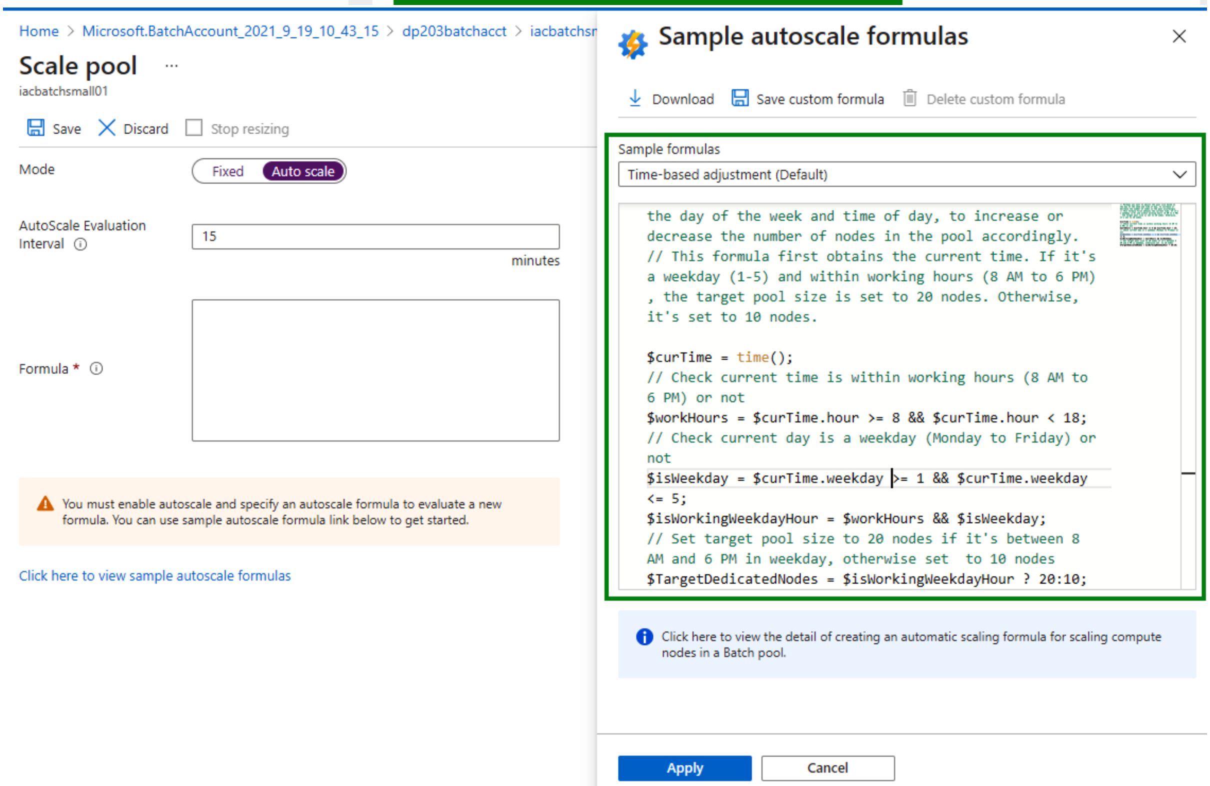

General
Source
Sink
Mapping
Settings
User properties
You will be charged # of used DIUs * copy duration * $0.25/DIU-hour. Local currency and
Data integration unit
Auto
Edit
Degree of copy parallelism
Edit
Data consistency verification
Fault tolerance
Enable logging
Logging settings
Storage connection name *
Enable staging
Select all
Skip incompatible rows
Skip missing files
Skip forbidden files
Skip files with invalid names
New

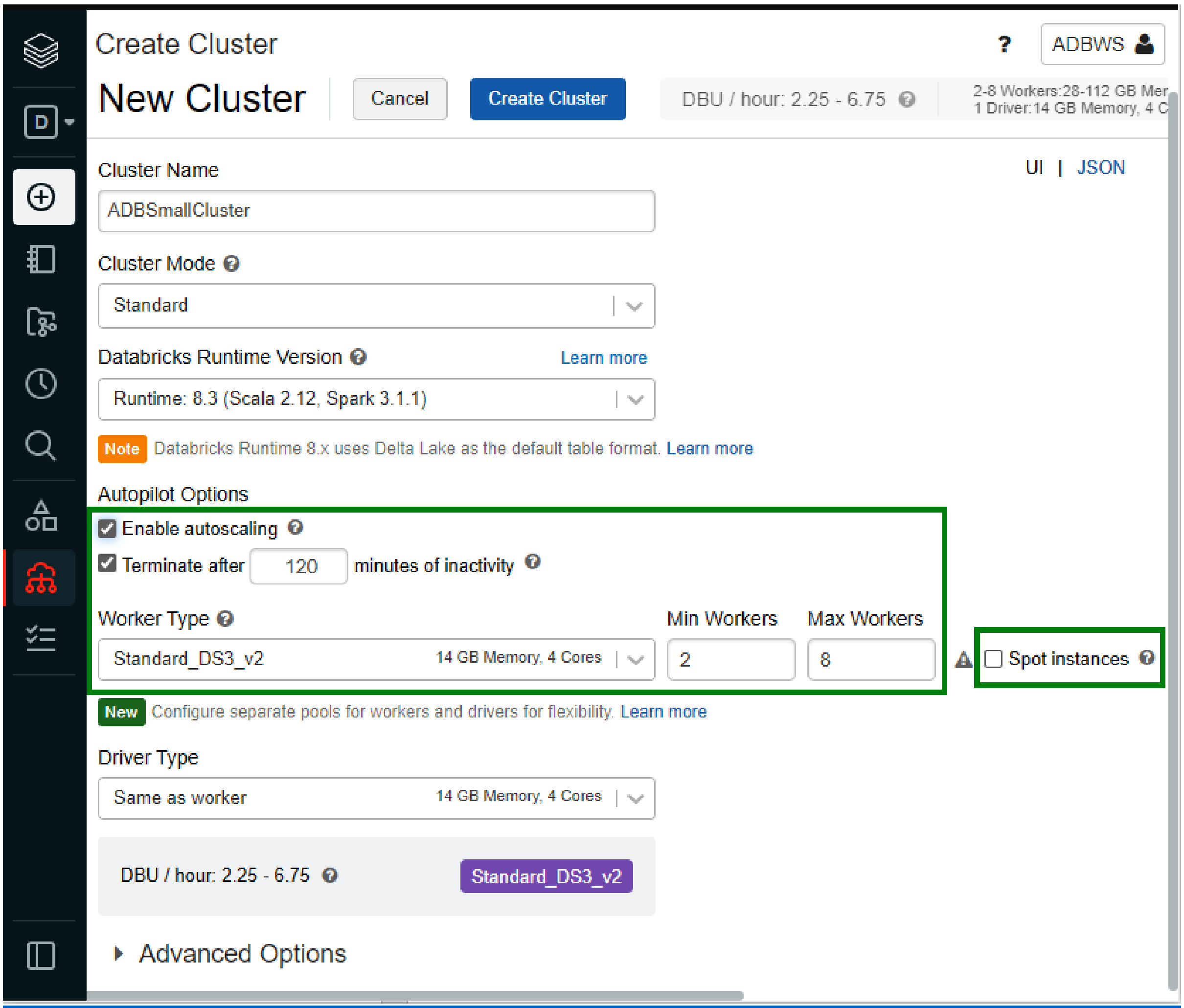

Data flow
FetchTripsDataFromBl
ob
Copy data
Copy Data to Store
Delete
Rollback - Delete
Incomplete copy

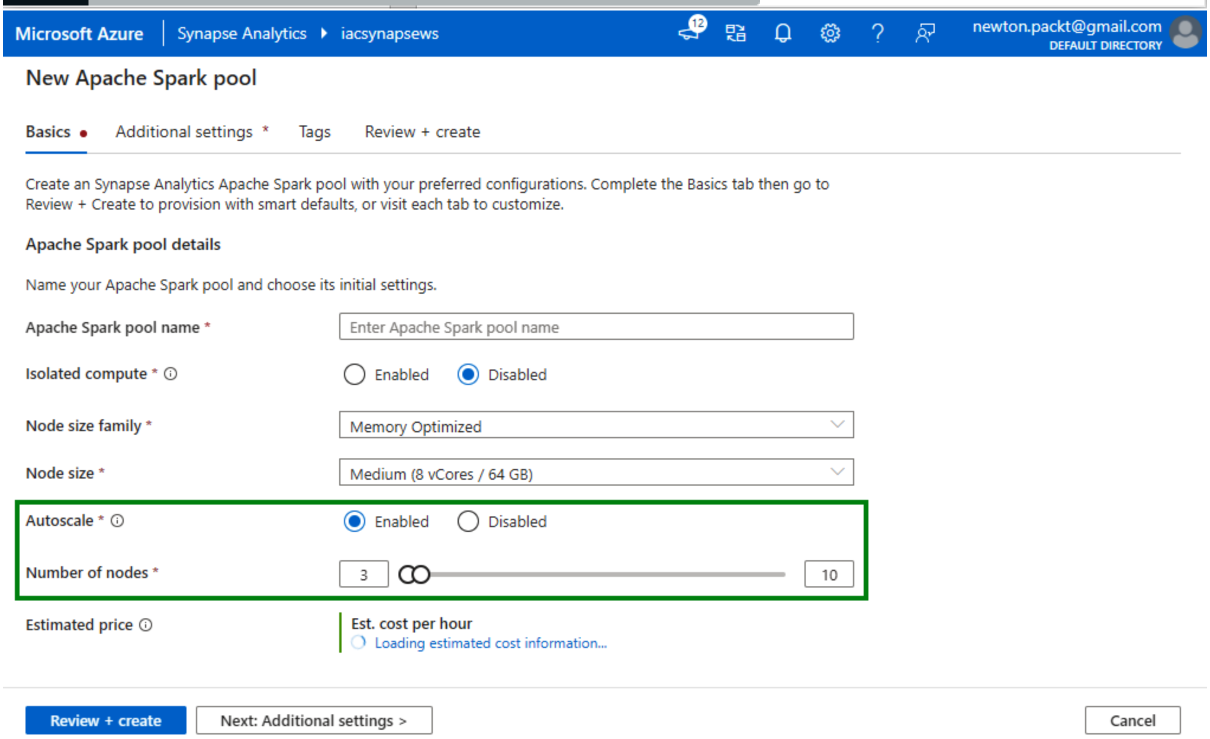

General
Source
Logging settings
User properties
Dataset *
CleanedDriverDataSet
Open
New
Preview data
Learn more
File path type
File path in dataset
Wildcard file path
List of files
Wildcard file name
toString(year(currentDate())) + "/" + to...
Start time (UTC)
End time (UTC)
Filter by last modified
Recursively
Max concurrent connections

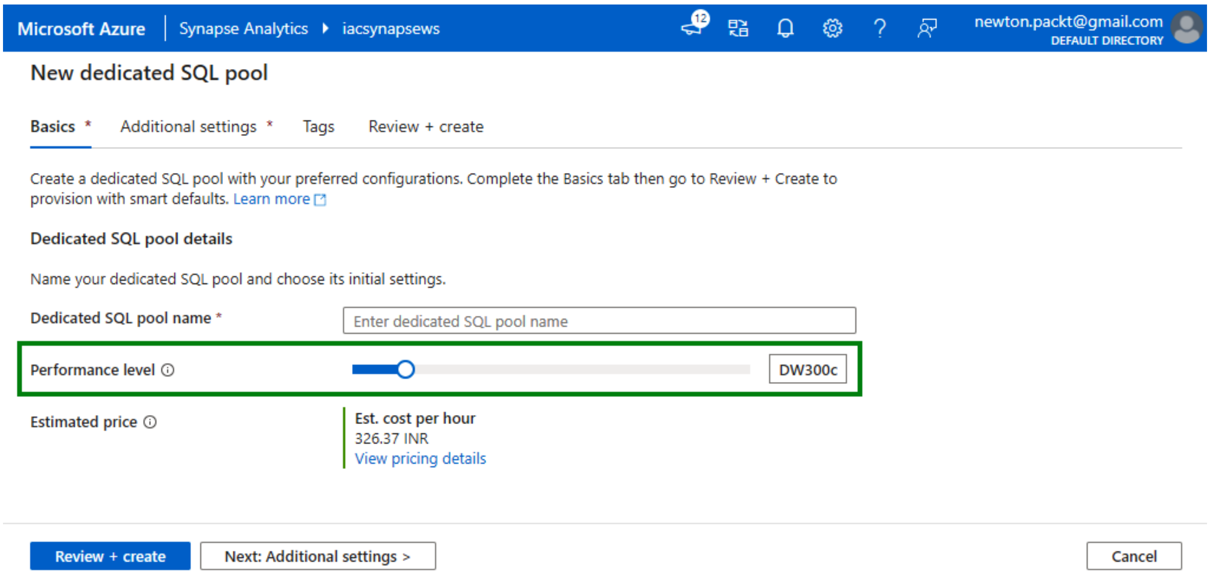

Home > Batch accounts >
dp203batchacct
Batch account
Search (Ctrl+/)
Refresh
Delete
Keys
Open in Batch Explorer
Essentials
JSON View
Resource group (change)
DP203SandboxRG
Status
Online
Location
Central US
Subscription (change)
Azure subscription 1
Subscription ID
[redacted]
Tags (change)
Click here to add tags
URL
https://[redacted].batch.azure.com
Identity type
None
Public network access
Enabled
Pool allocation mode
Batch service
Account usage
Click here to view Account usage
Quick start
Properties
Quotas
Identity
Encryption
Storage account
Keys
Authentication mode
Locks
Features
Applications
Pools
Jobs
Job schedules
Certificates
Monitoring
Alerts
Metrics
Diagnostic settings
Logs
Advisor recommendations
Show data for last:
1 hour
6 hours
1 day
7 days
30 days
Chart type
Line
Show UTC time
vCPU minutes
1
0.5
0
6 PM
Sep 19
6 AM
UTC+05:30
Dedicated Core Count...
dp203batchacct
0.02
LowPriority Core Cou...
dp203batchacct
0
Failed tasks
100
80
60
40
20
0
6 PM
Sep 19
6 AM
UTC+05:30
Task Fail Events (Sum)
dp203batchacct
--


Home > dp203batchacct
dp203batchacct | Quotas
Batch account
Search (Ctrl+/)
Overview
Activity log
Access control (IAM)
Tags
Settings
Quick start
Properties
Quotas
Identity
Encryption
Storage account
Keys
Authentication mode
Locks
Features
Applications
Pools
Jobs
Request quota increase
Active jobs and schedules
100
Pools
20
Batch accounts per region per subscription
1
Low-priority vCPUs
10
Both the total dedicated vCPU quota and the dedicated vCPU per VM Series quotas are enforced.
Total dedicated vCPUs
10
Dedicated vCPUs per VM Series
VM Series
Quota
A Series
0
A Series - compute intensive
0
Av2 Series
10
Basic A Series
0
D Series
0

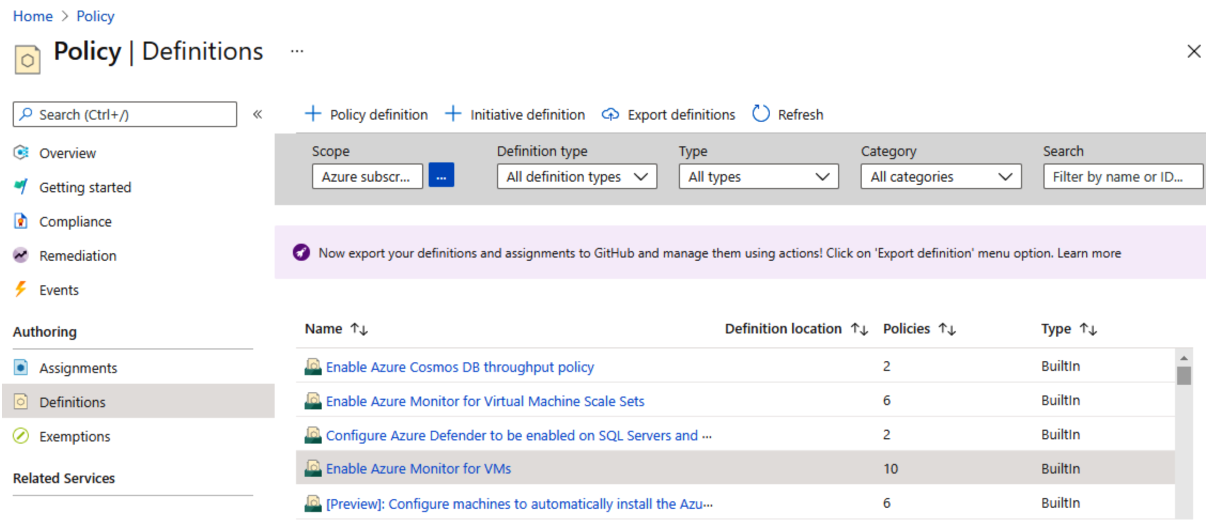

Home > Microsoft.BatchAccount_2021_9_19_10_43_15 > dp203batchacct > iacbatchsr
Scale pool
iacbatchsmall01
Save
Discard
Stop resizing
Mode
Fixed
Auto scale
AutoScale Evaluation Interval
15
minutes
Formula *
You must enable autoscale and specify an autoscale formula to evaluate a new formula. You can use sample autoscale formula link below to get started.
Click here to view sample autoscale formulas
Sample autoscale formulas
Download
Save custom formula
Delete custom formula
Sample formulas
Time-based adjustment (Default)
the day of the week and time of day, to increase or
decrease the number of nodes in the pool accordingly.
// This formula first obtains the current time. If it's
a weekday (1-5) and within working hours (8 AM to 6 PM)
, the target pool size is set to 20 nodes. Otherwise,
it's set to 10 nodes.
$curTime = time();
// Check current time is within working hours (8 AM to
6 PM) or not
$workHours = $curTime.hour >= 8 && $curTime.hour < 18;
// Check current day is a weekday (Monday to Friday) or
not
$isWeekday = $curTime.weekday >= 1 && $curTime.weekday
<= 5;
$isWorkingWeekdayHour = $workHours && $isWeekday;
// Set target pool size to 20 nodes if it's between 8
AM and 6 PM in weekday, otherwise set to 10 nodes
$TargetDedicatedNodes = $isWorkingWeekdayHour ? 20:10;
Click here to view the detail of creating an automatic scaling formula for scaling compute nodes in a Batch pool.
Apply
Cancel

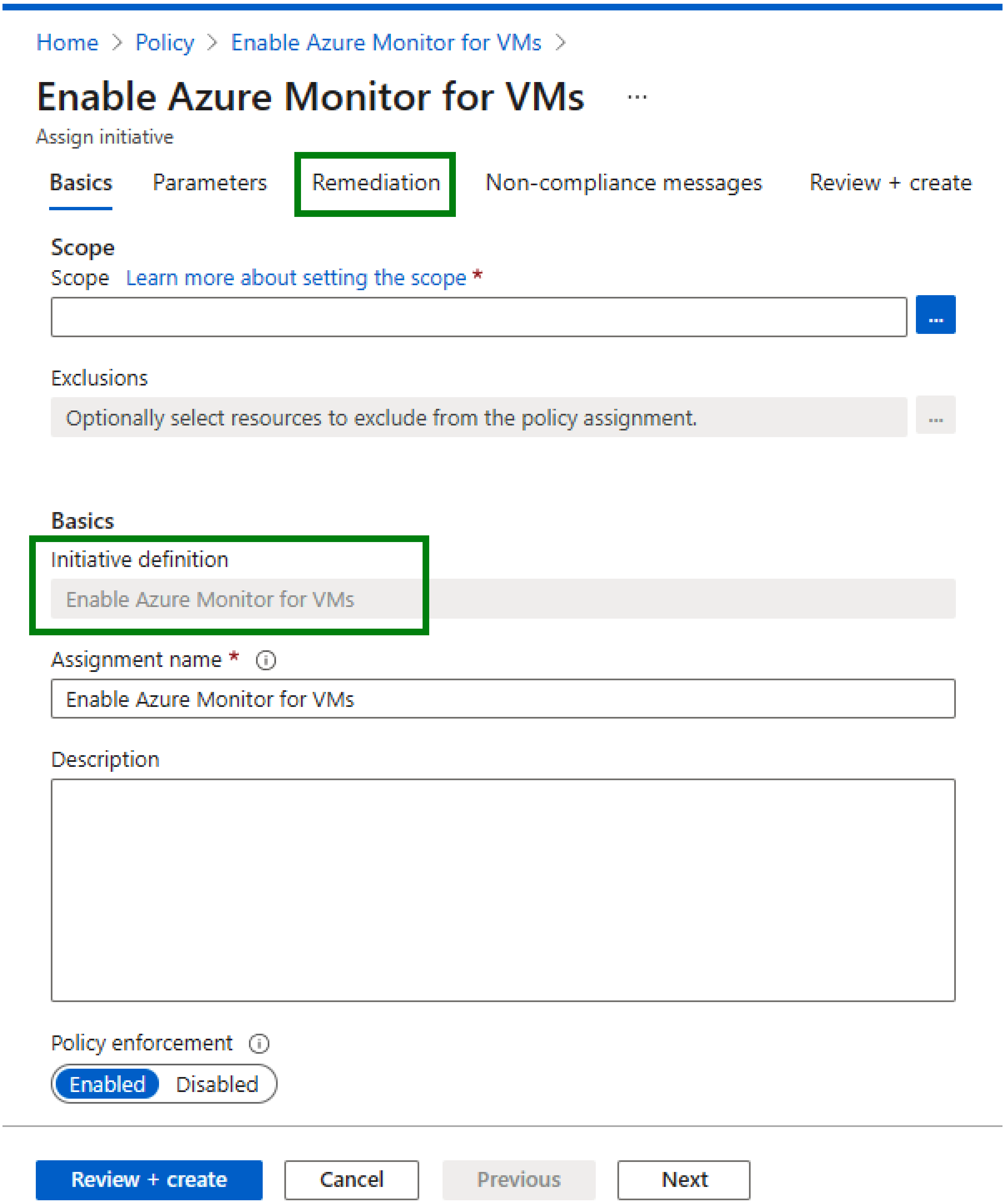

Create Cluster
?
ADBWS
New Cluster
Cancel
Create Cluster
DBU / hour: 2.25 - 6.75
2-8 Workers:28-112 GB Mer
1 Driver:14 GB Memory, 4 C
UI | JSON
Cluster Name
ADBSmallCluster
Cluster Mode
Standard
Databricks Runtime Version
Learn more
Runtime: 8.3 (Scala 2.12, Spark 3.1.1)
Note Databricks Runtime 8.x uses Delta Lake as the default table format. Learn more
Autopilot Options
Enable autoscaling
Terminate after 120 minutes of inactivity
Worker Type
Min Workers
Max Workers
Standard_DS3_v2
14 GB Memory, 4 Cores
2
8
Spot instances
New Configure separate pools for workers and drivers for flexibility. Learn more
Driver Type
Same as worker
14 GB Memory, 4 Cores
DBU / hour: 2.25 - 6.75
Standard_DS3_v2
Advanced Options


Microsoft Azure
Synapse Analytics
iacsynapsews
newton.packt@gmail.com
DEFAULT DIRECTORY
New Apache Spark pool
Basics
Additional settings *
Tags
Review + create
Create an Synapse Analytics Apache Spark pool with your preferred configurations. Complete the Basics tab then go to Review + Create to provision with smart defaults, or visit each tab to customize.
Apache Spark pool details
Name your Apache Spark pool and choose its initial settings.
Apache Spark pool name *
Enter Apache Spark pool name
Isolated compute *
Enabled
Disabled
Node size family *
Memory Optimized
Node size *
Medium (8 vCores / 64 GB)
Autoscale *
Enabled
Disabled
Number of nodes *
3
10
Estimated price
Est. cost per hour
Loading estimated cost information...
Review + create
Next: Additional settings >
Cancel

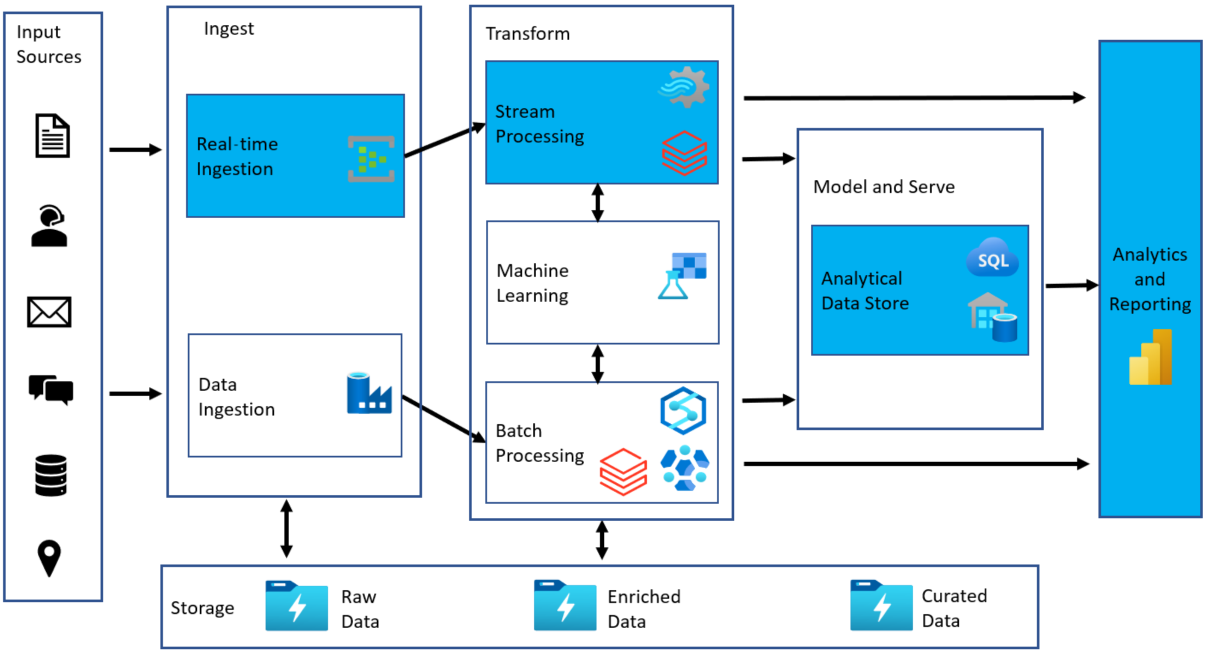


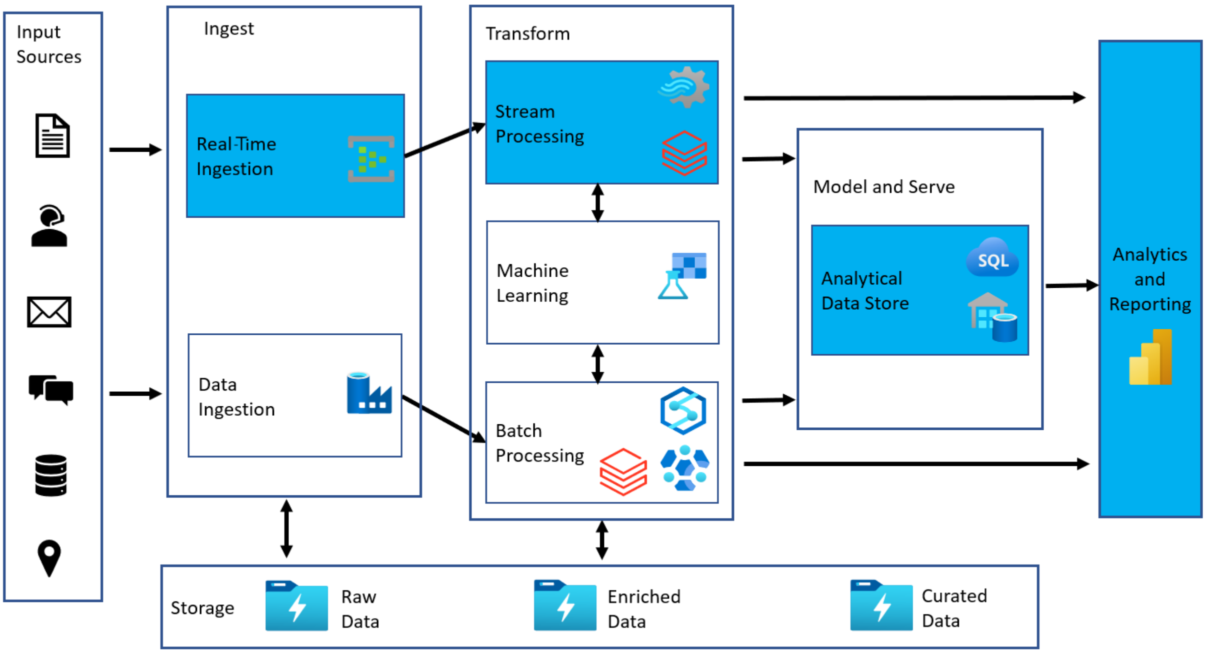
Error Code	Category	User Message
`AutoScalingFormulaSyntaxError`	BadRequest	The specified autoscaling formula has a syntax error.
`CommandLaunchFailed`	UserError	Failed to launch the specified command line.
`NodeBeingRebooted`	Conflict	The specified node is being rebooted.
`DiskFull`	ServerError	There is not enough disk space on the node that was selected to run the task.

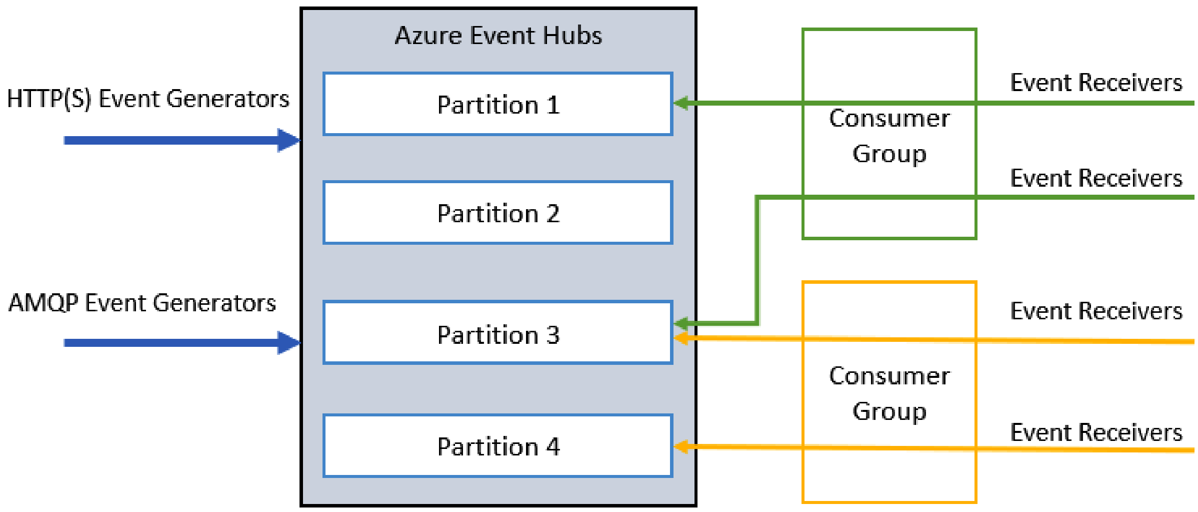

Home > Policy
Policy | Definitions
Search (Ctrl+/)
Overview
Getting started
Compliance
Remediation
Events
Authoring
Assignments
Definitions
Exemptions
Related Services
Policy definition
Initiative definition
Export definitions
Refresh
Scope
Azure subscr...
Definition type
All definition types
Type
All types
Category
All categories
Search
Filter by name or ID...
Now export your definitions and assignments to GitHub and manage them using actions! Click on 'Export definition' menu option. Learn more
Name
Definition location
Policies
Type
Enable Azure Cosmos DB throughput policy
2
BuiltIn
Enable Azure Monitor for Virtual Machine Scale Sets
6
BuiltIn
Configure Azure Defender to be enabled on SQL Servers and ...
2
BuiltIn
Enable Azure Monitor for VMs
10
BuiltIn
[Preview]: Configure machines to automatically install the Azu...
6
BuiltIn

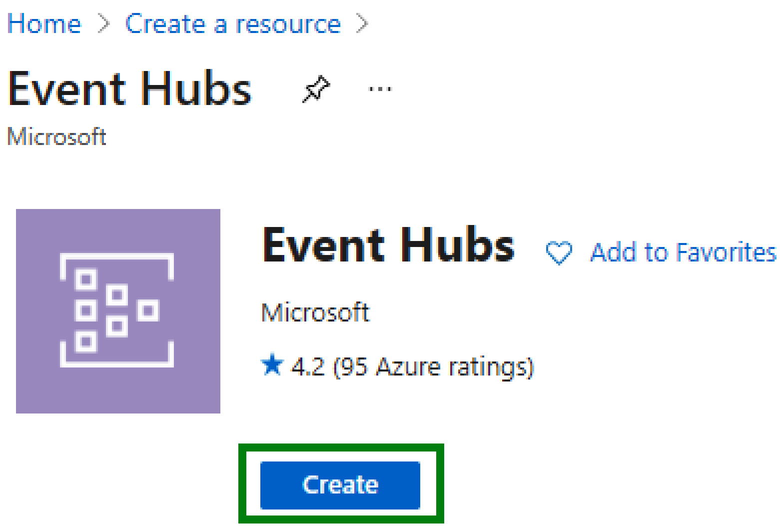

Home > Policy > Enable Azure Monitor for VMs >
Enable Azure Monitor for VMs ...
Assign initiative
Basics
Parameters
Remediation
Non-compliance messages
Review + create
Scope
Scope Learn more about setting the scope *
Exclusions
Optionally select resources to exclude from the policy assignment.
Basics
Initiative definition
Enable Azure Monitor for VMs
Assignment name *
Enable Azure Monitor for VMs
Description
Policy enforcement
Enabled
Disabled
Review + create
Cancel
Previous
Next

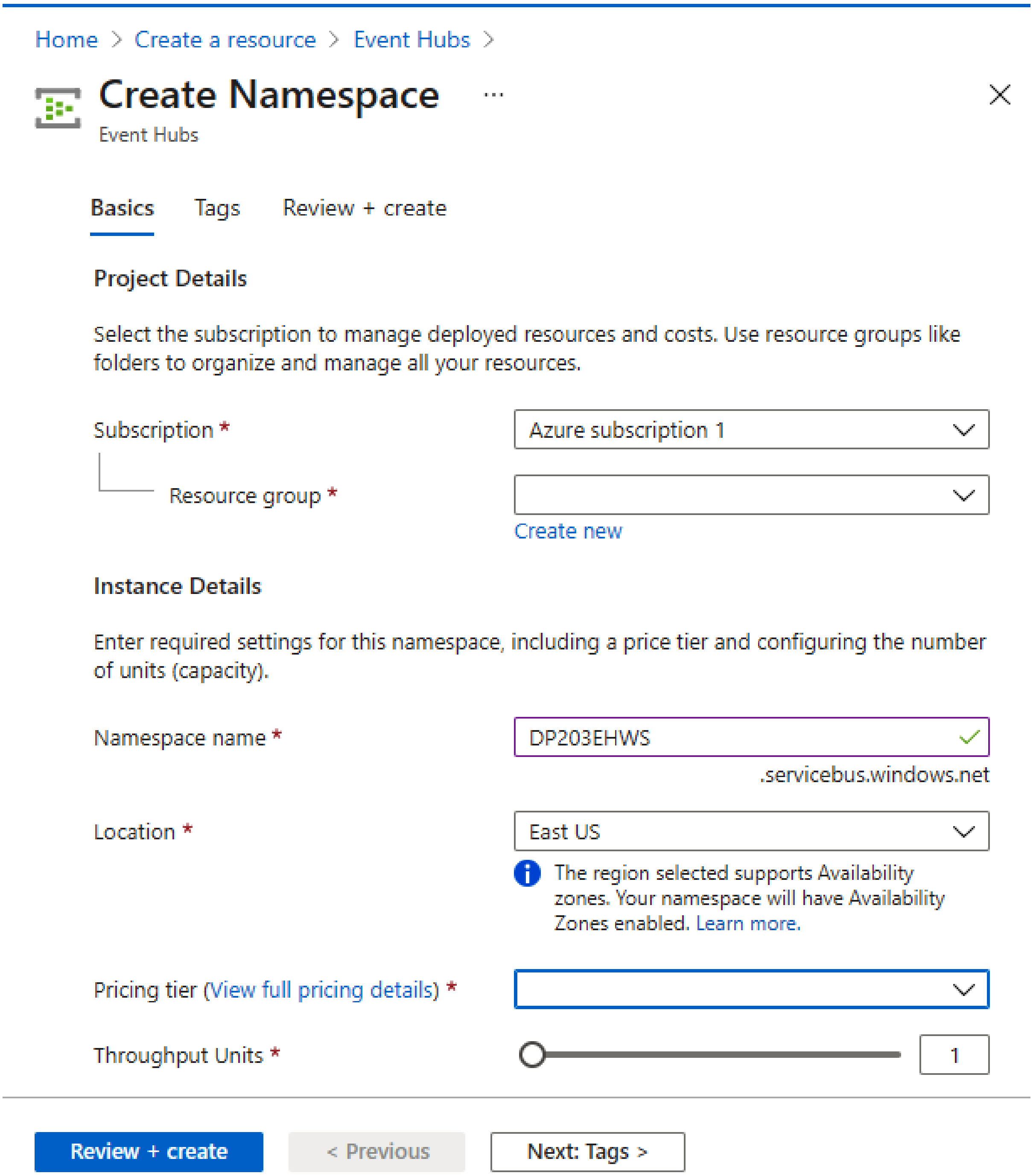
Chapter 10: Designing and Developing a Stream Processing Solution

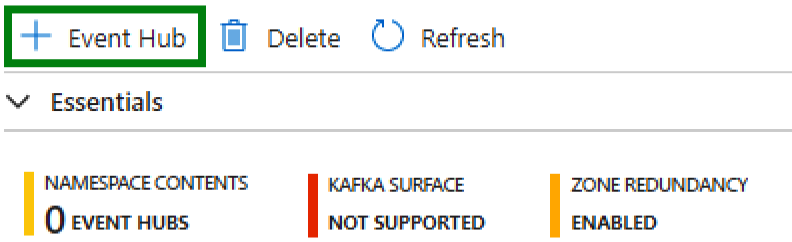


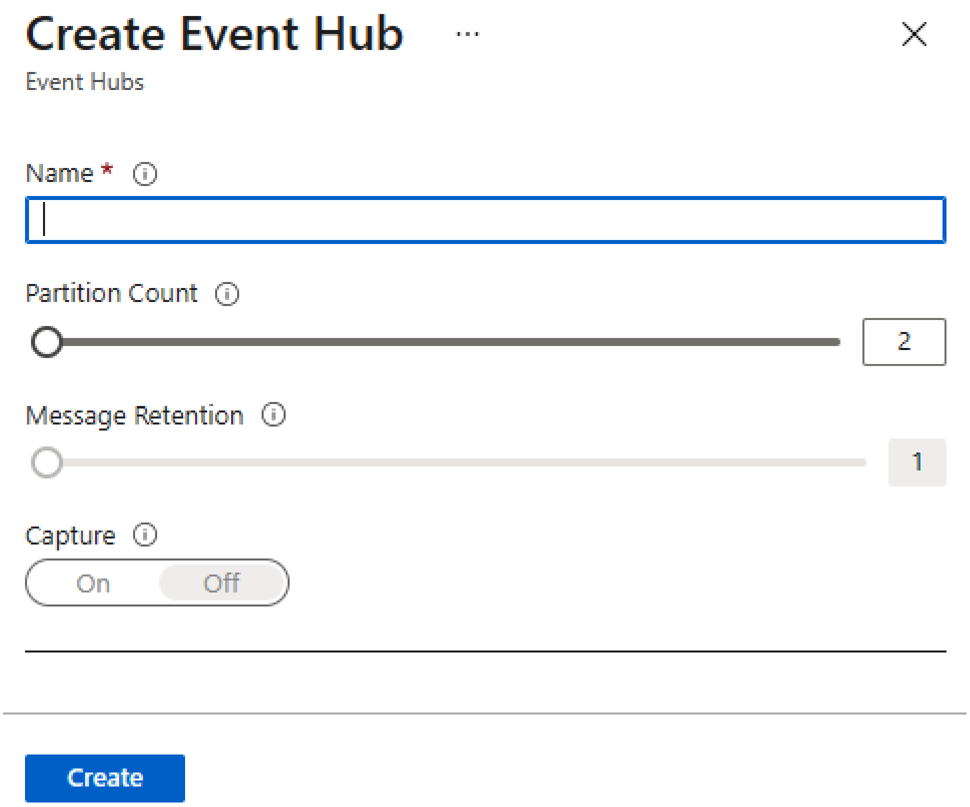


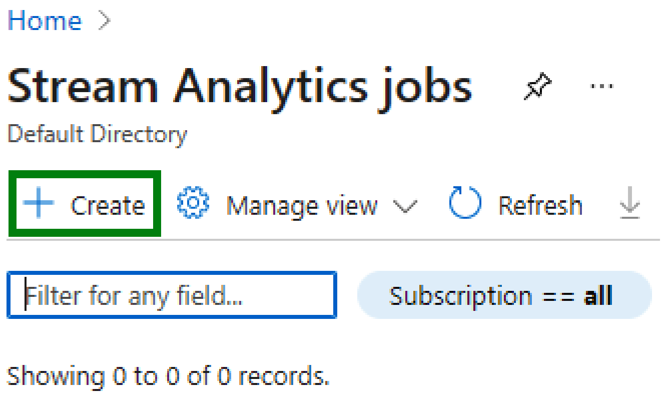

Azure Event Hubs
HTTP(S) Event Generators
AMQP Event Generators
Partition 1
Partition 2
Partition 3
Partition 4
Consumer Group
Consumer Group
Event Receivers
Event Receivers
Event Receivers
Event Receivers

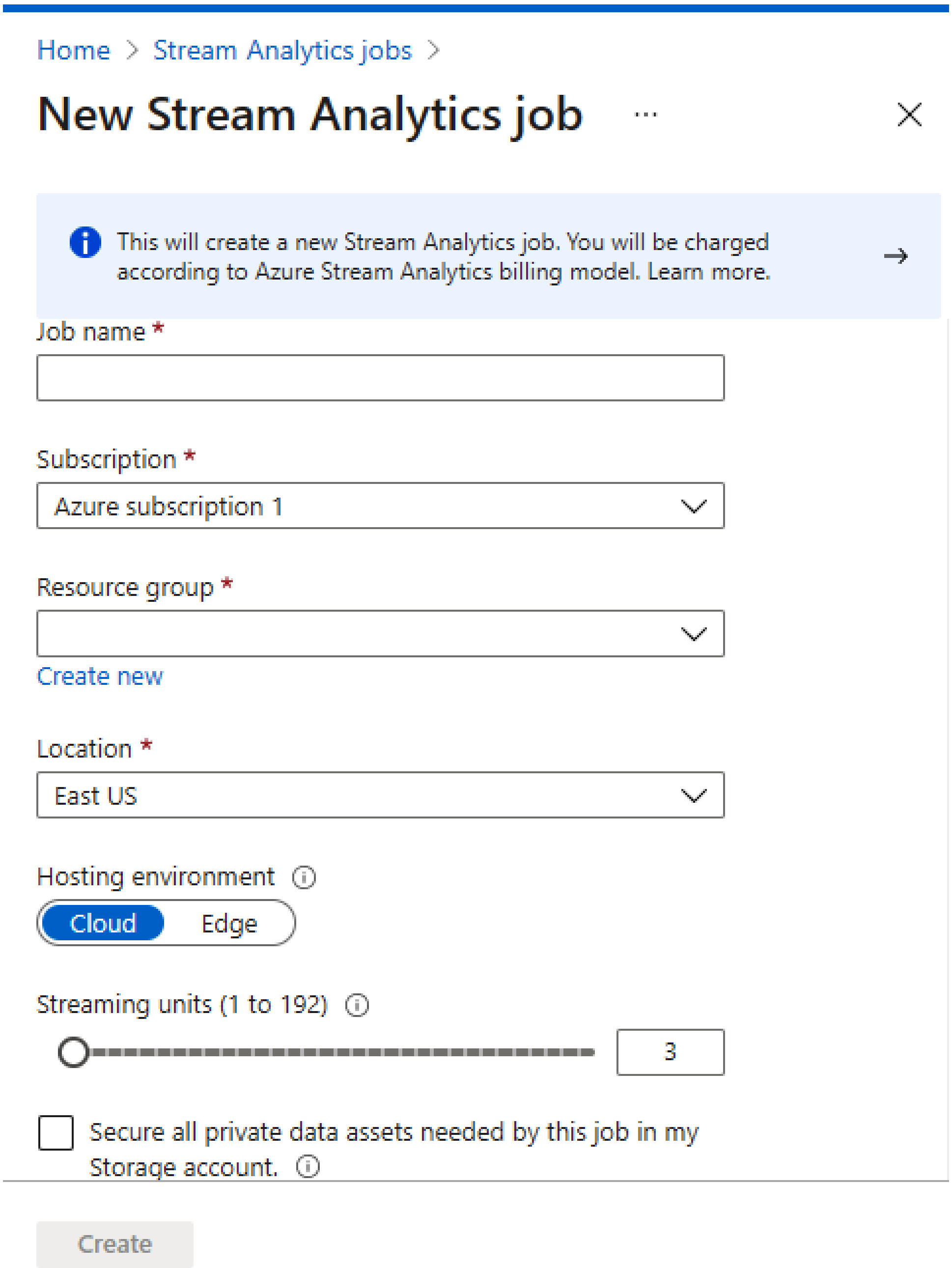

Home > Create a resource >
Event Hubs
Microsoft
Event Hubs
Add to Favorites
Microsoft
4.2 (95 Azure ratings)
Create

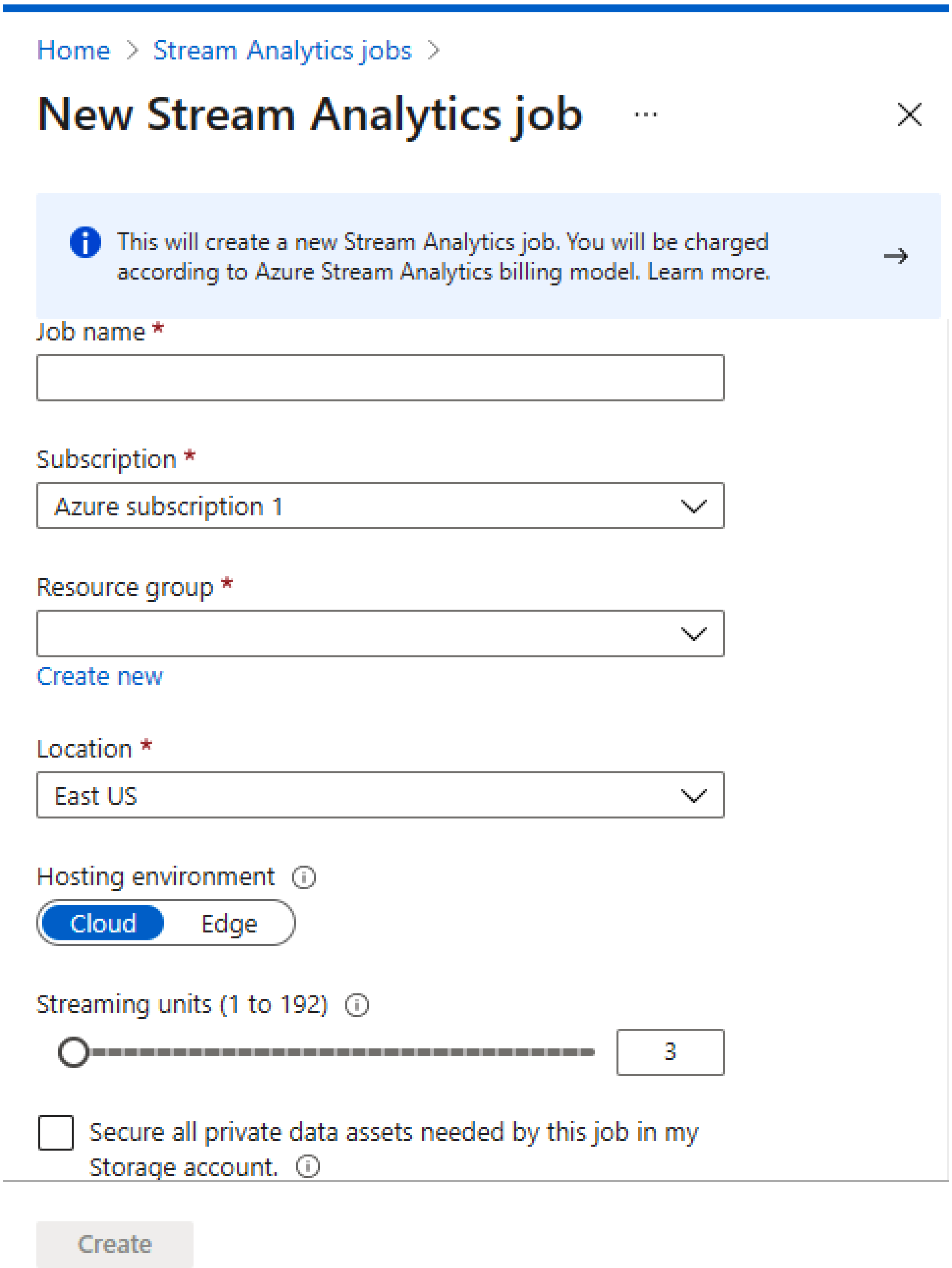

Home > Create a resource > Event Hubs >
Create Namespace ...
Event Hubs
Basics
Tags
Review + create
Project Details
Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.
Subscription *
Azure subscription 1
Resource group *
Create new
Instance Details
Enter required settings for this namespace, including a price tier and configuring the number of units (capacity).
Namespace name *
DP203EHWS
.servicebus.windows.net
Location *
East US
The region selected supports Availability zones. Your namespace will have Availability Zones enabled. Learn more.
Pricing tier (View full pricing details) *
Throughput Units *
1
Review + create
< Previous
Next: Tags >

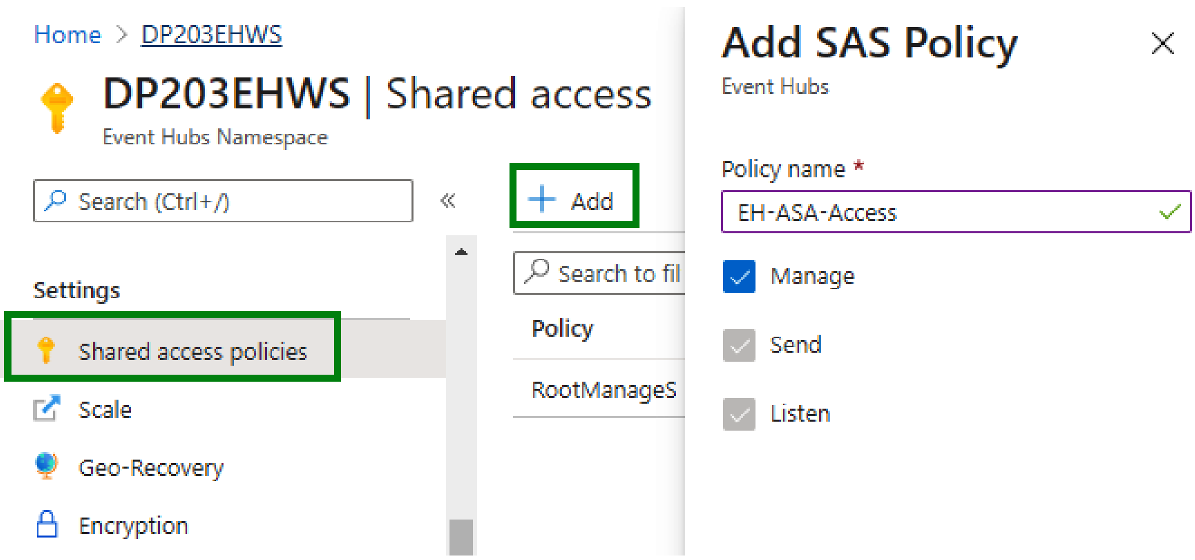

Event Hub
Delete
Refresh
Essentials
NAMESPACE CONTENTS
0 EVENT HUBS
KAFKA SURFACE
NOT SUPPORTED
ZONE REDUNDANCY
ENABLED

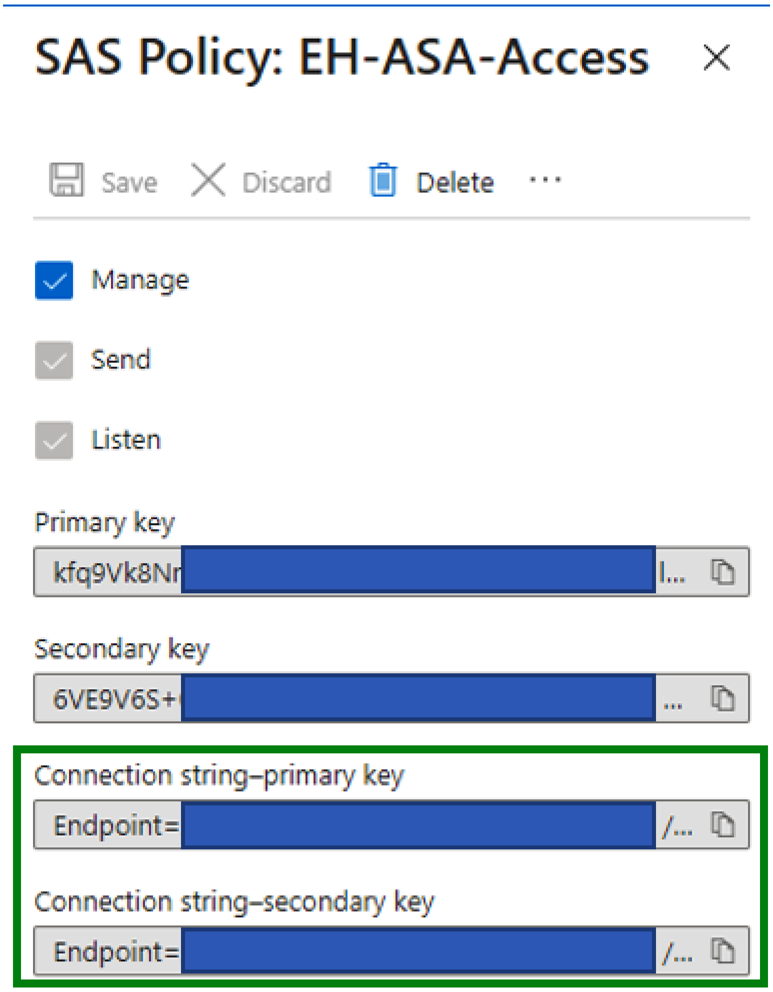

Create Event Hub
Event Hubs
Name *
Partition Count
2
Message Retention
1
Capture
On
Off
Create

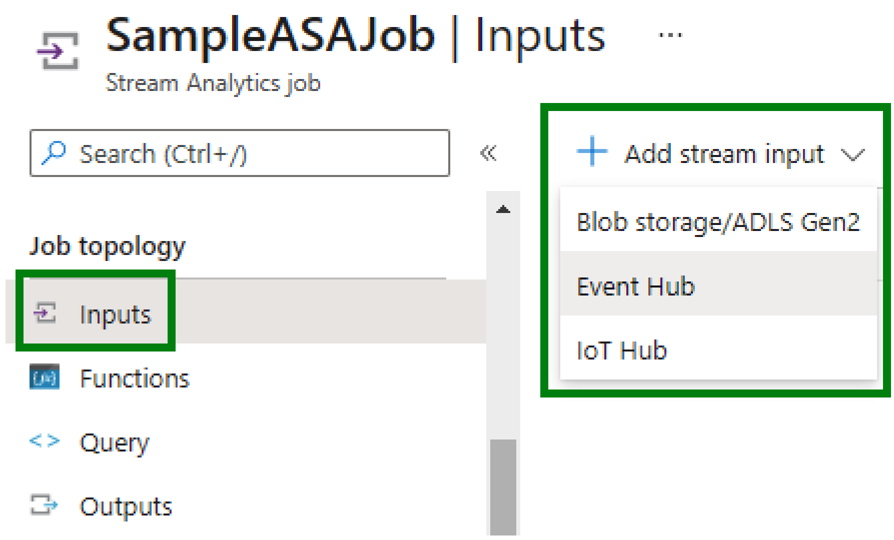

Home
Stream Analytics jobs
Default Directory
Create
Manage view
Refresh
Filter for any field...
Subscription == all
Showing 0 to 0 of 0 records.

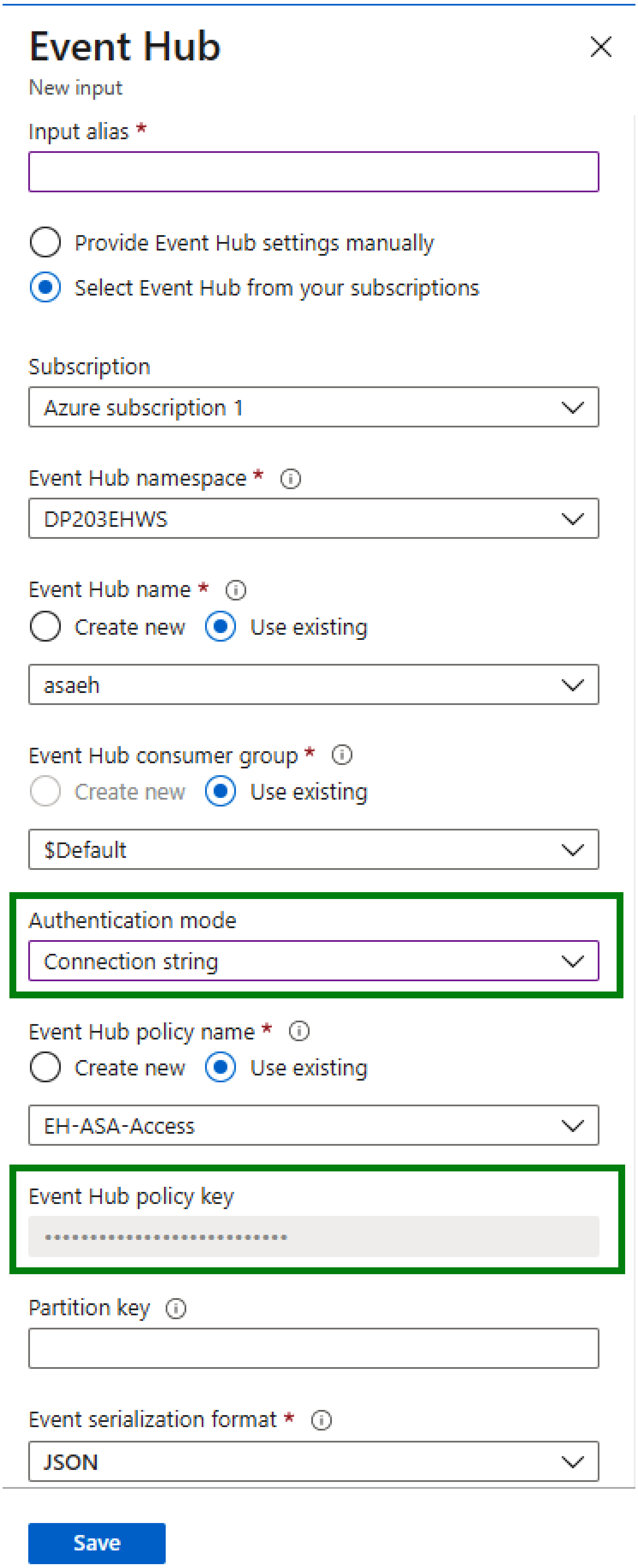

Home > Stream Analytics jobs >
New Stream Analytics job
This will create a new Stream Analytics job. You will be charged according to Azure Stream Analytics billing model. Learn more.
Job name *
Subscription *
Azure subscription 1
Resource group *
Create new
Location *
East US
Hosting environment
Cloud
Edge
Streaming units (1 to 192)
3
Secure all private data assets needed by this job in my Storage account.
Create

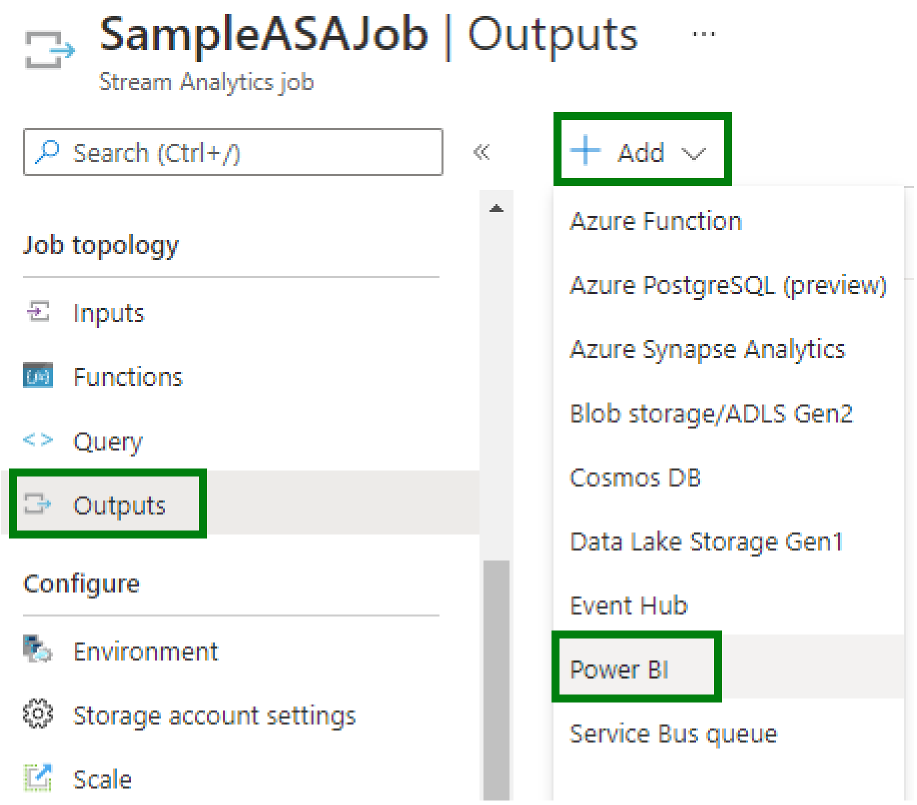

Home > Stream Analytics jobs >
New Stream Analytics job ...
This will create a new Stream Analytics job. You will be charged according to Azure Stream Analytics billing model. Learn more.
Job name *
Subscription *
Azure subscription 1
Resource group *
Create new
Location *
East US
Hosting environment
Cloud
Edge
Streaming units (1 to 192)
3
Secure all private data assets needed by this job in my Storage account.
Create

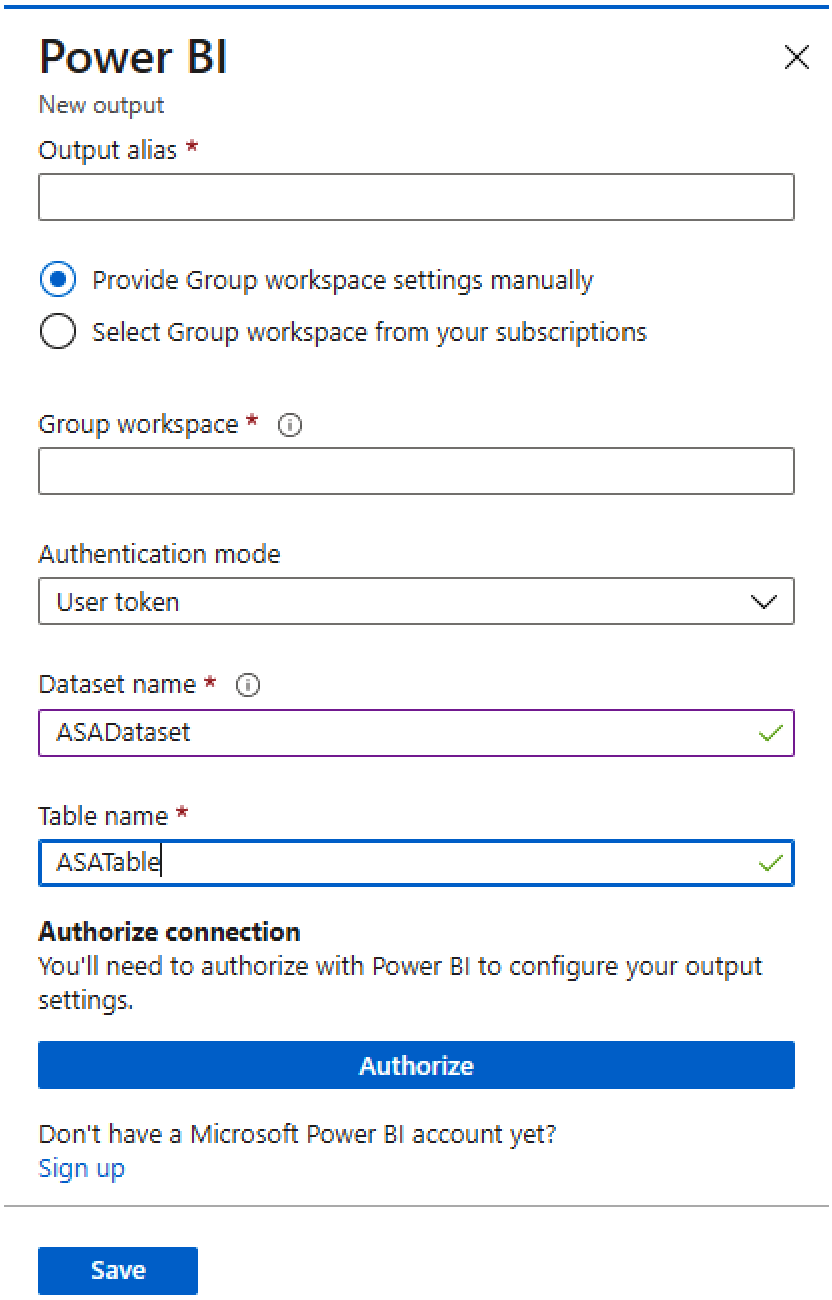

Home > DP203EHWS
DP203EHWS | Shared access
Event Hubs Namespace
Search (Ctrl+/)
Settings
Shared access policies
Scale
Geo-Recovery
Encryption
Add
Search to fil
Policy
RootManageS
Add SAS Policy
Event Hubs
Policy name *
EH-ASA-Access
Manage
Send
Listen

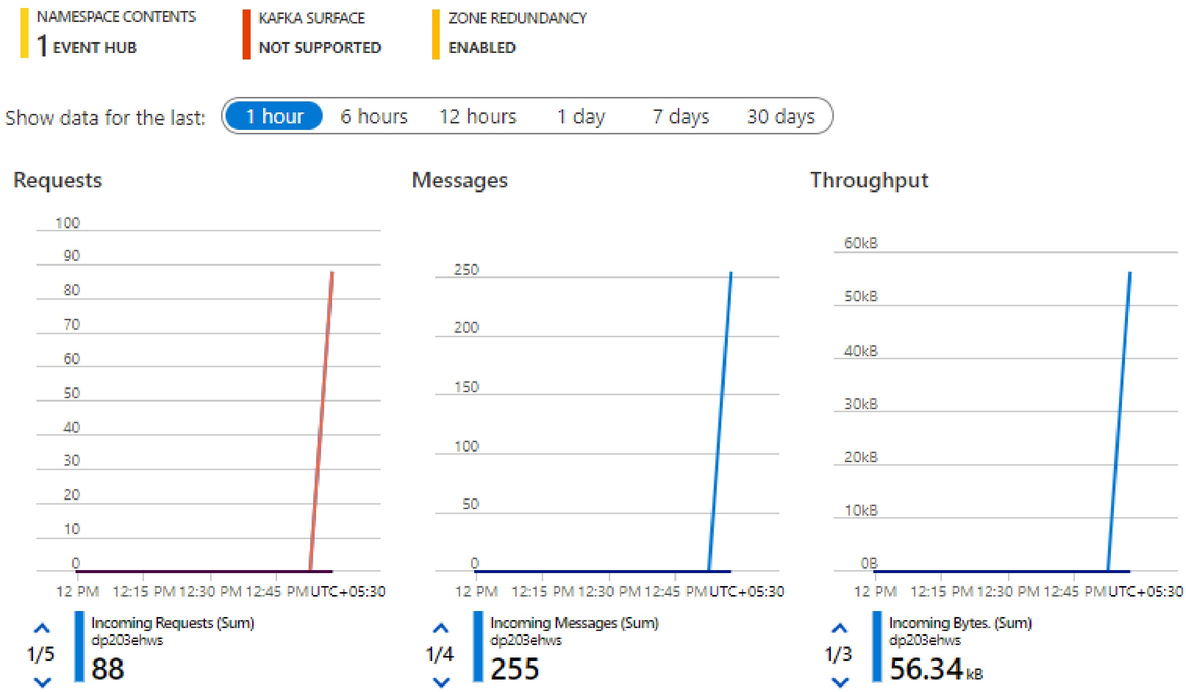

SAS Policy: EH-ASA-Access
Save
Discard
Delete
Manage
Send
Listen
Primary key
kfq9Vk8Nr [redacted] l...
Secondary key
6VE9V6S+ [redacted] ...
Connection string–primary key
Endpoint= [redacted] /...
Connection string–secondary key
Endpoint= [redacted] /...

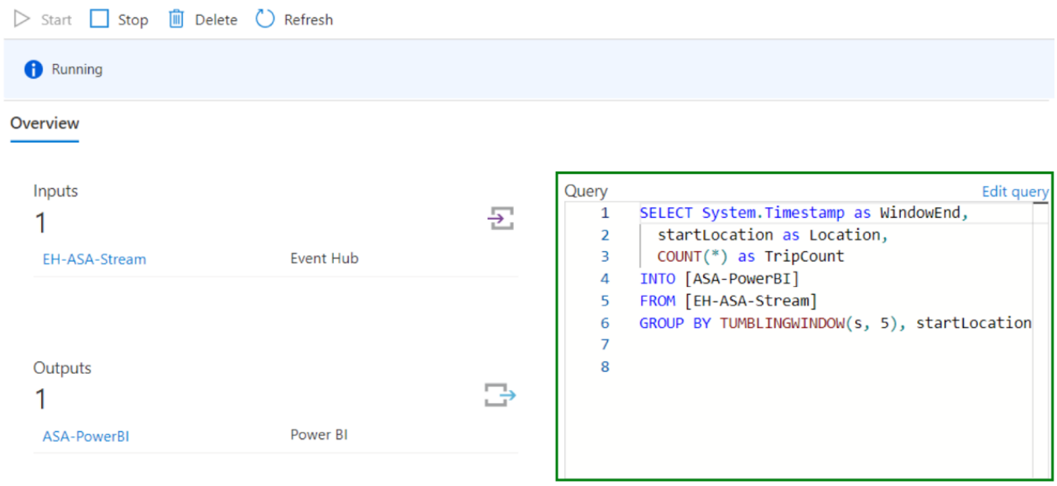

SampleASAJob | Inputs
Stream Analytics job
Search (Ctrl+/)
Job topology
Inputs
Functions
Query
Outputs
Add stream input
Blob storage/ADLS Gen2
Event Hub
IoT Hub

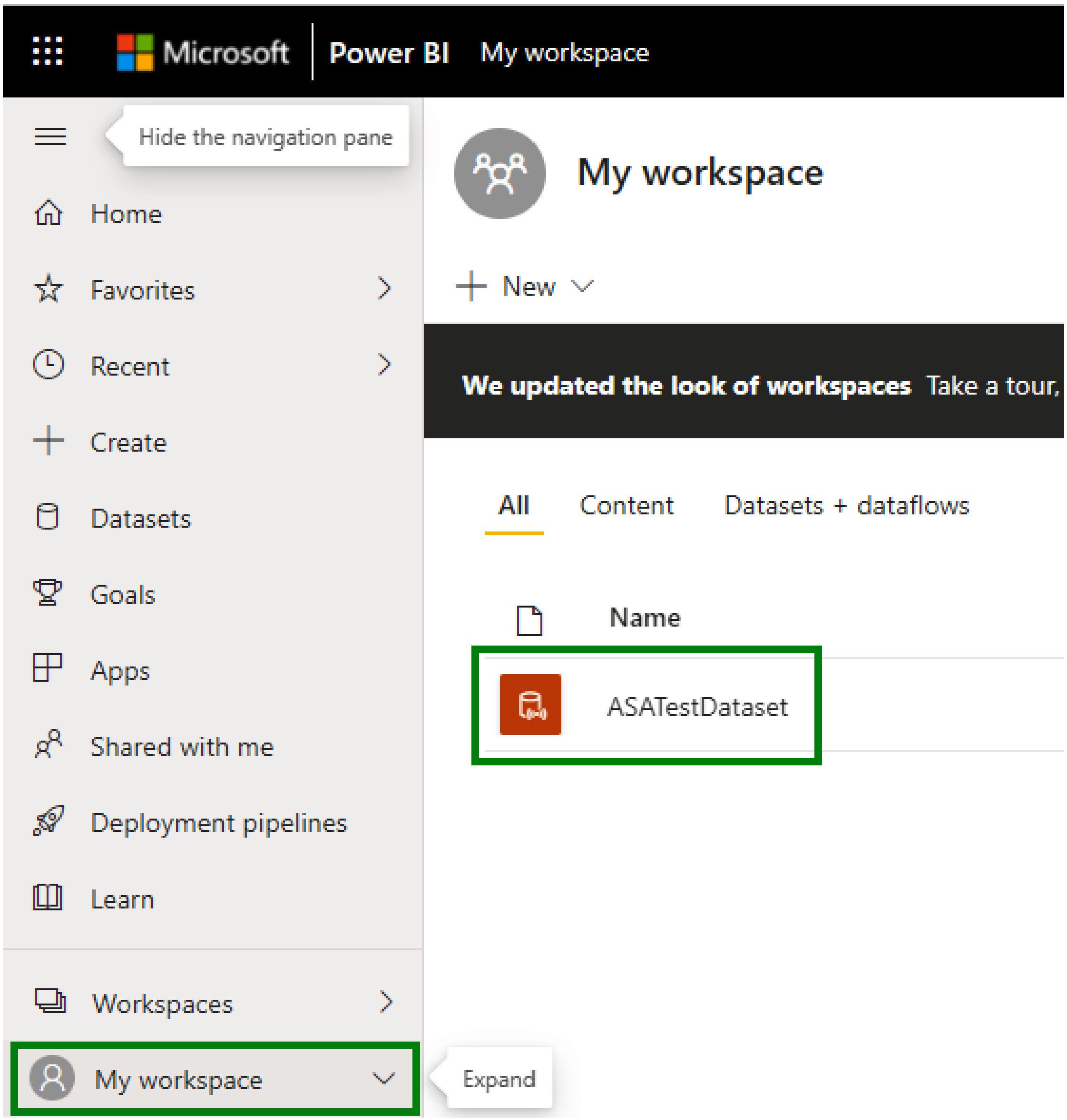

Event Hub
New input
Input alias *
Provide Event Hub settings manually
Select Event Hub from your subscriptions
Subscription
Azure subscription 1
Event Hub namespace *
DP203EHWS
Event Hub name *
Create new
Use existing
asaeh
Event Hub consumer group *
Create new
Use existing
$Default
Authentication mode
Connection string
Event Hub policy name *
Create new
Use existing
EH-ASA-Access
Event Hub policy key
Partition key
Event serialization format *
JSON
Save


SampleASAJob | Outputs
Stream Analytics job
Search (Ctrl+/)
Add
Job topology
Inputs
Functions
Query
Outputs
Configure
Environment
Storage account settings
Scale
Azure Function
Azure PostgreSQL (preview)
Azure Synapse Analytics
Blob storage/ADLS Gen2
Cosmos DB
Data Lake Storage Gen1
Event Hub
Power BI
Service Bus queue


Power BI
New output
Output alias *
Provide Group workspace settings manually
Select Group workspace from your subscriptions
Group workspace *
Authentication mode
User token
Dataset name *
ASADataset
Table name *
ASATable
Authorize connection
You'll need to authorize with Power BI to configure your output settings.
Authorize
Don't have a Microsoft Power BI account yet?
Sign up
Save


NAMESPACE CONTENTS
1 EVENT HUB
KAFKA SURFACE
NOT SUPPORTED
ZONE REDUNDANCY
ENABLED
Show data for the last:
1 hour
6 hours
12 hours
1 day
7 days
30 days
Requests
Messages
Throughput
12 PM 12:15 PM 12:30 PM 12:45 PM UTC+05:30
1/5
Incoming Requests (Sum)
dp203ehws
88
1/4
Incoming Messages (Sum)
dp203ehws
255
1/3
Incoming Bytes. (Sum)
dp203ehws
56.34 kB


Start
Stop
Delete
Refresh
Running
Overview
Inputs
1
EH-ASA-Stream
Event Hub
Outputs
1
ASA-PowerBI
Power BI
Query
Edit query
SELECT System.Timestamp as WindowEnd,
startLocation as Location,
COUNT(*) as TripCount
INTO [ASA-PowerBI]
FROM [EH-ASA-Stream]
GROUP BY TUMBLINGWINDOW(s, 5), startLocation


Microsoft
Power BI
My workspace
Hide the navigation pane
Home
Favorites
Recent
Create
Datasets
Goals
Apps
Shared with me
Deployment pipelines
Learn
Workspaces
My workspace
Expand
My workspace
New
We updated the look of workspaces Take a tour,
All
Content
Datasets + dataflows
Name
ASATestDataset

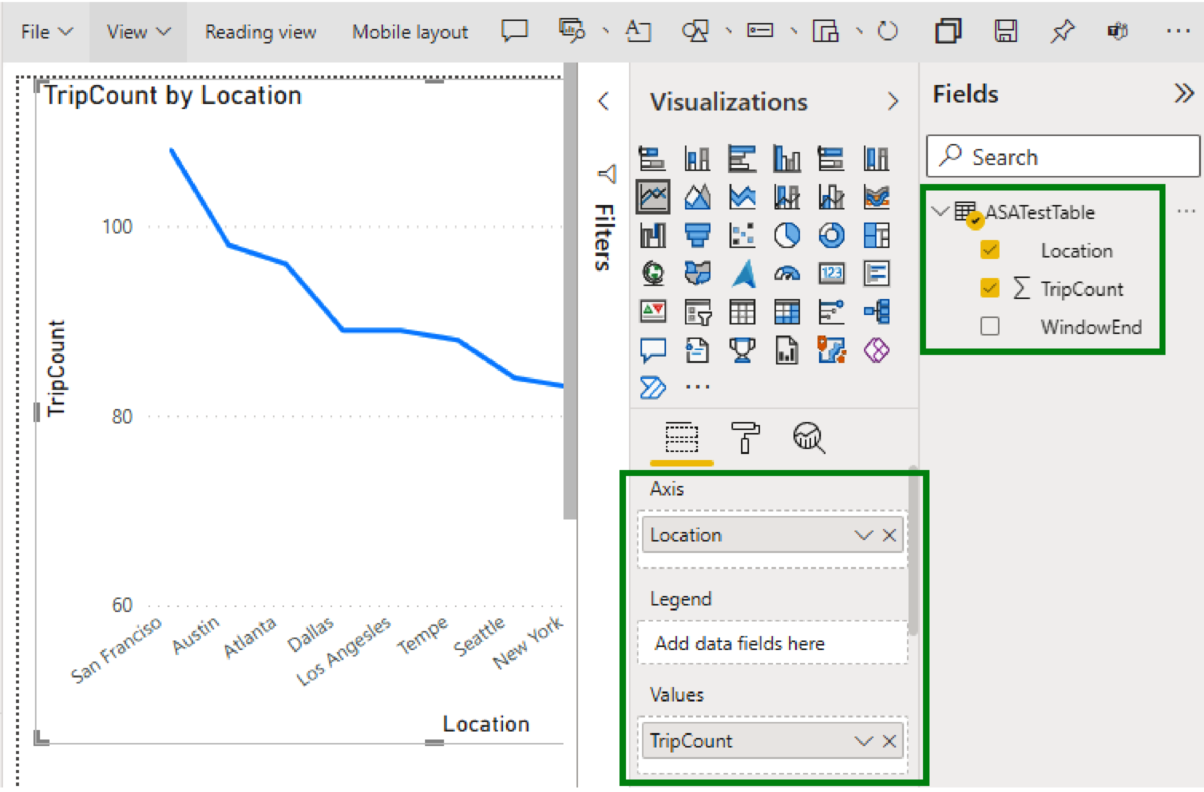


Table | Data Profile

	window	startLocation	count
1	{"start": "2021-12-06T11:52:00.000+0000", "end": "2021-12-06T11:53:00.000+0000"}	San Franciso	4
2	{"start": "2021-12-06T11:52:00.000+0000", "end": "2021-12-06T11:53:00.000+0000"}	Dallas	2
3	{"start": "2021-12-06T11:52:00.000+0000", "end": "2021-12-06T11:53:00.000+0000"}	Atlanta	4
4	{"start": "2021-12-06T11:52:00.000+0000", "end": "2021-12-06T11:53:00.000+0000"}	Tempe	8
5	{"start": "2021-12-06T11:52:00.000+0000", "end": "2021-12-06T11:53:00.000+0000"}	San Jose	1
6	{"start": "2021-12-06T11:52:00.000+0000", "end": "2021-12-06T11:53:00.000+0000"}	Denver	2

Showing all 14 rows.

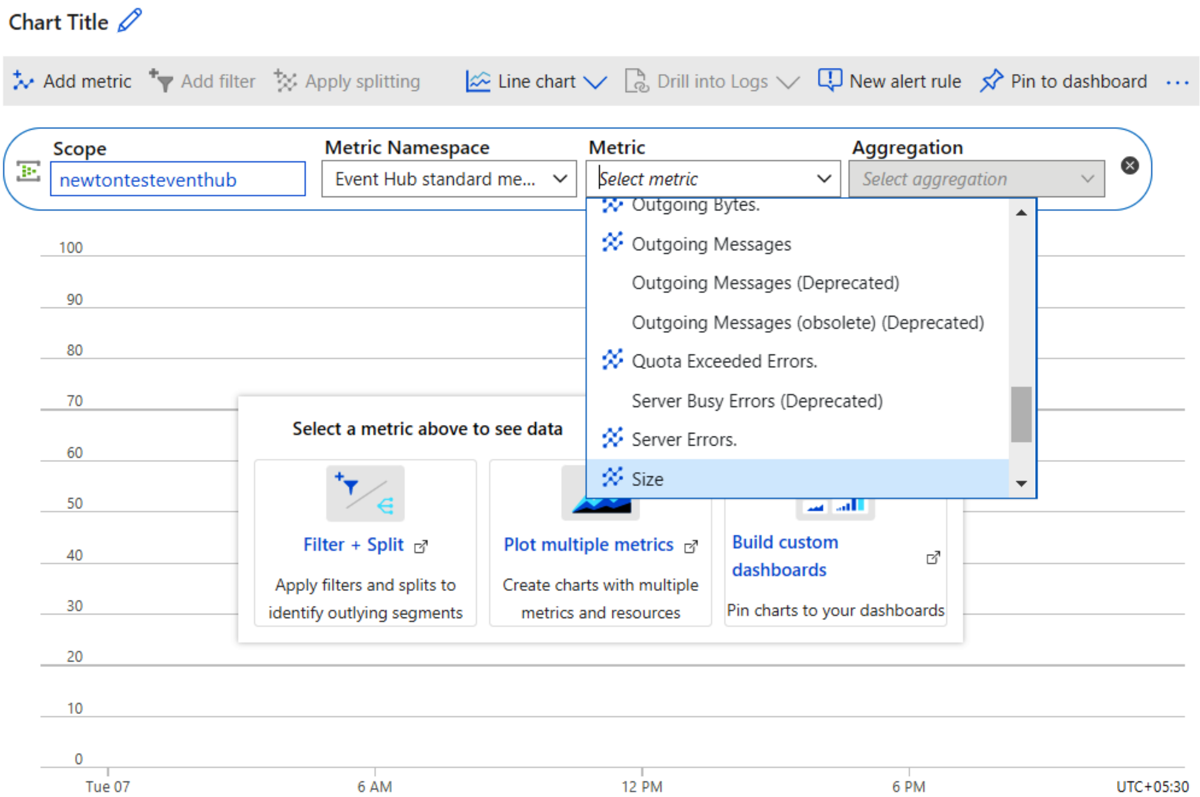
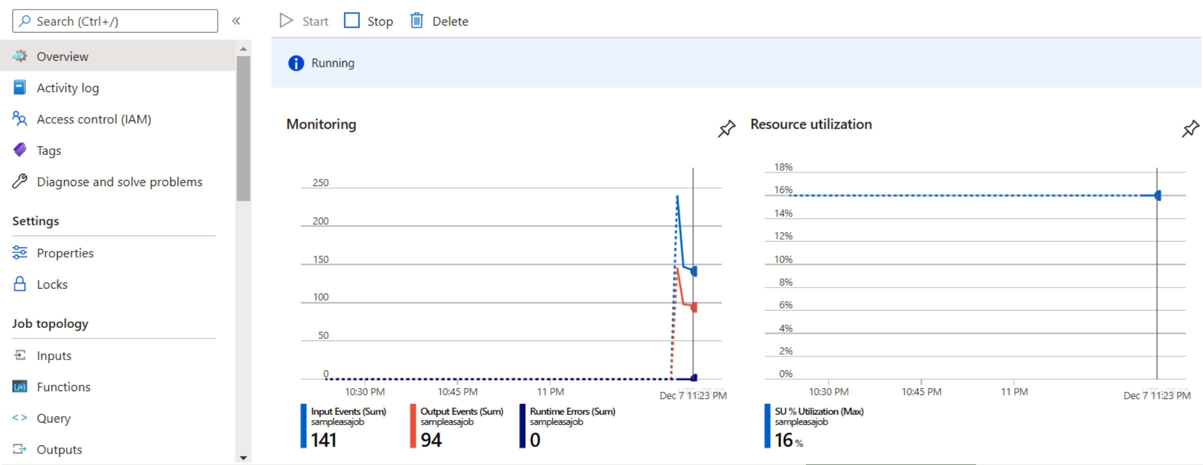
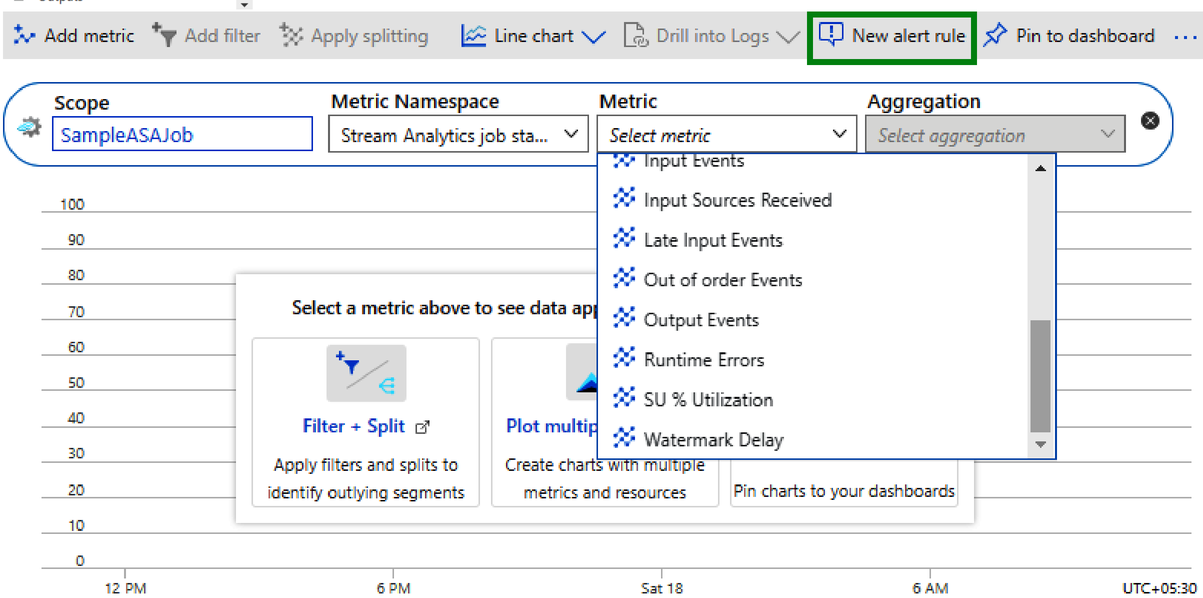
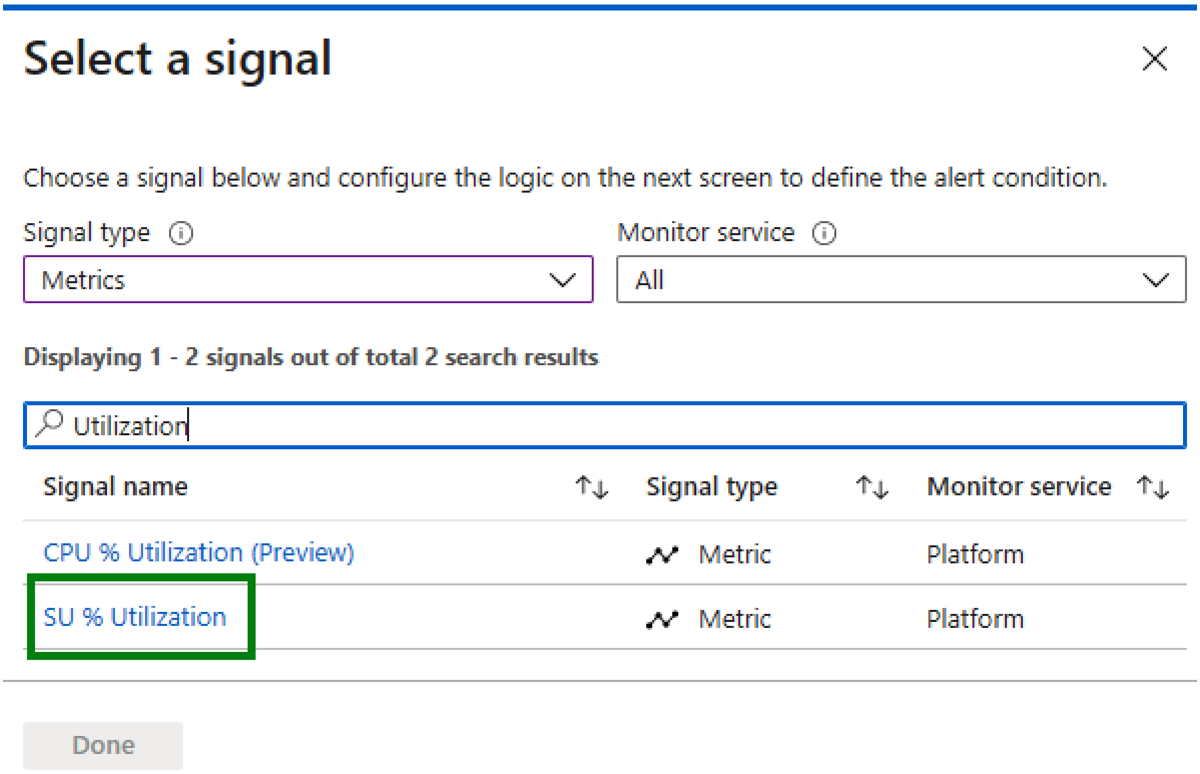
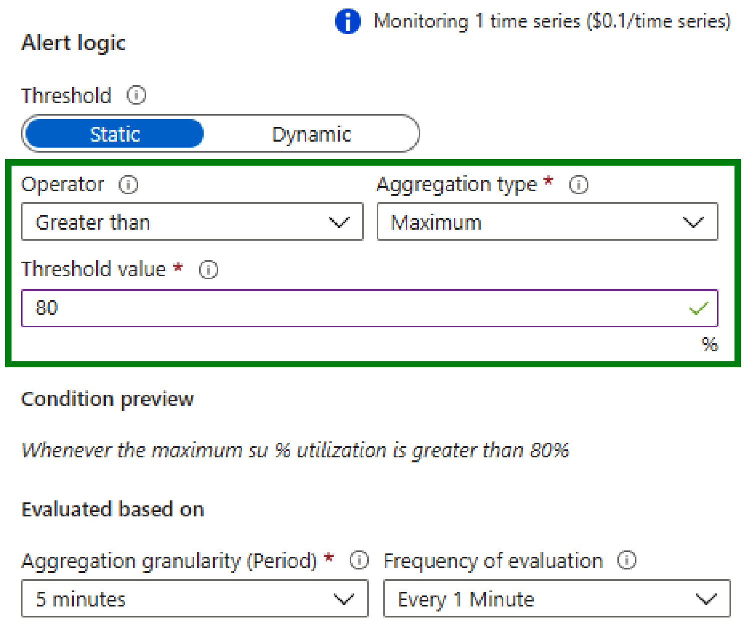


Chart Title
Add metric
Add filter
Apply splitting
Line chart
Drill into Logs
New alert rule
Pin to dashboard
Scope
newtontesteventhub
Metric Namespace
Event Hub standard me...
Metric
Select metric
Aggregation
Select aggregation
Outgoing Bytes.
Outgoing Messages
Outgoing Messages (Deprecated)
Outgoing Messages (obsolete) (Deprecated)
Quota Exceeded Errors.
Server Busy Errors (Deprecated)
Server Errors.
Size
Select a metric above to see data
Filter + Split
Apply filters and splits to identify outlying segments
Plot multiple metrics
Create charts with multiple metrics and resources
Build custom dashboards
Pin charts to your dashboards
Tue 07
6 AM
12 PM
6 PM
UTC+05:30

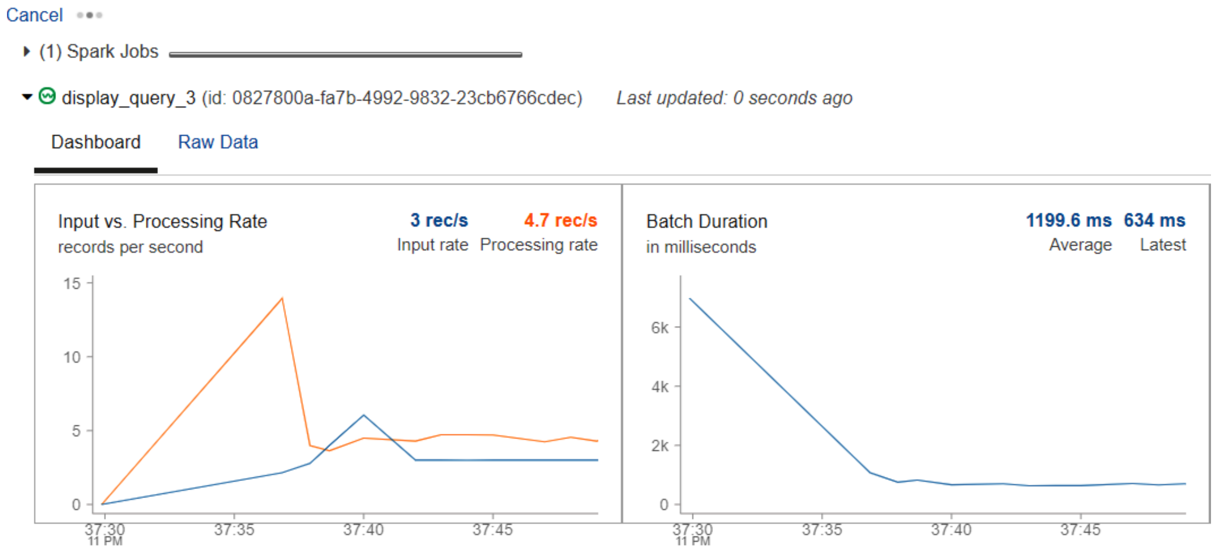

Search (Ctrl+/)
Start
Stop
Delete
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Settings
Properties
Locks
Job topology
Inputs
Functions
Query
Outputs
Running
Monitoring
Resource utilization
Input Events (Sum)
141
Output Events (Sum)
94
Runtime Errors (Sum)
0
SU % Utilization (Max)
16

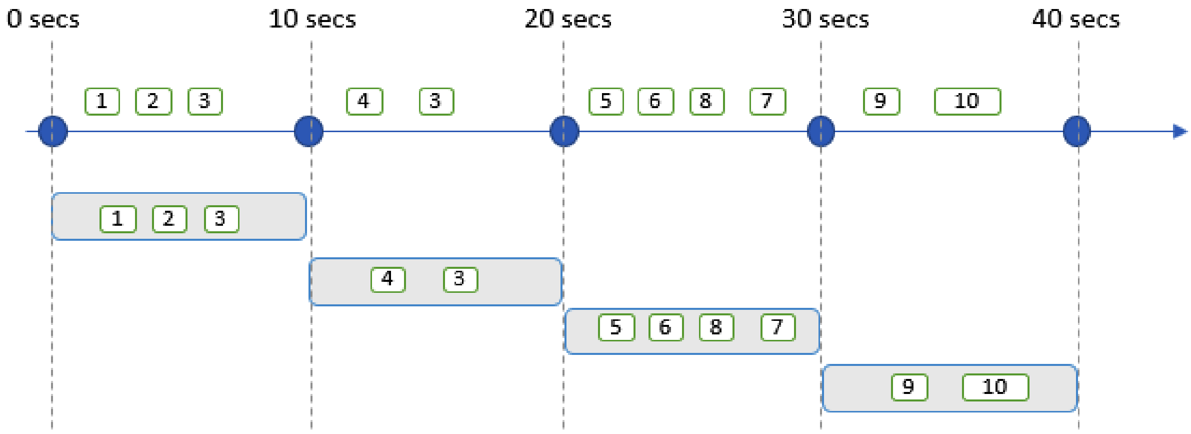

Add metric
Add filter
Apply splitting
Line chart
Drill into Logs
New alert rule
Pin to dashboard
Scope
SampleASAJob
Metric Namespace
Stream Analytics job sta...
Metric
Select metric
Aggregation
Select aggregation
Input Events
Input Sources Received
Late Input Events
Out of order Events
Output Events
Runtime Errors
SU % Utilization
Watermark Delay
Select a metric above to see data
Filter + Split
Apply filters and splits to identify outlying segments
Create charts with multiple metrics and resources
Pin charts to your dashboards
12 PM
6 PM
Sat 18
6 AM
UTC+05:30

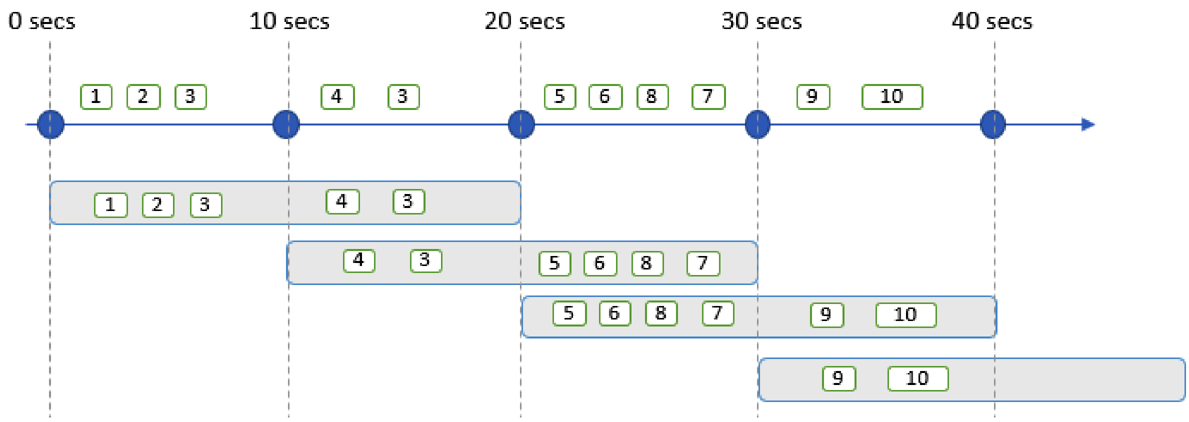

Select a signal
Choose a signal below and configure the logic on the next screen to define the alert condition.
Signal type
Metrics
Monitor service
All
Displaying 1 - 2 signals out of total 2 search results
Utilization
Signal name
Signal type
Monitor service
CPU % Utilization (Preview)
Metric
Platform
SU % Utilization
Metric
Platform
Done

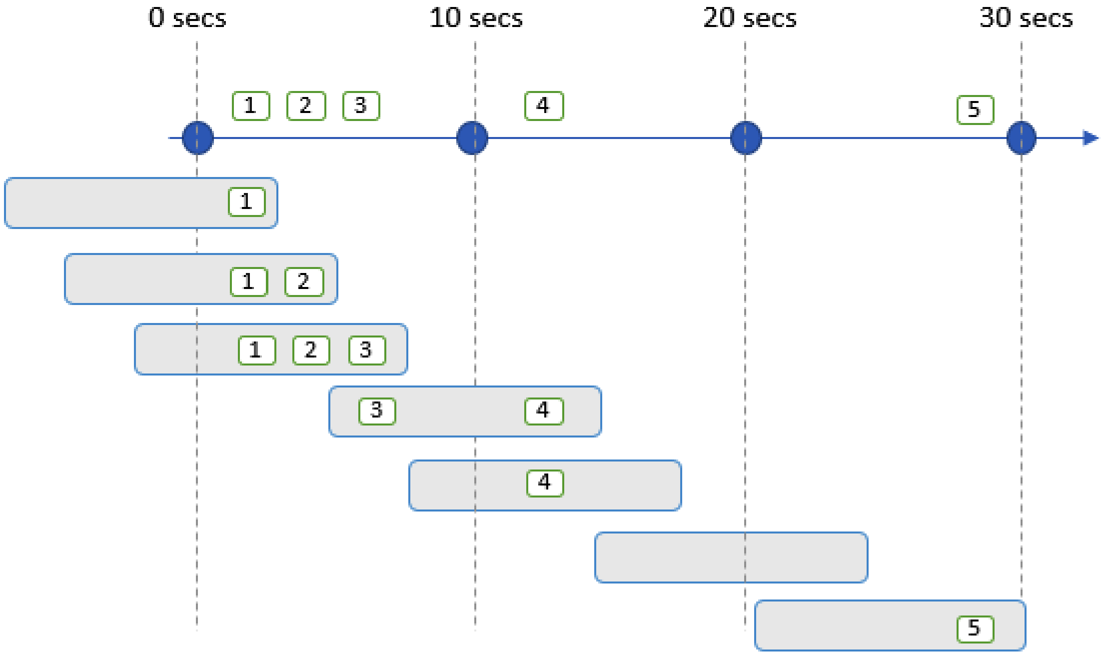

Monitoring 1 time series ($0.1/time series)
Alert logic
Threshold
Static
Dynamic
Operator
Greater than
Aggregation type *
Maximum
Threshold value *
80
%
Condition preview
Whenever the maximum su % utilization is greater than 80%
Evaluated based on
Aggregation granularity (Period) *
5 minutes
Frequency of evaluation
Every 1 Minute

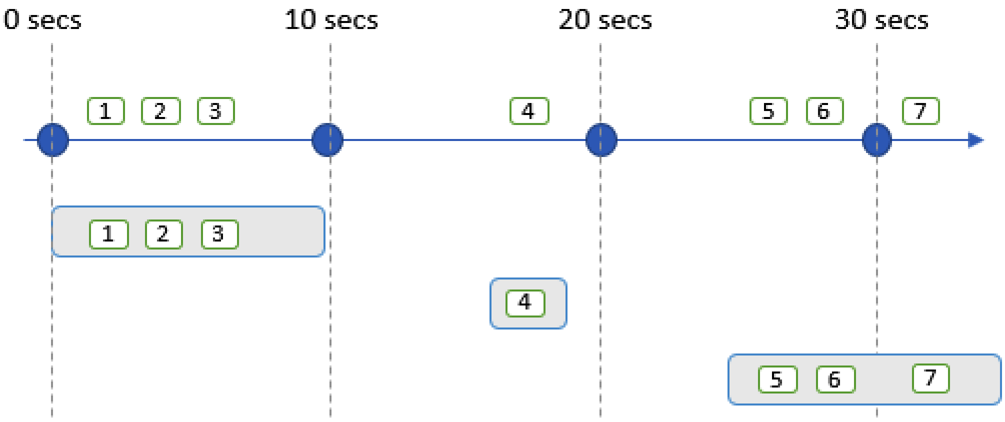
```
EHStreamJsonDF=newEventHubDf.select("tripjson.*")
display(EHStreamJsonDF)
```

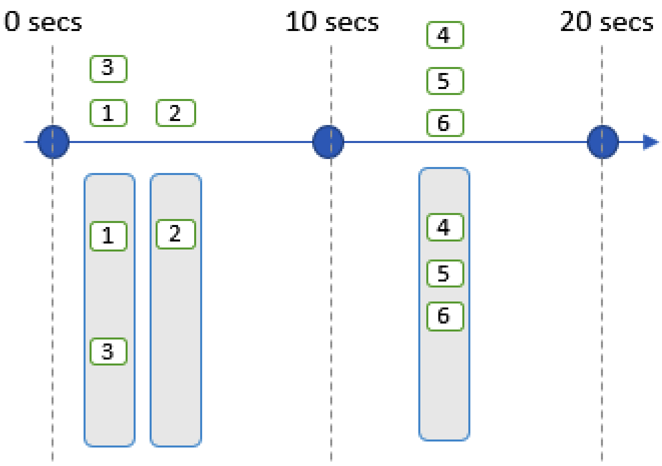


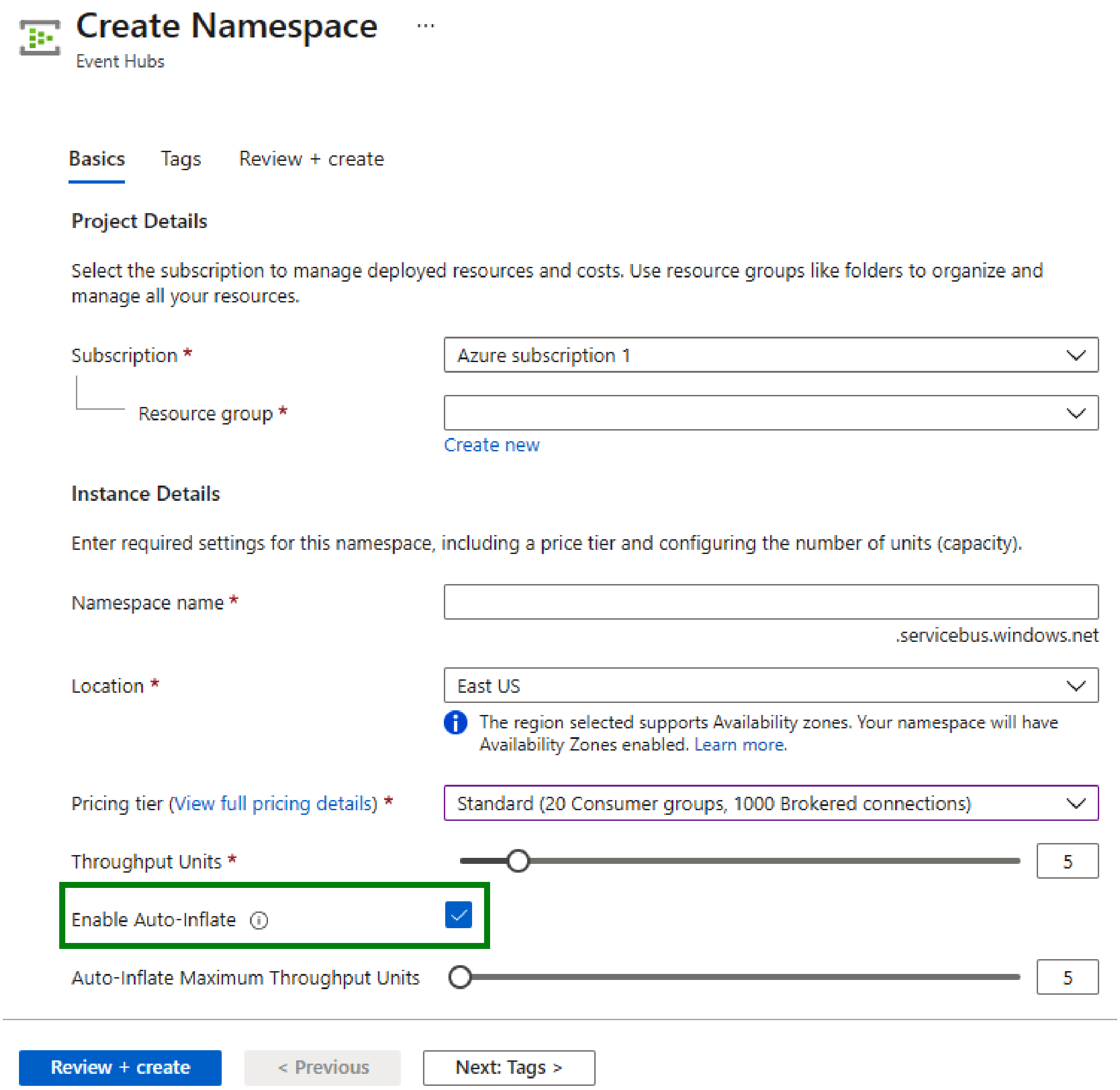


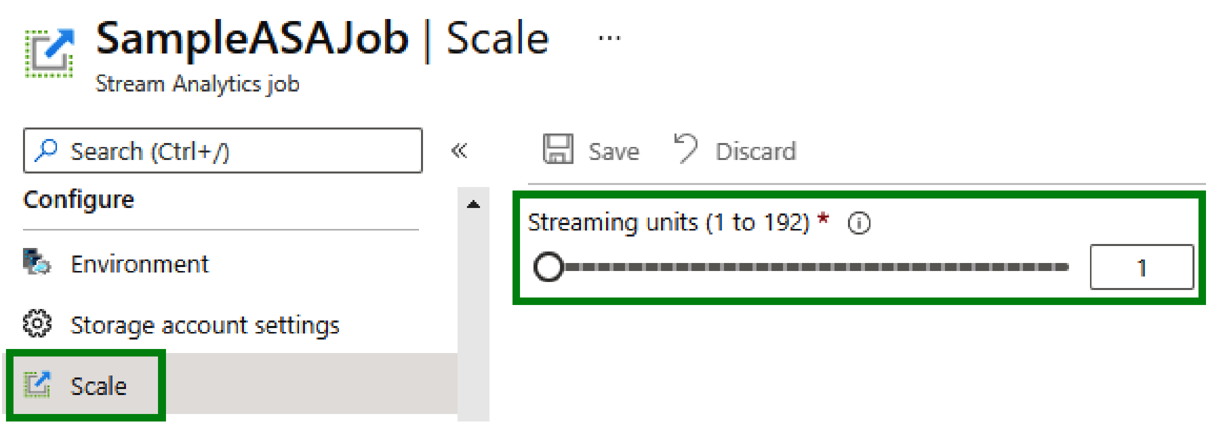


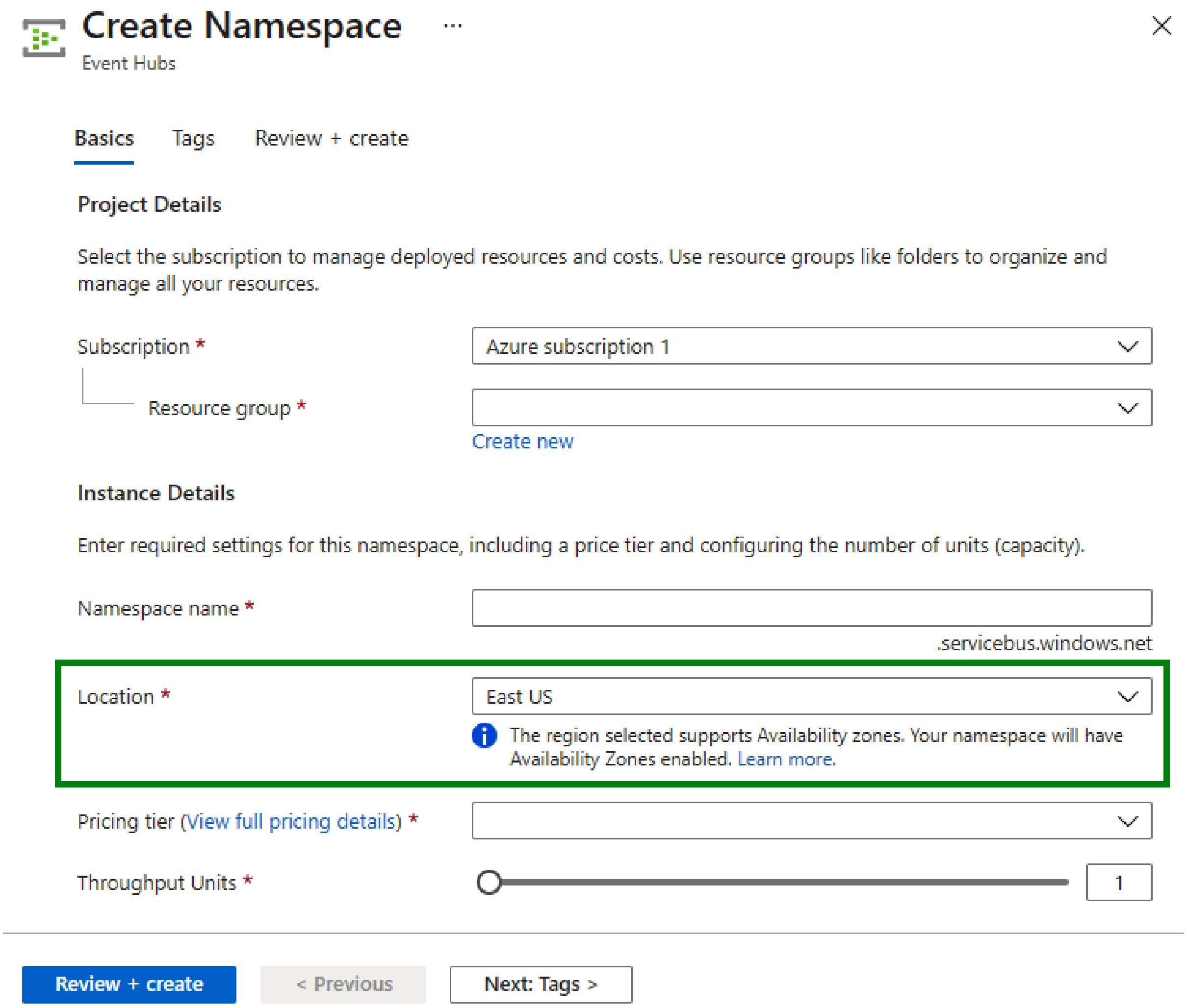

0 secs
10 secs
20 secs
30 secs

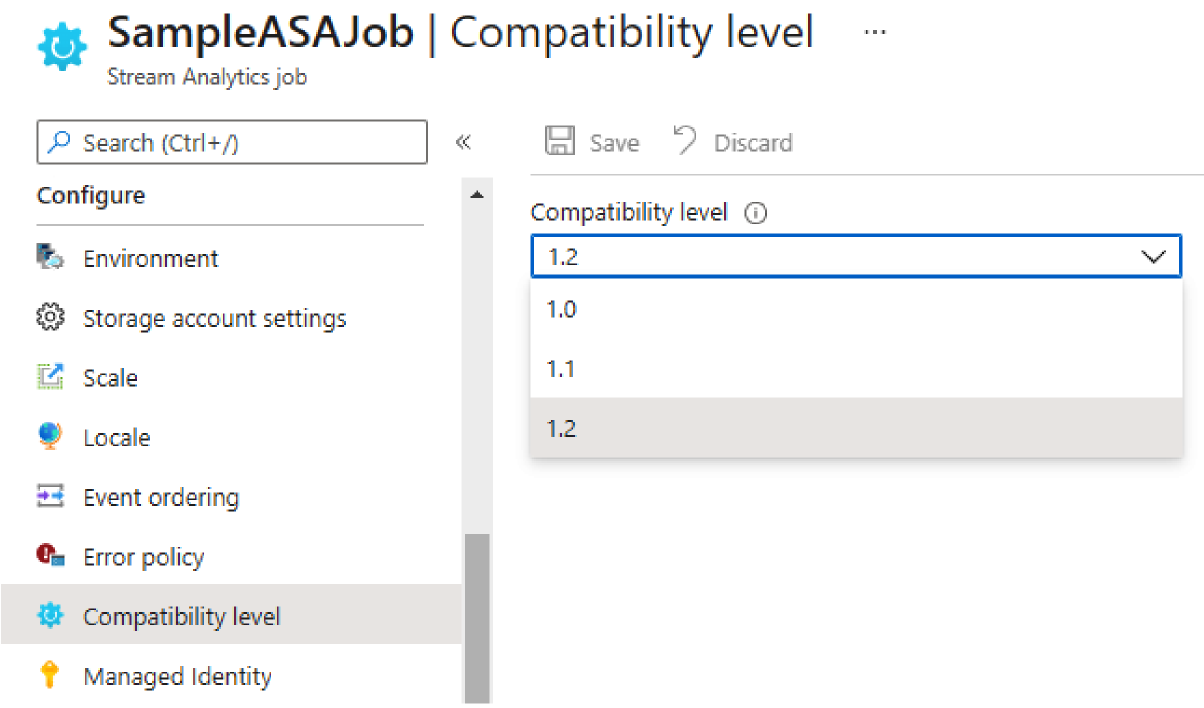

0 secs
10 secs
20 secs
30 secs

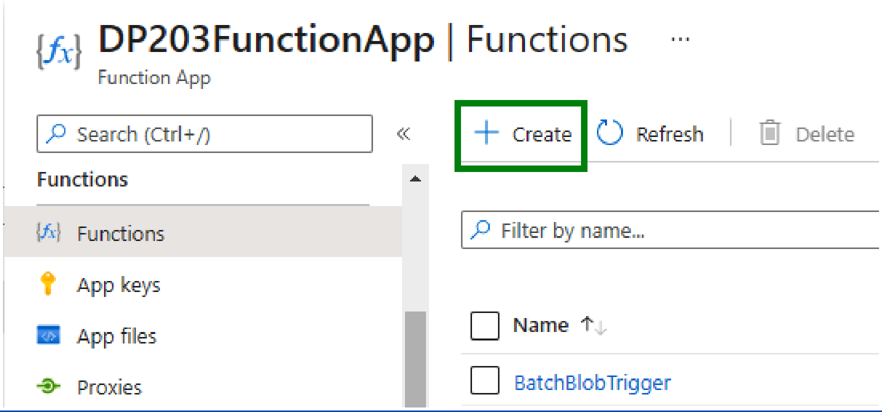

0 secs
10 secs
20 secs

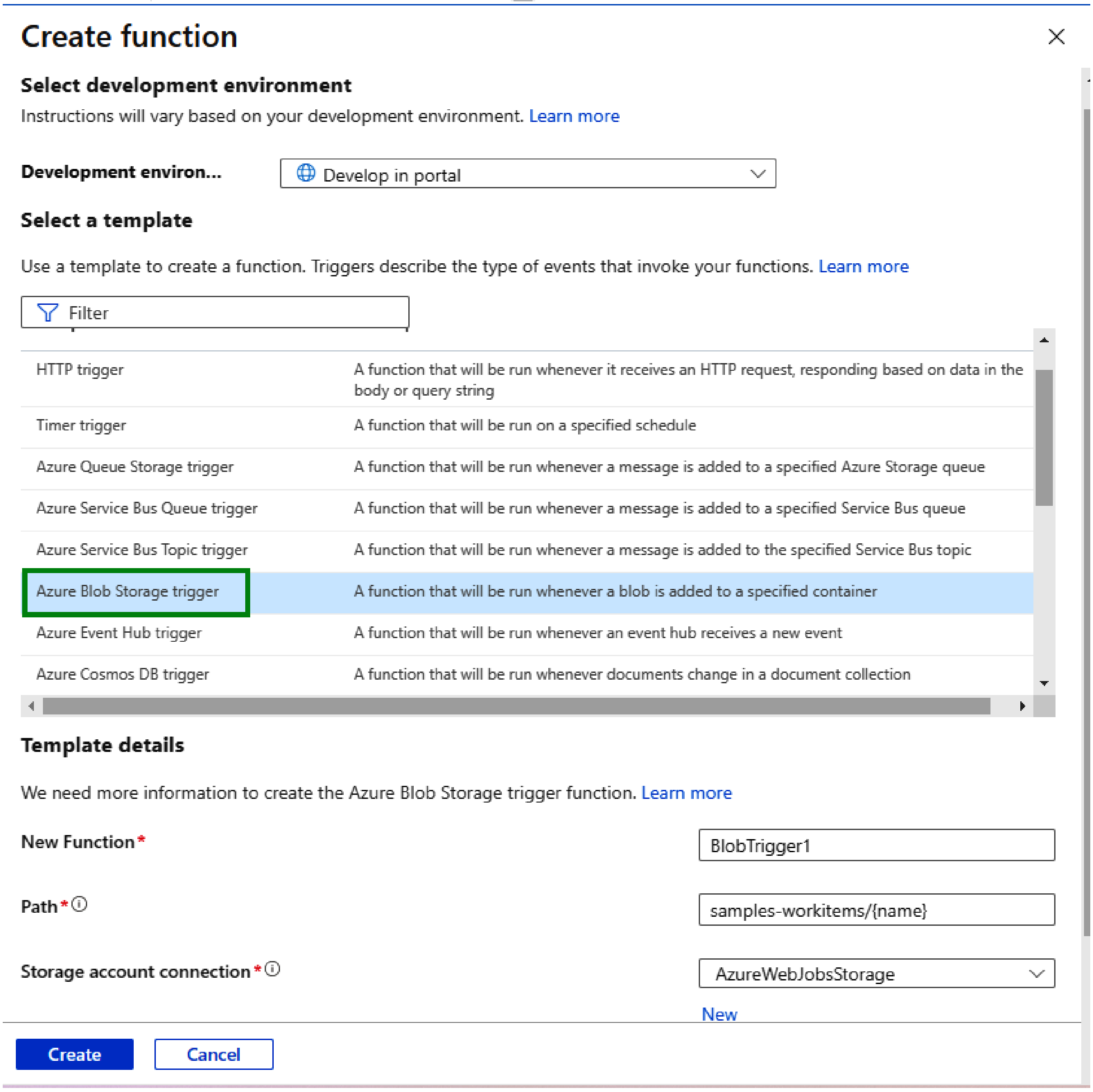

Create Namespace
Event Hubs
Basics
Tags
Review + create
Project Details
Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.
Subscription *
Azure subscription 1
Resource group *
Create new
Instance Details
Enter required settings for this namespace, including a price tier and configuring the number of units (capacity).
Namespace name *
.servicebus.windows.net
Location *
East US
The region selected supports Availability zones. Your namespace will have Availability Zones enabled. Learn more.
Pricing tier (View full pricing details) *
Standard (20 Consumer groups, 1000 Brokered connections)
Throughput Units *
5
Enable Auto-Inflate
Auto-Inflate Maximum Throughput Units
5
Review + create
< Previous
Next: Tags >

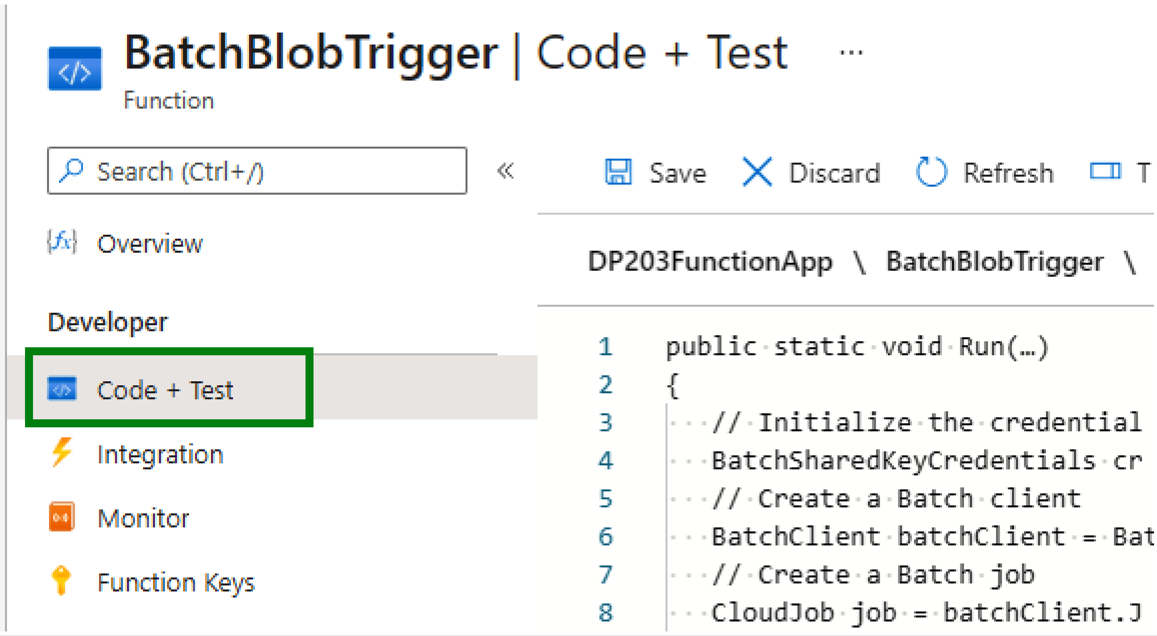

SampleASAJob | Scale
Stream Analytics job
Search (Ctrl+/)
Save
Discard
Configure
Environment
Storage account settings
Scale
Streaming units (1 to 192) *
1

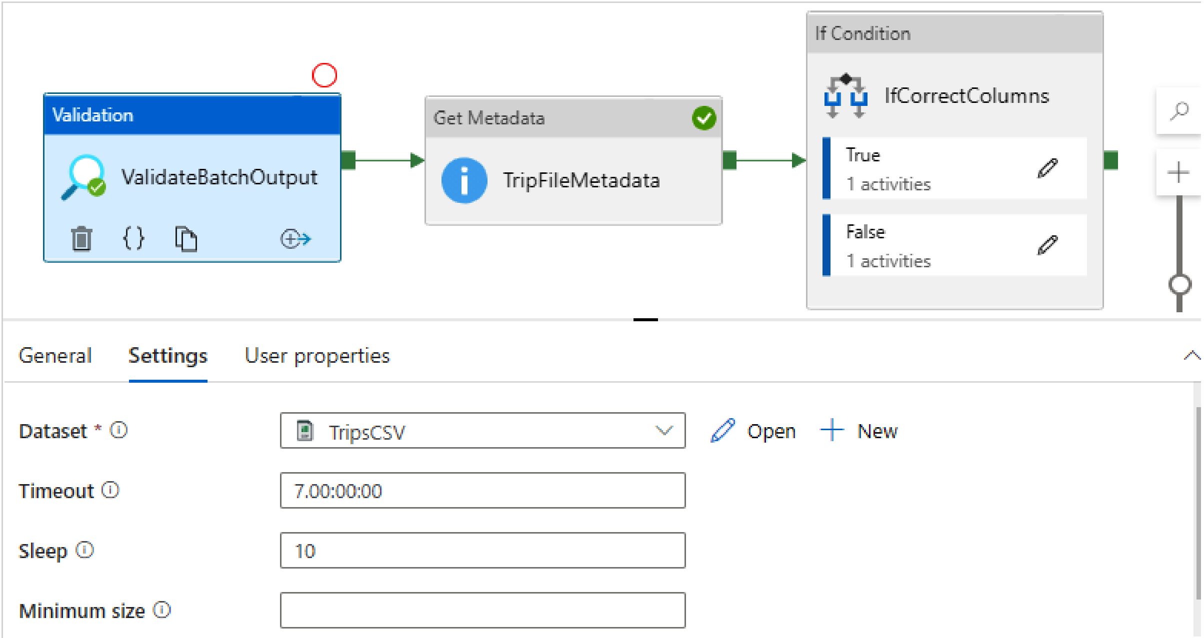

Create Namespace
Event Hubs
Basics
Tags
Review + create
Project Details
Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.
Subscription *
Azure subscription 1
Resource group *
Create new
Instance Details
Enter required settings for this namespace, including a price tier and configuring the number of units (capacity).
Namespace name *
.servicebus.windows.net
Location *
East US
The region selected supports Availability zones. Your namespace will have Availability Zones enabled. Learn more.
Pricing tier (View full pricing details) *
Throughput Units *
1
Review + create
< Previous
Next: Tags >

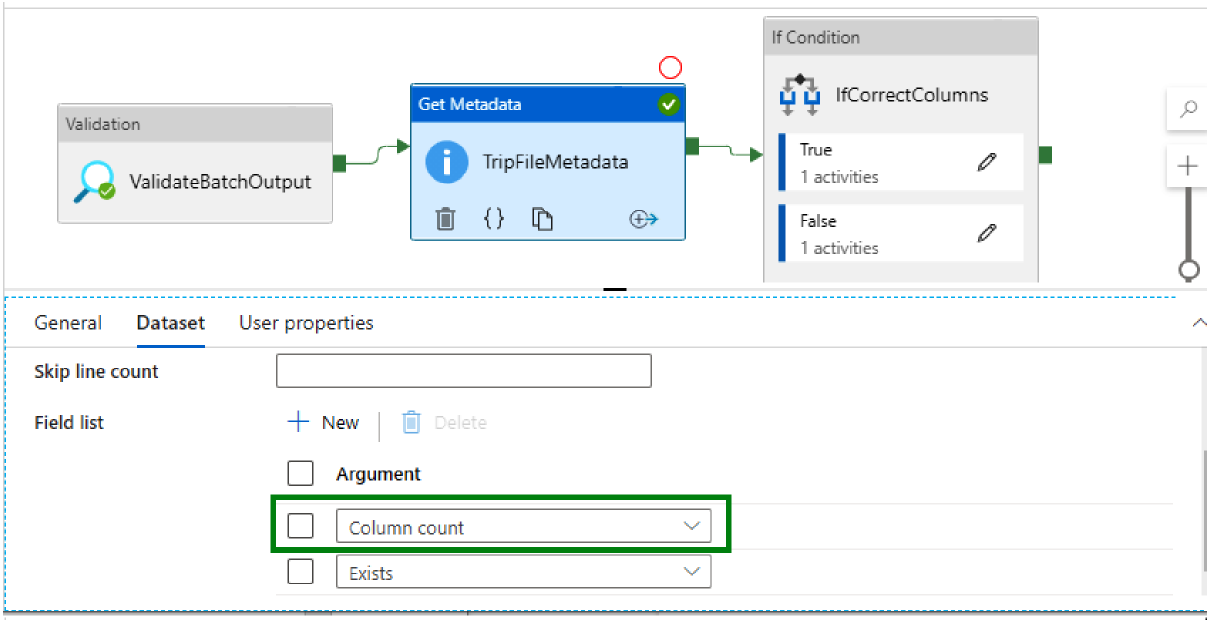

SampleASAJob | Compatibility level
Stream Analytics job
Search (Ctrl+/)
Configure
Environment
Storage account settings
Scale
Locale
Event ordering
Error policy
Compatibility level
Managed Identity
Save
Discard
Compatibility level
1.2
1.0
1.1
1.2

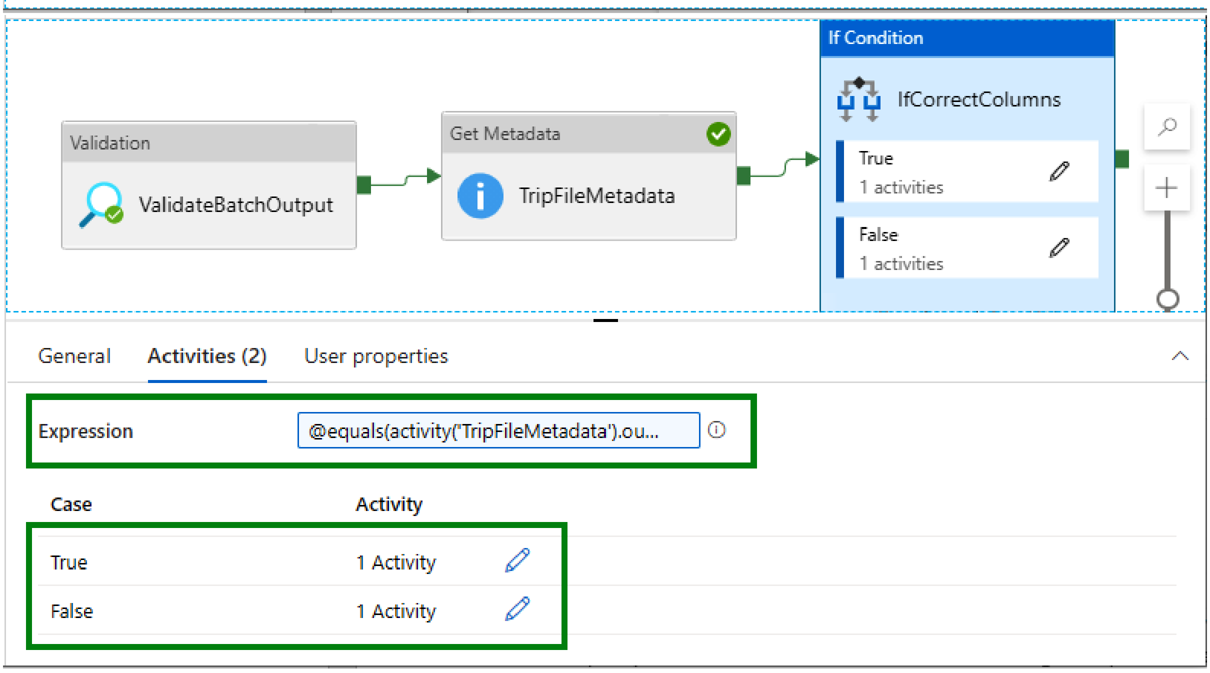

DP203FunctionApp | Functions
Function App
Search (Ctrl+/)
Create
Refresh
Delete
Functions
Functions
App keys
App files
Proxies
Filter by name...
Name
BatchBlobTrigger

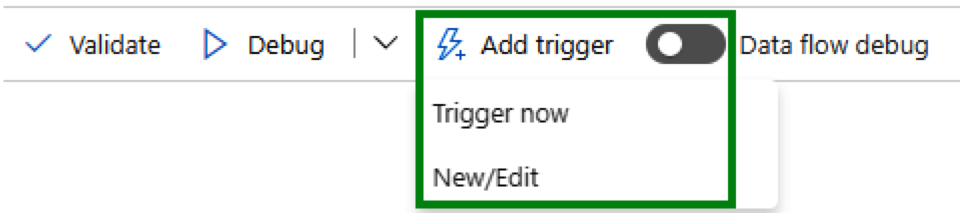

Create function
Select development environment
Instructions will vary based on your development environment. Learn more
Development environ...
Develop in portal
Select a template
Use a template to create a function. Triggers describe the type of events that invoke your functions. Learn more
Filter
HTTP trigger
A function that will be run whenever it receives an HTTP request, responding based on data in the body or query string
Timer trigger
A function that will be run on a specified schedule
Azure Queue Storage trigger
A function that will be run whenever a message is added to a specified Azure Storage queue
Azure Service Bus Queue trigger
A function that will be run whenever a message is added to a specified Service Bus queue
Azure Service Bus Topic trigger
A function that will be run whenever a message is added to the specified Service Bus topic
Azure Blob Storage trigger
A function that will be run whenever a blob is added to a specified container
Azure Event Hub trigger
A function that will be run whenever an event hub receives a new event
Azure Cosmos DB trigger
A function that will be run whenever documents change in a document collection
Template details
We need more information to create the Azure Blob Storage trigger function. Learn more
New Function *
BlobTrigger1
Path *
samples-workitems/{name}
Storage account connection *
AzureWebJobsStorage
New
Create
Cancel

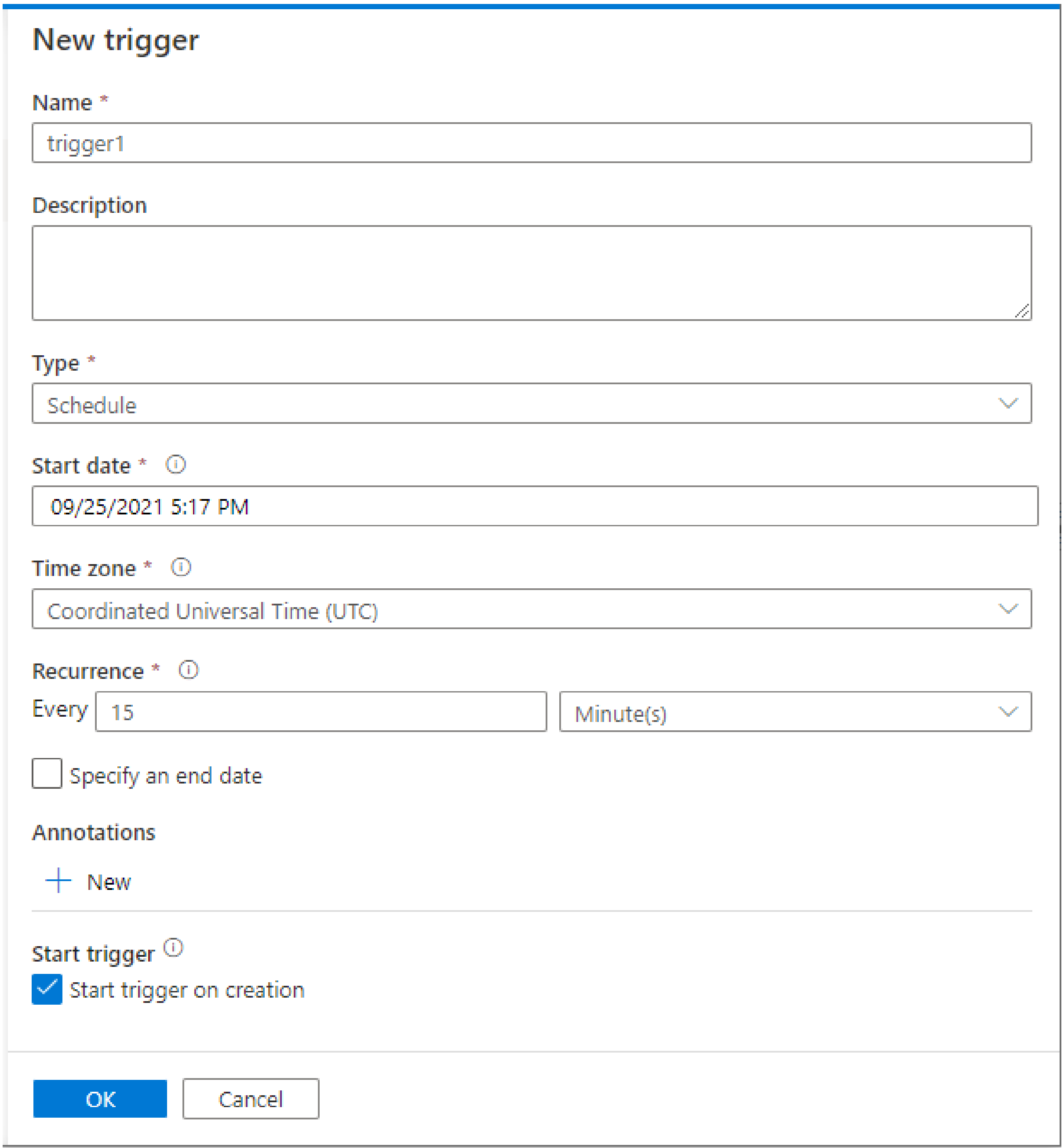

BatchBlobTrigger | Code + Test
Function
Search (Ctrl+/)
Save
Discard
Refresh
Overview
DP203FunctionApp \ BatchBlobTrigger \
Developer
Code + Test
Integration
Monitor
Function Keys
public static void Run(…)
{
// Initialize the credential
BatchSharedKeyCredentials cr
// Create a Batch client
BatchClient batchClient = Bat
// Create a Batch job
CloudJob job = batchClient.J

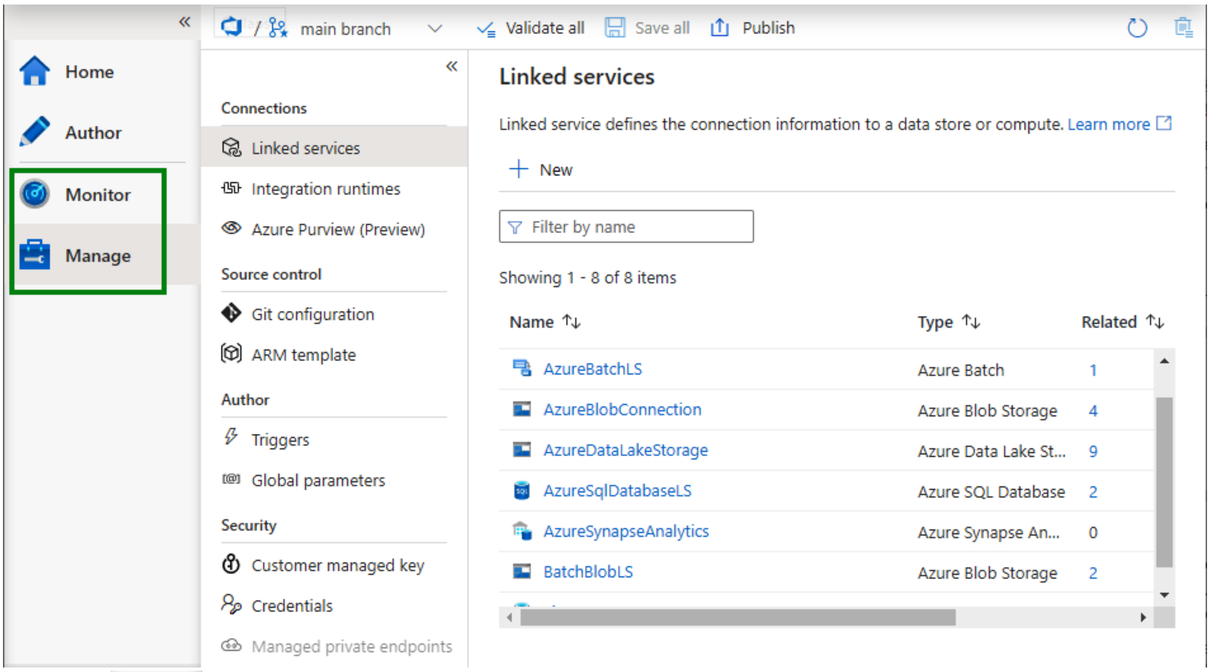

Validation
ValidateBatchOutput
Get Metadata
TripFileMetadata
If Condition
IfCorrectColumns
True
1 activities
False
1 activities
General
Settings
User properties
Dataset *
TripsCSV
Open
New
Timeout
7.00:00:00
Sleep
10
Minimum size

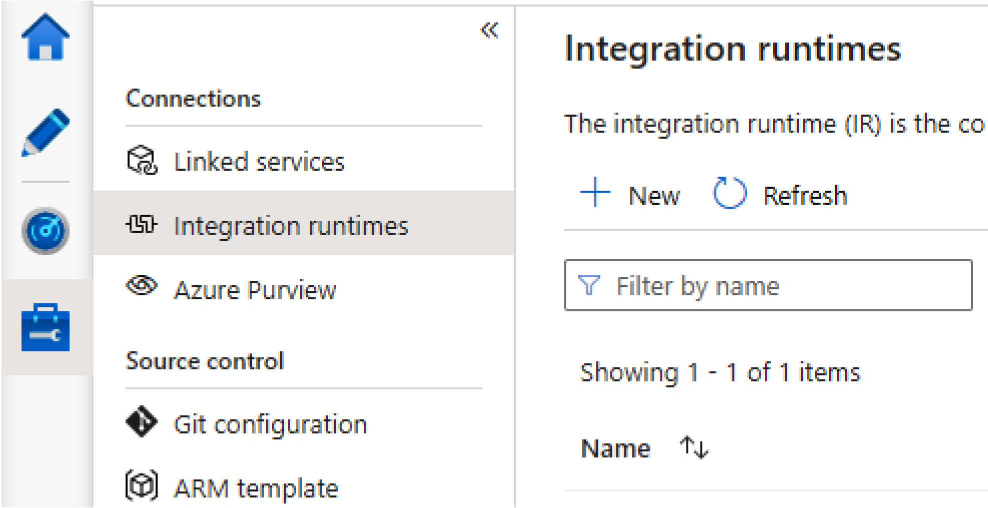

Validation
ValidateBatchOutput
Get Metadata
TripFileMetadata
If Condition
IfCorrectColumns
True
1 activities
False
1 activities
General
Dataset
User properties
Skip line count
Field list
New
Delete
Argument
Column count
Exists

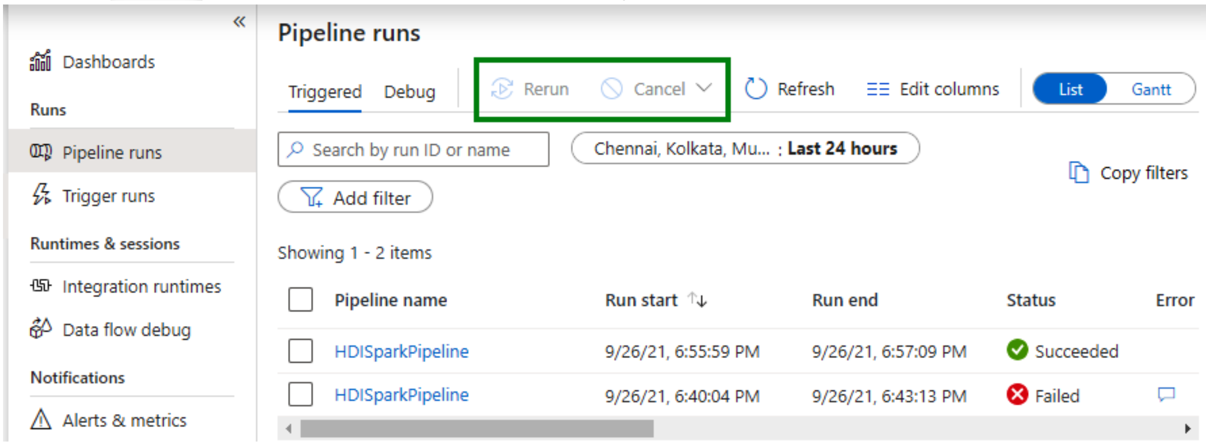

Validation
ValidateBatchOutput
Get Metadata
TripFileMetadata
If Condition
IfCorrectColumns
True
1 activities
False
1 activities
General
Activities (2)
User properties
Expression
@equals(activity('TripFileMetadata').ou...
Case
Activity
True
1 Activity
False
1 Activity

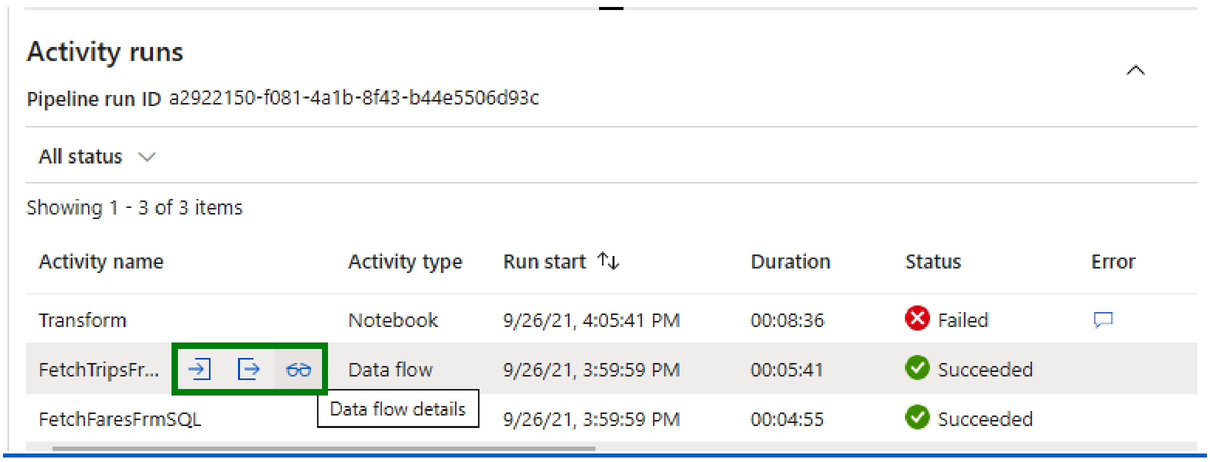

Validate
Debug
Add trigger
Data flow debug
Trigger now
New/Edit

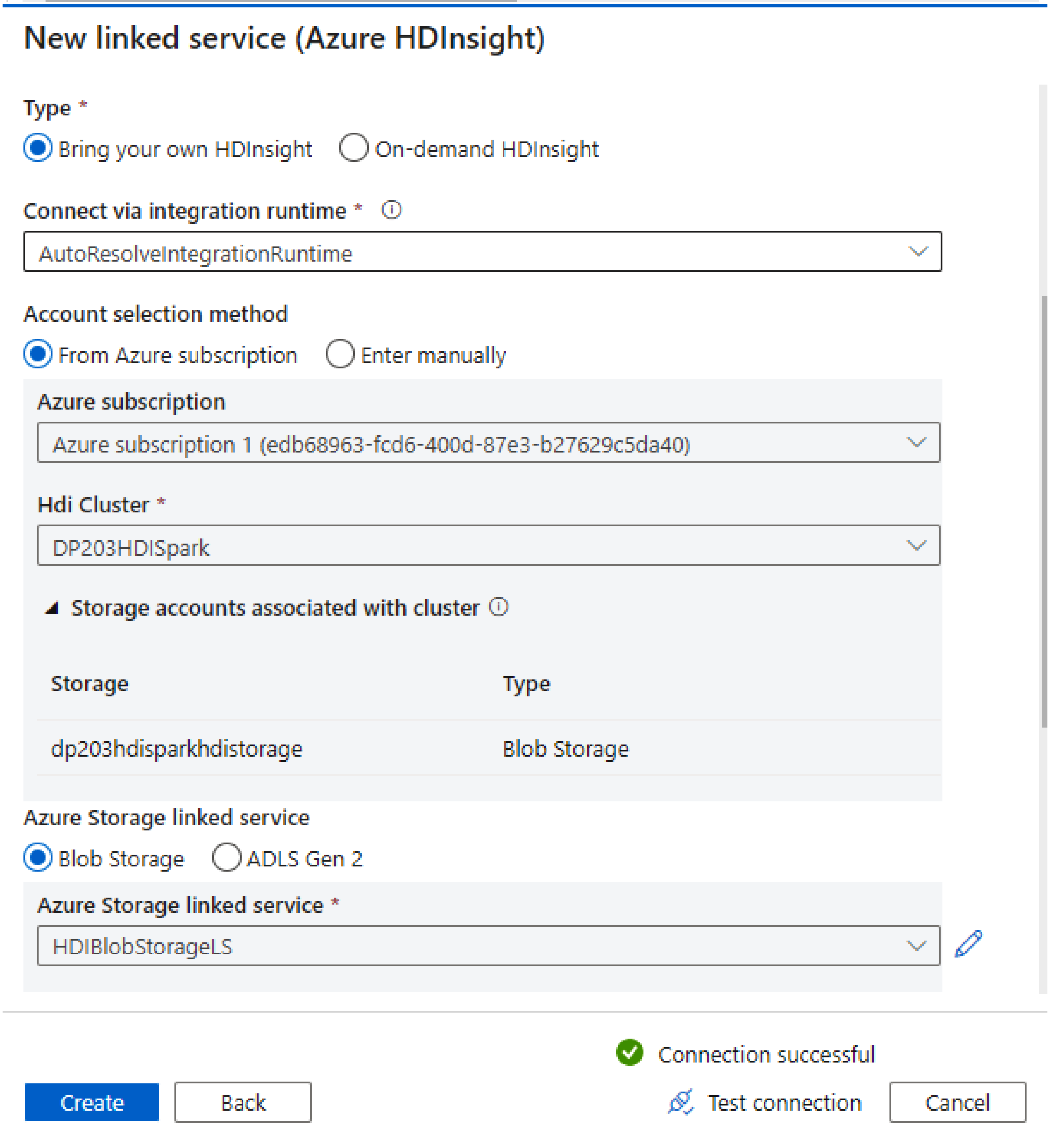

New trigger
Name *
trigger1
Description
Type *
Schedule
Start date *
09/25/2021 5:17 PM
Time zone *
Coordinated Universal Time (UTC)
Recurrence *
Every
15
Minute(s)
Specify an end date
Annotations
New
Start trigger
Start trigger on creation
OK
Cancel

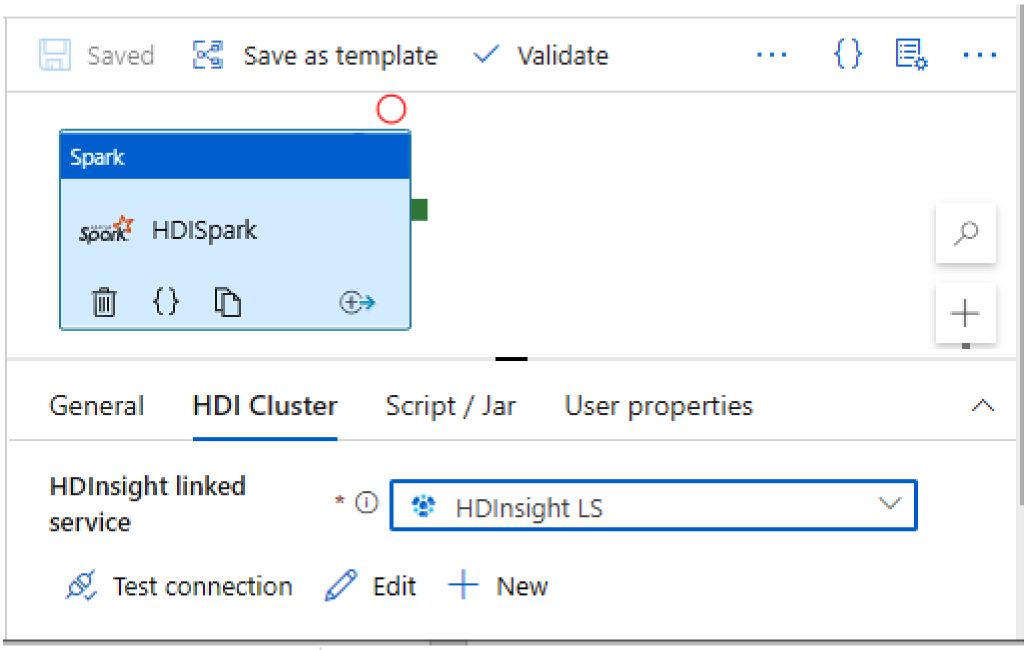

main branch
Validate all
Save all
Publish
Home
Author
Monitor
Manage
Connections
Linked services
Integration runtimes
Azure Purview (Preview)
Source control
Git configuration
ARM template
Author
Triggers
Global parameters
Security
Customer managed key
Credentials
Managed private endpoints
Linked services
Linked service defines the connection information to a data store or compute. Learn more
New
Filter by name
Showing 1 - 8 of 8 items
Name
Type
Related
AzureBatchLS
Azure Batch
1
AzureBlobConnection
Azure Blob Storage
4
AzureDataLakeStorage
Azure Data Lake St...
9
AzureSqlDatabaseLS
Azure SQL Database
2
AzureSynapseAnalytics
Azure Synapse An...
0
BatchBlobLS
Azure Blob Storage
2

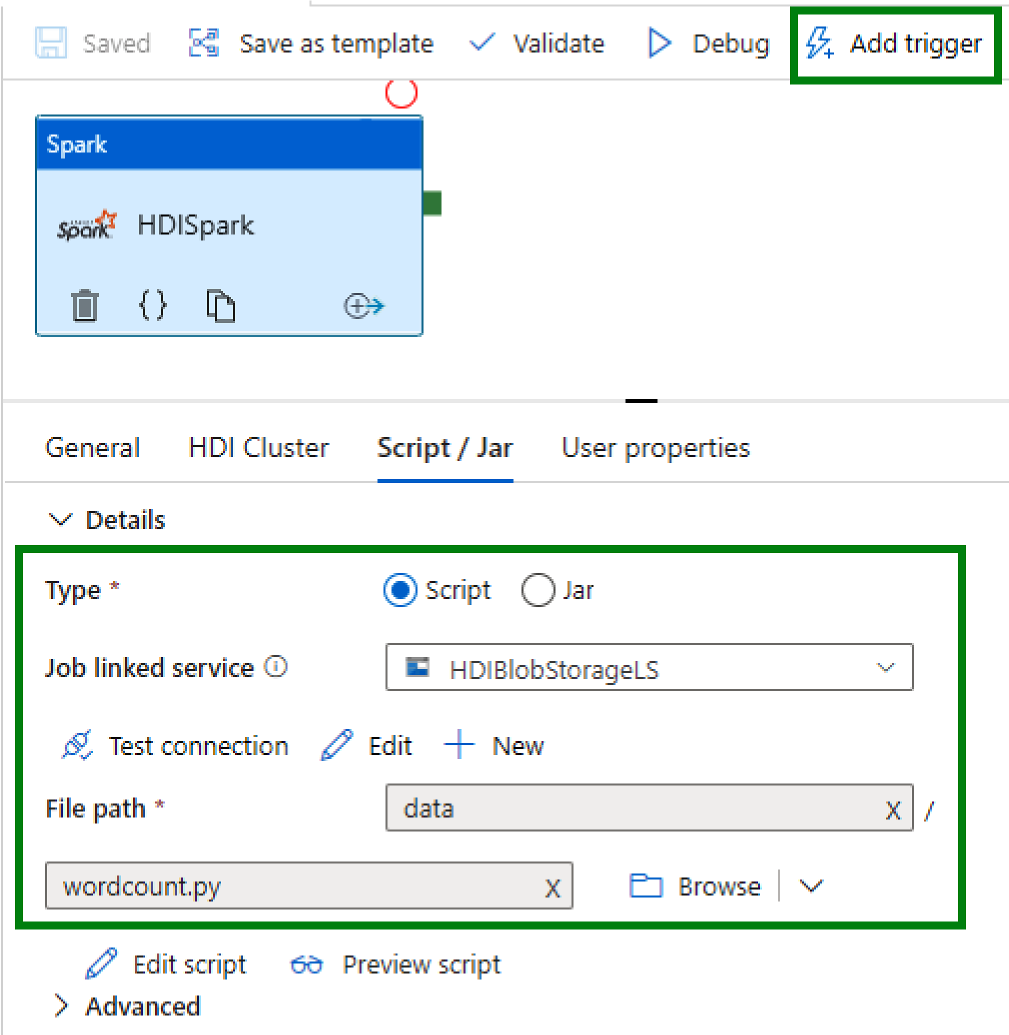

Connections
Linked services
Integration runtimes
Azure Purview
Source control
Git configuration
ARM template
Integration runtimes
The integration runtime (IR) is the co
New
Refresh
Filter by name
Showing 1 - 1 of 1 items
Name

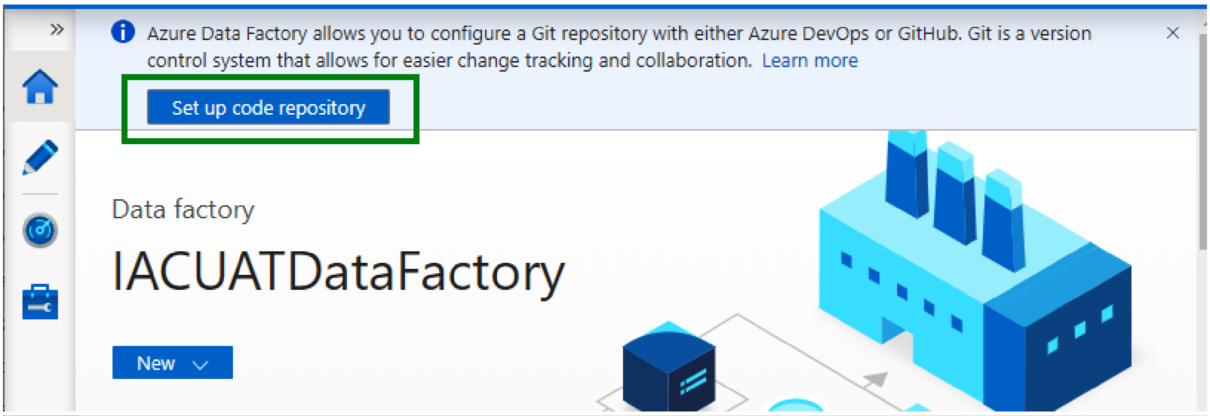

Dashboards
Runs
Pipeline runs
Trigger runs
Runtimes & sessions
Integration runtimes
Data flow debug
Notifications
Alerts & metrics
Pipeline runs
Triggered
Debug
Rerun
Cancel
Refresh
Edit columns
List
Gantt
Search by run ID or name
Chennai, Kolkata, Mu... : Last 24 hours
Copy filters
Add filter
Showing 1 - 2 items
Pipeline name
Run start
Run end
Status
Error
HDISparkPipeline
9/26/21, 6:55:59 PM
9/26/21, 6:57:09 PM
Succeeded
HDISparkPipeline
9/26/21, 6:40:04 PM
9/26/21, 6:43:13 PM
Failed

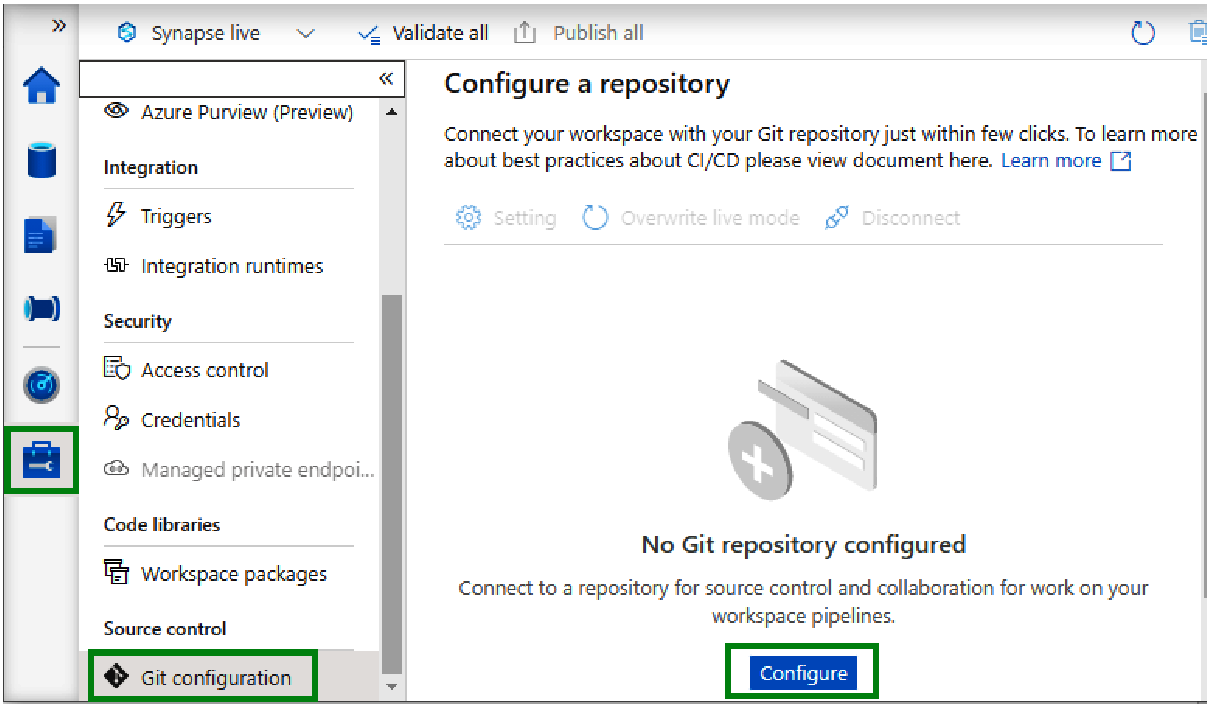

Activity runs
Pipeline run ID a2922150-f081-4a1b-8f43-b44e5506d93c
All status
Showing 1 - 3 of 3 items
Activity name
Activity type
Run start
Duration
Status
Error
Transform
Notebook
9/26/21, 4:05:41 PM
00:08:36
Failed
FetchTripsFr...
Data flow
9/26/21, 3:59:59 PM
00:05:41
Succeeded
Data flow details
FetchFaresFrmSQL
9/26/21, 3:59:59 PM
00:04:55
Succeeded

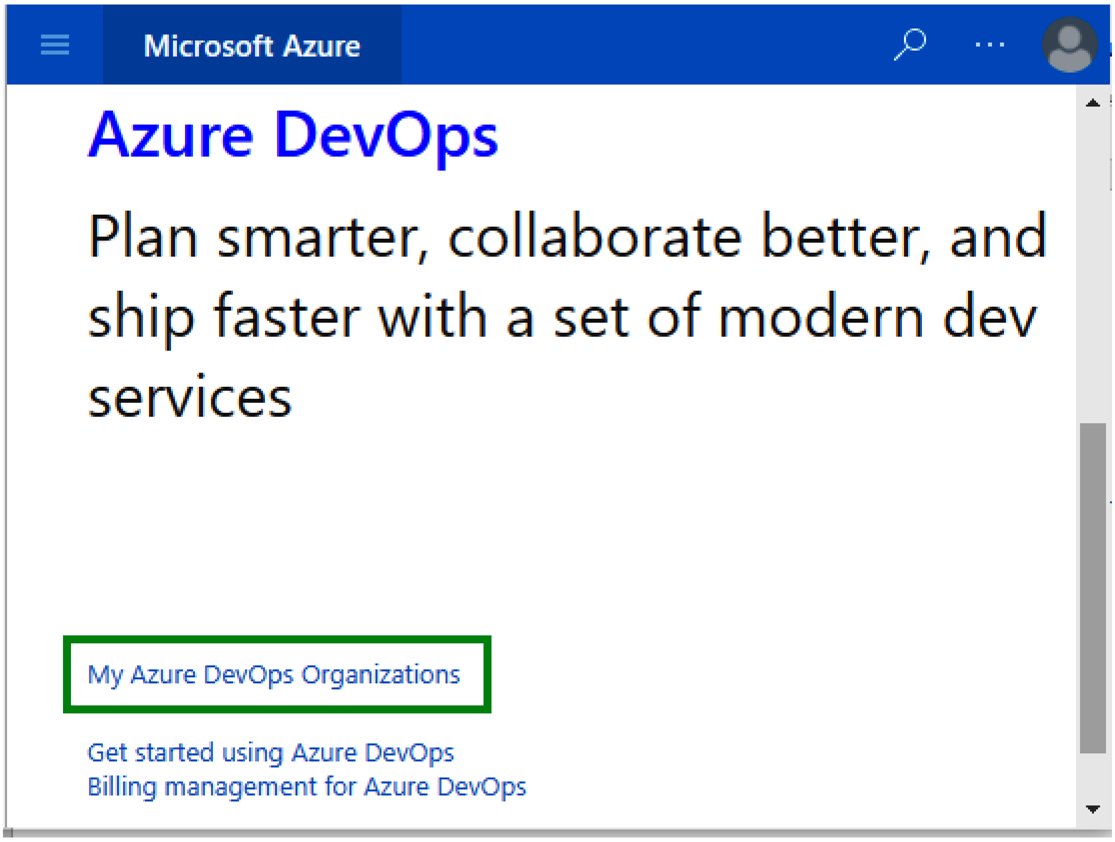

New linked service (Azure HDInsight)
Type *
Bring your own HDInsight
On-demand HDInsight
Connect via integration runtime *
AutoResolveIntegrationRuntime
Account selection method
From Azure subscription
Enter manually
Azure subscription
Azure subscription 1 (edb68963-fcd6-400d-87e3-b27629c5da40)
Hdi Cluster *
DP203HDISpark
Storage accounts associated with cluster
Storage
Type
dp203hdisparkhdistorage
Blob Storage
Azure Storage linked service
Blob Storage
ADLS Gen 2
Azure Storage linked service *
HDIBlobStorageLS
Connection successful
Create
Back
Test connection
Cancel

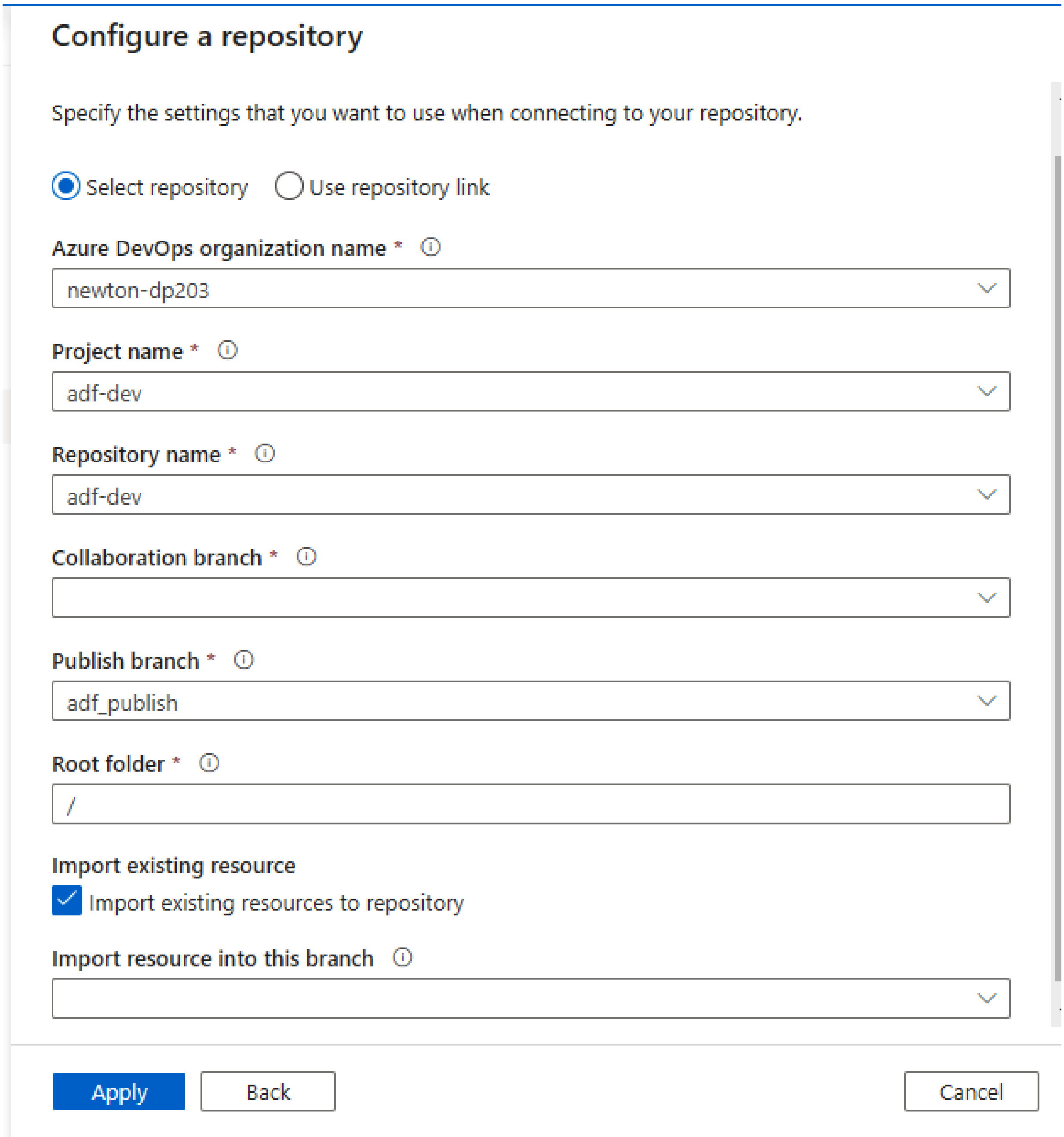

Saved
Save as template
Validate
Spark
HDISpark
General
HDI Cluster
Script / Jar
User properties
HDInsight linked service
HDInsight LS
Test connection
Edit
New

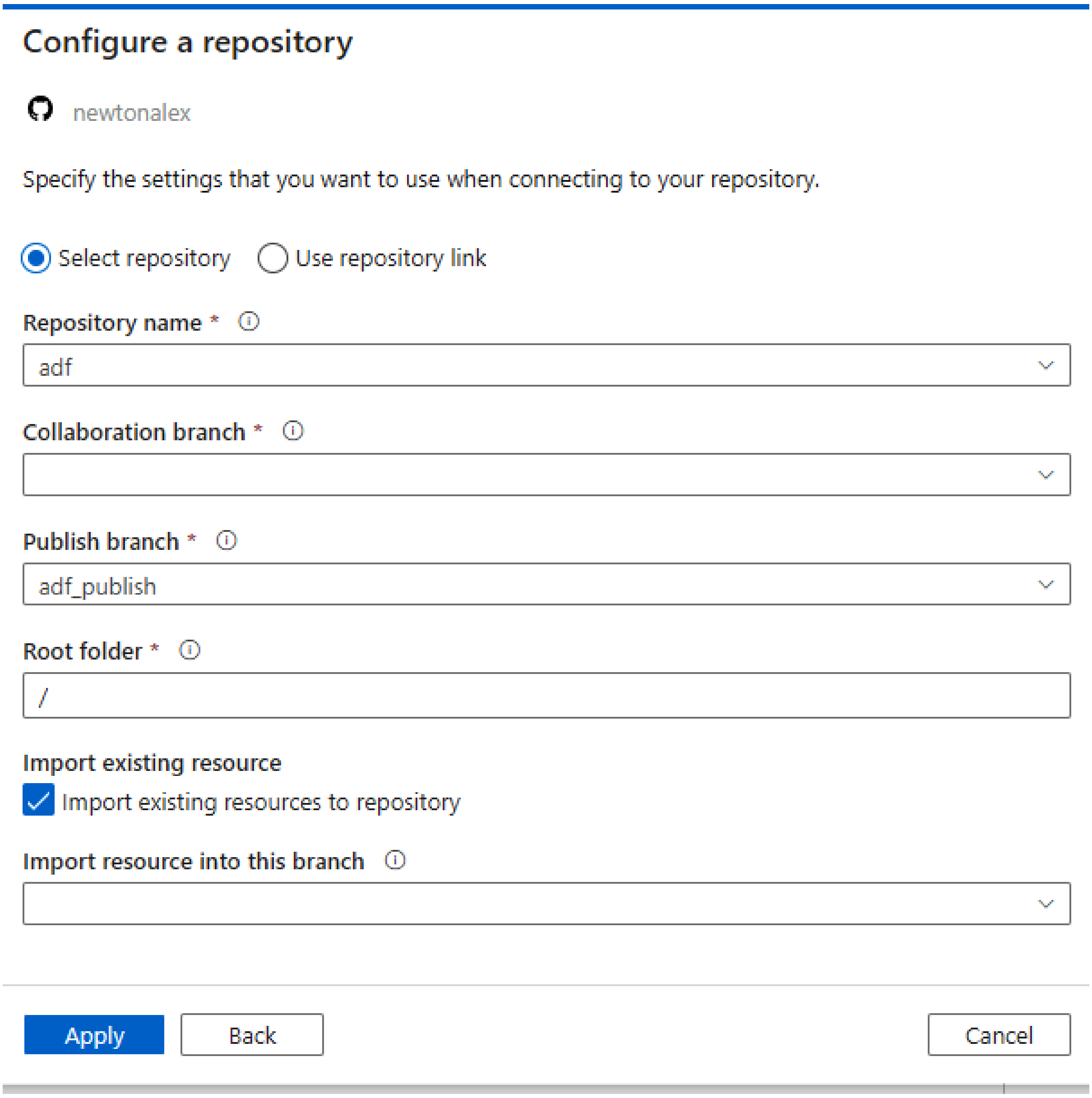

Saved
Save as template
Validate
Debug
Add trigger
Spark
HDISpark
General
HDI Cluster
Script / Jar
User properties
Details
Type *
Script
Jar
Job linked service
HDIBlobStorageLS
Test connection
Edit
New
File path *
data
wordcount.py
Browse
Edit script
Preview script
Advanced


Azure Data Factory allows you to configure a Git repository with either Azure DevOps or GitHub. Git is a version control system that allows for easier change tracking and collaboration. Learn more
Set up code repository
Data factory
IACUATDataFactory
New

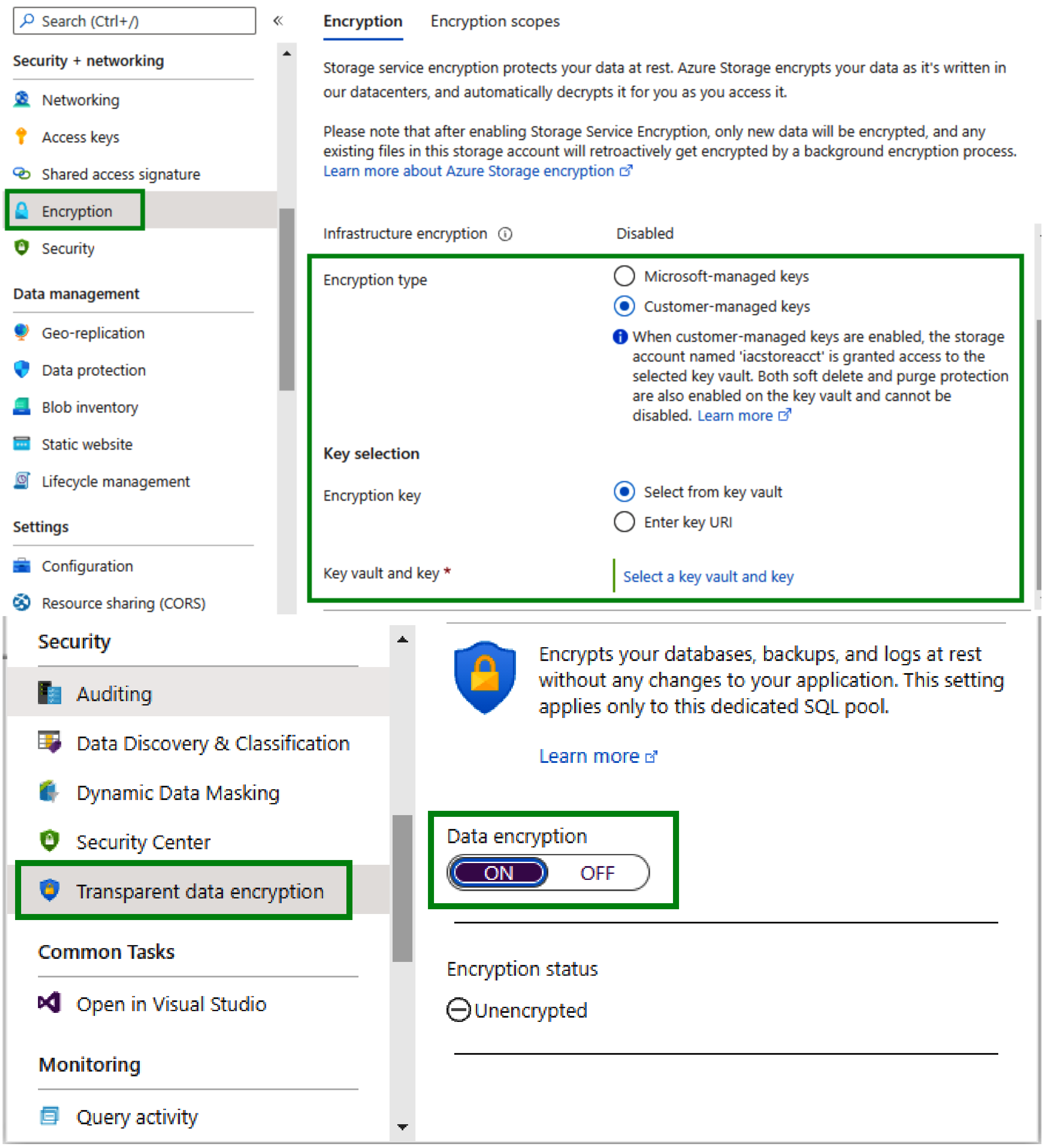

Synapse live
Validate all
Publish all
Azure Purview (Preview)
Integration
Triggers
Integration runtimes
Security
Access control
Credentials
Managed private endpoi...
Code libraries
Workspace packages
Source control
Git configuration
Configure a repository
Connect your workspace with your Git repository just within few clicks. To learn more about best practices about CI/CD please view document here. Learn more
Setting
Overwrite live mode
Disconnect
No Git repository configured
Connect to a repository for source control and collaboration for work on your workspace pipelines.
Configure

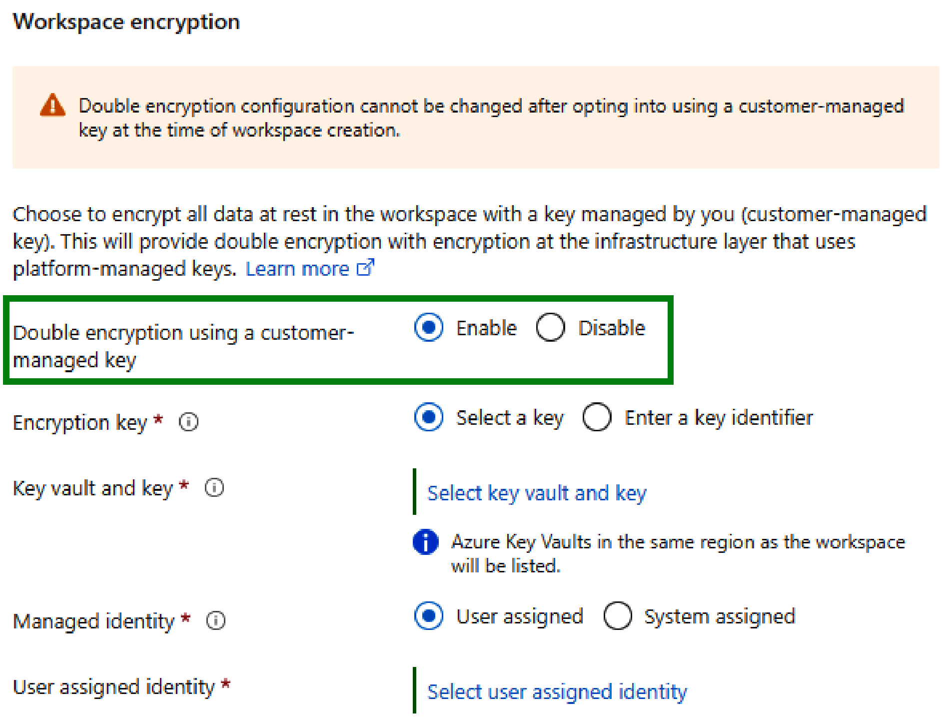

Microsoft Azure
Azure DevOps
Plan smarter, collaborate better, and ship faster with a set of modern dev services
My Azure DevOps Organizations
Get started using Azure DevOps
Billing management for Azure DevOps

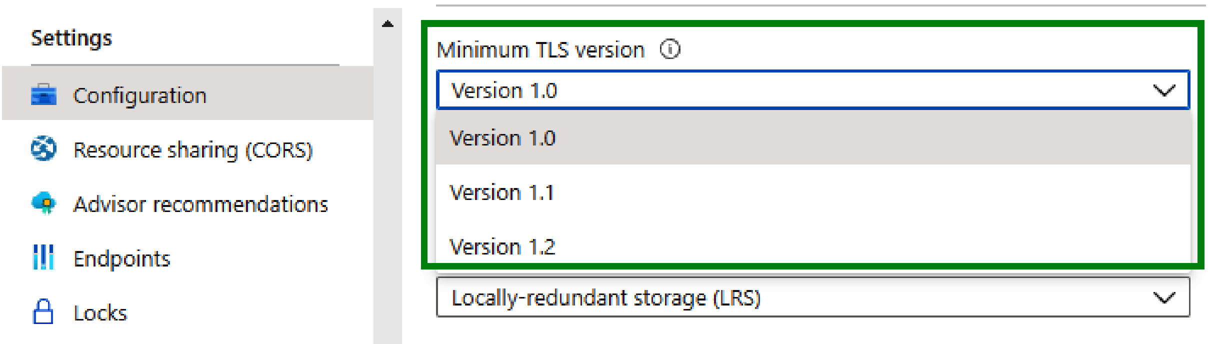

Configure a repository
Specify the settings that you want to use when connecting to your repository.
Select repository
Use repository link
Azure DevOps organization name *
newton-dp203
Project name *
adf-dev
Repository name *
adf-dev
Collaboration branch *
Publish branch *
adf_publish
Root folder *
/
Import existing resource
Import existing resources to repository
Import resource into this branch
Apply
Back
Cancel

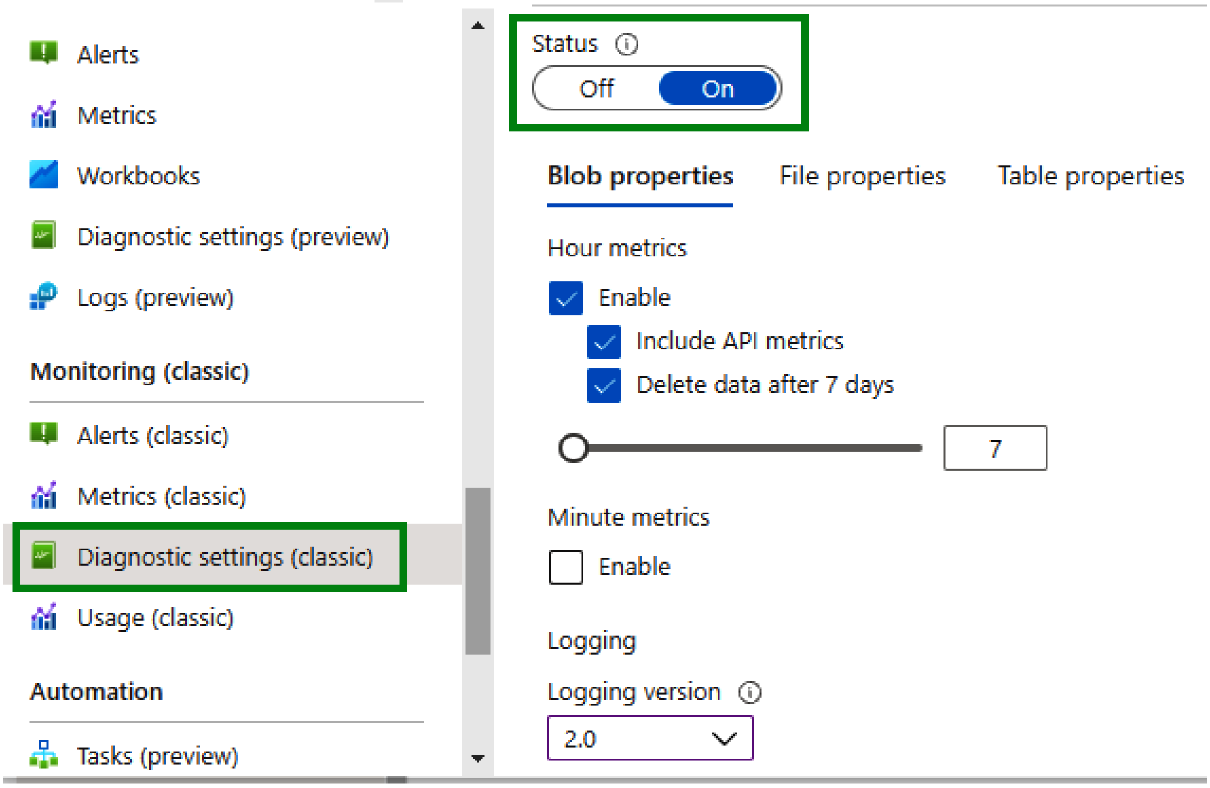

Configure a repository
newtonalex
Specify the settings that you want to use when connecting to your repository.
Select repository
Use repository link
Repository name *
adf
Collaboration branch *
Publish branch *
adf_publish
Root folder *
/
Import existing resource
Import existing resources to repository
Import resource into this branch
Apply
Back
Cancel

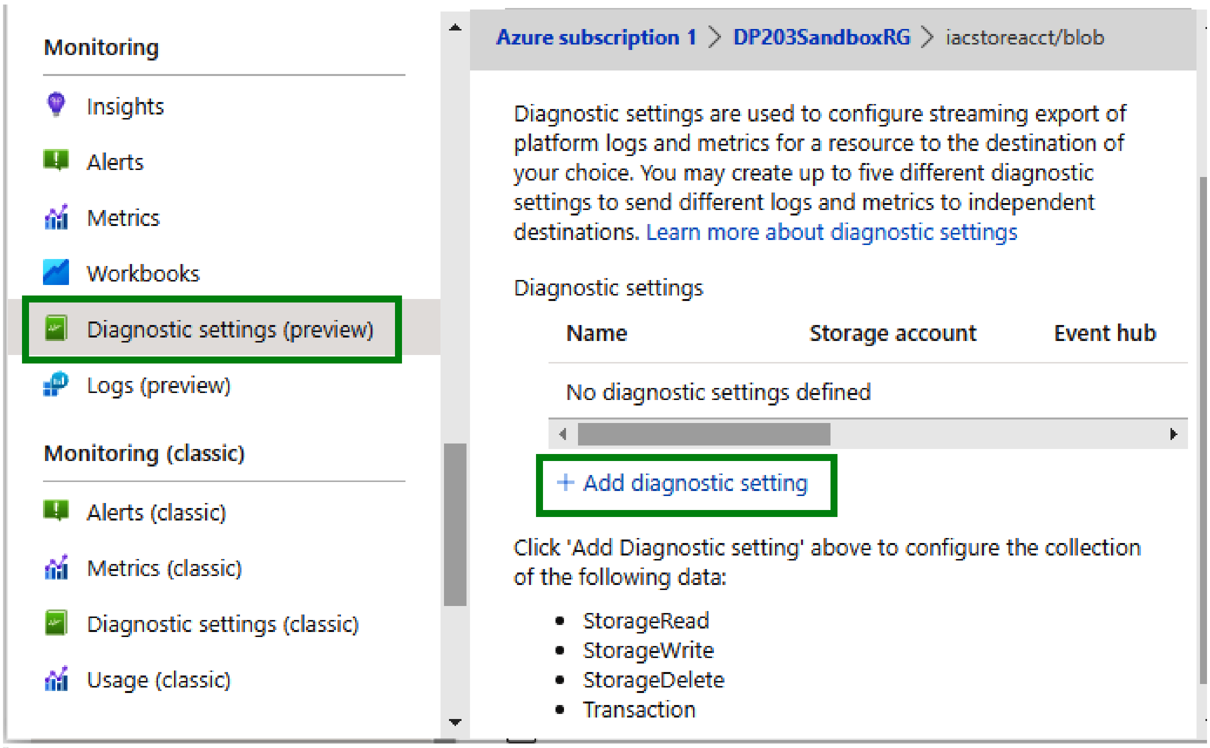
Chapter 12: Designing Security for Data Policies and Standards

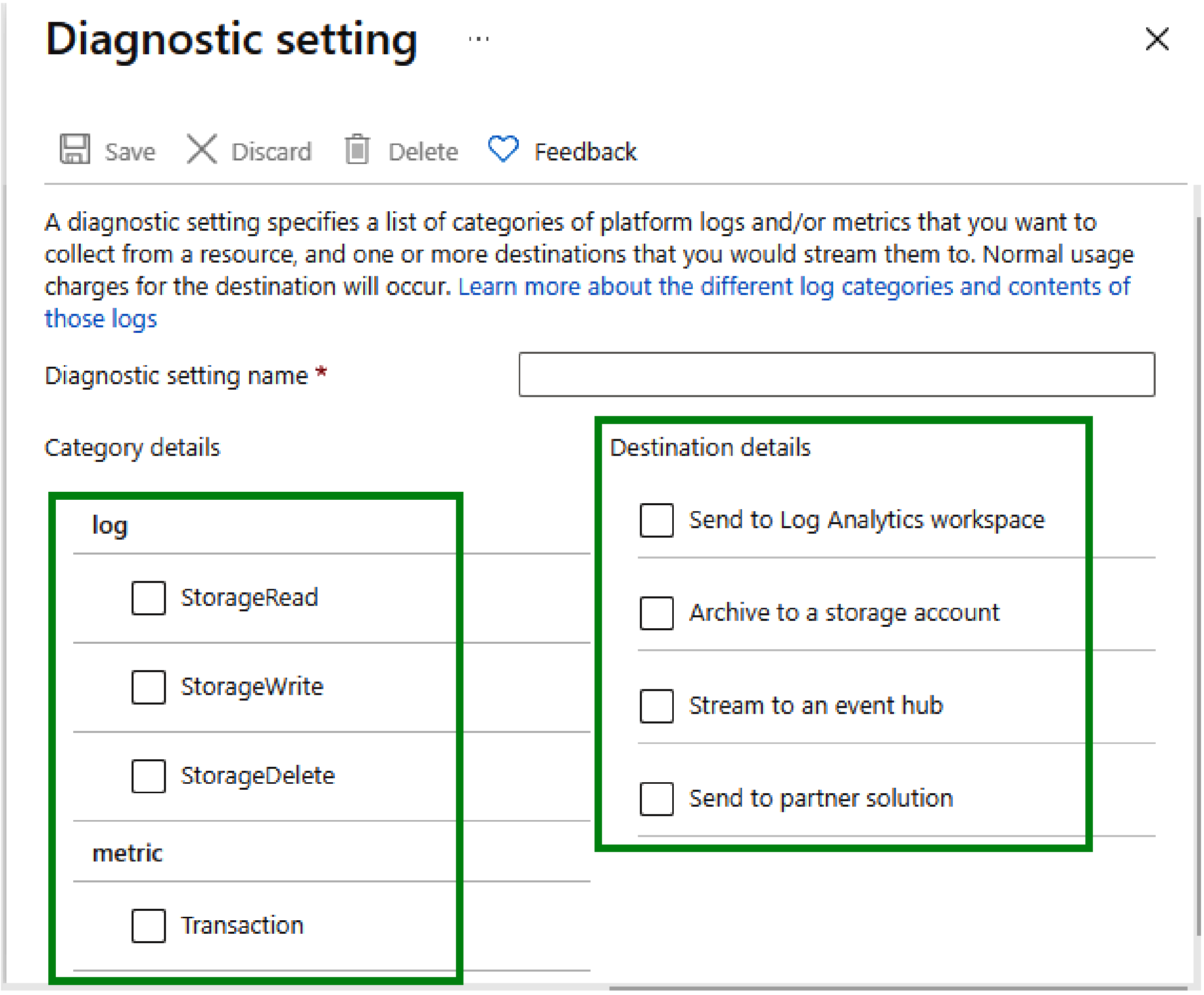


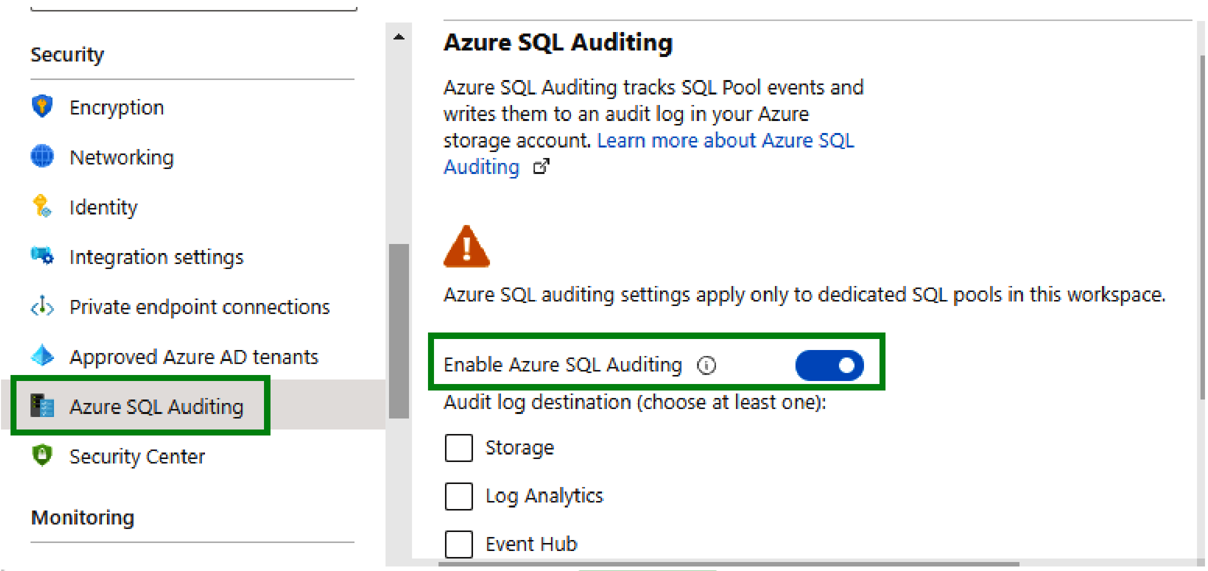

Workspace encryption
Double encryption configuration cannot be changed after opting into using a customer-managed key at the time of workspace creation.
Choose to encrypt all data at rest in the workspace with a key managed by you (customer-managed key). This will provide double encryption with encryption at the infrastructure layer that uses platform-managed keys. Learn more
Double encryption using a customer-managed key
Enable
Disable
Encryption key *
Select a key
Enter a key identifier
Key vault and key *
Select key vault and key
Azure Key Vaults in the same region as the workspace will be listed.
Managed identity *
User assigned
System assigned
User assigned identity *
Select user assigned identity

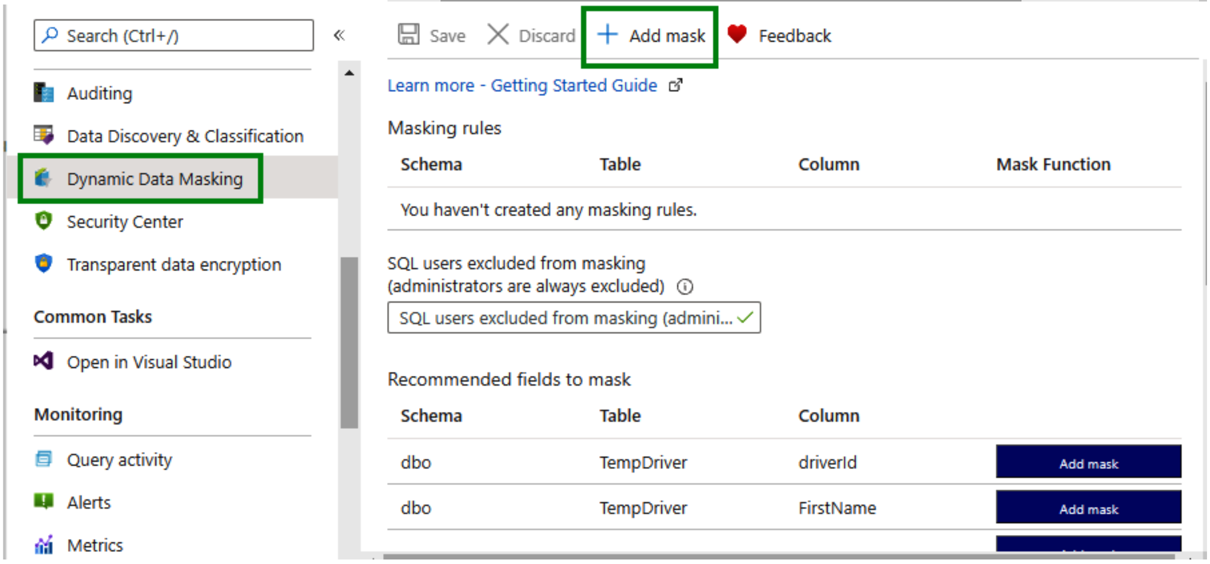

Settings
Configuration
Resource sharing (CORS)
Advisor recommendations
Endpoints
Locks
Minimum TLS version
Version 1.0
Version 1.0
Version 1.1
Version 1.2
Locally-redundant storage (LRS)

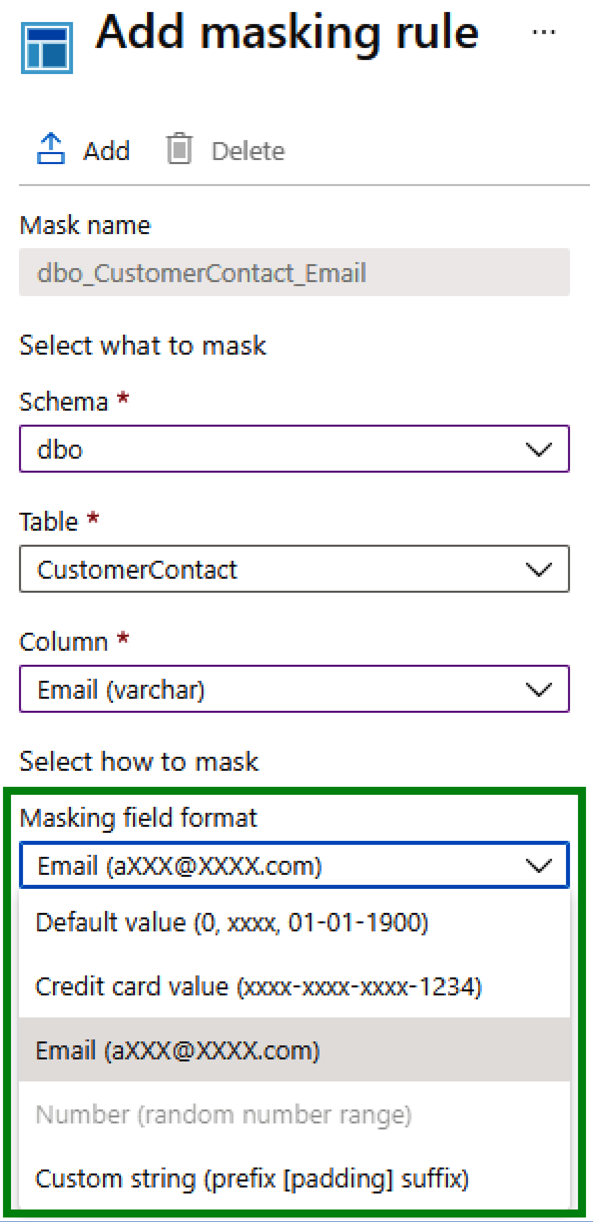

Alerts
Metrics
Workbooks
Diagnostic settings (preview)
Logs (preview)
Monitoring (classic)
Alerts (classic)
Metrics (classic)
Diagnostic settings (classic)
Usage (classic)
Automation
Tasks (preview)
Status
Off
On
Blob properties
File properties
Table properties
Hour metrics
Enable
Include API metrics
Delete data after 7 days
7
Minute metrics
Enable
Logging
Logging version
2.0

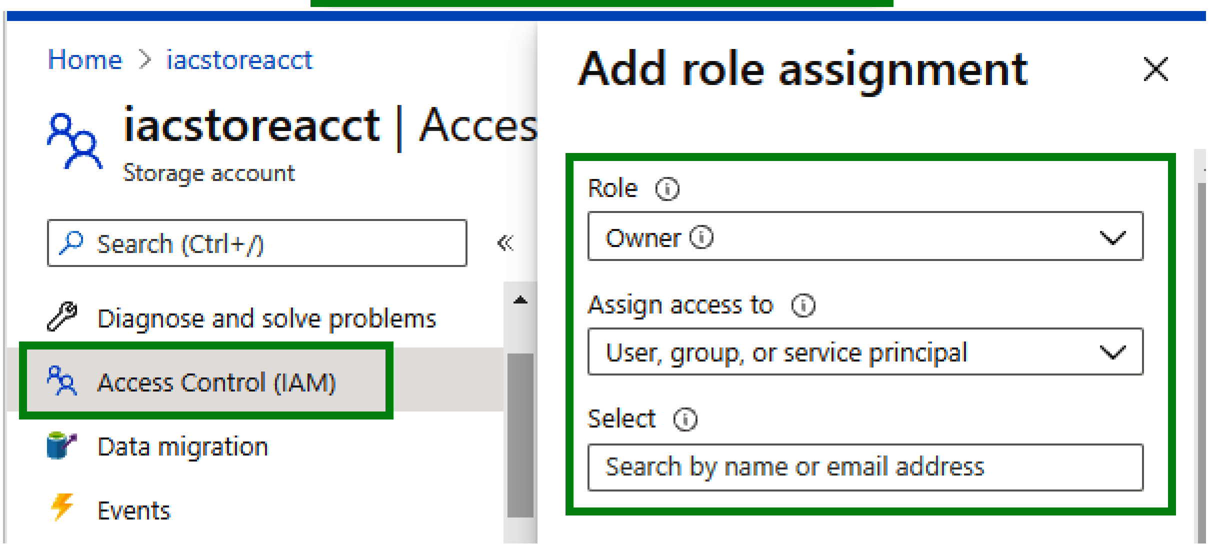

Monitoring
Insights
Alerts
Metrics
Workbooks
Diagnostic settings (preview)
Logs (preview)
Monitoring (classic)
Alerts (classic)
Metrics (classic)
Diagnostic settings (classic)
Usage (classic)
Azure subscription 1 > DP203SandboxRG > iacstoreacct/blob
Diagnostic settings are used to configure streaming export of platform logs and metrics for a resource to the destination of your choice. You may create up to five different diagnostic settings to send different logs and metrics to independent destinations. Learn more about diagnostic settings
Diagnostic settings
Name
Storage account
Event hub
No diagnostic settings defined
+ Add diagnostic setting
Click 'Add Diagnostic setting' above to configure the collection of the following data:
StorageRead
StorageWrite
StorageDelete
Transaction

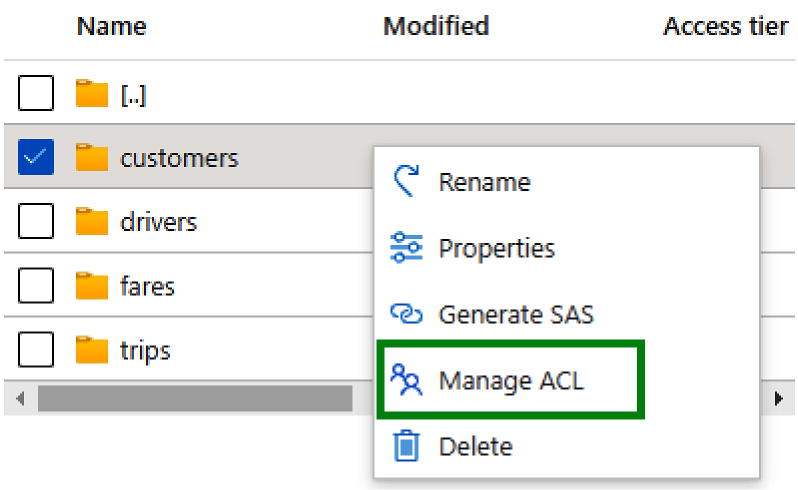

Diagnostic setting
Save
Discard
Delete
Feedback
A diagnostic setting specifies a list of categories of platform logs and/or metrics that you want to collect from a resource, and one or more destinations that you would stream them to. Normal usage charges for the destination will occur. Learn more about the different log categories and contents of those logs
Diagnostic setting name *
Category details
log
StorageRead
StorageWrite
StorageDelete
metric
Transaction
Destination details
Send to Log Analytics workspace
Archive to a storage account
Stream to an event hub
Send to partner solution

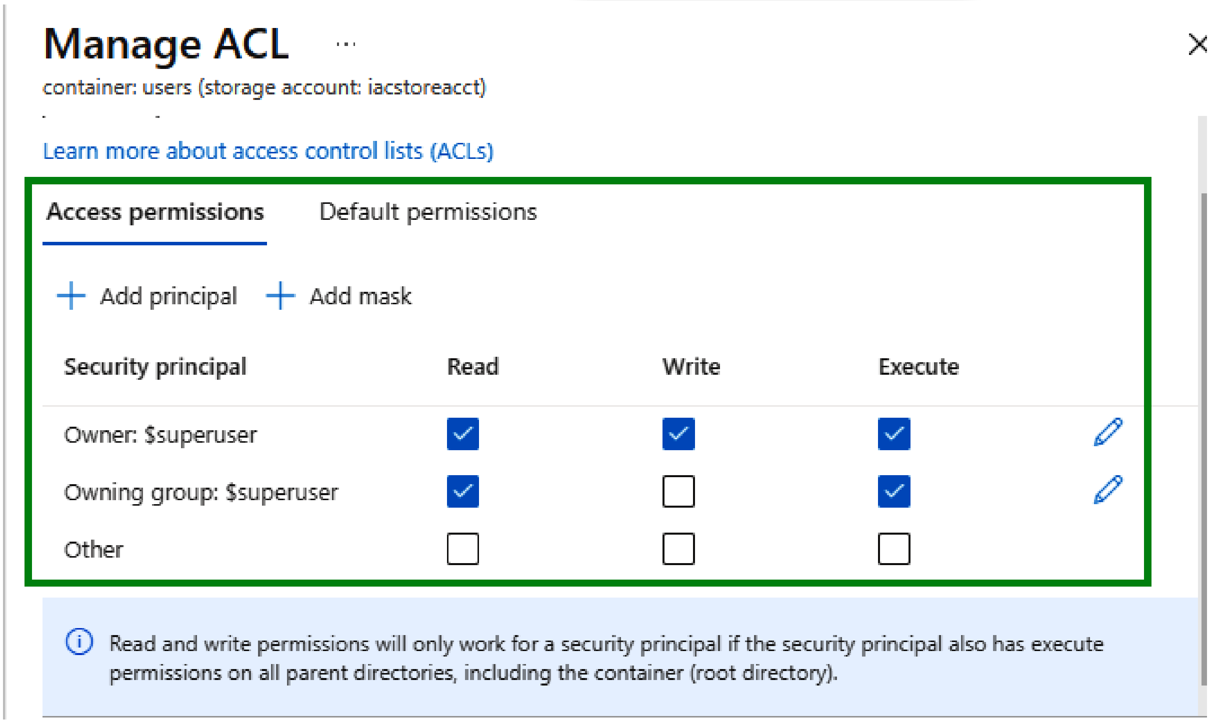

Security
Encryption
Networking
Identity
Integration settings
Private endpoint connections
Approved Azure AD tenants
Azure SQL Auditing
Security Center
Monitoring
Azure SQL Auditing
Azure SQL Auditing tracks SQL Pool events and writes them to an audit log in your Azure storage account. Learn more about Azure SQL Auditing
Azure SQL auditing settings apply only to dedicated SQL pools in this workspace.
Enable Azure SQL Auditing
Audit log destination (choose at least one):
Storage
Log Analytics
Event Hub

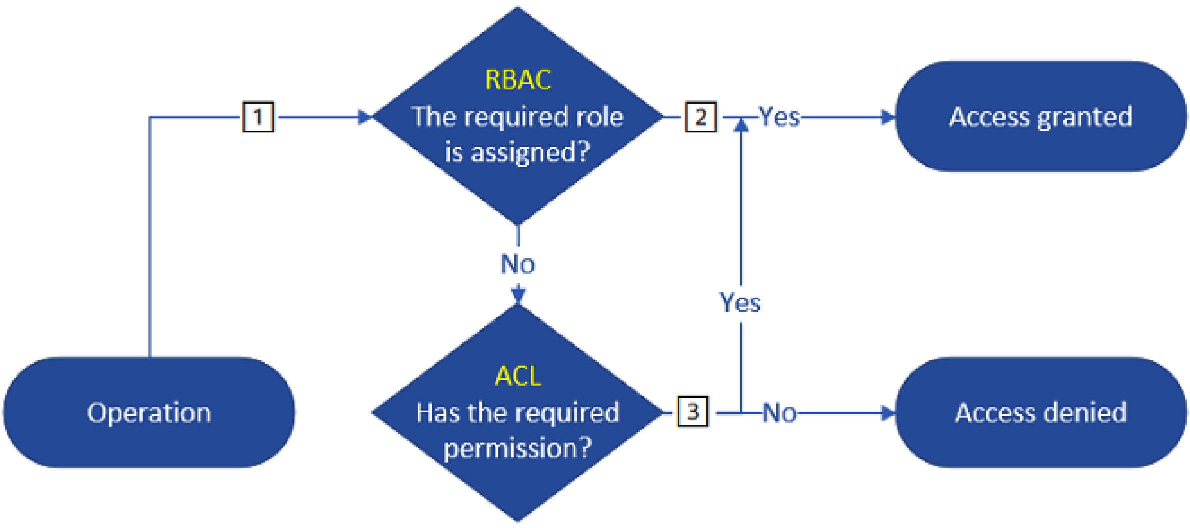

Search (Ctrl+/)
Auditing
Data Discovery & Classification
Dynamic Data Masking
Security Center
Transparent data encryption
Common Tasks
Open in Visual Studio
Monitoring
Query activity
Alerts
Metrics
Save
Discard
Add mask
Feedback
Learn more - Getting Started Guide
Masking rules
Schema
Table
Column
Mask Function
You haven't created any masking rules.
SQL users excluded from masking (administrators are always excluded)
SQL users excluded from masking (admini...
Recommended fields to mask
Schema
Table
Column
dbo
TempDriver
driverId
Add mask
dbo
TempDriver
FirstName
Add mask


Add masking rule
Add
Delete
Mask name
dbo_CustomerContact_Email
Select what to mask
Schema *
dbo
Table *
CustomerContact
Column *
Email (varchar)
Select how to mask
Masking field format
Email (aXXX@XXXX.com)
Default value (0, xxxx, 01-01-1900)
Credit card value (xxxx-xxxx-xxxx-1234)
Email (aXXX@XXXX.com)
Number (random number range)
Custom string (prefix [padding] suffix)


Home
iacstoreacct
iacstoreacct | Acces
Storage account
Search (Ctrl+/)
Diagnose and solve problems
Access Control (IAM)
Data migration
Events
Add role assignment
Role
Owner
Assign access to
User, group, or service principal
Select
Search by name or email address


Name
Modified
Access tier
[..]
customers
drivers
fares
trips
Rename
Properties
Generate SAS
Manage ACL
Delete


Manage ACL
container: users (storage account: iacstoreacct)
Learn more about access control lists (ACLs)
Access permissions
Default permissions
Add principal
Add mask
Security principal
Read
Write
Execute
Owner: $superuser
Owning group: $superuser
Other
Read and write permissions will only work for a security principal if the security principal also has execute permissions on all parent directories, including the container (root directory).


Operation
1
RBAC
The required role is assigned?
2
Yes
Access granted
No
ACL
Has the required permission?
3
Yes
No
Access denied

Results Messages

View Table Chart Export results

Search

tripId	driverId	customerID	tripDate	startLocation	endLocation
106	206	306	20220301	Atlanta	Chicago
102	202	302	20220101	Miami	Dallas
900	299	399	20220301	Pre-Launch	Pre-Launch
103	203	303	20220102	Phoenix	Tempe

00:00:02 Query executed successfully.

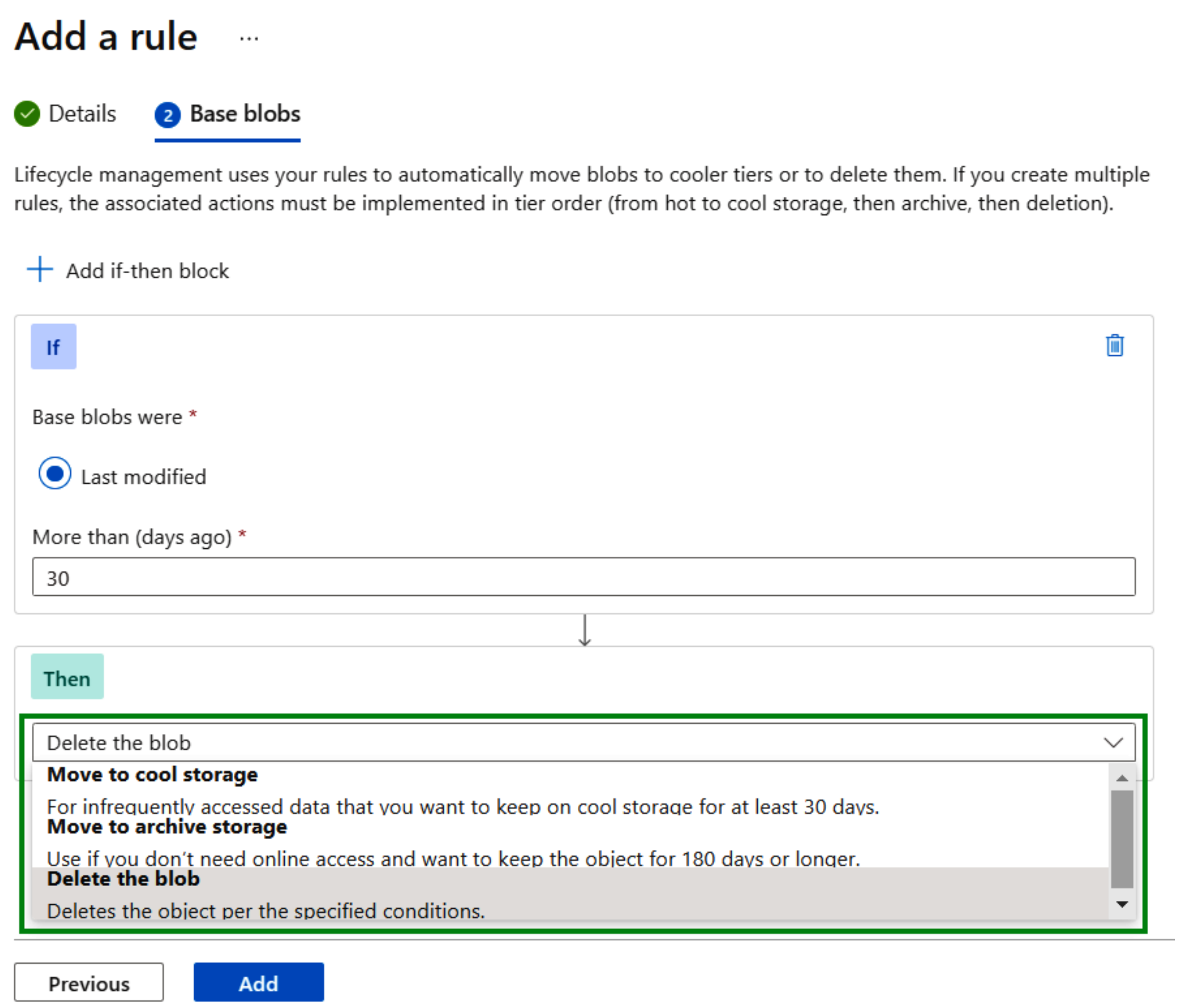
Results Messages

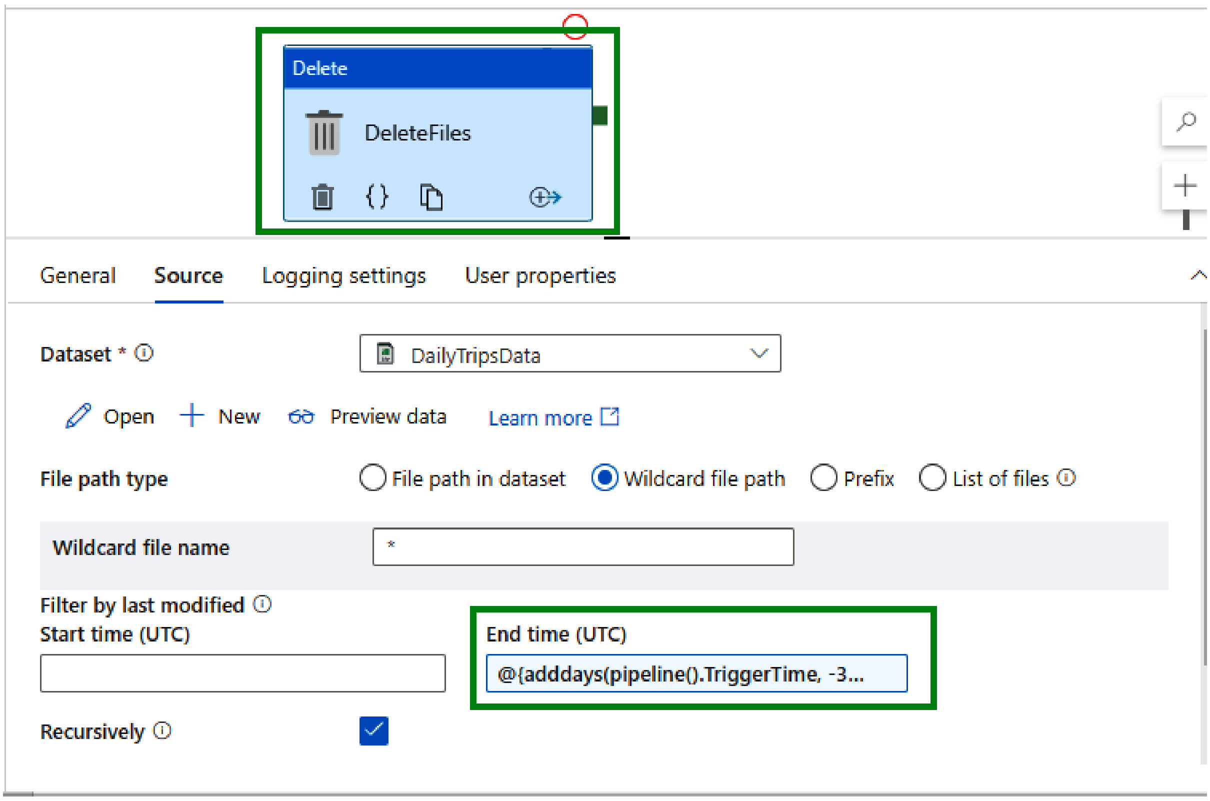
View Table Chart Export results

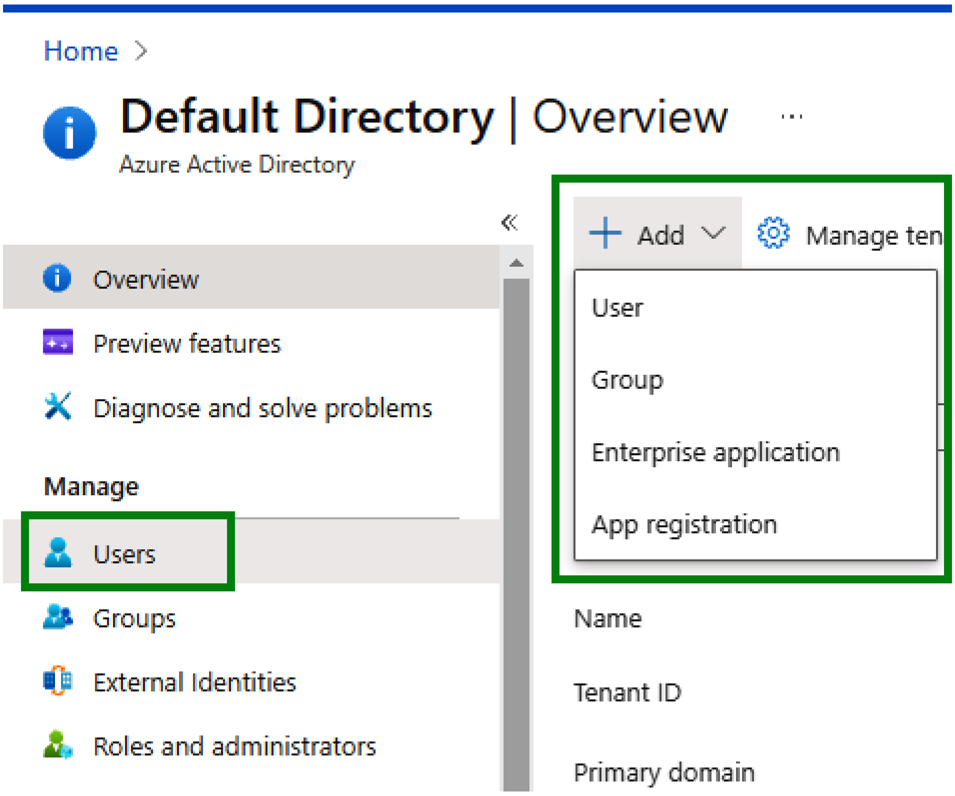
Search

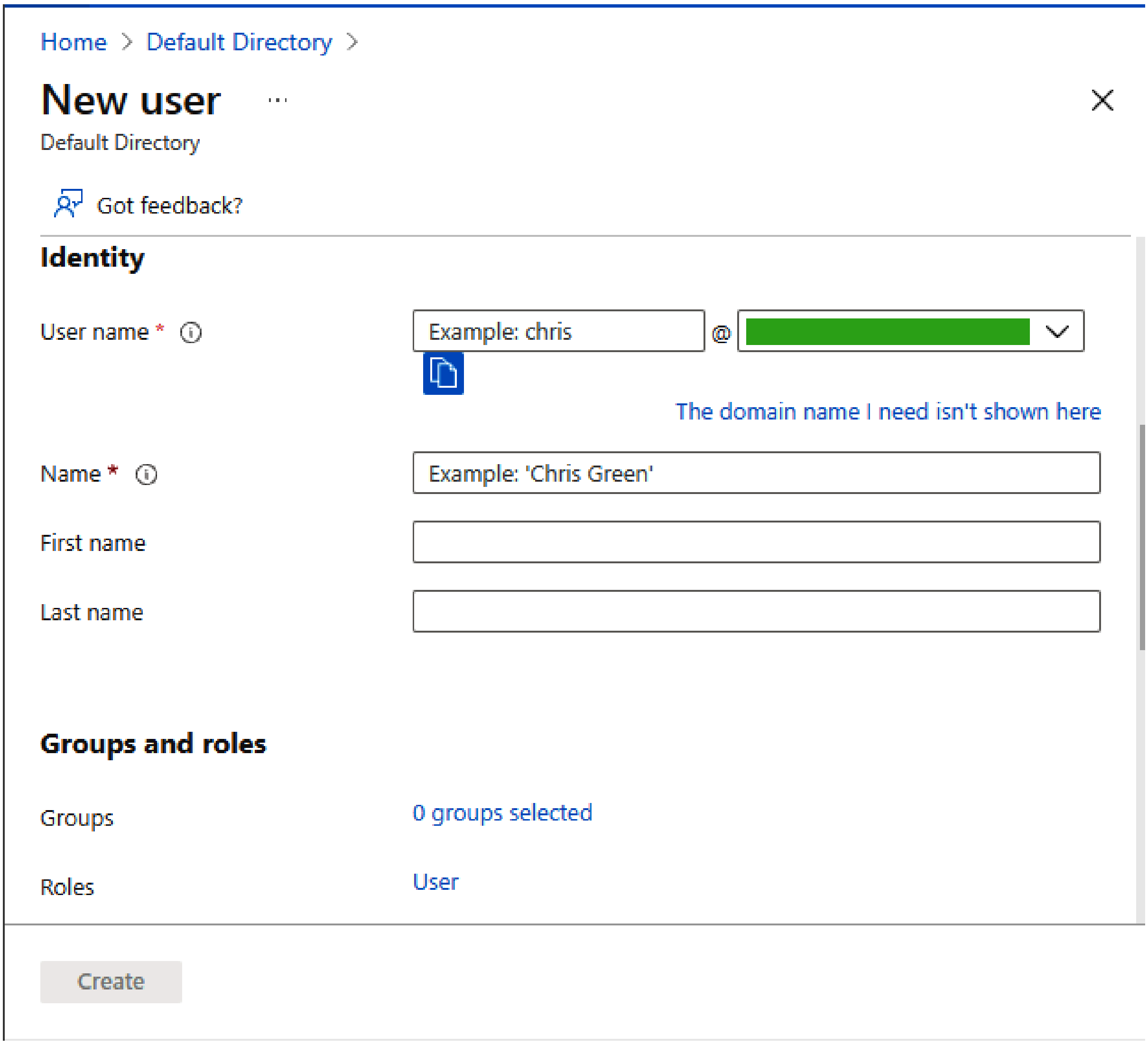
tripId	driverId	customerID	tripDate	startLocation	endLocation
106	206	306	20220301	Atlanta	Chicago
102	202	302	20220101	Miami	Dallas
103	203	303	20220102	Phoenix	Tempe
101	201	301	20220101	New York	New Jersey

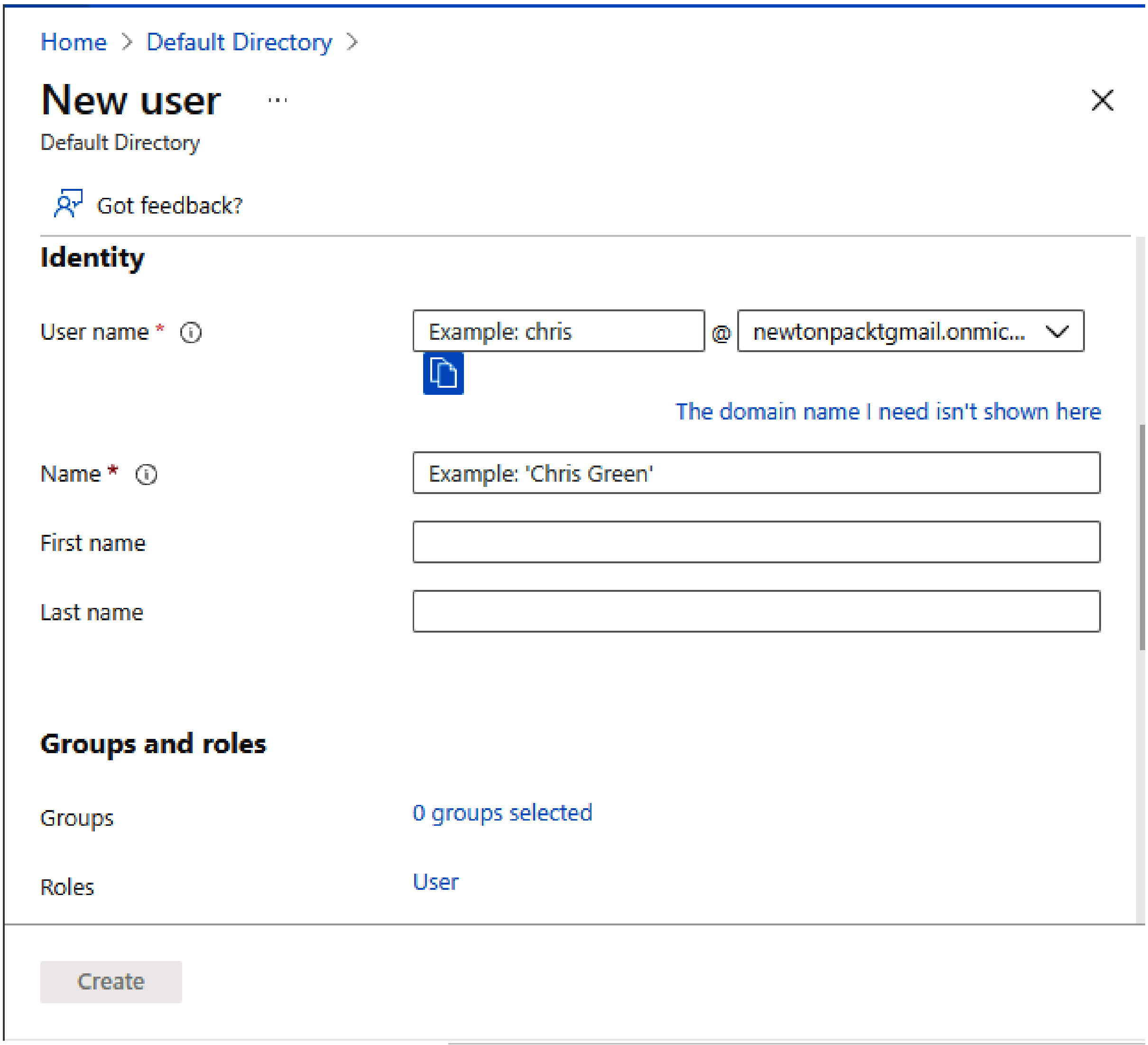
00:00:02 Query executed successfully.

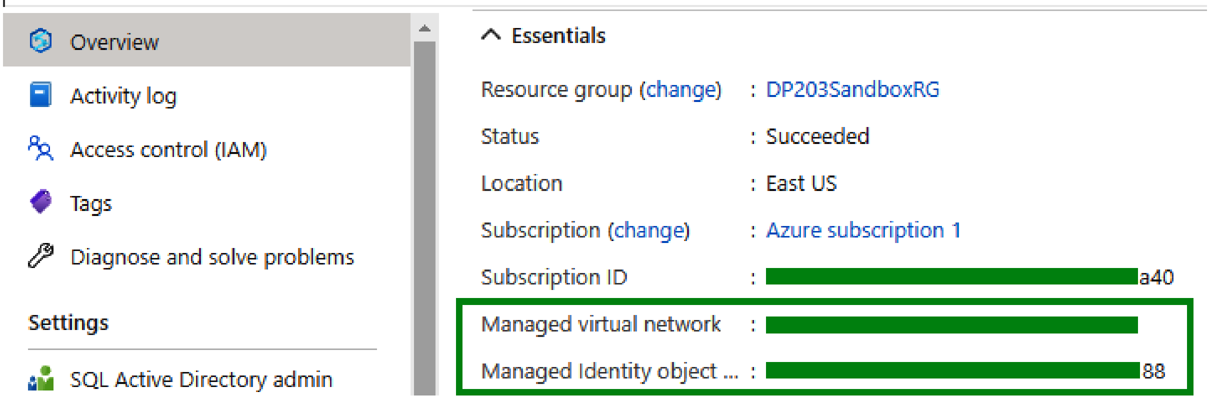

Add a rule
Details
2 Base blobs
Lifecycle management uses your rules to automatically move blobs to cooler tiers or to delete them. If you create multiple rules, the associated actions must be implemented in tier order (from hot to cool storage, then archive, then deletion).
Add if-then block
If
Base blobs were *
Last modified
More than (days ago) *
30
Then
Delete the blob
Move to cool storage
For infrequently accessed data that you want to keep on cool storage for at least 30 days.
Move to archive storage
Use if you don't need online access and want to keep the object for 180 days or longer.
Delete the blob
Deletes the object per the specified conditions.
Previous
Add

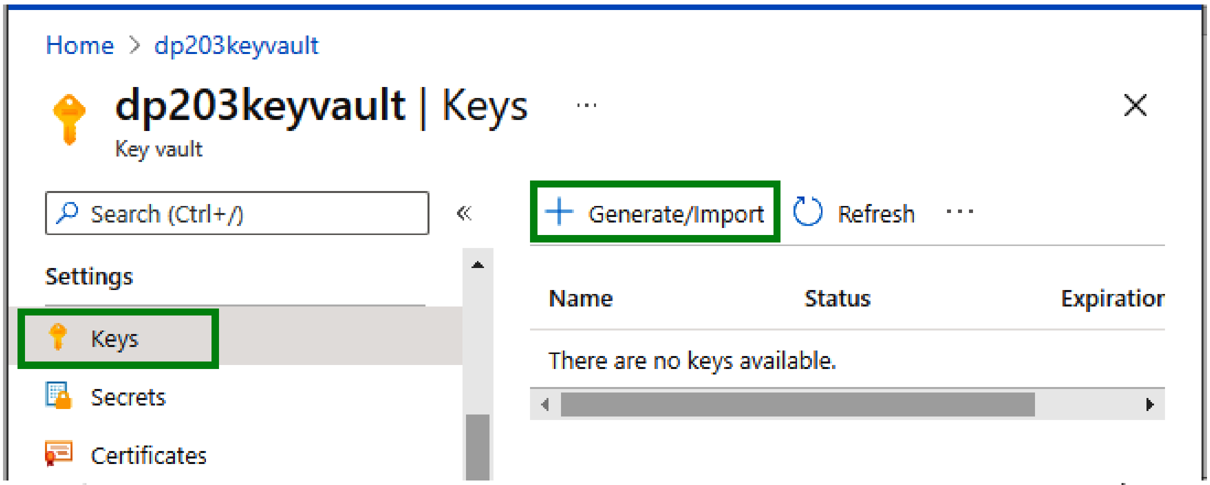

Delete
DeleteFiles
General
Source
Logging settings
User properties
Dataset *
DailyTripsData
Open
New
Preview data
Learn more
File path type
File path in dataset
Wildcard file path
Prefix
List of files
Wildcard file name
*
Filter by last modified
Start time (UTC)
End time (UTC)
@{adddays(pipeline().TriggerTime, -3...
Recursively

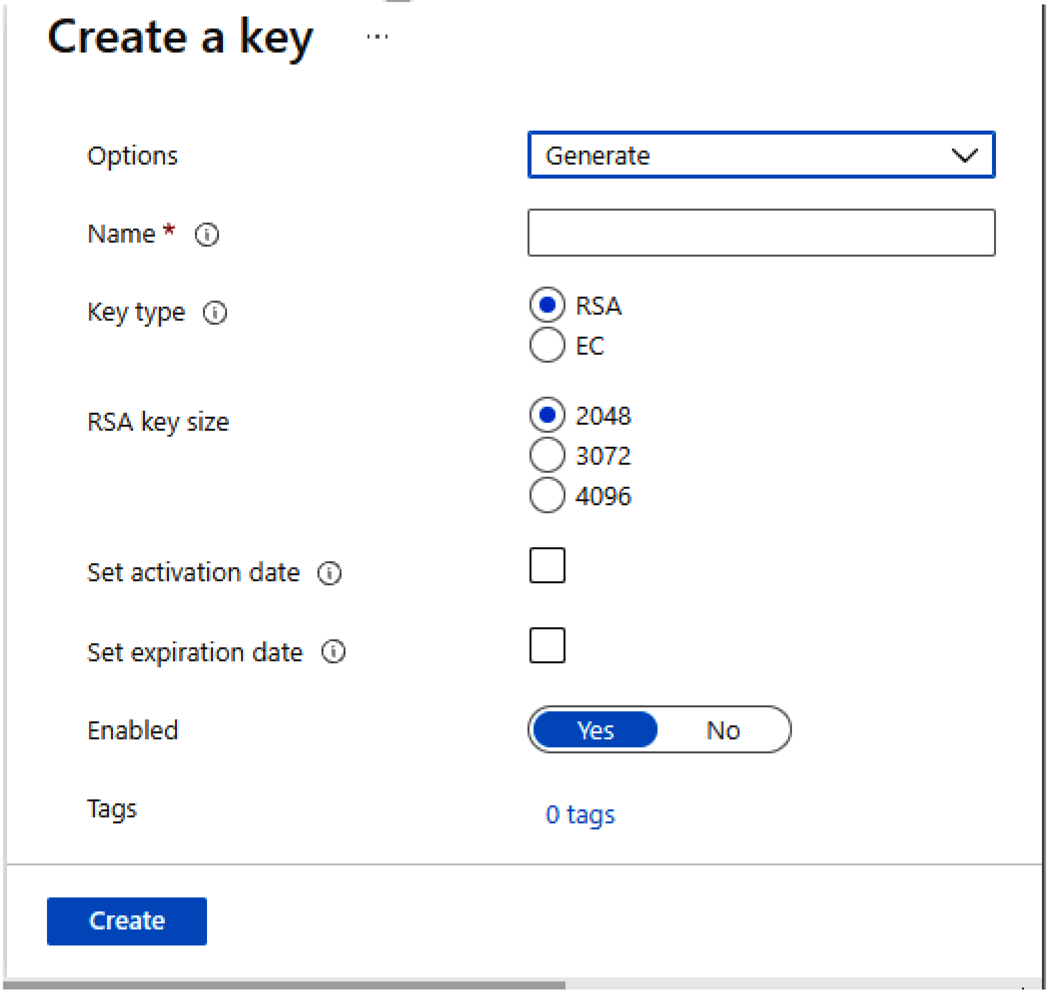

Home
Default Directory | Overview
Azure Active Directory
Overview
Preview features
Diagnose and solve problems
Manage
Users
Groups
External Identities
Roles and administrators
Add
Manage ten
User
Group
Enterprise application
App registration
Name
Tenant ID
Primary domain

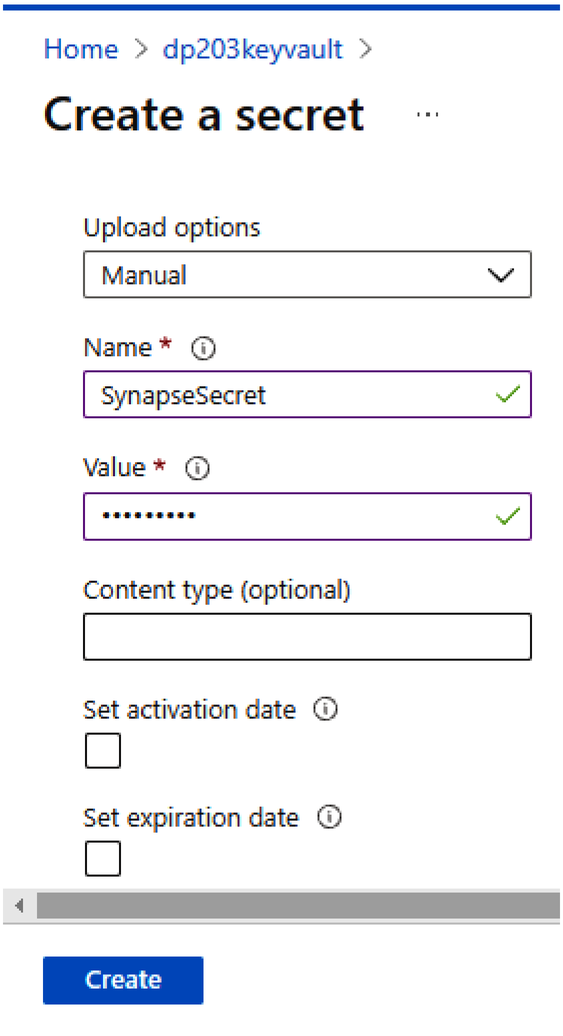

Home > Default Directory >
New user ...
Default Directory
Got feedback?
Identity
User name *
Example: chris
@
The domain name I need isn't shown here
Name *
Example: 'Chris Green'
First name
Last name
Groups and roles
Groups
0 groups selected
Roles
User
Create

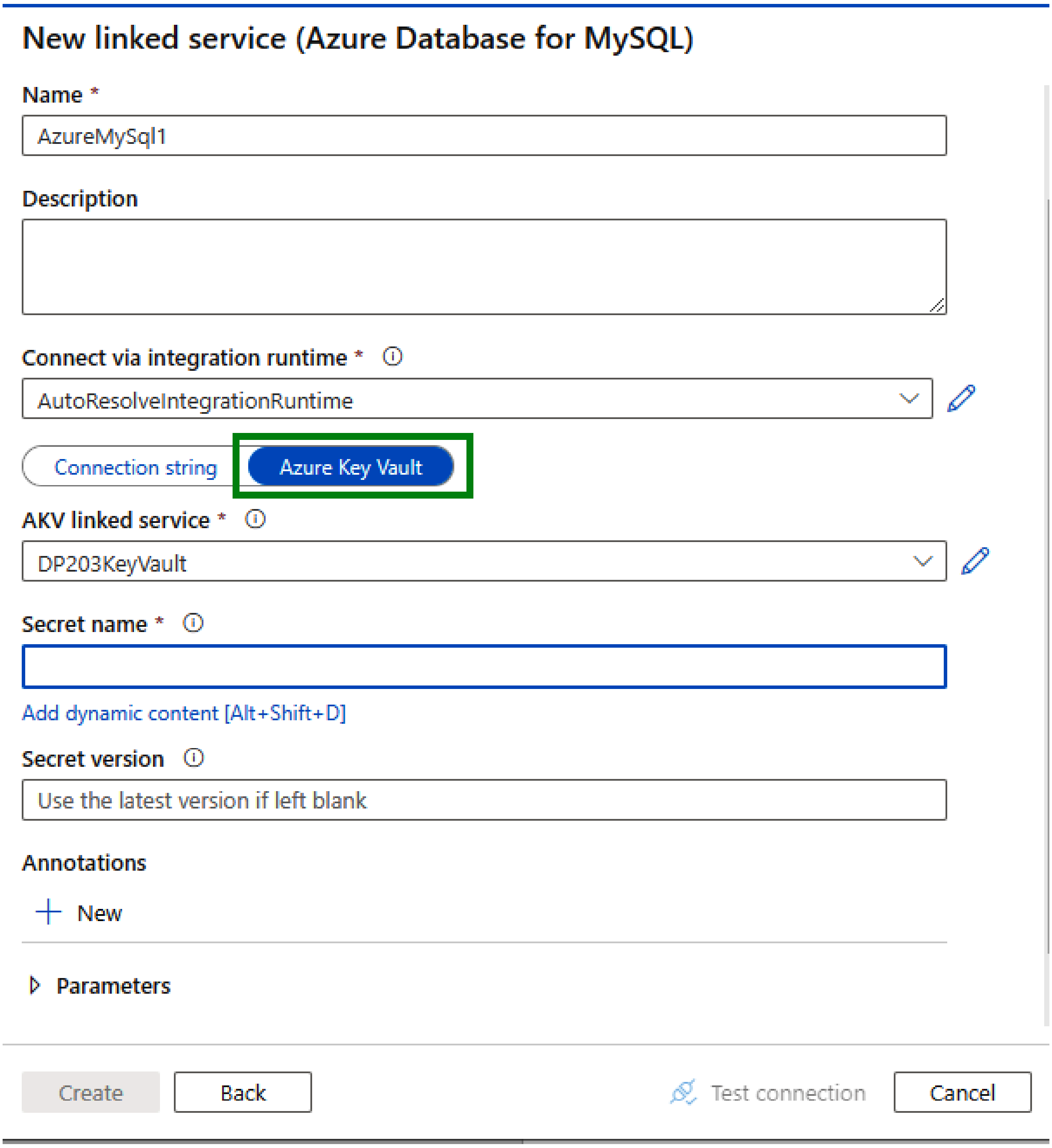

Home > Default Directory >
New user ...
Default Directory
Got feedback?
Identity
User name * ⓘ
Example: chris
@
newtonpacktgmail.onmic...
The domain name I need isn't shown here
Name * ⓘ
Example: 'Chris Green'
First name
Last name
Groups and roles
Groups
0 groups selected
Roles
User
Create

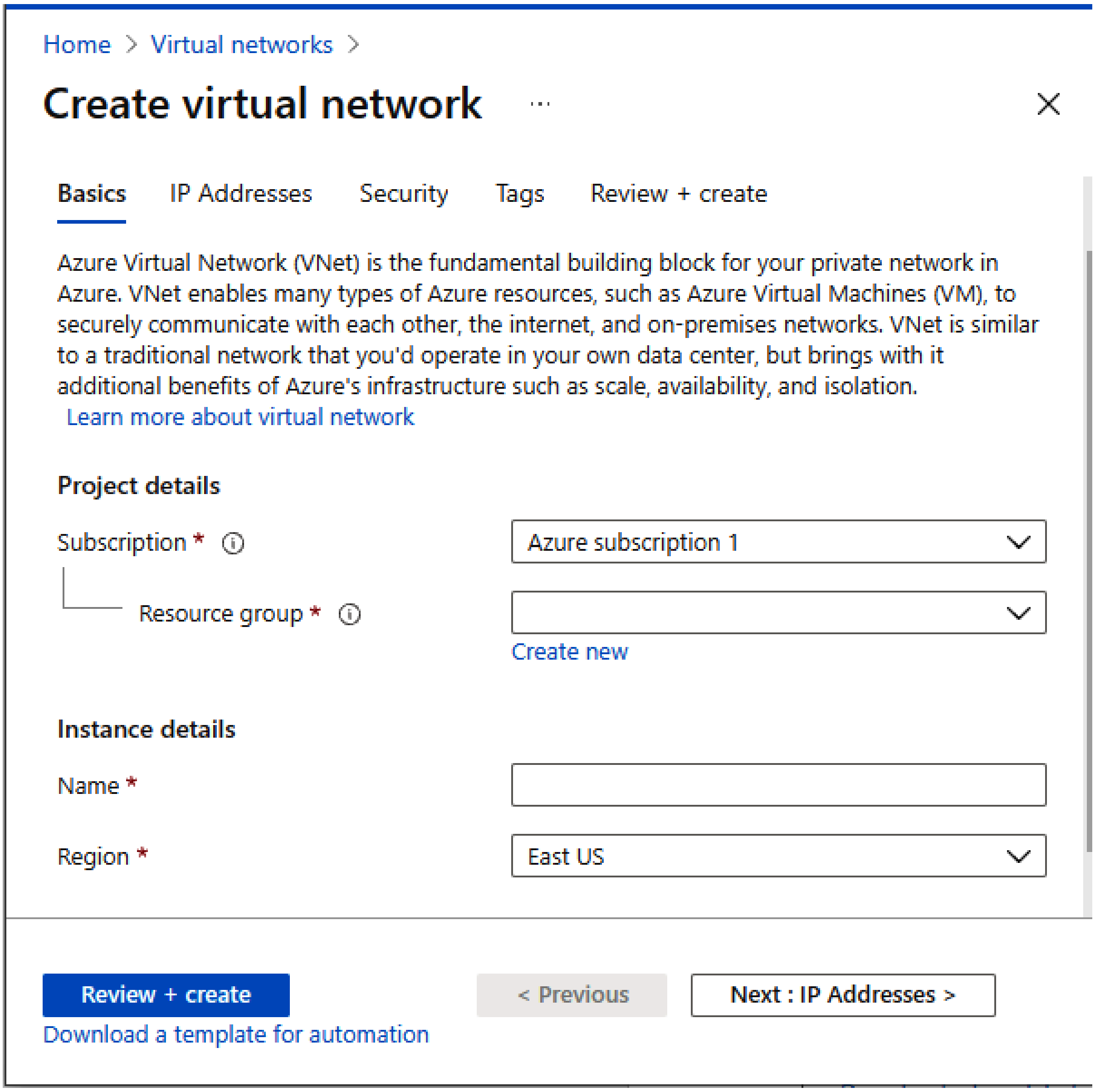

Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Settings
SQL Active Directory admin
Essentials
Resource group (change) : DP203SandboxRG
Status : Succeeded
Location : East US
Subscription (change) : Azure subscription 1
Subscription ID : [redacted]a40
Managed virtual network : [redacted]
Managed Identity object ... : [redacted]88

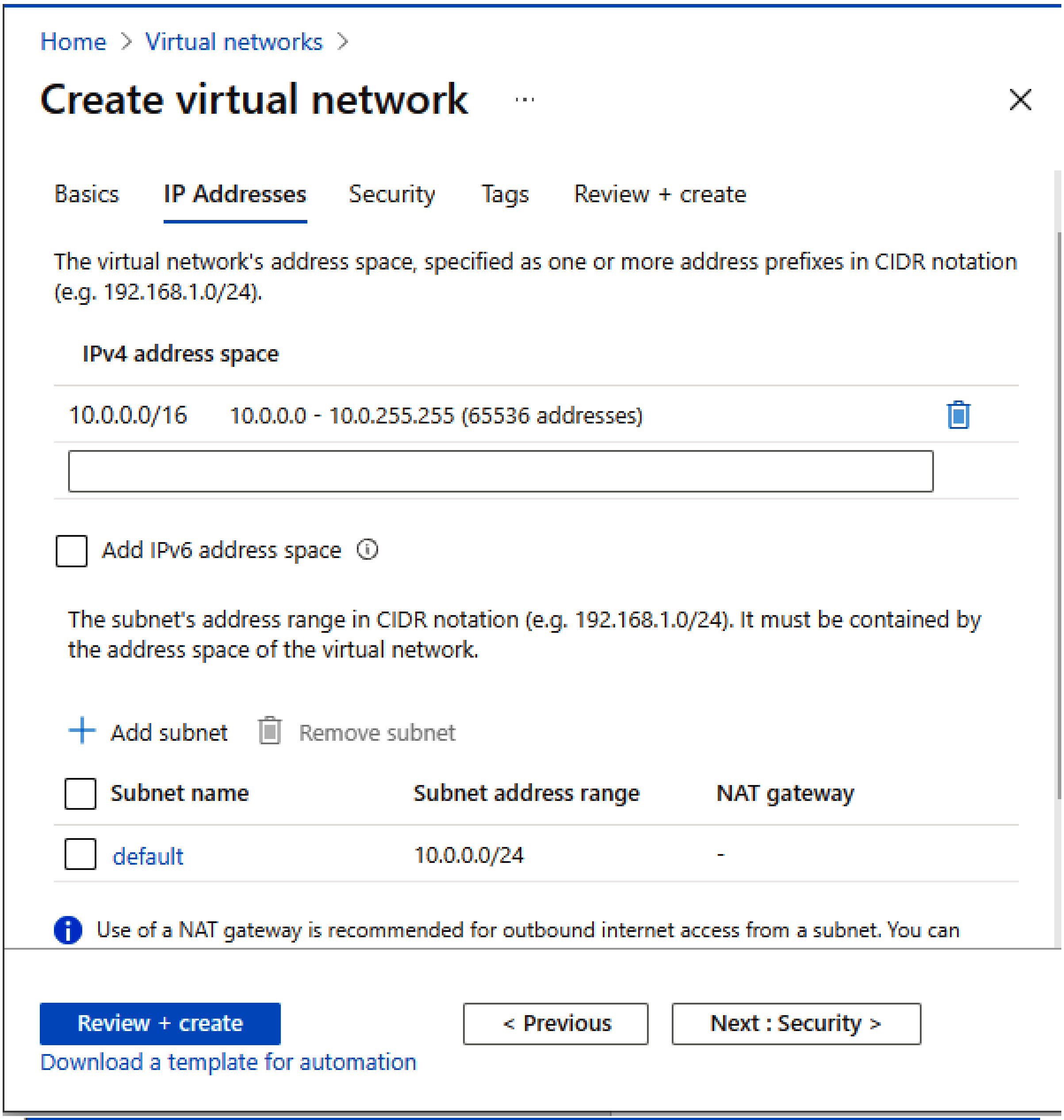

Home > dp203keyvault
dp203keyvault | Keys
Key vault
Search (Ctrl+/)
Generate/Import
Refresh
Settings
Name
Status
Expiration
Keys
There are no keys available.
Secrets
Certificates

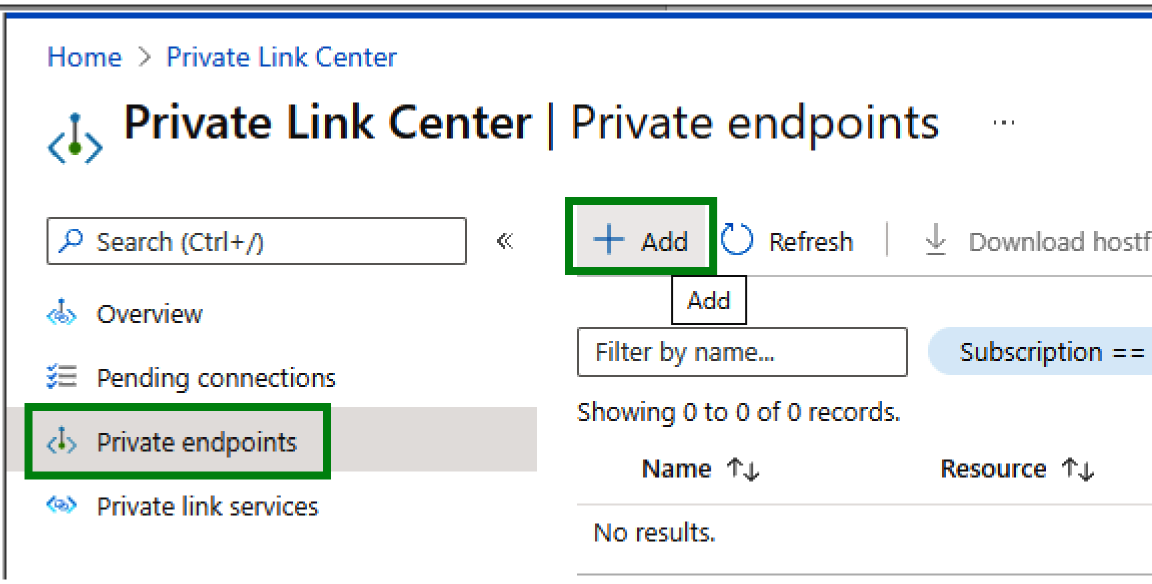

Create a key
Options
Generate
Name *
Key type
RSA
EC
RSA key size
2048
3072
4096
Set activation date
Set expiration date
Enabled
Yes
No
Tags
0 tags
Create

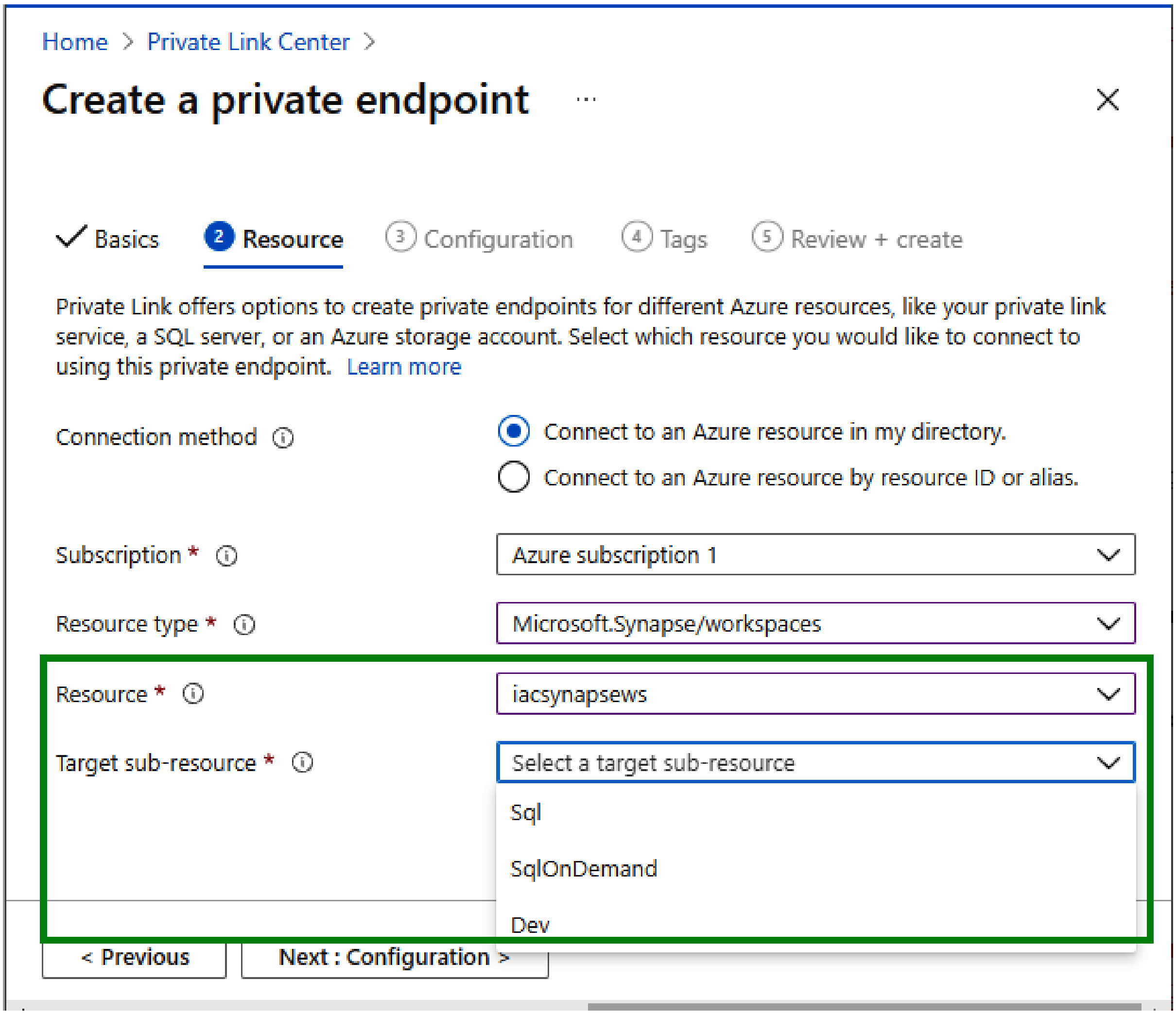

Home > dp203keyvault >
Create a secret
Upload options
Manual
Name *
SynapseSecret
Value *
Content type (optional)
Set activation date
Set expiration date
Create

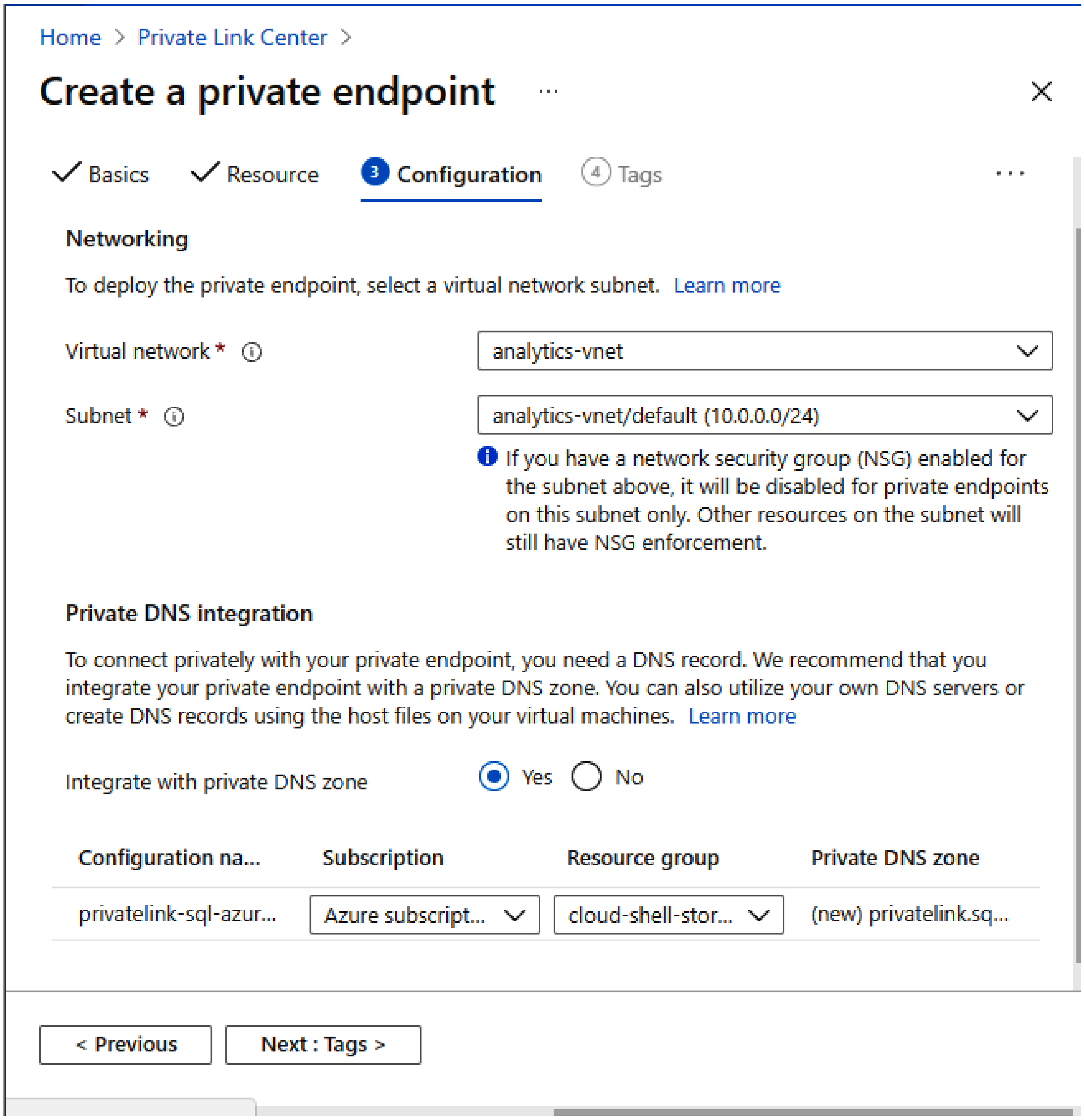

New linked service (Azure Database for MySQL)
Name *
AzureMySql1
Description
Connect via integration runtime *
AutoResolveIntegrationRuntime
Connection string
Azure Key Vault
AKV linked service *
DP203KeyVault
Secret name *
Add dynamic content [Alt+Shift+D]
Secret version
Use the latest version if left blank
Annotations
New
Parameters
Create
Back
Test connection
Cancel

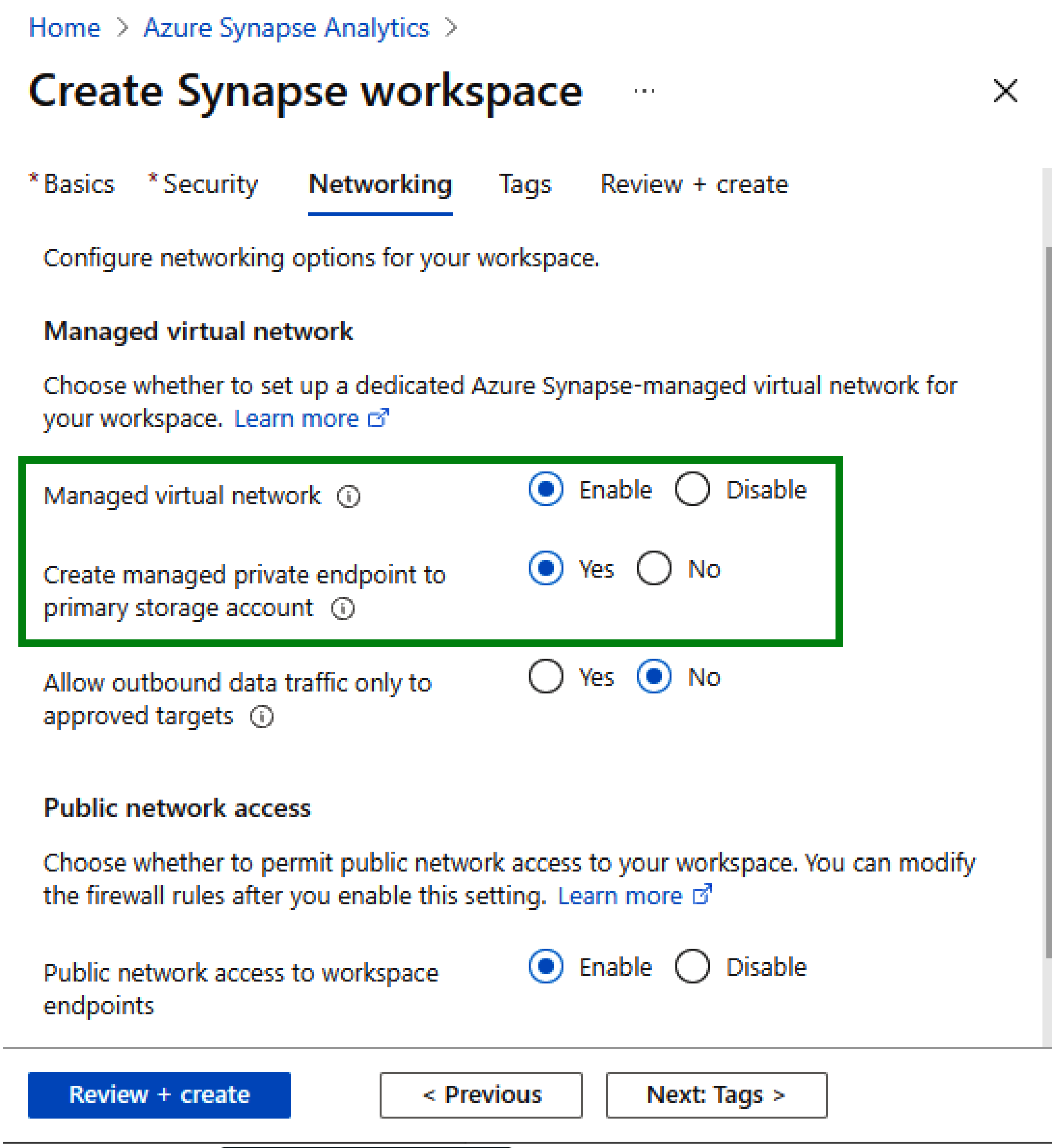

Home > Virtual networks >
Create virtual network ...
Basics
IP Addresses
Security
Tags
Review + create
Azure Virtual Network (VNet) is the fundamental building block for your private network in Azure. VNet enables many types of Azure resources, such as Azure Virtual Machines (VM), to securely communicate with each other, the internet, and on-premises networks. VNet is similar to a traditional network that you'd operate in your own data center, but brings with it additional benefits of Azure's infrastructure such as scale, availability, and isolation.
Learn more about virtual network
Project details
Subscription *
Azure subscription 1
Resource group *
Create new
Instance details
Name *
Region *
East US
Review + create
< Previous
Next : IP Addresses >
Download a template for automation

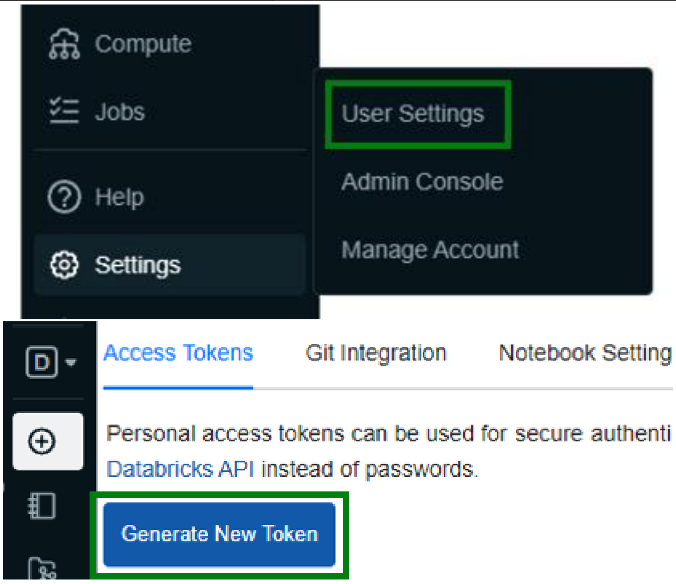

Home > Virtual networks >
Create virtual network ...
Basics
IP Addresses
Security
Tags
Review + create
The virtual network's address space, specified as one or more address prefixes in CIDR notation (e.g. 192.168.1.0/24).
IPv4 address space
10.0.0.0/16
10.0.0.0 - 10.0.255.255 (65536 addresses)
Add IPv6 address space
The subnet's address range in CIDR notation (e.g. 192.168.1.0/24). It must be contained by the address space of the virtual network.
Add subnet
Remove subnet
Subnet name
Subnet address range
NAT gateway
default
10.0.0.0/24
-
Use of a NAT gateway is recommended for outbound internet access from a subnet. You can
Review + create
< Previous
Next : Security >
Download a template for automation


Home > Private Link Center
Private Link Center | Private endpoints ...
Search (Ctrl+/)
Add
Refresh
Download hostf
Add
Overview
Filter by name...
Subscription ==
Pending connections
Showing 0 to 0 of 0 records.
Private endpoints
Name
Resource
Private link services
No results.


Home > Private Link Center >
Create a private endpoint ...
Basics
2 Resource
3 Configuration
4 Tags
5 Review + create
Private Link offers options to create private endpoints for different Azure resources, like your private link service, a SQL server, or an Azure storage account. Select which resource you would like to connect to using this private endpoint. Learn more
Connection method
Connect to an Azure resource in my directory.
Connect to an Azure resource by resource ID or alias.
Subscription *
Azure subscription 1
Resource type *
Microsoft.Synapse/workspaces
Resource *
iacsynapsews
Target sub-resource *
Select a target sub-resource
Sql
SqlOnDemand
Dev
< Previous
Next : Configuration >


Home > Private Link Center >
Create a private endpoint ...
Basics
Resource
3 Configuration
4 Tags
Networking
To deploy the private endpoint, select a virtual network subnet. Learn more
Virtual network *
analytics-vnet
Subnet *
analytics-vnet/default (10.0.0.0/24)
If you have a network security group (NSG) enabled for the subnet above, it will be disabled for private endpoints on this subnet only. Other resources on the subnet will still have NSG enforcement.
Private DNS integration
To connect privately with your private endpoint, you need a DNS record. We recommend that you integrate your private endpoint with a private DNS zone. You can also utilize your own DNS servers or create DNS records using the host files on your virtual machines. Learn more
Integrate with private DNS zone
Yes
No
Configuration na...
Subscription
Resource group
Private DNS zone
privatelink-sql-azur...
Azure subscript...
cloud-shell-stor...
(new) privatelink.sq...
< Previous
Next : Tags >


Home > Azure Synapse Analytics >
Create Synapse workspace
* Basics * Security Networking Tags Review + create
Configure networking options for your workspace.
Managed virtual network
Choose whether to set up a dedicated Azure Synapse-managed virtual network for your workspace. Learn more
Managed virtual network
Enable Disable
Create managed private endpoint to primary storage account
Yes No
Allow outbound data traffic only to approved targets
Yes No
Public network access
Choose whether to permit public network access to your workspace. You can modify the firewall rules after you enable this setting. Learn more
Public network access to workspace endpoints
Enable Disable
Review + create
< Previous
Next: Tags >


Compute
Jobs
Help
Settings
User Settings
Admin Console
Manage Account
Access Tokens
Git Integration
Notebook Setting
Personal access tokens can be used for secure authenti
Databricks API instead of passwords.
Generate New Token

Generate New Token

Comment

What's this token for?

Lifetime (days)

90

Cancel | Generate

	Name	SSN	email
1	Adam Smith	111-11-1111	james@james.com
2	Brenda Harman	222-22-2222	brenda@brenda.com
3	Carmen Pinto	333-33-3333	carmen@carmen.com

	Name	SSN
1	Adam Smith	gAAAAABhYbAciaaO-twCsrR2cRSxv8i5HSQcQI5nLDQPGXrDabn5UW5a5hGNkzooEEilPqmqrnlvxq8niDF1
2	Brenda Harman	gAAAAABhYbAdEYKEYdqSKyr3DG87EvVre2SMXK2_dfB2zZM4pt1Wlm2DKB-YI1kinKsdSVP4Fz_EshH7HdU6EHyvj5JU1OGBDA==
3	Carmen Pinto	gAAAAABhYbAcMjc-lj0rkAcMbB5t4A3dLouJxVKj4o_DbxTLnPENuaM6JsnKRMwnqKhZs3mNzdxjHgrxXDqQUfugDYdmg==

decrypted: pyspark.sql.dataframe.DataFrame = [Name: string, SSN: string

	Name	SSN	email
1	Adam Smith	111-11-1111	james@james.com
2	Brenda Harman	222-22-2222	brenda@brenda.com
3	Carmen Pinto	333-33-3333	carmen@carmen.com

Security

- Auditing
- Data Discovery & Classification
- Dynamic Data Masking
- Security Center
- Transparent data encryption

Common Tasks

- Open in Visual Studio

Monitoring

- Query activity

Overview | Classification

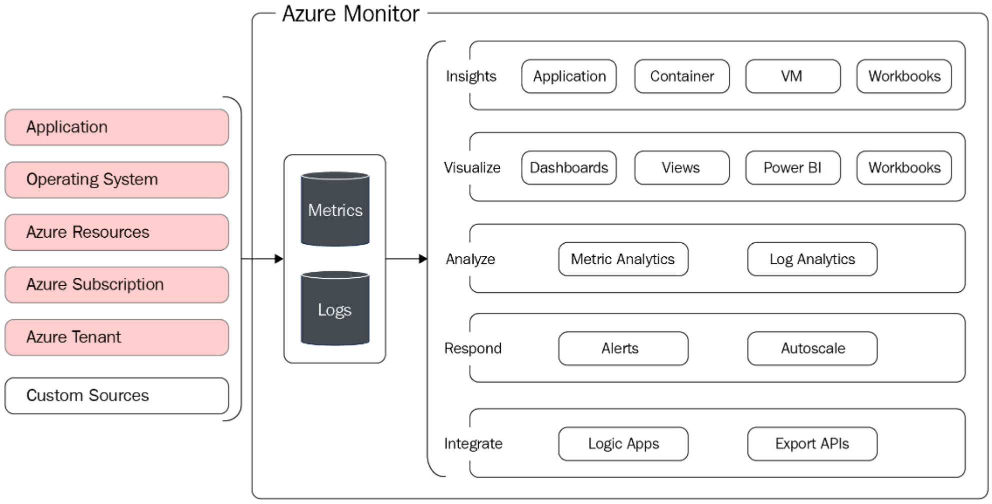
15 columns with classification recommendations (Click to minimize)

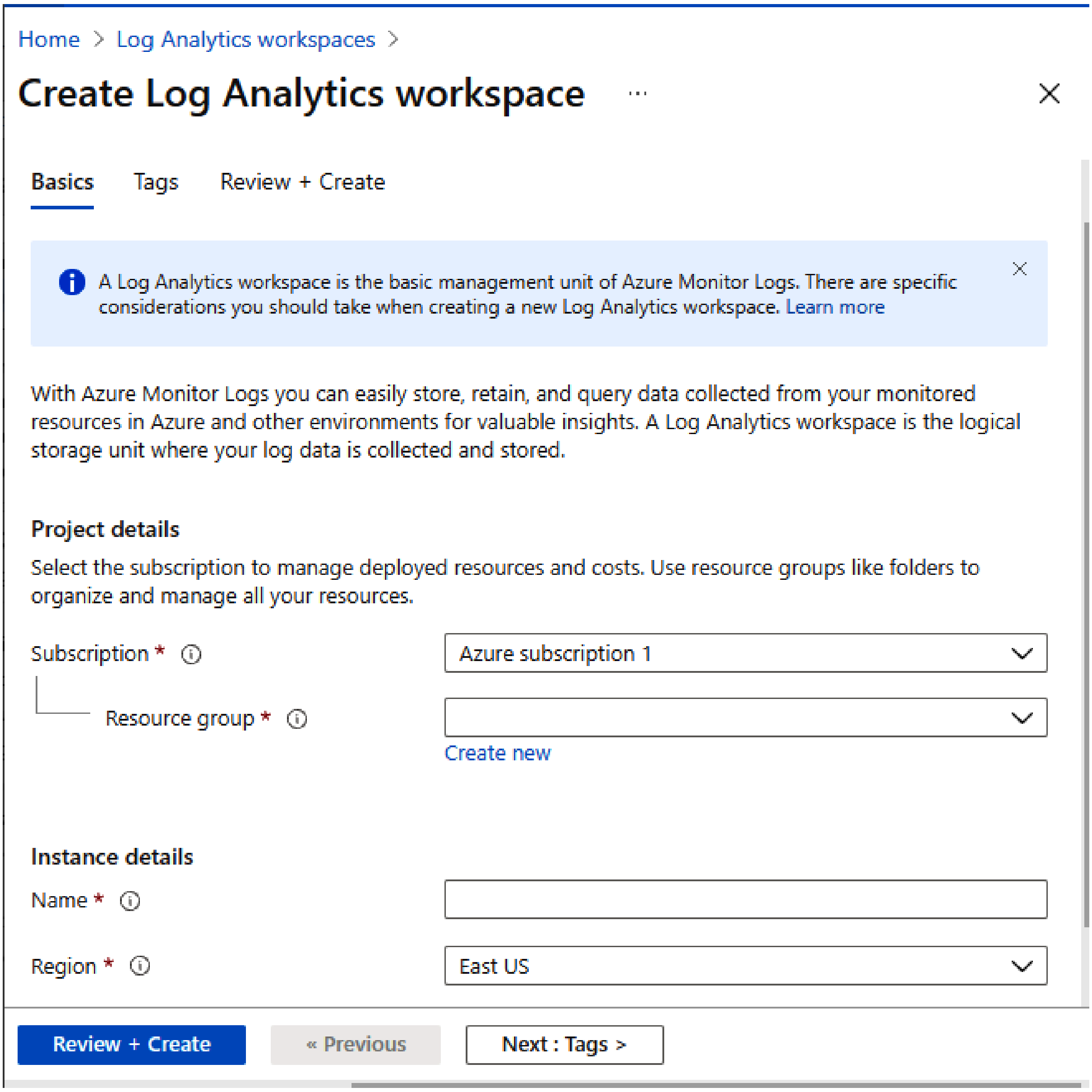
Accept selected recommendations | Dismiss selected recommendations | Show dismissed recommendations

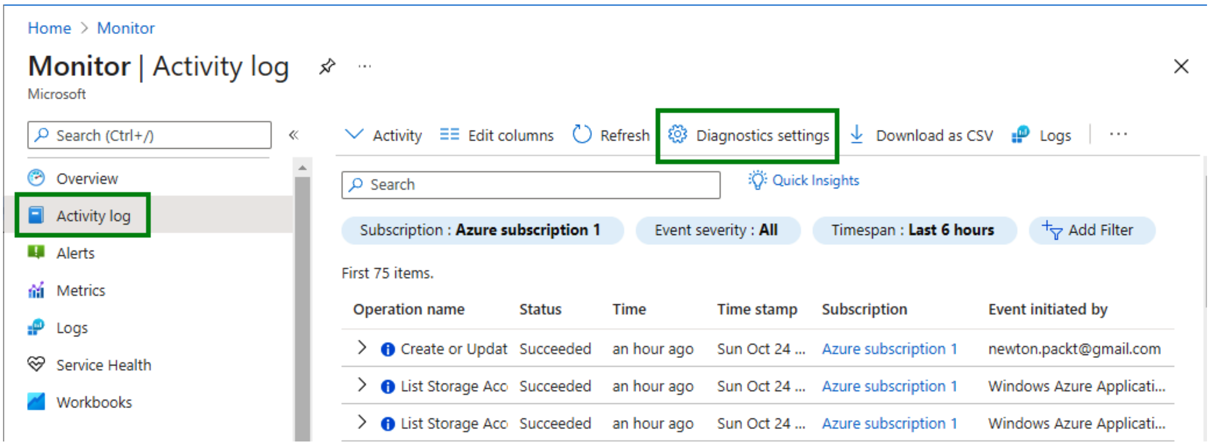
Select all | Schema: 1 selected | Table: 7 selected | Filter by column | Information type: 3... | Sensitivity label: 2 s...

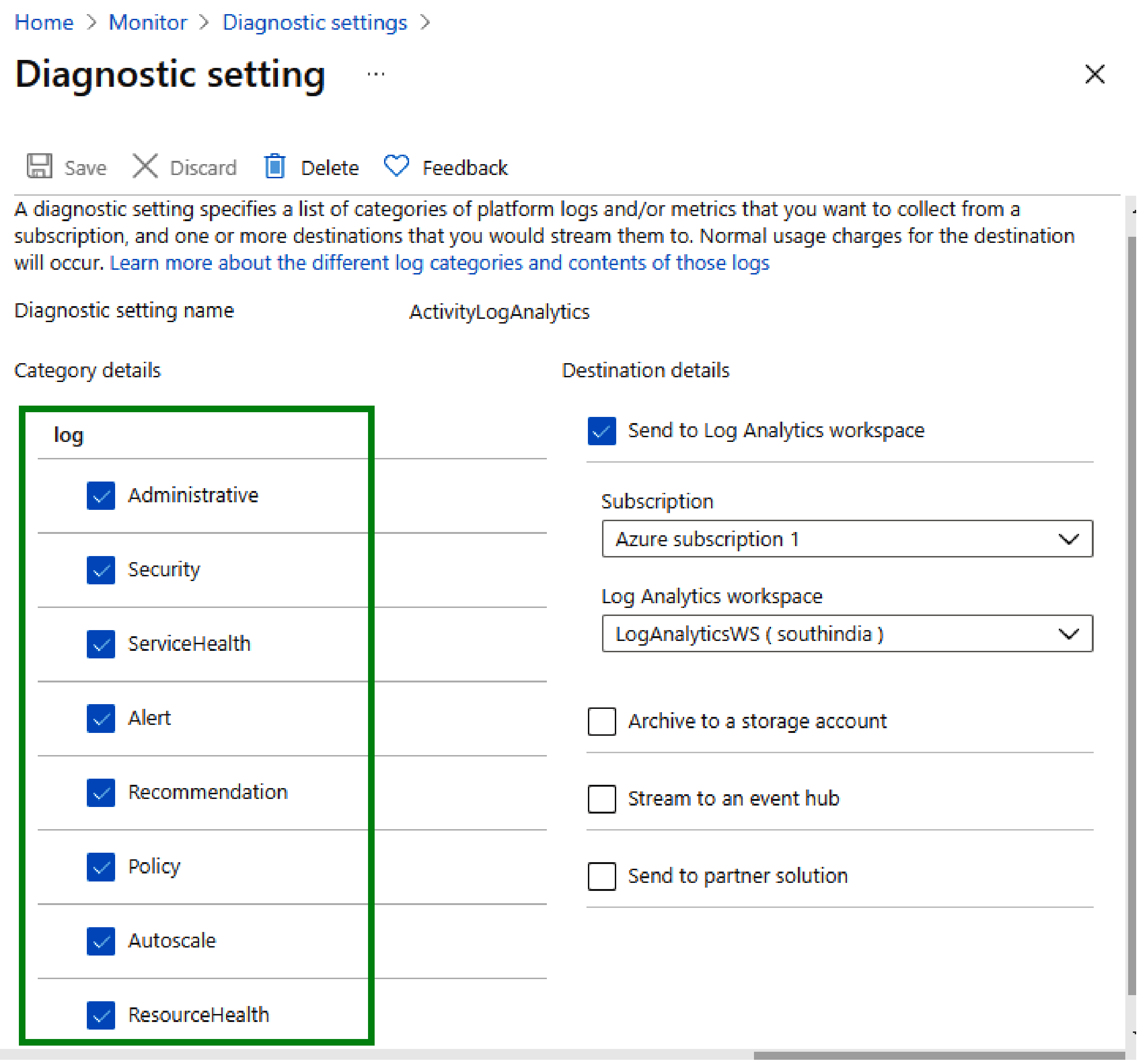
	Schema	Table	Column	Information type	Sensitivity label
☐	dbo	TempDriver	driverId	National ID	Confidential - GDPR
☐	dbo	TempDriver	FirstName	Name	Confidential - GDPR
☐	dbo	TempDriver	LastName	Name	Confidential - GDPR

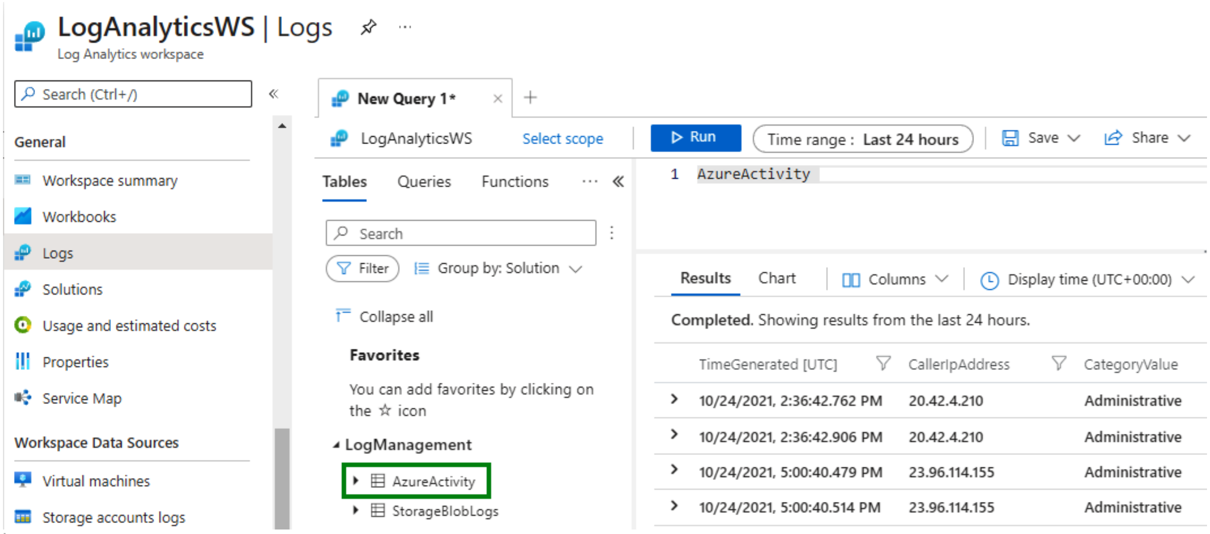
Chapter 13: Monitoring Data Storage and Data Processing

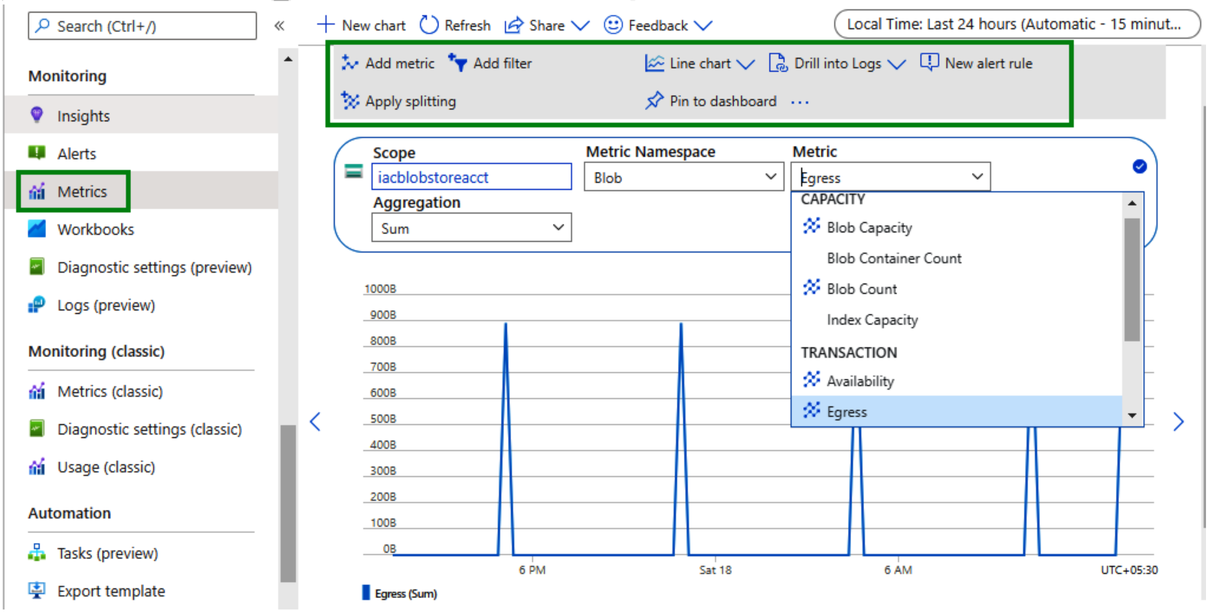


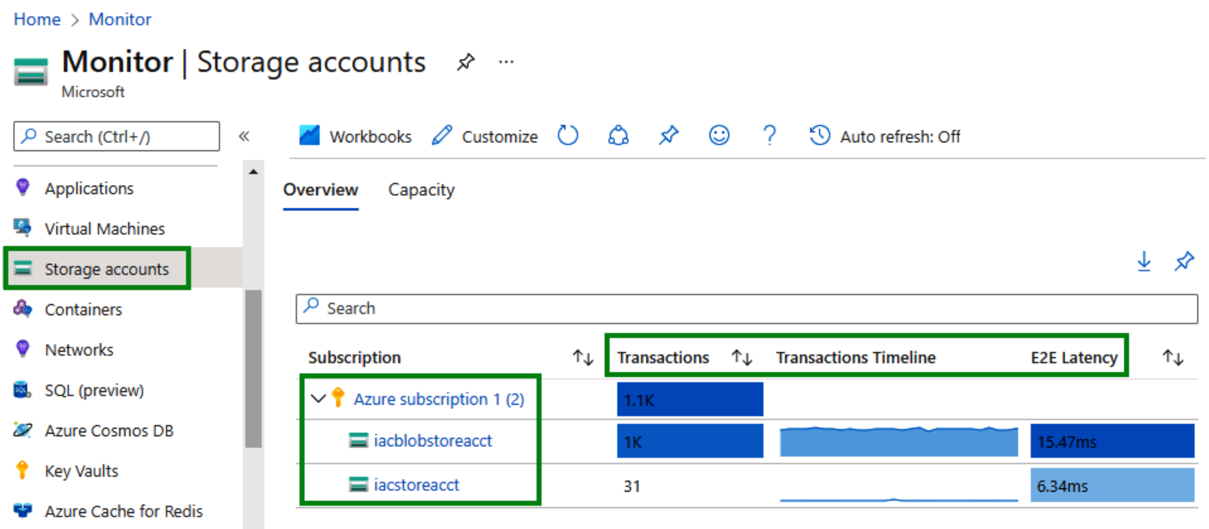


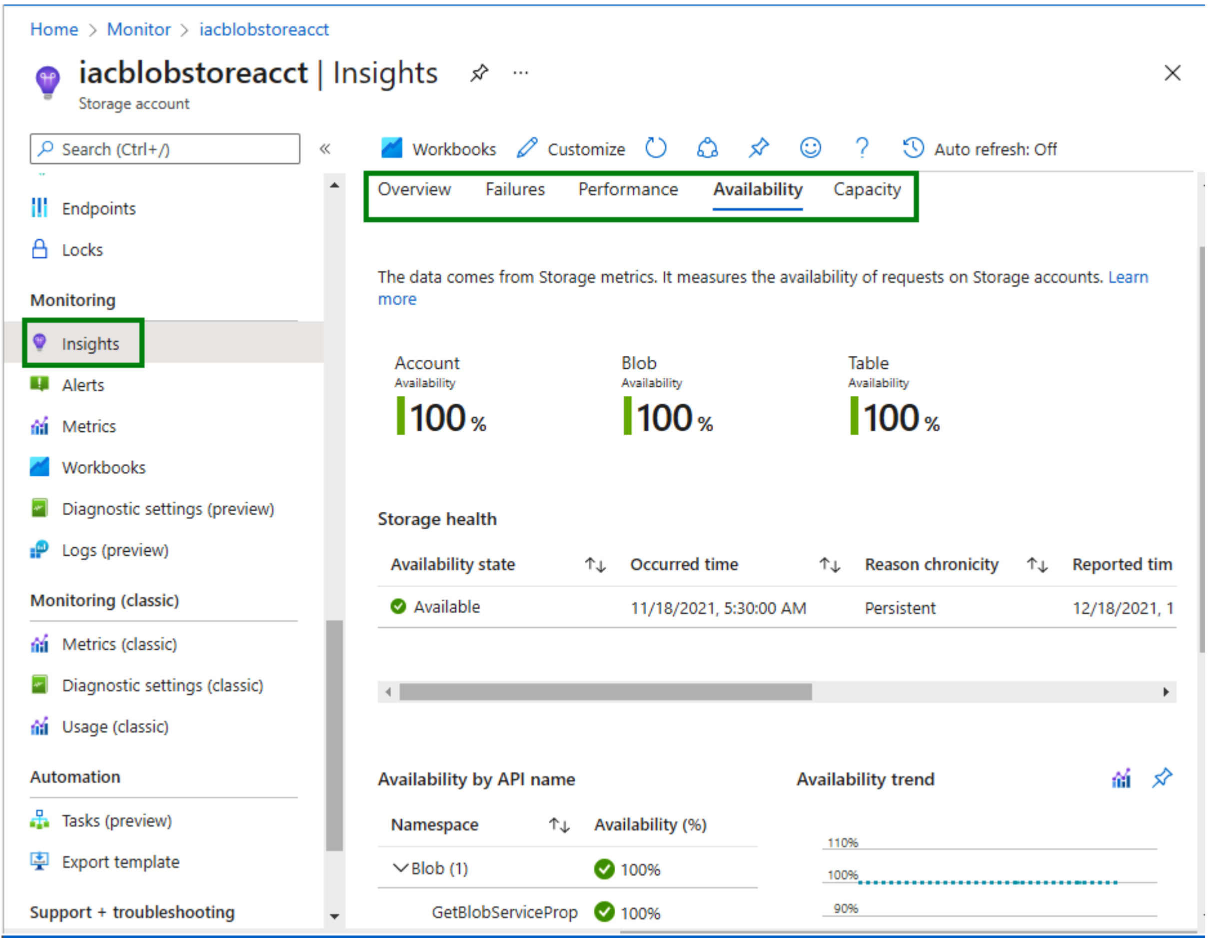

Home > Monitor
Monitor | Activity log
Microsoft
Search (Ctrl+/)
Overview
Activity log
Alerts
Metrics
Logs
Service Health
Workbooks
Activity
Edit columns
Refresh
Diagnostics settings
Download as CSV
Logs
Search
Quick Insights
Subscription : Azure subscription 1
Event severity : All
Timespan : Last 6 hours
Add Filter
First 75 items.
Operation name
Status
Time
Time stamp
Subscription
Event initiated by
Create or Updat Succeeded an hour ago Sun Oct 24 ... Azure subscription 1 newton.packt@gmail.com
List Storage Acc Succeeded an hour ago Sun Oct 24 ... Azure subscription 1 Windows Azure Applicati...
List Storage Acc Succeeded an hour ago Sun Oct 24 ... Azure subscription 1 Windows Azure Applicati...

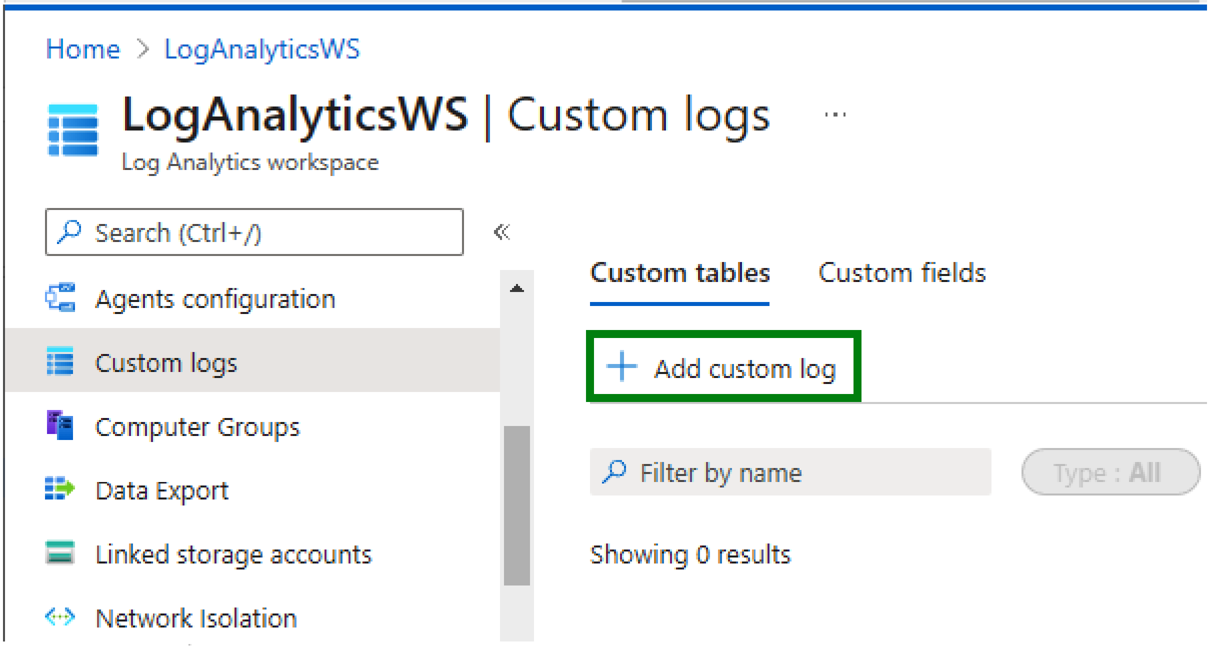

Home > Monitor > Diagnostic settings >
Diagnostic setting
Save
Discard
Delete
Feedback
A diagnostic setting specifies a list of categories of platform logs and/or metrics that you want to collect from a subscription, and one or more destinations that you would stream them to. Normal usage charges for the destination will occur. Learn more about the different log categories and contents of those logs
Diagnostic setting name
ActivityLogAnalytics
Category details
Destination details
log
Administrative
Security
ServiceHealth
Alert
Recommendation
Policy
Autoscale
ResourceHealth
Send to Log Analytics workspace
Subscription
Azure subscription 1
Log Analytics workspace
LogAnalyticsWS (southindia)
Archive to a storage account
Stream to an event hub
Send to partner solution

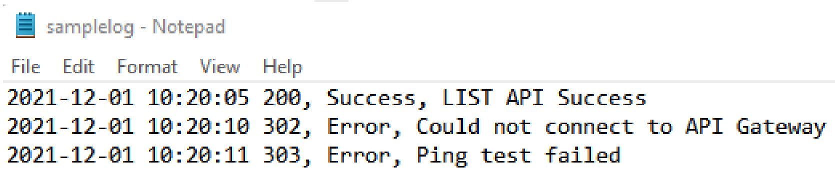

LogAnalyticsWS | Logs
Log Analytics workspace
Search (Ctrl+/)
General
Workspace summary
Workbooks
Logs
Solutions
Usage and estimated costs
Properties
Service Map
Workspace Data Sources
Virtual machines
Storage accounts logs
New Query 1*
LogAnalyticsWS
Select scope
Run
Time range : Last 24 hours
Save
Share
Tables
Queries
Functions
Search
Filter
Group by: Solution
Collapse all
Favorites
You can add favorites by clicking on the ☆ icon
LogManagement
AzureActivity
StorageBlobLogs
1 AzureActivity
Results
Chart
Columns
Display time (UTC+00:00)
Completed. Showing results from the last 24 hours.
TimeGenerated [UTC]
CallerIpAddress
CategoryValue
10/24/2021, 2:36:42.762 PM 20.42.4.210 Administrative
10/24/2021, 2:36:42.906 PM 20.42.4.210 Administrative
10/24/2021, 5:00:40.479 PM 23.96.114.155 Administrative
10/24/2021, 5:00:40.514 PM 23.96.114.155 Administrative

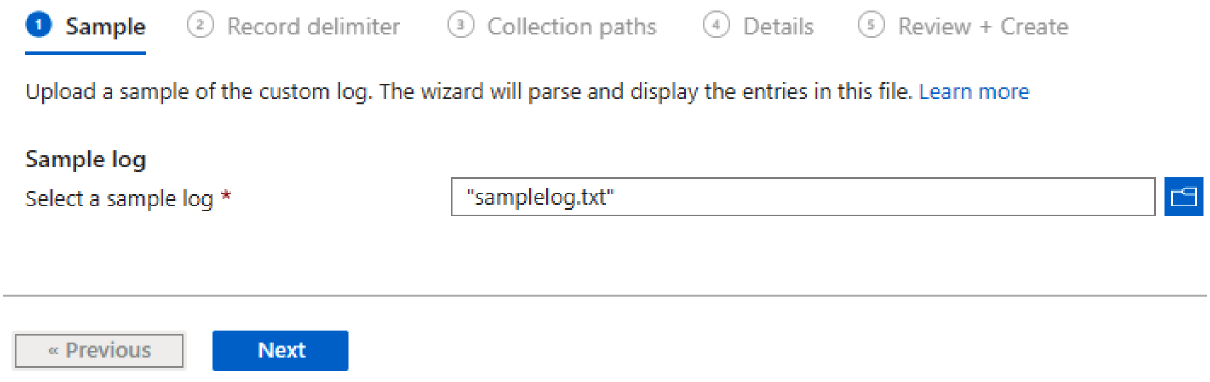

Search (Ctrl+/)
Monitoring
Insights
Alerts
Metrics
Workbooks
Diagnostic settings (preview)
Logs (preview)
Monitoring (classic)
Metrics (classic)
Diagnostic settings (classic)
Usage (classic)
Automation
Tasks (preview)
Export template
New chart
Refresh
Share
Feedback
Local Time: Last 24 hours (Automatic - 15 minut...
Add metric
Add filter
Line chart
Drill into Logs
New alert rule
Apply splitting
Pin to dashboard
Scope
iacblobstoreacct
Metric Namespace
Blob
Metric
Egress
Aggregation
Sum
CAPACITY
Blob Capacity
Blob Container Count
Blob Count
Index Capacity
TRANSACTION
Availability
Egress
1000B
900B
800B
700B
600B
500B
400B
300B
200B
100B
0B
6 PM
Sat 18
6 AM
UTC+05:30
Egress (Sum)

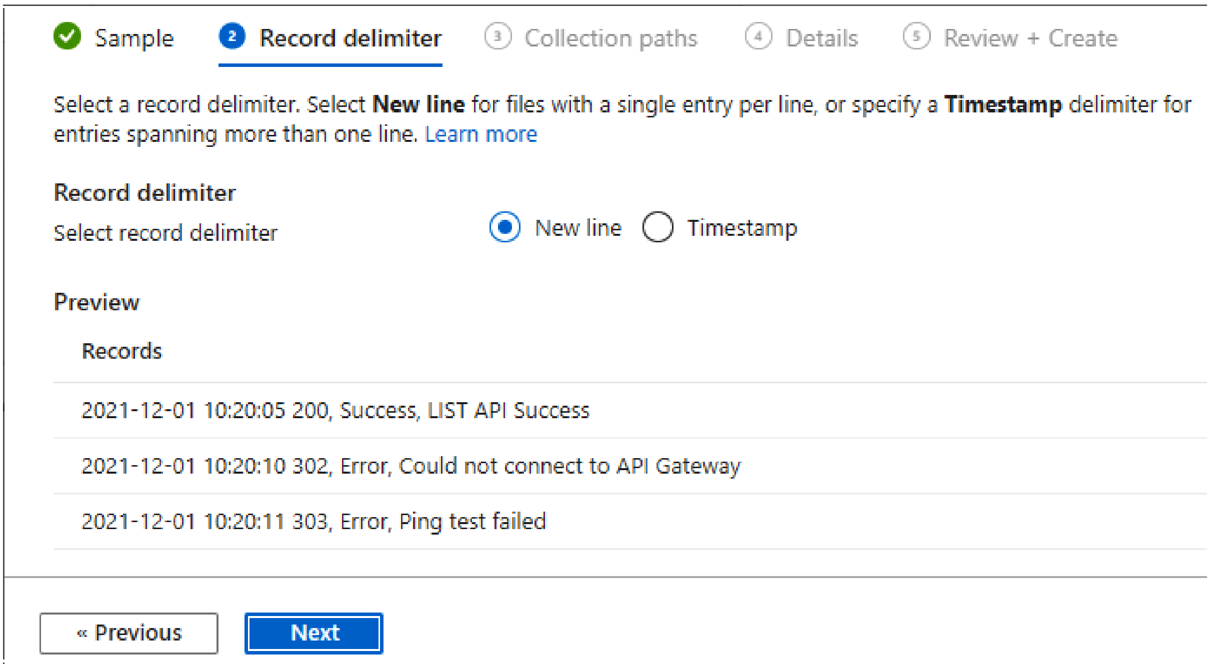

Home > Monitor
Monitor | Storage accounts
Microsoft
Search (Ctrl+/)
Workbooks
Customize
Auto refresh: Off
Applications
Virtual Machines
Storage accounts
Containers
Networks
SQL (preview)
Azure Cosmos DB
Key Vaults
Azure Cache for Redis
Overview
Capacity
Search
Subscription
Transactions
Transactions Timeline
E2E Latency
Azure subscription 1 (2)
1.1K
iacblobstoreacct
1K
15.47ms
iacstoreacct
31
6.34ms

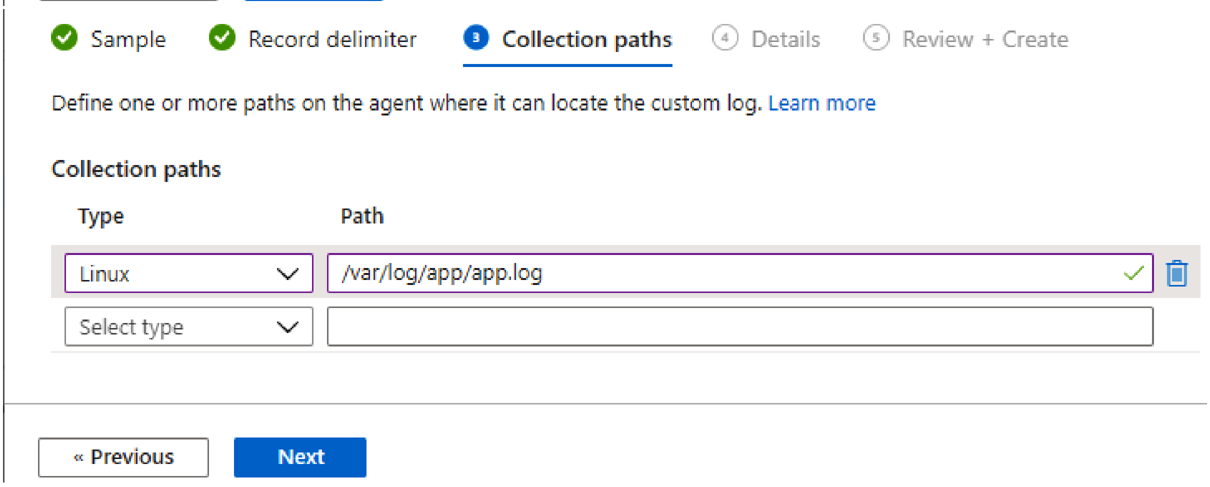

Home > Monitor > iacblobstoreacct
iacblobstoreacct | Insights
Storage account
Search (Ctrl+/)
Workbooks
Customize
Auto refresh: Off
Endpoints
Locks
Monitoring
Insights
Alerts
Metrics
Workbooks
Diagnostic settings (preview)
Logs (preview)
Monitoring (classic)
Metrics (classic)
Diagnostic settings (classic)
Usage (classic)
Automation
Tasks (preview)
Export template
Support + troubleshooting
Overview
Failures
Performance
Availability
Capacity
The data comes from Storage metrics. It measures the availability of requests on Storage accounts. Learn more
Account
Availability
100%
Blob
Availability
100%
Table
Availability
100%
Storage health
Availability state
Occurred time
Reason chronicity
Reported tim
Available
11/18/2021, 5:30:00 AM
Persistent
12/18/2021, 1
Availability by API name
Namespace
Availability (%)
Blob (1)
100%
GetBlobServiceProp
100%
Availability trend
110%
100%
90%

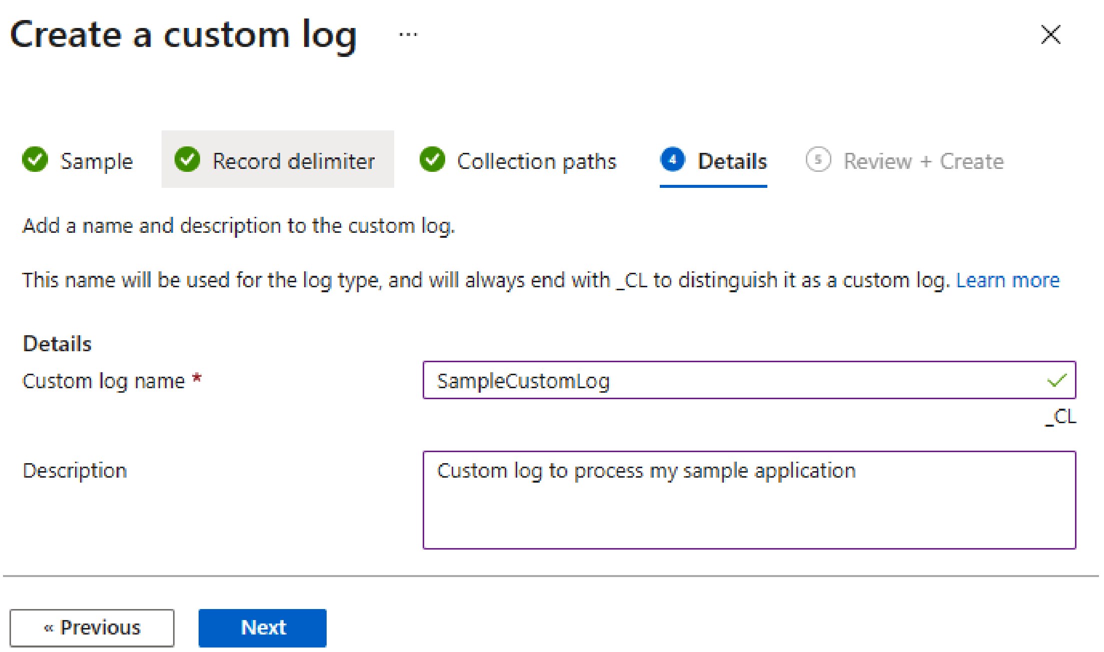

Home > LogAnalyticsWS
LogAnalyticsWS | Custom logs
Log Analytics workspace
Search (Ctrl+/)
Agents configuration
Custom logs
Computer Groups
Data Export
Linked storage accounts
Network Isolation
Custom tables
Custom fields
Add custom log
Filter by name
Type : All
Showing 0 results

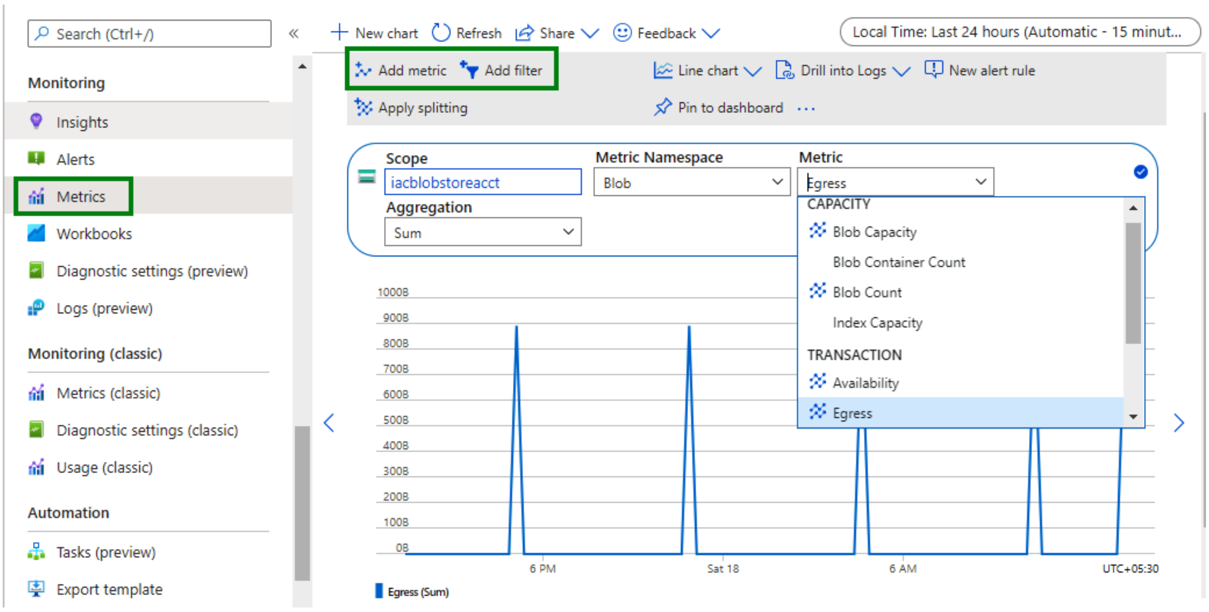

samplelog - Notepad
File
Edit
Format
View
Help
2021-12-01 10:20:05 200, Success, LIST API Success
2021-12-01 10:20:10 302, Error, Could not connect to API Gateway
2021-12-01 10:20:11 303, Error, Ping test failed

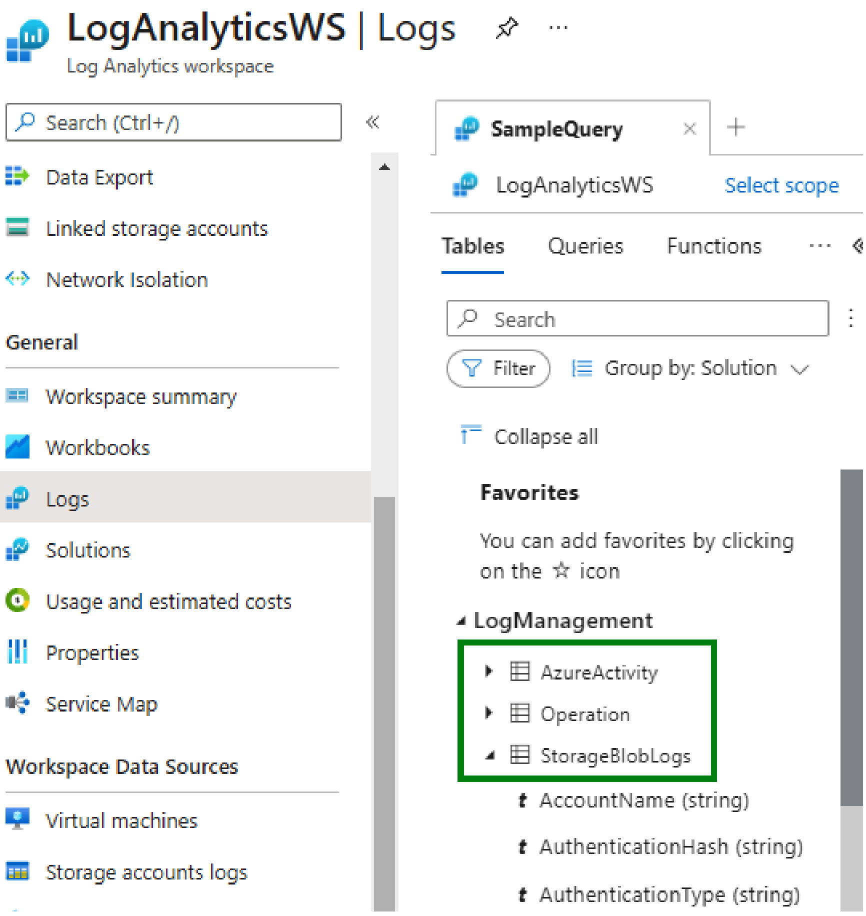

1 Sample
2 Record delimiter
3 Collection paths
4 Details
5 Review + Create
Upload a sample of the custom log. The wizard will parse and display the entries in this file. Learn more
Sample log
Select a sample log *
"samplelog.txt"
« Previous
Next

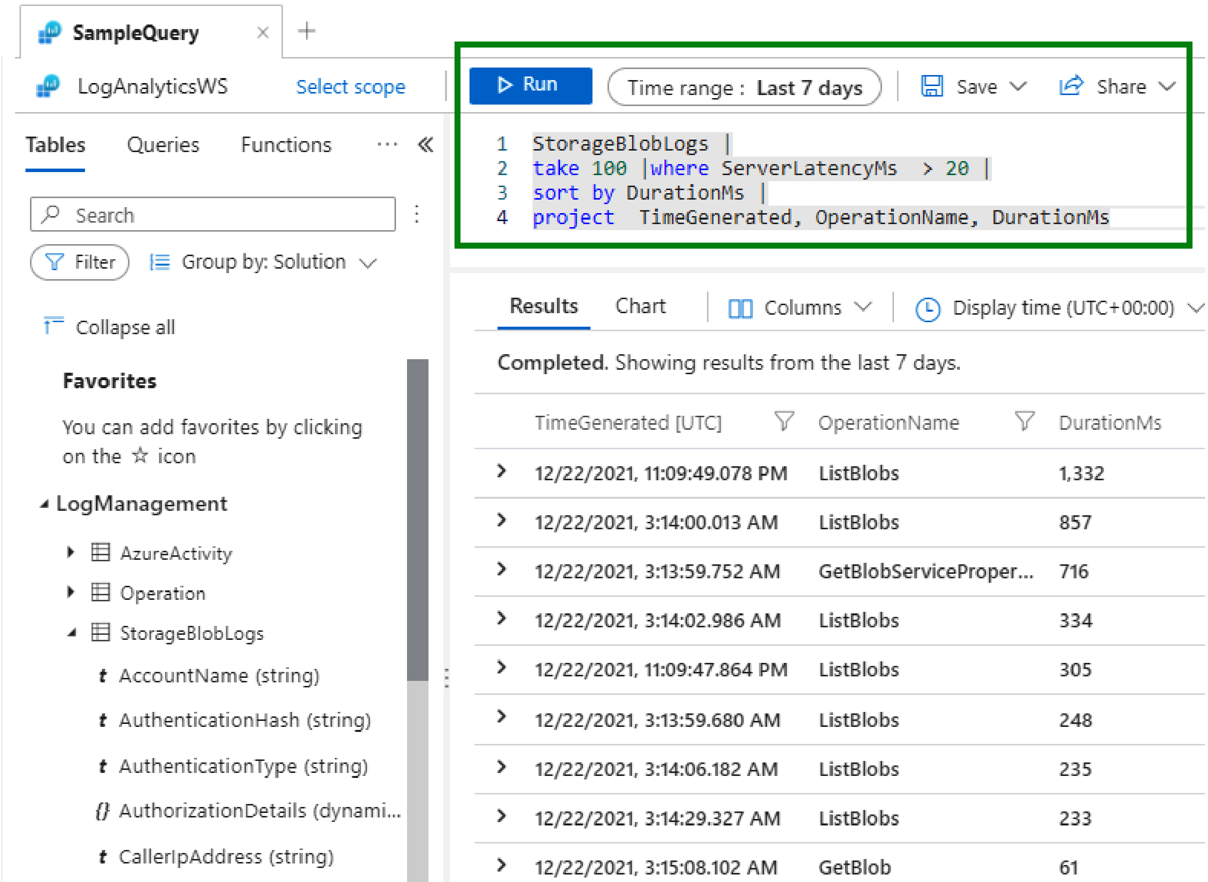

Sample
2 Record delimiter
3 Collection paths
4 Details
5 Review + Create
Select a record delimiter. Select New line for files with a single entry per line, or specify a Timestamp delimiter for entries spanning more than one line. Learn more
Record delimiter
Select record delimiter
New line
Timestamp
Preview
Records
2021-12-01 10:20:05 200, Success, LIST API Success
2021-12-01 10:20:10 302, Error, Could not connect to API Gateway
2021-12-01 10:20:11 303, Error, Ping test failed
« Previous
Next

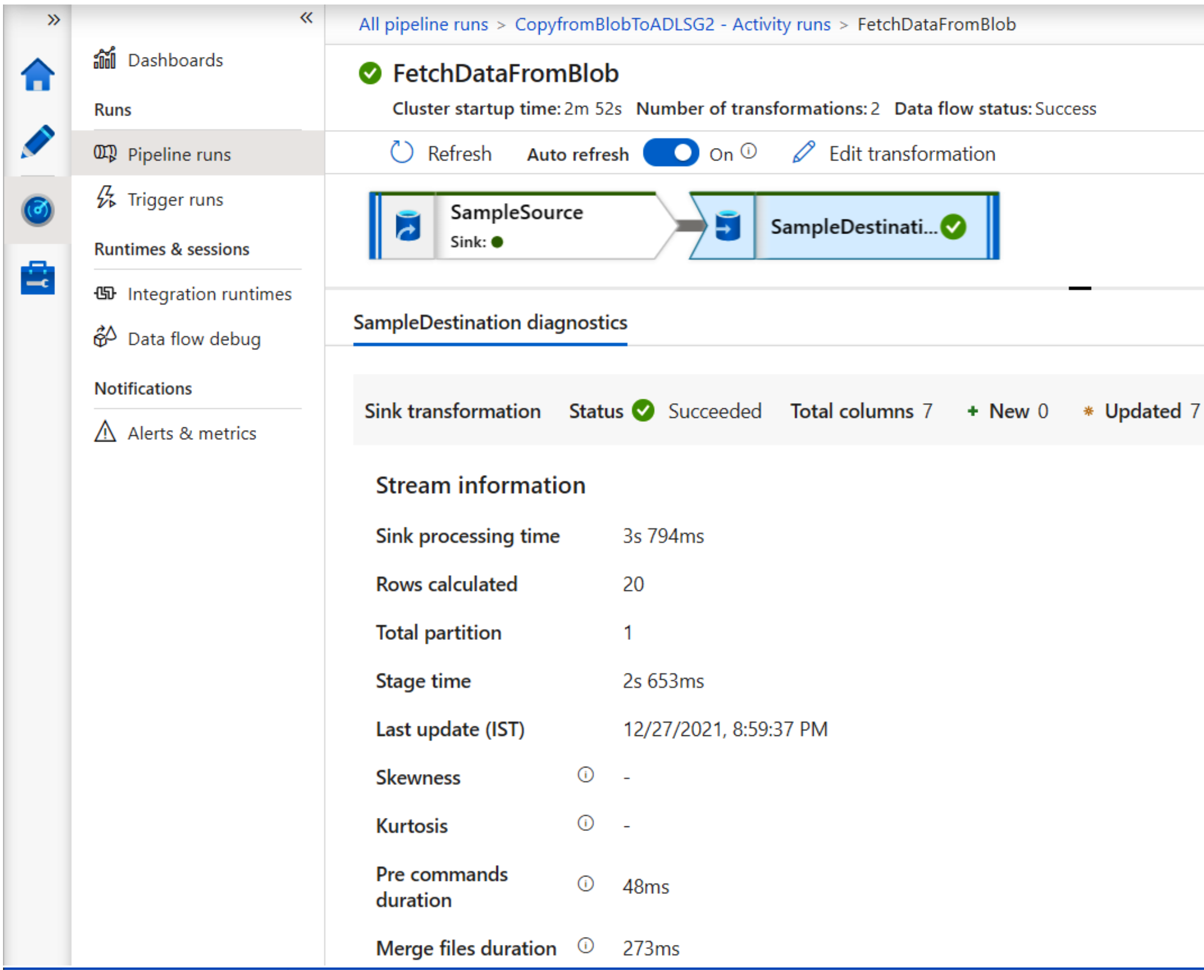
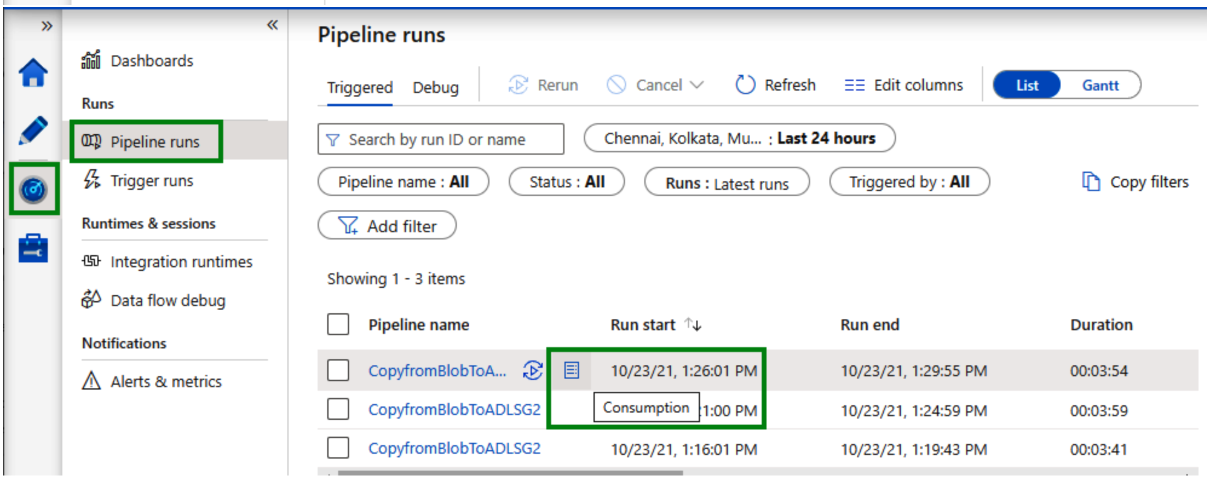

Sample
Record delimiter
3 Collection paths
4 Details
5 Review + Create
Define one or more paths on the agent where it can locate the custom log. Learn more
Collection paths
Type
Path
Linux
/var/log/app/app.log
Select type
« Previous
Next

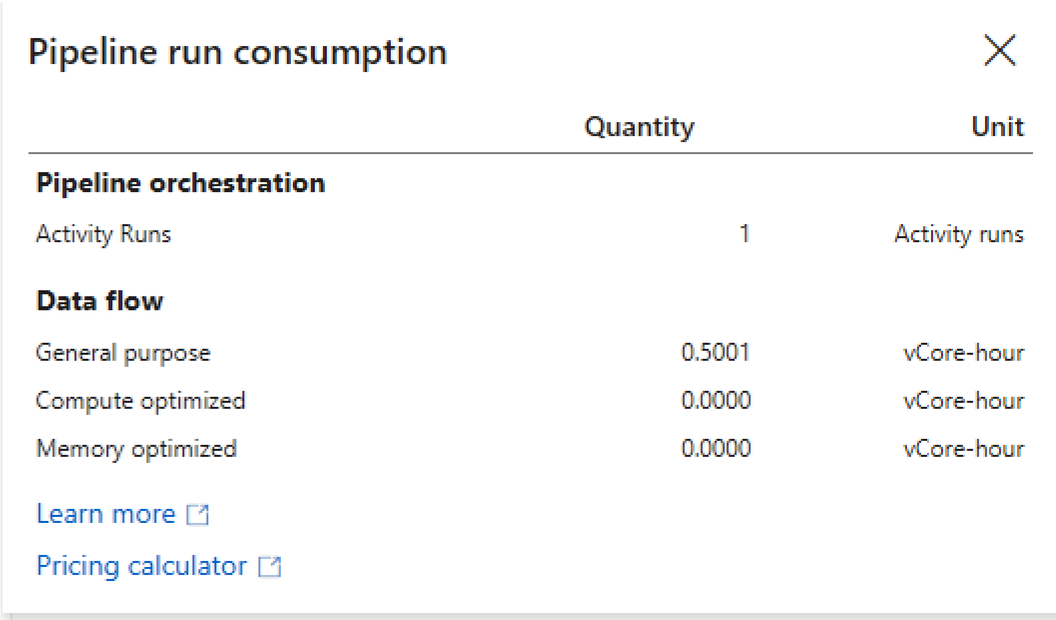

Create a custom log
Sample
Record delimiter
Collection paths
Details
Review + Create
Add a name and description to the custom log.
This name will be used for the log type, and will always end with _CL to distinguish it as a custom log. Learn more
Details
Custom log name *
SampleCustomLog
_CL
Description
Custom log to process my sample application
« Previous
Next

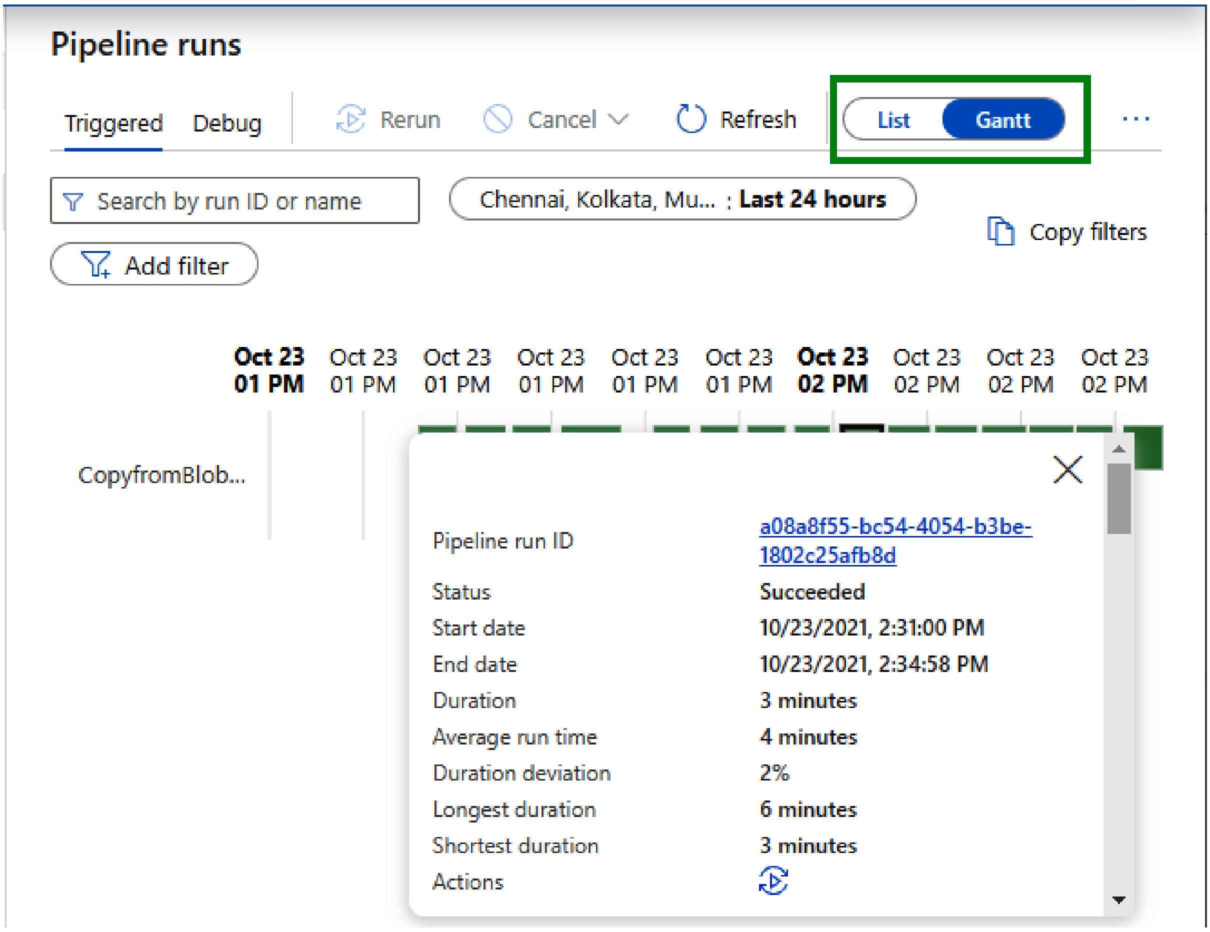

Search (Ctrl+/)
Monitoring
Insights
Alerts
Metrics
Workbooks
Diagnostic settings (preview)
Logs (preview)
Monitoring (classic)
Metrics (classic)
Diagnostic settings (classic)
Usage (classic)
Automation
Tasks (preview)
Export template
New chart
Refresh
Share
Feedback
Local Time: Last 24 hours (Automatic - 15 minut...
Add metric
Add filter
Line chart
Drill into Logs
New alert rule
Apply splitting
Pin to dashboard
Scope
iacblobstoreacct
Metric Namespace
Blob
Metric
Egress
Aggregation
Sum
CAPACITY
Blob Capacity
Blob Container Count
Blob Count
Index Capacity
TRANSACTION
Availability
Egress
1000B
900B
800B
700B
600B
500B
400B
300B
200B
100B
0B
6 PM
Sat 18
6 AM
UTC+05:30
Egress (Sum)

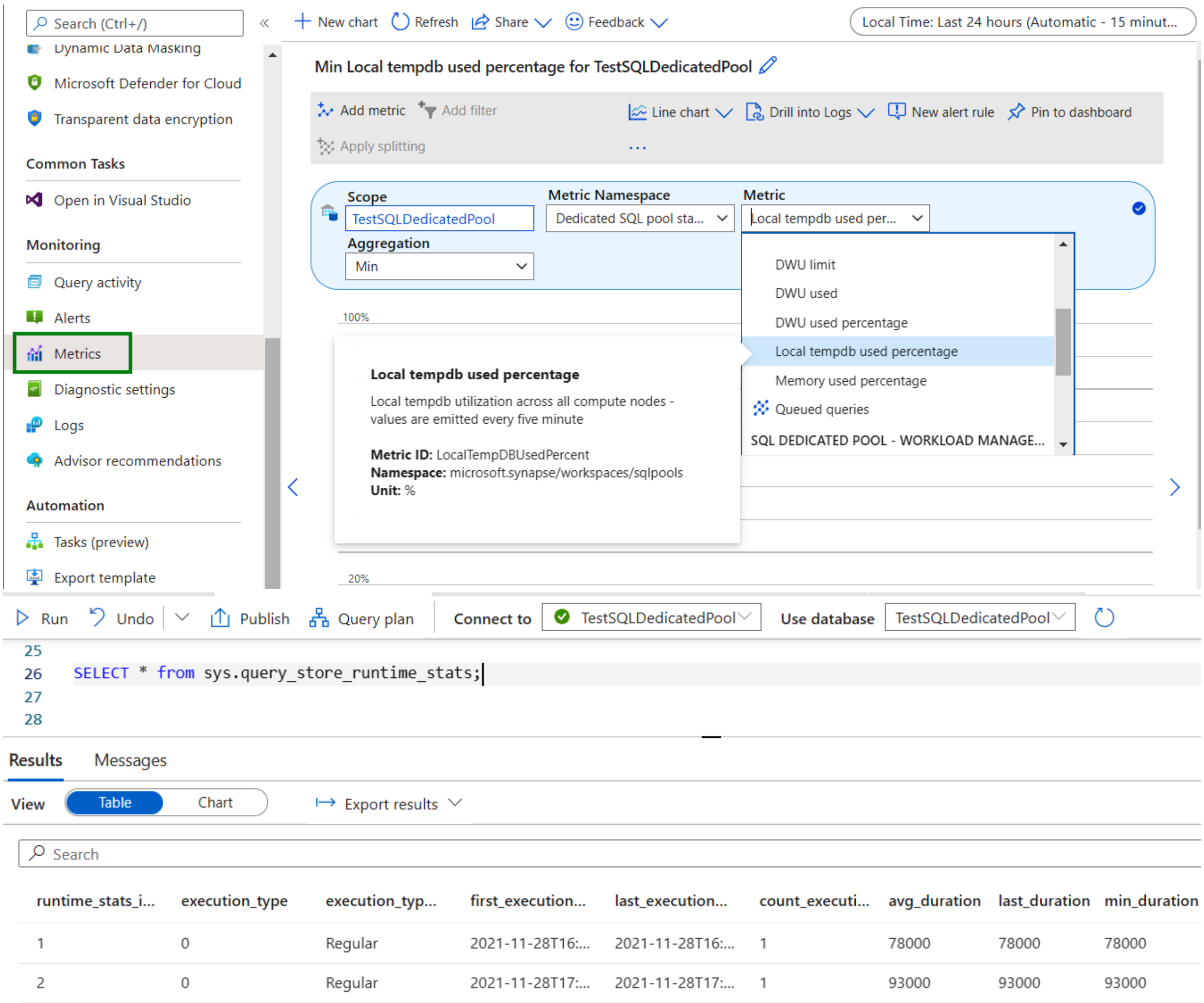

LogAnalyticsWS | Logs
Log Analytics workspace
Search (Ctrl+/)
Data Export
Linked storage accounts
Network Isolation
General
Workspace summary
Workbooks
Logs
Solutions
Usage and estimated costs
Properties
Service Map
Workspace Data Sources
Virtual machines
Storage accounts logs
SampleQuery
LogAnalyticsWS
Select scope
Tables
Queries
Functions
Search
Filter
Group by: Solution
Collapse all
Favorites
You can add favorites by clicking on the ☆ icon
LogManagement
AzureActivity
Operation
StorageBlobLogs
AccountName (string)
AuthenticationHash (string)
AuthenticationType (string)

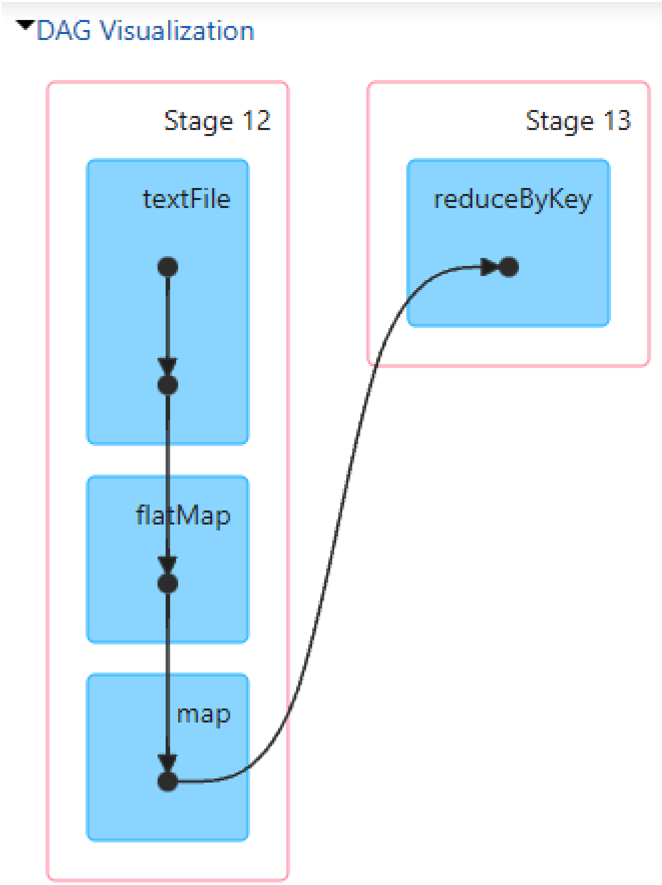

SampleQuery
LogAnalyticsWS
Select scope
Run
Time range : Last 7 days
Save
Share
1 StorageBlobLogs |
2 take 100 |where ServerLatencyMs > 20 |
3 sort by DurationMs |
4 project TimeGenerated, OperationName, DurationMs
Tables
Queries
Functions
Search
Filter
Group by: Solution
Collapse all
Favorites
You can add favorites by clicking on the ☆ icon
LogManagement
AzureActivity
Operation
StorageBlobLogs
AccountName (string)
AuthenticationHash (string)
AuthenticationType (string)
AuthorizationDetails (dynami...
CallerIpAddress (string)
Results
Chart
Columns
Display time (UTC+00:00)
Completed. Showing results from the last 7 days.
TimeGenerated [UTC]
OperationName
DurationMs
12/22/2021, 11:09:49.078 PM ListBlobs 1,332
12/22/2021, 3:14:00.013 AM ListBlobs 857
12/22/2021, 3:13:59.752 AM GetBlobServiceProper... 716
12/22/2021, 3:14:02.986 AM ListBlobs 334
12/22/2021, 11:09:47.864 PM ListBlobs 305
12/22/2021, 3:13:59.680 AM ListBlobs 248
12/22/2021, 3:14:06.182 AM ListBlobs 235
12/22/2021, 3:14:29.327 AM ListBlobs 233
12/22/2021, 3:15:08.102 AM GetBlob 61


All pipeline runs > CopyfromBlobToADLSG2 - Activity runs > FetchDataFromBlob
Dashboards
Runs
Pipeline runs
Trigger runs
Runtimes & sessions
Integration runtimes
Data flow debug
Notifications
Alerts & metrics
FetchDataFromBlob
Cluster startup time: 2m 52s Number of transformations: 2 Data flow status: Success
Refresh
Auto refresh
On
Edit transformation
SampleSource
Sink:
SampleDestinati...
SampleDestination diagnostics
Sink transformation
Status Succeeded
Total columns 7
New 0
Updated 7
Stream information
Sink processing time 3s 794ms
Rows calculated 20
Total partition 1
Stage time 2s 653ms
Last update (IST) 12/27/2021, 8:59:37 PM
Skewness -
Kurtosis -
Pre commands duration 48ms
Merge files duration 273ms

Dashboards
Runs
Pipeline runs
Trigger runs
Runtimes & sessions
Integration runtimes
Data flow debug
Notifications
Alerts & metrics
Pipeline runs
Triggered
Debug
Rerun
Cancel
Refresh
Edit columns
List
Gantt
Search by run ID or name
Chennai, Kolkata, Mu... : Last 24 hours
Pipeline name : All
Status : All
Runs : Latest runs
Triggered by : All
Copy filters
Add filter
Showing 1 - 3 items
Pipeline name
Run start
Run end
Duration
CopyfromBlobToA...
10/23/21, 1:26:01 PM
10/23/21, 1:29:55 PM
00:03:54
CopyfromBlobToADLSG2
Consumption
1:00 PM
10/23/21, 1:24:59 PM
00:03:59
CopyfromBlobToADLSG2
10/23/21, 1:16:01 PM
10/23/21, 1:19:43 PM
00:03:41

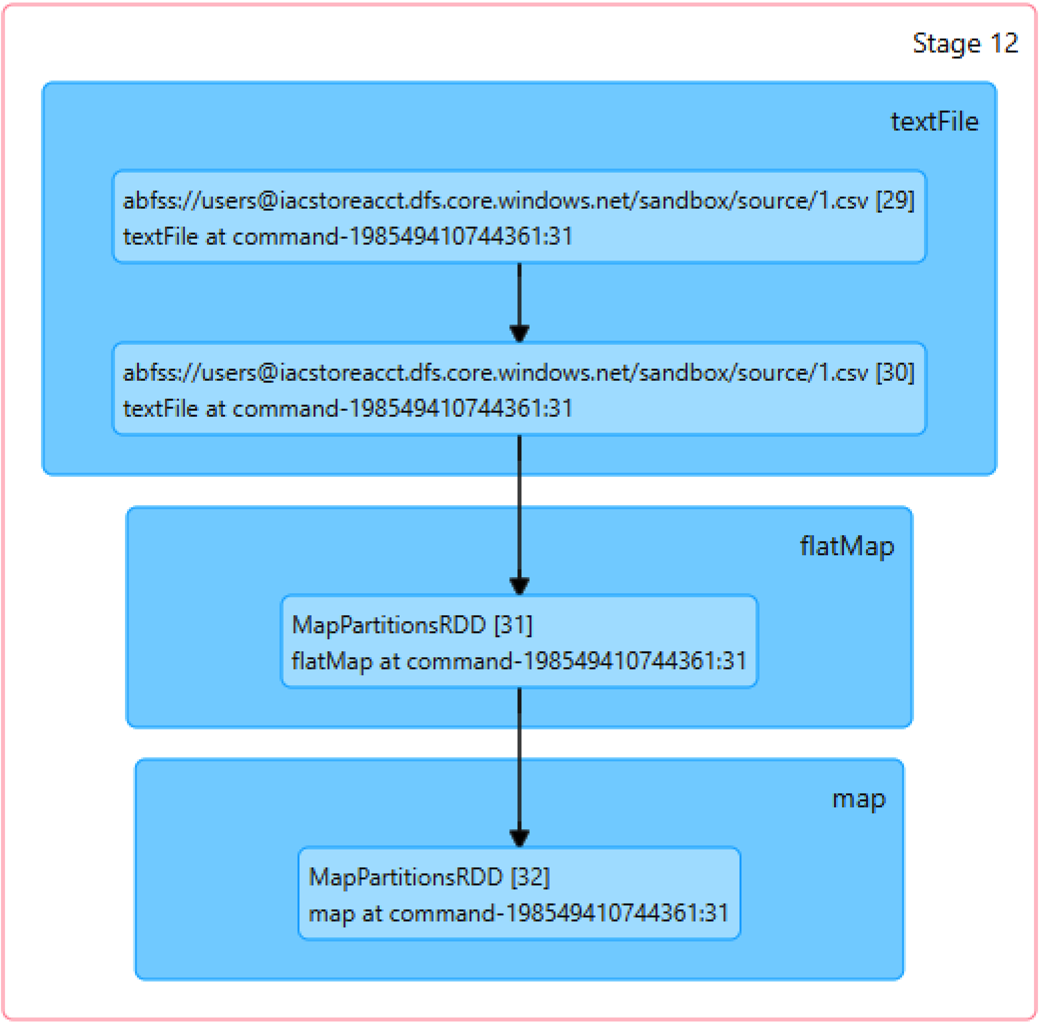

Pipeline run consumption
Quantity
Unit
Pipeline orchestration
Activity Runs
1
Activity runs
Data flow
General purpose
0.5001
vCore-hour
Compute optimized
0.0000
vCore-hour
Memory optimized
0.0000
vCore-hour
Learn more
Pricing calculator

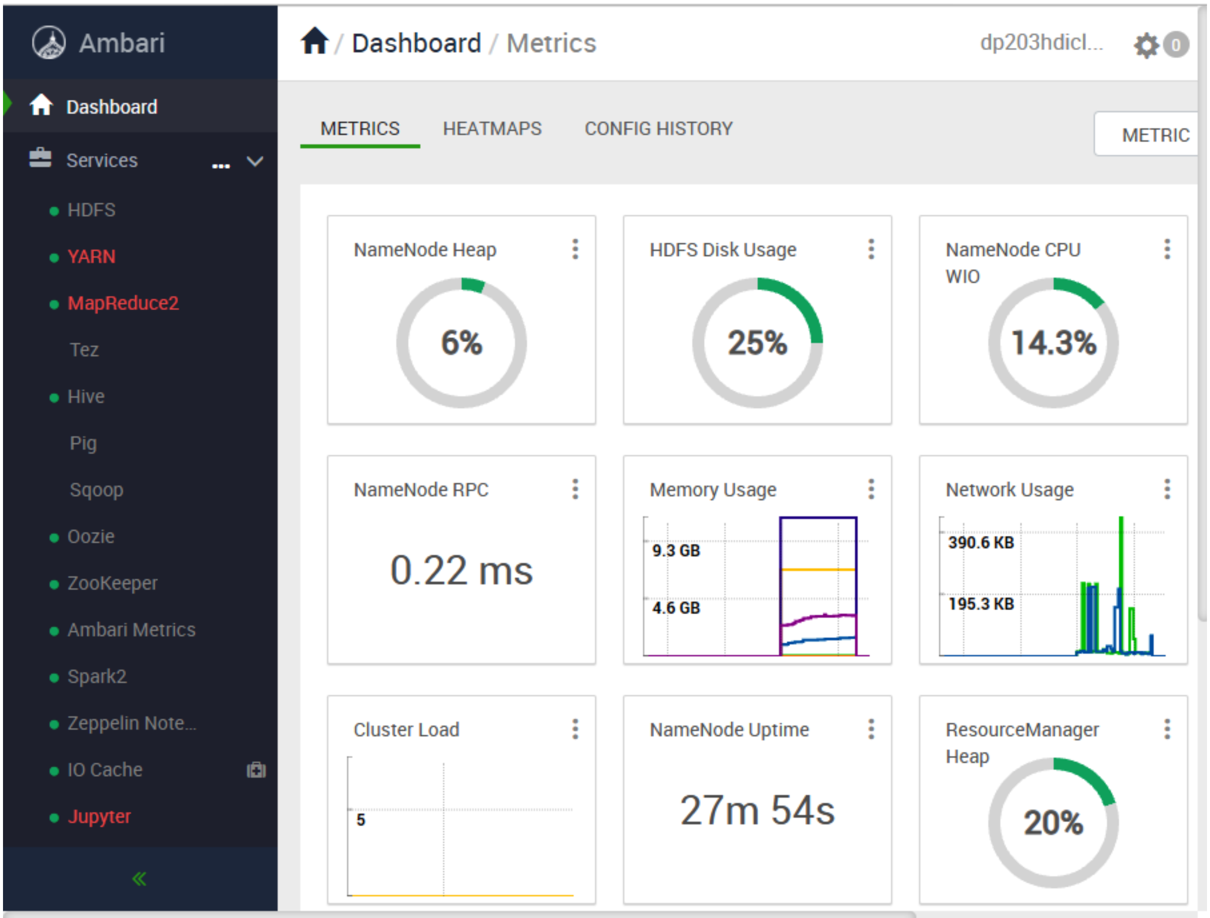

Pipeline runs
Triggered
Debug
Rerun
Cancel
Refresh
List
Gantt
Search by run ID or name
Chennai, Kolkata, Mu... : Last 24 hours
Copy filters
Add filter
Oct 23 01 PM
Oct 23 01 PM
Oct 23 01 PM
Oct 23 01 PM
Oct 23 01 PM
Oct 23 01 PM
Oct 23 02 PM
Oct 23 02 PM
Oct 23 02 PM
Oct 23 02 PM
CopyfromBlob...
Pipeline run ID
a08a8f55-bc54-4054-b3be-1802c25afb8d
Status
Succeeded
Start date
10/23/2021, 2:31:00 PM
End date
10/23/2021, 2:34:58 PM
Duration
3 minutes
Average run time
4 minutes
Duration deviation
2%
Longest duration
6 minutes
Shortest duration
3 minutes
Actions

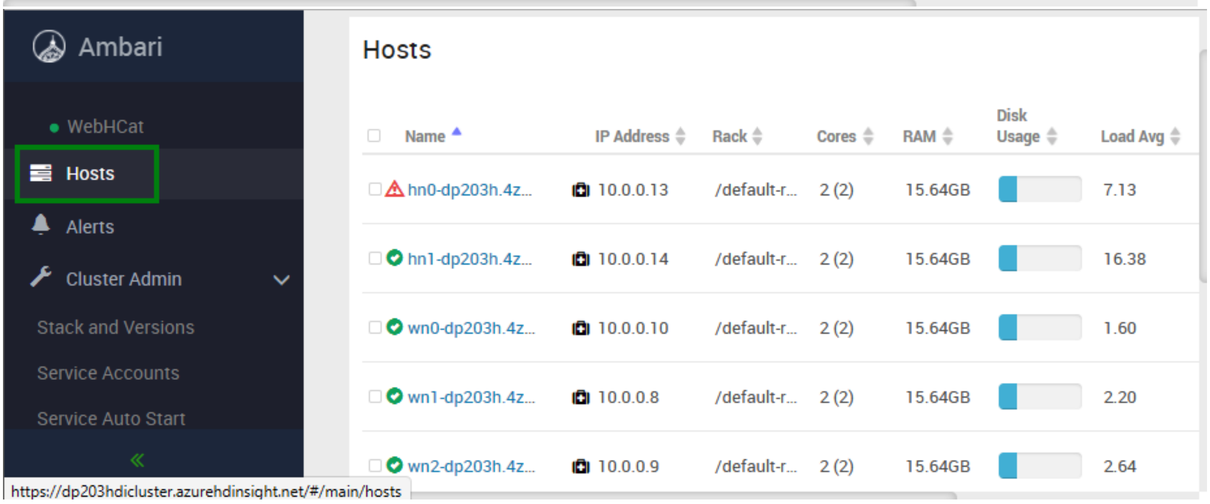

Search (Ctrl+/)
Dynamic Data Masking
Microsoft Defender for Cloud
Transparent data encryption
Common Tasks
Open in Visual Studio
Monitoring
Query activity
Alerts
Metrics
Diagnostic settings
Logs
Advisor recommendations
Automation
Tasks (preview)
Export template
New chart
Refresh
Share
Feedback
Local Time: Last 24 hours (Automatic - 15 minut...
Min Local tempdb used percentage for TestSQLDedicatedPool
Add metric
Add filter
Line chart
Drill into Logs
New alert rule
Pin to dashboard
Apply splitting
Scope
TestSQLDedicatedPool
Metric Namespace
Dedicated SQL pool sta...
Metric
Local tempdb used per...
Aggregation
Min
DWU limit
DWU used
DWU used percentage
Local tempdb used percentage
Memory used percentage
Queued queries
SQL DEDICATED POOL - WORKLOAD MANAGE...
100%
20%
Local tempdb used percentage
Local tempdb utilization across all compute nodes - values are emitted every five minute
Metric ID: LocalTempDBUsedPercent
Namespace: microsoft.synapse/workspaces/sqlpools
Unit: %
Run
Undo
Publish
Query plan
Connect to
TestSQLDedicatedPool
Use database
TestSQLDedicatedPool
SELECT * from sys.query_store_runtime_stats;
Results
Messages
View
Table
Chart
Export results
Search
runtime_stats_i... execution_type execution_typ... first_execution... last_execution... count_executi... avg_duration last_duration min_duration
1 0 Regular 2021-11-28T16:... 2021-11-28T16:... 1 78000 78000 78000
2 0 Regular 2021-11-28T17:... 2021-11-28T17:... 1 93000 93000 93000


DAG Visualization
Stage 12
textFile
flatMap
map
Stage 13
reduceByKey

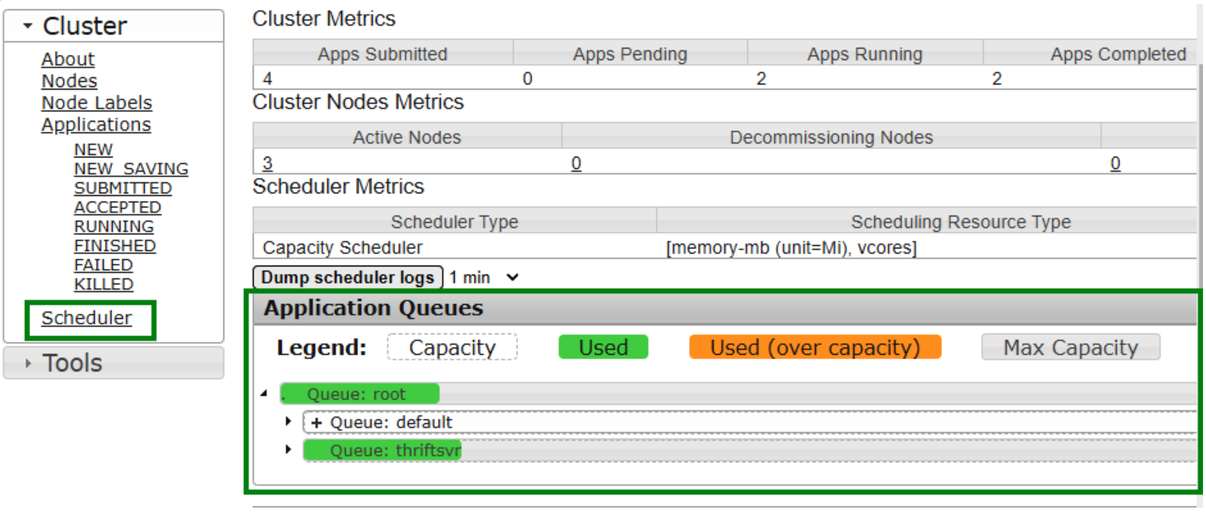
▼DAG Visualization

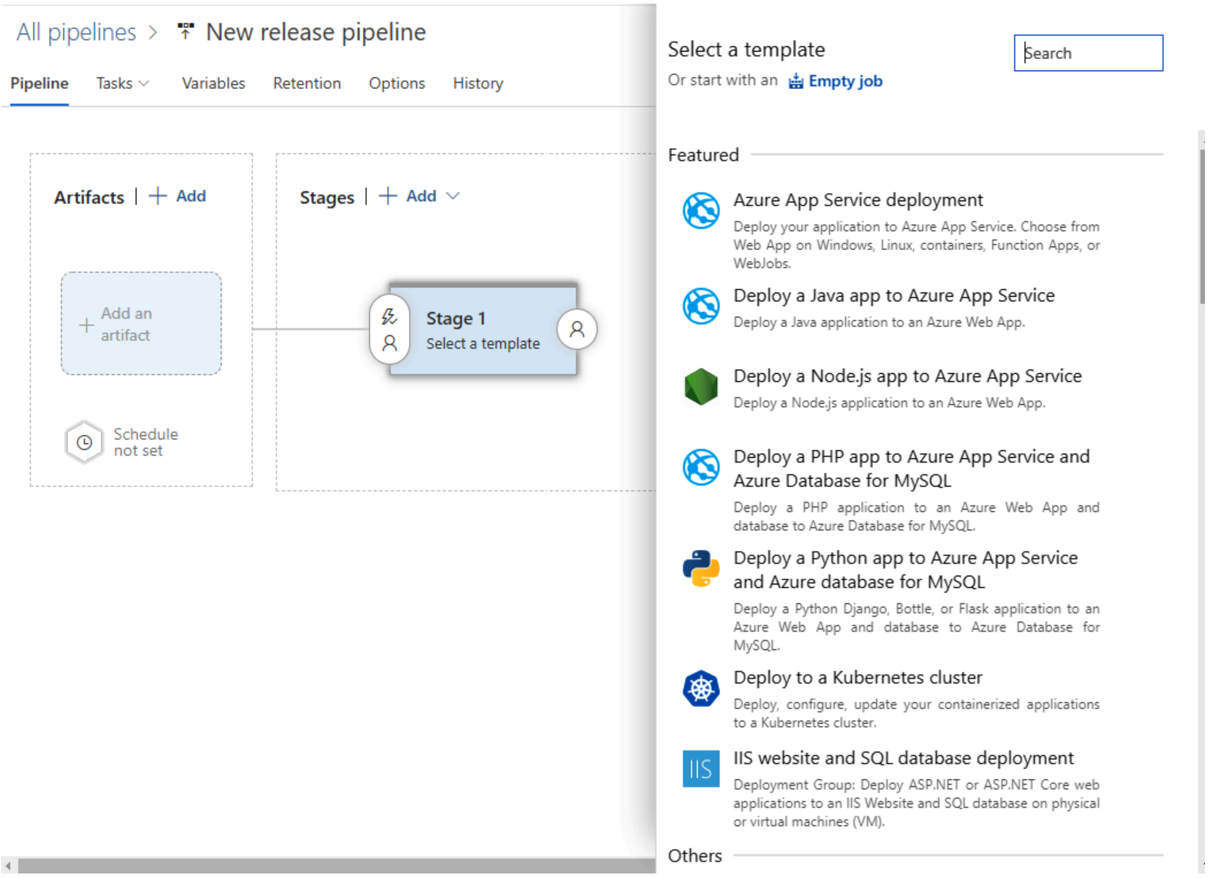


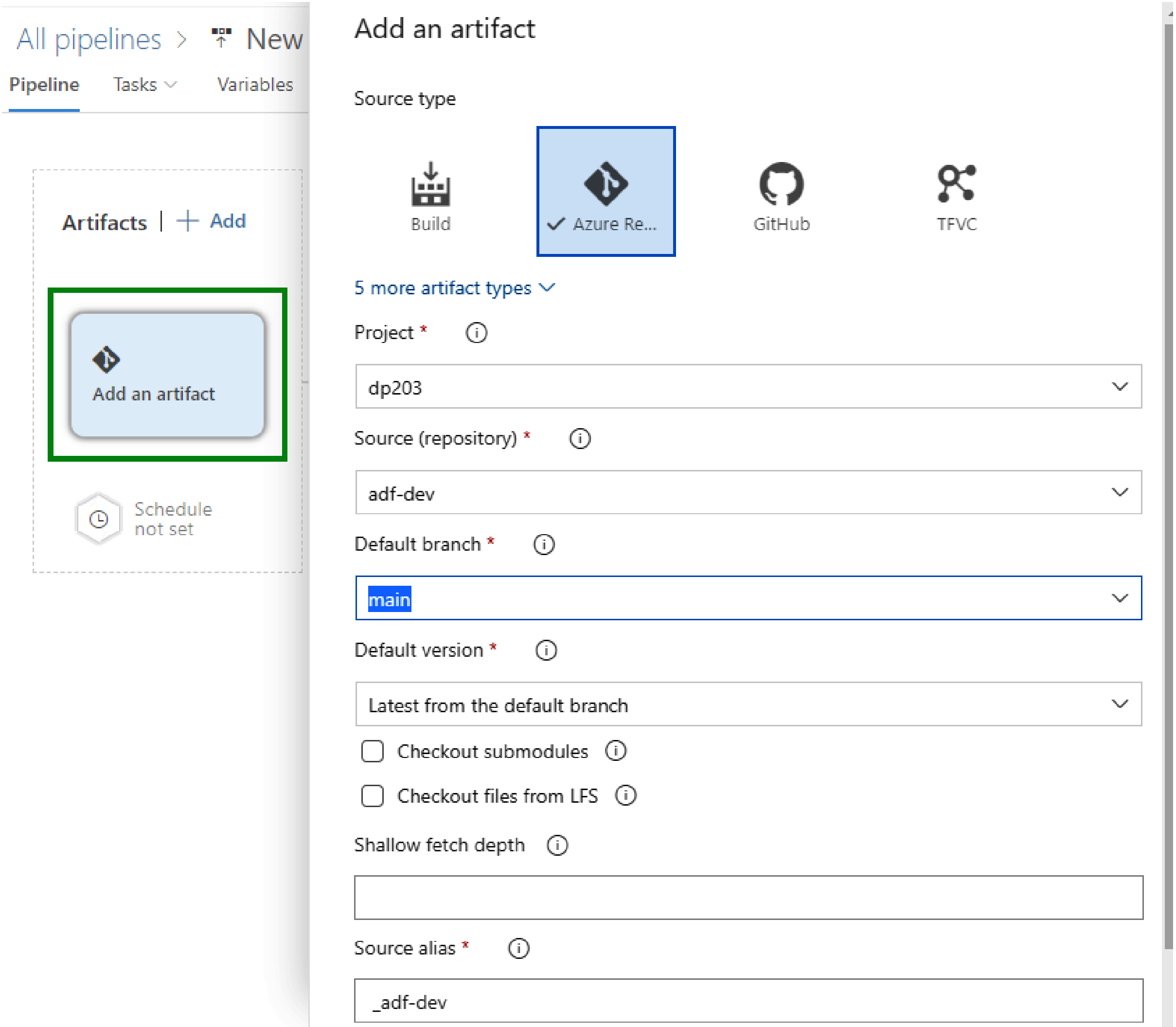

Ambari
Dashboard / Metrics
dp203hdicl...
Dashboard
Services
HDFS
YARN
MapReduce2
Tez
Hive
Pig
Sqoop
Oozie
ZooKeeper
Ambari Metrics
Spark2
Zeppelin Note...
IO Cache
Jupyter
METRICS
HEATMAPS
CONFIG HISTORY
METRIC
NameNode Heap
6%
HDFS Disk Usage
25%
NameNode CPU WIO
14.3%
NameNode RPC
0.22 ms
Memory Usage
9.3 GB
4.6 GB
Network Usage
390.6 KB
195.3 KB
Cluster Load
5
NameNode Uptime
27m 54s
ResourceManager Heap
20%

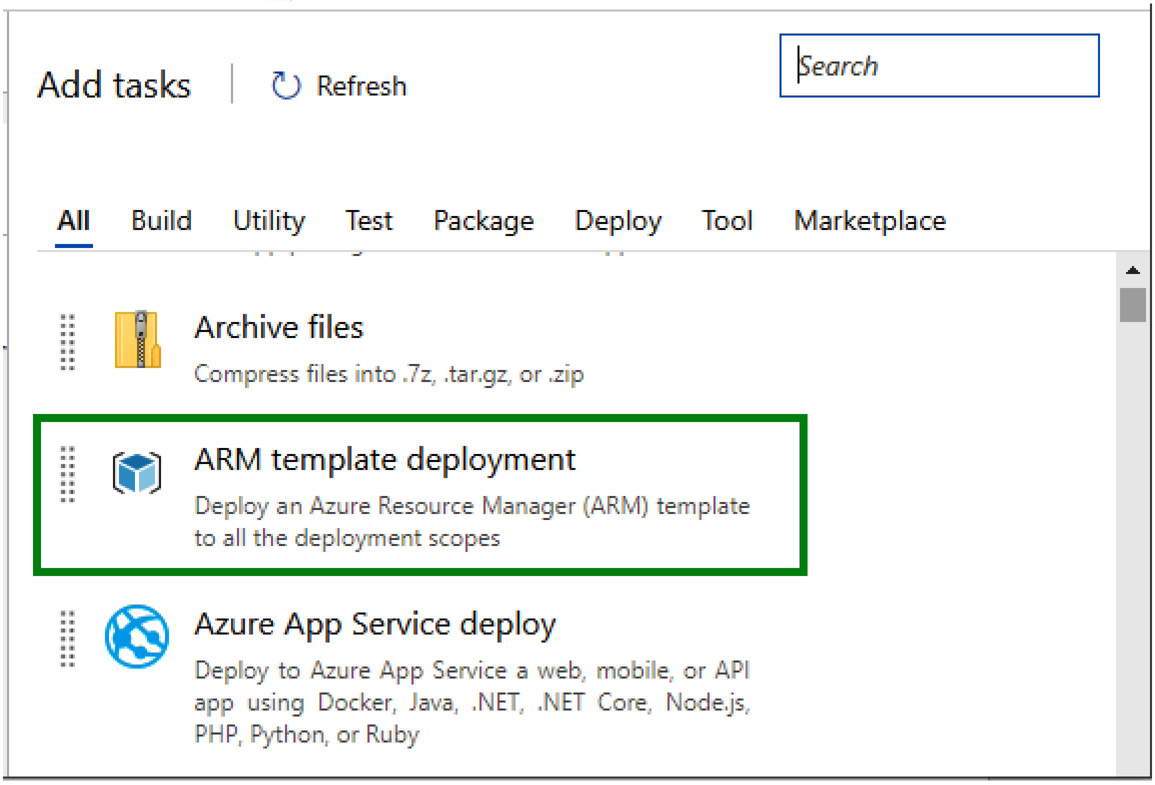

Ambari
WebHCat
Hosts
Alerts
Cluster Admin
Stack and Versions
Service Accounts
Service Auto Start
Hosts
Name
IP Address
Rack
Cores
RAM
Disk Usage
Load Avg
hn0-dp203h.4z... 10.0.0.13 /default-r... 2 (2) 15.64GB 7.13
hn1-dp203h.4z... 10.0.0.14 /default-r... 2 (2) 15.64GB 16.38
wn0-dp203h.4z... 10.0.0.10 /default-r... 2 (2) 15.64GB 1.60
wn1-dp203h.4z... 10.0.0.8 /default-r... 2 (2) 15.64GB 2.20
wn2-dp203h.4z... 10.0.0.9 /default-r... 2 (2) 15.64GB 2.64
https://dp203hdicluster.azurehdinsight.net/#/main/hosts

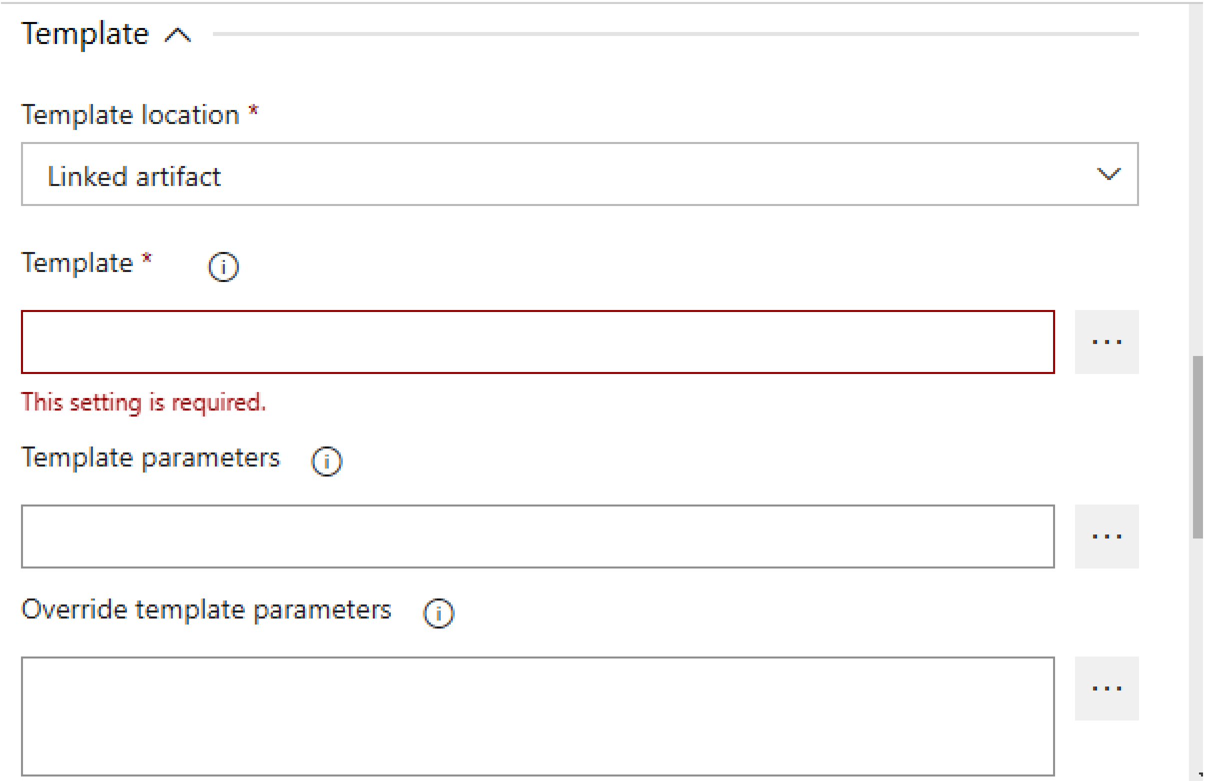

Host Metrics
LAST 1 HOUR
100%
50%
CPU Usage
139.6 GB
93.1 GB
46.5 GB
Disk Usage
20
10
Load
9.3 GB
4.6 GB
Memory Usage
3.8 MB
1.9 MB
Network Usage
200
100
Processes

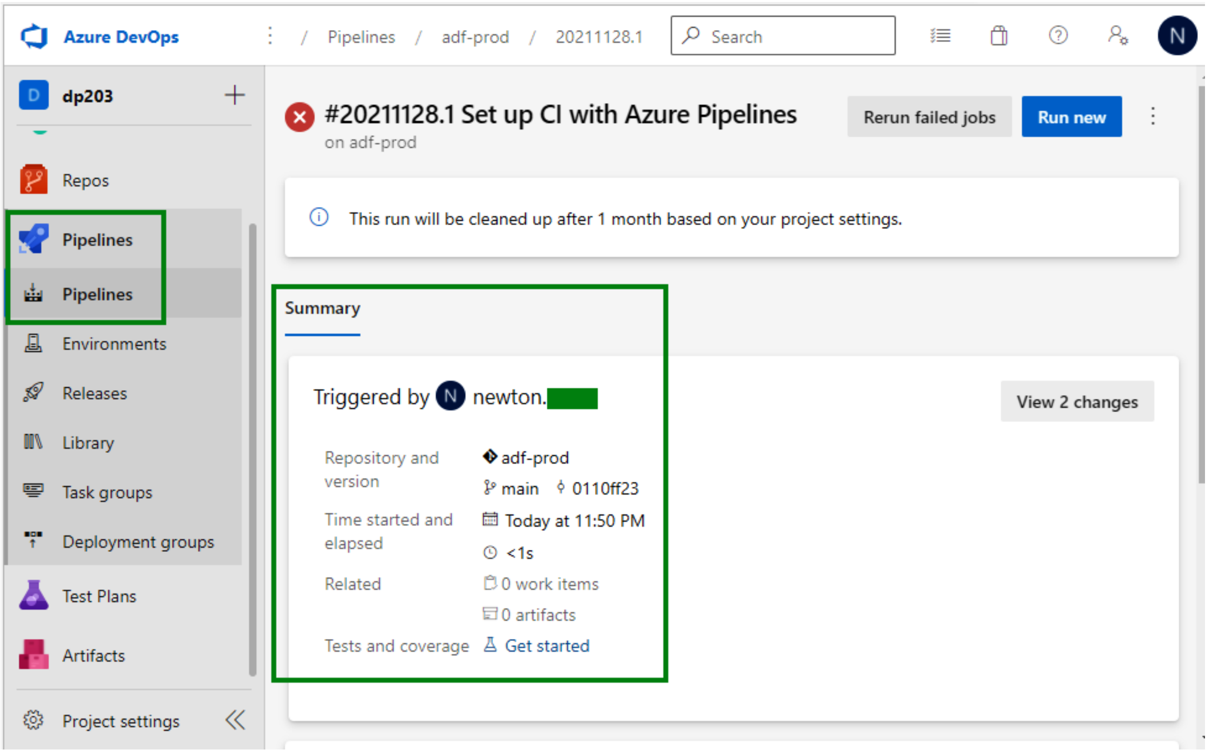

Cluster
About
Nodes
Node Labels
Applications
NEW
NEW_SAVING
SUBMITTED
ACCEPTED
RUNNING
FINISHED
FAILED
KILLED
Scheduler
Tools
Cluster Metrics
Apps Submitted
Apps Pending
Apps Running
Apps Completed
4
0
2
2
Cluster Nodes Metrics
Active Nodes
Decommissioning Nodes
3
0
0
Scheduler Metrics
Scheduler Type
Scheduling Resource Type
Capacity Scheduler
[memory-mb (unit=Mi), vcores]
Dump scheduler logs
1 min
Application Queues
Legend:
Capacity
Used
Used (over capacity)
Max Capacity
Queue: root
+ Queue: default
Queue: thriftsvr


All pipelines > New release pipeline
Pipeline
Tasks
Variables
Retention
Options
History
Artifacts | + Add
Add an artifact
Schedule not set
Stages | + Add
Stage 1
Select a template
Select a template
Or start with an Empty job
Search
Featured
Azure App Service deployment
Deploy your application to Azure App Service. Choose from Web App on Windows, Linux, containers, Function Apps, or WebJobs.
Deploy a Java app to Azure App Service
Deploy a Java application to an Azure Web App.
Deploy a Node.js app to Azure App Service
Deploy a Node.js application to an Azure Web App.
Deploy a PHP app to Azure App Service and Azure Database for MySQL
Deploy a PHP application to an Azure Web App and database to Azure Database for MySQL.
Deploy a Python app to Azure App Service and Azure database for MySQL
Deploy a Python Django, Bottle, or Flask application to an Azure Web App and database to Azure Database for MySQL.
Deploy to a Kubernetes cluster
Deploy, configure, update your containerized applications to a Kubernetes cluster.
IIS website and SQL database deployment
Deployment Group: Deploy ASP.NET or ASP.NET Core web applications to an IIS Website and SQL database on physical or virtual machines (VM).
Others

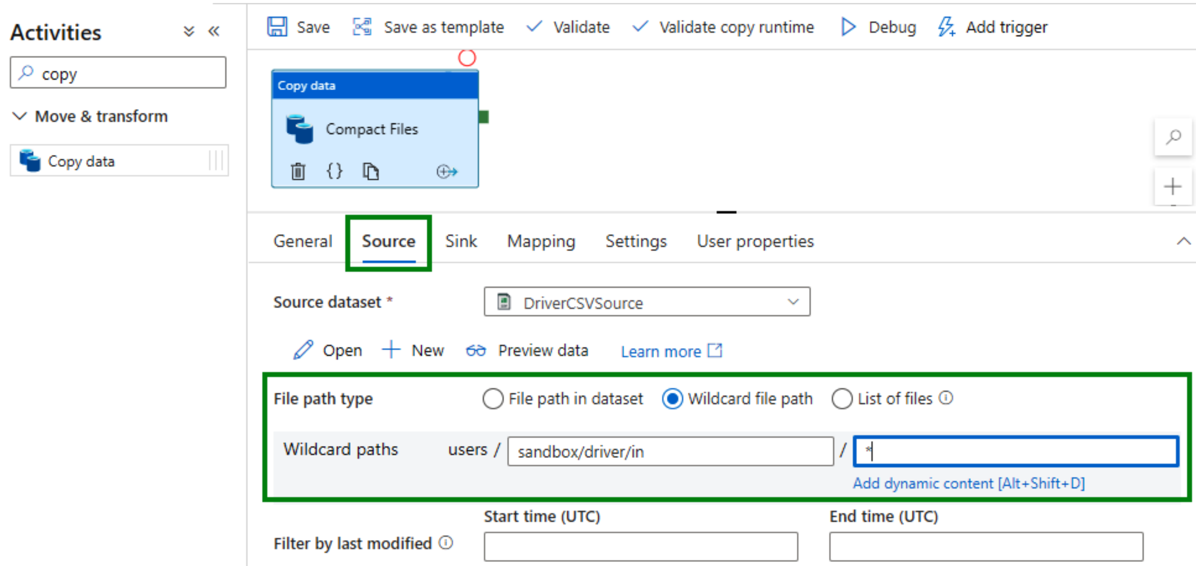

All pipelines > New
Pipeline
Tasks
Variables
Artifacts | + Add
Add an artifact
Schedule not set
Add an artifact
Source type
Build
Azure Re...
GitHub
TFVC
5 more artifact types
Project *
dp203
Source (repository) *
adf-dev
Default branch *
main
Default version *
Latest from the default branch
Checkout submodules
Checkout files from LFS
Shallow fetch depth
Source alias *
_adf-dev

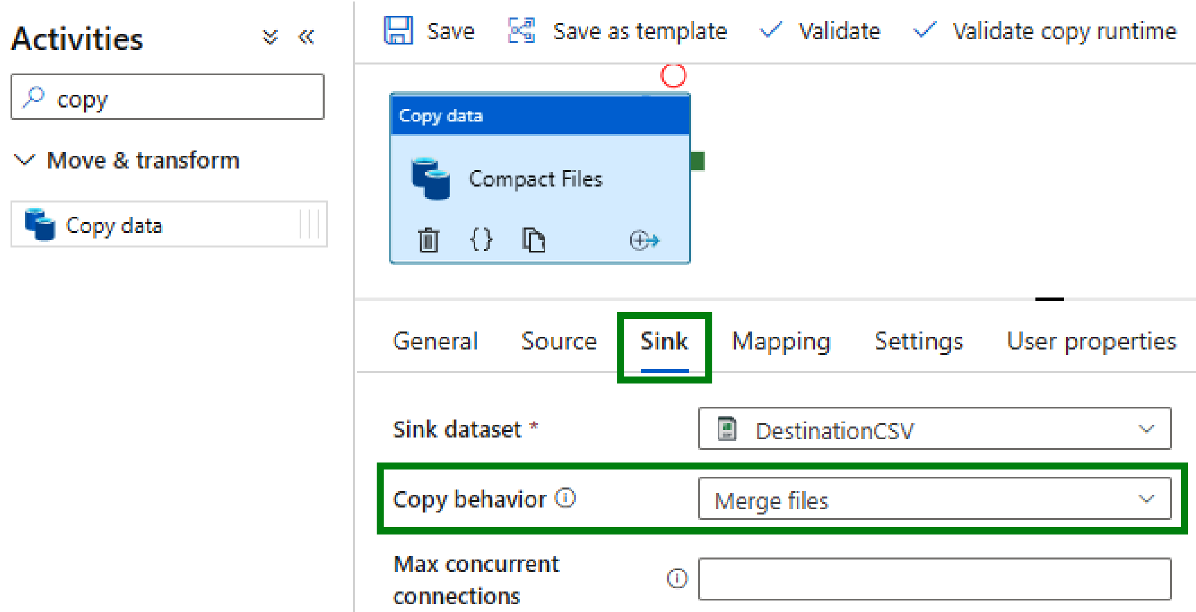

Add tasks
Refresh
Search
All
Build
Utility
Test
Package
Deploy
Tool
Marketplace
Archive files
Compress files into .7z, .tar.gz, or .zip
ARM template deployment
Deploy an Azure Resource Manager (ARM) template to all the deployment scopes
Azure App Service deploy
Deploy to Azure App Service a web, mobile, or API app using Docker, Java, .NET, .NET Core, Node.js, PHP, Python, or Ruby


Template
Template location *
Linked artifact
Template *
This setting is required.
Template parameters
Override template parameters


Azure DevOps
Pipelines / adf-prod / 20211128.1
Search
dp203
Repos
Pipelines
Pipelines
Environments
Releases
Library
Task groups
Deployment groups
Test Plans
Artifacts
Project settings
#20211128.1 Set up CI with Azure Pipelines
on adf-prod
Rerun failed jobs
Run new
This run will be cleaned up after 1 month based on your project settings.
Summary
Triggered by newton.[redacted]
View 2 changes
Repository and version
adf-prod
main 0110ff23
Time started and elapsed
Today at 11:50 PM
<1s
Related
0 work items
0 artifacts
Tests and coverage
Get started

Chapter 14: Optimizing and Troubleshooting Data Storage and Data Processing





```sql

-- View the rows in the table
SELECT * FROM dbo.CustomerContact;

```

Results Messages

View Table Chart Export results

Search

CustomerID	Name	Email
2	Bran	bryan@domain.com
4	Demin	demin@wrongdomain
1	Arielle	arielle
3	Cathy	cathy@domain.com
5	Ethan	ethan@domain.com

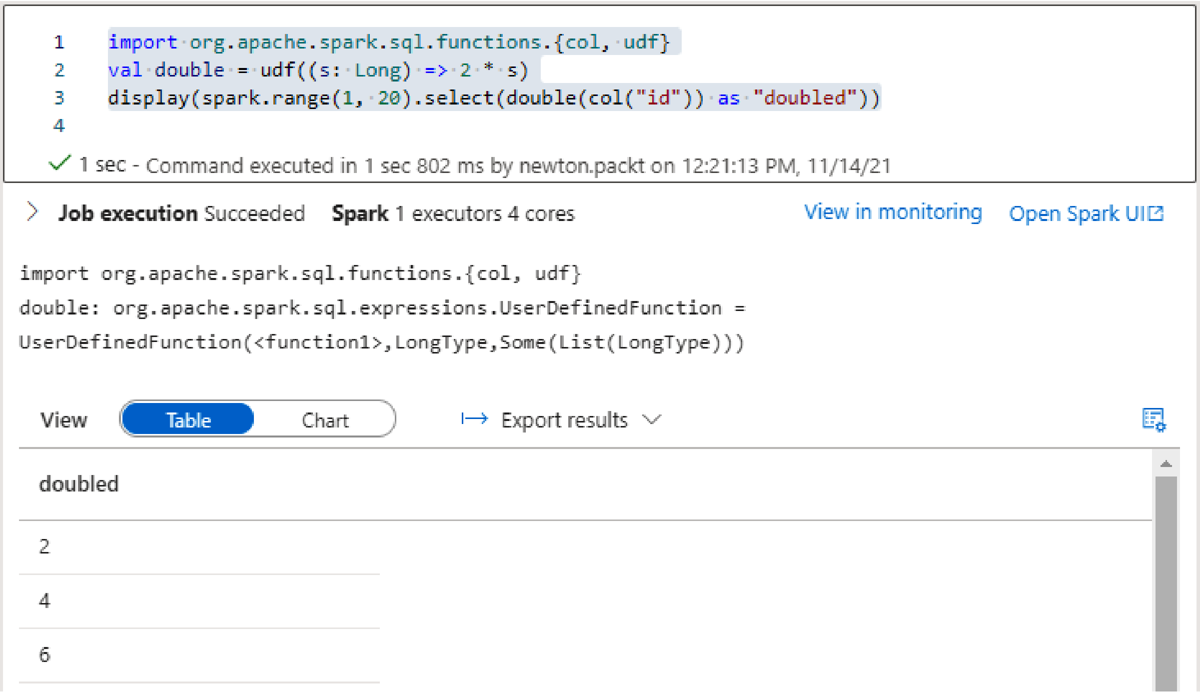
```sql
-- Try to use the UDF
SELECT CustomerID, Name, dbo.isValidEmail(Email) AS Email FROM dbo.CustomerContact;

```

Results Messages

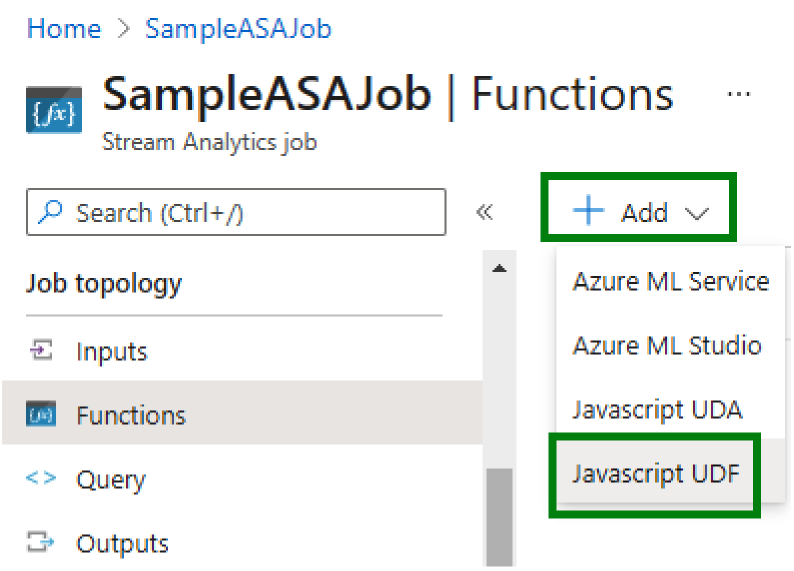
View Table Chart Export results

Search

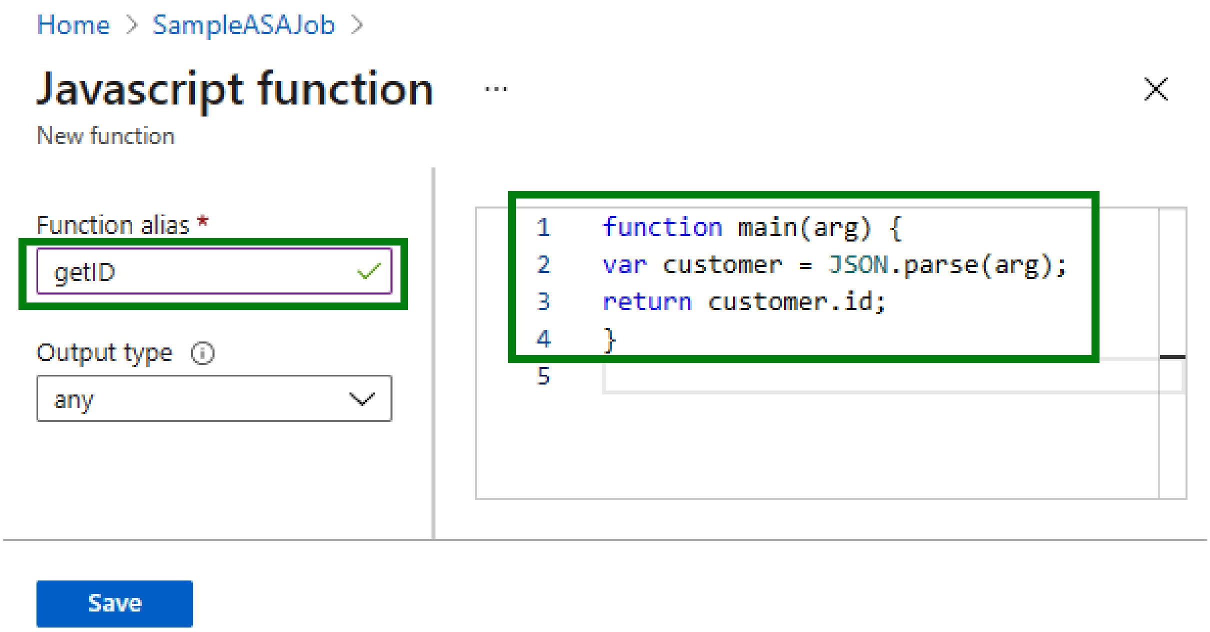
CustomerID	Name	Email
2	Bran	bryan@domain.com
4	Demin	Not Available
1	Arielle	Not Available
3	Cathy	cathy@domain.com
5	Ethan	ethan@domain.com

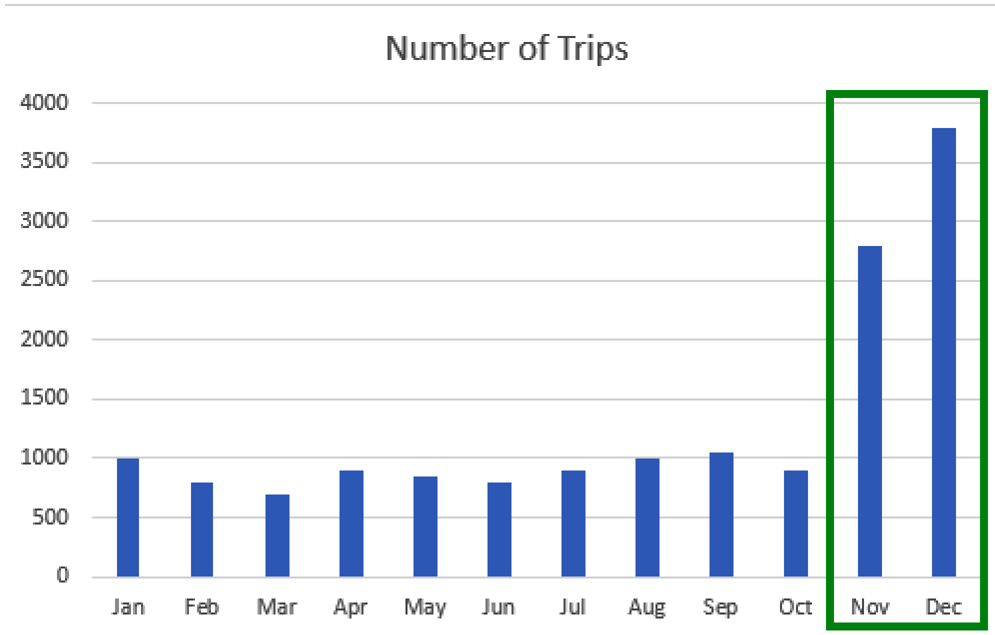


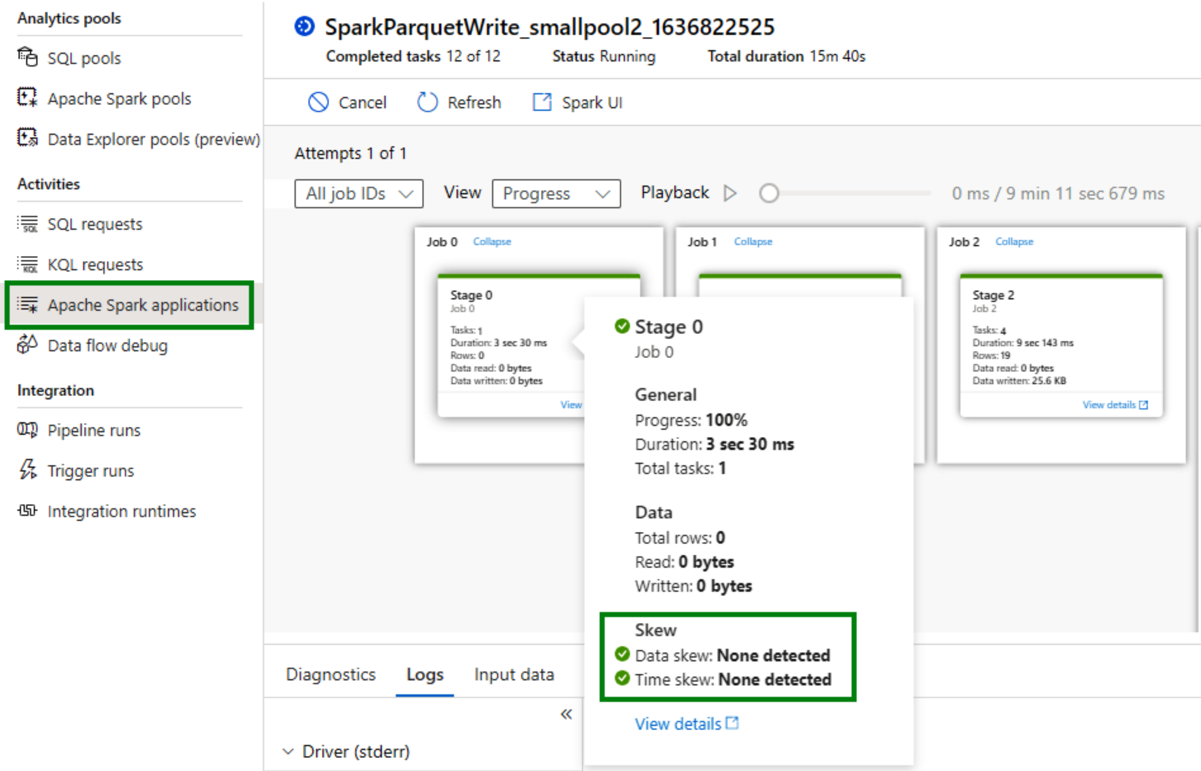
import org.apache.spark.sql.functions.{col, udf}
val double = udf((s: Long) => 2 * s)
display(spark.range(1, 20).select(double(col("id")) as "doubled"))
1 sec - Command executed in 1 sec 802 ms by newton.packt on 12:21:13 PM, 11/14/21
Job execution Succeeded Spark 1 executors 4 cores
View in monitoring
Open Spark UI
import org.apache.spark.sql.functions.{col, udf}
double: org.apache.spark.sql.expressions.UserDefinedFunction =
UserDefinedFunction(<function1>,LongType,Some(List(LongType)))
View
Table
Chart
Export results
doubled
2
4
6



Home > SampleASAJob
SampleASAJob | Functions
Stream Analytics job
Search (Ctrl+/)
Add
Azure ML Service
Azure ML Studio
Javascript UDA
Javascript UDF
Job topology
Inputs
Functions
Query
Outputs


Home > SampleASAJob >
Javascript function
New function
Function alias *
getID
Output type
any
1 function main(arg) {
2 var customer = JSON.parse(arg);
3 return customer.id;
4 }
5
Save


Number of Trips
4000
3500
3000
2500
2000
1500
1000
500
0
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec



Duration	GC Time	Spill (Disk)
76.0 ms		124 B / 4
76.0 ms		122 B / 4

```
29   SELECT * FROM dbo.DimDriver;
```

Results Messages

View Table Chart Export results

Search

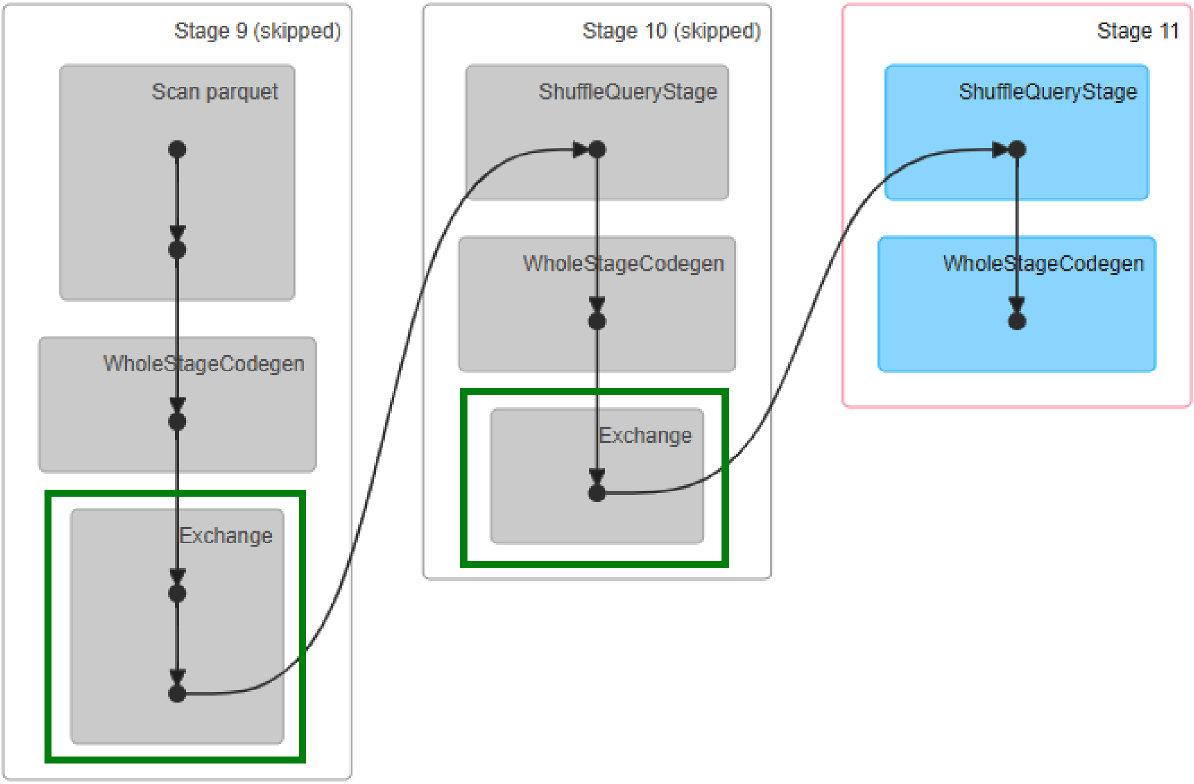
driverId	firstName	middleName	lastName	city	gender	salary
211	Brandon		Rhodes	New York	Male	3000
213	Dennis		Li	Florida	Male	2500
210	Alicia		Yang	New York	Female	2000
215	Maile		Green	Florida	Female	4000
212	Cathy		Mayor	California	Female	3000
214	Jeremey		Stilton	Arizona	Male	2500

Search

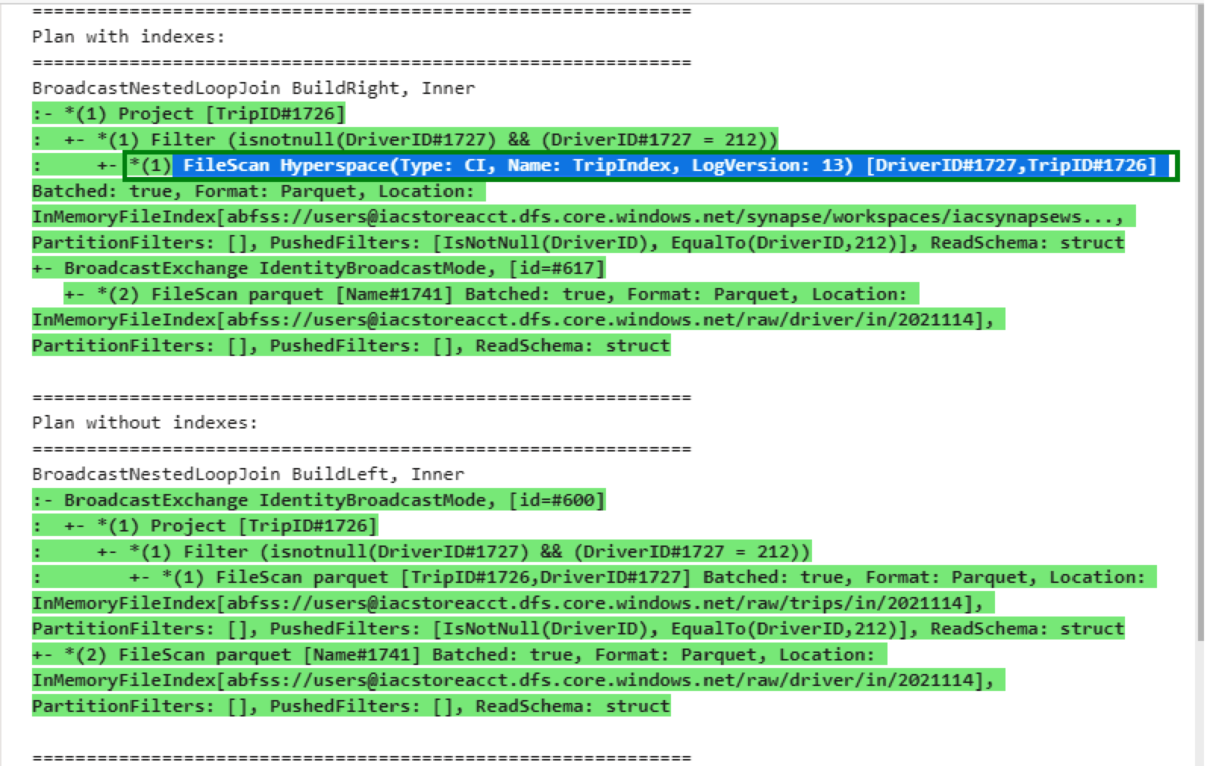
driverId	FirstName	MiddleNmae	LastName	City	Gender	Salary
215	Maile		Green	Florida	Female	4000
212	Cathy		Mayor	California	Female	3000
214	Jeremey		Stilton	Arizona	Male	2500

```xml
<?xml version="1.0" encoding="utf-8"?>
<dsql_query number_nodes="1" number_distributions="60" number_distributions_per_node="60">
    <sql>SELECT [gender], SUM([salary]) as Totalsalary FROM dbo.DimDriver GROUP BY [gender]
    </sql>
    <materialized_view_candidates>
    <dsql_operations total_cost="0.01056" total_number_operations="5">
        <dsql_operation operation_type="RND_ID">
        <dsql_operation operation_type="ON">
        <dsql_operation operation_type="SHUFFLE_MOVE">
            <operation_cost cost="0.01056" accumulative_cost="0.01056" average_rowsize="44"
            output_rows="31.6228" GroupNumber="8"/>
            <source_statement>SELECT [T1_1].[gender] AS [gender], [T1_1].[col] AS [col] FROM
            (SELECT SUM([T2_1].[salary]) AS [col], [T2_1].[gender] AS [gender] FROM
            [dp203dedicatedsql].[dbo].[DimDriver] AS T2_1 GROUP BY [T2_1].[gender]) AS T1_1
            OPTION (MAXDOP 1, MIN_GRANT_PERCENT = [MIN_GRANT], DISTRIBUTED_MOVE(N''))
            </source_statement>
            <destination_table>[TEMP_ID_5]</destination_table>
            <shuffle_columns>gender;</shuffle_columns>
        </dsql_operation>
        <dsql_operation operation_type="RETURN">
        <dsql_operation operation_type="ON">
    </dsql_operations>
</dsql_query>
```

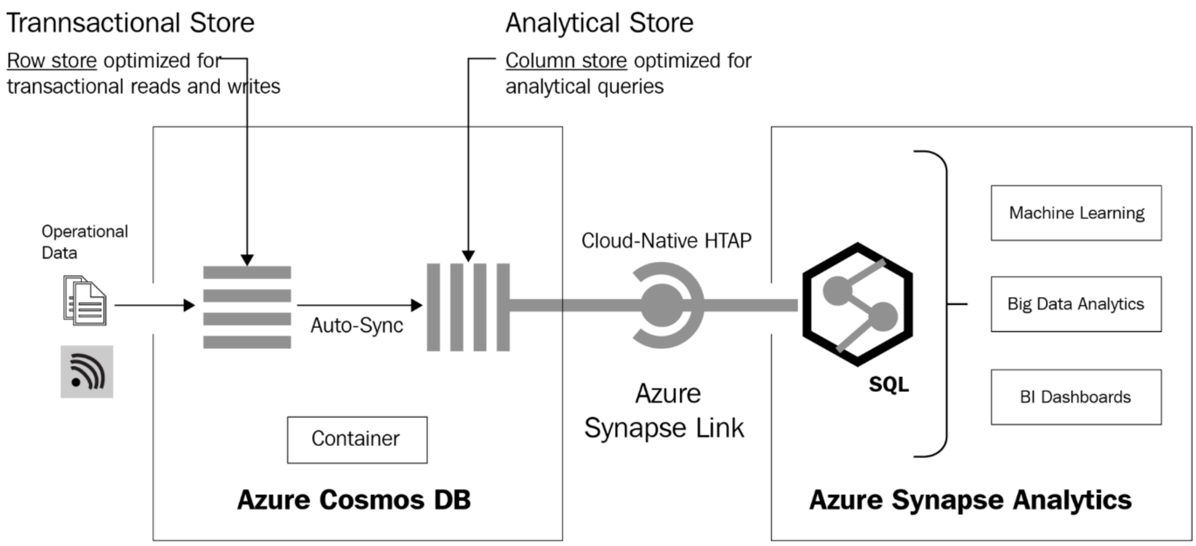
```xml
<?xml version="1.0" encoding="utf-8"?>
<dsql_query number_nodes="1"
            number_distributions="60"
            number_distributions_per_node="60">
   <sql>SELECT [Gender],SUM([Salary]) as TotalSalary FROM dbo.TempDriver GROUP BY [Gender]</sql>
   <materialized_view_candidates>
   <dsql_operations total_cost="0.00216"
                    total_number_operations="5">
      <dsql_operation operation_type="RND_ID">
      <dsql_operation operation_type="ON">
      <dsql_operation operation_type="SHUFFLE_MOVE">
         <operation_cost cost="0.00216"
                         accumulative_cost="0.00216"
                         average_rowsize="9"
                         output_rows="2"
                         GroupNumber="8"/>
         <source_statement>SELECT [T1_1].[Gender] AS [Gender], [T1_1].[col] AS [col] FROM (SELECT
         SUM([T2_1].[Salary]) AS [col], [T2_1].[Gender] AS [Gender] FROM
         [DedicatedSmall].[dbo].[TempDriver] AS T2_1 GROUP BY [T2_1].[Gender]) AS T1_1 OPTION
         (MAXDOP 1, MIN_GRANT_PERCENT = [MIN_GRANT], DISTRIBUTED_MOVE(N''))</source_statement>
         <destination_table>[TEMP_ID_8]</destination_table>
         <shuffle_columns>Gender;</shuffle_columns>
      </dsql_operation>
      <dsql_operation operation_type="RETURN">
      <dsql_operation operation_type="ON">
   </dsql_operations>
</dsql_query>
```

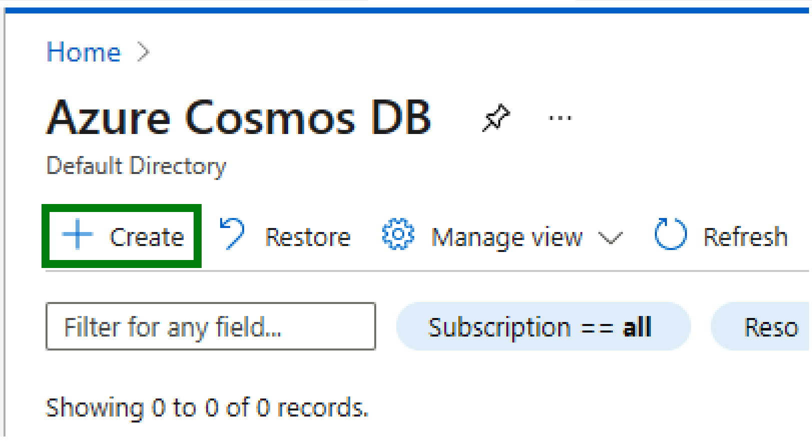


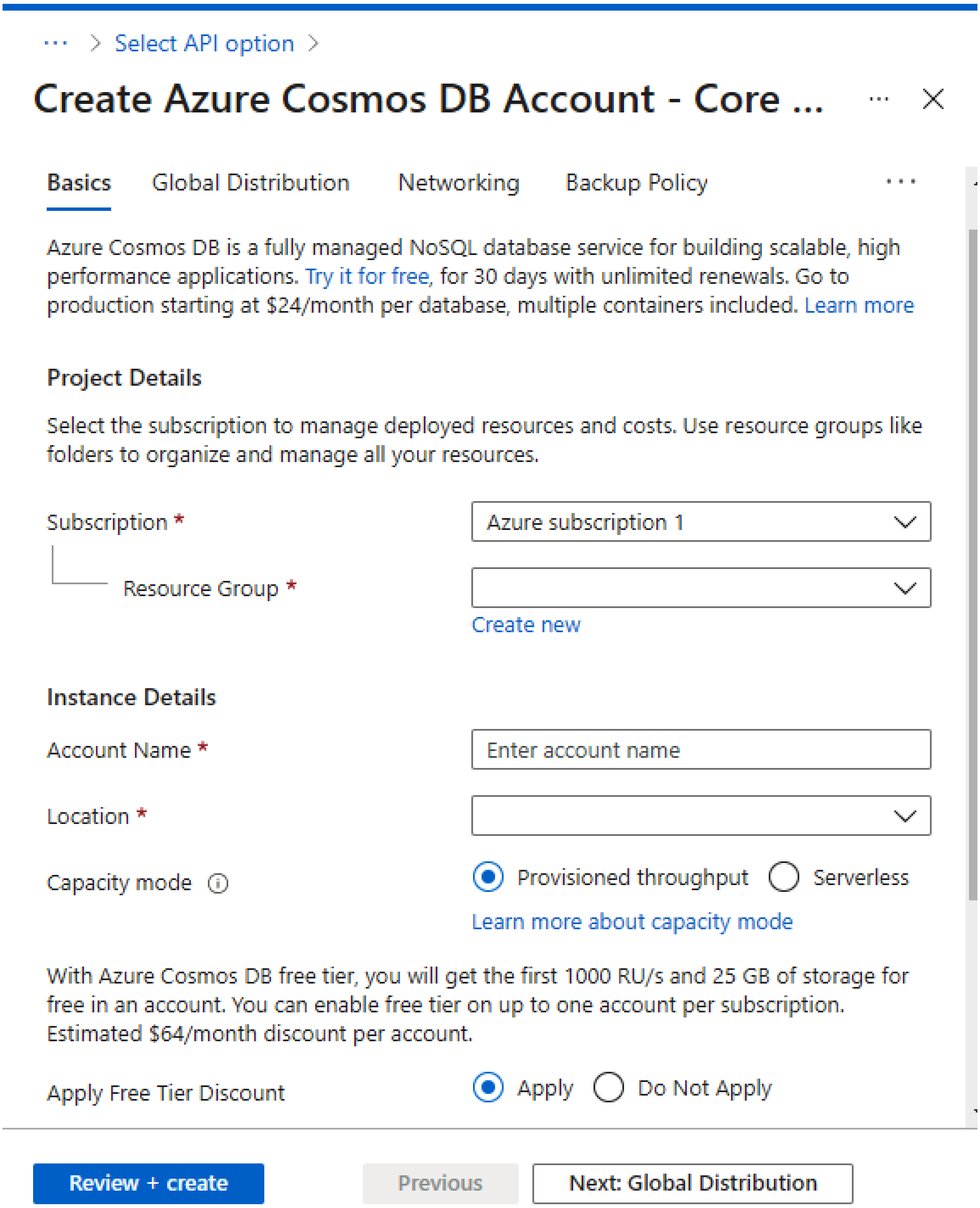

```
=============================================================
Plan with indexes:
=============================================================
Project [tripId#859, name#874]
+- BroadcastHashJoin [driverId#860], [driverId#873], Inner, BuildRight, false
   :- Filter isnotnull(driverId#860)
   :  +- ColumnarToRow
   :     +- FileScan Hyperspace(Type: CI, Name: TripIndex, LogVersion: 25) [driverId#860,tripId#859] Batched: true,
DataFilters: [isnotnull(driverId#860)], Format: Parquet, Location:
InMemoryFileIndex[abfss://sandbox@dp203teststorage.dfs.core.windows.net/synapse/workspaces/dp203t..., PartitionFilters: [],
PushedFilters: [IsNotNull(driverId)], ReadSchema: struct
   +- BroadcastExchange HashedRelationBroadcastMode(List(input[0, string, false]),false), [id=#857]
      +- *(1) Filter isnotnull(driverId#873)
         +- *(1) ColumnarToRow
            +- FileScan Hyperspace(Type: CI, Name: DriverIndex, LogVersion: 25) [driverId#873,name#874] Batched: true,
DataFilters: [isnotnull(driverId#873)], Format: Parquet, Location:
InMemoryFileIndex[abfss://sandbox@dp203teststorage.dfs.core.windows.net/synapse/workspaces/dp203t..., PartitionFilters: [],
PushedFilters: [IsNotNull(driverId)], ReadSchema: struct

=============================================================
Plan without indexes:
=============================================================
Project [tripId#859, name#874]
+- BroadcastHashJoin [driverId#860], [driverId#873], Inner, BuildRight, false
   :- Filter isnotnull(driverId#860)
   :  +- ColumnarToRow
   :     +- FileScan parquet [tripId#859,driverId#860] Batched: true, DataFilters: [isnotnull(driverId#860)], Format:
Parquet, Location: InMemoryFileIndex[abfss://sandbox@dp203teststorage.dfs.core.windows.net/hyperspace/trips],
PartitionFilters: [], PushedFilters: [IsNotNull(driverId)], ReadSchema: struct
   +- BroadcastExchange HashedRelationBroadcastMode(List(input[0, string, false]),false), [id=#810]
      +- *(1) Filter isnotnull(driverId#873)
         +- *(1) ColumnarToRow
            +- FileScan parquet [driverId#873,name#874] Batched: true, DataFilters: [isnotnull(driverId#873)], Format:
Parquet, Location: InMemoryFileIndex[abfss://sandbox@dp203teststorage.dfs.core.windows.net/hyperspace/driver],
PartitionFilters: [], PushedFilters: [IsNotNull(driverId)], ReadSchema: struct
```



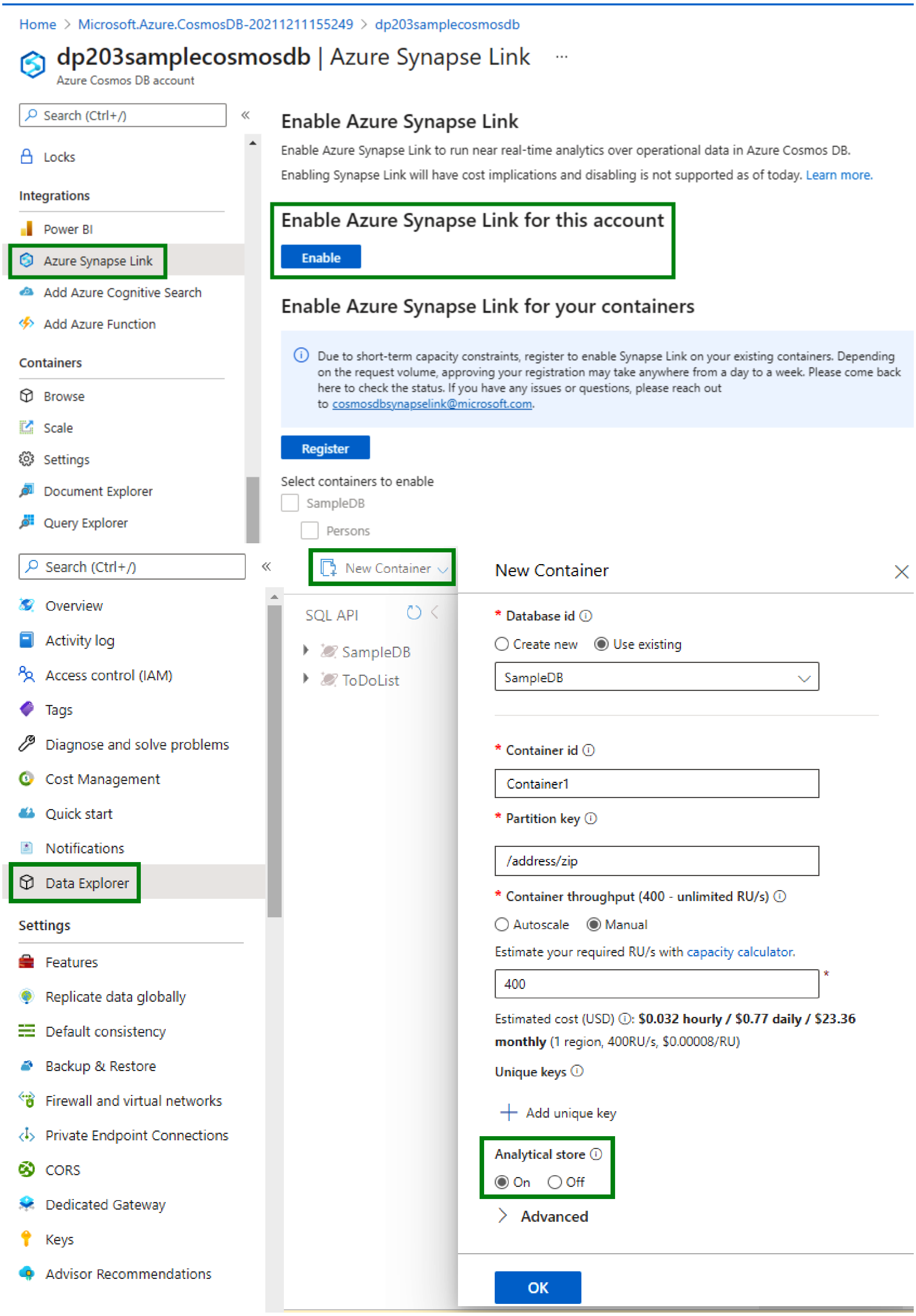
```
============================================================
Plan with indexes:
============================================================
BroadcastNestedLoopJoin BuildRight, Inner
:- *(1) Project [TripID#1726]
:  +- *(1) Filter (isnotnull(DriverID#1727) && (DriverID#1727 = 212))
:     +- *(1) FileScan Hyperspace(Type: CI, Name: TripIndex, LogVersion: 13) [DriverID#1727,TripID#1726]
Batched: true, Format: Parquet, Location:
InMemoryFileIndex[abfss://users@iacstoreacct.dfs.core.windows.net/synapse/workspaces/iacsynapsews...,
PartitionFilters: [], PushedFilters: [IsNotNull(DriverID), EqualTo(DriverID,212)], ReadSchema: struct
+- BroadcastExchange IdentityBroadcastMode, [id=#617]
   +- *(2) FileScan parquet [Name#1741] Batched: true, Format: Parquet, Location:
InMemoryFileIndex[abfss://users@iacstoreacct.dfs.core.windows.net/raw/driver/in/2021114],
PartitionFilters: [], PushedFilters: [], ReadSchema: struct

============================================================
Plan without indexes:
============================================================
BroadcastNestedLoopJoin BuildLeft, Inner
:- BroadcastExchange IdentityBroadcastMode, [id=#600]
:  +- *(1) Project [TripID#1726]
:     +- *(1) Filter (isnotnull(DriverID#1727) && (DriverID#1727 = 212))
:        +- *(1) FileScan parquet [TripID#1726,DriverID#1727] Batched: true, Format: Parquet, Location:
InMemoryFileIndex[abfss://users@iacstoreacct.dfs.core.windows.net/raw/trips/in/2021114],
PartitionFilters: [], PushedFilters: [IsNotNull(DriverID), EqualTo(DriverID,212)], ReadSchema: struct
+- *(2) FileScan parquet [Name#1741] Batched: true, Format: Parquet, Location:
InMemoryFileIndex[abfss://users@iacstoreacct.dfs.core.windows.net/raw/driver/in/2021114],
PartitionFilters: [], PushedFilters: [], ReadSchema: struct

============================================================
```

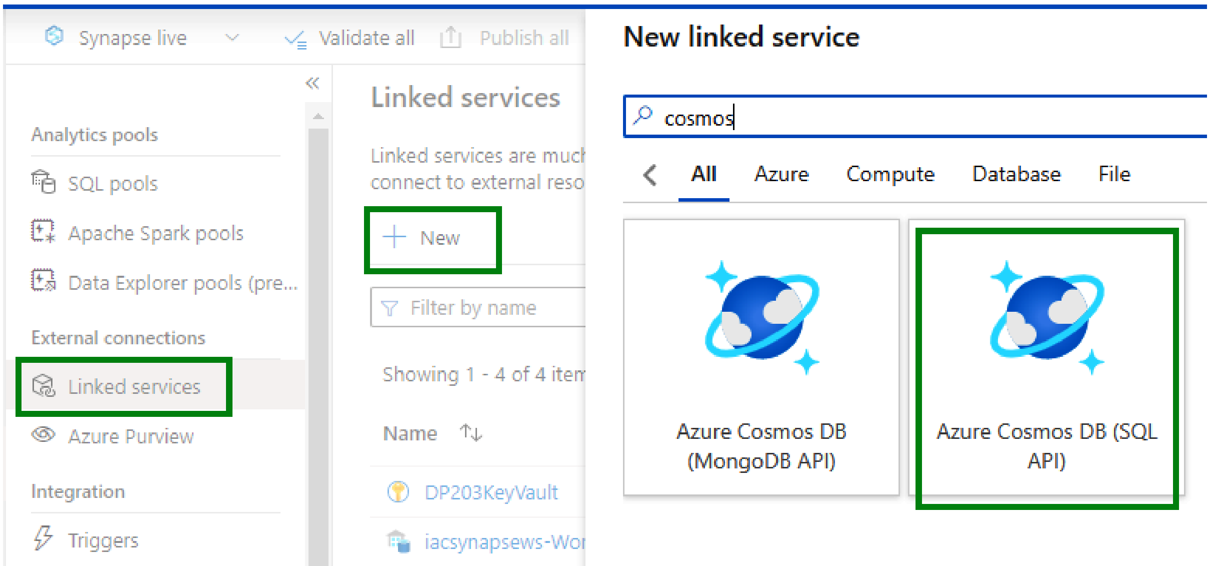


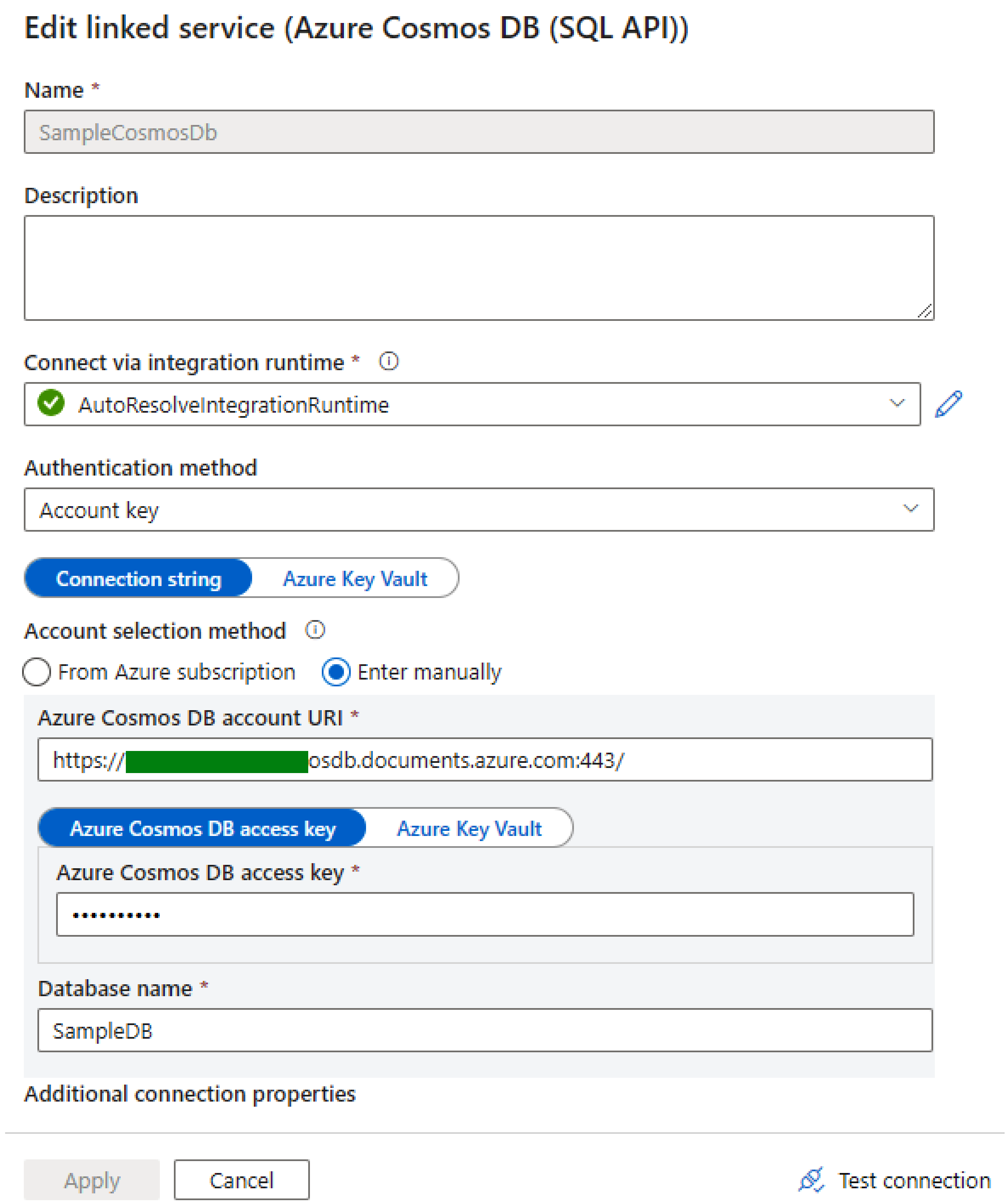


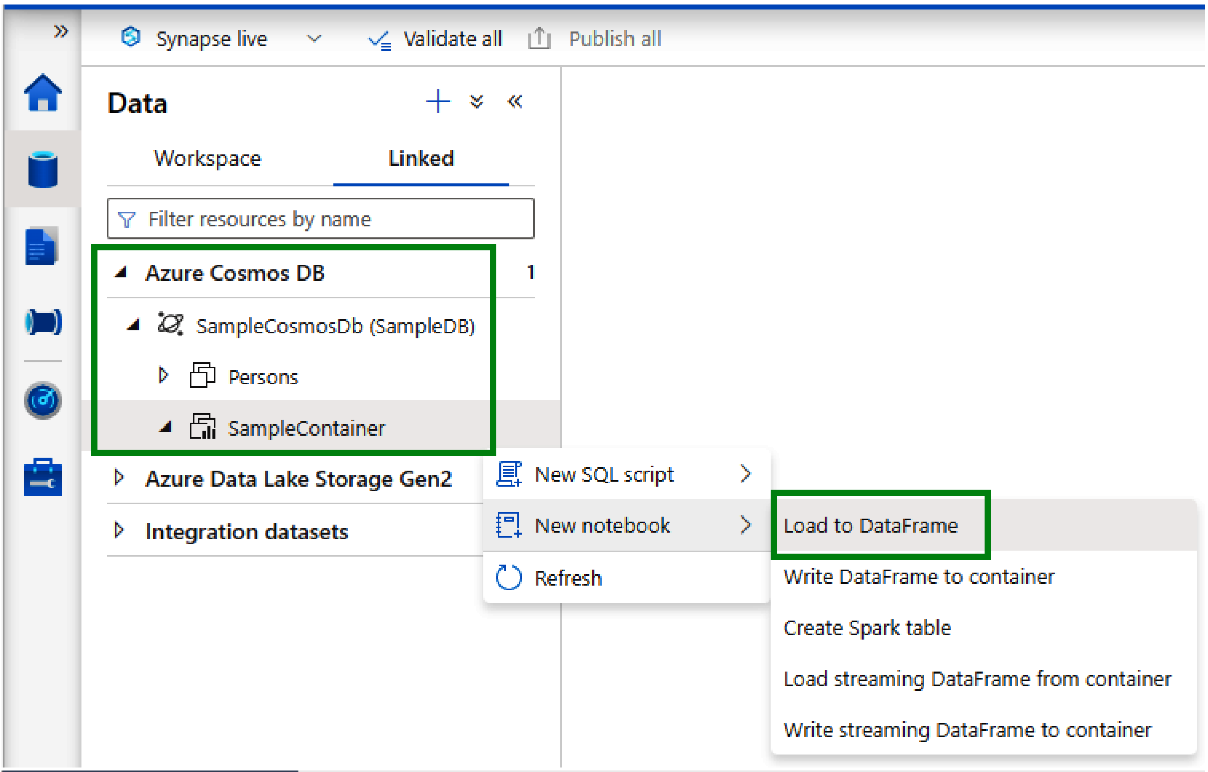


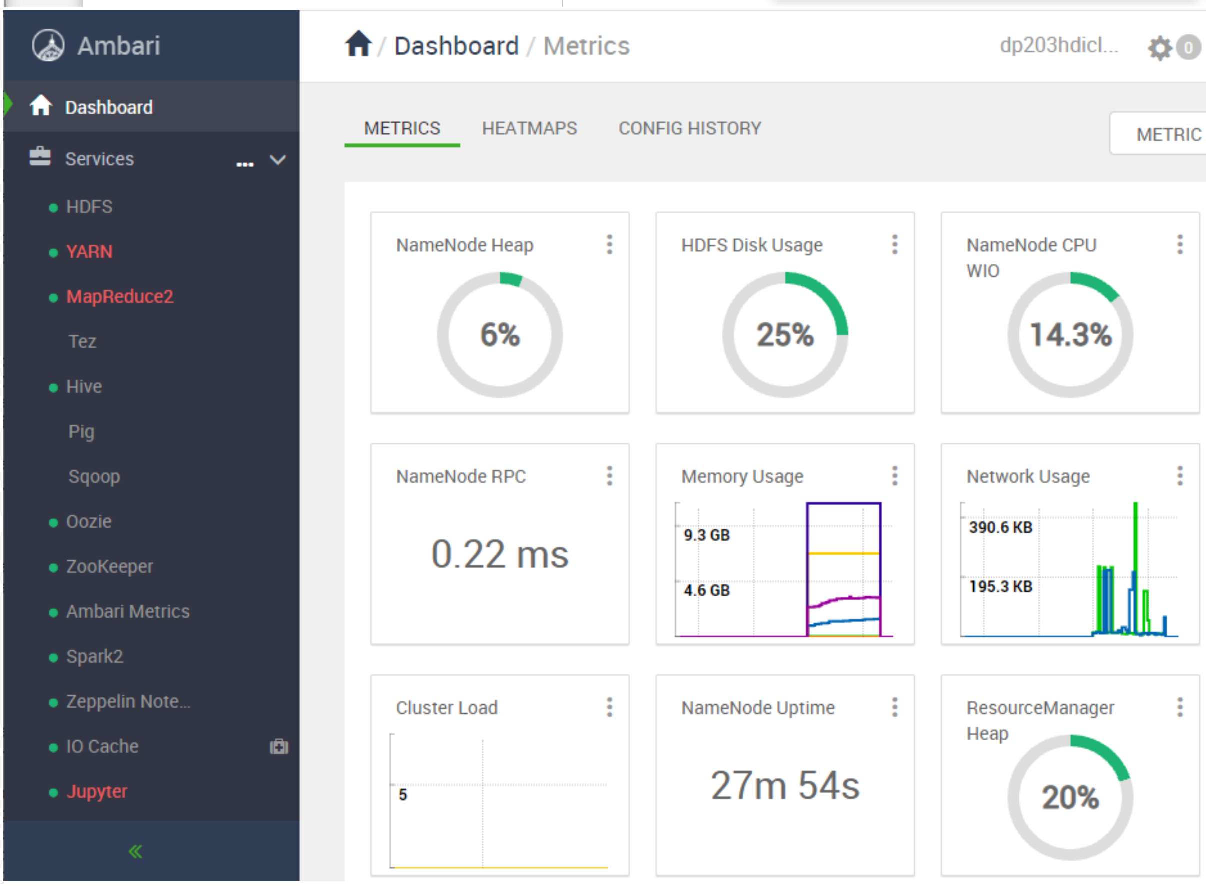
··· > Select API option >
Create Azure Cosmos DB Account - Core ...
Basics
Global Distribution
Networking
Backup Policy
Azure Cosmos DB is a fully managed NoSQL database service for building scalable, high performance applications. Try it for free, for 30 days with unlimited renewals. Go to production starting at $24/month per database, multiple containers included. Learn more
Project Details
Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.
Subscription *
Azure subscription 1
Resource Group *
Create new
Instance Details
Account Name *
Enter account name
Location *
Capacity mode
Provisioned throughput
Serverless
Learn more about capacity mode
With Azure Cosmos DB free tier, you will get the first 1000 RU/s and 25 GB of storage for free in an account. You can enable free tier on up to one account per subscription. Estimated $64/month discount per account.
Apply Free Tier Discount
Apply
Do Not Apply
Review + create
Previous
Next: Global Distribution

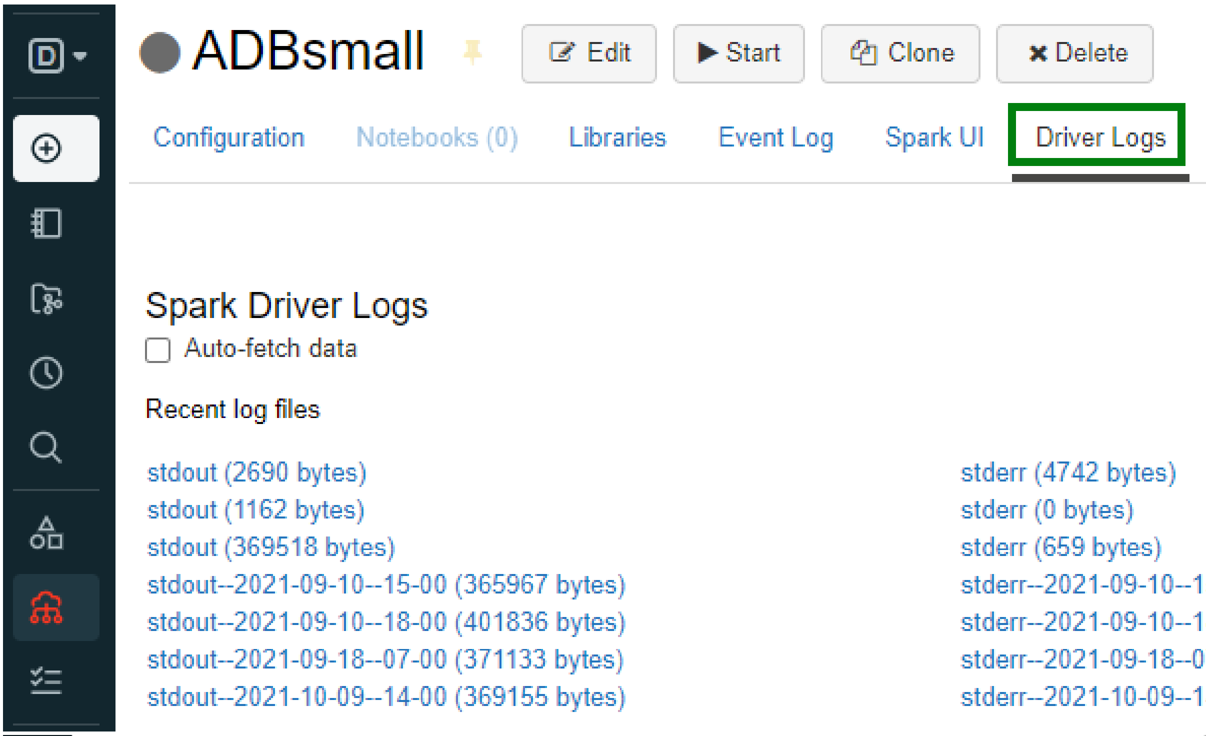

Home > Microsoft.Azure.CosmosDB-20211211155249 > dp203samplecosmosdb
dp203samplecosmosdb | Azure Synapse Link
Azure Cosmos DB account
Search (Ctrl+/)
Locks
Integrations
Power BI
Azure Synapse Link
Add Azure Cognitive Search
Add Azure Function
Containers
Browse
Scale
Settings
Document Explorer
Query Explorer
Enable Azure Synapse Link
Enable Azure Synapse Link to run near real-time analytics over operational data in Azure Cosmos DB.
Enabling Synapse Link will have cost implications and disabling is not supported as of today. Learn more.
Enable Azure Synapse Link for this account
Enable
Enable Azure Synapse Link for your containers
Due to short-term capacity constraints, register to enable Synapse Link on your existing containers. Depending on the request volume, approving your registration may take anywhere from a day to a week. Please come back here to check the status. If you have any issues or questions, please reach out to cosmosdbsynapselink@microsoft.com.
Register
Select containers to enable
SampleDB
Persons
Search (Ctrl+/)
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Cost Management
Quick start
Notifications
Data Explorer
Settings
Features
Replicate data globally
Default consistency
Backup & Restore
Firewall and virtual networks
Private Endpoint Connections
CORS
Dedicated Gateway
Keys
Advisor Recommendations
New Container
SQL API
SampleDB
ToDoList
New Container
* Database id
Create new
Use existing
SampleDB
* Container id
Container1
* Partition key
/address/zip
* Container throughput (400 - unlimited RU/s)
Autoscale
Manual
Estimate your required RU/s with capacity calculator.
400
Estimated cost (USD): $0.032 hourly / $0.77 daily / $23.36 monthly (1 region, 400RU/s, $0.00008/RU)
Unique keys
Add unique key
Analytical store
On
Off
Advanced
OK

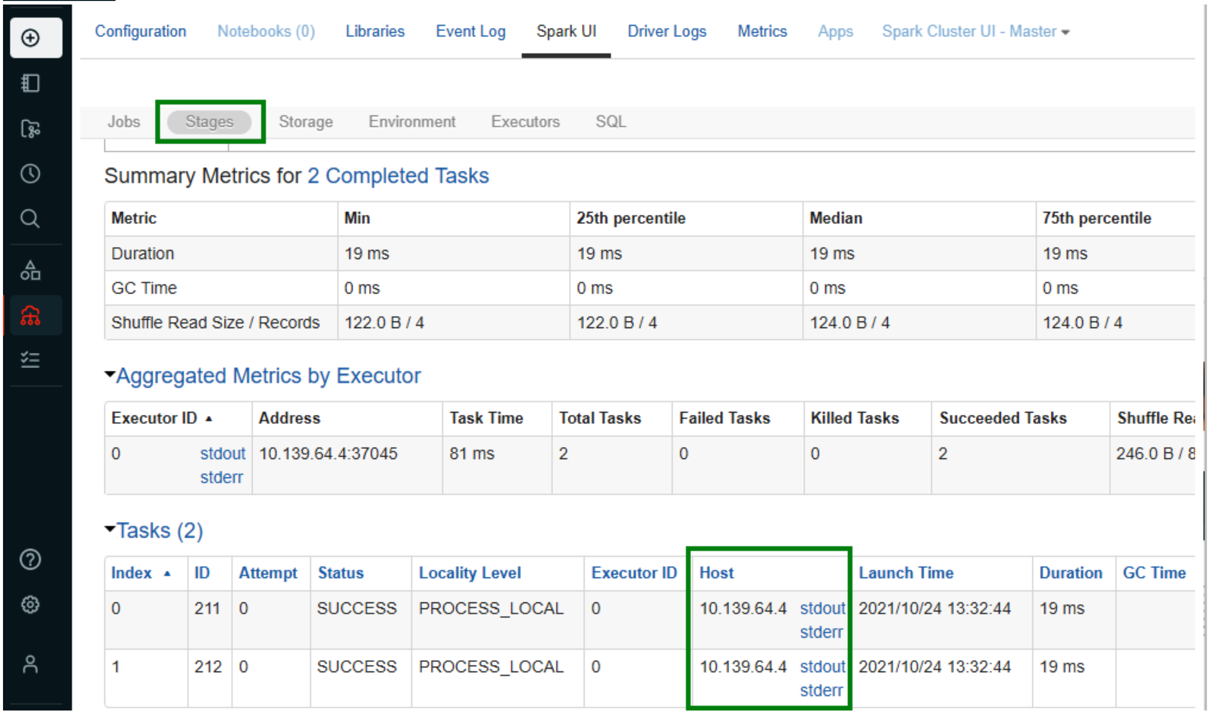

Synapse live
Validate all
Publish all
Analytics pools
SQL pools
Apache Spark pools
Data Explorer pools (pre...
External connections
Linked services
Azure Purview
Integration
Triggers
Linked services
Linked services are much
connect to external reso
New
Filter by name
Showing 1 - 4 of 4 item
Name
DP203KeyVault
iacsynapsews-Wor
New linked service
cosmos
All
Azure
Compute
Database
File
Azure Cosmos DB (MongoDB API)
Azure Cosmos DB (SQL API)

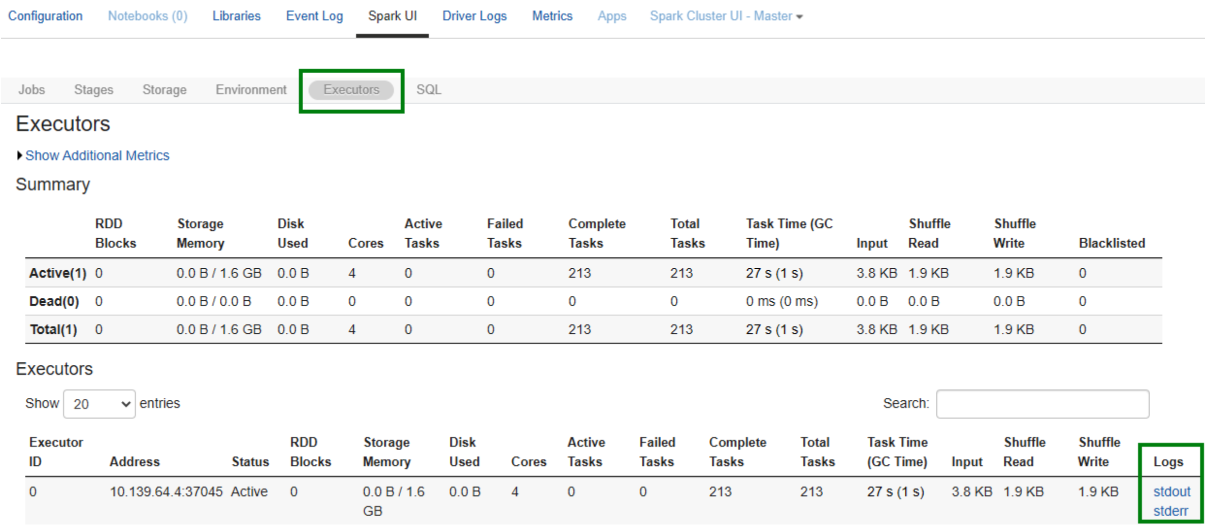

Edit linked service (Azure Cosmos DB (SQL API))
Name *
SampleCosmosDb
Description
Connect via integration runtime *
AutoResolveIntegrationRuntime
Authentication method
Account key
Connection string
Azure Key Vault
Account selection method
From Azure subscription
Enter manually
Azure Cosmos DB account URI *
https://[redacted]osdb.documents.azure.com:443/
Azure Cosmos DB access key
Azure Key Vault
Azure Cosmos DB access key *
••••••••••
Database name *
SampleDB
Additional connection properties
Apply
Cancel
Test connection

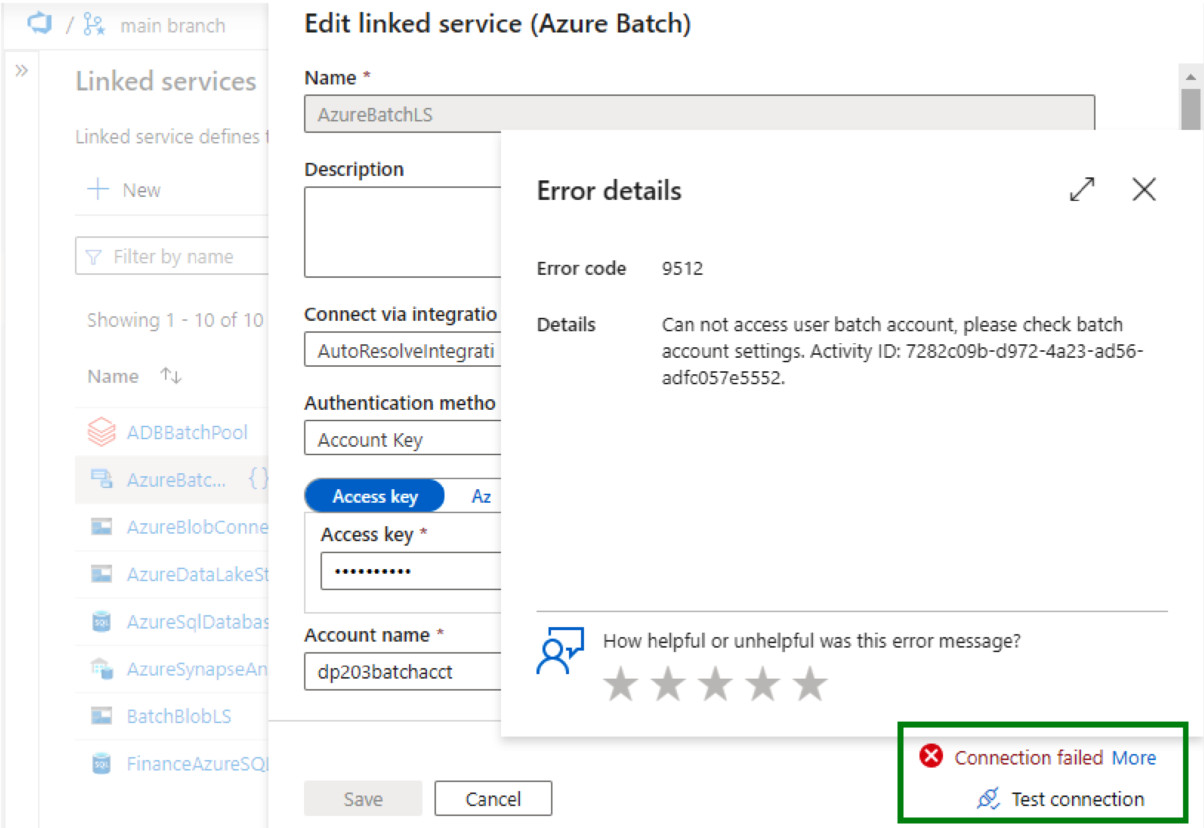

Synapse live
Validate all
Publish all
Data
Workspace
Linked
Filter resources by name
Azure Cosmos DB
1
SampleCosmosDb (SampleDB)
Persons
SampleContainer
Azure Data Lake Storage Gen2
Integration datasets
New SQL script
New notebook
Refresh
Load to DataFrame
Write DataFrame to container
Create Spark table
Load streaming DataFrame from container
Write streaming DataFrame to container

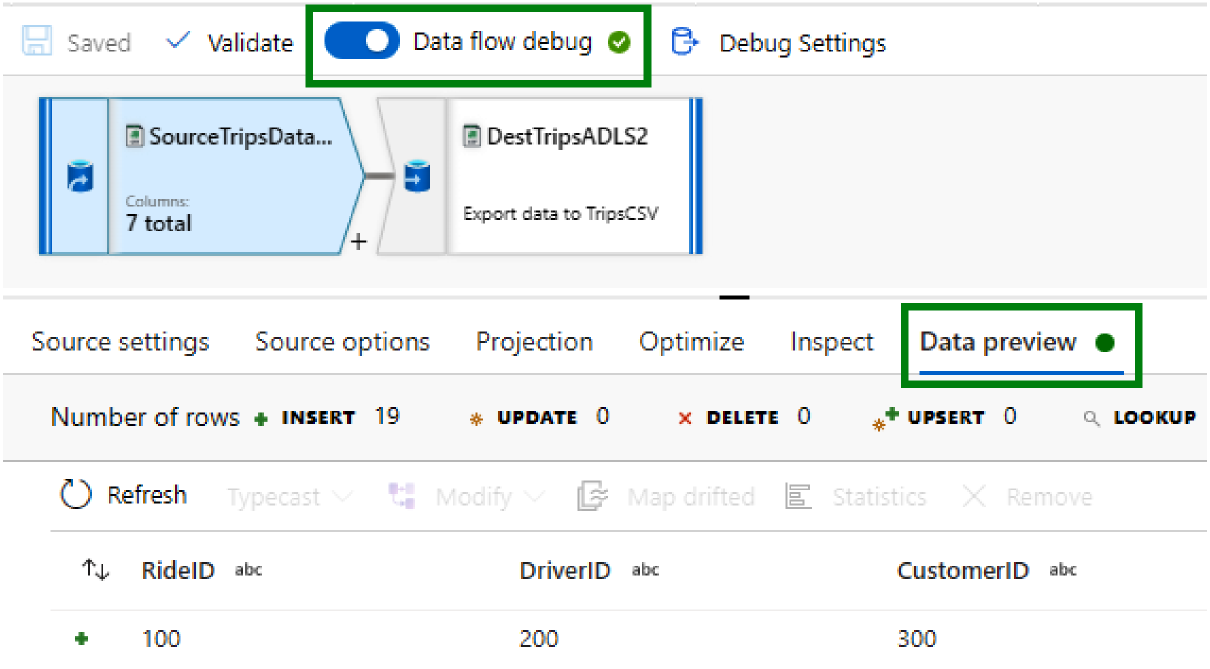

Ambari
Dashboard / Metrics
dp203hdicl...
Dashboard
Services
HDFS
YARN
MapReduce2
Tez
Hive
Pig
Sqoop
Oozie
ZooKeeper
Ambari Metrics
Spark2
Zeppelin Note...
IO Cache
Jupyter
METRICS
HEATMAPS
CONFIG HISTORY
METRIC
NameNode Heap
6%
HDFS Disk Usage
25%
NameNode CPU WIO
14.3%
NameNode RPC
0.22 ms
Memory Usage
9.3 GB
4.6 GB
Network Usage
390.6 KB
195.3 KB
Cluster Load
5
NameNode Uptime
27m 54s
ResourceManager Heap
20%

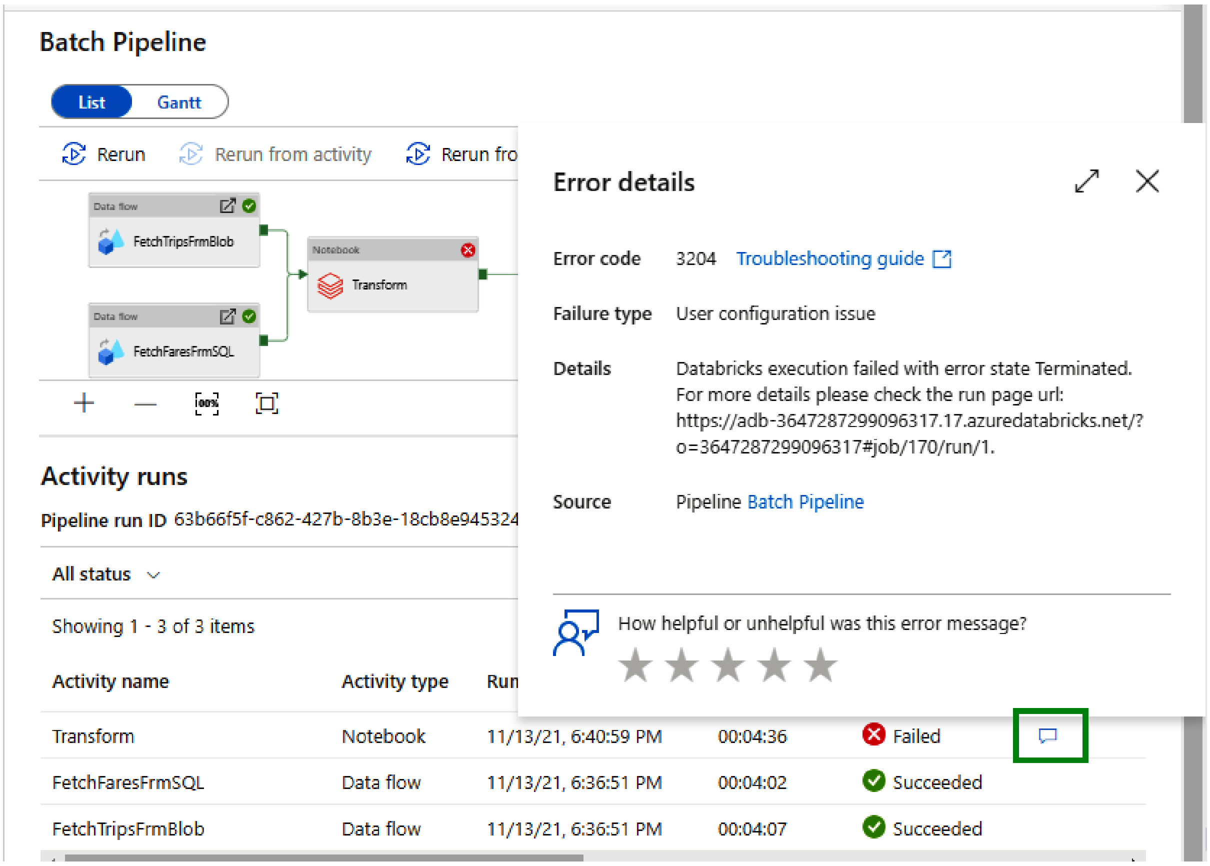

ADBsmall
Edit
Start
Clone
Delete
Configuration
Notebooks (0)
Libraries
Event Log
Spark UI
Driver Logs
Spark Driver Logs
Auto-fetch data
Recent log files
stdout (2690 bytes)
stdout (1162 bytes)
stdout (369518 bytes)
stdout--2021-09-10--15-00 (365967 bytes)
stdout--2021-09-10--18-00 (401836 bytes)
stdout--2021-09-18--07-00 (371133 bytes)
stdout--2021-10-09--14-00 (369155 bytes)
stderr (4742 bytes)
stderr (0 bytes)
stderr (659 bytes)
stderr--2021-09-10--1
stderr--2021-09-10--1
stderr--2021-09-18--0
stderr--2021-10-09--1


Configuration
Notebooks (0)
Libraries
Event Log
Spark UI
Driver Logs
Metrics
Apps
Spark Cluster UI - Master
Jobs
Stages
Storage
Environment
Executors
SQL
Summary Metrics for 2 Completed Tasks
Metric | Min | 25th percentile | Median | 75th percentile
Duration | 19 ms | 19 ms | 19 ms | 19 ms
GC Time | 0 ms | 0 ms | 0 ms | 0 ms
Shuffle Read Size / Records | 122.0 B / 4 | 122.0 B / 4 | 124.0 B / 4 | 124.0 B / 4
Aggregated Metrics by Executor
Executor ID | Address | Task Time | Total Tasks | Failed Tasks | Killed Tasks | Succeeded Tasks | Shuffle Re
0 stdout stderr | 10.139.64.4:37045 | 81 ms | 2 | 0 | 0 | 2 | 246.0 B / 8
Tasks (2)
Index | ID | Attempt | Status | Locality Level | Executor ID | Host | Launch Time | Duration | GC Time
0 | 211 | 0 | SUCCESS | PROCESS_LOCAL | 0 | 10.139.64.4 stdout stderr | 2021/10/24 13:32:44 | 19 ms
1 | 212 | 0 | SUCCESS | PROCESS_LOCAL | 0 | 10.139.64.4 stdout stderr | 2021/10/24 13:32:44 | 19 ms


Configuration
Notebooks (0)
Libraries
Event Log
Spark UI
Driver Logs
Metrics
Apps
Spark Cluster UI - Master
Jobs
Stages
Storage
Environment
Executors
SQL
Executors
Show Additional Metrics
Summary
RDD Blocks | Storage Memory | Disk Used | Cores | Active Tasks | Failed Tasks | Complete Tasks | Total Tasks | Task Time (GC Time) | Input | Shuffle Read | Shuffle Write | Blacklisted
Active(1) 0 0.0 B / 1.6 GB 0.0 B 4 0 0 213 213 27 s (1 s) 3.8 KB 1.9 KB 1.9 KB 0
Dead(0) 0 0.0 B / 0.0 B 0.0 B 0 0 0 0 0 0 ms (0 ms) 0.0 B 0.0 B 0.0 B 0
Total(1) 0 0.0 B / 1.6 GB 0.0 B 4 0 0 213 213 27 s (1 s) 3.8 KB 1.9 KB 1.9 KB 0
Executors
Show 20 entries
Search:
Executor ID | Address | Status | RDD Blocks | Storage Memory | Disk Used | Cores | Active Tasks | Failed Tasks | Complete Tasks | Total Tasks | Task Time (GC Time) | Input | Shuffle Read | Shuffle Write | Logs
0 10.139.64.4:37045 Active 0 0.0 B / 1.6 GB 0.0 B 4 0 0 213 213 27 s (1 s) 3.8 KB 1.9 KB 1.9 KB stdout stderr


main branch
Linked services
Linked service defines t
New
Filter by name
Showing 1 - 10 of 10
Name
ADBBatchPool
AzureBatc...
AzureBlobConne
AzureDataLakeSt
AzureSqlDatabas
AzureSynapseAn
BatchBlobLS
FinanceAzureSQl
Edit linked service (Azure Batch)
Name *
AzureBatchLS
Description
Connect via integratio
AutoResolveIntegrati
Authentication metho
Account Key
Access key
Az
Access key *
••••••••••
Account name *
dp203batchacct
Save
Cancel
Error details
Error code 9512
Details Can not access user batch account, please check batch account settings. Activity ID: 7282c09b-d972-4a23-ad56-adfc057e5552.
How helpful or unhelpful was this error message?
Connection failed More
Test connection


Saved
Validate
Data flow debug
Debug Settings
SourceTripsData...
Columns:
7 total
DestTripsADLS2
Export data to TripsCSV
Source settings
Source options
Projection
Optimize
Inspect
Data preview
Number of rows
INSERT 19
UPDATE 0
DELETE 0
UPSERT 0
LOOKUP
Refresh
Typecast
Modify
Map drifted
Statistics
Remove
RideID abc
DriverID abc
CustomerID abc
100
200
300


Batch Pipeline
List
Gantt
Rerun
Rerun from activity
Rerun fro
Data flow
FetchTripsFrmBlob
Notebook
Transform
Data flow
FetchFaresFrmSQL
Activity runs
Pipeline run ID 63b66f5f-c862-427b-8b3e-18cb8e945324
All status
Showing 1 - 3 of 3 items
Activity name
Activity type
Run
Transform
Notebook
11/13/21, 6:40:59 PM
00:04:36
Failed
FetchFaresFrmSQL
Data flow
11/13/21, 6:36:51 PM
00:04:02
Succeeded
FetchTripsFrmBlob
Data flow
11/13/21, 6:36:51 PM
00:04:07
Succeeded
Error details
Error code
3204
Troubleshooting guide
Failure type
User configuration issue
Details
Databricks execution failed with error state Terminated.
For more details please check the run page url:
https://adb-3647287299096317.17.azuredatabricks.net/?
o=3647287299096317#job/170/run/1.
Source
Pipeline Batch Pipeline
How helpful or unhelpful was this error message?

Chapter 15: Sample Questions with Solutions

No Images